

Tesis Doctoral

Probing the Dark Universe

Machine Learning and Statistical Approaches to
Gamma-Ray Dark Matter Searches



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preparada por

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Bryan Zaldívar Montero, Científico Titular del Consejo Superior de Investigaciones Científicas, **certifica:**

que la presente memoria **Probing the Dark Universe: Machine Learning and Statistical Approaches to Gamma-Ray Dark Matter Searches** ha sido realizada bajo su dirección en el Instituto de Física Corpuscular, centro mixto de la Universidad de Valencia y del CSIC, por **Aurelio Amerio**, y constituye su Tesis para optar al grado de Doctor en Física.

Y para que así conste, en cumplimiento de la legislación vigente, presenta en el Departamento de Física Teórica de la Universidad de Valencia la referida Tesis Doctoral, y firma el presente certificado.

Valencia, a 16 de Julio de 2026,

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ABSTRACT

The evidence for dark matter comes from many independent probes at different scales. Galactic rotation curves, the motions of galaxies within clusters, gravitational lensing, and the anisotropies of the cosmic microwave background all require a form of matter that outweighs ordinary baryons by roughly a factor of five and interacts with light only through gravity. These probes, however, are all gravitational in nature, and they say little about what dark matter actually is. If it is made of weakly interacting massive particles, it could annihilate or decay into Standard Model states, and gamma rays are especially attractive messengers: undeflected by magnetic fields, they carry the spectral signature of the interaction that produced them. For more than fifteen years the Large Area Telescope on board the Fermi Gamma-ray Space Telescope (Fermi-LAT) has surveyed the GeV sky where these signatures are expected, and its measurements are at the core of this thesis.

In this thesis, we argue that progress in indirect dark matter searches now depends on statistical and machine-learning methods capable of recovering signals in noise-dominated environments. The work follows an arc from the Galactic Center outward to the cosmic web, tracking the expected signal into regimes of progressively lower signal-to-noise ratio. To this end, we draw on traditional statistical methods, generative modeling, implicit likelihood methods, and deep learning, applied to Fermi-LAT data and to forecasts for future instruments.

The Galactic Center hosts the Galactic Center Excess (GCE), a surplus of GeV photons that may signal dark matter annihilation or an unresolved population of millisecond pulsars (MSPs). To test the pulsar hypothesis we measured the gamma-ray luminosity function of MSPs in Milky Way globular clusters, finding a mean luminosity $\langle L_\gamma \rangle \sim (1-8) \times 10^{33}$ erg/s between 0.1 and 100 GeV and a broad log-normal spread, $\sigma_L \sim 1.4-2.8$. Had the GCE been produced by pulsars with these properties, Fermi-LAT would already have resolved roughly 17 to 37 of them individually, whereas only three candidates are known — a tension that reignites the debate over whether the excess comes from pulsars or from dark matter.

We then turned to fainter targets: dark matter subhalos potentially hidden among the Fermi-LAT sources that lack an astrophysical counterpart. We built what is, to the best of our knowledge, the first generative model of these unassociated sources at latitudes above ten degrees: a statistical mixture of known astrophysical classes and a possible subhalo component, trained directly on this population and corrected for potential prior and covariate shifts. We found no significant subhalo contribution and set 95% confidence upper limits on the annihilation cross section in the $b\bar{b}$ channel for dark matter masses between 10 GeV and 1 TeV.

Below the Fermi-LAT detection threshold, individual sources blur together, and the object of study becomes the gamma-ray source population as a whole — astrophysical and potential dark matter sources alike. We trained a convolutional neural network on about 10^6 synthetic Fermi-LAT sky maps to reconstruct the source-count distribution dN/dS of extragalactic sources from fourteen years of data between 1 and 10 GeV. The reconstruction reaches fluxes a factor of about 50 below the Fermi-LAT threshold, agrees with existing catalogs in the resolved regime, and extends

as $dN/dS \propto S^{-2}$ down to roughly $5 \times 10^{-12} \text{ cm}^{-2} \text{ s}^{-1}$. Building on this distribution, we developed a probabilistic cataloging framework that compares the observed sky with simulations and assigns each direction a likelihood of hosting a sub-threshold source. This framework identifies roughly 50% more candidate source directions than the 4FGL-DR3 catalog, and delivers a public catalog particularly suited to cross-correlation studies.

At the largest scales, the annihilation signal follows the matter distribution itself, so cross-correlating gamma-ray anisotropies with galaxy positions can separate a cosmological dark matter component from isotropic backgrounds. We forecast the sensitivity of the Cherenkov Telescope Array Observatory (CTAO) to this signal. With a dense low-redshift galaxy catalog such as 2MASS and about 50 hours of observation, the cross-correlation reaches sensitivities to both annihilating and decaying dark matter comparable to those of dwarf-galaxy and galaxy-cluster analyses.

Taken together, these results trace a coherent path through the gamma-ray sky, from the brightest anticipated signal to the faint imprint of the cosmic web. None of these searches returns a confirmed detection, and the particle identity of dark matter remains an open problem. The next steps in indirect detection, we argue, will require better instruments, more data, and sharper statistical tools to read them, all advancing together. This thesis contributes to the third of these fronts, working toward new statistical tools for the field: population models that make sub-threshold sources statistically accessible and inference frameworks robust to the mismatch between simulations and real data, together with probabilistic catalogs and cross-correlation analyses. The hope is that this groundwork will help extend these searches as current gamma-ray surveys continue to grow and next-generation observatories come online.

RESUMEN

La evidencia de la materia oscura proviene de numerosas pruebas independientes a distintas escalas. Las curvas de rotación galácticas, los movimientos de las galaxias dentro de los cúmulos, las lentes gravitatorias y las anisotropías del fondo cósmico de microondas requieren todas ellas una forma de materia que supera a los bariones ordinarios aproximadamente en un factor cinco y que interactúa con la luz solo a través de la gravedad. Sin embargo, todas estas pruebas son de naturaleza gravitatoria y dicen poco sobre qué es realmente la materia oscura. Si está formada por partículas masivas de interacción débil, podría aniquilarse o desintegrarse en estados del Modelo Estándar, y los rayos gamma son mensajeros especialmente atractivos: al no ser desviados por los campos magnéticos, llevan consigo la firma espectral de la interacción que los produjo. Durante más de quince años, el Large Area Telescope a bordo del telescopio espacial de rayos gamma Fermi (Fermi-LAT) ha cartografiado el cielo del GeV, donde se esperan estas firmas, y sus medidas constituyen el núcleo de esta tesis.

En esta tesis defendemos que el progreso en las búsquedas indirectas de materia oscura depende hoy de métodos estadísticos y de aprendizaje automático capaces de recuperar señales en entornos dominados por el ruido. El trabajo sigue un arco que va del Centro Galáctico hacia el exterior, hasta la red cósmica, siguiendo la señal esperada en regímenes de relación señal-ruido progresivamente menor. Para ello recurrimos a métodos estadísticos tradicionales, al modelado generativo, a los métodos de verosimilitud implícita y al aprendizaje profundo, aplicados a los datos de Fermi-LAT y a predicciones para instrumentos futuros.

El Centro Galáctico alberga el Exceso del Centro Galáctico (GCE, por sus siglas en inglés), un excedente de fotones GeV que podría señalar la aniquilación de materia oscura o una población no resuelta de púlsares de milisegundo (MSPs). Para poner a prueba la hipótesis de los púlsares, medimos la función de luminosidad en rayos gamma de los MSPs en los cúmulos globulares de la Vía Láctea, y encontramos una luminosidad media $\langle L_\gamma \rangle \sim (1-8) \times 10^{33}$ erg/s entre 0.1 y 100 GeV y una dispersión log-normal ancha, $\sigma_L \sim 1.4-2.8$. Si el GCE hubiera sido producido por púlsares con estas propiedades, Fermi-LAT ya habría resuelto individualmente entre 17 y 37 de ellos, mientras que solo se conocen tres candidatos — una tensión que reaviva el debate sobre si el exceso procede de púlsares o de materia oscura.

Nos dirigimos después a objetivos más débiles: subhalos de materia oscura potencialmente ocultos entre las fuentes de Fermi-LAT que carecen de contrapartida astrofísica. Construimos el que es, hasta donde sabemos, el primer modelo generativo de estas fuentes no asociadas a latitudes superiores a diez grados: una mezcla estadística de clases astrofísicas conocidas y una posible componente de subhalos, entrenada directamente sobre esta población y corregida por posibles desplazamientos de covariables y de probabilidades a priori. No encontramos ninguna contribución significativa de subhalos y establecimos límites superiores al 95% de confianza sobre la sección eficaz de aniquilación en el canal $b\bar{b}$ para masas de materia oscura entre 10 GeV y 1 TeV.

Por debajo del umbral de detección de Fermi-LAT, las fuentes individuales se emborronan

entre sí, y el objeto de estudio pasa a ser la población de fuentes de rayos gamma en su conjunto — tanto las astrofísicas como las posibles fuentes de materia oscura. Entrenamos una red neuronal convolucional con alrededor de 10^6 mapas sintéticos del cielo de Fermi-LAT para reconstruir la distribución de recuento de fuentes dN/dS de las fuentes extragalácticas a partir de catorce años de datos entre 1 y 10 GeV. La reconstrucción alcanza flujos un factor 50 por debajo del umbral de Fermi-LAT, concuerda con los catálogos existentes en el régimen resuelto y se extiende como $dN/dS \propto S^{-2}$ hasta aproximadamente $5 \times 10^{-12} \text{ cm}^{-2} \text{ s}^{-1}$. Sobre esta distribución desarrollamos un marco de catalogación probabilística que compara el cielo observado con simulaciones y asigna a cada dirección una probabilidad de albergar una fuente sub-umbral. Este marco identifica aproximadamente un 50% más de direcciones candidatas a fuente que el catálogo 4FGL-DR3, y proporciona un catálogo público particularmente adecuado para estudios de correlación cruzada.

A las escalas más grandes, la señal de aniquilación sigue la propia distribución de materia, de modo que correlacionar las anisotropías de rayos gamma con las posiciones de las galaxias puede separar una componente cosmológica de materia oscura de los fondos isótropos. Predecimos la sensibilidad del Cherenkov Telescope Array Observatory (CTAO) a esta señal. Con un catálogo denso de galaxias a bajo corrimiento al rojo como 2MASS y unas 50 horas de observación, la correlación cruzada alcanza sensibilidades a la materia oscura tanto en aniquilación como en desintegración comparables a las de los análisis de galaxias enanas y cúmulos de galaxias.

En conjunto, estos resultados trazan un camino coherente a través del cielo de rayos gamma, desde la señal anticipada más brillante hasta la débil huella de la red cósmica. Ninguna de estas búsquedas devuelve una detección confirmada, y la identidad de la partícula de materia oscura sigue siendo un problema abierto. Los próximos pasos de la detección indirecta, defendemos, requerirán mejores instrumentos, más datos y herramientas estadísticas más afiladas para leerlos, avanzando todos a la vez. Esta tesis contribuye al tercero de estos frentes, trabajando en nuevas herramientas estadísticas para el campo: modelos de poblaciones que hacen estadísticamente accesibles las fuentes sub-umbral y marcos de inferencia robustos frente al desajuste entre simulaciones y datos reales, junto con catálogos probabilísticos y análisis de correlaciones cruzadas. La esperanza es que este trabajo de base ayude a extender estas búsquedas a medida que los sondeos actuales de rayos gamma sigan creciendo y los observatorios de próxima generación entren en funcionamiento.

RESUM

L'evidència de la matèria fosca prové de nombroses proves independents a diferents escales. Les corbes de rotació galàctiques, els moviments de les galàxies dins dels cúmuls, les lents gravitatòries i les anisotropies del fons còsmic de microones requereixen totes una forma de matèria que supera els barions ordinaris aproximadament en un factor cinc i que interactua amb la llum només a través de la gravetat. Aquestes proves, però, són totes de naturalesa gravitatòria, i diuen poc sobre què és realment la matèria fosca. Si està formada per partícules massives d'interacció feble, podria aniquilar-se o desintegrar-se en estats del Model Estàndard, i els rajos gamma són missatgers especialment atractius: com que no són desviats pels camps magnètics, porten la signatura espectral de la interacció que els va produir. Durant més de quinze anys, el Large Area Telescope a bord del telescopi espacial de rajos gamma Fermi (Fermi-LAT) ha cartografiat el cel en la banda dels GeV, on s'esperen aquestes signatures, i les seues mesures constitueixen el nucli d'aquesta tesi.

En aquesta tesi defenem que el progrés en les cerques indirectes de matèria fosca depèn hui de mètodes estadístics i d'aprenentatge automàtic capaços de recuperar senyals en entorns dominats pel soroll. El treball segueix un arc que va del Centre Galàctic cap a fora, fins a la xarxa còsmica, seguint el senyal esperat en règims de relació senyal-soroll progressivament menor. Amb aquest fi, recorrem a mètodes estadístics tradicionals, a la modelització generativa, als mètodes de versemblança implícita i a l'aprenentatge profund, aplicats a les dades de Fermi-LAT i a prediccions per a instruments futurs.

El Centre Galàctic alberga l'Excés del Centre Galàctic (GCE, per les seues sigles en anglés), un excedent de fotons amb energies de l'ordre dels GeV que podria assenyalar l'aniquilació de matèria fosca o una població no resolta de púlsars de mil·lisegon (MSPs). Per a provar la hipòtesi dels púlsars, vam mesurar la funció de lluminositat en rajos gamma dels MSPs als cúmuls globulars de la Via Làctia, i vam trobar una lluminositat mitjana $\langle L_\gamma \rangle \sim (1-8) \times 10^{33}$ erg/s entre 0.1 i 100 GeV i una dispersió log-normal ampla, $\sigma_L \sim 1.4-2.8$. Si el GCE haguera sigut produït per púlsars amb aquestes propietats, Fermi-LAT ja n'hauria resolt individualment entre 17 i 37, mentre que només es coneixen tres candidats — una tensió que reaviva el debat sobre si l'excés prové de púlsars o de matèria fosca.

Després ens vam dirigir a objectius més febles: subhalos de matèria fosca potencialment amagats entre les fonts de Fermi-LAT que no tenen contrapartida astrofísica. Vam construir el que és, fins on sabem, el primer model generatiu d'aquestes fonts no associades a latituds superiors a deu graus: una mescla estadística de classes astrofísiques conegudes i una possible component de subhalos, entrenada directament sobre aquesta població i corregida per possibles desplaçaments de covariables i de probabilitats a priori. No vam trobar cap contribució significativa de subhalos i vam establir límits superiors al 95% de confiança sobre la secció eficaç d'aniquilació en el canal $b\bar{b}$ per a masses de matèria fosca entre 10 GeV i 1 TeV.

Per davall del llindar de detecció de Fermi-LAT, les fonts individuals es difuminen entre si, i l'objecte d'estudi passa a ser la població de fonts de rajos gamma en conjunt — tant les

astrofísiques com les possibles fonts de matèria fosca. Vam entrenar una xarxa neuronal convolucional amb al voltant de 10^6 mapes sintètics del cel de Fermi-LAT per a reconstruir la distribució de recompte de fonts dN/dS de les fonts extragalàctiques a partir de catorze anys de dades entre 1 i 10 GeV. La reconstrucció assoleix fluxos un factor 50 per davall del llindar de Fermi-LAT, concorda amb els catàlegs existents en el règim resolt i s'estén com $dN/dS \propto S^{-2}$ fins a aproximadament $5 \times 10^{-12} \text{ cm}^{-2} \text{ s}^{-1}$. Sobre aquesta distribució, vam desenvolupar un marc de catalogació probabilística que compara el cel observat amb simulacions i assigna a cada direcció una probabilitat d'albergar una font per davall del llindar. Aquest marc identifica aproximadament un 50% més de direccions candidates a font que el catàleg 4FGL-DR3, i proporciona un catàleg públic particularment adequat per a estudis de correlació creuada.

A les escales més grans, el senyal d'aniquilació segueix la mateixa distribució de matèria, de manera que correlacionar les anisotropies de rajos gamma amb les posicions de les galàxies pot separar una component cosmològica de matèria fosca dels fons isòtrops. Vam predir la sensibilitat del Cherenkov Telescope Array Observatory (CTAO) a aquest senyal. Amb un catàleg dens de galàxies a baix desplaçament cap al roig com 2MASS i unes 50 hores d'observació, la correlació creuada assoleix sensibilitats a la matèria fosca tant en aniquilació com en desintegració comparables a les de les anàlisis de galàxies nanes i de cúmuls de galàxies.

En conjunt, aquests resultats tracen un camí coherent a través del cel de rajos gamma, des del senyal anticipat més brillant fins a la lleu empremta de la xarxa còsmica. Cap d'aquestes cerques retorna una detecció confirmada, i la identitat de la partícula de matèria fosca continua sent un problema obert. Els pròxims passos de la detecció indirecta, defenem, requeriran millors instruments, més dades i ferramentes estadístiques més esmolades per a llegir-les, avançant tots alhora. Aquesta tesi contribueix al tercer d'aquests fronts, treballant en noves ferramentes estadístiques per al camp: models de poblacions que fan estadísticament accessibles les fonts per davall del llindar i marcs d'inferència robustos davant del desajust entre simulacions i dades reals, juntament amb catàlegs probabilístics i anàlisis de correlació creuada. L'esperança és que aquest treball de base ajude a estendre aquestes cerques a mesura que els sondejos actuals de rajos gamma continuen creixent i els observatoris de pròxima generació entren en funcionament.

LIST OF PUBLICATIONS

This doctoral thesis is based on the following publications, presented here in the order in which they appear in the thesis:

1. *Millisecond pulsars in globular clusters and implications for the galactic center gamma-ray excess* [1]
A. Amerio, D. Hooper, and T. Linden.
JCAP **10** (2025) 106, [2412.05220]. (Chapter 4)
2. *Search for dark matter subhalos among unassociated Fermi-LAT sources in presence of dataset shift* [2]
A. Amerio, D. Malyshev, B. Zaldivar, V. Gammaldi, and M. Á. Sánchez-Conde.
arXiv:2503.14584 [astro-ph.HE]. (Chapter 5)
3. *Extracting the gamma-ray source-count distribution below the Fermi-LAT detection limit with deep learning* [3]
A. Amerio, A. Cuoco, and N. Fornengo.
JCAP **09** (2023) 029, [2302.01947]. (Chapter 6)
4. *Deepening gamma-ray point-source catalogues with sub-threshold information* [4]
A. Amerio, F. Calore, P. D. Serpico, and B. Zaldivar.
JCAP **03** (2024) 055, [2306.16483]. (Chapter 7)
5. *Dark matter and galaxy cross-correlations with the Cherenkov Telescope Array Observatory* [5]
E. Pinetti, V. Vodeb, **A. Amerio**, A. Cuoco, S. Camera, N. Fornengo, and G. Zaharijas.
Phys. Rev. D **112** (2025) 123010, [2505.20383]. (Chapter 8)

The following work was also completed during the doctorate but is *not* included in this thesis:

6. *GenSBI: Generative Methods for Simulation-Based Inference in JAX*
A. Amerio.
arXiv:2605.27499 [cs.LG]; software (v0.3.4): Zenodo, 10.5281/zenodo.20410084 (2026).

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Part I

Introduction

THE DARK MATTER PROBLEM

The nature of dark matter is an open question at the heart of fundamental physics. Over the past century, independent observations have established that the visible matter we can detect — stars, gas, dust — accounts for only a small fraction of the total matter content of the Universe. Flat rotation curves in spiral galaxies [6, 7], the dynamics and gravitational lensing of galaxy clusters [8], the detailed pattern of temperature anisotropies in the cosmic microwave background (CMB) [9], and the baryon acoustic oscillations imprinted on the large-scale distribution of galaxies [10] all point consistently toward a dominant, non-luminous matter component. These observations, spanning scales from individual galaxies to the observable Universe as a whole, converge on the Λ CDM model, in which approximately 85% of the matter density is composed of cold, non-baryonic dark matter [9]. The non-baryonic character of this component is independently confirmed by Big Bang nucleosynthesis (BBN), which constrains the baryon density to $\Omega_b h^2 \approx 0.022$ — far below the total matter density $\Omega_m h^2 \approx 0.14$ measured by the CMB [11].

If dark matter is not made of ordinary matter, then what is it? Proposed candidates span nearly ninety orders of magnitude in mass, from ultralight axions to primordial black holes, and none is singled out by the data alone. Among these, the weakly interacting massive particle (WIMP) has long occupied a privileged position because of the so-called “WIMP miracle”: a stable particle with mass and coupling strength at the electroweak scale, produced thermally in the early Universe, naturally yields a relic abundance consistent with the observed dark matter density [11, 12]. The same annihilation process that sets the relic abundance also predicts that WIMPs should annihilate today in regions of high dark matter density, producing Standard Model particles — gamma rays, neutrinos, positrons, and antiprotons — that can in principle be detected by astrophysical observatories [13]. This annihilation signature is what makes the WIMP a natural target for the gamma-ray searches developed in this thesis, and it is why we adopt it as our working hypothesis.

This chapter lays the groundwork for the searches presented in the rest of the thesis. We begin by reviewing the multi-scale evidence for dark matter, from galactic rotation curves to the CMB (Section 1.1). We then introduce the WIMP paradigm, deriving the thermal freeze-out mechanism and the resulting relic abundance calculation that defines the target cross-section for

experimental searches (Section 1.2). Section 1.3 surveys the three complementary experimental strategies — direct, collider, and indirect detection — and argues that indirect detection via gamma-rays provides the most direct test of the WIMP hypothesis. Finally, Section 1.4 develops the indirect detection formalism in detail — annihilation channels, spectral signatures, density profiles, and the J -factor/ D -factor framework — that connects particle physics parameters to observable gamma-ray fluxes. The observational targets and the current status of indirect searches, reviewed at the end of this chapter, provide the direct motivation for the advanced statistical and machine learning approaches that constitute the core of this thesis.

1.1 Evidence for Dark Matter

Evidence of the existence of dark matter spans a wide range of astrophysical scales. We follow the hierarchical classification of Cirelli et al. [11], organizing the evidence from galactic (“mini”) to cluster (“midi”) to cosmological (“maxi”) scales. The internal consistency of these measurements — obtained with fundamentally different techniques, probing different physical environments — is one of the strongest arguments in favor of a dark matter interpretation in terms of new physics.

1.1.1 Galactic Scale

Among the earliest and most compelling evidence for dark matter is the kinematics of galaxies. In a spiral galaxy, the circular velocity $v_c(r)$ of stars and gas at distance r from the center is determined by the enclosed mass through $v_c(r) = \sqrt{GM(r)/r}$. If the visible mass — concentrated in a central bulge — was the main source of the gravitational potential, the rotation curve should decline as $v_c \propto r^{-1/2}$ beyond the luminous extent of the bulge, following the standard Keplerian falloff.

Observations tell a different story. The first hints of anomalous galactic dynamics appeared with Babcock’s 1939 study of M31, though the measurements lacked the radial extent and precision to be conclusive [14]. The modern picture took shape in 1970, when Rubin and Ford [6] obtained the first high-precision optical rotation curve of M31, tracing approximately 70 H II regions out to ~ 22 kpc and finding no evidence of the expected Keplerian decline. In the same year, Freeman identified similar discrepancies in the disk dynamics of several spiral galaxies [15]. Over the following decades, radio observations of the 21-cm hydrogen line extended rotation curves well beyond the optical disk, establishing that $v_c(r)$ remains approximately constant out to the largest measurable radii in hundreds of systems [16, 17]. To maintain a constant circular velocity, the enclosed mass must grow linearly with radius, $M(r) \propto r$, which suggests an extended, roughly spherical halo of invisible matter with density $\rho(r) \propto r^{-2}$ at large radii. The comprehensive analysis of Persic, Salucci, and Stel [7] synthesized rotation curves across a wide range of galaxy luminosities into a “Universal Rotation Curve”, demonstrating that the dark-to-luminous mass ratio increases systematically with decreasing galaxy luminosity.

Modern data have only strengthened this picture. The Spitzer Photometry and Accurate Rotation Curves (SPARC) compilation [18] provides high-quality mass models for 175 disk galaxies spanning a broad range of surface brightnesses and morphologies. These data confirm the ubiquity of flat rotation curves and reveal tight empirical scaling relations — such as the baryonic Tully-Fisher relation linking asymptotic rotation velocity to total baryonic mass [19, 20] — that any successful dark matter model must reproduce. Figure 1.1 illustrates this evidence: the left panel shows the original optical rotation curve of M31 from Rubin and Ford [6], while the right panel presents a modern mass decomposition demonstrating the dominant contribution of the dark matter halo at large radii.

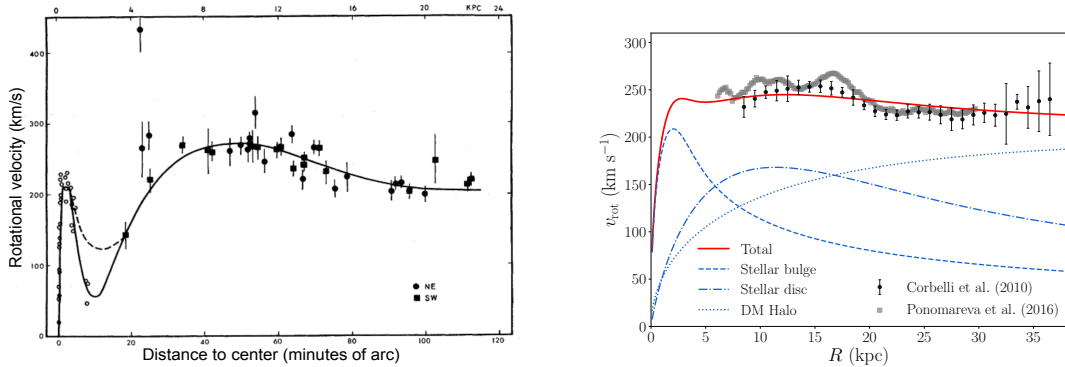


Fig. 1.1. Rotation curves and mass decomposition of M31. *Left:* the original optical rotation curve measured by Rubin and Ford [6], one of the first datasets to show that the observed velocities do not follow the expected Keplerian decline. *Right:* a modern mass decomposition from Afruni et al. [21], showing the contributions of the stellar disk, gas, and dark matter halo to the total rotation curve; the dark matter halo dominates at large radii.

An independent line of evidence comes from the satellite galaxies of the Milky Way. Dwarf spheroidal galaxies (dSphs) — small, gas-poor stellar systems such as Draco, Sculptor, and Segue 1 — orbit within the gravitational potential of our Galaxy and represent some of the smallest and faintest galaxies known. Unlike the disk galaxies discussed above, stars in dSphs move on essentially random orbits, making these systems pressure-supported rather than rotationally supported. The simple relation $v_c(r) = \sqrt{GM(r)/r}$ that governs rotating disks is therefore not directly applicable, and the enclosed mass is instead inferred from the line-of-sight velocity dispersion σ_* of the stellar population. Wolf et al. [22] showed that the mass enclosed within the three-dimensional half-light radius is robustly estimated by $M(< r_{1/2}) \approx 3\sigma_*^2 r_{1/2}/G$, a result that is largely insensitive to the unknown velocity anisotropy. Applying this estimator reveals extreme mass-to-light ratios: from $M/L \sim 10 M_\odot/L_\odot$ in the classical dwarfs, to several hundred in Draco ($M/L \sim 300$), to well over a thousand in ultra-faint systems like Segue 1 [22, 23]. These ratios cannot be explained by baryonic mechanisms alone, establishing dSphs as the most dark-matter-dominated objects known. Their low astrophysical backgrounds also make them prime targets for indirect dark matter searches, a point to which we return in Section 1.4.5.

1.1.2 Galaxy Cluster Scale

Moving to larger scales, the dynamics of galaxy clusters provide another class of evidence for dark matter. The earliest quantitative argument dates back to 1933, when Fritz Zwicky studied the radial velocities of galaxies in the Coma Cluster [24]. Applying the virial theorem — which relates the time-averaged kinetic energy of a gravitationally bound system to its potential energy, $2\langle K \rangle + \langle U \rangle = 0$ — he estimated the cluster’s total mass and found a mass-to-light ratio of ~ 400 , far exceeding what could be accounted for by the luminous galaxies alone. Zwicky concluded that the cluster must contain a large amount of unseen matter, to which he referred as “dunkle materie” — dark matter.

Modern X-ray observations have placed these early dynamical estimates on much firmer ground. Galaxy clusters contain massive reservoirs of ionized gas — the intracluster medium (ICM) — heated to temperatures of 10^7 – 10^8 K as it falls into the cluster potential well. This gas radiates thermally through bremsstrahlung emission, observed in the X-ray band [25]. Under the

assumption of hydrostatic equilibrium, the outward gas pressure gradient balances the inward gravitational pull, so the total gravitating mass can be reconstructed from the observed gas temperature and density profiles. These analyses consistently show that the visible galaxies plus the X-ray-emitting gas account for only $\sim 15\%$ of the total cluster mass [11], with the rest attributable to a dark, non-luminous component.

The Sunyaev-Zel'dovich (SZ) effect [26] offers a complementary probe. When CMB photons traverse a cluster, they undergo inverse-Compton scattering off the hot ICM electrons, gaining energy and producing a characteristic spectral distortion [27]. The amplitude of this distortion is proportional to the integrated electron pressure along the line of sight and, unlike X-ray surface brightness, does not diminish with redshift. Combined with independent constraints on the baryon density from BBN and the CMB (where $\Omega_b \approx 0.05$), SZ measurements of the cluster baryon fraction $f_b \approx \Omega_b/\Omega_m$ yield a total matter density $\Omega_m \approx 0.3$, well above the baryonic contribution [9].

Gravitational lensing provides a direct method for mapping the total mass distribution of clusters, independent of assumptions about dynamical equilibrium. Strong lensing in the dense cluster core produces highly distorted arcs and multiple images of background galaxies, while weak lensing measures the statistical shearing of background galaxy shapes over larger angular scales [27]. Both regimes reconstruct the total mass distribution — visible and dark — from the geometry of light paths alone.

A striking application of gravitational lensing is the Bullet Cluster (1E 0657-558), analyzed by Clowe et al. [8]. This system consists of two galaxy clusters that passed through each other roughly 150 million years ago. X-ray observations reveal that the collisional baryonic gas from the two clusters interacted strongly, producing a bow shock and lagging behind the galaxies. Weak-lensing mass maps, however, show that the bulk of the total mass is spatially offset from the gas, coinciding instead with the collisionless galaxies. This spatial separation is visible in Figure 1.2: the green mass contours reconstructed through weak lensing are displaced from the X-ray-emitting gas, demonstrating that the dominant mass component is collisionless. The observed separation matches the behavior expected of collisionless dark matter and contradicts modified gravity theories such as modified Newtonian dynamics (MOND) [28], in which the lensing signal should trace the baryonic gas. Harvey et al. [29] extended this analysis to 72 colliding clusters, confirming the separation of dark matter from baryonic gas at greater than 7σ significance and setting stringent upper limits on the dark matter self-interaction cross-section.

1.1.3 Cosmological Scale

At the largest scales, precision cosmological observations provide the tightest constraints on dark matter and its non-baryonic nature.

The CMB radiation is the thermal relic of the hot, dense early Universe. Roughly 380,000 years after the Big Bang, the expanding cosmos cooled sufficiently for protons and electrons to combine into neutral hydrogen — an epoch known as *recombination*. Before this transition, baryonic matter existed as a hot plasma tightly coupled to photons through Thomson scattering; after recombination, photons streamed freely, and this “last scattering surface” is what we observe today as the CMB. Its temperature is remarkably uniform at ~ 2.725 K, but carries tiny anisotropies at the level of $\sim 10^{-5}$ that encode detailed information about the matter-energy content of the cosmos.

These anisotropies are conventionally decomposed into angular scales via the temperature power spectrum C_ℓ , which quantifies the variance of temperature fluctuations at each angular multipole ℓ — with larger ℓ corresponding to smaller angular separations on the sky. Peaks

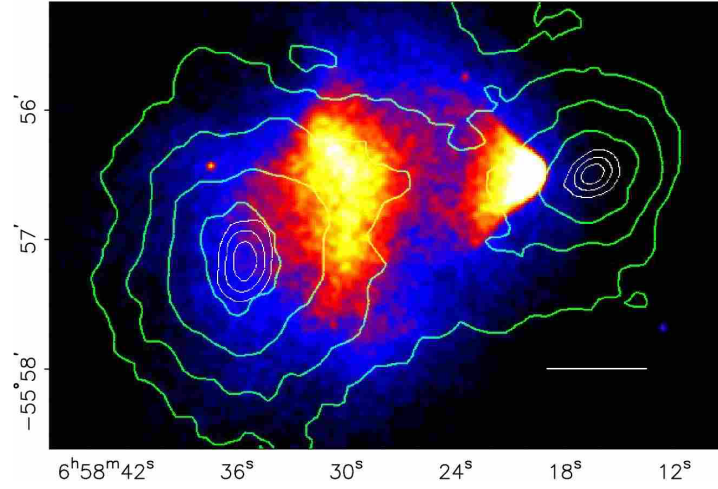


Fig. 1.2. The Bullet Cluster (1E 0657-558). The color map shows the X-ray surface brightness of the hot intracluster gas from *Chandra* observations, while the green contours trace the total mass distribution reconstructed through weak gravitational lensing. The clear spatial offset between the two demonstrates that the dominant mass component is collisionless. The white bar corresponds to 200 kpc. Credit: Cirelli et al. [11], after Clowe et al. [8].

in C_ℓ correspond to sound waves in the pre-recombination baryon-photon plasma that were caught at maximum compression or rarefaction at the moment of decoupling; the positions and heights of these peaks encode the composition and geometry of the Universe [9, 11]. Baryons load the oscillating fluid, enhancing the compressional (odd-numbered) peaks relative to the rarefaction (even-numbered) ones, so that the ratio of the first to the second peak directly constrains the physical baryon density. Dark matter, which does not couple to the photon pressure, provides the gravitational scaffolding within which the oscillations occur: its abundance sets the overall depth of the potential wells and modulates the relative heights of successive peaks. In particular, a universe without cold dark matter would produce a strongly suppressed third peak (cf. Figure 1.3), in clear contradiction with the observed spectrum. The Planck 2018 analysis yields $\Omega_c h^2 = 0.1200 \pm 0.0012$ for the cold dark matter density and $\Omega_b h^2 = 0.02237 \pm 0.00015$ for the baryon density [9], suggesting that dark matter is roughly five times more abundant than ordinary matter. Figure 1.3 illustrates the contrast between the observed spectrum and the prediction for a baryon-only universe.

The observed large-scale structure of the Universe delivers a complementary argument for dark matter. In a purely baryonic universe, structure formation would have been severely hindered: before recombination, baryons were tightly coupled to photons through Thomson scattering, and their density perturbations were erased on small scales by photon diffusion — a process known as *Silk damping* [33]. Cold dark matter, which does not interact electromagnetically, is immune to this damping and began to cluster gravitationally during the matter-dominated era, well before recombination [34]. The potential wells established by dark matter then acted as seeds into which baryons fell after decoupling, seeding the formation of galaxies and clusters (see Section 5.1 for a more detailed treatment of dark matter halo formation and substructure). Without this pre-existing dark matter framework, the matter power spectrum $P(k)$ cannot be reproduced; Figure 1.4 illustrates the striking difference in predicted large-scale structure between a universe with and without cold dark matter.

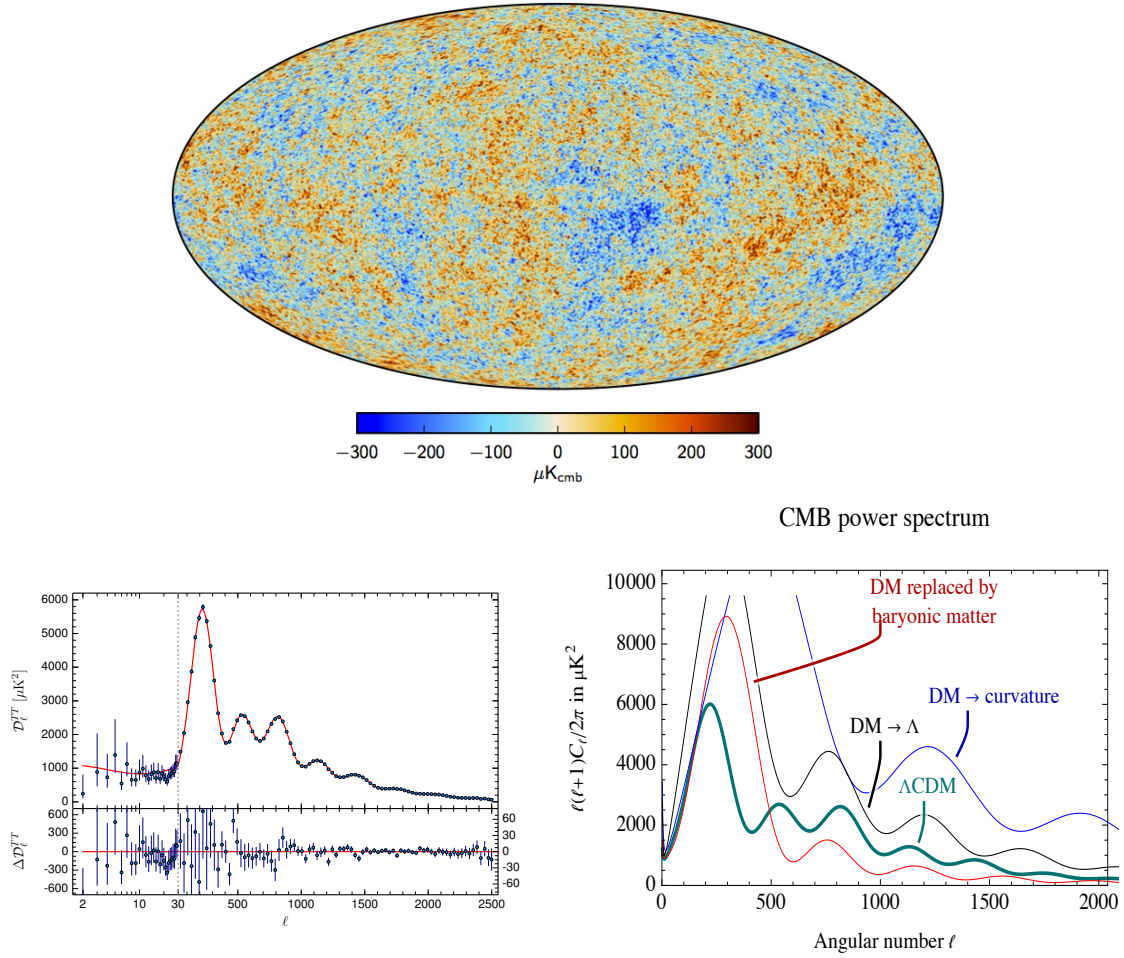


Fig. 1.3. The imprint of dark matter on the CMB. *Top:* the Planck CMB temperature anisotropy map [9], showing the $\sim 10^{-5}$ fluctuations in the otherwise uniform 2.725 K radiation field. *Bottom left:* the observed angular power spectrum C_ℓ^{TT} of the CMB temperature anisotropies, measured by the Planck satellite [9]. *Bottom right:* the same quantity computed in several different cosmological scenarios. The green line represents the best-fit Λ CDM model, while the red line shows the prediction for a universe without cold dark matter, which fails to reproduce the observed third acoustic peak. Adapted from the Planck Collaboration [9] and Cirelli et al. [11].

Beyond the overall shape of the matter power spectrum, the primordial sound waves left a second, sharper imprint on the distribution of matter. Baryon acoustic oscillations (BAO) preserve this imprint in the large-scale galaxy distribution. The same pressure waves that produced the acoustic peaks in the CMB power spectrum were also imprinted on the baryonic matter: when the baryon-photon plasma decoupled at recombination, the outward-propagating sound shells froze in place, leaving a characteristic comoving scale of ~ 150 Mpc in the spatial correlations of galaxies. This scale appears as a well-defined bump in the two-point correlation function and provides a “standard ruler” for measuring the expansion history of the Universe. Its first detection in the galaxy distribution by Eisenstein et al. [10] delivered independent confirmation of the Λ CDM parameters. BAO measurements of the expansion rate, combined with distance measurements from Type Ia supernovae — which established the accelerated expansion of the Universe [35, 36]

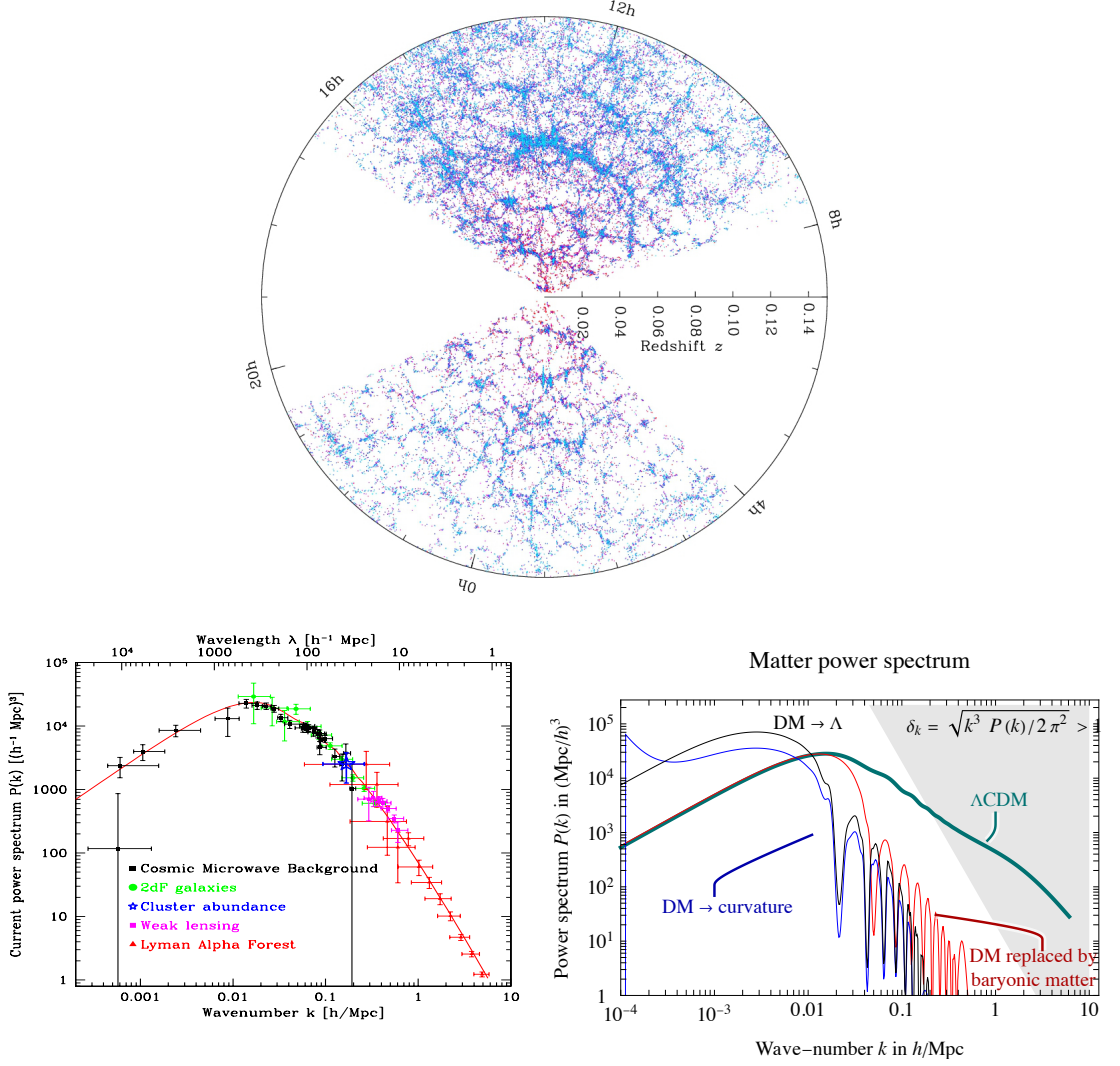


Fig. 1.4. The imprint of dark matter on large-scale structure. *Top*: the large-scale distribution of galaxies from the Sloan Digital Sky Survey (SDSS) [30]. *Bottom left*: the observed matter power spectrum $P(k)$ [31]. *Bottom right*: the same quantity computed in several different cosmological scenarios. The green line represents the best-fit Λ CDM model, while the red line shows the prediction for a universe without cold dark matter, in which power on small (large- k) scales is strongly suppressed — reflecting the absence of small-scale structure [32]. Adapted from the SDSS Collaboration [30], Tegmark and Zaldarriaga [31], and Cirelli et al. [11].

— and CMB data, constrain the dark energy density to $\Omega_\Lambda \approx 0.7$ and the total matter density to $\Omega_m \approx 0.3$ [9, 37].

So far we have pointed out that the total matter density is $\Omega_m h^2 \approx 0.14$, while the CMB independently constrains the baryon density to $\Omega_b h^2 \approx 0.022$ [9] — so the dark, supposedly non-baryonic component outweighs ordinary matter by roughly a factor of five. Could this missing matter simply be a form of ordinary matter that happens to be dark: cold molecular gas clouds, compact stellar remnants, or some as-yet-undetected baryonic component hiding in the intergalactic medium? BBN decisively rules out this possibility. The production of light

elements during the first few minutes after the Big Bang is highly sensitive to the baryon-to-photon ratio, and the observed primordial abundances of deuterium, ^4He , and ^7Li independently constrain the baryon density to $\Omega_b h^2 \approx 0.022$ [38, 39] — in agreement with the CMB, but through entirely different physics. Crucially, this constraint applies to the *total* baryon density, not just the visible fraction: any baryonic matter, however dark or diffuse, would have participated in nucleosynthesis and altered the primordial element yields. Whatever dark matter is, it must therefore be predominantly *non-baryonic*¹ — a new form of matter beyond the Standard Model.

In the next section we introduce and explore the WIMP paradigm, one of the most popular and well-motivated particle physics candidate for dark matter.

1.2 The WIMP Paradigm

The evidence discussed in the previous section suggests that dark matter constitutes the dominant form of matter in the Universe. Furthermore, as established in the previous section, and setting aside the exotic possibility of primordial black holes, it cannot be accounted for by Standard Model particles alone. This conclusion opens a vast parameter space of candidates. In this section, we narrow our focus to the candidate class most relevant to Chapters 4 through 8 of this thesis: *weakly interacting massive particles* (WIMPs), whose appeal is largely due to the so-called *WIMP miracle*, discussed in Section 1.2.2.

Section 1.2.1 surveys the dark matter candidate landscape to place the WIMP focus in context. Section 1.2.2 describes the thermal freeze-out mechanism and identifies the thermally averaged annihilation cross section $\langle\sigma v_{\text{rel}}\rangle$ as the central observable linking early-universe cosmology to present-day signals. Section 1.2.3 reviews the theoretical bounds that bracket the allowed WIMP mass range, detailing why most WIMPs are expected to lie within the MeV–100 TeV thermal window targeted by gamma-ray experiments. Section 1.2.4 discusses the refinements that complicate the mapping between the freeze-out cross section and the present-day annihilation rate: velocity dependence, Sommerfeld enhancement, co-annihilation, and resonances. The thermal-relic cross section, established by the freeze-out calculation, is the primary benchmark for the indirect-detection analyses throughout this thesis.

1.2.1 The Landscape of Candidates

Having ruled out every Standard Model particle as the dominant matter component — dark matter outweighs baryons by roughly a factor of five (Section 1.1.3), while species that were relativistic at decoupling, such as Standard Model neutrinos, free-stream out of the forming potential wells that observations require [27] — the search turns to new physics. As Figure 1.5 illustrates, the range of possibilities is enormous: from ultra-light bosonic fields near 10^{-22} eV, below which the de Broglie wavelength would exceed the size of observed dwarf galaxies [40], up to macroscopic primordial black holes of tens of solar masses, and interaction strengths range from purely gravitational to just below current direct-detection sensitivity. A separate, quantum-mechanical floor on fermionic candidates comes from the *Tremaine–Gunn bound* [41, 42]: Pauli exclusion limits the phase-space density that a fermionic species can achieve, requiring $m \gtrsim 0.4\text{--}1$ keV to match the inner profiles observed in dwarf-spheroidal halos. Dark matter must

¹The one notable exception is *primordial black holes*, which form from the gravitational collapse of large density perturbations before nucleosynthesis: having decoupled from the baryon budget so early, they behave as effectively non-baryonic dark matter, even though they invoke no new particle species (cf. Section 1.2.1). In the remainder of this thesis, whenever we state that dark matter cannot be composed of Standard Model particles, this exception is left implicit.

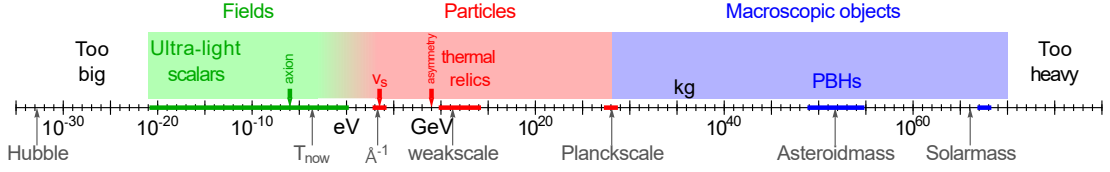


Fig. 1.5. The dark-matter mass landscape: viable candidate masses span roughly ninety orders of magnitude, from ultra-light bosonic fields to macroscopic primordial black holes. Each class is characterized by a distinct production mechanism and probed by a complementary set of experiments. Credit: adapted from Cirelli et al. [11].

therefore be a new, non-baryonic particle: either *cold* (non-relativistic at decoupling, negligible free-streaming length) or at most *warm* (semi-relativistic, free-streaming length intermediate between neutrino-like relics and cold candidates, suppressing small-scale structure without erasing it). The open question is which microphysical realization Nature has chosen. We survey the main candidate classes in order of increasing mass, reading Figure 1.5 from left to right, closing with the question of whether dark matter is truly cold and collisionless.

At the light end of the spectrum sit *axions*, which have drawn growing attention from the community as direct-detection limits on WIMPs have tightened. The axion was originally proposed by Peccei and Quinn [43] to solve the strong CP problem of QCD [44, 45]. It acquires a small mass at the QCD phase transition, and an initial misalignment between its field value and the potential minimum sources a coherent oscillating condensate that today behaves as cold dark matter [46–48]. Because axions are bosons, they evade the Tremaine–Gunn bound and can behave as a classical field. The phenomenologically viable range of axion masses is broad, spanning roughly 10^{-12} to 10^7 eV once astrophysical, cosmological, and laboratory bounds are combined [49, 50].

Not every candidate was ever in thermal contact with the primordial plasma (like in the freeze-out scenario). Feebly interacting massive particles (FIMPs) couple so weakly to the thermal bath that they never reach equilibrium [51]. Their abundance is instead built up gradually through processes like rare decays and $2 \rightarrow 2$ scatterings, and the final relic density grows with the coupling rather than shrinking — the inverse of the freeze-out logic described in Section 1.2.2. This production mechanism is known as *freeze-in*. The viable FIMP mass range is broad. Some realizations occupy the keV-scale window: for example sterile (right-handed) neutrinos, which do not couple to the Standard Model weak interaction but mix with active neutrinos. Two freeze-in production channels are commonly considered: the Dodelson–Widrow scenario of non-resonant oscillations in the early Universe [52], and the Shi–Fuller mechanism in which a primordial lepton asymmetry drives resonant oscillations and shifts the produced spectrum to colder momenta [53]. Resonant production is the more flexible of the two because it evades the most stringent free-streaming bounds from Lyman- α forest data and Milky Way satellite counts, which otherwise push the allowed mass above 20–30 keV [42]. Sterile neutrinos are quasi-stable but decay radiatively as $N \rightarrow \nu\gamma$, yielding a monochromatic X-ray line that provides a distinctive observational signature [42, 54].

Several orders of magnitude heavier, in the GeV–TeV range, sit the WIMPs that are the primary focus of this thesis and one of the most popular cold dark matter candidates; they interact with Standard Model particles through weak-scale couplings. Unlike FIMPs, they reach full thermal equilibrium with the primordial plasma and are produced through thermal freeze-out

in the early Universe, a mechanism we describe in detail in Section 1.2.2. The *WIMP miracle* (Section 1.2.2) has kept them among the dominant paradigms in the field so far. Concrete realizations include the lightest neutralino of supersymmetric extensions of the Standard Model [55] and *minimal dark matter* scenarios [56].

Primordial black holes (PBHs) offer a qualitatively different option in which the dark matter is not a new particle at all, but a population of macroscopic objects formed from the collapse of large density perturbations before BBN. Because PBHs are not made of baryons after BBN, they bypass the baryon-budget constraint of Section 1.1.3. Microlensing surveys, dynamical bounds on wide stellar binaries and ultra-faint dwarfs, and CMB spectral-distortion limits together exclude PBHs as the dominant dark-matter component across most of the mass range [57, 58]. The single surviving window for a monochromatic PBH population to constitute all of the dark matter is the asteroid-mass range, $10^{-16} M_{\odot} \lesssim M \lesssim 3 \times 10^{-12} M_{\odot}$, where Hawking-evaporation bounds from below and microlensing limits from above leave a narrow gap that next-generation experiments aim to close.

A question independent of the specific candidate is whether dark matter departs at all from the cold, collisionless scenario on which Λ CDM is built. Several persistent tensions at sub-galactic scales motivate this possibility: the cusp/core problem in dwarf galaxies [59, 60], the diversity of rotation curves at fixed maximum circular velocity [61], and the too-big-to-fail puzzle of the Milky Way satellites [62, 63] all point to fewer dense, cuspy halos than collisionless N -body simulations predict [27] (see Section 1.4.7 for more information). One alternative is *warm dark matter* (WDM), a kinematic classification encompassing any candidate with a free-streaming length intermediate between hot and cold relics; the resulting suppression of small-scale power softens these tensions while leaving the large-scale successes of Λ CDM intact. Resonantly produced keV sterile neutrinos are one concrete realization, but WDM as a classification is broader, and the astrophysical tests that constrain it are reviewed in [64]. A second possibility is *self-interacting dark matter* (SIDM), originally proposed by Spergel and Steinhardt [65] to thermalize the inner regions of halos through elastic scattering and so produce cored rather than cuspy density profiles. The review of Tulin and Yu [66] summarizes the status of the program, and the ETHOS framework [67] provides a systematic way to integrate non-standard dark-matter interactions into cosmological hydrodynamical simulations for direct comparison with dwarf-galaxy data.

These examples of candidates make clear how little the existing evidence constrains the microphysics of the dark sector. We narrow our attention now to the WIMP paradigm — the candidate class probed by the gamma-ray indirect-detection analyses developed in the later chapters of this thesis. The appeal of this framework is that its phenomenology is governed by a single quantity, the velocity-averaged annihilation cross section $\langle\sigma v\rangle$, which simultaneously fixes the cosmological relic abundance, the elastic-scattering rate on nuclei, the production cross section at colliders, and — under some assumptions — the present-day annihilation rate in galactic halos. The following subsection reviews the freeze-out mechanism that explains how $\langle\sigma v\rangle$ governs the relic density, and why it defines a natural benchmark for the experimental program that follows.

1.2.2 The Freeze-Out Mechanism

Consider a hypothetical species χ held in chemical equilibrium with the Standard Model plasma of the early Universe, sustained by pair-annihilation reactions $\chi\chi \leftrightarrow \text{SM SM}$ that proceed in both directions at equal rates. As the Universe expands and cools, two processes run in competition: the cosmological expansion, which dilutes everything and lowers the temperature, and the annihilation rate, which depends on the density of χ particles still available to find each other.

Eventually the expansion wins: the reactions become too slow to maintain equilibrium, the comoving number density stops tracking its equilibrium value, and whatever population remains is locked in as the relic abundance we observe today. The purpose of this subsection is to describe this process, the freeze-out mechanism, which is responsible for setting the relic abundance of a thermal dark-matter candidate, and to show that the main quantity that controls the process is the velocity-averaged annihilation cross section $\langle\sigma v\rangle$.

As long as chemical equilibrium holds, the abundance of χ is dictated by the temperature of the bath alone: creation and annihilation proceed at equal rates, and the number density tracks the equilibrium value $n_{\text{eq}}(T)$, obtained by integrating the thermal phase-space distribution of the species. Since the mass M_χ sets the only relevant scale, it is convenient to measure the temperature through the dimensionless ratio $x \equiv M_\chi/T$, which grows as the Universe cools and therefore plays the role of time. The equilibrium density takes simple forms in the two opposite regimes,

$$n_{\text{eq}} \propto T^3 \quad (x \ll 1), \quad n_{\text{eq}} = g_\chi \left(\frac{M_\chi T}{2\pi} \right)^{3/2} e^{-M_\chi/T} \quad (x \gg 1), \quad (1.2.1)$$

with g_χ the number of internal degrees of freedom of χ . At high temperatures ($x \ll 1$) the species is relativistic and roughly as abundant as the photons. Once the temperature drops below the rest mass ($x \gtrsim 1$), however, only the exponential tail of the thermal bath carries enough energy to create a χ pair, and the equilibrium abundance collapses as e^{-x} . Were equilibrium maintained indefinitely, the dark matter would be exponentially wiped out; a relic population survives precisely because equilibrium eventually fails.

A quantitative description of this competition follows the dark-matter number density through a variation of the Boltzmann equation — here in the reduced, number-density form [68],²

$$\dot{n}_\chi + 3Hn_\chi = -\langle\sigma v_{\text{rel}}\rangle (n_\chi^2 - n_{\text{eq}}^2), \quad (1.2.2)$$

where H is the Hubble expansion rate, n_χ the dark matter number density, and n_{eq} the equilibrium density of Eq. (1.2.1). $3Hn_\chi$ is the dilution due to Hubble expansion; $-\langle\sigma v_{\text{rel}}\rangle n_\chi^2$ is the depletion of χ by annihilation into Standard Model final states; and $+\langle\sigma v_{\text{rel}}\rangle n_{\text{eq}}^2$ is the inverse process, in which χ pairs are recreated from the thermal bath at the rate dictated by detailed balance. The expansion rate H is itself fixed by the Friedmann equation, which ties it to the total energy density of the Universe. During the radiation-dominated era that density is set by the relativistic species of the plasma, $\rho \propto g_* T^4$, with g_* the effective number of relativistic degrees of freedom, so that $H \sim T^2/M_{\text{Pl}}$. Chemical equilibrium persists roughly as long as the annihilation rate $\Gamma = n_\chi \langle\sigma v_{\text{rel}}\rangle$ remains above this value.

It is convenient to factor out the trivial expansion dilution by passing from n_χ to the *comoving abundance* $Y \equiv n_\chi/s$, where $s = (2\pi^2/45)g_{*s}T^3$ is the entropy density of the thermal bath, with g_{*s} the effective number of entropic degrees of freedom. Because the total entropy in a comoving volume sa^3 is conserved during adiabatic expansion, Y remains constant whenever the dark-matter number per comoving volume is also conserved; variations of Y thus track only

²Equation (1.2.2) rests on several simplifying assumptions [68]. The Universe is treated as homogeneous — primordial inhomogeneities are of order 10^{-5} , so the number density carries no appreciable spatial dependence. Rapid elastic scattering off the thermal bath keeps the dark matter in kinetic equilibrium: its momentum distribution stays thermal, a single equation for the integrated density $n_\chi(t)$ suffices, and $\langle\sigma v_{\text{rel}}\rangle$ is averaged over that distribution. The distribution functions take their classical Maxwell–Boltzmann form, appropriate in the non-relativistic regime, and no particle–antiparticle asymmetry is present. Finally, a single species dominates, with $2 \rightarrow 2$ annihilation as the only relevant process (co-annihilations are deferred to Section 1.2.4), and detailed balance fixes the creation term to $\langle\sigma v_{\text{rel}}\rangle n_{\text{eq}}^2$.

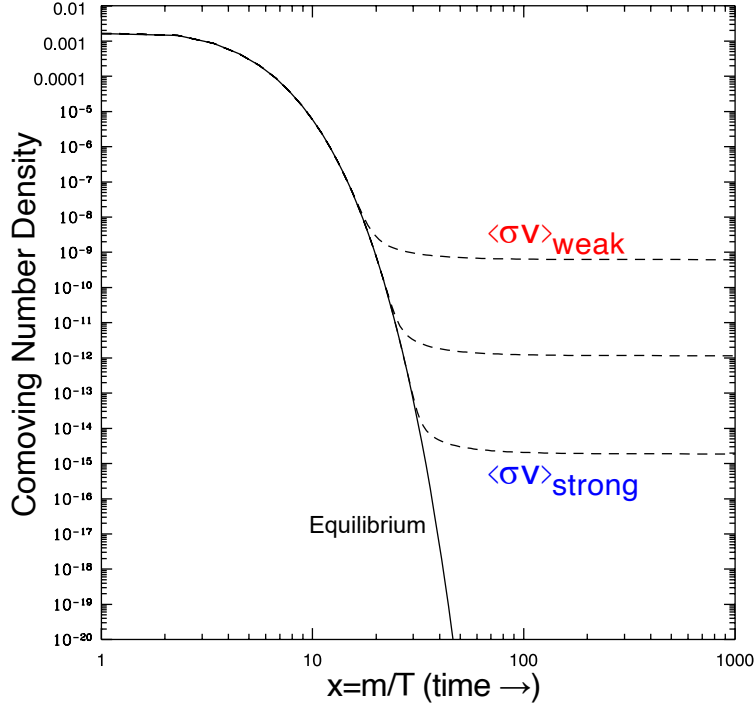


Fig. 1.6. Freeze-out of a thermal relic. The comoving abundance Y is shown against the dimensionless inverse temperature $x \equiv M_\chi/T$, which grows as the Universe expands and cools. The solid curve is the equilibrium abundance Y_{eq} , exponentially Boltzmann-suppressed once the temperature falls below the dark-matter mass ($x \gtrsim 1$). The dashed curves show the actual abundance for three values of the annihilation cross section: as long as annihilations outpace the Hubble expansion the abundance tracks equilibrium, and once they become inefficient it freezes out at a constant comoving value, with $Y_\infty \propto 1/\langle\sigma v\rangle$. A larger cross section (lower curve) keeps the particle in equilibrium longer and leaves fewer relics than a smaller one (upper curve). Credit: adapted from Hooper [69].

annihilation and creation, rather than the cosmological dilution shared by every species. The same rescaling applied to n_{eq} defines the equilibrium abundance $Y_{\text{eq}} \equiv n_{\text{eq}}/s$.

Figure 1.6 shows the behaviour of the solutions of Eq. (1.2.2) in these variables. At high temperatures the actual abundance follows the equilibrium curve Y_{eq} ; once the temperature drops below the rest mass, Y_{eq} collapses under the Boltzmann suppression of Eq. (1.2.1). The actual abundance tracks this decline only as long as the annihilation rate Γ exceeds the Hubble rate H ; once Γ drops below H , annihilations can no longer keep up with expansion, Y moves away from Y_{eq} , and the comoving abundance plateaus at its asymptotic value Y_∞ . This departure is gradual: the dark-matter abundance begins to move away from the equilibrium curve around $x \sim 10$ – 15 , while the plasma is still at a temperature of order a tenth of M_χ , and only settles onto its frozen plateau somewhat later. For a weakly interacting candidate at the electroweak scale, the *freeze-out* completes at $T_{\text{fo}} \sim M_\chi/(20\text{--}25)$ — when the particles are already non-relativistic, but the Universe is still deep in radiation domination.

A particle that annihilates strongly enough stays in chemical equilibrium for a long time (lowest dashed curve in Figure 1.6): it follows the equilibrium population as the latter drops

exponentially, and only departs from it when the equilibrium abundance has already become small. Such a particle leaves behind few relics. A particle that annihilates weakly (uppermost curve), by contrast, loses contact with the bath earlier, while its equilibrium abundance is still relatively large, and therefore freezes out at a higher residual density. In the thermal-relic context, the weakest particle wins. The relic abundance is thus set to first approximation by how much a candidate annihilates — by the velocity-averaged annihilation cross section $\langle\sigma v_{\text{rel}}\rangle$ — while the mass enters only at second order, logarithmically, through the temperature at which decoupling occurs.

After freeze-out both annihilation and creation become negligible, and the comoving abundance retains its plateau value Y_∞ ; the present-day mass density in dark matter is then $\rho_\chi = M_\chi s_0 Y_\infty$, with s_0 the entropy density today. Dividing by the critical density gives the relic abundance

$$\Omega_\chi = \frac{M_\chi s_0 Y_\infty}{\rho_{\text{crit}}}, \quad \rho_{\text{crit}} = \frac{3H_0^2}{8\pi G}, \quad (1.2.3)$$

which, evaluated with the measured entropy density and expansion rate, can be recast numerically as $Y_\infty \simeq (0.44 \text{ eV}/M_\chi)(\Omega_\chi h^2/0.12)$. The observed abundance therefore fixes the product $M_\chi Y_\infty$: a heavier candidate must survive in smaller numbers to supply the same mass density.

A subtlety worth flagging is the meaning of $\langle\sigma v_{\text{rel}}\rangle$, since the dark-matter literature uses the term “cross section” for two physically distinct objects. The σv_{rel} that enters Eq. (1.2.2) (in units of cm^3/s) is the cross section for the process $\chi\chi \rightarrow \text{SMSM}$ multiplied by the relative velocity of the incoming pair, then averaged over a Maxwell–Boltzmann distribution at temperature T . The cross section quoted in direct-detection experiments (in units of cm^2), by contrast, is the elastic cross section $\sigma_{\chi N}$ for the recoil process $\chi + N \rightarrow \chi + N$ on a target nucleon, involving no thermal average. The two thus refer to different reactions, different kinematics, and different epochs in the dark-matter history. Throughout this thesis the symbol $\langle\sigma v\rangle$ always denotes the annihilation case. One further subtlety is that the thermal average entering the Boltzmann equation is evaluated at freeze-out velocities of order the thermal scale, $v_{\text{rel}} \sim \sqrt{T_{\text{fo}}/M_\chi} \approx 0.2c$, whereas dark-matter particles in present-day galactic halos move at velocities set by the gravitational potential, $v_{\text{rel}} \sim 10^{-3}c$ — roughly two orders of magnitude smaller. Whether $\langle\sigma v\rangle$ depends on the relative velocity therefore controls the mapping between the cosmological cross section that fixes the relic abundance and the annihilation rate accessible to indirect detection. To keep track of this dependence we expand the annihilation cross section in partial waves, and defer its observational consequences to Section 1.2.4.

Non-relativistic two-body annihilation admits the partial-wave expansion

$$\sigma v_{\text{rel}} \simeq \sigma_0 + \sigma_1 v_{\text{rel}}^2 + \mathcal{O}(v_{\text{rel}}^4), \quad (1.2.4)$$

where σ_0 is the s -wave ($\ell = 0$) contribution and $\sigma_1 v_{\text{rel}}^2$ the p -wave ($\ell = 1$) contribution [11]. Averaging this expansion over the thermal distribution of relative velocities gives $\langle\sigma v_{\text{rel}}\rangle = \sigma_0 + (6T/M_\chi)\sigma_1 + \dots$; at freeze-out, with $x_{\text{fo}} \equiv M_\chi/T_{\text{fo}}$, the p -wave term is therefore suppressed by a factor $6/x_{\text{fo}}$ relative to the s -wave piece.

Solving Eq. (1.2.2) through the freeze-out transition places the decoupling at $x_{\text{fo}} \approx 25$, consistent with the value $T_{\text{fo}} \sim M_\chi/(20\text{--}25)$ anticipated above; this number depends only logarithmically on the dark-matter mass and couplings, so candidates spanning a broad range of masses freeze out at comparable x . The surviving comoving abundance then scales as

$$Y_\infty \propto \frac{x_{\text{fo}}}{M_{\text{Pl}} M_\chi (\sigma_0 + 3\sigma_1/x_{\text{fo}})},$$

inversely proportional to the annihilation strength. The p -wave coefficient enters here as $3\sigma_1/x_{f_0}$, rather than the $6\sigma_1/x_{f_0}$ of the instantaneous thermal average: annihilations continue to deplete the abundance after freeze-out, so the velocity-dependent term is averaged over the subsequent evolution rather than sampled at T_{f_0} alone. Inserting this into Eq. (1.2.3) yields the relic density in terms of the partial-wave coefficients [12],

$$\frac{\Omega_\chi h^2}{0.12} \approx \frac{x_{f_0}}{25} \frac{2.2 \times 10^{-26} \text{ cm}^3/\text{s}}{\sigma_0 + 3\sigma_1/x_{f_0}}. \quad (1.2.5)$$

In the limit of pure s -wave annihilation the coefficient σ_1 vanishes, the thermal average becomes the constant $\langle\sigma v_{\text{rel}}\rangle = \sigma_0$, and Eq. (1.2.5) reduces, for $x_{f_0} \approx 25$, to

$$\frac{\Omega_\chi h^2}{0.12} \approx \frac{2.2 \times 10^{-26} \text{ cm}^3/\text{s}}{\langle\sigma v_{\text{rel}}\rangle}, \quad (1.2.6)$$

where the relic abundance is essentially the inverse of the annihilation cross section, with the mass entering only through logarithmic corrections to the prefactor. The numerator is the canonical *thermal cross section*, $\langle\sigma v\rangle_{\text{cosmo}} \approx 2.2 \times 10^{-26} \text{ cm}^3/\text{s}$: any dark-matter candidate that annihilates at this rate reproduces, to within a few percent for masses above roughly 10 GeV (cf. Figure 1.7), the observed cosmological abundance. For an annihilation amplitude characterized by a coupling g , with $\alpha \equiv g^2/(4\pi)$, the natural size of the cross section is $\langle\sigma v\rangle \sim \alpha^2/M_\chi^2$. Plugging in the weak-interaction value $\alpha \sim \alpha_{\text{weak}} \sim 0.03$ and demanding a match to $\langle\sigma v\rangle_{\text{cosmo}}$ returns a mass M_χ of order a few hundred GeV to a few TeV — the electroweak scale. This result is known as the *WIMP miracle*: a stable particle with mass and coupling strength at the electroweak scale, produced thermally in the early Universe, naturally yields a relic abundance consistent with the observed dark matter density [12]. The inverse relation of Eq. (1.2.6) is illustrated in Figure 1.7: the orange band marks the cross section that reproduces the measured abundance, while smaller (larger) values freeze out too early (too late) and overproduce (underproduce) dark matter.

In the same s -wave limit, the cross section fixed by Eq. (1.2.6) also sets the rate at which dark matter annihilates in present-day astrophysical environments. This connection between cosmological abundance and indirect-detection signals further motivates the gamma-ray searches of Chapters 4–8.

1.2.3 Theoretical Bounds on the WIMP Mass

In this section we explore the viable mass range for WIMPs. Consider first a particle that interacts with the Standard Model only through the weak interaction, with no dark mediator of its own. At freeze-out energies, using the Fermi approximation, the annihilation cross section for such a particle, dominated by exchange of W and Z bosons, scales as $\sigma_0 \approx G_F^2 M_\chi^2/2\pi$, where G_F is the Fermi constant [70]. This is an *s-wave* process (cf. Eq. (1.2.4)), meaning the cross section is independent of the relative velocity — a noteworthy property, since it distinguishes whether or not the same $\langle\sigma v\rangle$ that fixes the relic abundance also drives present-day annihilations in galactic halos. Because σ_0 scales as M_χ^2 in this approximation, lowering the candidate mass reduces the annihilation cross section quadratically; combined with the inverse dependence of the relic abundance on $\langle\sigma v\rangle$ in Eq. (1.2.6), this means a sufficiently light particle annihilates too weakly to thin itself out and ends up overproducing dark matter. The original Lee–Weinberg analysis [71], contemporaneous with the closely related arguments of Hut [72] and Vysotsky, Dolgov, and Zeldovich [73], obtained $M_\chi \gtrsim 2 \text{ GeV}$ for a heavy Dirac neutrino. Modern evaluations with updated effective-degree-of-freedom counts give [11, 70]

$$M_\chi \gtrsim 3 \text{ GeV} \quad (\text{Lee–Weinberg bound}), \quad (1.2.7)$$

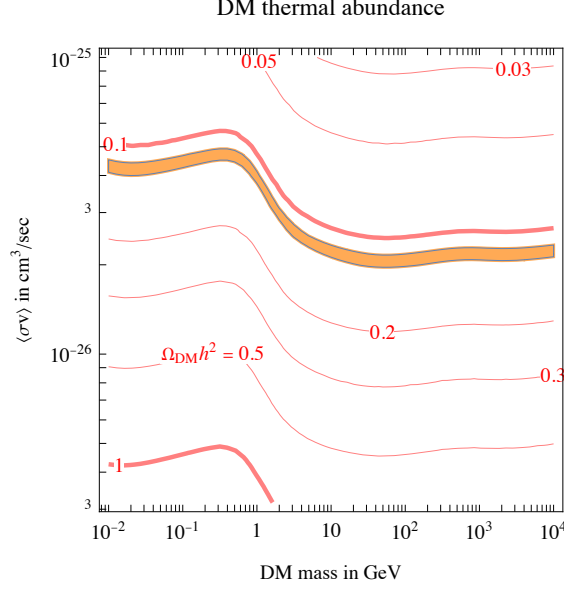


Fig. 1.7. Contours for the dark matter freeze-out abundance $\Omega_{\text{DM}} h^2$ as a function of DM mass and annihilation cross-section $\langle\sigma v\rangle$ for Majorana dark matter. The measured cosmological abundance $\Omega_{\text{DM}} h^2 = 0.120 \pm 0.001$ is reproduced within the orange band at 3σ . From this plot it is possible to appreciate how the annihilation cross-section needed to match the observed abundance varies with mass in the $1 - 3 \times 10^{-26} \text{ cm}^3/\text{s}$ range. Credit: Cirelli et al. [11].

below which a thermal relic coupled to the Standard Model weak interaction is cosmologically forbidden. This GeV-scale floor is specific to annihilation mediated by W and Z bosons. When that assumption is dropped, the much weaker BBN bound discussed below takes over, extending the allowed lower edge into the MeV regime.

It is important to stress the coupling with the Standard Model weak interaction. The Lee–Weinberg bound says nothing about models in which a new mediator — a dark photon, a light scalar, or a hidden-sector gauge boson — carries the annihilation. With such ingredients the relic abundance can be reproduced down to the sub-GeV and even keV regimes, which is precisely why a parallel low-mass direct-detection program (SENSEI, DAMIC, and related semiconductor experiments [74, 75]) has grown alongside the conventional GeV–TeV searches.

The upper end of the mass window has a more universal origin, which is often referred to as the *unitarity bound*. Quantum mechanics requires probabilities to sum to one; in scattering theory this *unitarity* condition bounds the cross section for any two-body process from above by a quantity set purely by the available phase space, $\sigma \leq 4\pi/k^2$, where $k = M_\chi v/2$ is the center-of-mass momentum [76]. For non-relativistic dark-matter annihilation this translates directly to $\langle\sigma v\rangle_{\text{max}} \propto 1/M_\chi^2$, a ceiling that any model must respect regardless of its coupling structure. As the candidate mass increases, the unitarity ceiling drops; at some mass it sinks below the thermal benchmark $\langle\sigma v\rangle_{\text{cosmo}}$, and beyond that point even a maximally coupled candidate must be overabundant. The classic estimate of Griest and Kamionkowski places this crossover at [76]

$$M_\chi \lesssim 100 \text{ TeV}, \quad (\text{unitarity bound}) \quad (1.2.8)$$

above which thermal-relic dark matter is no longer viable as a single-particle, two-body annihilation problem and requires either bound-state formation, multi-step annihilation cascades, or a

non-thermal cosmological history.

An independent lower bound, insensitive to which mediator carries the annihilation, comes from BBN [77]. A dark-matter particle still in chemical contact with the plasma at $T \sim 1$ MeV would contribute to the radiation energy density during nucleosynthesis and inject electromagnetic or hadronic energy through residual annihilations, both of which spoil the otherwise excellent agreement between predicted and observed primordial light-element abundances. Requiring freeze-out to complete before BBN enforces $M_\chi \gtrsim 3$ MeV, a floor that holds for any thermal relic regardless of its couplings.

Taken together, these arguments delineate a mass window

$$3 \text{ GeV} \lesssim M_\chi \lesssim 100 \text{ TeV}, \quad (1.2.9)$$

for WIMPs that annihilate through Standard Model weak interactions, with the BBN floor extending the lower edge into the MeV regime for candidates with non-Standard-Model mediators. Within this window the thermal benchmark $\langle\sigma v\rangle_{\text{cosmo}} \approx 2.2 \times 10^{-26} \text{ cm}^3/\text{s}$ is the natural target for indirect-detection experiments: any instrument that reaches this sensitivity directly tests the thermal freeze-out hypothesis. The Fermi Large Area Telescope already reaches the benchmark in the $b\bar{b}$ channel up to masses of several tens of GeV looking at dSphs, an analysis we revisit in Chapter 5 when studying dark matter subhalos. The Cherenkov Telescope Array Observatory, with its larger collection area and lower energy threshold relative to current imaging atmospheric Cherenkov arrays, will extend indirect-detection sensitivity into the TeV domain, which is the subject of Chapter 8.

1.2.4 Beyond s-wave annihilation

The bounds derived in Section 1.2.3 assumed the simplest possible freeze-out scenario: a single stable species, s -wave-dominated two-body annihilation, and weak-scale couplings. Several physically motivated effects modify this picture and, more importantly, alter the mapping between the cosmological cross section $\langle\sigma v\rangle_{\text{cosmo}}$ that fixes the relic abundance and the present-day annihilation rate probed by indirect detection.

p -wave suppression. The starting point is the partial-wave expansion of Eq. (1.2.4) from Section 1.2.2. When the s -wave piece is dominant, the cross section is approximately constant from freeze-out velocities ($v_{\text{rel}} \sim 0.2c$) down to typical halo velocities ($v_{\text{rel}} \sim 10^{-3}c$), and $\langle\sigma v\rangle_{\text{cosmo}}$ remains a meaningful benchmark for indirect-detection searches. When the leading contribution is p -wave, in contrast, the present-day rate is suppressed by $v_{\text{rel}}^2 \sim 10^{-6}$ relative to its value at freeze-out, and the channel becomes effectively invisible to instruments that rely on annihilation in cold galactic halos.

Sommerfeld enhancement. A second class of corrections arises when dark matter couples to a mediator with mass $M_V \ll M_\chi$. The long-range attractive force distorts the incoming-pair wave function and enhances the annihilation probability at low velocities, a non-perturbative effect known as the *Sommerfeld enhancement* [78, 79]. The enhancement factor S grows as the relative velocity falls, roughly as $S \propto 1/v_{\text{rel}}$, and can be resonantly amplified for particular mediator-to-dark-matter mass ratios. Halo velocities $v_{\text{rel}} \sim 10^{-3}c$ therefore probe a regime in which S can reach $\mathcal{O}(10^2\text{--}10^3)$ [78], while at freeze-out velocities $v_{\text{rel}} \sim 0.2c$ the enhancement is only moderate. The Sommerfeld effect is essential for heavy electroweak WIMPs — the pure wino at $M_\chi \sim 3$ TeV, the higgsino near $M_\chi \sim 1.1$ TeV, and Minimal Dark Matter candidates more generally — where it shifts the thermal-relic mass from hundreds of GeV to multi-TeV scales. We

return to this point in Section 1.4 when discussing J -factors and the loop-suppressed channels of Minimal Dark Matter. For the pure wino in particular, the enhanced halo rate at $v_{\text{rel}} \sim 10^{-3} c$ is large enough that current gamma-ray constraints effectively rule the candidate out for cuspy halo profiles. For the GeV-scale candidates relevant to the Galactic Center excess analyzed in Chapter 4, by contrast, the standard thermal cross section fits the observed signal without invoking Sommerfeld enhancement, as Daylan et al. [80] showed explicitly for a $\sim 35\text{--}50\text{ GeV}$ WIMP annihilating at $\langle\sigma v\rangle \approx (1\text{--}3) \times 10^{-26} \text{ cm}^3/\text{s}$.

Bound-state formation. The same light mediator that produces the Sommerfeld enhancement supports a second non-perturbative effect: two dark matter particles can emit a mediator and fall into a bound state, in close analogy with hydrogen recombination [81]. Because the constituents of the bound state annihilate rapidly, this capture process opens an additional depletion channel during freeze-out, which lowers the relic abundance and pushes the thermal mass of heavy candidates to larger values [82]. At halo velocities the formation rate is enhanced much like the Sommerfeld factor, and the mediator emitted during capture carries away the binding energy, producing a sharp spectral feature that indirect searches can target [83].

Co-annihilation and resonance. Two further refinements deserve mention because they directly affect the indirect detection prospects. The first is *co-annihilation*: if the dark sector contains partner particles nearly degenerate with χ , they stay abundant during freeze-out and help deplete the dark-matter density [55, 84]. Since these partners later decay into χ , the present-day annihilation cross section of the surviving species can differ substantially from the effective freeze-out value, weakening the link between $\Omega_\chi h^2$ and the indirect-detection signal. The second is *resonant annihilation*: when dark matter annihilates through an s -channel mediator with mass near threshold, $m_{\text{med}} \approx 2M_\chi$, the cross section acquires a sharp velocity dependence of Breit–Wigner form [68, 84], so the present-day rate may be enhanced relative to freeze-out. Both effects introduce model dependence into the mapping from $\langle\sigma v\rangle_{\text{cosmo}}$ to the observable annihilation rate.

These refinements lead to several distinct theoretical scenarios, of which the standard case — s -wave annihilation — is the simplest and the least model-dependent. In that case, the freeze-out cross section also fixes the galactic-halo annihilation rate (Section 1.2.2), providing the cleanest target for indirect-detection experiments. In richer scenarios the mapping remains calculable but model-dependent: Sommerfeld enhancement can boost halo rates by orders of magnitude, p -wave suppression or off-resonance Breit–Wigner effects can lower them, and co-annihilation can shift the relic-abundance target relative to the halo signal. Throughout this thesis we adopt the standard thermal-relic value of $\langle\sigma v\rangle$ as the primary benchmark when quoting cross-section bounds, while flagging explicitly the regimes in which these refinements would shift the interpretation. With this theoretical setting in place, we turn to the experimental strategies for testing the WIMP hypothesis.

1.3 Searching for Dark Matter

The same coupling that governs dark matter annihilation in the early universe also mediates scattering off nuclei in underground detectors and pair-production at particle colliders. We therefore explore three complementary strategies for detecting dark matter: direct detection, collider searches, and indirect detection — see Figure 1.8.

Direct and indirect detection both probe dark matter in its natural habitat: the galactic halo, where particles drift at non-relativistic velocities of $v \sim 10^{-3} c$. Colliders, by contrast, collide

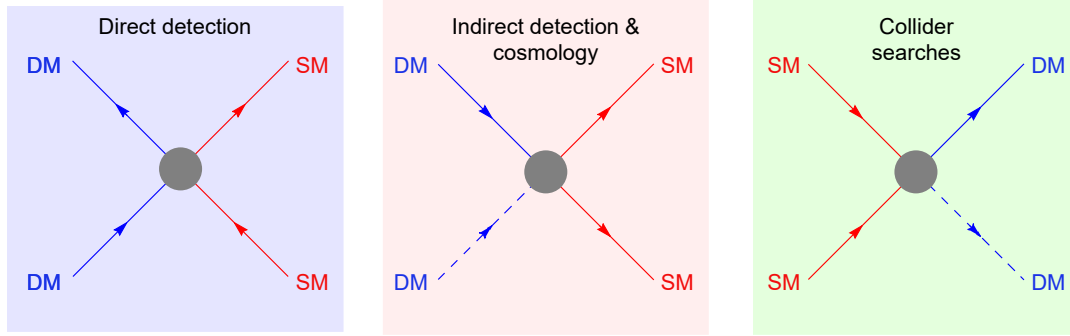


Fig. 1.8. The detection triangle: three complementary strategies for detecting dark matter, each corresponding to a different reading of the same underlying interaction vertex. *Left*: direct detection — dark matter scatters off a Standard Model target in a laboratory. *Center*: indirect detection — dark matter annihilates or decays into Standard Model particles in astrophysical environments. In decay, the process involves a single dark matter particle, thus the dashed line. *Right*: collider production — Standard Model particles produce dark matter at accelerators. A dashed line indicates that the process might involve production of just one dark matter particle. Credit: Cirelli et al. [11].

particles at energies potentially exceeding the dark matter mass, accessing a relativistic regime. The same dark matter–Standard Model coupling can yield very different cross-sections depending on the kinematic regime: velocity-dependent or momentum-dependent suppressions that are negligible at collider energies can reduce the non-relativistic rate probed by direct and indirect detection by orders of magnitude. An interaction that produces a strong signal in one channel can therefore be entirely invisible in another. A signal in one experiment combined with a non-detection in another would narrow the type of interaction at play, and together they carry complementary information that no single channel could yield.

This section reviews direct detection and collider searches in turn, with emphasis on the physical principles rather than experimental details, and closes with the complementary constraints that cosmology places on the annihilation cross-section. We defer a more detailed discussion of indirect detection — which is particularly relevant for this thesis — to Section 1.4.

1.3.1 Direct Detection

Direct detection experiments search for the faint nuclear recoil produced when a dark matter particle from the galactic halo scatters off an atomic nucleus inside a deep-underground detector. A WIMP with mass M_χ and velocity $v \sim 10^{-3}c$ striking a xenon nucleus of comparable mass transfers a recoil energy of order 20 keV [85] — small enough that detecting it demands extremely low backgrounds, but large enough to be within reach of modern detectors. For spin-independent interactions at small momentum transfer, the scattering amplitudes of the individual nucleons add coherently, and the cross-section scales as the square of the atomic mass number, $\sigma_{\text{SI}} \propto A^2$ [86]; this coherent enhancement is the main reason the most sensitive experiments employ heavy targets such as xenon ($A \approx 131$).

The field is currently led by dual-phase liquid xenon time projection chambers — LZ [87], XENONnT [88], and PandaX-4T [89] — which have pushed the upper limit on the spin-independent WIMP–nucleon cross-section in a mass-dependent manner, for example down to $\sigma_{\text{SI}} \lesssim 10^{-47} \text{ cm}^2$

at $M_\chi \sim 30$ GeV [90]. These experiments have now reached in sensitivity the *neutrino fog*: Standard Model neutrinos — solar, atmospheric, and from the diffuse supernova background — scatter coherently off nuclei and produce recoils kinematically indistinguishable from a WIMP signal, an irreducible background whose onset is already visible in current data through the observation of solar ^8B coherent neutrino–nucleus scattering [90–93]. Within the fog, sensitivity improves only as the inverse square root of the accumulated exposure rather than inversely [94], so progress will come slowly even for the multi-ton successors now being planned [95].

Direct detection is most sensitive when the WIMP mass is comparable to that of the target nucleus — roughly tens of GeV for xenon: at lower masses the recoil energy falls below the experimental threshold, while at higher masses the local number density decreases as $n \propto 1/M_\chi$, suppressing the event rate. These mass-dependent limitations, together with the entry into the neutrino fog, motivate complementary search strategies. For comprehensive treatments of direct detection — experimental techniques, current bounds, and the path through the neutrino fog — we refer the reader to the reviews by Schumann [96], Cirelli et al. [11], and Hooper [86].

1.3.2 Collider Searches

If dark matter couples to quarks or gluons, it can be pair-produced in high-energy collisions at the Large Hadron Collider. The challenge is detection: dark matter particles are in general expected to escape the detector without leaving a trace, just as neutrinos do [11]. For this reason, they have to be searched for by looking at the missing transverse energy (E_T^{miss}).

Pure pair-production of dark matter ($pp \rightarrow \chi\chi$) produces no visible particles and goes entirely unnoticed. A detectable signal requires that an additional Standard Model particle — a jet, a photon, or a W/Z boson — is radiated from the initial state, recoiling against the invisible dark matter pair. These are the mono-X searches: mono-jet, mono-photon, mono- Z , and mono-Higgs, each paying a cross-section penalty of roughly $g^2/(4\pi)^2 \sim 10^{-3}$ for the additional emission. A complementary channel probes **invisible Higgs decays**: if the dark matter is lighter than half the Higgs boson mass ($M_\chi < M_h/2$), the Higgs can decay directly into a dark matter pair. Current LHC measurements constrain the invisible branching ratio to $\text{BR}_{\text{inv}} \lesssim 0.11$ at 95% C.L., essentially excluding Higgs-portal thermal relic dark matter below $M_h/2 \approx 63$ GeV [97]. These results are interpreted within **simplified models** that include an explicit mediator and its couplings to quarks and dark matter, enabling direct comparison with direct and indirect detection bounds [98]. Current data exclude s -channel mediators in roughly the 0.1–4 TeV range and push the lower bound on colored mediators (squarks, gluinos) to 1–2 TeV [86].

Despite their power, collider experiments face a fundamental limitation: even if a new invisible particle is produced and characterized, the LHC cannot establish that it is cosmologically stable or that it constitutes the dark matter observed in gravitational systems.

For a comprehensive treatment of dark matter searches at colliders, we refer the reader to the review by Boveia and Doglioni [98], and to the broader reviews by Cirelli et al. [11] and Hooper [86].

1.3.3 Cosmological Probes: the CMB

An additional methodology to constrain the dark matter annihilation cross-section comes from cosmology [99]. During the cosmic *dark ages* — the interval between recombination and the formation of the first stars — dark matter annihilation injects electrons, positrons, and photons that ionize and heat the neutral intergalactic gas; the resulting extra free electrons broaden the last-scattering surface, damping the high-multipole acoustic peaks and enhancing large-scale

polarization [100]. Because the imprint responds only to the total energy deposited around $z \approx 600$, the CMB constrains a single degenerate combination, $p_{\text{ann}} = f_{\text{eff}} \langle \sigma v_{\text{rel}} \rangle / M_\chi$, where f_{eff} is the fraction of the released energy absorbed by the gas [100]. Planck 2018 data bound this combination to $p_{\text{ann}} < 3.2 \times 10^{-28} \text{ cm}^3 \text{ s}^{-1} \text{ GeV}^{-1}$ [9], a limit that excludes the thermal relic cross-section for s -wave annihilation at masses below roughly 30 GeV in leptonic channels and 10–15 GeV in hadronic ones [99]. The bound involves only the homogeneous background dark matter density and well-understood atomic and radiative physics, independent of halo profiles and cosmic-ray propagation modelling, and is therefore among the most robust in the field. That robustness comes at a price: the CMB measures a single amplitude rather than a spectrum, so even a detection would fix neither the dark matter mass nor the annihilation channel without further assumptions, as mass and annihilation channel are degenerate for CMB probes. The constraint moreover weakens sharply for p -wave annihilation, whose rate is suppressed by the low dark matter velocities of the dark ages [11]. Taken together, these strategies answer different questions. Direct detection and collider searches probe the particle nature of dark matter — its mass and its couplings to the Standard Model — yet neither can establish its cosmological identity [11]. A collider discovery, as noted above, would demonstrate stability only over the time the particle takes to escape the detector, not over the age of the universe. A recoil signal underground fixes only the product of the local density and the scattering cross-section, so it could equally be produced by a subdominant halo component with an enhanced coupling to nuclei. The CMB has the opposite limitation: it probes the annihilation of the cosmological dark matter itself, but its single degenerate amplitude cannot resolve the particle behind the signal. Astrophysical indirect detection bridges the two. An annihilation signal would originate from the dark matter populating galactic halos and cosmological structures — the same population whose gravitational effects are reviewed in Section 1.1. Its intensity is set by the cross-section $\langle \sigma v_{\text{rel}} \rangle$ that determined the relic abundance, while its energy spectrum would discriminate the dark matter mass and annihilation channel. Since it is the main strategy pursued throughout this thesis, we devote the next section entirely to it.

1.4 Indirect Detection via Gamma-Rays

We now turn to indirect detection searches. Indirect detection via gamma-rays directly probes the annihilation cross-section — the quantity that governs the thermal relic abundance and the one quantity that underground laboratories cannot access. This section derives the relevant equations for the gamma-rays flux, which is key to every analysis presented in this thesis.

We first consider the microphysics of annihilation and decay and the gamma-ray spectra they produce (§1.4.1–§1.4.2), then introduce the dark matter density profiles that determine how the signal varies across the sky (§1.4.3). These ingredients combine into the J -factor and D -factor formalism, which factorizes the predicted flux into a particle-physics term and an astrophysical term (§1.4.4). We then survey the observational targets that define the thesis programme (§1.4.5–§1.4.7).

1.4.1 Annihilation and Decay Physics

In the Λ CDM framework, structure forms hierarchically from cold, non-relativistic dark matter, and the particles that populate present-day galactic halos retain typical velocities of order $v \sim 10^{-3} c$ [86]. They therefore annihilate or decay essentially at rest. This kinematic fact has two immediate consequences. First, the total center-of-mass energy available in an annihilation event today is roughly $\sqrt{s} \sim 2m_\chi$, so the most energetic photon that can be produced carries

energy $E_\gamma = m_\chi$. For decay, the available energy is instead m_χ , and momentum conservation among at least two daughter particles limits the maximum photon energy to $E_\gamma = m_\chi/2$. These hard kinematic endpoints are among the most distinctive features of a dark matter signal, since known astrophysical gamma-ray sources do not generically produce spectra with such a sharp cutoff tied to a single mass scale.

The second consequence is more profound and concerns the spatial morphology of the expected signal. Annihilation is a two-body process and its rate per unit volume is therefore proportional to the square of the local dark matter density:

$$\frac{d\Gamma_{\text{ann}}}{dV} = \frac{\langle\sigma v_{\text{rel}}\rangle}{2} \frac{\rho^2}{m_\chi^2}, \quad (1.4.1)$$

where $\langle\sigma v_{\text{rel}}\rangle$ is the thermally averaged annihilation cross-section and the factor of 1/2 avoids double-counting for self-conjugate particles [86]. For Dirac dark matter, where particle and antiparticle are distinct, the factor becomes 1/4, but the thermal cross-section required for the correct relic abundance is correspondingly larger, leaving the observable rate unchanged. Decay, by contrast, is a single-particle process with a rate proportional to the first power of the density:

$$\frac{d\Gamma_{\text{dec}}}{dV} = \frac{\rho}{m_\chi \tau}, \quad (1.4.2)$$

where τ is the dark matter lifetime. Current gamma-ray constraints place τ some ten orders of magnitude beyond the age of the universe (the numbers are collected in Section 1.4.7), making decay an intrinsically rare process whose detection requires enormous integrated exposures. The ρ^2 versus ρ distinction has direct observational consequences: annihilation signals are sharply concentrated in the densest regions of a halo — the inner kiloparsec of the Galactic Center, or the cores of dwarf galaxies — while decay signals are spread diffusely across the entire halo volume, producing a much flatter angular profile. This morphological difference provides a powerful diagnostic: an excess that tracks the ρ^2 profile of the inner Galaxy but vanishes at high latitudes points toward annihilation, while a persistent, broadly distributed signal favours decay.

The stable particles emerging from an annihilation or decay event — photons, neutrinos, electrons, protons, antimatter — are determined by the primary channel into which the dark matter annihilates. In general, one parametrizes the annihilation as $\chi\chi \rightarrow f\bar{f}$ for a specific Standard Model final state f (quarks, leptons, or gauge bosons), and the total gamma-ray spectrum is given by a sum over channels weighted by branching ratios [11, 101, 102]:

$$\frac{dN_\gamma}{dE} = \sum_f \text{BR}_f \frac{dN_\gamma^{(f)}}{dE}. \quad (1.4.3)$$

Quite often, limits are derived for one channel at a time — setting $\text{BR}_f = 1$ for a chosen final state — so each "benchmark channel" maps to a distinctive spectral shape. The spectra $dN_\gamma^{(f)}/dE$ for each channel are computed using Monte Carlo event generators such as Pythia, with the results tabulated in standard libraries such as PPPC 4 DM ID [101] or the more recent CosmiXs [102]. These spectra exhibit qualitatively different shapes depending on f . In hadronic channels such as $b\bar{b}$ or $t\bar{t}$, the quarks hadronize into jets containing many neutral pions, which decay almost instantly via $\pi^0 \rightarrow \gamma\gamma$. The resulting gamma-ray spectrum is broad and soft, peaking at $E_\gamma \sim m_\chi/20$ in $E^2 dN/dE$ units [11]. The $b\bar{b}$ channel has become the standard benchmark in Fermi-LAT analyses for two complementary reasons: the resulting gamma-ray spectrum depends

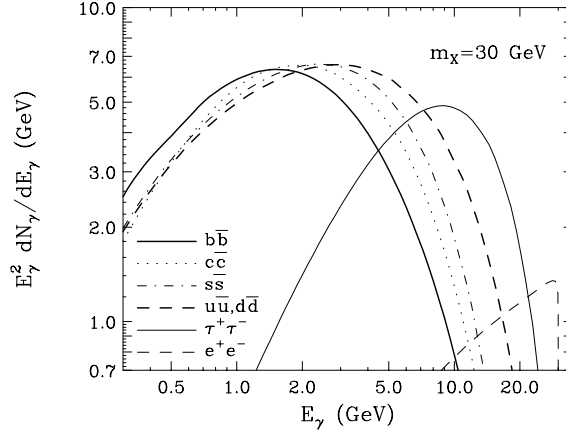


Fig. 1.9. Prompt gamma-ray spectrum dN_γ/dE_γ produced per dark matter annihilation for a 30 GeV WIMP, shown for a variety of Standard Model final states. Hadronic channels ($b\bar{b}$, $c\bar{c}$) produce soft, broad spectra peaked at $E_\gamma \sim m_\chi/20$, while leptonic channels ($\tau^+\tau^-$, $\mu^+\mu^-$) yield harder spectra concentrated near the kinematic endpoint. Figure from Daylan et al. [80].

dominantly on neutral pion decay in the hadronization cascade and is therefore representative of nearly all hadronic annihilation channels; and for typical WIMP masses ($\sim 30\text{--}60$ GeV), the $b\bar{b}$ spectrum peaks at $\sim 1\text{--}3$ GeV — the energy range where Fermi-LAT exhibits its greatest sensitivity [86].

Leptonic channels behave differently. The $\tau^+\tau^-$ final state undergoes hadronic decays that produce fewer but more energetic pions, yielding a harder spectrum peaking at $E_\gamma \sim m_\chi/3$. Lighter leptons (e^+e^- , $\mu^+\mu^-$) do not produce pions at tree level — their gamma-ray spectrum is dominated by final-state radiation and carries far less total flux, concentrated near the kinematic endpoint. Gauge boson channels (W^+W^- , ZZ) produce a combination of both behaviours: the hadronic decays of the W and Z generate a broad low-energy component similar to $b\bar{b}$, while leptonic decays and final-state radiation add a harder tail extending toward $E_\gamma \sim m_\chi$ [86].

The more hadrons produced in the cascade, the softer the spectrum, because the available energy is shared among a larger multiplicity of pions. Channels that produce fewer, more energetic particles yield harder spectra with sharper features. In practice, the distinction between the $b\bar{b}$ and $\tau^+\tau^-$ benchmarks brackets the range of spectral shapes relevant for most thermal relic models, and are often used for reference. These qualitative differences are illustrated in Figure 1.9.

1.4.2 Spectral Features and Signatures

The continuum spectra described above are broad, featureless apart from a cutoff at m_χ , and not unique to dark matter — astrophysical gamma-ray sources produce similar power-law-like spectra through conventional particle acceleration (cf. Section 2.1). The discriminating power of gamma-ray searches therefore lies in spectral features that have no plausible astrophysical counterpart; and unlike charged cosmic rays, whose spectral structures are smeared by propagation through galactic magnetic fields, sharp features in the gamma-ray spectrum propagate intact from source to detector and can be searched for directly [103].

Monochromatic lines. The cleanest such feature is a gamma-ray line at $E_\gamma = m_\chi$, produced when two dark matter particles annihilate directly into a pair of photons, $\chi\chi \rightarrow \gamma\gamma$ [11]. Because dark matter carries no electric charge, this process cannot occur at tree level; it must proceed

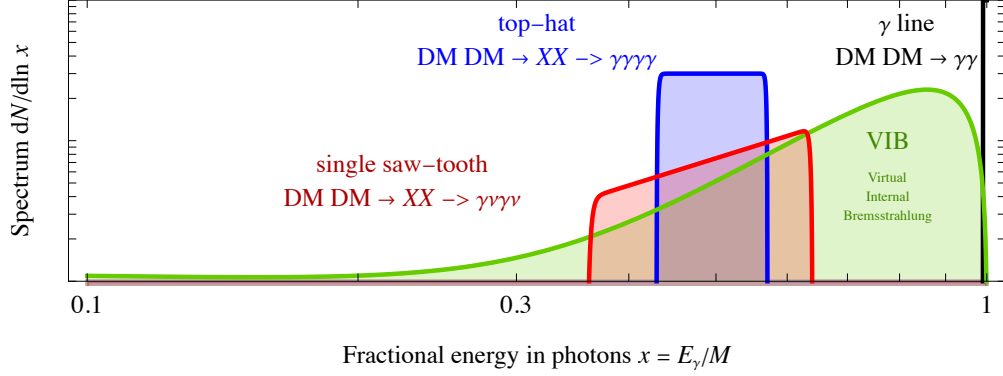


Fig. 1.10. Notable sharp gamma-ray spectral features from dark matter annihilation, as a function of the fractional energy carried by the photon, $x = E_\gamma/m_\chi$. From left to right: the single saw-tooth from $\chi\chi \rightarrow XX \rightarrow \gamma\gamma\gamma$ (red), the “top-hat” (box-shaped) spectrum from cascade decays $\chi\chi \rightarrow XX \rightarrow \gamma\gamma\gamma$ (blue), the “saw-tooth” shape of virtual internal bremsstrahlung (VIB, green), and the monochromatic line from $\chi\chi \rightarrow \gamma\gamma$ at $E_\gamma = m_\chi$ (black). Unlike the case for charged cosmic rays, these sharp features propagate intact to the detector and can be searched for directly. Credit: Cirelli et al. [11].

through quantum loops of charged particles, suppressing the cross-section by a factor of order α^2 relative to annihilation into Standard Model final states [13]. The expected flux is correspondingly small, yet no known astrophysical mechanism produces a sharp line superimposed on a smooth continuum — the monochromatic signal has long been regarded as a “smoking gun” for dark matter [13, 104]. Related processes such as $\chi\chi \rightarrow \gamma Z$ or γh may also produce lines at lower energies, since part of the available centre-of-mass energy goes into the mass of the Z or Higgs boson.

Virtual internal bremsstrahlung (VIB). When a photon is radiated from a charged particle mediating the annihilation, the resulting spectrum acquires a characteristic *saw-tooth* shape: rising steeply toward a sharp cutoff at $E_\gamma \lesssim m_\chi$ [105]. In models where two-body annihilation into light fermions is suppressed, this radiative three-body channel can become the dominant process, making the saw-tooth feature the leading spectral signature [105].

These spectral features are illustrated schematically in Figure 1.10.

Electroweak corrections. When the dark matter mass lies far above the electroweak scale ($m_\chi \gg M_W$), the annihilation products are energetic enough to radiate W and Z bosons readily — much as a fast-moving charge emits photons — and this radiation can no longer be neglected for heavy candidates [106]. Its effect is twofold. First, the subsequent decays of the radiated bosons smear sharp features such as monochromatic lines and add a soft continuum of lower-energy gamma-rays. Second, they feed final states that would otherwise be absent, so that even purely leptonic channels ($\chi\chi \rightarrow e^+e^-, \mu^+\mu^-$) end up producing antiprotons and every annihilation channel contributes to every messenger. These are corrections to the *final state*, reshaping the spectrum produced per annihilation. They should not be confused with the Sommerfeld enhancement of Section 1.2.4, which in this regime is likewise of electroweak origin: for $m_\chi \gg M_W$, the W and Z mediate a long-range force between the slow incoming pair. That effect, however, acts on the initial state and scales the annihilation rate rather than the spectrum.

Prompt versus secondary emission. The gamma-ray production mechanisms discussed

so far — pion decay, final-state radiation, lines, VIB — all occur promptly after their parent annihilation or decay event, and are thus referred to as *prompt emission* [11, 107]. Prompt photons propagate freely through the Galaxy and arrive at the detector with their spectral shape intact, because the Galaxy is effectively transparent to gamma-rays below a few hundred TeV [86].³

Different contributions arise when the energetic electrons and positrons produced by dark matter propagate through the interstellar medium. These leptons upscatter ambient low-energy photons to gamma-ray energies via *inverse Compton scattering* (ICS; see Section 2.1.2) [108]. The target photon field — the interstellar radiation field (ISRF) — consists of three components: the cosmic microwave background, optical starlight, and dust-reprocessed infrared light [109]. As a representative example, a 1 TeV electron upscatters a CMB photon to ~ 1.5 GeV, or a UV starlight photon to ~ 150 GeV [11]. The propagating $e^\pm$ also emit *bremsstrahlung* radiation when deflected by the electric fields of interstellar gas nuclei, a process that dominates their energy losses in the 1–10 GeV range near the Galactic plane where gas densities are highest [103, 108]. These processes produce *secondary* gamma-ray emission that is spatially extended and spectrally softer than the prompt component, and whose morphology depends on the magnetic field structure, gas distribution, and radiation fields of the host galaxy [109]. The same $e^\pm$ population also emits *synchrotron* radiation as it gyrates in the Galactic magnetic field, but for weak-scale dark matter candidates this component peaks at radio and microwave frequencies — a 10 GeV electron in the local $\sim 5 \mu\text{G}$ field radiates near 8 GHz — and therefore does not contribute directly to gamma-ray searches. Secondary emission therefore depends not only on the dark matter distribution but also on the astrophysical environment — magnetic fields, radiation fields, gas densities — that reprocesses the primary electrons and positrons. Despite this additional layer of modelling uncertainty, secondary emission plays a central role in the interpretation of the Galactic Center excess discussed in Chapter 4, where the identification of the signal relies on subtracting the diffuse gamma-ray background dominated by cosmic-ray ICS and bremsstrahlung in the dense inner Galaxy.

1.4.3 Dark Matter Density Profiles

The annihilation and decay rates presented in Section 1.4.1 depend directly on the dark matter density $\rho(r)$ — and for annihilation, on $\rho(r)^2$. The predicted gamma-ray flux from any astrophysical target is therefore directly related to the assumed density profile. The same measurement constraining the particle physics simultaneously probes the dark matter distribution.

The standard profiles come from cosmological N -body simulations which follow the gravitational collapse of billions of dark matter particles from primordial density perturbations to the present-day cosmic web. These simulations show that the spherically averaged density $\rho(r)$ of collapsed halos follows a logarithmic slope $s(r) \equiv d \ln \rho / d \ln r$ that steepens with radius, converging to $s \approx -3$ in the outskirts. The behaviour in the inner region, by contrast, remains an open question — the *cuspy-core problem* discussed below — with different profiles predicting central slopes that range from a steep cusp to a constant-density core (Figure 1.11). These results can be captured by several fitting functions, each with a characteristic density ρ_s and a scale radius r_s .

³This transparency is specific to propagation *within* the Galaxy, where pair production ($\gamma\gamma \rightarrow e^+e^-$) on the interstellar radiation field and the CMB only becomes significant at PeV energies. Over cosmological distances, by contrast, the interaction length on the extragalactic background light drops below the Hubble length already at energies $\gtrsim 100$ GeV, defining an effective gamma-ray horizon for extragalactic sources [86].

NFW (Navarro–Frenk–White). The most widely adopted benchmark has a density that diverges as $\rho \propto r^{-1}$ toward the center (a “cusp”) and falls off as $\rho \propto r^{-3}$ in the outer halo [110]. It was introduced in 1997 by Navarro, Frenk, and White as a fitting function for dark matter halos in N -body simulations of cold dark matter:

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{\frac{r}{r_s} \left(1 + \frac{r}{r_s}\right)^2}, \quad (1.4.4)$$

A common generalization replaces the fixed inner slope with a free parameter γ , giving the “generalized NFW” (gNFW) profile $\rho \propto (r/r_s)^{-\gamma}(1+r/r_s)^{\gamma-3}$ [11]. The standard NFW corresponds to $\gamma = 1$; steeper values ($\gamma > 1$) describe adiabatically contracted halos, while $\gamma < 1$ accommodates shallower cusps. This freedom is directly relevant for the Galactic Center excess analysis in Chapter 4, where the morphology of the signal constrains γ .

Einasto. Originally proposed by Einasto (1965) [111] and later identified as a fitting function for simulated halos, this profile has a logarithmic slope that varies continuously with radius, without converging to a fixed power law at the center. It is controlled by a shape parameter α_{Ein} :

$$\rho_{\text{Ein}}(r) = \rho_s \exp \left\{ -\frac{2}{\alpha_{\text{Ein}}} \left[\left(\frac{r}{r_s} \right)^{\alpha_{\text{Ein}}} - 1 \right] \right\}. \quad (1.4.5)$$

High-resolution simulations seem to favour Einasto over NFW as a fitting function, with $\alpha_{\text{Ein}} \approx 0.17$ providing a good average fit across a range of halo masses [11]. Nonetheless, both NFW and Einasto are commonly used as standard benchmarks.

Burkert (cored). A profile with a constant-density core at small radii, it is motivated by observed rotation curves of dwarf and low-surface-brightness galaxies rather than N -body simulations [27, 112]:

$$\rho_{\text{Bur}}(r) = \frac{\rho_s}{\left(1 + \frac{r}{r_s}\right) \left(1 + \left(\frac{r}{r_s}\right)^2\right)}. \quad (1.4.6)$$

These three profiles are compared in Figure 1.11, which shows both the radial density dependence and the fitted parameters for the Milky Way. At large radii (around $r_\odot$) all profiles agree, but they diverge by orders of magnitude in the inner kiloparsec.

When fit to the Milky Way, the parameters of these profiles are fixed by requiring consistency with the total Galactic mass ($\sim 4.7 \times 10^{11} M_\odot$) and the local dark matter density at the Sun’s position ($r_\odot \simeq 8.33$ kpc) [11, 107]. The most recent determinations place the local density at $\rho_\odot \approx 0.40 \pm 0.10$ GeV/cm³, though many analyses continue to use the historical benchmark of 0.3 GeV/cm³. Characteristic Milky Way parameters are $r_s \approx 14.5$ kpc and $\rho_s \approx 0.57$ GeV/cm³ for NFW, and $r_s \approx 13.6$ kpc with $\rho_s \approx 0.15$ GeV/cm³ for Einasto.

At the solar radius, all three profiles agree by construction. They differ sharply in the inner kiloparsec, where no direct observational data constrain the dark matter profile and the resolution of current simulations reaches its limit (roughly a fraction of a kiloparsec for a Milky Way–sized halo) [27, 110]. The NFW and Einasto profiles predict densities at $r \sim 100$ pc that exceed the Burkert core prediction by one to two orders of magnitude. This is one of the main sources of uncertainty in indirect detection, and the choice of profile propagates directly into the predicted flux, as quantified in Section 1.4.4.

The disagreement between cuspy simulations and cored observations is known as the **cusp–core problem** [11, 27]. Dark matter–only N -body simulations consistently produce cusps with

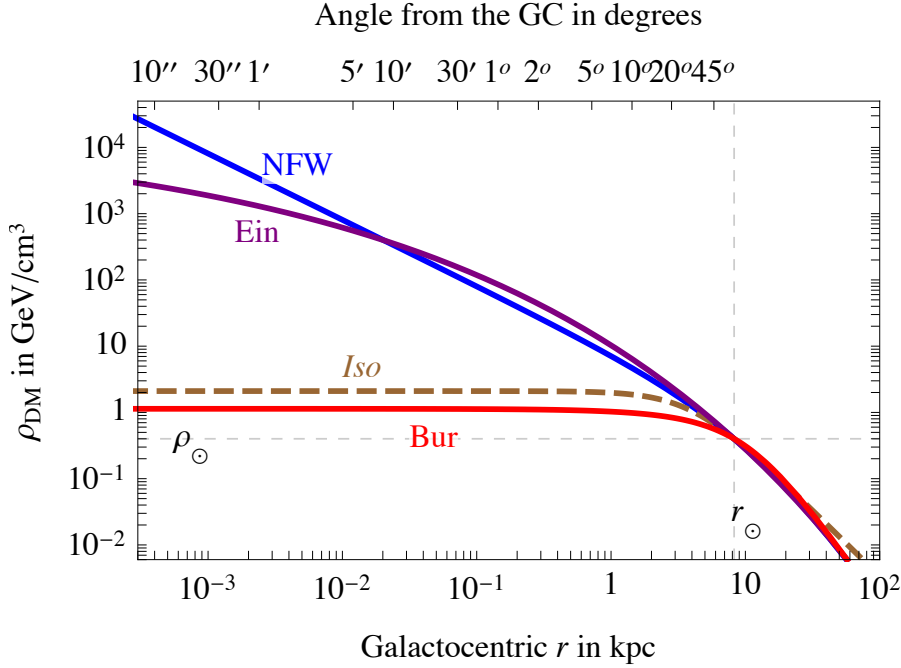


Fig. 1.11. Dark matter density profiles for the Milky Way: NFW (blue), Einasto (purple), Burkert (red), and Isothermal (brown) as a function of galactocentric distance. The profiles are normalized to agree at $r \sim r_\odot$ but diverge by orders of magnitude in the inner kiloparsec. Credit: Cirelli et al. [11].

inner slopes $\gamma \approx 1\text{--}1.5$, while rotation curve measurements — particularly in dwarf galaxies — often favour constant-density cores ($\gamma \approx 0\text{--}0.5$) [27]. Several solutions to this discrepancy have been proposed. From the particle physics side, self-interacting dark matter (introduced in Section 1.2.1), with $\sigma/m \approx 0.5\text{--}10 \text{ cm}^2/\text{g}$, transfers heat to halo centers and produces cores of the observed size [27, 65, 66], while ultralight “fuzzy” dark matter generates solitonic cores through quantum pressure on scales comparable to the de Broglie wavelength [40]. From the astrophysical side, modern hydrodynamical simulations show that repeated bursts of supernova-driven outflows can rapidly alter the central gravitational potential, pushing dark matter outward and flattening a primordial cusp into a core. The inclusion of baryonic feedback has thus considerably relaxed the original tension — cores of the observed size now emerge within plain ΛCDM , without invoking new dark-sector physics — although the problem is not regarded as fully settled, since the predicted core sizes depend on phenomenological sub-grid feedback prescriptions and differ between simulation codes [27].

The effectiveness of baryonic feedback depends non-monotonically on the stellar-to-halo mass ratio $M_\star/M_{\text{vir}}$: core formation peaks at intermediate ratios ($M_\star/M_{\text{vir}} \approx 5 \times 10^{-3}$, the regime of bright dwarfs), where supernovae are frequent enough to redistribute dark matter, but is negligible in ultrafaint dwarfs (too few stars) and in massive galaxies where adiabatic contraction recreates a cusp [27]. For the Milky Way ($M_\star/M_{\text{vir}} \approx 0.08$), this would place the expected inner profile in the cuspy regime, although the argument rests on the assumption that feedback and contraction have been correctly modelled, which remains a field of active research with large uncertainties.

Substructure and boost factors. Real dark matter halos are not smooth. N -body simula-

tions reveal a rich hierarchy of gravitationally bound subhalos embedded within the main halo, with a mass function $dn/dM \propto M^{-2}$ extending from cluster masses down to the resolution limit [11, 113, 114]. Because annihilation scales as ρ^2 , the clumpy density field produces a higher total signal than a smooth halo of the same average density. This follows from Jensen’s inequality applied to the convex function ρ^2 : for any non-uniform density distribution, the volume-averaged squared density satisfies $\langle \rho^2 \rangle > \langle \rho \rangle^2$, with equality only for a perfectly homogeneous halo. This enhancement is parametrized by a multiplicative boost factor $B \equiv \langle \rho^2 \rangle / \langle \rho \rangle^2$.

Early estimates of the boost factor ranged from modest ($B \sim 10$) to extreme ($B \sim 10^9$) [115–118], but modern analyses find much smaller values [119, 120]. The survival of subhalos is governed by tidal stripping from the host halo’s gravitational field, gravitational “disk shocking” as subhalos cross the dense baryonic disk, and encounters with individual stars — all of which preferentially destroy subhalos in the inner Galaxy [120]. When these effects are included, the boost factor for the Galaxy as a whole is $B \lesssim \mathcal{O}(10) - \mathcal{O}(30)$ [120, 121]. Toward the Galactic Center, where subhalos are most heavily depleted by tidal forces, the boost is expected to be negligible [120]. The outer halo, by contrast, retains a richer subhalo population, since tidal forces weaken with galactocentric distance. These surviving subhalos — seen preferentially at high galactic latitudes — are the targets of the searches described in Chapter 5.

The subhalo hierarchy does not extend to arbitrarily small masses. It is truncated at a minimum mass $M_{\min}$ set by the kinetic decoupling of the dark matter particle from the thermal plasma in the early universe. After kinetic decoupling, dark matter particles free-stream over a characteristic length λ_{fs} , erasing density perturbations below the corresponding mass scale $M_{\min} \simeq (4\pi/3)\rho_{\text{DM}}\lambda_{\text{fs}}^3$. For a 100 GeV WIMP with weak-scale couplings, $M_{\min} \sim 10^{-6} M_{\odot}$ — roughly an Earth mass [11]. This scale lies far below the resolution of any current simulation and represents a fundamental uncertainty in the predicted total annihilation rate, since the integral over the subhalo mass function is dominated by the smallest masses.

The density profiles and substructure properties introduced here feed directly into the J -factor and D -factor formalism of Section 1.4.4.

1.4.4 The Flux Factorization: J -Factor and D -Factor

Combining the particle physics of Section 1.4.1 with the density profiles of Section 1.4.3, we can now write the differential gamma-ray flux from dark matter annihilation observed from a direction ψ relative to the Galactic Center [11]:

$$\frac{d\Phi_{\gamma}}{dE d\Omega} = \underbrace{\frac{1}{4\pi} \frac{\langle \sigma v_{\text{rel}} \rangle}{2m_{\chi}^2} \frac{dN_{\gamma}}{dE}}_{\text{Particle Physics}} \times \underbrace{J(\psi)}_{\text{Astrophysics}}. \quad (1.4.7)$$

The prefactor $1/(4\pi)$ reflects the isotropic emission of photons over the full sphere, and the factor of $1/2$ avoids double-counting for self-conjugate (Majorana) particles, as discussed in Section 1.4.1. The J -factor is the line-of-sight integral of the density squared:

$$J(\psi) = \int_{\text{l.o.s.}} \rho^2(r(s, \psi)) ds, \quad (1.4.8)$$

where s parametrizes the distance along the line of sight and r is the galactocentric radius, related to s and the observation angle ψ by $r = (r_{\odot}^2 + s^2 - 2r_{\odot}s \cos \psi)^{1/2}$, with $r_{\odot} \simeq 8.33$ kpc the Sun–Galactic Center distance. In this dimensionful convention, J carries units of $\text{GeV}^2 \text{ cm}^{-5}$. An alternative dimensionless definition, in which ρ and s are normalized to $\rho_{\odot}$ and $r_{\odot}$ respectively, is also often used in the literature.

For decaying dark matter, the analogous expression is [107]:

$$\frac{d\Phi_\gamma}{dE d\Omega} = \frac{1}{4\pi} \frac{\Gamma}{m_\chi} \frac{dN_\gamma}{dE} \times D(\psi), \quad (1.4.9)$$

where $\Gamma = 1/\tau$ is the decay rate. No symmetry factor appears, since decay is a single-particle process. The D -factor is:

$$D(\psi) = \int_{\text{l.o.s.}} \rho(r(s, \psi)) ds, \quad (1.4.10)$$

and carries units of GeV cm^{-2} . Because D depends on ρ rather than ρ^2 , it is far less sensitive to the assumed density profile than the J -factor, making decay searches less model-dependent than their annihilation counterparts. Beyond this technical difference, the two processes are interesting for different reasons. Annihilation is the characteristic signature of thermal-relic WIMPs, whose present-day cross-section is tied to the observed relic abundance (Section 1.2) [70], and the ρ^2 weighting makes the densest concentrations of dark matter — the Galactic Center and individual subhalos — the most rewarding targets. Decay, by contrast, is the natural channel for feebly-coupled or non-thermal candidates such as sterile neutrinos [42, 52] and axion-like particles [122], whose only requirement is a lifetime exceeding the age of the universe; the linear ρ weighting instead favours large, diffuse targets that maximise the enclosed mass — the outer Galactic halo, galaxy clusters, the isotropic background [123], and even cosmic-web filaments, where an annihilation signal would be suppressed [107].

The particle physics factor — containing m_χ , $\langle \sigma v_{\text{rel}} \rangle$ (or τ), and the spectral shape dN_γ/dE — encodes the dark matter model we wish to test. The astrophysical factor — J or D — encodes the distribution of dark matter along the line of sight, determined by the density profile of the host halo and constrained independently through kinematic measurements. A non-detection translates directly into an upper limit on $\langle \sigma v_{\text{rel}} \rangle$ for annihilation; and an upper limit on Γ (equivalently, a lower limit on τ) for decay.

When applied to an extended region of the sky, the integrated flux is obtained by integrating J (or D) over the solid angle of interest. For comparing different targets, one typically quotes the J -factor integrated over a cone of half-angle $\theta_\star$ centered on the target:

$$J_{\text{int}}(\theta_\star) = \int_0^{\theta_\star} J(\psi) 2\pi \sin \psi d\psi. \quad (1.4.11)$$

For dwarf spheroidal galaxies observed by Fermi-LAT, typical integrated J -factors (within $\theta_\star = 0.5^\circ$) range from $\sim 10^{16}$ to $\sim 10^{19.5} \text{ GeV}^2 \text{ cm}^{-5}$, depending on the system's mass, distance, and the quality of the kinematic data used to reconstruct its density profile [86, 124]. The choice of integration angle reflects the angular extent of the target: dSphs are resolved by Fermi-LAT and extended emission can be recovered within $\sim 0.5^\circ$, while dark matter subhalos at typical Galactic distances are unresolved and effectively point-like, so their J -factors are integrated within $\theta_\star \sim 0.1^\circ$, matching the instrument's point-spread function. These values are orders of magnitude smaller than the Galactic Center J -factor (cf. the signal-versus-background comparison in Section 1.4.5).

Several caveats affect the simple factorization above. First, it assumes a *velocity-independent* annihilation cross-section. If $\langle \sigma v_{\text{rel}} \rangle$ depends on the dark matter velocity — as it does for p -wave annihilation ($\sigma v \propto v^2$) or Sommerfeld-enhanced models ($\sigma v \propto 1/v$) — the cross-section can no longer be factored out of the line-of-sight integral [11]. In such cases, one must compute an *effective J -factor* that folds the full, position-dependent velocity distribution into the integral, and the clean particle-physics/astrophysics separation is lost.

Second, even for velocity-independent cross-sections, the J -factor depends quadratically on the density profile, making it extremely sensitive to the uncertainties discussed in Section 1.4.3. For the Galactic Center, the J -factor computed with an NFW profile can exceed the Burkert value by orders of magnitude; even within the NFW paradigm, varying the inner slope from $\gamma = 0.7$ to $\gamma = 1.3$ changes the predicted flux by more than an order of magnitude [86]. The astrophysical factor is therefore the dominant systematic uncertainty in annihilation searches for a given dark matter model.

1.4.5 Observational Targets for Gamma-Ray Searches

To maximise the gamma-ray signal from dark matter, one should observe targets with large J -factors (for annihilation) or D -factors (for decay) and low astrophysical backgrounds. No single target satisfies both criteria simultaneously, and the field has consequently developed a multi-target observation programme in which the complementary strengths of different targets compensate each other's weaknesses. We now review the principal targets, noting their advantages, limitations, and connection to the analyses developed in this thesis.

The Galactic Center. The center of the Milky Way offers the largest J -factor of any astronomical source, owing to its proximity (~ 8 kpc) and the expected steep rise of the dark matter density at small radii [86]. In the inner $40^\circ \times 40^\circ$ region, the physical J -factor is approximately $J \sim 10^{23} \text{ GeV}^2 \text{ cm}^{-5} \text{ sr}$ [125]. For an NFW profile with $\gamma = 1$ and a local density $\rho_\odot = 0.4 \text{ GeV/cm}^3$, the predicted flux from the inner 10° exceeds that from any individual dwarf galaxy by more than three orders of magnitude. If a steeper inner slope ($\gamma \approx 1.3$) is assumed, the flux increases by an additional factor of ~ 5 ; for a shallower slope ($\gamma \approx 0.7$) it decreases by ~ 3 . In either case, the Galactic Center remains the brightest expected source of dark matter annihilation products on the sky.

The advantage is offset by severe astrophysical contamination. The inner few degrees of the Galaxy feature a bright diffuse emission from cosmic-ray protons and electrons scattering off interstellar gas (through pion production and bremsstrahlung) and ambient radiation fields (through inverse Compton scattering), together with numerous point sources — supernova remnants, pulsars, and the central supermassive black hole [86]. Models of this diffuse foreground (see Section 2.1.3) are constructed from gas maps and the locally measured cosmic-ray spectrum, and while they capture the broad features of the observed emission, they cannot account for its detailed spectral and morphological structure. This imperfect background knowledge is the primary obstacle to extracting a dark matter signal from the Galactic Center.

Despite these difficulties, the Fermi-LAT has detected a statistically significant excess of gamma-rays from the inner Galaxy at energies between ~ 0.5 and 5 GeV — the so-called Galactic Center excess (GCE) [80, 126, 127]. The spectrum and approximately spherical morphology of this excess are consistent with the predictions for a ~ 35 – 50 GeV dark matter particle annihilating into $b\bar{b}$ quarks at a rate close to the thermal relic value [80]. The dark matter interpretation is compelling, but an unresolved population of millisecond pulsars in the Galactic bulge provides an equally plausible astrophysical explanation. Multiple statistical analyses have investigated the point-source versus diffuse origin of the excess, reaching contradictory conclusions. The morphological debate — whether the excess traces a spherical dark matter halo or the stellar distribution of the Galactic bulge — remains unresolved. This persistent ambiguity motivates the dedicated analysis in Chapter 4, which examines the millisecond pulsar hypothesis by constraining the pulsar luminosity function from globular cluster observations.

Dwarf spheroidal galaxies. The Milky Way's satellite dwarf galaxies occupy the opposite end of the signal-versus-background trade-off. Dwarf spheroidals are among the most dark

matter-dominated objects known (Section 1.1.1): their shallow gravitational potential wells allowed early supernova feedback to expel most of the baryonic gas, halting star formation and leaving behind systems with extreme mass-to-light ratios. The scarcity of stars, gas, and other active astrophysical objects means that their gamma-ray backgrounds are subdominant [11]. The expected dark matter flux from any individual dwarf is far smaller than from the Galactic Center, as quantified above, but the absence of much competing emission makes them the cleanest known targets available.

Eight “classical” dwarf spheroidals (Sculptor, Fornax, Leo I and II, Ursa Minor, Draco, Carina, and Sextans) have been known since before the 1990s. The advent of digital surveys — SDSS, DES, and Gaia — has brought the total to about fifty confirmed ultrafaint dwarfs, with roughly ten more candidates awaiting spectroscopic confirmation [128]. This growing census is expected to continue with the Rubin Observatory.

By combining observations of dozens of these systems in a stacked analysis, the Fermi-LAT collaboration has derived some of the tightest constraints on the annihilation cross-section to date [129, 130]. These stacked analyses exclude the thermal relic cross-section for dark matter masses below ~ 100 GeV in the $b\bar{b}$ channel; the current numbers, and their evolution from 15 to 50 dwarfs, are collected in Section 1.4.7 [129, 130]. The dominant systematic uncertainty lies in the J -factors themselves, which must be inferred from the kinematics of the dwarfs’ constituent stars. For ultrafaint systems, the number of kinematic tracers can be extremely small, and removing or adding a single star from the sample can shift the reconstructed J -factor by orders of magnitude [11]. Additional systematics arise from the cusp–core ambiguity in the inner density profile and from the choice of integration angle $\theta_\star$. Despite these challenges, dwarfs remain the gold standard for model-independent constraints. Chapter 5 of this thesis searches for dark matter subhalos — which may include unassociated dwarf satellites — among the unidentified Fermi-LAT sources.

Dark subhalos. Λ CDM simulations predict far more gravitationally bound subhalos within the Milky Way’s halo than the ~ 50 known satellite galaxies can account for [27, 113]. Most of these subhalos are expected to be entirely dark — devoid of stars and gas — yet some of them may be sufficiently massive to produce a detectable gamma-ray flux if the annihilation cross-section is near the thermal relic value. Such dark subhalos would appear in the Fermi-LAT data as unassociated point sources with no counterpart at other wavelengths. Their positions are unknown a priori, turning the search into a classification problem: distinguishing rare dark matter subhalos from the much larger population of unidentified astrophysical sources (primarily pulsars and faint blazars). This challenge is addressed in Chapter 5, where machine learning methods are applied to search for unassociated Fermi-LAT sources in the presence of dataset shift between simulations and real data.

The unresolved gamma-ray background. The cumulative, nearly isotropic gamma-ray flux that remains after subtracting resolved point sources and Galactic diffuse emission constitutes the unresolved gamma-ray background (UGRB) [131]. Chapters 6 and 7 develop methods to characterize the unresolved source populations that compose the UGRB. This unresolved background receives contributions from unresolved blazars (BL Lac objects and flat-spectrum radio quasars), misaligned active galactic nuclei, star-forming galaxies, and millisecond pulsars (see Section 2.2) [131]. Unresolved BL Lac objects alone can account for the majority of the UGRB above ~ 100 GeV, while at lower energies other populations are required [131]. A dark matter component — originating from annihilation or decay in all halos and subhalos across cosmic history — may also be present, but it is expected to be subdominant in the total intensity spectrum [131].

Because the dark matter signal is potentially hidden beneath dominant astrophysical foregrounds, its extraction requires methods that go beyond the energy spectrum alone. The angular anisotropies of the UGRB carry additional information: different source classes imprint different patterns of spatial fluctuations because their clustering properties, redshift distributions, and luminosity functions differ. Cross-correlating these anisotropies with independent gravitational tracers of the large-scale structure — galaxy catalogs, cosmic shear from weak lensing surveys, or 21 cm intensity mapping of neutral hydrogen — can isolate the component that traces the dark matter distribution [107, 131]. The approach is particularly powerful because dark matter annihilation scales as ρ^2 and has a redshift evolution distinct from that of astrophysical gamma-ray emitters, providing a spectral and tomographic handle for separation. This cross-correlation technique forms the basis of the analysis in Chapter 8, where the sensitivity of next-generation instruments to the dark matter cross-correlation signal is forecast.

The complementary strengths of these targets motivate the multi-target programme developed in Parts II through IV of this thesis. Each target exposes different aspects of the dark matter parameter space and is sensitive to different systematic uncertainties.

1.4.6 Multi-Messenger Context

We argued that, if dark matter annihilates or decays, it could produce stable Standard Model particles, not only gamma-rays. Each messenger would open a complementary window onto the dark sector, with its own advantages and limitations.

Neutrinos share the key advantage of photons — having no electrical charge — and therefore propagate in straight lines, preserving directional information [11]. They can also escape environments opaque to photons: dark matter captured in the solar core, for instance, annihilates into neutrinos that are the only annihilation products to reach the surface, making high-energy solar neutrinos a probe of the dark matter–nucleon scattering cross-section. Searches by IceCube, ANTARES, and Super-Kamiokande have found no signal, but for dark matter masses below a few hundred GeV they set bounds on the spin-dependent cross-section roughly one order of magnitude stronger than direct detection experiments [132]. For galactic targets, however, neutrino constraints remain generally weaker than the corresponding gamma-ray bounds, becoming competitive only above roughly 1–10 TeV depending on the annihilation channel.

Charged cosmic rays — electrons, positrons, antiprotons, and heavier anti-nuclei — suffer from a fundamental limitation: galactic magnetic fields randomize their arrival directions, erasing all spatial information and leaving only the energy spectrum as the main observable [11, 107]. The positron fraction measured by PAMELA and AMS-02 rises steeply above 10 GeV [133], but fitting it with dark matter requires a cross-section three orders of magnitude above the thermal relic value, in severe tension with gamma-ray and CMB constraints; the excess is now widely attributed to nearby pulsars. AMS-02 antiproton data show a tentative excess at 10–20 GeV compatible with an ~ 80 GeV dark matter particle annihilating into $b\bar{b}$ at roughly the thermal cross-section [134]. These parameters overlap with those invoked for the Galactic Center excess (Chapter 4), but the significance drops below $\sim 2\sigma$ once systematic uncertainties in the propagation model and production cross-sections are included. Low-energy anti-deuterons and anti-helium have generally low astrophysical backgrounds, so their detection would be a qualitatively different signature. AMS-02 and the upcoming GAPS experiment may reach the sensitivity needed to probe the thermal relic cross-section [86].

CMB — An independent and increasingly powerful constraint comes from the CMB, whose physics we introduced in Section 1.3: dark matter annihilation during and after recombination

injects energy into the photon–baryon plasma [99], distorting the temperature and polarisation anisotropies in a way that is tightly bounded by Planck data [9]. For s -wave annihilation, these bounds exclude the thermal relic cross-section below roughly 30 GeV in leptonic channels and 10–15 GeV in hadronic ones [9, 99] — limits that are largely independent of astrophysical modelling and therefore highly complementary to gamma-ray searches.

1.4.7 The Status of the Field

Despite decades of increasingly sensitive searches across all three detection strategies, no unambiguous signal of dark matter has been found. The current status is best summarized as a set of progressively tightening bounds on the cross-section and decay lifetime, with a few anomalies whose dark matter interpretations have been challenged by plausible astrophysical alternatives.

Gamma-ray constraints. The tightest indirect detection limits come from the Fermi-LAT stacked analysis of Milky Way dwarf spheroidal galaxies. The original combined analysis of 15 dSphs with six years of data excluded the thermal relic cross-section for dark matter masses in the range ~ 10 –100 GeV in the $b\bar{b}$ channel [129]; below ~ 10 GeV the Fermi-LAT effective area drops steeply and the constraints weaken. The most recent legacy analysis extends the sample to 50 dSphs and 14 years of Fermi-LAT exposure, confirming and slightly strengthening these constraints [130]; no significant excess is found in the combined likelihood analysis, though a marginal (2 – 3σ local) excess near $M_\chi \approx 150$ –230 GeV in the $b\bar{b}$ channel has motivated projections for future observations [130]. For a generic annihilation channel, marginalizing over branching ratios yields a model-independent lower limit of $m_\chi \gtrsim 20$ GeV [11]. Constraints on the dark matter lifetime, derived primarily from the isotropic gamma-ray background and Milky Way halo observations, reach $\tau \gtrsim 10^{27}$ – 10^{28} seconds depending on the decay channel and dark matter mass — about ten orders of magnitude longer than the age of the universe [135, 136].

Direct detection and collider constraints. The current generation of dual-phase liquid xenon detectors — LZ, XENONnT, and PandaX-4T — has pushed the spin-independent WIMP–nucleon cross-section below 10^{-46} cm² for masses near the electroweak scale (see Section 1.3 for the experiments and bounds). At the LHC, mono-jet and mono-photon searches have found no evidence for new physics, excluding strongly interacting superpartners (squarks, gluinos) up to masses of 1–2 TeV and charged superpartners up to several hundred GeV.

Unresolved anomalies. Several persistent anomalies keep the possibility of discovery alive. The Galactic Center excess, first reported in 2009, remains the most debated (its spectrum and morphology are described in Section 1.4.5) [11, 80]. Chapter 4 of this thesis contributes directly to this debate by challenging the leading astrophysical interpretation of the excess as an unresolved population of millisecond pulsars. Other excesses, by contrast, admit a more satisfying astrophysical explanation. The charged cosmic-ray anomalies discussed in Section 1.4.6 — the AMS-02 antiproton signal and the rise in the positron fraction — remain consistent with conventional astrophysical sources once systematic uncertainties are fully accounted for [86]. A similar pattern extends beyond gamma-rays: the 511 keV positron-annihilation line from the Galactic bulge admits light dark-matter interpretations alongside plausible astrophysical ones; the 3.5 keV X-ray line, once attributed to sterile-neutrino decay, is now disfavoured; and the DAMA/LIBRA annual modulation remains unconfirmed by experiments with the same NaI target [11].

Small-scale structure tensions. A different class of problem concerns the small-scale predictions of Λ CDM itself. Alongside the cusp–core problem of Section 1.4.3 and the “missing satellites” deficit — now largely resolved by the low galaxy-formation efficiency of small halos and the discovery of ultrafaint dwarfs — several tensions remain open: the observed bright Milky Way

satellites are less centrally dense than the most massive subhalos predicted by dark-matter-only simulations, which, being too big for star formation to fail, should host precisely such galaxies (*too-big-to-fail*); dwarfs of similar maximum circular velocity display remarkably different inner profiles (the *diversity of rotation curves*); and the satellites of the Milky Way and Andromeda align in thin, apparently corotating planes that simulations struggle to reproduce. These tensions motivate self-interacting (Section 1.2.1), but remain entangled with baryonic feedback modelling: for now they constrain the nature of dark matter rather than reveal it [27, 66].

The methodological case for this thesis. Taken together, these results define the landscape in which this thesis is situated. The viable parameter space for the simplest thermal WIMP models has narrowed substantially — what Cirelli et al. describe as a field “at a turning point” [11]. The standard approach to indirect detection — searching for a bright, spectrally distinctive excess from a single target above the astrophysical background — has not yet delivered an unambiguous discovery. Nonetheless, the thermal relic benchmark is not the prediction of a specific model but a cosmological target: if dark matter is a thermal relic, the annihilation cross-section must have the value that produces the observed abundance, regardless of the underlying theory. Current exclusions cover only a portion of the mass range and a subset of annihilation channels; heavier candidates, leptonic final states, and velocity-dependent cross-sections remain largely unexplored. Extracting these signals demands methods that go beyond simple signal-over-background thresholding.

The Galactic Center excess demands careful modelling of the interstellar emission and the unresolved point-source population — distinguishing dark matter annihilation from millisecond pulsars requires statistical tools, not a single spectral fit (Chapter 4). Individual dark matter subhalos among the Fermi-LAT unassociated sources cannot be identified by classification alone; a fully probabilistic model that accounts for both covariate and prior probability shifts between associated and unassociated sources is needed to derive statistically rigorous constraints (Chapter 5). Below the point-source detection threshold, deep learning applied to gamma-ray sky maps can recover the source-count distribution of faint populations down to fluxes a factor of 50 below the Fermi-LAT sensitivity limit, while probabilistic catalogues extend spatial source information into the sub-threshold regime (Chapters 6 and 7). Finally, at the largest scales, the dark matter contribution to the unresolved gamma-ray background can potentially be isolated through its cross-correlation with independent tracers of the large-scale structure (Chapter 8).

THE GAMMA-RAY SKY AND FERMI-LAT

The gamma-ray sky is rich and complex, dominated by diffuse emission that traces the interstellar gas of the Milky Way, punctuated by thousands of individual point sources, and underlaid by a faint, isotropic extragalactic glow. For indirect searches for dark matter, these astrophysical gamma rays constitute the primary background. Extracting a faint dark matter signal—whether from annihilation at the Galactic Center or from the collective emission of unresolved subhalos—requires us to understand where these background photons originate and how our instruments record them.

Standard methodologies successfully classify the brightest objects and map the prominent diffuse components, but they struggle with overlapping populations and instrumental constraints such as finite spatial resolution and detection thresholds. These challenges obscure the faintest dark matter signatures and the broader population of unresolved sources. Distinctly isolating a dark matter signal from this background therefore requires novel statistical and machine learning techniques capable of modeling complex systematic uncertainties.

In this chapter we establish the astrophysical and instrumental framework on which the analyses in the remainder of this thesis is based. Section 2.1 reviews the fundamental production mechanisms of high-energy gamma rays, detailing the hadronic and leptonic processes that generate the Galactic foreground. Section 2.2 surveys the dominant source populations, progressing from Galactic entities such as millisecond pulsars to extragalactic classes like blazars, and formalizes the source-count distribution and its connection to the isotropic background. Finally, Section 2.3 describes the *Fermi* Large Area Telescope, focusing on the instrumental response functions, detection thresholds, and data products that define the practical limits of current observations. The physical complexity and instrumental limitations introduced here constitute the primary systematic uncertainties that we address using probabilistic methods and simulation-based inference in the following chapters.

2.1 Gamma-Ray Production Mechanisms

This section reviews the standard gamma-ray production mechanisms responsible for the Galactic diffuse gamma-ray emission. These processes fall mainly into two broad categories: hadronic in-

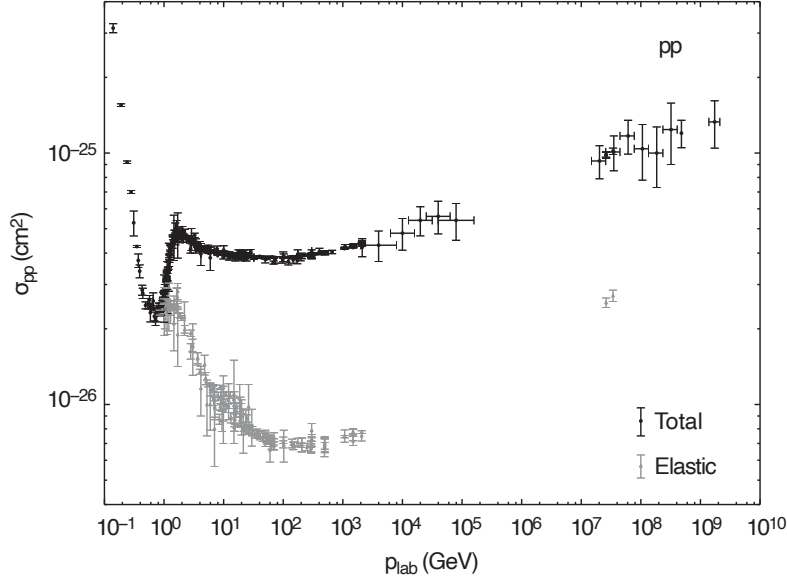


Fig. 2.1. Measurements of the total and inelastic proton-proton cross section, where the so-called pion bump is clearly visible. Adapted from [39].

interactions involving cosmic-ray nucleons colliding with interstellar gas, and leptonic interactions involving energetic electrons and positrons scattering off radiation fields and gas.

Both families of processes operate within the *interstellar medium* (ISM), the mix of gas, dust, radiation, magnetic fields, and cosmic rays that fills the space between stars. The ISM is mainly made of gas—roughly three-quarters hydrogen and one-quarter helium, with heavier elements and dust particles together contributing less than a percent. This gas supplies the collision targets for the hadronic interactions and bremsstrahlung discussed below, while the *interstellar radiation field*—starlight, infrared emission, and the CMB—provides the seed photons for inverse Compton scattering.

2.1.1 Hadronic Emission

High-energy cosmic-ray protons and heavier nuclei interact with the interstellar medium through the strong nuclear force, primarily colliding with ambient hydrogen gas. These inelastic proton-proton collisions produce secondary particles, most notably pions:

$$p + p \rightarrow \pi^0 + \pi^\pm + X. \quad (2.1.1)$$

The reaction is subject to a strict kinematic threshold. A cosmic-ray proton colliding with a stationary target proton must have a total energy of at least $E_p > m_p + 2m_\pi + m_\pi^2/2m_p \approx 1.2$ GeV in order to produce one or more pions [86]. Once this threshold is crossed, the total inelastic cross section increases only mildly with energy, rising from $\sigma_{\text{inel}}^{pp} \approx 30\text{--}40$ mb at $E_p \lesssim 10$ GeV to $\sigma_{\text{inel}}^{pp} \approx 100$ mb near $E_p \sim 1$ EeV. Because a significant fraction of the proton's kinetic energy is transferred in each collision (inelasticity $K_{pp} \sim 0.5$), only a few successive interactions suffice to extract most of the cosmic ray's original energy [103].

While charged pions ($\pi^\pm$) decay into muons and neutrinos, neutral pions decay electromagnetically with a mean lifetime of only $\tau_{\pi^0} \approx 8.4 \times 10^{-17}$ s:

$$\pi^0 \rightarrow \gamma\gamma. \quad (2.1.2)$$

In the rest frame of the π^0 , this decay produces two photons of equal energy $E_\gamma = m_{\pi^0}/2 \approx 67.5$ MeV. In the laboratory frame, however, a pion moving with energy E_π generates a gamma-ray spectrum that is flat between the kinematic bounds $(E_\pi - \sqrt{E_\pi^2 - m_\pi^2})/2$ and $(E_\pi + \sqrt{E_\pi^2 - m_\pi^2})/2$, and vanishes outside this interval [86]. This flat distribution has a remarkable consequence: regardless of the spectral shape of the parent pion population, the resulting gamma-ray spectrum—when plotted as $E^2 dN/dE$ —always peaks at $m_{\pi^0}/2 \approx 67.5$ MeV. This universal spectral feature, known as the *pion bump*, is the characteristic spectral signature of hadronic cosmic-ray interactions with gas and has been observed in both supernova remnants and the Galactic Plane emission [137], see Figure 2.1.

Hadronic emission dominates the Galactic Diffuse Emission along the Galactic Plane, where the column density of interstellar gas is highest. As a result, it constitutes the primary astrophysical background for any dark matter search targeting the inner Galaxy, with consequences we quantify in Section 2.1.3. While hadronic emission dominates wherever gas is abundant, leptonic processes—inverse Compton scattering, bremsstrahlung, and synchrotron radiation—provide the complementary diffuse components, particularly at high Galactic latitudes.

2.1.2 Leptonic Emission

Relativistic electrons and positrons produce high-energy photons through three primary mechanisms: inverse Compton scattering, bremsstrahlung, and synchrotron radiation. These processes operate across different energy regimes and target environments, and their relative importance depends on the local astrophysical conditions along each line of sight.

Inverse Compton scattering. Relativistic electrons can up-scatter low-energy ambient photons—the Cosmic Microwave Background (CMB), infrared dust emission, and starlight—into the X-ray and gamma-ray bands [108]. The behavior of this process depends critically on the ratio of the electron energy to the target photon energy ϵ . In the Thomson regime ($E_e \ll m_e^2/\epsilon$), a given electron up-scatters photons to a maximum energy of $E_\gamma^{\max} \approx 4\epsilon E_e^2/m_e^2$. The quadratic dependence on E_e implies that higher-energy electrons are vastly more efficient radiators, which steepens the resulting photon spectrum relative to the parent electron population. Quantitatively, a power-law electron spectrum with index α ($dN_e/dE_e \propto E_e^{-\alpha}$) produces a gamma-ray spectrum with a softer index $-(\alpha + 1)/2$ [86]. In this regime, the ICS energy loss rate takes a form mathematically identical to that of synchrotron radiation:

$$\left. \frac{dE_e}{dt} \right|_{\text{ICS}}^{\text{Thomson}} = -\frac{4}{3} \sigma_T \rho_{\text{rad}} \gamma_e^2 c, \quad (2.1.3)$$

where ρ_{rad} is the energy density of the target photon field.

At highly relativistic energies ($E_e \gg m_e^2/\epsilon$), the interaction transitions into the Klein-Nishina regime [86]. Here the scattering cross section drops below the Thomson value, and the energy loss rate acquires an additional suppression factor of $\sim m_e^4/(E_e^2 T^2)$ relative to the Thomson prediction, where T is the effective temperature of the target radiation field. Consequently, the scattered gamma rays carry away nearly all the incoming electron's energy ($E_\gamma \approx E_e$), and the differential cross section peaks sharply just below E_e rather than being spread across a broad range. This transition from the Thomson to the Klein-Nishina regime is illustrated in Figure 2.2.

In practice, cosmic-ray electrons scatter off starlight, infrared dust emission, and the CMB. At higher electron energies, Klein-Nishina suppression progressively reduces the efficiency of scattering off higher-frequency photons, shifting the dominant ICS target toward the CMB.

Bremsstrahlung. In gas-rich regions of the interstellar medium, electrons undergo bremsstrahlung as they decelerate in the Coulomb fields of neutral and ionized atomic gas. The

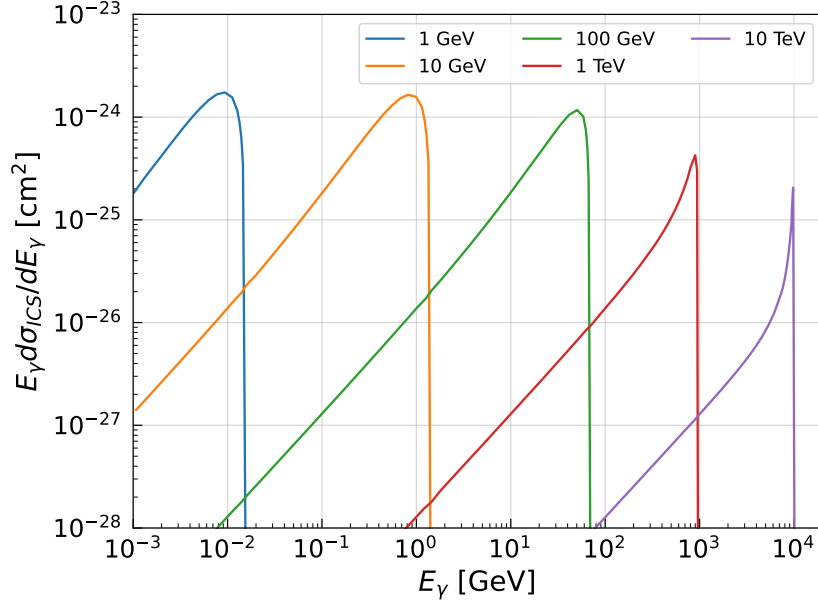


Fig. 2.2. The differential cross section for inverse Compton scattering of a high-energy electron with a 1 eV target photon, for several values of E_e . At high energies ($E_e \gg m_e^2/c$), the process enters the Klein-Nishina regime: the cross section is severely suppressed and the scattered photon carries nearly all the electron’s energy.

bremsstrahlung cross section scales with the square of the target nuclear charge ($\propto Z^2$), and hydrogen and helium are the dominant targets in the ISM. A distinctive feature of bremsstrahlung is that its energy loss rate scales linearly with the electron energy at leading order ($b_{\text{brem}} \propto E_e$) [107], resulting in an energy loss timescale that is effectively independent of E_e . As a consequence, the gamma-ray spectrum produced through bremsstrahlung closely mirrors the spectral shape of the parent electron population [86]. This process is the dominant cooling mechanism for cosmic-ray electrons in the MeV to GeV range in gas-dense environments such as the Galactic Plane, though it is broadly subdominant to hadronic emission in the same regions.

Synchrotron radiation. In the presence of the Galactic magnetic field ($B \sim \text{few } \mu\text{G}$), ultra-relativistic electrons gyrate around magnetic field lines and emit synchrotron radiation. This process primarily generates photons in the radio and microwave bands, but it traces the same underlying electron populations that produce gamma rays via ICS and bremsstrahlung. The synchrotron energy loss rate scales as $b_{\text{syn}} \propto B^2 E_e^2$ [86], mirroring the E_e^2 dependence of Thomson-regime ICS. Which of the two mechanisms dominates depends on the local ratio of magnetic field energy density to radiation field energy density ($\rho_{\text{mag}}/\rho_{\text{rad}}$). At the highest electron energies, Klein-Nishina suppression of ICS breaks this symmetry: the ICS loss rate falls below its Thomson-regime prediction, leaving synchrotron radiation as the primary cooling channel for the most energetic cosmic-ray electrons.

These three leptonic processes, together with the hadronic pion-decay mechanism described in the preceding section, constitute the main set of fundamental interactions that generate the diffuse gamma-ray sky. The following section describes how these individual components are combined into a global model of the astrophysical Galactic Diffuse Emission.

2.1.3 The Galactic Diffuse Emission Model

The individual production mechanisms described in the preceding sections combine to produce the Galactic Diffuse Emission (GDE)—the brightest feature in the gamma-ray sky, accounting for roughly 80% of all photons detected by the *Fermi*-LAT [131, 138]. This emission arises from the integrated interactions of propagating cosmic rays with the interstellar gas and radiation fields of the Milky Way.

To model this complex, spatially structured foreground, astrophysicists rely on numerical cosmic-ray propagation codes, of which GALPROP [109] and DRAGON [139] are two of the most established. Starting from a set of assumptions about cosmic-ray source distributions (primarily supernova remnants), injection spectra, and diffusion parameters, GALPROP solves the steady-state diffusion-loss equation for all relevant cosmic-ray species as they propagate through the Galactic magnetic field and interstellar medium. The output is a set of sky maps predicting the gamma-ray intensity from each emission component: a π^0 -decay map that traces the gas column density (dominated along the Galactic Plane), an inverse Compton map that traces the interstellar radiation field (producing an extended halo at high latitudes), and a bremsstrahlung map that follows the gas distribution but with a different spectral shape.

The resulting spatial and spectral morphology is highly structured and energy-dependent. Along the Galactic Plane, where atomic and molecular gas is dense, hadronic pion decay dominates the high-energy emission, while bremsstrahlung and ionization govern the cooling of low-energy cosmic-ray electrons. At high Galactic latitudes, the gas density drops while the interstellar radiation field remains significant. There, inverse Compton scattering overtakes bremsstrahlung as the dominant energy-loss mechanism for electrons above a few tens of MeV, producing an extended, diffuse gamma-ray glow [107].

When analyzing *Fermi*-LAT data, the standard procedure is to model the observed sky through template decomposition [138]. The total photon counts in each spatial and energy bin are fit to a linear combination of: the GALPROP-derived π^0 -decay, ICS, and bremsstrahlung templates; a catalog of individually resolved point sources; spatial templates for large-scale structures such as Loop I and the Fermi Bubbles; and a spatially uniform isotropic component representing the combined extragalactic background and residual instrumental contamination. The normalizations of these Galactic templates are typically left as free parameters and fitted independently in each energy bin, allowing the data to correct for imperfections in the GALPROP predictions. The Fermi collaboration applies precisely this procedure to produce the latest official Galactic interstellar emission model `gll_iem_v07`, which serves as the standard foreground template in *Fermi*-LAT analyses. One subtlety is worth flagging: in `gll_iem_v07` the Fermi Bubbles and Loop I are not modeled from dedicated physical templates but absorbed into an *ad hoc* component—the *patch*—built by smoothing the large-scale residuals left over after the physical templates are fit, and adding them back into the model [140, 141]. Since the patch soaks up any extended residual, including the Galactic Center Excess, the official Pass 8 models are by construction ill-suited to extended-emission studies—a caveat we return to in Chapter 4. Despite the physical sophistication of GALPROP and the flexibility of the template-fitting procedure, the GDE cannot currently be modeled perfectly. Imperfect modeling of the interstellar emission remains the largest single source of systematic uncertainty in gamma-ray astrophysics, introducing errors between 15% and 30% depending on the energy range [131, 138]. Uncertainties in the gas column densities (particularly the conversion from CO emission to molecular hydrogen), the interstellar radiation field, and the cosmic-ray propagation parameters all contribute. Handling the Galactic foreground properly is a necessary condition for any analysis

of the gamma-ray sky: without adequate foreground control, both spectral characterization and source detection are biased. This challenge recurs throughout this thesis—in the extraction of the Galactic Center Excess (Chapter 4), the search for dark matter subhalos (Chapter 5), and the reconstruction of the extragalactic source-count distribution (Chapter 6). The probabilistic and simulation-based methods developed in later chapters are designed to operate robustly in the presence of these foreground uncertainties, rather than to eliminate them.

2.2 Astrophysical Gamma-Ray Sources

The high-energy sky observed by the Fermi Large Area Telescope (Fermi-LAT) is a complex superposition of processes spanning our local Galactic neighborhood and the broader universe. Phenomenologically, this emission falls into two domains. The Galactic sky is dominated by the diffuse foreground of cosmic-ray interactions and populated by Galactic gamma-ray sources, most notably pulsars. The extragalactic sky, in contrast, is characterized by an isotropic background arising primarily from unresolved active galactic nuclei and star-forming galaxies. A detailed understanding of the astrophysical processes behind these conventional emitters is necessary for dark matter indirect detection, where every ordinary astrophysical signal is a potential background.

2.2.1 Galactic Sources

The gamma-ray emission from the Milky Way comprises two broad components: a diffuse foreground and a diverse population of individual point sources. The Galactic Diffuse Emission dominates the observed sky and, as discussed in Section 2.1.3, its template-based modeling remains the largest source of systematic uncertainty in Fermi-LAT analyses. Alongside this diffuse foreground, the Galactic sky is populated by individual energetic sources, among which pulsars are the most prominent gamma-ray emitters.

Young pulsars are rapidly spinning neutron stars with large surface magnetic fields (10^{11} – 10^{13} G), emitting gamma rays primarily through curvature radiation in their magnetospheric gaps. This emission typically continues over millions of years as the pulsar gradually decelerates (spins down). Over longer timescales, neutron stars in binary systems can undergo a recycling process: the older neutron star accretes matter and angular momentum from its companion, spinning back up to periods of 2–10 milliseconds [142]. These millisecond pulsars (MSPs) possess weaker magnetic fields (10^8 – 10^9 G) but maintain their rapid rotation for gigayears. While young pulsars remain tightly clustered near their birthplaces in the Galactic disk, MSPs have orbited the Galaxy numerous times. Their natal kicks and prolonged dynamical history broaden their spatial distribution, causing MSP populations to extend to higher Galactic latitudes and to densely populate the Galactic bulge [143].

Their concentration in the Galactic bulge places MSPs squarely in the regions most targeted by dark matter searches, but the challenge extends beyond spatial overlap: curvature radiation produces a characteristic power-law spectrum with an exponential cutoff at a few GeV. This spectral shape is basically indistinguishable from the expected photon yield of a ~ 50 GeV WIMP annihilating into $b\bar{b}$ quarks [11] — a topic we will extensively explore in Chapters 4 and 5.

Among the remaining Galactic gamma-ray sources, supernova remnants and pulsar wind nebulae are routinely detected by the Fermi-LAT but represent subdominant backgrounds for the dark matter searches undertaken in this thesis.

2.2.2 Extragalactic Sources

The high-latitude sky is dominated by the Extragalactic Gamma-Ray Background (EGB), which encompasses the total gamma-ray emission originating outside our Galaxy. The EGB combines both individually resolved extragalactic sources and the unresolved glow of fainter objects [131, 144]. The Unresolved Gamma-Ray Background (UGRB) is the residual, almost isotropic emission that remains after subtracting the Galactic Diffuse Emission and all cataloged point sources [131, 138]. It is composed by the cumulative emission of all the currently unresolved gamma-ray sources, plus a residual truly diffuse isotropic background. The Fermi-LAT has measured the UGRB from 100 MeV to 820 GeV, and characterizing its composition is essential for identifying the origins of high-energy cosmic radiation and for constraining dark matter annihilation.

The dominant contributors to the extragalactic sky are blazars, a subclass of active galactic nuclei (AGNs) with relativistic jets oriented along the observer's line of sight. Accretion onto a supermassive black hole powers these jets, producing a bimodal Spectral Energy Distribution (SED) with a low-frequency synchrotron peak and a high-frequency inverse-Compton peak [131]. Blazars are classified into two populations: Flat-Spectrum Radio Quasars (FSRQs) and BL Lacertae objects (BL Lacs). FSRQs exhibit broad optical emission lines and typically have lower-frequency synchrotron peaks, resulting in softer gamma-ray spectra that contribute most to the background below 1 GeV. BL Lacs lack strong emission lines, have harder energy spectra, and frequently exhibit high-frequency synchrotron peaks. As a result, nearby, low-luminosity BL Lacs produce the majority of the EGB emission above 100 GeV [144]. At these energies, gamma rays interact with the Extragalactic Background Light (EBL) to produce electron-positron pairs, an attenuation effect that naturally explains the exponential cutoff observed in the EGB spectrum above roughly 250 GeV. As a unified class, resolved and unresolved blazars account for approximately $50^{+12}_{-11}\%$ of the total EGB photons [144]. Because they define both the resolved catalogs and the upper end of the unresolved background, the classification and modeling of blazars are central to the research presented in Chapters 5, 6, and 7.

The remainder of the extragalactic gamma-ray emission is primarily due to misaligned AGNs and star-forming galaxies (SFGs). Misaligned AGNs, including Fanaroff-Riley radio galaxies, possess jets oriented away from our line of sight [131]. Without strong Doppler boosting, they are intrinsically fainter than blazars but far more abundant, contributing an estimated 20% to 25% of the UGRB [131, 145]. Star-forming galaxies, on the other hand, produce gamma rays through hadronic interactions as described in Section 2.1.1: cosmic-ray protons accelerated by supernovae collide with interstellar gas to produce neutral pions that decay into gamma rays. While their individual luminosities are low, SFGs are ubiquitous and their aggregate emission closely tracks the cosmic star-formation history. Current models suggest they account for 10% to 30% of the EGB [144, 146], potentially dominating among the faintest source populations. Detailed population models demonstrate that the combined emission from blazars, misaligned AGNs, and SFGs can account for the entire amplitude and spectral shape of the measured EGB (Figure 2.3), establishing tight constraints on any dark matter contribution [144].

The connection between resolved source catalogs and the unresolved background is formalized by the source-count distribution, dN/dS , which describes the differential number density of sources per unit solid angle as a function of their gamma-ray flux S . Integrating the dN/dS distribution from zero flux up to the telescope's point-source detection threshold yields the intensity of the UGRB. While the bright end of the dN/dS is directly anchored by catalog measurements, probing the distribution below the detection threshold requires statistical techniques that extract

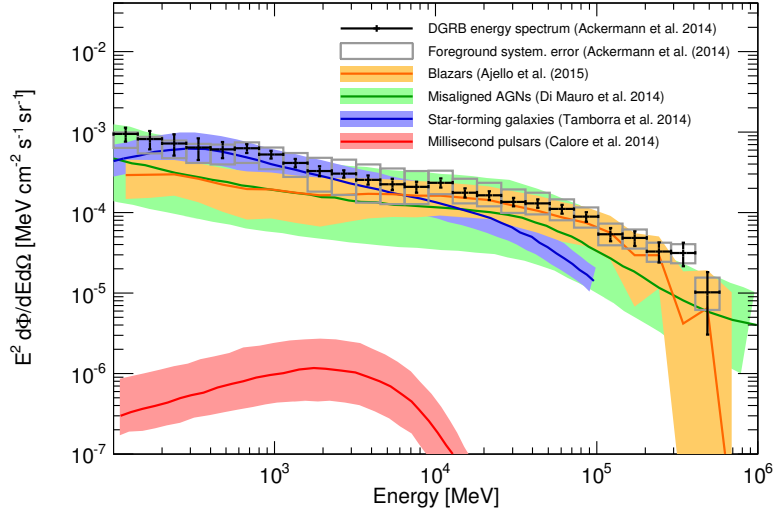


Fig. 2.3. The energy spectrum of the Diffuse Gamma-Ray Background and its expected composition. The black points represent the Fermi-LAT measurement, with gray bands indicating systematic uncertainty from the Galactic diffuse foreground subtraction. The colored lines and bands illustrate the predicted contributions from unresolved blazars (orange), misaligned AGNs (green), star-forming galaxies (blue), and millisecond pulsars (red). Credit: Figure 9 from [131].

population-level information from the collective imprint of unresolved sources on the photon-count map. The formal definition of dN/dS , its connection to the underlying luminosity function, and the methods developed to reconstruct it in the unresolved regime are the subject of Chapter 6.

2.3 The Fermi Large Area Telescope

The overlapping source populations, diffuse foregrounds, and the faint unresolved emission described in the previous section place specific demands on both the detecting instrument and the data analysis pipeline. To interpret the astrophysical gamma-ray sky, we must first understand the telescope that observes it. Launched in 2008, the *Fermi* Large Area Telescope (LAT) has provided the most sensitive all-sky survey in the GeV domain to date, operating continuously for over fifteen years. While the detailed pipelines for data extraction and analysis are discussed directly within the subsequent chapters, it is necessary to establish the basic instrumental parameters and data products that define the practical limits of current observations. In this section, we review the fundamental design of the LAT, the energy-dependent instrumental response functions that govern its spatial and energy resolution, and the resulting catalogs of point sources that serve as the foundation for both individual target searches and statistical population studies.

2.3.1 Instrument Overview

The *Fermi* Large Area Telescope (LAT) is a pair-conversion telescope designed to measure cosmic gamma rays. Because high-energy gamma rays cannot be reflected or refracted, they must be detected indirectly through their interaction with matter. Fermi-LAT is therefore structured around four primary subsystems that convert incident photons into charged particles, track their

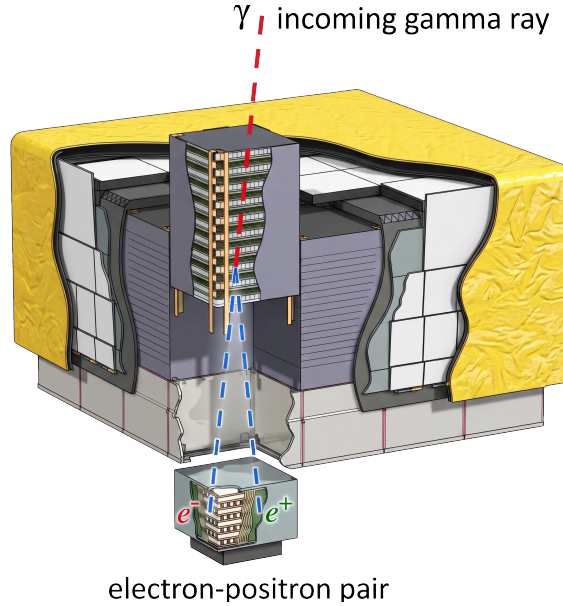


Fig. 2.4. Schematic diagram of the Large Area Telescope, showing the tracker modules, calorimeter, and the surrounding Anticoincidence Detector. The telescope’s dimensions are $1.8 \text{ m} \times 1.8 \text{ m} \times 0.72 \text{ m}$. Adapted from Figure 1 from [147].

trajectories, measure their energy, and reject the overwhelming background of cosmic rays [147]. Figure 2.4 illustrates the instrument components.

The conversion process takes place in the precision tracker, which consists of a 4×4 array of 16 modules. Each module contains 18 x, y tracking planes of single-sided silicon microstrip detectors, interleaved with high- Z tungsten conversion foils. An incoming gamma ray preferentially undergoes pair production within the tungsten, yielding an electron-positron (e^+e^-) pair. The silicon strip detectors then record the passage of these charged particles through successive layers. By precisely tracing the particle tracks back to their common vertex, the instrument reconstructs the arrival direction of the original photon.

Below the tracker lies the calorimeter, composed of 96 cesium iodide (CsI(Tl)) scintillating crystals per module, arranged in a configuration of 8 layers with alternating orientation. The calorimeter measures the total energy deposited by the electromagnetic shower and images its longitudinal and transverse development profile. This three-dimensional shower imaging is essential not only for energy reconstruction—allowing measurements to extend beyond the calorimeter’s physical depth up to TeV energies—but also for background rejection, as it separates the narrow electromagnetic showers from photon-induced events from the broader hadronic cascades produced by cosmic rays.

The entire tracker array is covered by the Anticoincidence Detector (ACD), a segmented shield of 89 plastic scintillator tiles read out by wavelength-shifting fibers and photomultiplier tubes. The ACD provides a charged-particle detection efficiency exceeding 0.9997 averaged over its area, enabling rejection of the charged cosmic-ray background that outnumber gamma rays by several orders of magnitude [147]. The segmentation of the ACD is necessary to suppress the “backsplash” effect: at high energies, secondary particles from electromagnetic showers in the calorimeter can scatter back into the shield, creating false veto signals. By considering only the ACD segment nearest the candidate photon’s entry point, the instrument avoids falsely rejecting

valid high-energy gamma-ray events.

This hardware configuration enables the LAT to achieve substantial improvements over previous-generation instruments. The LAT operates across an energy range spanning from 20 MeV to over 300 GeV. Its compact aspect ratio grants a large field of view of approximately 2.4 sr at 1 GeV, ensuring that nearly all pair-conversion events initiated in the tracker pass into the calorimeter for energy measurement. In its primary survey mode, the spacecraft rocks alternately by $\pm 35^\circ$ from the zenith on consecutive orbits, achieving nearly uniform exposure of the entire sky every two orbits—approximately three hours. Combined with a peak effective area of $\sim 9,500 \text{ cm}^2$ at normal incidence and a dead time of only 26.5 μs per event readout, these capabilities result in a sensitivity approximately 25 times greater than EGRET, enabling the detection and characterization of thousands of faint source populations [147].

2.3.2 Instrument Response Functions

The physical characteristics of the LAT—its tracking efficiency, multiple scattering within the conversion foils, and incomplete shower containment—are encapsulated in the Instrument Response Functions (IRFs). The IRFs map the true energy, direction, and conversion location of incident gamma rays to their reconstructed quantities. Three IRF components are particularly relevant for the analyses in this thesis: the Point Spread Function, the energy dispersion, and the effective area.

The Point Spread Function (PSF) describes the angular resolution of the telescope, quantifying the probability distribution of reconstructed directions around the true source position. At energies below a few GeV, the resolution is dominated by the multiple Coulomb scattering of the newly produced electron-positron pair as it traverses the tungsten conversion foils. Consequently, the PSF is strongly energy-dependent, as shown in Figure 2.5: at 100 MeV, the 68% containment angle spans approximately 3.5° to 5° ; it tightens to roughly 0.6° at 1 GeV and reaches an asymptotic limit of less than 0.15° above 10 GeV [147]. Off-axis, the containment angle degrades by up to a factor of 1.7 at 55° incidence. The broad PSF at lower energies creates a severe source confusion limit: because the spatial profiles of nearby point sources overlap with each other and with the intense Galactic diffuse emission, it becomes statistically difficult to disentangle individual faint objects. This confusion directly limits the depth of point-source catalogs and motivates the use of statistical population methods—such as those developed in Chapters 6 and 7—rather than standard source-by-source extraction.

The energy dispersion quantifies the accuracy of the reconstructed photon energy. It is determined by the calorimeter’s ability to contain and measure the electromagnetic shower. For on-axis photons, the equivalent Gaussian 1σ energy resolution ranges from 9% to 15% in the 100 MeV–1 GeV band, improves to 8%–9% between 1 and 10 GeV, and then degrades to $\sim 18\%$ near 300 GeV as shower leakage from the finite calorimeter depth becomes significant [147]. At large off-axis angles ($> 60^\circ$), where the effective calorimeter depth increases, the resolution can improve to $\sim 6\%$ above 10 GeV. For most analyses in this thesis, the energy resolution is sufficiently precise that energy dispersion is not a dominant systematic.

Not every gamma ray that enters the LAT produces a usable event: some pass through without converting, others are rejected during background filtering. The effective area captures this by expressing the instrument’s photon-collecting power as the area of an ideal, perfectly efficient detector that would record the same number of events. At normal incidence, it rises steeply from $\sim 100 \text{ cm}^2$ at 20 MeV to a broad plateau of $\sim 8,000$ – $9,500 \text{ cm}^2$ between 1 GeV and several hundred GeV, before declining at the highest energies [147]. See Figure 2.6 for the energy and incidence angle dependence of the effective area. For comparison, the LAT’s overall

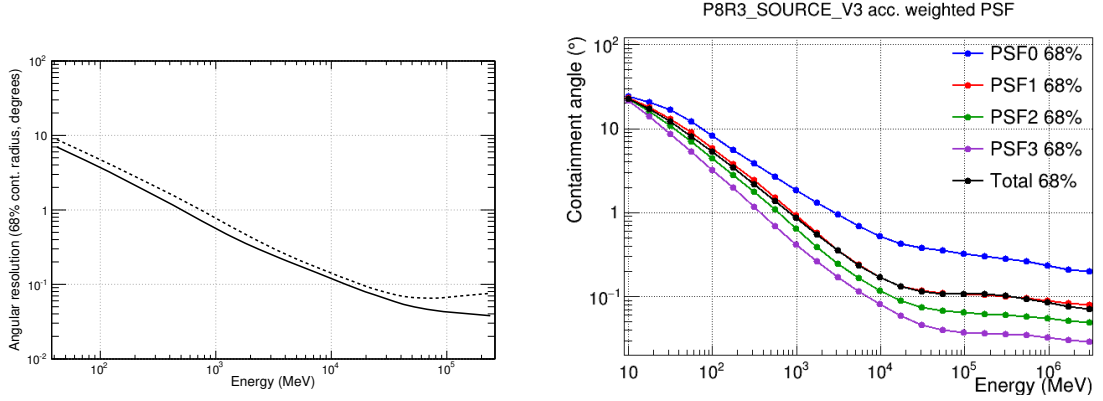


Fig. 2.5. *Left:* The 68% containment angle of the Point Spread Function as a function of energy at normal incidence (solid) and at a 60° off-axis incidence angle (dashed), for photons converting in the thin *front* section of the tracker. Photons converting in the thick *back* section have a PSF approximately twice as wide at 1 GeV. *Right:* the 68% containment angle of the PSF as a function of energy for the four PSF event types (PSF0–PSF3, from broadest to narrowest). Credit: Left from [147] and right from [148].

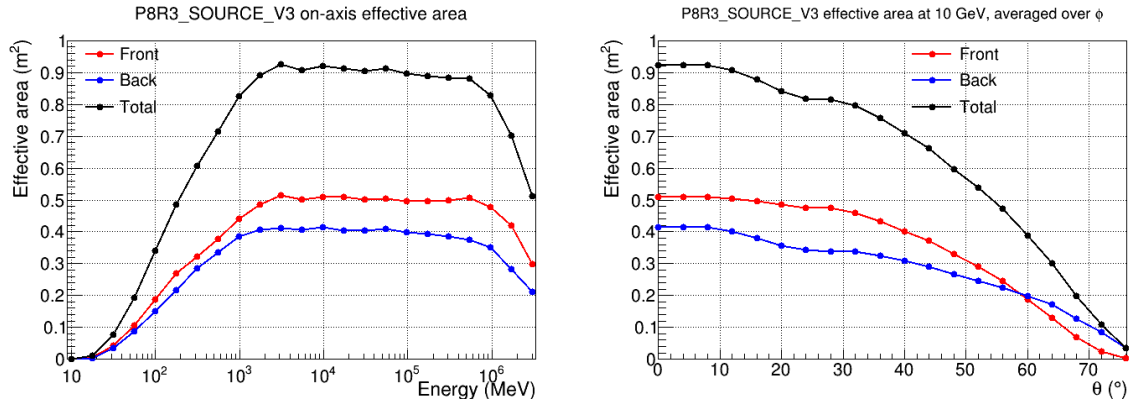


Fig. 2.6. *Left:* Effective area as a function of the energy of the incident photon. *Right:* Effective area at 10 GeV as a function of the angle of incidence. Credit: Fermi-LAT collaboration [148].

footprint is $1.8 \times 1.8 \text{ m}^2$; the peak effective area of $\sim 9,500 \text{ cm}^2$ —roughly a third of the geometric area—reflects the compounded probabilities of pair conversion in the tungsten, successful track reconstruction, and survival of background-rejection cuts. For a survey instrument, however, the relevant figure of merit is not the on-axis effective area alone, but the *acceptance*—the effective area integrated over all incident directions. While the effective area quantifies the detection efficiency for photons arriving from a single direction, the acceptance captures the instrument’s total photon-collecting power across the full field of view simultaneously. The LAT’s acceptance peaks at $\sim 2.5 \text{ m}^2 \text{ sr}$ above 1 GeV [147], a direct consequence of the large 2.4 sr field of view enabled by the tracker’s compact aspect ratio. Combined with the scanning survey mode described in Section 2.3.1, this large acceptance allows the LAT to achieve nearly uniform full-sky exposure every three hours, operating as a continuous all-sky monitor rather than a pointed observatory.

The accuracy of the IRFs relies directly on the underlying event reconstruction algorithms. The Pass 8 analysis framework is a complete reworking of the LAT event-level reconstruction

[149]. A major improvement in Pass 8 was the mitigation of “ghost signals”—instrumental pile-up and out-of-time residual hits from previous events that previously degraded the reconstruction. This enabled a significant expansion of the effective area, particularly below 100 MeV and above 1 TeV, while simultaneously reducing background contamination. Pass 8 also introduced event subclassifications that partition data by the quality of the reconstructed direction (PSF event types: PSF0–PSF3, from broadest to narrowest) and the quality of the energy measurement (EDISP event types). This partitioning allows analyses to select event classes optimized for their specific scientific goals—angular resolution for morphological studies, or photon statistics for spectral analyses.

Subsequent iterations, most notably the P8R3 release [150], further refined the residual background rejection, producing a spatially uniform and clean dataset that serves as the baseline for all modern *Fermi*-LAT analyses, including the work presented in this thesis.

2.3.3 Data Products and Catalogs

With the IRFs established, we now turn to how the LAT data are translated into astrophysical catalogs of gamma-ray sources. Converting raw photon events into measurements of individual sources requires disentangling finite objects from the structured diffuse backgrounds. The *Fermi*-LAT collaboration achieves this through a maximum likelihood framework [140, 151]. The observed sky is modeled through the template decomposition introduced in Section 2.1.3—the structured Galactic diffuse emission plus a spatially uniform isotropic component—augmented with individual point sources, each parametrized by a position and a spectral model. The normalizations and spectral parameters of all components are simultaneously adjusted to maximize the Poisson likelihood function against the observed photon counts across the sky.

To evaluate the significance of a candidate point source, the framework computes the Test Statistic (TS), defined as

$$TS = 2 \ln \frac{\mathcal{L}_1}{\mathcal{L}_0},$$

where $\mathcal{L}_1$ is the maximum likelihood of the model with the candidate source included and $\mathcal{L}_0$ is the maximum likelihood of the null hypothesis without it [151]. By Wilks’ theorem, TS is approximately distributed as χ^2 with a number of degrees of freedom equal to the additional free parameters of the source model. For a point source with four free parameters (two spatial and two spectral), the catalog detection threshold of $TS > 25$ corresponds to a significance of just over 4σ [140, 152]. As a practical rule of thumb, for an isolated source with known position (one spectral degree of freedom), the significance is well approximated by $\sqrt{TS} \sigma$. This threshold defines the boundary between what is considered a “resolved” source entering the catalog and what remains part of the unresolved diffuse background.

The results of these analyses have been systematically compiled into a series of catalogs of increasing depth. The progression from the First *Fermi*-LAT Catalog (1FGL, based on 11 months of data) through the Third Catalog (3FGL, 4 years, 3,033 sources) to the Fourth Catalog (4FGL, 8 years, 5,064 sources) [140, 152] reflects both the accumulation of exposure and successive improvements in event reconstruction and background modeling. The most recent published incremental release, 4FGL-DR4, extends the dataset to 14 years and contains 7,195 sources [153], with more recent data releases continuing to grow the catalog. Throughout this evolution, the detection threshold has remained at $TS > 25$, so the growing source count is driven primarily by longer exposure, improvements in the galactic foreground modeling, and improved sensitivity from Pass 8 event reconstruction.

Each catalog entry reports the source localization (with 95% confidence ellipse), the integrated energy flux in multiple bands, the best-fit spectral model, a variability index, and multi-wavelength associations. The spectral models employed include the simple power law, the log-parabola (commonly preferred for blazars), and the power law with an exponential cut-off (standard for pulsars). These spectral characterizations—in particular the flux, spectral index, and curvature from the log-parabola fits—form the feature space used in Chapter 5 for a population-level study of unassociated sources.

The association procedure matches each gamma-ray source against catalogs at radio, infrared, optical, and X-ray wavelengths using Bayesian and likelihood-ratio methods. Sources receiving a high-confidence match are labeled as “associated”; those with firm identification through periodicity, correlated variability, or resolved angular extent are labeled as “identified.” In the 4FGL-DR4, 4022 sources are identified or associated with extragalactic objects and 610 with Galactic source classes. However, a substantial fraction—2563 sources—lacks any plausible multi-wavelength counterpart and remains unassociated [153]. These unassociated sources are concentrated along the Galactic plane, where the dense background complicates counterpart searches, and at high latitudes, where their average spectral properties resemble those of blazars of unknown type.

This population likely consists of faint blazars and pulsars below the sensitivity of current multi-wavelength surveys, but it is also the primary search pool for exotic gamma-ray emitters such as dark matter subhalos. As mentioned in Chapter 1 and detailed further in Chapter 5, Λ CDM cosmology predicts a large population of low-mass Galactic subhalos that are too low in mass to retain gas or form stars, and are therefore invisible at all wavelengths except gamma rays, where WIMP annihilation could produce a faint, spectrally soft signal closely resembling an unassociated *Fermi*-LAT point source. Separating such candidates from the astrophysical sources within the 4FGL catalog is the central challenge addressed in Chapter 5.

STATISTICAL METHODS FOR NOISE-DOMINATED REGIMES

The gamma-ray sky described in the previous chapter presents a formidable data analysis challenge. Faint signals from potential dark matter annihilation or decay are embedded in bright, spatially structured astrophysical backgrounds; source populations overlap in both position and spectral shape; and instrumental effects—finite angular resolution, energy-dependent sensitivity, detection thresholds—leave a large fraction of the emitting population unresolved or unclassified. Extracting new physics from this environment requires moving beyond standard signal-over-background methods toward a more comprehensive statistical toolkit.

The analyses presented in this thesis draw on techniques spanning classical statistics, modern machine learning, and domain-specific tools developed for astrophysical inference. Each method addresses a different aspect of the extraction problem. Frequentist and Bayesian approaches provide the language for parameter estimation and hypothesis testing. Simulation-based inference bypasses the intractable likelihoods that arise in complex forward models, while neural networks learn directly from high-dimensional data when explicit physical models are unavailable. Domain adaptation corrects for the systematic mismatches between simulated training data and real observations. Finally, angular cross-correlations exploit the connection between gamma-ray emission and the large-scale distribution of matter, accessing collective signals invisible to single-instrument analyses.

This chapter introduces the mathematical framework used throughout the thesis. Each method is presented at a conceptual level, with forward references to the chapters where the full technical details appear. We begin with the foundations of frequentist and Bayesian inference in Section 3.1, establishing the likelihood function, maximum likelihood estimation, Bayesian posteriors, and hypothesis testing. Section 3.2 introduces the simulation-based inference paradigm for problems where the likelihood cannot be evaluated analytically. Section 3.3 covers the machine learning tools—convolutional neural networks and density estimation—used for regression and classification tasks on gamma-ray data. Section 3.4 formalizes the dataset shift problem that arises when training and target distributions differ. Finally, Section 3.5 develops the angular cross-correlation formalism that connects gamma-ray observations to gravitational tracers of dark matter structure at cosmological scales.

3.1 Frequentist and Bayesian Inference

Extracting physical information from noisy gamma-ray data requires a well-defined statistical language. The analyses presented in this thesis draw on two complementary paradigms—frequentist and Bayesian inference—each providing distinct tools for parameter estimation, uncertainty quantification, and hypothesis testing. This section introduces both frameworks from the shared foundation of the likelihood function, develops the key concepts used throughout this thesis, and discusses when each paradigm is most appropriate (according to personal bias).

3.1.1 The Inference Problem

The starting point for any quantitative analysis in physics is the problem of inference: given a set of observed data $\mathbf{x}$, and a theoretical model $\mathcal{M}$ governed by parameters θ , what can we learn about θ from $\mathbf{x}$? In gamma-ray astrophysics, $\mathbf{x}$ might be a photon-count map recorded by the Fermi LAT, while θ encodes the physical quantities of interest: source fluxes, spectral indices, or the dark matter annihilation cross-section $\langle\sigma v\rangle$. The model $\mathcal{M}$ connects these parameters to the expected data through a forward process that may include astrophysical emission templates, instrumental response functions, and Poisson sampling of photon counts.

The connection between parameters and data is formalized by the likelihood function $\mathcal{L}(\theta) = p(\mathbf{x}|\theta)$, which quantifies how well a given parameter choice explains the observed data [154]. This function is the shared foundation of both major inference paradigms. In the frequentist approach, the parameters θ are treated as fixed but unknown quantities, and inference proceeds by identifying the parameter values that maximize $\mathcal{L}$. In the Bayesian approach, θ are themselves treated as random variables with a prior distribution $p(\theta)$, and the goal is to compute the posterior $p(\theta|\mathbf{x})$ via Bayes’ theorem [155, 156].

These two philosophies are not in competition; they are complementary tools, each with distinct strengths suited to different aspects of the inference problem. In the analyses presented in this thesis, both are used—sometimes on the same dataset—to extract physical information and quantify uncertainty. The following two subsections develop each approach, starting from a common notation and building toward the specific formulations employed in later chapters.

3.1.2 Frequentist Inference

In the frequentist framework, the model parameters θ are fixed but unknown. The data $\mathbf{x}$ are treated as one realization of a random process, and all statistical statements are made in terms of the long-run frequency properties of that process. The central object is the likelihood function, which quantifies the probability of the observed data as a function of the model parameters [156].

Maximum Likelihood Estimation. The most common frequentist estimator is the maximum likelihood estimator (MLE), which identifies the parameter values that maximize the likelihood:

$$\hat{\theta}_{\text{MLE}} = \underset{\theta}{\operatorname{argmax}} \mathcal{L}(\theta). \quad (3.1.1)$$

As a concrete example, consider counting N photons from a gamma-ray source. If the counts follow a Poisson distribution with expected rate μ , the likelihood is $\mathcal{L}(\mu) = e^{-\mu} \mu^N / N!$, and the MLE is simply $\hat{\mu} = N$. While this example is trivial, the Poisson likelihood extends naturally to the pixel-by-pixel analysis of gamma-ray sky maps, where the joint likelihood is the product of Poisson terms over all pixels:

$$\mathcal{L}(\theta) = \prod_{a=1}^{N_{\text{pix}}} \frac{\lambda_a(\theta)^{k_a} e^{-\lambda_a(\theta)}}{k_a!}, \quad (3.1.2)$$

with k_a the observed counts and $\lambda_a(\boldsymbol{\theta})$ the model prediction in pixel a . This product-of-Poisson construction is the standard likelihood used in Fermi-LAT analyses [151].

Nuisance parameters and the profile likelihood. In realistic analyses, the parameter vector $\boldsymbol{\theta}$ contains both the parameters of interest $\boldsymbol{\psi}$ and a potentially large set of nuisance parameters $\boldsymbol{\eta}$ (background normalizations, instrumental systematics, source model parameters). The profile likelihood isolates the dependence on $\boldsymbol{\psi}$ by maximizing over the nuisance parameters at each fixed value of $\boldsymbol{\psi}$:

$$\mathcal{L}_{\text{prof}}(\boldsymbol{\psi}) = \max_{\boldsymbol{\eta}} \mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\eta}). \quad (3.1.3)$$

This construction preserves the frequentist interpretation of the likelihood without requiring any prior distribution on $\boldsymbol{\eta}$. An alternative approach is to integrate over the nuisance parameters within the likelihood framework itself, producing an integrated (or marginal) likelihood:

$$\mathcal{L}_{\text{int}}(\boldsymbol{\psi}) = \int \mathcal{L}(\boldsymbol{\psi}, \boldsymbol{\eta}) g(\boldsymbol{\eta}) d\boldsymbol{\eta}, \quad (3.1.4)$$

where $g(\boldsymbol{\eta})$ is a weight function derived from independent auxiliary measurements or physical constraints on the nuisance parameters. Although this resembles Bayesian marginalization in form, the integrated likelihood is interpreted differently: it is a tool for removing $\boldsymbol{\eta}$ from the likelihood rather than a posterior probability statement about $\boldsymbol{\psi}$.

Hypothesis testing and confidence intervals. To compare competing hypotheses—for instance, whether a gamma-ray source exists at a given position—we construct a likelihood-ratio test statistic (TS):

$$\text{TS} = -2 \ln \frac{\mathcal{L}_0}{\mathcal{L}_1}, \quad (3.1.5)$$

where $\mathcal{L}_0$ is the likelihood under the null hypothesis and $\mathcal{L}_1$ the likelihood under the alternative (or, more commonly, the globally maximized likelihood) [151]. By Wilks' theorem, under mild regularity conditions the TS is asymptotically distributed as a χ^2 random variable with degrees of freedom equal to the difference in free parameters between the two hypotheses [157], which lets us convert a TS value into a statistical significance without resorting to explicit Monte Carlo simulations. Wilks' theorem is, strictly speaking, a result about hypothesis testing. In the context of this thesis, however, hypothesis testing mostly means setting confidence intervals or upper bounds on a parameter of interest. These are obtained by inverting the test: rather than fixing a single threshold and asking whether the null is rejected, we scan the parameter and retain the range over which the profile-likelihood TS stays below the appropriate χ^2 quantile.

Bootstrap confidence intervals. The χ^2 calibration of the TS rests on the regularity conditions of Wilks' theorem, and these can fail in situations that recur in gamma-ray analyses: when a parameter lies on a physical boundary—a flux or a cross-section that cannot be negative—or when the sample is small enough that the asymptotic regime has not yet set in. In such cases the bootstrap provides a general-purpose strategy [158]: we repeatedly refit the estimator, either on datasets resampled from the observed data (the *nonparametric* bootstrap, or "with replacement") or on synthetic datasets drawn from the fitted model (the *parametric* bootstrap), and read the confidence interval directly from the spread of the re-fitted estimates—the containment interval of their empirical distribution. In Chapter 5, the uncertainty band on the dark matter annihilation cross-section bounds is obtained using a parametric bootstrap.

Goodness of fit. A different question arises when we wish to assess the absolute compatibility of the data with a single model, without testing it against any specific alternative. Two general-purpose goodness-of-fit tests serve this purpose [159]: Pearson’s χ^2 test and the Kolmogorov–Smirnov (KS) test [160]. Pearson’s χ^2 test compares binned counts with model expectations and refers the sum of squared normalized residuals to a χ^2 distribution. The KS test instead compares data and model at the level of their cumulative distributions: its test statistic is the supremum of the absolute difference between the empirical cumulative distribution function of the data and the cumulative distribution predicted by the model. Because it operates on the unbinned data directly, the KS test is additionally independent of any binning choice. Moreover, when the null model is fully specified, the null distribution of its statistic does not depend on the model being tested. In practice, the model prediction is often available only as a finite simulated sample rather than an analytic cumulative distribution. The same statistic computed between the two empirical cumulative distribution functions is then known as the *two-sample* form of the KS test. The KS test is therefore a convenient distribution-free diagnostic when no specific alternative is preferred.

Non-nested model comparison. The Wilks’ likelihood-ratio test presumes that the two hypotheses are nested, with one model recovered as a special case of the other; when this is not so, we need a criterion able to rank models of different structure. Information criteria fill this gap. The Akaike Information Criterion [161] is defined as

$$\text{AIC} = -2\ln \mathcal{L}_{\max} + 2k, \quad (3.1.6)$$

where $\mathcal{L}_{\max}$ is the maximum value of the likelihood and k the number of free parameters. The AIC provides an estimator of the relative Kullback–Leibler divergence between each candidate model and the data-generating process, with the $2k$ term penalising additional parameters and thus discouraging overfitting. Model selection then proceeds by minimising the AIC across the candidates; the differences in AIC quantify how strongly each model is disfavoured relative to the best one.

Application preview. The frequentist framework developed above is used extensively in this thesis:

- In Chapter 4, the MSP luminosity function is constrained via MLE applied to an integrated likelihood, and model comparison is performed through $-2\Delta\ln\mathcal{L}$ values.
- In Chapter 5, a profile likelihood test statistic is used with Wilks’ theorem to set 95% CL upper bounds on the dark matter annihilation cross-section, with the uncertainty band on those bounds derived from a parametric bootstrap; the AIC is additionally used to compare competing models for the unassociated source population.
- In Chapter 7, a two-sample KS test compares the cumulative pixel-wise TS distributions of the real *Fermi*-LAT sky against synthetic realizations drawn from the source-count distribution, providing the statistical criterion that defines sub-threshold source candidates.
- In Chapter 8, a $\Delta\chi^2$ test statistic compares astrophysical-only and astrophysical-plus-DM hypotheses for the cross-correlation angular power spectrum.

3.1.3 Bayesian Inference

The Bayesian framework treats the model parameters θ not as fixed unknowns but as random variables associated with a probability distribution. Before observing any data, our state of knowledge about θ is encoded in a prior distribution $p(\theta)$. After observing data $\mathbf{x}$, this knowledge is updated through Bayes' theorem:

$$p(\theta | \mathbf{x}) = \frac{p(\mathbf{x} | \theta) p(\theta)}{p(\mathbf{x})}, \quad (3.1.7)$$

where $p(\mathbf{x} | \theta) = \mathcal{L}(\theta)$ is the same likelihood function used in the frequentist approach, and $p(\mathbf{x}) = \int p(\mathbf{x} | \theta') p(\theta') d\theta'$ is the evidence, a normalization constant that integrates the likelihood weighted by the prior over the full parameter space [155]. Within the assumed model, the posterior $p(\theta | \mathbf{x})$ is the complete answer to the inference problem: it encodes not only the best-fit values but also the full uncertainty structure, including correlations between parameters.

Prior, posterior, and evidence. The choice of prior $p(\theta)$ is one of the defining features of the Bayesian approach. Informative priors can incorporate physical constraints—for example, requiring fluxes to be non-negative or restricting spectral indices to physically motivated ranges—and can improve inference when data are sparse. Uninformative or weakly informative priors are used when prior knowledge is limited, though the choice is never fully objective [156]. For complex models, the evidence $p(\mathbf{x})$ is typically intractable to compute, as it requires integration over the full parameter space, but it plays an important role in Bayesian model comparison, where the ratio of evidences between competing models (the Bayes factor) provides a natural criterion for model selection [162].

Credible intervals versus confidence intervals. In the Bayesian framework, uncertainty is quantified through credible intervals: a 95% credible interval is a region of parameter space containing 95% of the posterior probability. This contrasts with the frequentist confidence interval, which guarantees that 95% of intervals constructed from repeated experiments will contain the true parameter value. The distinction is subtle in practice—for well-constrained problems, the two often agree numerically—but the conceptual difference matters: a credible interval makes a direct probability statement about the parameter given the data, while a confidence interval makes a statement about the procedure.

Marginalization over nuisance parameters. As in the frequentist case, realistic Bayesian analyses involve nuisance parameters. The Bayesian approach handles them by marginalizing the joint posterior:

$$p(\boldsymbol{\psi} | \mathbf{x}) = \int p(\boldsymbol{\psi}, \boldsymbol{\eta} | \mathbf{x}) d\boldsymbol{\eta}, \quad (3.1.8)$$

which integrates out the nuisance parameters $\boldsymbol{\eta}$ to yield the marginal posterior for the parameters of interest $\boldsymbol{\psi}$. This is conceptually analogous to the frequentist integrated likelihood (see Section 3.1.2), but the probabilistic interpretation differs: the Bayesian marginal posterior is a fully coherent probability distribution over $\boldsymbol{\psi}$, conditioned on both the data and the prior.

When the likelihood is intractable—as is often the case for complex forward models involving high-dimensional simulations—neural networks offer a route to Bayesian inference through implicit likelihood methods, in which a density estimator learns the posterior directly from simulated data–parameter pairs. These approaches are developed in Section 3.2.

Application preview. In Chapter 6, the Bayesian framework is implemented via simulation-based inference to reconstruct the source-count distribution dN/dS from gamma-ray sky maps, with posterior uncertainty estimates validated against an independent frequentist cross-check (Section 3.1.4).

3.1.4 Frequentist versus Bayesian

For most inference tasks encountered in practice—parameter estimation, upper-limit setting, model comparison—both the frequentist and Bayesian frameworks can reach similar scientific conclusions. Neither is inherently superior to the other. Both provide uncertainty quantification—confidence intervals constructed via the profile likelihood or the bootstrap on one side, posterior credible intervals on the other. Both can incorporate external constraints. In the Bayesian framework these enter through the prior—restricting fluxes to non-negative values, for example, or spectral indices to physically motivated ranges. A frequentist analysis imposes the same requirements through bounded parameter ranges or constrained fits. Bounding a parameter is not entirely free of subtleties, however: when the true value may lie close to the boundary, standard interval constructions can undercover or return empty intervals. The unified approach of Feldman and Cousins [163] resolves this by transitioning smoothly between upper limits and two-sided intervals while guaranteeing correct coverage. The choice between them is therefore pragmatic rather than fundamental, driven by the structure of the problem, computational considerations, and—above all—the conventions of the relevant research community. High-energy particle physics and the Fermi-LAT collaboration have traditionally favored frequentist methods: source detection via the TS, upper limits via the profile likelihood and Wilks’ theorem, catalog pipelines built on maximum-likelihood fitting [151]. Cosmology and large-scale-structure analyses, by contrast, have largely adopted Bayesian Markov Chain Monte Carlo (MCMC) methods [164]. Neither convention reflects a judgment on the foundations of probability; both reflect what works well for the typical dimensionality and structure of the problems each field encounters most often. In this thesis, we follow the convention of the community relevant to each analysis.

In the limit of abundant, constraining data, the distinction fades altogether: the likelihood dominates any reasonable prior, and Bayesian credible intervals coincide numerically with frequentist confidence intervals. The choice becomes consequential precisely when data are sparse, priors are informative, or the parameter space is high-dimensional. In the analyses presented in this thesis, both paradigms are used, and in some cases they are applied to the same problem as independent cross-checks. In Chapter 6, a simulation-based inference approach provides Bayesian posterior estimates on the reconstructed dN/dS . These are validated against a frequentist analysis of the same data, in which confidence intervals are extracted from the empirical distribution of residuals across the validation set.

3.2 The Simulation-Based Inference Paradigm

The inference framework presented in Section 3.1 assumes that the likelihood function can be evaluated explicitly. This assumption holds for many standard analyses, but it breaks down when the data-generating process involves a complex simulator whose internal stochastic structure prevents the likelihood from being written in closed form. Such simulators are increasingly common in physics: as experiments gather ever richer data, the models built to describe detectors and observations have grown steadily in fidelity and complexity [154]. At this level of complexity, simple approximate likelihoods no longer describe the simulator output faithfully.

Simulation-based inference (SBI) provides a principled alternative: rather than writing down an explicit approximation of the likelihood, one trains machine learning models on simulated data to approximate the posterior, the likelihood, or the likelihood-to-evidence ratio. This section introduces the SBI paradigm, the neural estimation strategies behind it, and the validation techniques required to ensure the reliability of the results.

3.2.1 The Intractable Likelihood Problem

The inference methods presented in Section 3.1 share a common requirement: the ability to evaluate the likelihood function $p(\mathbf{x}|\boldsymbol{\theta})$ for a given set of parameters. In many problems of interest, however, the data-generating process is described by a complex computer simulation, and the results of such simulations are formally samples from the implicit likelihood. Such models are known as *implicit models*, in contrast to *prescribed models* for which the likelihood can be written in closed form [154, 165]. The resulting inference problem has historically been called *likelihood-free inference*, although the term is somewhat misleading: these methods typically estimate the intractable likelihood rather than dispensing with it entirely. The more descriptive term *implicit likelihood methods* is now preferred [154], and *simulation-based inference* (SBI) is a term often used as an equally valid synonym. For convenience, and to align with the machine learning literature, we will refer to simulation-based inference throughout this thesis.

The source of the difficulty is straightforward to state. A typical simulator takes as input a vector of physical parameters $\boldsymbol{\theta}$, samples a series of internal stochastic states or latent variables z_i ,¹ and produces synthetic data $\mathbf{x}$ as output. The exact likelihood of this output requires marginalizing over the joint density of data and latent variables:

$$p(\mathbf{x}|\boldsymbol{\theta}) = \int p(\mathbf{x}, \mathbf{z}|\boldsymbol{\theta}) d\mathbf{z}, \quad (3.2.1)$$

where the integral runs over every possible latent variable configuration of the simulator [154]. For realistic simulations involving billions of latent variables, such as particle transport through a detector, the stochastic evolution of astrophysical populations, or the formation of large-scale structure, this high-dimensional integral is computationally intractable.

The asymmetry between the forward and inverse problems is a defining feature of the SBI setting. Running the simulator to produce synthetic observations from a given $\boldsymbol{\theta}$ is computationally doable: one simply executes the program. Inverting this process to obtain $p(\mathbf{x}|\boldsymbol{\theta})$, however, would require enumerating all possible internal configurations that could have led to the same observable output. This forward–inverse asymmetry motivates an alternative strategy: rather than evaluating the likelihood directly, one can generate large numbers of simulated datasets and use them to learn the relationship between parameters and observables.

A concrete astrophysical example arises in the measurement of the gamma-ray source-count distribution dN/dS below the Fermi-LAT detection threshold. As discussed in detail in Chapter 6, the observed photon-count map results from the superposition of thousands of unresolved point sources. Writing a closed-form likelihood for this full sky map would require integrating over every possible spatial configuration, flux, and number of the unresolved sources. Generating a synthetic map from a given set of dN/dS parameters, by contrast, amounts to a sequence of well-defined sampling steps: draw source fluxes from the dN/dS , scatter them on the sky, and

¹In the context of implicit likelihood methods, the latent variables z_i represent the internal stochastic states of the simulator that are not directly observable. For example, in the context of the work of Chapter 6, that would be the position of the individual gamma-ray sources.

apply additional well-understood sources of noise. The forward simulation is straightforward; the inverse problem is not. SBI addresses exactly this kind of problem, and the following subsections describe how the inversion is accomplished.

3.2.2 Simulation-Based Inference

Given the impossibility of evaluating the likelihood directly, SBI replaces explicit likelihood computation with learned surrogates trained on simulated data. The idea has a long history: Diggle and Gratton [165] proposed using kernel density estimation to approximate intractable likelihoods from simulator output as early as 1984, while Approximate Bayesian Computation (ABC) developed in parallel as an entirely sampling-based alternative (see [154] for a review). Both approaches, however, suffer from the curse of dimensionality: they require hand-crafted summary statistics to compress the data into low-dimensional representations, inevitably discarding information about the parameters of interest. Modern neural SBI methods overcome these limitations by using deep neural networks as flexible density estimators that potentially operate directly on high-dimensional data [154]. Three main neural strategies have emerged, distinguished by which statistical quantity the neural network learns to approximate.

Neural Posterior Estimation (NPE) trains a conditional density estimator—typically a discrete or continuous normalizing flow [166, 167]—to directly approximate the posterior distribution $p(\theta|\mathbf{x})$ [168]. By conditioning the density estimator on the observed data $\mathbf{x}$, the network learns to output the posterior density for any input observation. Because NPE directly provides the posterior, it is possibly the most straightforward approach for Bayesian inference. However, because the prior enters the training, changing the prior requires retraining.

Neural Likelihood Estimation (NLE) instead approximates the likelihood function $p(\mathbf{x}|\theta)$, conditioning the density estimator on θ rather than on $\mathbf{x}$ [154]. Once the likelihood surrogate is trained, Bayesian inference proceeds through standard means, for example through MCMC sampling with any choice of prior. This separation of the likelihood from the prior provides additional flexibility. The downside is that approximating a high-dimensional likelihood is generally more expensive than targeting the posterior directly, and posterior samples are not immediately available: they must be drawn from the trained likelihood surrogate through the additional MCMC step.

Neural Ratio Estimation (NRE) takes a different approach altogether: rather than learning a probability density, a classifier is trained via supervised learning to discriminate between synthetic datasets generated at different parameter points. This is attractive because classification is sometimes a more tractable learning problem than density estimation, especially in high-dimensional spaces. The classifier output can be converted into the likelihood-to-evidence ratio $r(\mathbf{x}, \theta) = p(\mathbf{x}|\theta)/p(\mathbf{x})$, which is proportional to the posterior up to the prior [154, 169]. Once the classifier is trained, evaluating the posterior density at any point is fast, but drawing samples still requires MCMC. This is possibly the main drawback of NRE, as full posterior sampling and calibration are significantly more expensive than in NPE.

All three strategies support *amortized* variants: the computationally expensive simulations are performed once, and the trained model can then sample the posterior for any new observation without additional calls to the simulator [154]. This reuse of the upfront simulation cost across arbitrarily many observations is what *amortization* properly refers to; as an added practical benefit, the simulated dataset itself can also be reused to train multiple neural networks, which is particularly useful for hyperparameter tuning and model selection. A further advantage is that SBI by construction implicitly marginalizes over the vast latent space of the simulator, so the trained surrogate can retain information that hand-crafted summary statistics would

discard [154, 170]. This is one of the properties that makes SBI particularly interesting for field-level inference, as demonstrated in cosmological analyses of galaxy clustering and weak gravitational lensing [171–174].

Application preview. In this thesis, NPE is the SBI strategy most directly relevant to our work. In Chapter 6, a convolutional neural network trained on synthetic gamma-ray sky maps estimates the source-count distribution dN/dS and its uncertainty for any input map, without evaluating an explicit likelihood. The approach operates in the spirit of NPE: the network directly maps an observed sky map to posterior summaries (the mean and variance for each flux bin). Once trained, the inference is amortized: the network can be applied to the real Fermi-LAT sky map to quickly obtain the source-count distribution without requiring new simulations.

3.2.3 Validation of Simulation-Based Inference

A neural density estimator trained on finite simulations can, in principle, produce posteriors that are poorly calibrated: the reported credible intervals may be systematically too narrow or too wide. Validating SBI results is therefore an integral part of any simulation-based analysis, and the simulator itself provides the means for doing so. We notice that validation would in principle also be required even when performing inference with a known likelihood, but it is sometimes overlooked in practice as it is often computationally intractable for standard MCMC methods.

The standard validation strategy is *coverage testing*, and the procedure is conceptually simple. One samples parameters θ^* from the prior, generates synthetic data $\mathbf{x}^* \sim p(\mathbf{x}|\theta^*)$ using the simulator, and then applies the trained estimator to obtain the approximate posterior $\hat{p}(\theta|\mathbf{x}^*)$. Because the true parameters are known by construction, one can check whether the posterior places the correct amount of probability mass around θ^* . Repeating this test across hundreds or thousands of independent simulations yields a statistical picture of the estimator’s calibration: if the $(1 - \alpha)$ credible intervals contain the true parameters in exactly a fraction $(1 - \alpha)$ of trials, the estimator is well-calibrated. Comparing the empirical coverage with the nominal level across a range of credibility thresholds produces a coverage-vs-credibility plot; departures from the diagonal indicate over- or underdispersion of the posterior, and the discrepancy is formally quantified through the empirical calibration error [154, 175]. When perfect calibration cannot be achieved, a slightly underconfident model (above the diagonal) is generally preferable in physics, since it yields conservative error bars.

Coverage testing is particularly well-suited to amortized SBI estimators, since evaluating the posterior for each new synthetic observation is generally fast (particularly for NPE) once the network has been trained. For non-amortized methods, the computational cost of fitting a new posterior for each of the hundreds or thousands of required test datapoints would be prohibitive.

Passing a coverage test is, however, a necessary but not a sufficient condition for proper calibration. Because the test parameters are drawn from the prior, an estimator that ignores the data entirely and simply returns the prior passes the test perfectly [175, 176]. Its credible regions are simply those of the prior, identical for every observation, and a parameter drawn from the prior falls within the prior’s own $(1 - \alpha)$ credible region with probability exactly $(1 - \alpha)$,

at every credibility level.² Moreover, since the coverage is averaged over many observations, overconfident and underconfident regions of the parameter space can compensate each other [175]. Several complementary diagnostics are therefore often employed alongside it. *Simulation-based calibration* (SBC) sharpens the coverage test by requiring that the rank of each true parameter among the corresponding posterior samples be uniformly distributed, marginal by marginal [177]. *Tests of accuracy with random points* (TARP) assess joint coverage through credible regions centered on random reference points, which extends the diagnostic to generative estimators whose density cannot be evaluated explicitly [176]. Local diagnostics such as the *local classifier two-sample test* (L-C2ST) train a classifier to distinguish true from estimated posterior samples for individual observations, exposing failures that the global averages above wash out [178]. Most importantly, *posterior predictive checks* validate the inference in data space: parameters drawn from the estimated posterior are fed back through the simulator, and the resulting synthetic observations are compared with the observed data. While possibly the most complete test, posterior predictive checks are also the most computationally expensive, since they require additional simulator calls for each posterior sample.

In practice, many SBI analyses supplement these general calibration checks with problem-specific diagnostics.

3.3 Machine Learning in Astrophysics

Section 3.2 introduced simulation-based inference as a framework for problems with intractable likelihoods; this section presents the concrete machine-learning architectures and techniques that implement it, and that support data-driven inference in astrophysics and cosmology more broadly. In practice, the data that these methods must process arrive at one of two levels. *Field-level data*—images, sky maps of photon counts, light curves—preserve the full spatial or temporal structure of the signal and call for architectures that can exploit that structure directly. *Summary statistics*—energy spectra, variability indicators, flux ratios—compress the raw signal into a low-dimensional feature set that is easier to model but inevitably discards some information in the process. The two approaches are complementary: field-level methods retain more information at the cost of architectural complexity, while summary statistics trade some information for a physically motivated, lower-dimensional input space. Both appear in this thesis: Chapter 6 works at the field level, feeding gamma-ray sky maps into a convolutional network to reconstruct the source-count distribution, while Chapter 5 operates on spectral and variability features to search for a dark matter subhalo population among unassociated Fermi-LAT sources.

This section introduces the machine-learning concepts and architectures underlying both analyses: the canonical learning tasks (Section 3.3.1); neural networks, their training, and uncertainty estimation (Section 3.3.2); and convolutional architectures and their spherical extensions for sky-map data (Section 3.3.3).

²Formally, the expected coverage is the probability, under the joint distribution $p(\boldsymbol{\theta}, \mathbf{x})$, that $\boldsymbol{\theta}$ falls within the $(1 - \alpha)$ highest-density region $\Omega_{\hat{p}}(1 - \alpha)$ of the estimated posterior. When $\hat{p}(\boldsymbol{\theta}|\mathbf{x}) = p(\boldsymbol{\theta})$, this region does not depend on $\mathbf{x}$, so the likelihood integrates out and the coverage reduces to $\int_{\Omega_{\hat{p}}(1 - \alpha)} p(\boldsymbol{\theta}) d\boldsymbol{\theta} = 1 - \alpha$, which is the definition of the credible region. More generally, the test probes only the marginal consistency condition $\int \hat{p}(\boldsymbol{\theta}|\mathbf{x}) p(\mathbf{x}) d\mathbf{x} = p(\boldsymbol{\theta})$: averaged over observations, the estimated posteriors must reproduce the prior. The true posterior satisfies this condition by construction, but so does the prior itself, trivially.

3.3.1 Learning Tasks

Machine learning provides additional tools for extracting information from data when explicit physical models are either unavailable or computationally intractable [155, 156]. In Astrophysics, machine learning problems are often divided into three common learning tasks.

The first is *density estimation*: approximating the probability distribution from which the observed data are drawn. Rather than answering a single question about each input, a density estimator reconstructs the distribution as a whole—“what does the overall population look like?”—and any quantity of interest can then be computed from it. Two analyses in this thesis rely on this methodology explicitly. In Chapter 5, Kernel Density Estimation models the feature distributions of associated gamma-ray sources, and a generative mixture model built on these densities infers the energy spectrum composition of the unassociated Fermi-LAT population. Because the method reconstructs full distributions rather than decision boundaries, it can accommodate populations absent from the training set—a property the dark matter subhalo search exploits. In Chapter 6, density estimation appears in its neural, conditional form: the neural posterior estimation framework of Section 3.2, in which a network approximates the posterior distribution of the source-count parameters given a gamma-ray sky map.

The second is *classification*: assigning a discrete label to an input based on its observed features. In the gamma-ray context, this amounts to determining the astrophysical nature of a detected source—“what class is this source?”—rather than characterising the population as a whole. Roughly one-third of the sources in Fermi-LAT catalogs lack firm multi-wavelength associations [179], and supervised classification algorithms—trained on populations of identified AGNs and pulsars—are used to predict the class membership of these unassociated objects. The input features are quantities derived from the gamma-ray signal itself: spectral curvature significance, variability index, hardness ratios, and energy flux [179].

The third task is *regression*: predicting a continuous quantity from observational data. Although the analysis of Chapter 6 is best understood as neural posterior estimation, a network similar to the one adopted there can be used for a regression task, mapping gamma-ray sky maps directly to the maximum likelihood estimation of the source-count distribution dN/dS .

It is important to note that when the full distribution is accessible, the other two tasks can be performed as special cases. Class labels can be assigned by comparing the estimated densities of the candidate populations, which is explored as a secondary result in Chapter 5. A point prediction can likewise be read off as a summary of the distribution—its mean, for instance—which is what we do in Chapter 6 when deriving the frequentist confidence intervals. In this sense, density estimation is arguably the most fundamental of the three tasks: it retains the richest description of the data, from which discrete labels and point estimates can be derived, when needed. This generality has a cost—estimating a full distribution is harder than answering a single question about each input—so direct classification or regression remains the practical choice for many applications when a solid understanding of the uncertainties is not required.

3.3.2 Neural Networks and Deep Learning

A neural network is a parametric function $f_{\mathbf{w}} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ composed of a sequence of affine transformations, each followed by a non-linear activation function. In a network with L layers, the output of layer ℓ is $\mathbf{h}^{(\ell)} = \sigma(\mathbf{W}^{(\ell)} \mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)})$, where $\mathbf{W}^{(\ell)}$ and $\mathbf{b}^{(\ell)}$ are trainable weight matrices and bias vectors, and σ is a non-linear function such as the rectified linear unit $\text{ReLU}(x) = \max(0, x)$, among others. Given a sufficiently large number of parameters, neural networks can approximate any continuous function to arbitrary accuracy [155, 180].

Training consists of adjusting the weights $\mathbf{w} = (\mathbf{W}^{(1)}, \mathbf{b}^{(1)}, \dots, \mathbf{W}^{(L)}, \mathbf{b}^{(L)})$ to minimize a cost function $\mathcal{J}(\mathbf{w})$ using gradient-based optimisation. Backpropagation provides the necessary gradients by applying the chain rule recursively through the network layers, yielding the partial derivative of $\mathcal{J}$ with respect to every weight in a single backward pass. In practice, gradients are estimated from mini-batches (typically 32 to 128 samples from the dataset, but sometimes even higher) rather than the full dataset, giving rise to stochastic gradient descent (SGD) and its modern adaptive variants such as Adam and AdamW [181]. The empirical success of deep learning over the past decade rests on algorithmic advances together with network depth, large datasets, and GPU-accelerated compute [182].

The choice of cost function determines what and how the network learns. For example, for a standard regression problem, the mean squared error $\mathcal{J} = \sum_i (y_i^{\text{true}} - y_i^{\text{pred}})^2$ penalises deviations between predictions and targets uniformly and coincides, up to constants, with the negative log-likelihood of the targets under a homoscedastic Gaussian noise model. For classification, the cross-entropy loss measures the divergence between predicted class probabilities and one-hot labels and is equivalently the negative log-likelihood of the observed labels under a categorical model. Density estimation in machine learning is often cast as minimising the Kullback–Leibler (KL) divergence, in its forward or reverse form. The KL divergence quantifies the mismatch between the data distribution and the parametric model: it is non-negative and vanishes only when the two distributions coincide, so smaller values indicate a better fit. The KL divergence is not the only possible measure of discrepancy between distributions, and other notions of distance exist, see [167, 183].

When the noise level varies across the output space, the Gaussian variance can itself be parameterised as a network output, yielding the heteroscedastic Gaussian cost function [184]; the formalism is presented in Section 6.5.3.4, where Chapter 6 puts it to use.

Beyond a point estimate, predictions carry two distinct sources of error: *aleatoric* (statistical) uncertainty, intrinsic to the data (irreducible noise that more training data cannot remove), and *epistemic* (broadly speaking systematic) uncertainty, reflecting the model’s ignorance about which parameter values best describe the data (reducible in the limit of infinite data and capacity). Chapter 6 presents one practical solution to model both uncertainties, in which MC dropout and a Gaussian distribution are combined to produce per-output mean and variance estimates; the technical formalism is detailed there (Section 6.5.3.4).

3.3.3 Convolutional Neural Networks

For image-domain analyses, the input is a pixel grid—or, in astrophysics, a HEALPix sky map—and convolutional neural networks (CNNs) exploit this spatial structure directly. CNNs have been applied to astrophysical problems such as galaxy morphology classification [185], strong gravitational lens identification [186], and electromagnetic transient detection. In this thesis, Chapter 6 adopts a CNN to reconstruct the source-count distribution from gamma-ray sky maps.

A convolutional layer applies a set of learnable filters (kernels) to local patches of the input, producing feature maps that respond to spatial patterns at specific scales. For a two-dimensional input with pixels identified by indices (i, j) , the discrete convolution of a filter g with an input f is

$$(f * g)[i, j] = \sum_m \sum_n f[m, n] g[i - m, j - n]. \quad (3.3.1)$$

Because the same filter is applied everywhere across the input, the number of parameters per layer is independent of the input size, and the response to a given pattern is the same regardless of its location—a property known as translation equivariance.

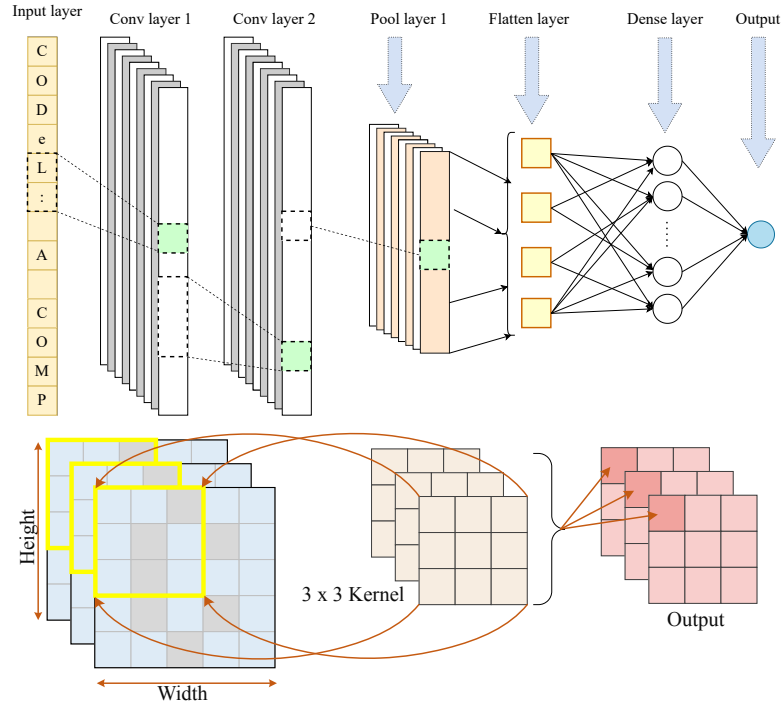


Fig. 3.1. *Upper:* Generic (1D) convolutional neural network architecture: a kernel slides over the input image producing feature maps; successive convolution and pooling layers reduce spatial extent while increasing channel depth, before a fully connected head maps the learned features to the output. *Lower:* schematic portraying how a 3x3 kernel slides over a 2D input image to produce a feature map. Credit: Younesi et al. [187]

Stacking convolutional layers builds progressively larger receptive fields: shallow layers detect local features such as edges and peaks, while deeper layers combine these into more abstract representations. Pooling operations (max or average over local windows) downsample the feature maps between successive convolutional stages, providing approximate translation invariance at the cost of spatial resolution [180, 182]. This hierarchy of local-to-global features allows CNNs to capture information distributed across spatial scales, and is summarized in Figure 3.1.

Modern CNNs build on a few architectural ideas: VGG-style stacks of small convolutions [188] and the residual connections introduced by ResNet [189], which enabled training of networks hundreds of layers deep. Contemporary computer-vision architectures often combine convolutional building blocks with attention mechanisms borrowed from transformers [190], as in vision transformers [191] and hybrid models.

A specific challenge arises when the data live on the sphere rather than on a flat grid. Sky maps live on S^2 , not $\mathbb{R}^2$: standard CNNs assume Euclidean translation symmetry, whereas the sphere has rotational symmetry instead, and any flat parametrisation introduces distortion that breaks weight sharing. Three broad strategies have been developed to extend convolutional architectures to spherical data.

The first strategy performs convolutions in the spherical harmonic domain. Because the spherical harmonics are the natural eigenbasis of the Laplace–Beltrami operator on S^2 , a spectral convolution can be defined in exact analogy with the classical convolution theorem on the real

line. DeepSphere [192] implements this idea using a graph-based approximation, representing the sphere as a graph whose nodes are HEALPix pixels and whose edges encode pixel adjacencies; spectral graph convolutions then replace the standard spatial filters. This approach preserves the global topology of the sphere and handles irregular pixelisations.

The second strategy applies convolutions to local projections of the sphere. Rather than operating on the curved manifold directly, these methods project spherical patches onto tangent planes or polyhedral faces, where standard 2D convolutions apply without modification. Icosahedral projections [193] decompose the sphere into 20 equilateral triangular faces that cover it nearly uniformly, limiting the distortion inherent in any flat projection.

The third strategy, and the one most directly relevant to this thesis, operates on the HEALPix pixelisation itself. NNHealpix [194] exploits the fixed neighbourhood structure of HEALPix pixels by unrolling each pixel and its eight nearest neighbours into a nine-element vector and applying standard 1D convolutions with stride nine (Figure 3.2). This preserves the full topology of the sphere at the cost of higher computational overhead and sensitivity to the orientation of pixel neighbourhoods. The map2patch algorithm developed in Chapter 6 subdivides the sphere into the 12 equal-area base pixels of the HEALPix scheme and processes them jointly as a stack of flat tiles, exploiting the exact equal-area property of the base faces so that standard 2D convolutions apply without resampling distortion. The spherical topology is preserved across patch boundaries through a padding scheme that borrows pixels from the adjacent base faces, so that no spatial information is lost where neighbouring tiles meet. The construction preserves the exact area and photon content of every pixel; details of the implementation are deferred to Chapter 6. For statistically isotropic, small-scale signals such as the point-source population studied in Chapter 6, map2patch and NNHealpix produce compatible results, while map2patch achieves more than an order of magnitude improvement in execution speed. More recently, HEAL-SWIN [195] adapted the Swin Vision Transformer to the HEALPix sphere: shifted-window attention operates within local HEALPix patches, and the window boundaries are shifted between successive transformer blocks so that the model captures both local and global spherical structure while retaining the computational efficiency of window-based attention.

Chapter 6 combines the map2patch tiling with concrete dropout and a heteroscedastic Gaussian negative log-likelihood to produce both the reconstructed source-count distribution and calibrated uncertainty estimates; the full architecture and loss formulation are presented in Section 6.5.3.4.

3.4 The Domain Shift Challenge

The machine learning methods introduced in Section 3.3 are built on the implicit assumption that the training data faithfully represent the conditions under which the model will be deployed. When this assumption breaks down—as it does for several astrophysical classification tasks—the standard approaches described in the previous sections require modification. We now turn to the problem of *dataset shift*, a systematic mismatch between training and target distributions that, if left unaddressed, can bias inference results. We first formalize the problem (Section 3.4.1) and then describe its principal types and the strategies used to correct them (Section 3.4.2).

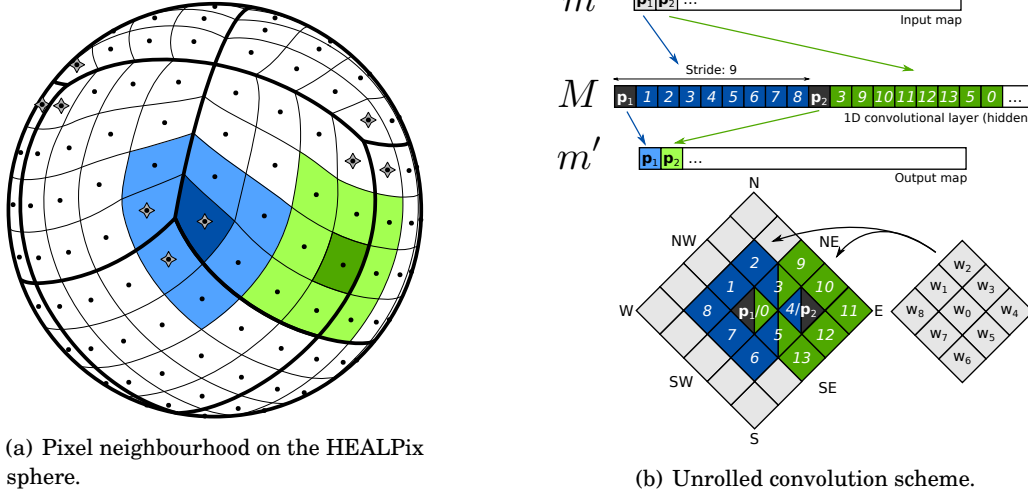


Fig. 3.2. Convolution on the HEALPix sphere. *Left:* each pixel and its nearest neighbours define a local window that can be mapped to a flat patch for convolution. *Right:* in the NNHealpix scheme, each pixel and its eight neighbours are unrolled into a nine-element vector, and a standard one-dimensional convolution with stride nine is applied. Credit: Figs. 2 and 3 of [194].

3.4.1 The Problem Statement

Supervised classification algorithms rely on the assumption that the training data and the target data share the same joint probability distribution of input features $\mathbf{x}$ and output classes k :

$$p_{\text{train}}(\mathbf{x}, k) = p_{\text{target}}(\mathbf{x}, k). \quad (3.4.1)$$

When this condition holds, a classifier trained on labeled examples can be applied directly to unlabeled data with the expectation that its performance will generalize. In practice, however, this assumption is sometimes violated. A mismatch between the training and target distributions—known as *dataset shift* [196]—invalidates the direct application of the trained model and can produce systematically biased estimates of class prevalences and, consequently, incorrect statistical constraints on physical parameters.

This problem is directly relevant for astrophysical source classification. In the Fermi-LAT catalogs, sources are divided into *associated* and *unassociated* categories based on whether a counterpart at other wavelengths has been identified. Machine learning classifiers are typically trained on associated sources—whose class labels (e.g., pulsar or active galactic nucleus) are known from their multi-wavelength counterparts—and then applied to unassociated sources to infer their nature (cf. Section 3.3.1). However, several observational biases break the equal-distribution assumption. Bright sources at high Galactic latitudes are easier to associate with known counterparts, so the associated sample is not a representative subsample of the full source population [197]. As a consequence, the spectral parameter distributions—flux, spectral index α , and spectral curvature β —differ measurably between associated and unassociated sources, as illustrated in Figure 3.3 (see Chapter 5 for the full analysis). Whether this discrepancy arises from observational selection effects, from genuinely different class compositions, or from a combination of both, cannot be determined a priori; we formalize the relevant types of dataset shift in Section 3.4.2. Any analysis that ignores this mismatch risks producing biased class

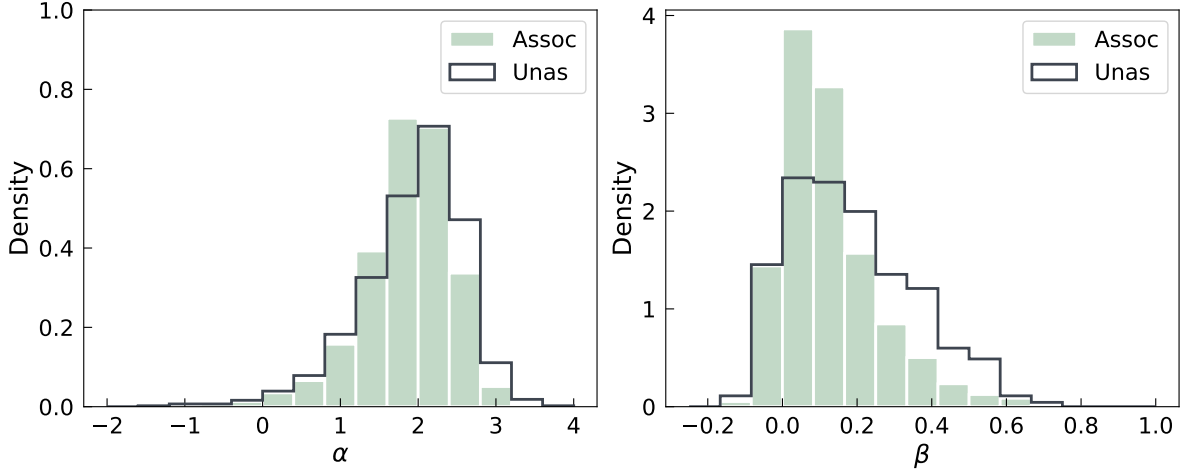


Fig. 3.3. Distributions of the spectral index α and spectral curvature β for associated (green, filled) and unassociated (gray outline) Fermi-LAT sources from the 4FGL-DR4 catalog. The visible mismatch between the two populations illustrates the dataset shift between the associated and unassociated samples discussed in Section 3.4.1. The full analysis is presented in Chapter 5.

prevalence estimates and, in the context of dark matter searches, unreliable upper bounds on the annihilation cross-section.

3.4.2 Types of Dataset Shift

The joint distribution of features and class labels can be decomposed in two equivalent ways:

$$p(\mathbf{x}, k) = p(k|\mathbf{x})p(\mathbf{x}) = p(\mathbf{x}|k)p(k). \quad (3.4.2)$$

These two factorizations give rise to two distinct types of dataset shift, depending on which factor changes between the training and target domains [196]. Each type has distinct statistical consequences and correction strategies.

Covariate shift. Covariate shift is most naturally understood through the first factorization, $p(\mathbf{x}, k) = p(k|\mathbf{x})p(\mathbf{x})$. Under this type of shift, the posterior class probabilities given the features may remain unchanged between the training and target distributions:

$$p_{\text{train}}(k|\mathbf{x}) = p_{\text{target}}(k|\mathbf{x}), \quad (3.4.3)$$

while the marginal distribution of the features differs: $p_{\text{train}}(\mathbf{x}) \neq p_{\text{target}}(\mathbf{x})$ [196]. The class-conditional relationship between features and labels is preserved, but the relative abundance of feature values changes. Covariate shift typically arises from selection effects or observational biases that influence which data points enter the training sample. In the Fermi-LAT context, the association procedure preferentially identifies sources in low-background regions far from the Galactic plane, where multi-wavelength counterparts are easier to match. Proximity to the Galactic plane is, in fact, the dominant source of association bias: the high density of overlapping sources and diffuse emission near the plane causes multi-wavelength identification methods to fail more frequently [197]. The ratio $C(\mathbf{x}) = p_{\text{target}}(\mathbf{x})/p_{\text{train}}(\mathbf{x})$ between the target and training feature distributions characterizes the covariate shift [196]. When this ratio is known or can be modeled, the training-set predictions can be corrected by importance weighting.

In practice, $C(\mathbf{x})$ is rarely known analytically, but it can be approximated parametrically. If the ratio is monotonic in each feature dimension, a product of sigmoid functions provides a physically constrained model that can still accommodate a range of shift shapes:

$$\tilde{C}(\mathbf{x}; \boldsymbol{\theta}_{\text{cov}}) = \prod_{d=1}^D f(x_d; b_d, c_d), \quad f(x_d; b_d, c_d) = \frac{1}{1 + e^{-b_d(x_d - c_d)}}, \quad (3.4.4)$$

where b_d controls the steepness and c_d the midpoint of the modulation along feature dimension d (see Chapter 5 for the derivation and application). These parameters are optimized jointly with the rest of the model by maximizing the likelihood of the target data.

Prior shift. The prior shift is most clearly seen in the second factorization, $p(\mathbf{x}, k) = p(\mathbf{x}|k)p(k)$. Here, the class-conditional feature distributions are invariant:

$$p_{\text{train}}(\mathbf{x}|k) = p_{\text{target}}(\mathbf{x}|k), \quad (3.4.5)$$

but the class prevalences differ: $p_{\text{train}}(k) \neq p_{\text{target}}(k)$ [196]. This type of shift is common when the population composition changes between the training and deployment environments. Among Fermi-LAT sources, for instance, the fraction of Galactic sources is considerably larger in the unassociated sample than in the associated one, because a higher proportion of unassociated sources lie near the Galactic plane [197]. Standard classifiers trained with balanced or representative class fractions from the associated data will therefore assign incorrect class probabilities when applied to the unassociated population. The quantification learning literature [198] addresses this problem by fitting the mixture weights π_k of a generative model directly to the target data, rather than relying on the class fractions of the training set [199, 200].

Combined shift. In general, covariate and prior shifts act simultaneously, and separating their effects is difficult. Changes in the marginal class prevalences $p(k)$ produce corresponding changes in the marginal feature distribution $p(\mathbf{x})$, and vice versa, so that the same observed mismatch between training and target distributions can be explained by either mechanism or by both acting together [196]. Resolving this degeneracy requires additional physical constraints or a model flexible enough to fit both effects at once. A purely covariate-shift correction, for example, might absorb prevalence changes into the reweighting function, while a purely prior-shift model would attribute selection biases to changes in class fractions. Neither gives the correct answer when both effects are present.

Application preview. Addressing this challenge requires a model that treats both effects simultaneously. In Chapter 5, we construct the first such statistical model for Fermi-LAT sources. The unassociated source distribution is modeled as a mixture of Galactic and extragalactic astrophysical components, $p(\mathbf{x}) = \sum_k \pi_k p(\mathbf{x}|k)$, whose class-conditional densities $p_{\text{assoc}}(\mathbf{x}|k)$ are estimated from the associated sources via kernel density estimation, plus a possible dark matter subhalo component derived from Monte Carlo simulations. The prior shift is handled by fitting the class prevalences π_k to the unassociated data, while the covariate shift is corrected by the sigmoid modulation functions $\tilde{C}(\mathbf{x}; \boldsymbol{\theta}_{\text{cov}})$ described above. This approach draws on the classical dataset shift taxonomy of Moreno-Torres et al. [196] and the quantification learning methods developed by González et al. [198] and Moreo et al. [200], combining them into a unified framework that can be used both for the probabilistic classification of unassociated sources—its primary astrophysical application—and for setting statistically rigorous upper bounds on the dark matter annihilation cross-section. The full analysis is presented in Chapter 5.

3.5 Cross-Correlations as a Complementary Probe

The methods discussed so far in this chapter—frequentist and Bayesian inference, simulation-based approaches, machine learning, domain adaptation in presence of a dataset shift—all operate on data from a single instrument, extracting information about source populations from the gamma-ray sky alone. We consider now a qualitatively different approach. The cross-correlation technique correlates a signal associated with the particle nature of dark matter—for instance, gamma rays from annihilation or decay—with an independent gravitational tracer of the matter distribution, such as a galaxy catalog. A positive correlation would favor an interpretation of DM in terms of new physics, since modified gravity theories do not predict the production of gamma-rays. Cross-correlations are powerful because they exploit the fact that dark matter is the dominant source of gravitational clustering. If dark matter annihilates or decays into gamma rays, the resulting emission must trace the same large-scale structure (cf. Section 1.1.3) mapped by galaxy surveys.

We introduce the cross-correlation formalism in two steps. Section 3.5.1 defines the angular power spectrum, the Limber approximation that projects three-dimensional clustering into two-dimensional observables, and the window functions that encode the redshift distribution of each signal. Section 3.5.2 presents the signal-to-noise ratio and the $\Delta\chi^2$ test statistic—used to forecast the sensitivity of a given experiment to dark matter. The full application of this formalism, including the halo model computation and the detailed sensitivity forecasts for the Cherenkov Telescope Array Observatory, is deferred to Chapter 8.

3.5.1 The Angular Power Spectrum

The observable sky maps produced by gamma-ray telescopes and galaxy surveys are two-dimensional projections of the three-dimensional matter distribution in the Universe. To study the spatial correlations between these projected fields, we expand them in spherical harmonics $Y_{\ell m}(\hat{n})$. The angular power spectrum (APS) C_ℓ quantifies the amplitude of these correlations as a function of the angular scale, parameterized by the multipole ℓ .

We begin by considering a generic intensity field $I_g(\hat{n})$ along the direction $\hat{n}$, which can be written as an integral over the radial comoving distance $\chi(z)$ [201, 202]:

$$I_g(\hat{n}) = \int d\chi g(\chi, \hat{n}) W(\chi), \quad (3.5.1)$$

where $g(\chi, \hat{n})$ is the density field of the source and $W(\chi)$ is the window function encoding the redshift-dependent emissivity or source distribution. The intensity fluctuations $\delta I_g(\hat{n}) \equiv I_g(\hat{n}) - \langle I_g \rangle$ can then be expanded in spherical harmonics as

$$\delta I_g(\hat{n}) = \sum_{\ell m} a_{\ell m} Y_{\ell m}(\hat{n}), \quad (3.5.2)$$

where the coefficients $a_{\ell m}$ encode the angular structure of the fluctuation field [201].

For two intensity fields labelled i and j (for instance, the gamma-ray flux and the galaxy number counts), the angular power spectrum is defined as the ensemble average of the product of their expansion coefficients [201, 202]:

$$C_\ell^{ij} = \frac{1}{2\ell + 1} \left\langle \sum_{m=-\ell}^{\ell} a_{\ell m}^{(i)} a_{\ell m}^{(j)*} \right\rangle. \quad (3.5.3)$$

When $i = j$, the expression gives the auto-correlation APS, which measures the intrinsic clustering of a single field. When $i \neq j$, we obtain the cross-correlation APS, which measures the spatial correlation between two distinct observables. By construction, $C_\ell^{ij} = C_\ell^{ji}$.

To connect the measured two-dimensional APS to the underlying three-dimensional power spectrum $P_{ij}(k, \chi)$, which encodes the spatial clustering of the correlated fields, we apply the Limber approximation [203]. This approximation, valid for multipoles $\ell \gg 1$, replaces the oscillatory integrals over spherical Bessel functions with a single line-of-sight integral by identifying the wavenumber $k = \ell/\chi$. The result is a compact expression for the angular power spectrum [201, 202]:

$$C_\ell^{ij} = \int \frac{d\chi}{\chi^2} W_i(\chi) W_j(\chi) P_{ij}\left(k = \frac{\ell}{\chi}, \chi\right). \quad (3.5.4)$$

The two ingredients play complementary roles. The window functions $W_i(\chi)$ act as line-of-sight weights that fix how each observable is distributed in redshift, whereas $P_{ij}(k, \chi)$ describes the spatial clustering of the fields within a shell at comoving distance χ , its χ -dependence tracking the cosmological evolution of the clustering. Since the integrand is a product of the two window functions, a significant cross-correlation requires that they overlap—that both observables trace the same large-scale structure at similar redshifts [201]. This idea of correlating a gamma-ray signal with a gravitational tracer of the dark matter distribution was pioneered by Camera et al. [202], who first showed that the cross-correlation of gamma-ray emission with weak-lensing cosmic shear provides a viable channel for indirect dark matter searches.

The window functions are key ingredients of this formalism, as they encode how each observable is distributed along the line of sight. Their specific form depends on the physical origin of the signal. For a galaxy catalog acting as a gravitational tracer of the large-scale structure, the window function is simply the normalized redshift distribution of the surveyed galaxies, $W_g(z) = dN_g/dz$ [201]. The angular power spectrum from annihilating dark matter was first formulated by Ando and Komatsu [204], and the formalism was later extended to multi-wavelength and multi-tracer cross-correlations by Fornengo and Regis [201].

For dark matter annihilation, the gamma-ray window function takes the form [201, 204]

$$W_{\gamma, \text{ann}}(E, z) = \frac{(\Omega_{\text{DM}} \rho_c)^2}{4\pi} \frac{\langle \sigma v \rangle}{2m_\chi^2} (1+z)^3 \Delta^2(z) \frac{dN_a}{dE} [E(1+z)] e^{-\tau(E, z)}, \quad (3.5.5)$$

where $\langle \sigma v \rangle$ is the velocity-averaged annihilation cross section, m_χ is the dark matter particle mass, $\Delta^2(z) = \langle \rho^2 \rangle / \bar{\rho}^2$ is the clumping factor accounting for the enhancement due to substructure within halos, dN_a/dE is the photon spectrum per annihilation event, and $\tau(E, z)$ is the optical depth for absorption on the extragalactic background light. For decaying dark matter, the window function scales linearly with the density rather than quadratically, and the clumping factor does not appear [201]. The dark matter window function is strongly peaked at low redshifts, where the local dark matter density is highest and the clumping factor most effective. By contrast, the *unresolved* astrophysical source populations that contribute to the diffuse gamma-ray background have markedly different redshift distributions: virtually all blazars at $z \lesssim 0.1$ have already been individually detected and cataloged, so the redshift distribution of the remaining unresolved population drops sharply toward $z \sim 0$, peaking instead at moderate redshifts. This separation in redshift is what makes the cross-correlation with low-redshift galaxy catalogs particularly powerful for isolating a dark matter signal: the dark matter window function peaks precisely where the unresolved astrophysical background is most suppressed [201].

The three-dimensional power spectrum $P_{ij}(k, \chi)$ in the Limber approximation describes the spatial clustering of the correlated fields. It is computed within the halo model framework, which

partitions all matter in the Universe into discrete, collapsed dark matter halos [205]. Under this assumption, the power spectrum decomposes into two contributions [201, 205]:

$$P_{ij}(k, z) = P_{ij}^{1h}(k, z) + P_{ij}^{2h}(k, z). \quad (3.5.6)$$

The one-halo term P^{1h} accounts for correlations between two points within the same individual halo; it probes the internal mass distribution of halos and dominates on small physical scales (large k). The two-halo term P^{2h} captures correlations between points in two distinct halos; it traces the underlying linear matter power spectrum modulated by the halo bias factors and dominates on large scales (small k). Both terms are expressed as integrals over the halo mass function dn/dM and the Fourier-transformed density profiles of the relevant tracers [201, 205]. The detailed computation of these terms for the specific gamma-ray and galaxy populations considered in this thesis is deferred to Chapter 8, where the full cross-correlation analysis is presented.

3.5.2 Statistical Sensitivity

Given the cross-correlation angular power spectrum $C_\ell^{\gamma g}$ defined in the previous section, we now turn to the statistical tools used to assess how well a given experiment can detect or constrain a dark matter signal. Two quantities are central to this assessment: the signal-to-noise ratio (SNR) and the $\Delta\chi^2$ test statistic.

The detectability of the cross-correlation signal from a given source population (whether astrophysical or dark matter) is quantified by the signal-to-noise ratio [201]:

$$\text{SNR} = \sqrt{\sum_{\ell, i} \left(\frac{C_{\ell, i}^{\gamma g}}{\Delta C_{\ell, i}^{\gamma g}} \right)^2}, \quad (3.5.7)$$

where the sum runs over the multipoles ℓ included in the analysis and the energy bins i . We note that the statistical formalism presented here is tailored to cross-correlations with galaxy catalogs, which is the gravitational tracer relevant for this thesis. Cross-correlations with other tracers, such as weak-lensing cosmic shear [202, 206–208] or HI intensity maps [209], involve analogous expressions but with modified noise terms and window functions that reflect the specifics of each probe and instrument. The variance of the cross-correlation spectrum for a fixed energy bin i , assuming Gaussian-distributed fluctuations, is given by [201]

$$(\Delta C_\ell^{\gamma g})^2 = \frac{1}{(2\ell + 1)f_{\text{sky}}} \left[(C_\ell^{\gamma g})^2 + \left(C_\ell^{\gamma\gamma} + \frac{C_N}{B_\ell^2} \right) (C_\ell^{gg} + C_{N_g}) \right], \quad (3.5.8)$$

where f_{sky} is the observed sky fraction, $C_\ell^{\gamma\gamma}$ and C_ℓ^{gg} are the gamma-ray and galaxy auto-correlation spectra, C_N is the photon noise of the gamma-ray detector, C_{N_g} is the shot noise of the galaxy catalog, and B_ℓ is the beam window function encoding the finite angular resolution (point-spread function) of the instrument. No beam correction enters the galaxy term, as the angular resolution of galaxy catalogs (typically arcseconds) is negligible at the scales relevant for this analysis (Chapter 8).

The structure of the variance formula reveals a fundamental advantage of cross-correlation over auto-correlation for dark matter searches. The corresponding auto-correlation variance is [201]

$$(\Delta C_\ell^{\gamma\gamma})^2 = \frac{2}{(2\ell + 1)f_{\text{sky}}} \left(C_\ell^{\gamma\gamma} + \frac{C_N}{B_\ell^2} \right)^2, \quad (3.5.9)$$

where the photon noise C_N now enters quadratically and adds directly to the signal. In an auto-correlation measurement, $C_\ell^{\gamma\gamma} + C_N$ must be measured and the noise accurately estimated and subtracted to recover the true clustering spectrum. In cross-correlation, by contrast, the noise sources in the gamma-ray map and in the galaxy catalog are physically independent: their cross-noise vanishes on average. The photon noise C_N and galaxy shot noise C_{N_g} therefore contribute only to the statistical uncertainty (the error bars) of the cross-correlation measurement, not to the signal itself. This allows faint correlated signals—such as a subdominant dark matter contribution—to be extracted even in noisy datasets where the auto-correlation would be completely dominated by instrumental noise [201, 202]. A quantitative comparison of auto- and cross-correlation SNR for CTAO observations is given in Table 8.2.

To assess the sensitivity to a dark matter signal specifically, we perform a nested-model comparison in the spirit of Wilks’ theorem [201]. The null hypothesis includes only astrophysical sources, while the alternative hypothesis adds a dark matter component for a fixed mass and parameterized by a single additional degree of freedom—the annihilation cross section $\langle\sigma v\rangle$ or, in the decay case, the lifetime τ_χ . The test statistic is

$$\Delta\chi^2 = \sum_{\ell,i} \left(\frac{C_{\ell,i}^{\gamma_{\text{astro}}+\text{DM } g} - C_{\ell,i}^{\gamma_{\text{astro}} g}}{\Delta C_{\ell,i}^{\gamma_{\text{astro}}+\text{DM } g}} \right)^2 \quad (3.5.10)$$

where $C_\ell^{\gamma_{\text{astro}} g}$ is the predicted cross-correlation signal under the null hypothesis (astrophysical sources only) and $C_\ell^{\gamma_{\text{astro}}+\text{DM } g}$ includes an additional dark matter component. The test statistic measures how much the inclusion of a dark matter signal improves the fit relative to the astrophysical-only model. In other words, we are comparing the hypothesis that the true unresolved gamma-ray emission is purely astrophysical against the alternative that it contains also a dark matter contribution. In Chapter 8, this formalism is applied to forecast the sensitivity of the Cherenkov Telescope Array Observatory (CTAO) to dark matter annihilation and decay, by cross-correlating the predicted gamma-ray emission with the 2MASS galaxy catalog.

Part II

Galactic Dark Matter

THE GALACTIC CENTER GAMMA-RAY EXCESS

Since 2009, analyses of Fermi-LAT data have consistently revealed a bright, statistically significant excess of gamma-ray emission from the region surrounding the Galactic Center [80, 126, 127, 210]. This signal peaks at energies between roughly 1 and 3 GeV, extends out to at least 10 degrees from Sgr A*, and persists across all existing models of the astrophysical diffuse emission [210]. Its spectrum, angular distribution, and overall intensity are each consistent with the signal predicted from the annihilation of dark matter particles distributed according to a contracted Navarro–Frenk–White profile, with best-fit masses that vary from analysis to analysis within roughly 30–50 GeV [11, 80, 210]. The corresponding annihilation cross section, $\langle\sigma v\rangle \sim (1\text{--}3) \times 10^{-26} \text{ cm}^3/\text{s}$, is compatible with the canonical thermal relic value [80, 210].

The leading astrophysical alternative attributes the excess to a large, unresolved population of millisecond pulsars (MSPs) in the Inner Galaxy [211]. MSPs produce gamma-ray emission with a spectral shape similar to the GCE, and by the late 2010s the combination of photon-count statistics, wavelet analyses, and morphological fits tracing the stellar bulge had persuaded much of the community that an astrophysical origin was favored [212–215].

The robustness of the point-source evidence was called into question [216, 217] and defended in turn [218–220], while recent analyses incorporating energy information seem to weaken the preference for point sources [221]. At the same time, the MSP hypothesis faces independent observational challenges: few pulsars have been detected in the Inner Galaxy, and their expected number based on known luminosity functions substantially exceeds the handful of candidates identified in current catalogs [1, 222, 223].

After more than fifteen years, the origin of the GCE remains an open question. No consensus has emerged: the literature contains active, well-argued cases both for and against each interpretation, and the dark matter hypothesis is arguably regaining credibility recently. This chapter traces the observational history of the excess and its consistency with dark matter annihilation (Section 4.1), examines the rise of the millisecond pulsar hypothesis and the challenges it faces (Section 4.2), and describes recent works that have reopened the question (Section 4.3). We then motivate an independent, orthogonal approach: constraining the MSP luminosity function through observations of the Milky Way’s globular cluster system (Section 4.4), a measurement that further sharpens the tension with the MSP interpretation as it has been formulated so far.

Section 4.5 then presents this analysis in full, following my published study of MSP gamma-ray luminosity functions in globular clusters and its implications for the GCE [1].

4.1 Discovery and Characterization of the GCE

This section traces the observational history of the Galactic Center Excess, from its first identification in early Fermi-LAT data through the systematic studies that refined its spectral and morphological properties. By the mid-2010s, these studies had converged on a set of observational properties that would frame the interpretive debate discussed in subsequent sections. We close by examining the consistency of these properties with dark matter annihilation.

4.1.1 First Identification

The possibility that dark matter annihilation could produce a detectable gamma-ray signal from the Galactic Center had been discussed extensively in the theoretical literature well before the launch of the Fermi Gamma-Ray Space Telescope. The high dark matter density predicted by N-body simulations in the inner kiloparsec of the Milky Way, combined with the ρ^2 dependence of the annihilation rate, had made this region one of the most promising targets for indirect detection (see Section 1.4).

The first observational evidence for an anomalous gamma-ray component from this region was reported by Goodenough and Hooper in 2009 [126], using roughly one year of Fermi-LAT data. By modeling the angular distribution and energy spectrum of gamma rays collected from the direction of the Galactic Center, they identified a distinctive spectral feature — a bump peaking between approximately 1 and 5 GeV — that could not be absorbed by known diffuse emission templates or cataloged point sources. The angular distribution of this excess emission was centered on the dynamical center of the Milky Way and extended across the inner few degrees. Goodenough and Hooper found that both the spectrum and the spatial distribution were well described by the annihilation of a relatively light dark matter particle with mass $m_\chi \approx 25\text{--}30$ GeV, annihilating primarily into $b\bar{b}$ quarks. The explored halo profile was a Navarro–Frenk–White distribution, which once fit displayed a mildly steepened inner slope, $\rho(r) \propto r^{-\gamma}$ with $\gamma = 1.1$, consistent with the adiabatic contraction expected from baryonic dynamics in the inner Galaxy (see Section 1.4.3). The corresponding annihilation cross section, $\langle\sigma v\rangle \approx 9 \times 10^{-26}$ cm³/s, was within a factor of a few of the value predicted for a thermally produced relic. A subsequent, refined analysis by Hooper and Goodenough [127], incorporating additional data and improved background modeling, confirmed these findings and sharpened the parameter estimates.

While these early results were suggestive, the authors were careful to note that an astrophysical origin for the excess — such as emission from a concentrated population of unresolved sources with a similar spectral shape — could not be excluded. Distinguishing between these possibilities would require both improved angular resolution and a systematic understanding of the diffuse background uncertainties, a challenge taken up by several groups in the following years.

4.1.2 Improved Characterization

The initial detection by Goodenough and Hooper prompted several independent groups to revisit the Galactic Center with more sophisticated analyses. Three studies, in particular, established the core observational properties of the excess and defined the systematic landscape that would dominate the debate for the following decade.

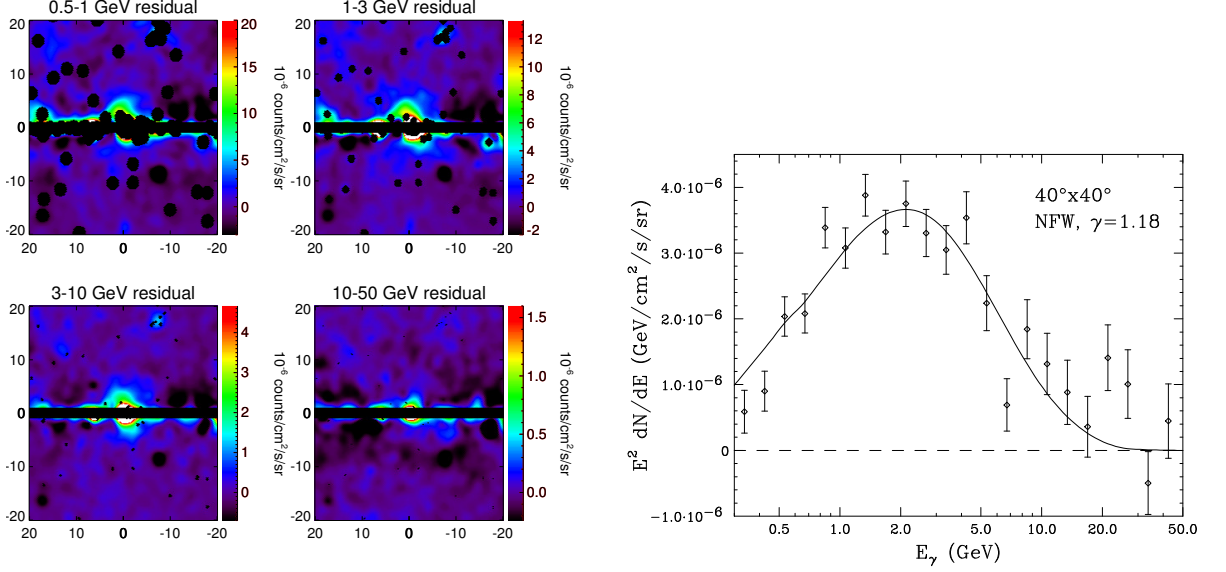


Fig. 4.1. *Left:* Residual gamma-ray intensity maps (in Galactic coordinates) after subtracting the best-fit Galactic diffuse model, Fermi bubbles, and isotropic templates. The dark-matter-like excess is clearly visible around the Galactic Center at energies between ~ 0.5 and 5 GeV. *Right:* Spectrum of the excess for a generalized NFW template with inner slope $\gamma = 1.18$, compared with the prediction for a 43 GeV dark matter particle annihilating to $b\bar{b}$ (solid line). Credit: Daylan et al. [80].

Daylan et al. [80] exploited a then-new Fermi event parameter, CTBCORE, which quantifies the reliability of each photon’s directional reconstruction. By selecting only the top two quartiles of front-converting Ultraclean events ranked by this parameter, they suppressed the tails of the point spread function substantially, particularly at low energies where the PSF is broadest. The resulting high-resolution maps (Figure 4.1, left) mitigated the leakage of bright astrophysical emission from the Galactic Plane into the regions of interest, and allowed a cleaner separation of the various gamma-ray components. Within these maps, Daylan et al. confirmed the excess at high statistical significance and measured its angular distribution to be approximately spherically symmetric, centered within $\sim 0.05^\circ$ of Sgr A^{*}, and extending to at least $\sim 10^\circ$ from the Galactic Center. The improved angular resolution also shifted the best-fit dark matter mass upward: the data favored a 36–51 GeV particle annihilating into $b\bar{b}$ with a cross section of $\langle\sigma v\rangle = (1-3) \times 10^{-26}$ cm³/s, normalized to a local dark matter density of 0.4 GeV/cm³ (Figure 4.1, right). The spatial profile was consistent with a contracted generalized NFW distribution.

Taking a complementary approach, Abazajian et al. [224] focused on constructing empirical models of the diffuse gamma-ray background toward the Galactic Center. Recognizing that the standard Fermi diffuse model might miss localized features, they included a spatial template tracing molecular gas as mapped by 20 cm radio emission. This template accounted for nonthermal bremsstrahlung produced by cosmic-ray electrons interacting with dense molecular clouds in the region. While the inclusion of this additional component confirmed that the extended excess remained highly significant, it also demonstrated a strong dependence of the inferred low-energy spectrum on the choice of background model. In particular, the molecular gas template absorbed some emission that had previously been attributed to the central point source Sgr A^{*}, substantially altering its apparent spectral shape. For the excess itself, Abazajian et al. found the

$b\bar{b}$ and $\tau^+\tau^-$ annihilation channels to provide statistically equivalent fits, with the $b\bar{b}$ channel yielding a best-fit mass of $m_\chi = 39.4^{+3.7}_{-2.9}$ (stat.) ± 7.9 (sys.) GeV.

The most comprehensive assessment of background model uncertainties was carried out by Calore, Cholis, and Weniger [210], who evaluated the excess against 60 distinct Galactic diffuse emission models built using the GALPROP cosmic-ray propagation code. These models spanned a deliberately wide range of physical parameters — cosmic-ray source distributions, gas maps, diffusion coefficients, re-acceleration, convection, interstellar radiation field normalizations, and magnetic field configurations — to bracket the theoretical uncertainty envelope.

Two results from this study proved particularly important. First, the spectral shape of the excess was relatively stable regardless of the adopted diffuse model, a robustness that we quantify together with the other established properties of the signal in Section 4.1.3. Second, when the large correlated uncertainties from empirical residuals were folded into spectral fits, the data were equally well described by a $49^{+6.4}_{-5.4}$ GeV dark matter particle annihilating into $b\bar{b}$, and by a generic broken power-law with break energy $E_{\text{break}} = 2.1 \pm 0.2$ GeV. Because such a broken power-law can readily parameterize astrophysical emission — from unresolved millisecond pulsars, for instance — this analysis made clear that spectral information alone could not distinguish between dark matter and astrophysical origins.

The Fermi-LAT Collaboration’s own analysis of the inner Galaxy [225] essentially confirmed these findings, although the appearance of similar residuals elsewhere along the Galactic plane raised questions about the completeness of existing diffuse models [210].

4.1.3 Established Properties of the Excess

By 2015, the combined results of these and other analyses had established a set of properties for the excess that are broadly agreed upon, even by groups that differ on the question of its origin.

Figure 4.1 displays the residual intensity maps which highlighted the GCE as well as the energy spectrum fit of the excess. The spectrum of the GCE displays a pronounced peak at energies of 1–3 GeV. Below the peak, the emission rises steeply, with a spectral index harder than ~ 2 down to the lowest energies accessible to the analysis (~ 300 MeV), though the precise shape of this sub-GeV component depends on the adopted diffuse model [210]. Above the peak, the spectrum follows a power-law tail with index ~ -2.7 [80, 210]. The behavior at energies above ~ 10 GeV remains the subject of some debate: early analyses suggested that the spectrum might drop sharply to zero at these energies, but the systematic study by Calore, Cholis, and Weniger [210] showed that this conclusion was not robust, and that a finite, model-dependent flux persists into the tens-of-GeV range. The behavior of the spectrum above 10 GeV would later become relevant to distinguishing between astrophysical and dark matter interpretations, as discussed in Section 4.1.4.

The spatial morphology of the excess is approximately spherically symmetric. The data strongly disfavor any elongation along or perpendicular to the Galactic Plane by more than $\sim 20\%$, and the centroid of the emission lies within $\sim 0.05^\circ$ of the position of Sgr A* [80]. The excess has been consistently identified in concentric rings out to at least $\sim 10^\circ$ (corresponding to roughly 1.5 kpc) from the Galactic Center, beyond which systematic uncertainties in the diffuse model become dominant [80, 210]. Some background models support an extension to $\sim 15^\circ$ [210]. The radial fall-off of the emission is consistent with a generalized NFW profile with an inner slope in the range $\gamma \approx 1.1\text{--}1.3$ [80, 210, 224, 226], with different analyses finding specific best-fit values of $\gamma = 1.12 \pm 0.05$ [224], 1.18 [80], and 1.26–1.28 over larger regions of interest [80]. The steepening beyond the canonical $\gamma = 1.0$ NFW slope is qualitatively consistent with the adiabatic

contraction of the dark matter profile due to the baryonic potential in the inner Galaxy (see Section 1.4.3).

The robustness of these properties is one of the most striking features of the excess. The $\sim 1\text{--}3$ GeV spectral peak persists across all 60 diffuse models tested by Calore, Cholis, and Weniger [210], with the flux above 1 GeV varying by less than a factor of a few. No known combination of astrophysical diffuse emission processes — including cosmic-ray interactions with gas and radiation fields, emission from the Fermi Bubbles, or resolved catalogued point source populations — reproduces the observed spherically symmetric morphology [80]. The spherical symmetry, in particular, does not trace any standard astrophysical component (gas, dust, starlight, or star formation), all of which are concentrated along the Galactic Plane. While the signal cannot by itself identify its origin, the existence of a genuine, spatially extended, spectrally peaked excess centered on the Galactic Center was firmly established as a consensus result by 2015.

4.1.4 Consistency with Dark Matter Annihilation

The observational properties established above align closely with the predictions of WIMP dark matter annihilation, a coincidence that has sustained interest in this interpretation for over a decade. The strength of the dark matter case rests on three independent lines of agreement—spectral, morphological, and normalization—each of which is tightly constrained by a small number of parameters.

The expected gamma-ray flux from dark matter annihilation in the Galactic halo takes the factorized form

$$\frac{d\Phi_\gamma}{d\Omega dE} = \frac{\langle\sigma v\rangle}{8\pi m_\chi^2} \frac{dN_\gamma}{dE} \times \int_{\text{l.o.s.}} \rho^2(r) dl, \quad (4.1.1)$$

where the particle physics content—the annihilation cross section $\langle\sigma v\rangle$, the WIMP mass m_χ , and the photon yield per annihilation dN_γ/dE —is cleanly separated from the astrophysical J-factor, the line-of-sight integral of the squared dark matter density (see Section 1.4 for a detailed treatment). Because annihilation occurs essentially at rest¹, the observed spectrum is fixed by the mass and annihilation channel, while the angular distribution on the sky is determined entirely by the density profile. With only three free parameters— m_χ , the annihilation channel, and the inner slope γ of the halo profile—the dark matter model must simultaneously reproduce both the energy spectrum and the spatial morphology of the excess [80]. Additional parameters of the halo profile—the scale radius r_s and the local dark matter density $\rho_\odot$ —are conventionally fixed using dynamical constraints external to the gamma-ray data [80, 224], but their uncertainties propagate into the inferred cross section, as discussed below.

This minimal model can accommodate all three observables simultaneously. The GCE energy spectrum (Section 4.1.2) and its spatial morphology (Section 4.1.3) are independently consistent with annihilation of a $\sim 30\text{--}50$ GeV WIMP into $b\bar{b}$ final states, distributed according to a contracted NFW profile with inner slope $\gamma \approx 1.1\text{--}1.3$. The term “morphology” here encompasses the full set of spatial properties established above: the approximate spherical symmetry of the emission, its radial fall-off, and its extension out to at least $\sim 10^\circ$ from the Galactic Center. No tuning of separate parameters is required to accommodate different observables: the same mass and channel that fit the spectrum also predict the correct spatial distribution, and the same halo profile that matches the morphology yields the observed flux level.

A notable feature of the dark matter fit is its normalization. The annihilation cross section required to produce the observed flux falls in the range $\langle\sigma v\rangle \sim (1\text{--}3) \times 10^{-26} \text{ cm}^3/\text{s}$ [80, 210], which

¹WIMPs are a cold dark matter candidate and are therefore non-relativistic.

is consistent with the canonical thermal relic value of $\langle\sigma v\rangle_{\text{cosmo}} \approx 2.2 \times 10^{-26} \text{ cm}^3/\text{s}$ [11]. In other words, the very cross section needed to explain the GCE brightness today is the same cross section that would cause a WIMP to freeze out in the early universe with an abundance equal to the observed cosmological dark matter density. No Sommerfeld enhancement, substructure boost factors, or non-thermal production mechanisms are required [80]. The fact that three independent observables—spectrum, morphology, and flux normalization—could be described by a simple, internally consistent dark matter model with parameters close to the thermal relic prediction has kept the GCE among the leading candidate signals in indirect detection for over a decade. This agreement should, however, be interpreted with care. The inferred cross section is degenerate with the assumed local dark matter density, since the annihilation flux scales as $\rho_\odot^2$; varying $\rho_\odot$ within its observationally allowed range (0.3–0.4 GeV/cm³) shifts $\langle\sigma v\rangle$ by a factor of ~ 1.8 [80, 224].

A more robust independent test of the annihilation hypothesis comes from the Milky Way dwarf spheroidal galaxies, which are dark-matter-dominated and nearly free of astrophysical gamma-ray backgrounds. If the excess is annihilation, the same particle must annihilate in these systems, at a rate fixed by the same cross section. The combined Fermi-LAT analysis of 15 dwarfs with six years of Pass 8 data excluded the thermal relic value for the $b\bar{b}$ and $\tau^+\tau^-$ channels up to $m_\chi \sim 100 \text{ GeV}$ [129], and the later extension to the roughly 45 confirmed and candidate satellites discovered by DES strengthened the bound [227]. Taken at face value, these limits cover the entire GCE best-fit region.

The comparison is not decisive, however, because it inherits large systematics on both sides. On the dwarf side, the J -factors rest on the stellar kinematics of a handful of member stars; non-spherical halos weaken the limits by a factor of ~ 2 , and more conservative priors on the density profile by up to a factor of ~ 7 [228]. On the GCE side, the inferred cross section scales with the same local density and inner slope γ that already loosen the thermal-relic coincidence noted above. When the Milky Way halo normalization is bracketed self-consistently, Di Mauro and Winkler found the GCE best-fit region to sit at about the 95% CL dwarf limit for the median profile — a tension of roughly 2σ — while remaining fully compatible for the bracketing choices [229]. The Fermi-LAT collaboration itself concluded that its dwarf limits did not yet confirm or refute the dark matter interpretation of the excess [227]. The tension is therefore present, but it is not an outright exclusion. Still, the absence of a corroborating signal in the cleanest dark matter targets, at the very cross section the excess demands, weakens the case that dark matter is the only viable explanation and sharpens the motivation to examine astrophysical alternatives.

4.2 The Millisecond Pulsar Hypothesis

The observational properties of the GCE are consistent with dark matter annihilation (Section 4.1.4), but they also allow for an astrophysical explanation. The leading alternative attributes the excess to a large population of unresolved millisecond pulsars (MSPs) concentrated in the Galactic bulge. Unlike the dark matter interpretation, which requires a new particle, this hypothesis invokes a known class of astrophysical source; the question is whether a sufficiently large and concentrated population exists in the Galactic Center region. This section reviews the MSP hypothesis, its supporting evidence, and the independent constraints that challenge it.

4.2.1 The Hypothesis and Its Initial Evidence

First proposed by Abazajian in 2011 [211], the MSP hypothesis builds on the observation that MSPs produce gamma-ray spectra naturally similar to the GCE: a stacked analysis of 61 Fermi-

LAT-detected MSPs yields a spectrum well described by a power law with spectral index $\alpha \approx 1.57$ and an exponential cutoff at $E_{\text{cut}} \approx 3.78$ GeV, parameters that fall within the 99% confidence contours of the GCE spectral fit [210].

The spatial distribution of these hypothetical MSPs presents the first challenge. The known gamma-ray MSP population follows a disk-like distribution, which would produce a signal elongated along the Galactic plane with an axis ratio of $\sim 1:6$ —incompatible with the observed spherical morphology of the GCE [80]. A standard disk MSP population can account for at most 5–10% of the excess flux [210]. To reproduce the concentrated, roughly spherical GCE morphology, the MSP number density would need to increase steeply toward the Galactic Center as approximately $r^{-2.5}$, requiring a distinct “bulge” population rather than the familiar disk population [210]. Such a bulge component is not unreasonable—the high stellar densities in the inner Galaxy could produce MSPs through the same dynamical channels operating in globular clusters—but it cannot easily account for the signal’s extension to $\sim 10\text{--}15^\circ$ from the Galactic Center [210].

Two independent statistical analyses published in 2016 provided the strongest evidence favoring the point-source interpretation. Lee et al. developed the Non-Poissonian Template Fitting (NPTF) method, which generalizes standard template fitting to account for the photon-count statistics expected from unresolved point sources [212]. While a smooth dark matter halo produces Poissonian photon counts in each pixel (the variance equals the mean), a population of faint point sources introduces additional variance through localized photon clusters. By modeling the source-count distribution dN/dS as a broken power law, Lee et al. found that point sources distributed according to an NFW-squared spatial template fully absorbed the GCE, with a Bayes factor of $\sim 10^6$ favoring the point-source model over smooth dark matter emission. Their inferred source-count function rose sharply just below Fermi’s detection threshold, with the break corresponding to a luminosity of $L_\gamma \sim 2\text{--}3 \times 10^{34}$ erg/s. Extrapolating this distribution, they estimated that ~ 400 near-threshold sources could account for the entire excess within 10° of the Galactic Center [212].

In a complementary approach, Bartels, Krishnamurthy & Weniger applied a wavelet decomposition to the Fermi-LAT maps, which is sensitive to small-scale spatial fluctuations produced by unresolved point sources. Their analysis independently detected excess power at small angular scales in the inner Galaxy, consistent with a population of sub-threshold sources contributing to the GCE [213].

A separate line of evidence came from morphological studies. Macias et al. argued that the GCE spatial distribution is better described by the stellar mass of the Milky Way’s boxy/X-shaped bulge than by a spherically symmetric NFW profile [214, 230]. Using near-infrared maps to trace the three-dimensional structure of the central stellar bar, they found that a bulge template absorbed the excess emission with a statistically better fit than the NFW template. Because MSPs are expected to trace the old stellar population that forms the bulge, this morphological preference was interpreted as further evidence for an astrophysical origin. Similar conclusions were reached by Bartels et al. [215], who found that the GCE traces the stellar mass distribution in the inner Galaxy, suggesting a close link between the excess and the Galactic bulge population.

By the late 2010s, the NPTF photon statistics, wavelet power, and stellar bulge morphology appeared to have settled the question in favor of the MSP interpretation. As we will see in Section 4.3, however, subsequent studies exposed weaknesses in each of these analyses, casting doubt on their conclusions and setting off a series of rebuttals and refinements.

4.2.2 Independent Challenges

Despite the statistical evidence from NPTF and wavelet analyses, several independent lines of reasoning have challenged the plausibility of the MSP interpretation. These arguments do not rely on the photon statistics of the GCE itself, but instead constrain the expected properties of any MSP population large enough to produce the excess. These constraints rest on our current understanding of MSP formation and evolution, in the Galactic center and elsewhere, rather than on the GCE data itself; revisions to that understanding could weaken them.

The first constraint comes from scaling the observed relationship between low-mass X-ray binaries (LMXBs) and MSPs. Because MSPs evolve directly from LMXBs, stellar populations of similar age should contain a predictable ratio between the two classes. Cholis, Hooper & Linden exploited this connection by measuring the gamma-ray output of 16 Fermi-detected globular clusters hosting a total of 5 bright ($L > 10^{36}$ erg/s) LMXBs, finding that their combined luminosity amounts to only 4.8% of the GCE flux within the inner 5° [231]. If MSPs were responsible for the entire excess, the inner Galaxy should contain ~ 100 bright LMXBs; INTEGRAL observations have detected only 7 candidates in this region. Even without correcting for selection biases that inflate the globular cluster gamma-ray yield, this comparison limits the MSP contribution to $\sim 4\text{--}11\%$ of the GCE. After accounting for these biases, the authors estimated that only $\sim 1\text{--}5\%$ of the excess can reasonably be attributed to unresolved MSPs [231].

A second challenge arises from the dynamics of MSP formation. Neutron stars receive natal velocity kicks of order $\sim 10^2$ km/s during their formation, and while the kicks received by recycled pulsars are typically smaller than those of young isolated pulsars, they remain large enough to significantly broaden the spatial distribution of the resulting MSP population over gigayear timescales [232]. Brandt & Kocsis showed that this kick energy should disperse a locally formed MSP population well beyond the concentrated profile inferred for the GCE, particularly in the innermost $\sim 1^\circ$ (~ 150 pc) where the observed emission requires a steep volumetric density scaling as $\rho \propto r^{-2.4}$ to $r^{-2.6}$ [232]. A kicked population would produce a significantly flatter, more extended profile, inconsistent with the observed concentration of the signal.

The third and perhaps most direct argument relies on the luminosity function of known MSP populations. If the pulsars responsible for the GCE share the luminosity function of MSPs observed in either the Galactic plane or in globular clusters, then the bright tail of this distribution should extend above Fermi's detection threshold, producing individually resolved sources. Hooper & Linden first quantified this constraint using the globular cluster MSP luminosity function, finding that Fermi should have detected a significant number of individual pulsars from the GCE population [222]. A complementary analysis by Holst & Hooper, adopting the luminosity function of Galactic plane MSPs, reached a similar conclusion: Fermi should have already detected dozens of MSPs from the GCE population, yet only a handful of candidates have been identified with directions and distances compatible with an inner Galaxy origin [223]. This discrepancy implies that the GCE can originate from MSPs only if those pulsars are systematically fainter than any MSP population observed so far. Such dimming is not implausible: unlike the disk and globular cluster populations, which are continuously replenished with freshly recycled pulsars, a bulge population that stopped forming new MSPs long ago would have faded through spin-down. Evolutionary models have indeed been used to argue for a fainter bulge population on these grounds [233–235]. How much fainter the inner-Galaxy pulsars could realistically be, however, remains an open question. This “missing bright pulsars” problem is one of the strongest empirical challenges to the MSP hypothesis. In Section 4.5, we present a dedicated analysis based on [1] that updates and extends this line of argument.

4.3 The Systematics Stalemate

The evidence reviewed in the preceding section, both for and against the MSP hypothesis, was obtained through template-based analyses that share a common vulnerability: their conclusions depend on modeling choices for which no unique, model-independent prescription exists. The choice of interstellar emission model, the treatment of the Fermi Bubbles at low latitudes, the masking of known point sources, and the parameterization of the source-count function at the faint end each contribute irreducible systematic uncertainties that propagate into the final answer. Over the past decade, a series of exchanges has demonstrated that these uncertainties are large enough to shift the conclusion between “predominantly point sources” and “predominantly smooth emission” depending on the analysis setup. This section traces how the debate reached an impasse and how recent methodological advances are beginning to chart a path forward.

4.3.1 Challenges to NPTF

The evidence for point sources presented in Section 4.2.1 rested largely on the Non-Poissonian Template Fitting (NPTF) framework, which exploits the difference in photon statistics between smooth (Poissonian) emission and populations of unresolved point sources. In 2019, a pair of analyses by Leane and Slatyer called the reliability of these results into question.

The first challenge came from a conceptually simple diagnostic: injecting a known, artificially generated dark matter signal into the real Fermi-LAT data and testing whether the NPTF pipeline could recover it [216]. The injected signal was drawn from a Poisson distribution tracing a contracted NFW profile, exactly as expected from WIMP annihilation. When the standard NPTF analysis was applied to this modified dataset, the injected dark matter flux was completely absorbed by the NFW point-source template, with the reconstructed dark matter component remaining consistent with zero. This misattribution persisted even when the injected signal was scaled to a flux roughly five times larger than the GCE itself. Only at injection levels approaching an order of magnitude above the observed excess did the fit begin to recover a dark matter component, and even then it returned only a fraction of the true injected value.

To understand the origin of this failure, Leane and Slatyer constructed a proof-of-principle simulation containing both a smooth dark matter signal and a population of point sources tracing the Fermi Bubbles. When they deliberately excluded the Fermi Bubbles point-source template from the fit, leaving this population unmodeled, the NFW point-source template absorbed the mismodeled emission. The simultaneously present dark matter signal was driven to zero, despite being the dominant component in the simulation. This demonstrated that any unmodeled source population with a spatial morphology not fully degenerate with existing templates could fatally bias the NPTF, causing it to absorb and mask a genuine dark matter signal.

A deeper investigation followed in 2020, when Leane and Slatyer discovered a pronounced north-south asymmetry in the GCE [217, 236]. Within a 10° region of interest, the smooth excess in the northern hemisphere was found to be substantially brighter than in the south, with the data preferring this asymmetric model at enormous statistical significance (Bayes factors up to $\sim 10^{27}$ in favor of asymmetry). As shown analytically in their companion paper [217], an overly rigid symmetric template that cannot capture this spatial variance in the data will generate excess pixel-to-pixel scatter that the NPTF interprets as point-source emission. When the authors subdivided the GCE template into independent northern and southern components and allowed both to float freely, the flux attributed to the point-source templates dropped to values consistent with zero. The Bayes factor in favor of point sources, which had stood at $\sim 10^{15}$

under the assumption of a symmetric GCE, collapsed by fourteen orders of magnitude to less than 10, rendering the preference for point sources statistically negligible.

These results did not go unchallenged. Buschmann et al. [218] argued that the signal injection failures reported by Leane and Slatyer were driven primarily by the use of the older p6v11 Galactic diffuse emission model, which suffers from severe over-subtraction in the inner Galaxy. In this model, the diffuse template absorbs more flux than it should, driving the smooth GCE normalization to unphysical negative values and creating a mathematical “hole” that swallows any injected dark matter signal before it can be recovered. To address this, Buschmann et al. introduced several improvements: a state-of-the-art hydrodynamic diffuse model (Model O) that provided a substantially better fit to the data; a spherical-harmonic marginalization technique that corrects for large-scale mismodeling in a data-driven manner without erasing small-scale structure; and a truncated source-count function that conservatively excludes point sources below the one-photon detection threshold, thereby breaking the mathematical degeneracy between ultra-faint sources and smooth emission [237]. With these modifications, the signal injection test worked correctly (injected dark matter signals were properly recovered), yet the data still yielded a Bayes factor of $\sim 10^{3.4}$ in favor of point sources. Similar conclusions were reached independently by Calore et al. [219] using the 1-point Probability Density Function (1pPDF) technique. The 1pPDF is an analytical framework mathematically equivalent in its goals to the NPTF, leveraging pixel-count statistics to reconstruct the source-count distribution. By combining 1pPDF with adaptive template fitting, they demonstrated that point sources constitute a robust sub-threshold component of the GCE, provided the diffuse emission is accurately modeled. A recent work by the same group [220] further enhanced the analysis with an improved estimation of the Galactic Diffuse Emission adopting the SkyFACT framework, further strengthening the evidence for point sources in the GCE.

Ultimately, this exchange exposed an underlying tension: results from analysis based on NPTF-like methods depend sensitively on analysis choices that remain a matter of debate. No analysis to date has achieved a resolution that all groups accept. The NPTF—and specifically the 1pPDF technique—remain valid statistical techniques; what part of the community questions is whether, given the sensitivity to the modeling choices discussed above, the conclusions reached so far on smooth versus point-source emission in the inner Galaxy can be regarded as definitive.

4.3.2 The Morphological Ambiguity

Beyond the statistical question of smooth versus point-source emission, the spatial morphology of the GCE itself has become a second axis of contention. As discussed in Section 4.1.3, the excess was initially characterized as approximately spherically symmetric, centered on Sgr A*, and consistent with a generalized NFW profile with inner slope $\gamma \approx 1.1\text{--}1.3$. This morphology has no obvious astrophysical counterpart among known Galactic structures and was therefore regarded as a strong indicator of dark matter annihilation. However, the stellar-bulge template analyses of Macias et al. [214, 230] and Bartels et al. [215], introduced in Section 4.2.1, challenged this picture by finding that the boxy, X-shaped stellar bulge fits the GCE better than a spherically symmetric NFW halo.

The significance of this claim, however, depends sensitively on the analysis setup. Whether the GCE prefers an NFW or bulge morphology can shift depending on the same diffuse-emission systematics discussed at the opening of this section [230]. Because the true diffuse background in the inner Galaxy is not known to the precision required, the template fitting procedure is inherently subject to systematic uncertainties that propagate directly into the morphological

inference. Different groups, using equally defensible analysis choices, have reached opposite conclusions: some find a strong preference for the bulge template, while others recover a spherically symmetric signal fully consistent with dark matter. No interstellar emission model achieves statistical fluctuation-level agreement with the data in the inner Galaxy; persistent large-scale residuals remain regardless of the morphological template adopted for the GCE. As a result, all morphological conclusions drawn from current analyses should be regarded as provisional.

Beyond the morphological debate, which remains unresolved at the spatial level, an independent spectral observation further complicates the astrophysical interpretation. Cholis et al. [238] examined the high-energy behavior of the GCE spectrum above ~ 10 GeV and found emission extending well beyond the energies at which millisecond pulsars are expected to contribute. The gamma-ray spectra of known MSPs exhibit exponential cutoffs at a few GeV, a characteristic feature of curvature radiation in pulsar magnetospheres. The presence of a power-law tail at higher energies is difficult to reconcile with a purely MSP origin and suggests that either a second emission component contributes at high energies or that the GCE is not entirely astrophysical. This spectral argument operates independently of the morphological debate and was initially cited as evidence against a complete MSP explanation.

However, this high-energy argument has recently been directly challenged. Manconi et al. [239] extended the 1pPDF technique to the > 10 GeV regime, combining it with adaptive template fitting to evaluate both the morphology and the photon-count statistics of the high-energy tail. They found strong evidence for the GCE at these elevated energies, noting that the emission strongly prefers a bulge morphology over a dark matter template. Crucially, their 1pPDF analysis reconstructed a significant population of faint, sub-threshold point sources at these high energies. Manconi et al. interpret this as evidence that the high-energy tail of the GCE can be explained by a population of point sources—possibly MSPs with harder spectra or a distinct sub-population—without requiring a dark matter contribution or an exotic secondary component.

4.3.3 Recent Revival of the Dark Matter Hypothesis

While the previous sections would seem to suggest that the GCE is now firmly established as a population of unresolved point sources, the situation is more nuanced. A recent analysis has reframed the question entirely by incorporating energy information into the point-source inference for the first time. While [220] argued that including energy information does not substantially alter the conclusions of the 1pPDF, List et al. [221] found that it has a dramatic effect on the inferred source-count distribution and the fraction of emission that can be attributed to point sources. List et al. [221] developed a CNN-based simulation-based inference framework that simultaneously fits the spatial and spectral properties of the inner Galaxy across ten logarithmically spaced energy bins from 2 to 20 GeV. Unlike current iterations of the 1pPDF technique, which discarded all spectral information and neglected pixel-to-pixel correlations from the finite point-spread function, this approach exploits the fact that the GCE has a distinct energy spectrum from the dominant diffuse backgrounds, providing an additional lever arm to separate signal from astrophysical contamination (see Chapter 3 for background on simulation-based inference).

The inclusion of energy information substantially alters the inferred source-count distribution (SCD). Earlier analyses, both the 2016 NPTF [212] and energy-independent CNN approaches [240, 241], had placed the hypothetical point-source population near or just below the Fermi detection threshold, typically predicting populations of a few hundred sources — from ~ 260 when resolved catalog sources are masked to the ~ 400 quoted in Section 4.2.1 when they are not [212]. The

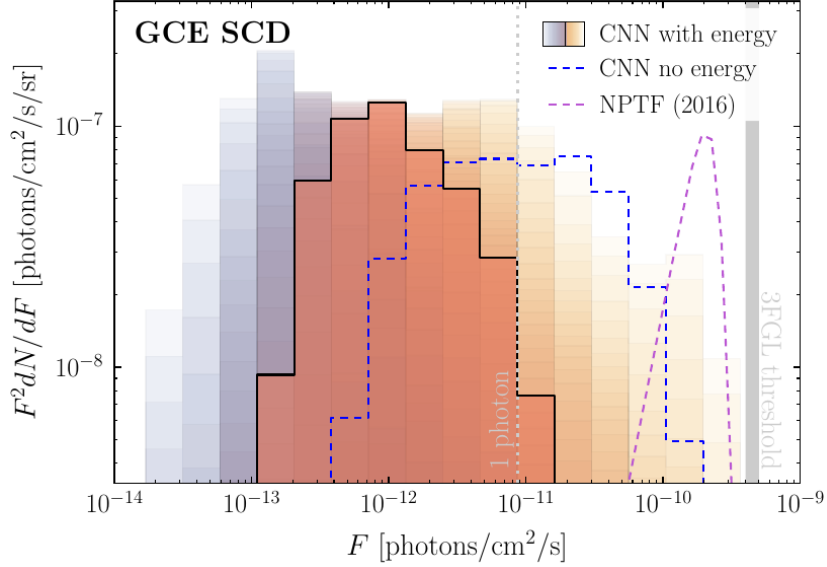


Fig. 4.2. Source-count distribution (SCD) of the GCE inferred via three approaches. The NPTF (dashed purple) placed the emission just below the Fermi 3FGL detection threshold; an energy-independent CNN (dashed blue) preferred a somewhat fainter population. Including energy information (solid black, with quantile bands from purple to orange) drives the SCD below the one-photon line. At this level, at 95% confidence, only 3% of the emission can be excluded as Poissonian. From [221], Figure 1 (right panel).

same energy-integrated picture emerges from the pixel-count analyses of Section 4.3.1: the SCD reconstructed with the 1pPDF by Calore et al. [219] is comparable to the energy-independent distributions shown in Figure 4.2. Manconi et al. [220] likewise report that their updated measurement remains compatible with that earlier result. The energy-dependent CNN drives the SCD to far fainter fluxes: the inferred distribution resides almost entirely below the one-photon threshold, meaning that a typical source would not even produce a single detected photon, see Figure 4.2. To account for the total observed GCE flux with such faint sources, the median prediction requires approximately 200,000 sources, with a 90% confidence lower bound of roughly 35,000 [221].

This result has a direct consequence. A population of point sources so faint that individual members produce fewer than one detected photon is mathematically indistinguishable from smooth, Poissonian emission, which is exactly the spatial signature predicted by dark matter annihilation. The distinction between the point-source and dark matter hypotheses is a nested one: an infinite number of infinitely dim sources produces exactly Poisson statistics. To quantify how much of the GCE can actually be attributed to point sources, List et al. trained a dedicated neural network to distinguish Poissonian from point-source emission given the inferred SCD. For the median SCD prediction, at 95% confidence only 3% of the GCE emission can be excluded as having a Poissonian origin [221]. The remaining 97% is fully consistent with the smooth emission expected from dark matter. This represents a marked shift from the earlier energy-independent CNN analysis, which had been able to exclude 34% of the emission as Poissonian at the same confidence level. This consistency reopens the dark matter hypothesis, but it does not demonstrate it: the result removes an obstacle to a smooth-emission origin rather than establishing one.

If the results of this work are confirmed by independent studies, the implications would be twofold. First, the key historical evidence that the GCE originates from point sources, including the NPTF’s strong Bayes factor and the inferred SCD peaking near the detection threshold, is substantially weakened when spectral information is properly incorporated. Second, if the GCE is nonetheless produced by astrophysical point sources, those sources must be extraordinarily faint and numerous, forming a population with no direct precedent among known Galactic source classes.

4.4 Breaking the Stalemate: Independent Constraints from Globular Clusters

The analyses discussed in the preceding sections share a common vulnerability: they all operate directly within the complex gamma-ray environment of the Galactic Center, where the results depend sensitively on the diffuse-emission systematics discussed in Section 4.3. Whether one uses template fitting, non-Poissonian statistics, or neural network-based inference, every conclusion about the nature of the GCE inherits these uncertainties.

To make further progress, independent constraints from environments that do not suffer from these systematics are needed.

Globular clusters are well suited for this purpose. Globular clusters host large, dynamically formed populations of millisecond pulsars, and the MSP content of a given cluster correlates with macroscopic properties — the stellar encounter rate and the visible-band luminosity — that are measured independently of the gamma-ray data (Section 4.5) [1]. Because globular clusters are discrete, spatially isolated systems far from the confused Galactic plane, their gamma-ray emission can be measured cleanly, without the severe interstellar emission model uncertainties that affect analyses of the GCE.

The MSP hypothesis requires the pulsars responsible for the excess to be systematically fainter than those observed in the Galactic Center, since otherwise Fermi-LAT should have already resolved many of them individually (cf. Section 4.2.2). Evolutionary models have been used to argue for a fainter bulge population on these grounds [233–235]. How much fainter the inner-Galaxy pulsars could be in practice, however, and the actual age distribution of MSPs in the bulge remain open questions.

Building on an earlier study by Hooper and Linden [222], in Amerio, Hooper, and Linden [1] we measured the gamma-ray luminosity function of MSPs across the Milky Way’s globular cluster system, using the full Fermi-LAT dataset to date. The luminosity function we obtain is consistent with that measured independently for MSPs close to the Galactic Center [223]: although the age distribution of the Inner Galaxy pulsars could in principle differ from that of globular clusters, the two environments seem to empirically yield similar luminosity functions.

If the pulsars responsible for the excess indeed share this luminosity function, the implications for the GCE are direct: Fermi-LAT’s source catalogs should already contain $N_{\text{MSP}} \sim 17\text{--}37$ individually resolved pulsars from the Inner Galaxy population. Only three MSP candidates have been identified in this region [223], and systematic radio follow-up campaigns targeting unassociated Fermi sources have not so far uncovered a hidden population. An MSP origin for the GCE would therefore require a pulsar population unlike any observed so far — a requirement for which the hypothesis, as currently formulated, offers no explanation.

This independent constraint connects directly to the recent findings of List et al. [221]. As discussed in Section 4.3.3, their energy-dependent CNN analysis pushes the source-count distribution of any GCE point sources to extremely faint levels, requiring on the order of 2×10^5 sources — each producing far less than one detected photon — to account for the excess. Such

an ultra-faint population would naturally explain the absence of individually resolved pulsars. The tension identified from globular clusters and the tension from the energy-dependent SCD analysis thus converge from independent lines of evidence. The remainder of this chapter is based on [1] and presents the analysis in full.

4.5 Millisecond Pulsars in Globular Clusters and Implications for the Galactic Center Gamma-Ray Excess

In this study, we revisit a previous work [222] attempting to constrain the gamma-ray luminosity function of MSPs in globular clusters. To this end, we use the publicly available Fermi data to measure or constrain the gamma-ray emission from each of the Milky Way’s 157 globular clusters (for related work, see Refs. [153, 242–249]). We detect statistically significant gamma-ray emission from 56 of these globular clusters, 8 of which are not contained in previous gamma-ray source catalogs. We then use this data, in combination with the visible luminosities and previously calculated stellar encounter rates of the globular clusters, to constrain the gamma-ray luminosity function of the MSPs in these systems. We find that this MSP population has a luminosity function that is consistent with that previously reported for MSPs in the Galactic Plane, featuring a mean luminosity of $\langle L_\gamma \rangle \sim (1 - 8) \times 10^{33}$ erg/s. If the GCE is generated by pulsars with the same gamma-ray luminosity function and other characteristics as those found in globular clusters, Fermi’s source catalogs would have contained $N_{\text{MSP}} \sim 17 - 37$ pulsars associated with this population. From this, we conclude that the GCE could be generated by pulsars only if they are significantly lower in luminosity or otherwise more difficult to identify than those in globular clusters (or in the Galactic Plane).

4.5.1 Fermi Data Analysis

In this section, we describe our procedure for measuring the spectrum and intensity of the gamma-ray emission from each of the Milky Way’s 157 globular clusters [250]. We employ the python package, Fermipy [251], as provided by the Fermi collaboration, to process the data collected by the Fermi-LAT. We work with HEALPix maps of order 11, corresponding to approximately 0.03° bins, and utilize 15.8 years of data.² We consider photons in the range of 100 MeV to 100 GeV, divided into 5 logarithmic bins per decade. We adopt the standard event selection for gamma-ray point sources, using the SOURCE event class and FRONT + BACK event types. We also employ a zenith angle cut of 90° and a rocking angle cut of 52° in order to reduce the contribution from the Earth’s limb. We require data quality=1 and only consider data that was taken while in survey mode. We do not use data that was collected while passing through the South Atlantic Anomaly. These and other aspects of our analysis are summarized in Table 4.1.

For each globular cluster, we study a $14^\circ \times 14^\circ$ region-of-interest (ROI) centered around the nominal position of the source. We then fit the energy spectrum of the globular cluster, adopting a parameterization in the form of a power law with an exponential cut-off:

$$\phi(E) = N_0 E^{-\alpha} e^{-E/E_c}. \quad (4.5.1)$$

In fitting these sources, we follow the standard procedure used in constructing Fermi-LAT source catalogs. We include in our fit any sources in the 4FGL-DR4 catalog that are located within in ROI [153], excluding any globular cluster, pulsar, or unassociated gamma-ray source that is

²MET range: 239557414 - 733276805

HEALPix map order	11
Weeks	9 – 826
Emin	100 MeV
Emax	100 GeV
Instrument Response Functions (IRFs)	P8R3_SOURCE_V3
EVCLASS	128 (Source)
EVTYPE	3 (Front + Back)
ZMAX	90°
Rocking angle cut	52°
Galactic diffuse emission template	gll_iem_v07
Isotropic background template	iso_P8R3_SOURCE_V3_v1
Source catalog	gll_psc_v35

Table 4.1. Fermi Science Tools settings used in this analysis.

located within 0.2° of the fiducial position of the globular cluster in question. We then add a new point source in the position of the globular cluster, as specified in Ref. [250], and reoptimize the model of the known sources within the ROI. Following standard procedure, we run the source localization pipeline, readjust the centroid of the globular cluster, and look for new point sources with a power-law spectrum. We then move to the characterization of the globular cluster, performing a 3D likelihood scan in the (α, E_c, N_0) parameter space. During this procedure, we model the Galactic diffuse emission and isotropic background contributions according to the templates provided by the Fermi collaboration (see Table 4.1). We allow their normalization to float, as well as the energy spectrum normalization of any other resolved source within 3 degrees. This allows us to identify, for each value of the integrated flux, the combination of these parameters that yields the greatest likelihood.

In Table 4.4, which we report in Appendix 4.A, we list each of the 56 globular clusters for which our analysis identified a statistically significant test statistic, $TS > 25$, along with the best-fit values for α , E_c , and the total flux (integrated between 0.1 and 100 GeV). Note that our previous study, which utilized 7 years of Fermi data, detected only 25 globular clusters at a level of $TS > 25$ [222]. We have compared our results to those reported in the 4FGL catalog [153] and find them to be in overall good agreement. The only significant differences are found in the cases of NGC 5139, Palomar 6, Terzan 5, and NGC 6218, for which our analysis yields integrated fluxes which are larger than those given in the 4FGL catalog by $2 - 3\sigma$. This appears to be due to the spectral parameterization we have adopted, as these four sources are fit in the 4FGL to a log-parabola or power-law parameterization, rather than that of a power-law with an exponential cutoff.

In addition to the 56 sources with $TS > 25$ listed in Table 4.4, we found an additional 7 sources which yielded $TS > 25$, but that had their highest TS values in bins below 0.4 GeV. Due to the large point spread function and related uncertainties that are encountered at these low energies, we do not consider these sources (which are listed in Table 4.5) to be unambiguous detections, and only report upper limits on their flux.

Finally, in Table 4.6, we report upper limits on the gamma-ray fluxes from each of the remaining globular clusters considered in our analysis, each of which yielded $TS < 25$.

4.5.2 Millisecond Pulsars in Globular Clusters

Due to the high stellar densities found within globular clusters, dynamical interactions between stars take place in these systems much more frequently than they do among stars in the Galactic Plane. As a result, large numbers of low-mass X-ray binaries (LMXBs) and millisecond pulsars (MSPs) are each found within globular clusters. At present, a total of 340 MSPs have been

identified within 45 different Milky Way globular clusters (see <https://www3.mpifr-bonn.mpg.de/staff/pfreire/GCpsr.html>, and Refs. [252, 253]). The majority of these pulsars have only been observed at radio wavelengths, as they are too distant to be detected by Fermi. In fact, Fermi’s Third Pulsar Catalog [253] contains only four MSPs that are located within a globular cluster: PSR J1717+4308A in NGC 6341, PSR J1823-3021A in NGC 6624, J1824-2452A in NGC 6626, and PSR J1835-3259B in NGC 6652.

The number of MSPs within a given globular cluster is correlated with that cluster’s overall visible-band luminosity, L_V (a proxy for stellar mass), and stellar encounter rate, Γ_e . The stellar encounter rate within a globular cluster is given by [254, 255]

$$\Gamma_e = \frac{4\pi}{\sigma_c} \int \rho^2(r) r^2 dr, \quad (4.5.2)$$

where $\rho(r)$ is the stellar density profile of the cluster and σ_c is the velocity dispersion near its core. Throughout this analysis, we make use of the stellar encounter rates tabulated in Ref. [256] (and as we list in Table 4.7).

In this study, we will consider the following five hypotheses for estimating the number of MSPs that will, on average, be found within a given globular cluster:

1. A case in which the mean number of MSPs in a globular cluster is proportional to its stellar encounter rate, $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e$. This approximate scaling might be expected if MSPs in globular clusters are formed largely through stellar interactions.
2. A case in which the relationship between the stellar encounter rate and the expected number of pulsars within a globular cluster is not linear, but instead scales as $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7}$. Empirically speaking, this hypothesis is preferred over that of a strictly linearly proportional relationship [247, 257].
3. A case in which the mean number of MSPs in a globular cluster is proportional to its visible luminosity, $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} L_V$.
4. A case in which the mean number of MSPs in a globular cluster is related to a combination of its stellar encounter rate and visible luminosity, $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e + R'_{\text{MSP}} L_V$.
5. A case in which the mean number of MSPs in a globular cluster is related to a combination of a non-linear scaling to its stellar encounter rate and its visible luminosity, $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$.

For further discussion regarding the dynamics of globular clusters and empirical relationships between their observed parameters, see Refs. [235, 247, 257, 258].

Throughout the remainder of this study, we will focus on the 87 globular clusters that have either $L_V > 10^5 L_\odot$ or $\Gamma_e > 0.01$ (in units such that $\Gamma_e = 1$ for the case of NGC 104). In Figure 4.3, we plot the number of MSPs that are known to exist in each of these 87 globular clusters, as a function of their stellar encounter rate, visible luminosity, or combinations of these parameters. We also show bands which represent the predicted relationships between these quantities, for selected values of $R_{\text{MSP}}^{\text{Radio}}$ ³. The inner (outer) bands are calculated such that they cover 68.27%

³While we take R_{MSP} and R'_{MSP} to be related to the predicted number of gamma-ray MSPs in a given globular cluster, the same quantities with the superscript “Radio” correspond to the predicted number of radio MSPs. Due to the different beaming angles of pulsars in the gamma-ray and radio bands, these quantities are not necessarily the same.

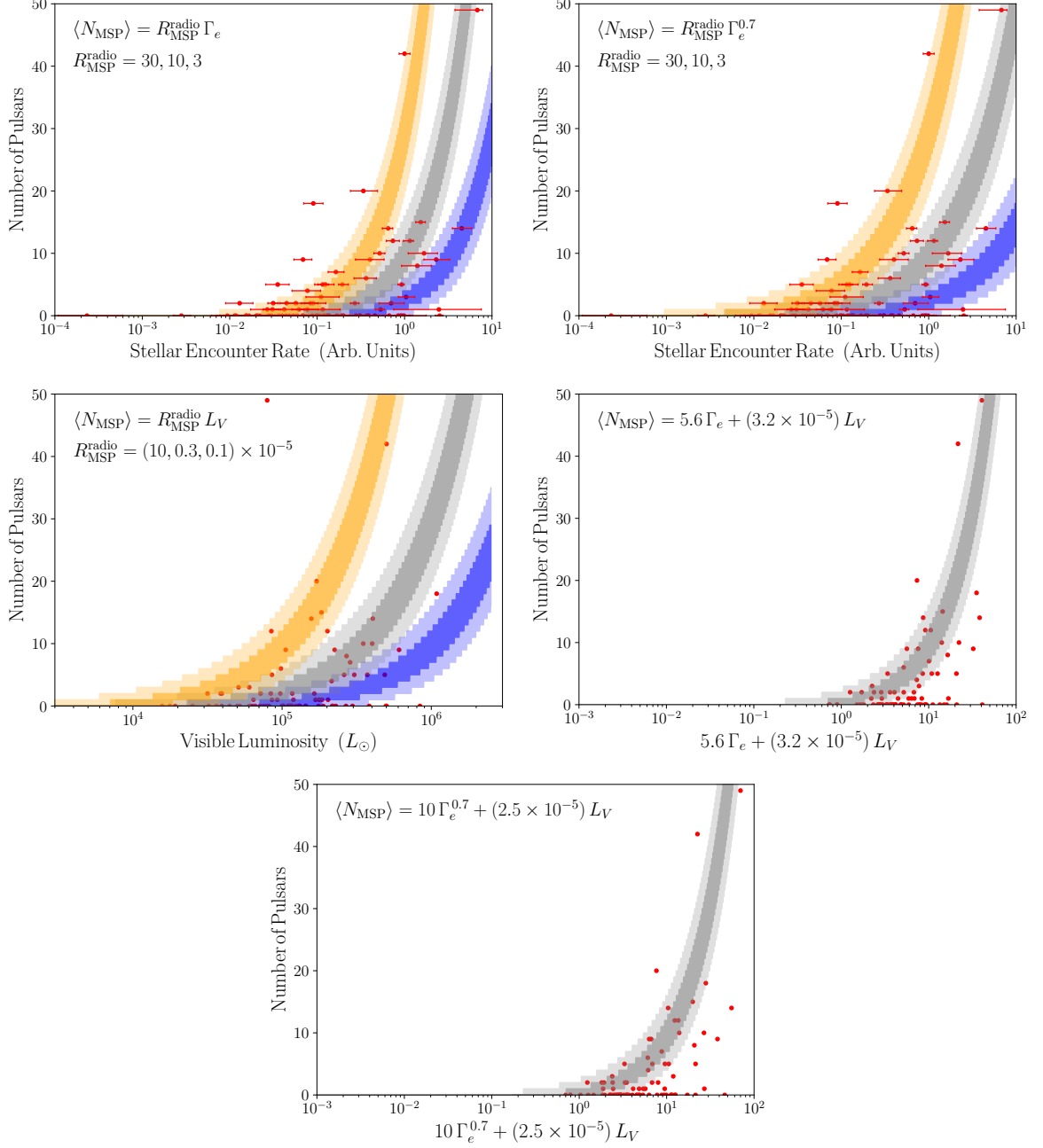


Fig. 4.3. The number of known millisecond pulsars in each of the 87 globular clusters considered in this study (those with $L_V > 10^5 L_\odot$ or $\Gamma_e > 0.01$) as a function of combinations of the stellar encounter rate, Γ_e , and visible luminosity, L_V (in units of solar luminosities). Also shown are colored bands representing the predicted number of MSPs (68.27% and 95.45% containment) in each case, for selected values of $R_{\text{MSP}}^{\text{Radio}}$. The stellar encounter rate is normalized such that $\Gamma_e = 1$ for NGC 104.

(95.45%) of a Poisson distribution for a mean value of $\langle N_{\text{MSP}}^{\text{Radio}} \rangle$. Even by eye, it is clear that the correlation is less tight for the case of $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e$ than for $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7}$, or for

$\langle N_{\text{MSP}} \rangle = L_V$. The tightest correlations are found for combinations of Γ_e and L_V (as shown in the final two frames). One should bear in mind that the current catalog of the MSPs in globular clusters is incomplete, limiting the utility of any quantitative comparisons. That being said, the results presented in Figure 4.3 provide us with motivation to consider the five hypotheses described above for the anticipated number of MSPs in globular clusters.

4.5.3 The Millisecond Pulsar Gamma-Ray Luminosity Function

For the gamma-ray luminosity function of MSPs, we adopt a log-normal parameterization:

$$\frac{dP}{dL_\gamma}(L_\gamma) = \frac{1}{L_\gamma \sigma_L \sqrt{2\pi}} \exp\left(-\frac{(\ln L_\gamma - \ln L_0)^2}{2\sigma_L^2}\right), \quad (4.5.3)$$

where L_0 and σ_L characterize the center and width of this distribution, respectively. We will often refer to this luminosity function in terms of the mean luminosity, which is given by $\langle L_\gamma \rangle = L_0 e^{\sigma_L^2/2}$.

The luminosity function of a globular cluster that contains a single MSP is simply equal to the expression given in Eq. 4.5.3,

$$\frac{dP}{dL_{\text{tot}}}(L_{\text{tot}}, N_{\text{MSP}} = 1) = \frac{dP}{dL_1}(L_1 = L_{\text{tot}}). \quad (4.5.4)$$

For a cluster with two MSPs, $N_{\text{MSP}} = 2$, it is instead given by

$$\frac{dP}{dL_{\text{tot}}}(L_{\text{tot}}, N_{\text{MSP}} = 2) = \int_0^{L_{\text{tot}}} \frac{dP}{dL_1}(L_1) \frac{dP}{dL_2}(L_2 = L_{\text{tot}} - L_1) dL_1, \quad (4.5.5)$$

and for $N_{\text{MSP}} = 3$,

$$\frac{dP}{dL_{\text{tot}}}(L_{\text{tot}}, N_{\text{MSP}} = 3) = \int_0^{L_{\text{tot}}-L_1} \int_0^{L_{\text{tot}}} \frac{dP}{dL_1}(L_1) \frac{dP}{dL_2}(L_2) \frac{dP}{dL_3}(L_3 = L_{\text{tot}} - L_1 - L_2) dL_1 dL_2, \quad (4.5.6)$$

and so on.

For a globular cluster containing a given mean number of expected MSPs, $\langle N_{\text{MSP}} \rangle$, its gamma-ray luminosity function will be given by

$$\frac{dP}{dL_{\text{GC}}}(L_{\text{GC}}, \langle N_{\text{MSP}} \rangle) = \sum_{N_{\text{MSP}}=0}^{\infty} P(N_{\text{MSP}}) \frac{dP}{dL_{\text{tot}}}(L_{\text{tot}}, N_{\text{MSP}}), \quad (4.5.7)$$

where

$$P(N_{\text{MSP}}, \langle N_{\text{MSP}} \rangle) = \frac{\langle N_{\text{MSP}} \rangle^{N_{\text{MSP}}} e^{-\langle N_{\text{MSP}} \rangle}}{N_{\text{MSP}}!} \quad (4.5.8)$$

is the Poisson probability that the globular cluster will contain N_{MSP} pulsars.

In order to constrain the gamma-ray luminosity function of MSPs in globular clusters, we closely follow the analysis from Ref. [222]. We repeat these steps here for clarity:

1. For a single globular cluster, i , we fit the energy spectrum to the Fermi-LAT data, as detailed in Section 4.5.1. This gives us the observational likelihood profile for the flux, ϕ_i , which we can write as $\mathcal{L}_{\text{obs}}(\phi_i)$. This function quantifies the measurement uncertainty of the flux.

2. We define the theoretical model for the gamma-ray luminosity function as in Eq. 4.5.7. This is a probability density function for the luminosity, $L_{\text{GC},i}$, which depends on a set of model parameters which we call θ . We write this dependence explicitly as $p(L_{\text{GC},i}|\theta)$.
3. Given the relation between the flux and luminosity, $\phi_i = L_{\text{GC},i}/4\pi d_i^2$, where d_i for each globular cluster is given in Table 4.7, it is possible to rewrite the luminosity function as function of the flux: $p_i(\phi_i|\theta)$. While the parameters θ would not depend on the specific globular cluster, the distance d_i does. For this reason, given a gamma ray luminosity function, $p_i(\phi_i|\theta)$ will be different for each globular cluster.
4. We combine the likelihood derived for a specific observation with the probability to observe a certain flux, and marginalize (integrate) over the flux in order to obtain the likelihood as a function of the parameter θ : $\mathcal{L}_{\text{obs}}^i(\theta) = \int_0^\infty \mathcal{L}_{\text{obs}}(\phi_i) p_i(\phi_i|\theta) d\phi_i$
5. We notice how each observation is independent from each other. For this reason, the joint likelihood on the entire set of globular clusters adopted in the analysis will be given by $\mathcal{L}_{\text{obs}}(\theta) = \prod_i \mathcal{L}_{\text{obs}}^i(\theta)$.
6. Given the likelihood of the entire dataset $\mathcal{L}_{\text{obs}}(\theta)$, it is now possible to derive a maximum likelihood estimation of the parameters θ .

In Figure 4.4, we show several examples of the results we obtain following this procedure. In each frame, we plot the measured gamma-ray luminosities of the 87 globular clusters considered in this portion of our study (those with $L_V > 10^5 L_\odot$ or $\Gamma_e > 0.01$), compared to their stellar encounter rates, visible luminosities, or selected combinations thereof. Then, as dark (light) grey bands, we plot the 68.27% (95.45%) range of the total gamma-ray luminosity from a given cluster that is predicted by the model in question. In each frame, we show the results as found adopting the best-fit values for $\langle L_\gamma \rangle$, σ_L , R_{MSP} , and/or R'_{MSP} (see Table 4.2).

In Figure 4.5, we plot the constraints on the luminosity function parameters, $\langle L_\gamma \rangle$ and σ_L , after marginalizing over the values of R_{MSP} and/or R'_{MSP} . The best-fit point in each frame is shown by a dot, while the contours reflect the regions that are preferred at the levels of 1, 2 and 3σ confidence. We show the results for each of the five hypotheses described in Section 4.5.2 for the relationship between the value of $\langle N_{\text{MSP}} \rangle$ and those of L_V and/or Γ_e , finding each case to yield qualitatively similar results. Namely, this procedure consistently favors scenarios with $\langle L_\gamma \rangle \sim (1 - 8) \times 10^{33}$ erg/s and $\sigma_L \sim 1.4 - 2.7$. In Table 4.2, we show the values of the parameters favored by our analysis, including those for R_{MSP} and R'_{MSP} . These parameters are directly related to the total number of gamma-ray MSPs contained within this sample of 87 globular clusters: $N_{\text{MSP}}^{\text{tot}} = 42.4 R_{\text{MSP}} + 1.55 \times 10^7 R'_{\text{MSP}}$ (for cases with $\langle N_{\text{MSP}}^{\text{tot}} \rangle \propto \Gamma_e$), or $N_{\text{MSP}} = 41.7 R_{\text{MSP}}^{0.7} + 155 \times 10^7 R'_{\text{MSP}}$ (for cases with $\langle N_{\text{MSP}} \rangle \propto \Gamma_e^{0.7}$). For the parameters favored by our analysis, these 87 globular clusters contain a total of $\sim 400 - 1500$ gamma-ray emitting MSPs.

We note that the different hypotheses we have considered for the relationship between the value of $\langle N_{\text{MSP}} \rangle$ and those of L_V and/or Γ_e can yield significantly different values for the overall likelihood. In Table 4.3, we show the difference between these resulting likelihoods (for the best-fit parameters in each case). Note that these hypotheses have different numbers of degrees-of-freedom, and this should be taken in to account when comparing their relative likelihoods.

So far, we have assumed that the mean number of MSPs in a given globular cluster is determined entirely by its luminosity and/or stellar encounter rate, without any departures

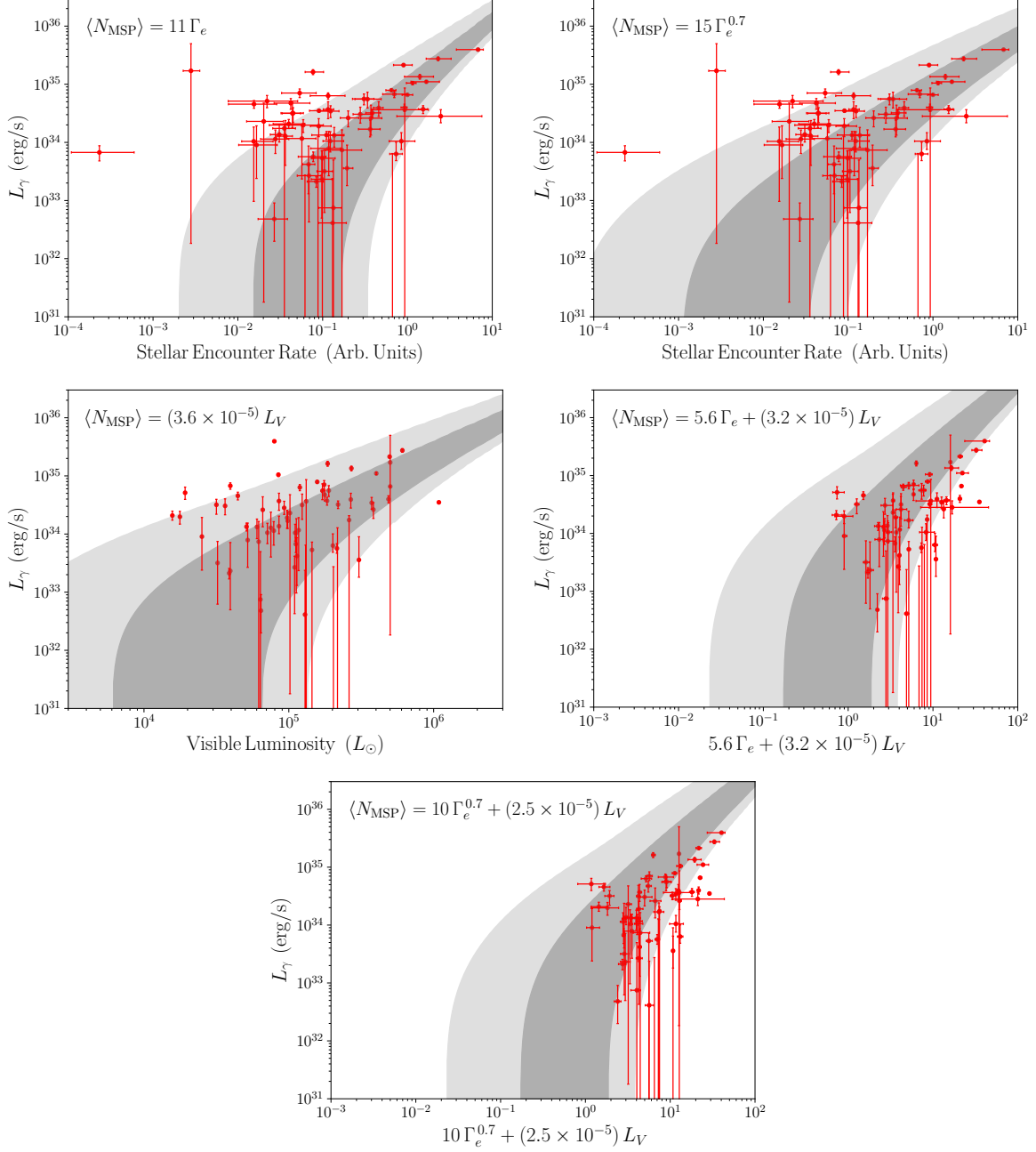


Fig. 4.4. The measured gamma-ray luminosities of the 87 globular clusters considered in this study (those with $L_V > 10^5 L_\odot$ or $\Gamma_e > 0.01$), compared to combinations of their stellar encounter rates and/or visible luminosities. As dark (light) grey bands, we show the 68.27% (95.45%) confidence interval predicted by the model for the total gamma-ray luminosity. In each frame, we show the results that were obtained using the best-fit values for $\langle L_\gamma \rangle$, σ_L , R_{MSP} , and/or R'_{MSP} (see Table 4.2).

from this prediction (other than the scatter inherent in each cluster's draw from the Poisson distribution for N_{MSP}). In reality, our models for determining $\langle N_{\text{MSP}} \rangle$ cannot possibly contain all

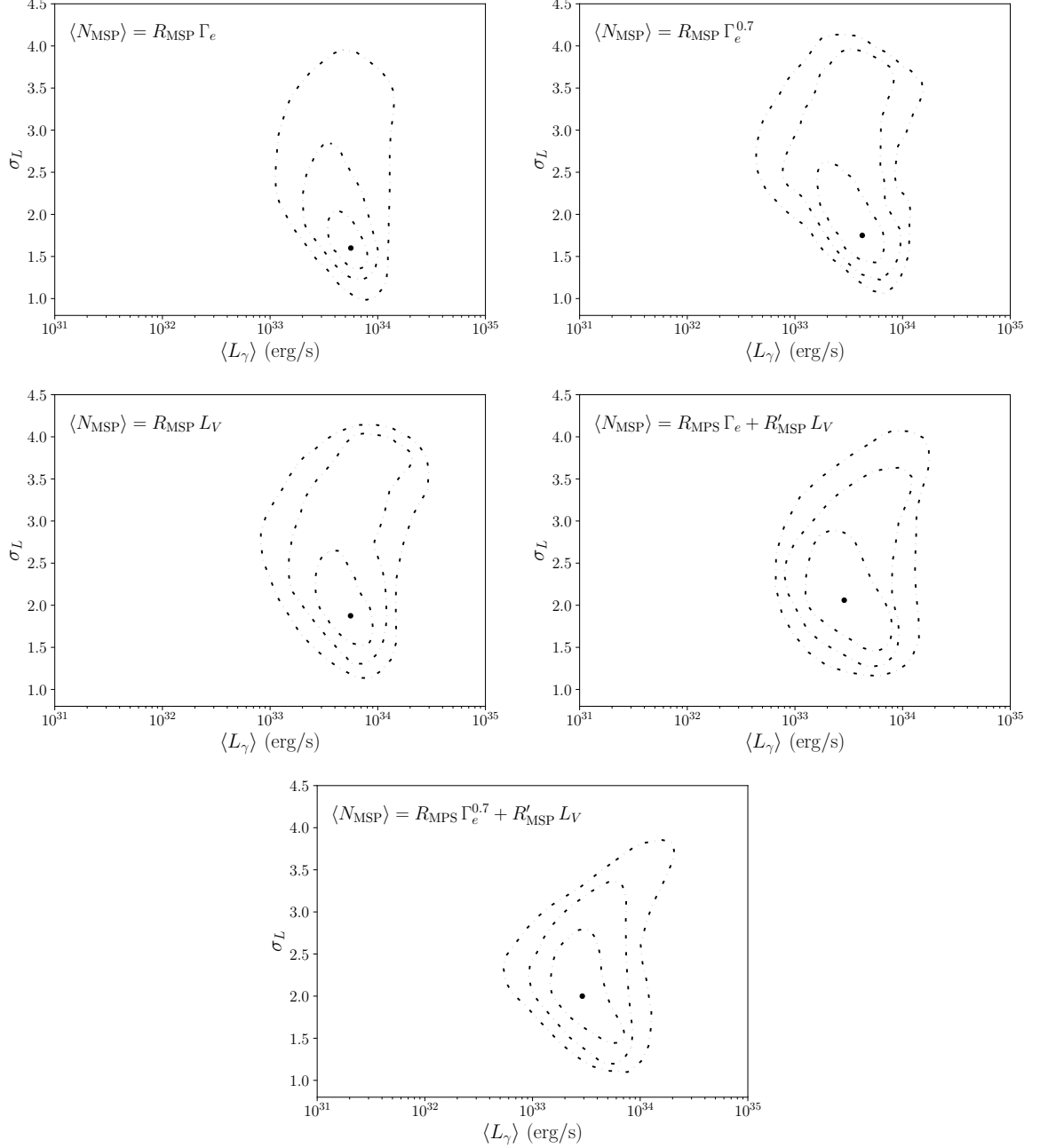


Fig. 4.5. Constraints on the MSP gamma-ray luminosity function parameters, after marginalizing over R_{MSP} and/or R'_{MSP} . The best-fit point in each frame is shown as a dot, while the contours reflect the regions that are preferred at the levels of 1, 2 and 3 σ confidence.

of the relevant information and, therefore, should not be perfectly predictive. This leads us to expect that there should be some degree of variation in the mean number of MSPs in given cluster, beyond that which can be predicted by the values of L_V and Γ_e alone.

To assess the impact of such cluster-to-cluster variation, we take the mean number of expected MSPs to have an intrinsic degree of scatter, which we model according to a log-normal distribution

$\langle N_{\text{MSP}} \rangle$	Parameter	Best Fit	1σ Range	2σ Range	3σ Range
$R_{\text{MSP}} \Gamma_e$	$\langle L_\gamma \rangle$	5.6×10^{33}	$(3.4 - 8.1) \times 10^{33}$	$(2.0 - 10) \times 10^{33}$	$(1.1 - 13) \times 10^{33}$
	σ_L	1.6	1.4 – 2.0	1.2 – 2.8	1.0 – 3.9
	R_{MSP}	11	9.5 – 15	7.6 – 32	6.4 – 110
$R_{\text{MSP}} \Gamma_e^{0.7}$	$\langle L_\gamma \rangle$	4.2×10^{33}	$(1.8 - 6.6) \times 10^{33}$	$(0.8 - 9.2) \times 10^{33}$	$(0.4 - 15) \times 10^{33}$
	σ_L	1.8	1.4 – 2.6	1.1 – 3.9	1.0 – 4.1
	R_{MSP}	15	11 – 36	8.5 – 130	7.1 – 220
$R_{\text{MSP}} L_V$	$\langle L_\gamma \rangle$	5.6×10^{33}	$(2.7 - 8.7) \times 10^{33}$	$(1.4 - 21) \times 10^{33}$	$(0.8 - 29) \times 10^{33}$
	σ_L	1.9	1.5 – 2.7	1.3 – 4.0	1.1 – 4.1
	R_{MSP}	3.6×10^{-5}	$(2.7 - 6.5) \times 10^{-5}$	$(1.8 - 22) \times 10^{-5}$	$(1.4 - 35) \times 10^{-5}$
$R_{\text{MSP}} \Gamma_e + R'_{\text{MSP}} L_V$	$\langle L_\gamma \rangle$	2.9×10^{33}	$(1.3 - 5.5) \times 10^{33}$	$(0.9 - 8.4) \times 10^{33}$	$(0.7 - 14) \times 10^{33}$
	σ_L	2.1	1.7 – 2.6	1.3 – 3.5	1.2 – 4.0
	R_{MSP}	5.6	3.2 – 14	1.8 – 21	1.2 – 39
	R'_{MSP}	3.2×10^{-5}	$(1.8 - 8.0) \times 10^{-5}$	$(1.4 - 11) \times 10^{-5}$	$(1.1 - 18) \times 10^{-5}$
$R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$	$\langle L_\gamma \rangle$	2.9×10^{33}	$(1.4 - 7.2) \times 10^{33}$	$(0.9 - 8.7) \times 10^{33}$	$(0.5 - 21) \times 10^{33}$
	σ_L	2.0	1.4 – 2.8	1.2 – 3.4	1.1 – 3.9
	R_{MSP}	10	6.4 – 23	3.9 – 66	2.9 – 110
	R'_{MSP}	2.5×10^{-5}	$(1.6 - 5.7) \times 10^{-5}$	$(1.1 - 8.0) \times 10^{-5}$	$(0.8 - 11) \times 10^{-5}$

Table 4.2. The parameter values favored by our analysis. Values for the average MSP gamma-ray luminosity, $\langle L_\gamma \rangle$, are given in erg/s.

$\langle N_{\text{MSP}} \rangle$	$-2\Delta \ln \mathcal{L}$
$R_{\text{MSP}} \Gamma_e$	146.0
$R_{\text{MSP}} \Gamma_e^{0.7}$	45.4
$R_{\text{MSP}} L_V$	37.7
$R_{\text{MSP}} \Gamma_e + R'_{\text{MSP}} L_V$	13.5
$R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$	0.0

Table 4.3. The change in twice the log-likelihood of our analysis, relative to the best-fit case shown in the bottom row of this table.

with a width, σ_N . This more complex model represents a potentially more conservative approach, and allows us to accounting for potential unmodeled systematic uncertainties. By rerunning our analysis, we find that in the case of our best fit model, $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$, the overall fit is improved by a maximum of $-2\Delta \ln \mathcal{L} \approx 8$ for $\sigma_N \approx 0.5$. In this new best fit, the values of $\langle L_\gamma \rangle$ and σ_L are shifted modestly from the case without this intrinsic scatter, from $\langle L_\gamma \rangle \approx 2.9 \times 10^{33}$ erg/s and $\sigma_L \approx 2.0$ to $\langle L_\gamma \rangle \approx 1.8 \times 10^{33}$ erg/s and $\sigma_L \approx 1.4$.

While an improvement of $-2\Delta \ln \mathcal{L} \approx 8$ corresponds to a statistical preference between 2σ and 3σ , this more complex model also introduces new modeling uncertainties. Given this, we

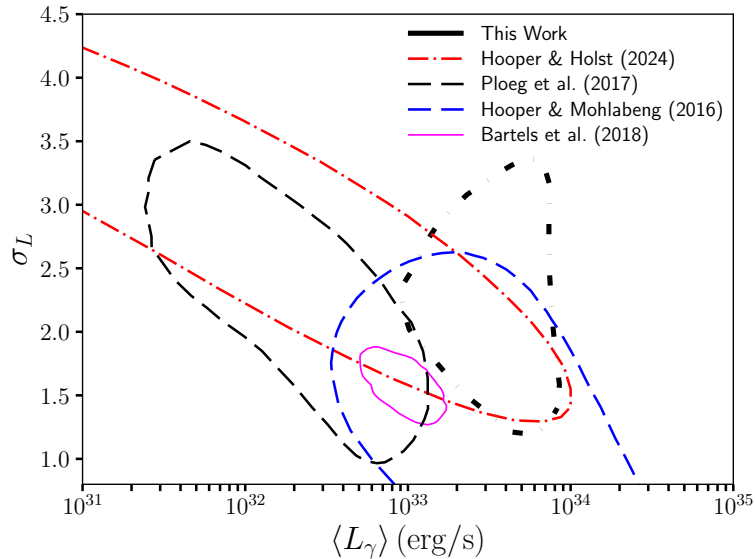


Fig. 4.6. A comparison of the 2σ constraints on the MSP gamma-ray luminosity found in this study (for the best-fitting case of $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$) to those published in previous studies of MSPs in the Galactic Plane. The luminosity function preferred by our analysis is in good agreement with those derived by Holst and Hooper [223], and by Hooper and Mohlabeng [222], but favor somewhat larger values for the average MSP luminosity than Ploeg *et al.* [261], or Bartels *et al.* [262].

cannot exclude that this improvement is due to statistical fluctuations or unknown systematic uncertainties. We therefore refrain from adopting the model with cluster-to-cluster variations as our fiducial baseline, and instead opt for the simpler fitting formula without variations, which has a stronger basis in the literature.

Before moving on, we will use these results to comment on the previous studies that have interpreted the gamma-ray emission observed from NGC 5139 [259] and NGC 104 [260] to be possible signals of annihilating dark matter. Although this interpretation cannot be easily ruled out, our results in Figure 4.4 demonstrate why no such exotic emission is required. In fact, these clusters have values of Γ_e and L_V that are entirely compatible with their observed gamma-ray luminosities. No dark matter annihilation or other exotic contributions are required.

4.5.4 Comparisons with Previous Studies

A number of previous studies have attempted to measure or constrain the gamma-ray luminosity function of MSPs [222, 223, 234, 248, 261–263]). In Figure 4.6, we compare the results of this study (for the best-fitting case of $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$) with those published in previous studies of the MSP population within the Galactic Plane. The luminosity function preferred by our analysis is in good agreement with those recently derived by Holst and Hooper [223], and previously by Hooper and Mohlabeng [222]. Our results, however, favor somewhat larger values of the average MSP luminosity, $\langle L_\gamma \rangle$, than the analyses of Ploeg *et al.* [261] or Bartels *et al.* [262].

We note that the analysis of Bartels *et al.* [262] made use of pulsar distance measurements, which are often subject to significant and difficult to quantify systematic uncertainties. For most pulsars, distance estimates are derived from measurements of their dispersion measure (ie., the

frequency-dependent time delay of the radio pulses, which can be used to determine the column density of free electrons along the pulsar’s line-of-sight). When the dispersion measure of a pulsar is combined with a model of the free electron distribution, one can obtain an estimate of its distance. Such estimates are known to vary significantly, however, depending on the electron distribution model that is adopted [264, 265], and sometimes disagree with more reliable parallax determinations by factors of up to ~ 3 [253].

It is possible that the MSPs population found within globular clusters could feature a gamma-ray luminosity function that is different from those pulsars in the Galactic Plane. This could be motivated by the fact that globular clusters are known to contain an old stellar population, and their pulsars could plausibly have already lost much of rotational kinetic energy, compared to that of more recently formed MSPs. This difference, however, would lead us to expect MSPs in globular clusters to be systematically *fainter* than those in the Galactic Plane, which is opposite to that found in the studies of Ploeg *et al.* [261] and Bartels *et al.* [262].

We can more directly compare our results to those of the 2016 analysis by Hooper and Linden [222], which also studied MSPs in globular clusters, using in their case 7 years of Fermi data (compared to the 15.8 years of data used in this analysis). The 2016 study favored a MSP gamma-ray luminosity function with $\langle L_\gamma \rangle \sim (0.7 - 5) \times 10^{34}$ erg/s and $\sigma_L \sim 1 - 3$. While the results presented here are statistically consistent with those of that study (at the approximately $\sim 2\sigma$ level), our analysis tends to prefer somewhat smaller values of $\langle L_\gamma \rangle$ and larger values of R_{MSP} , by a factor of $\sim 2 - 8$. For a more detailed comparison, see Appendix 4.A.1.

4.5.5 Implications for the Galactic Center Gamma-Ray Excess

As stated in the introduction, very few pulsars have been detected in the vicinity of the Inner Galaxy. If the GCE is generated by MSPs, this fact requires the MSPs responsible for this signal to be faint enough to have gone undetected. This, in turn, would require the MSPs in the Inner Galaxy to be systematically lower in luminosity than those found in other environments [223] (see also, Refs. [222, 234, 248, 261–263, 266]).

The possibility that the Inner Galaxy could contain a large population of low-luminosity pulsars has been motivated by the large proportion of old stars that are found in the Galactic Bulge relative to those in the Galactic Plane. The stellar populations found in globular clusters, however, are comparably old to those in the Bulge. By measuring the gamma-ray luminosity function of MSPs in globular clusters, we can hope to shed some light on the question of whether the Inner Galaxy might contain a very large abundance of very low-luminosity MSPs.

In Figure 4.7, we plot the number of MSPs that should have been detected by Fermi and contained within their source catalogs [253] if they were responsible for producing the GCE, as a function of the luminosity function parameters [223]. If the GCE is generated by MSPs with the same luminosity function and other characteristics as those found in globular clusters, Fermi’s source catalogs would have contained $N_{\text{MSP}} \sim 17 - 37$ individual members of this population. Given that only three millisecond pulsars have been detected with a direction and distance that does not preclude it from being part of an Inner Galaxy population, the luminosity function derived here is incompatible with pulsar interpretations of the GCE, unless a large number of unassociated Fermi sources are, in fact, unidentified MSPs. Given systematic efforts that have been conducted to search for radio pulsations from unassociated Fermi sources, this possibility seems unlikely. Alternatively, the observed gamma-ray excess could be generated by MSPs if those pulsars are significantly less luminous, on average, than those located in globular clusters or in the Galactic Plane.

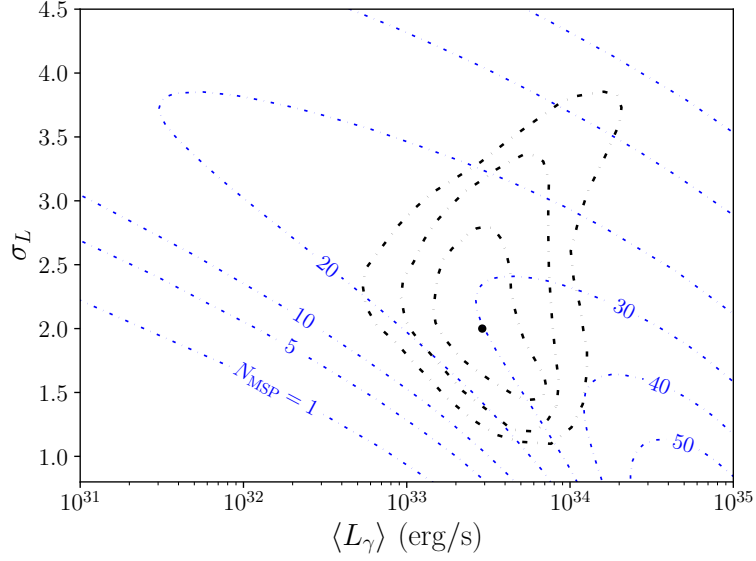


Fig. 4.7. The constraints on the MSP gamma-ray luminosity found in this study (shown for the case of $\langle N_{\text{MSP}} \rangle = R_{\text{MSP}} \Gamma_e^{0.7} + R'_{\text{MSP}} L_V$), shown along with contours representing the number of Inner Galaxy MSPs that should appear in the Fermi source catalogs, N_{MSP} , assuming that the Galactic Center Gamma-Ray Excess (GCE) is generated by MSPs [223]. As there are only three Inner Galaxy pulsars candidates in the 3PC, MSPs can generate the excess gamma-ray emission only if that source population consists of members that are significantly less luminous than those present in globular clusters.

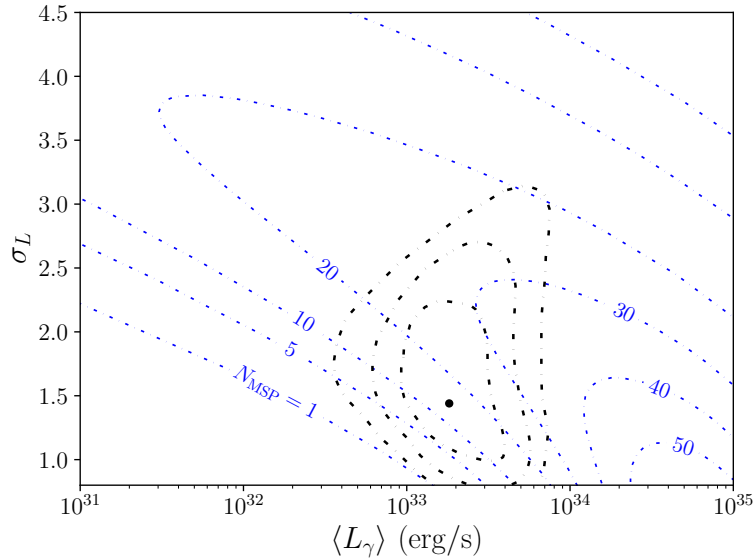


Fig. 4.8. As in Figure 4.7, but for the best fit model including cluster-to-cluster variations. The results are compatible with the estimates in Figure 4.7 within the 1σ band, although the model with cluster-to-cluster variations favors lower values for $\langle L_\gamma \rangle$ and σ_L .

As mentioned in Section 4.5.3, our models for the mean number of millisecond pulsars in a given globular cluster are subject to intrinsic uncertainties. To account for this, we also considered a more flexible model that includes an intrinsic cluster-to-cluster variation in the expected number of MSPs. The purpose of this approach is to derive a more conservative lower bound on the number of pulsars that should have been detected in the Inner Galaxy if they are responsible for the Galactic Center Excess. In Figure 4.8, we show the results of this analysis. By allowing for a wider range of pulsar populations, particularly those that are fainter on average, this model lowers the number of expected detections to the range of $N_{\text{MSP}} \approx 3 - 30$. Although this bound is less constraining by construction, the results nonetheless remain in tension at a level of at least 2σ with the three identified pulsar candidates.

4.5.6 Summary and Conclusions

In this study, we have analyzed the gamma-ray emission from the Milky Way’s globular clusters, using 15.8 years of publicly available data collected by the Fermi Gamma-Ray Space Telescope. We report the robust detection of 56 globular clusters with a statistically significant test statistic of $\text{TS} > 25$, 8 of which are not contained in previous gamma-ray source catalogs.

Making use of previously established correlations between the mean number of millisecond pulsars in a globular cluster and the stellar encounter rate or visible luminosity of that cluster, we perform a determination of the gamma-ray luminosity function of the millisecond pulsars in the Milky Way’s globular cluster system. To this end, we have considered 87 globular clusters with high stellar encounter rates and/or visible luminosities, and combined this information with our measurements of their gamma-ray luminosities. We then fitted the data to a log-normal luminosity function, finding it to favor a mean pulsar luminosity of $\langle L_\gamma \rangle \sim (1 - 8) \times 10^{33} \text{ erg/s}$ (integrated between 0.1 and 100 GeV), and a width of $\sigma_L \sim 1.4 - 2.7$. Our analysis also indicates that the Milky Way’s globular cluster system contains a total of $\sim 400 - 1500$ gamma-ray emitting millisecond pulsars. The luminosity function favored by this analysis is consistent with that exhibited by millisecond pulsars found in the Galactic Plane [223].

The leading astrophysical interpretation of the Galactic Center Gamma-Ray Excess has long been that this signal could be generated by large population of centrally-located millisecond pulsars. Very few pulsars have been detected in the Inner Galaxy, however, forcing us to place strong constraints on the gamma-ray luminosity function of any pulsar population that could be responsible for the excess emission. In particular, such a pulsar population would have to be systematically less bright than those found in the Galactic Plane [223].

It has been suggested that the Inner Galaxy pulsar population could be older and therefore less luminous, on average, than pulsars found in the Galactic Plane. The stars in globular clusters are also very old, however. Thus if the Inner Galaxy pulsar population is very faint, one would expect the same to be true among the pulsars found in globular clusters. The results of this study indicate that this is not the case. This result poses a challenge to pulsar interpretations of the Galactic Center Gamma-Ray Excess and, by extension, potentially bolsters dark matter interpretations of this long-standing signal.

4.A Additional Tables

In this section, we provide several tables which contain information regarding each of the globular clusters considered in our study, as well as the results of our gamma-ray analysis for each of these sources.

Table 4.4

Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	α	E_c (GeV)	TS
NGC 104	47 Tuc	$2.773^{+0.041}_{-0.051} \times 10^{-11}$	1.37	2.85	10801.3
Terzan 5	Terzan 11	$9.245^{+0.215}_{-0.181} \times 10^{-11}$	1.75	4.24	5955.9
NGC 6266	M62	$1.984^{+0.091}_{-0.097} \times 10^{-11}$	1.44	2.79	1504.2
NGC 6388		$1.823^{+0.091}_{-0.090} \times 10^{-11}$	1.41	2.31	1399.8
NGC 6626	M28	$2.170^{+0.111}_{-0.118} \times 10^{-11}$	1.52	2.35	1346.0
NGC 5139	Omega Centauri	$1.100^{+0.056}_{-0.054} \times 10^{-11}$	1.21	2.11	1168.9
NGC 6624		$1.405^{+0.086}_{-0.090} \times 10^{-11}$	1.60	4.75	902.9
GLIMPSE01		$4.305^{+0.256}_{-0.238} \times 10^{-11}$	1.55	4.22	686.5
NGC 6441		$1.710^{+0.112}_{-0.120} \times 10^{-11}$	1.85	3.91	675.9
NGC 6440		$1.586^{+0.141}_{-0.142} \times 10^{-11}$	1.90	6.41	399.2
NGC 6752		$3.954^{+0.444}_{-0.419} \times 10^{-12}$	1.07	1.46	341.5
NGC 6316		$1.248^{+0.115}_{-0.119} \times 10^{-11}$	1.86	5.12	336.2
GLIMPSE02		$2.568^{+0.233}_{-0.238} \times 10^{-11}$	1.78	2.81	203.7
NGC 6652		$5.664^{+0.694}_{-0.676} \times 10^{-12}$	1.87	4.78	191.8
2MS-GC01	2MASS-GC01	$2.666^{+0.226}_{-0.238} \times 10^{-11}$	1.66	1.86	191.8
NGC 2808		$3.625^{+0.489}_{-0.484} \times 10^{-12}$	1.69	4.06	182.1
NGC 6541		$4.747^{+0.666}_{-0.670} \times 10^{-12}$	1.85	4.80	154.8
NGC 6093	M80	$4.402^{+0.857}_{-0.807} \times 10^{-12}$	1.67	6.78	146.0
NGC 6569		$4.984^{+0.821}_{-0.779} \times 10^{-12}$	1.62	6.64	133.8
NGC 7078		$3.608^{+0.488}_{-0.517} \times 10^{-12}$	2.04	2.19	127.4
NGC 6717	Palomar 9	$3.454^{+0.619}_{-0.560} \times 10^{-12}$	1.29	3.26	121.2
NGC 6656	M22	$4.557^{+0.882}_{-0.784} \times 10^{-12}$	0.82	1.00	120.5
NGC 1904	M79	$3.176^{+0.373}_{-0.362} \times 10^{-12}$	2.81	56.20	100.9
Palomar 6		$1.137^{+0.173}_{-0.162} \times 10^{-11}$	2.20	≥ 100	94.9
Terzan 1	HP 2	$6.774^{+0.862}_{-0.871} \times 10^{-12}$	0.00	1.61	91.0
NGC 6139		$4.583^{+0.957}_{-0.911} \times 10^{-12}$	1.64	3.65	84.5
NGC 6254	M10	$2.295^{+0.407}_{-0.332} \times 10^{-12}$	0.00	0.92	79.7
Terzan 2	HP 3	$7.670^{+1.844}_{-1.672} \times 10^{-12}$	1.97	17.66	76.4
NGC 1851		$2.144^{+0.312}_{-0.314} \times 10^{-12}$	2.45	≥ 100	76.2
NGC 6544		$1.280^{+0.184}_{-0.193} \times 10^{-11}$	1.46	0.65	76.2
NGC 6341	M92	$1.334^{+0.340}_{-0.307} \times 10^{-12}$	1.38	1.95	73.7
NGC 6402	M14	$3.379^{+0.775}_{-0.706} \times 10^{-12}$	1.52	2.54	68.0

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Table 4.4 – continued from previous page

Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	α	E_c (GeV)	TS
Terzan 9		$5.727^{+1.324}_{-0.947} \times 10^{-12}$	0.00	0.58	67.5
NGC 6397		$3.363^{+0.671}_{-0.653} \times 10^{-12}$	2.08	5.21	67.3
NGC 5904	M5	$1.552^{+0.432}_{-0.374} \times 10^{-12}$	1.15	2.30	66.8
Terzan 6	HP 5	$5.145^{+1.434}_{-1.163} \times 10^{-12}$	0.38	1.21	59.4
NGC 6273	M19	$2.853^{+0.837}_{-0.808} \times 10^{-12}$	1.88	21.03	53.1
Whiting 1		$1.156^{+0.298}_{-0.277} \times 10^{-12}$	1.37	15.60	52.4
Liller 1		$9.263^{+2.229}_{-2.224} \times 10^{-12}$	1.63	5.60	51.1
NGC 6528		$4.068^{+1.252}_{-1.128} \times 10^{-12}$	1.08	1.20	47.2
UKS 1		$1.068^{+0.199}_{-0.193} \times 10^{-11}$	0.23	0.25	45.6
NGC 362		$7.162^{+3.977}_{-1.624} \times 10^{-13}$	0.35	1.27	44.6
NGC 6637	M 69	$2.061^{+0.678}_{-0.675} \times 10^{-12}$	1.45	5.22	43.8
NGC 6723		$2.529^{+0.623}_{-0.612} \times 10^{-12}$	1.75	5.50	43.6
NGC 6218	M12	$3.024^{+0.657}_{-0.730} \times 10^{-12}$	2.39	≥ 100	42.9
NGC 6539		$6.541^{+1.290}_{-1.355} \times 10^{-12}$	2.66	≥ 100	39.0
NGC 5286		$2.418^{+0.660}_{-0.657} \times 10^{-12}$	2.23	18.15	35.8
NGC 6304		$2.557^{+1.101}_{-0.853} \times 10^{-12}$	0.92	1.69	34.8
NGC 6342		$3.671^{+0.874}_{-1.082} \times 10^{-12}$	2.32	≥ 100	33.1
NGC 6712		$2.381^{+1.123}_{-0.561} \times 10^{-12}$	0.00	0.67	31.4
NGC 6380	Ton 1	$2.617^{+0.886}_{-0.773} \times 10^{-12}$	0.75	1.77	29.9
NGC 6517		$4.119^{+1.208}_{-1.183} \times 10^{-12}$	1.76	2.06	29.2
IC 1257		$1.529^{+0.657}_{-0.468} \times 10^{-12}$	0.58	2.33	29.0
NGC 6522		$2.389^{+0.968}_{-0.588} \times 10^{-12}$	0.00	0.75	28.1
NGC 6540	Djorg 3	$2.841^{+1.163}_{-0.937} \times 10^{-12}$	1.16	1.95	28.1
NGC 6352		$2.532^{+0.596}_{-0.635} \times 10^{-12}$	0.00	0.25	27.9

Table 4.4. The gamma-ray flux (integrated between 0.1 – 100 GeV) and the best-fit spectral parameters for each of the 56 globular clusters that were detected in our analysis with TS > 25.

Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	TS
NGC 6838	M 71	$< 5.65 \times 10^{-12}$	62.84
Lynga 7	BH184	$< 7.18 \times 10^{-12}$	54.87
AM 1	E 1	$< 1.47 \times 10^{-12}$	36.99
AM 4		$< 2.88 \times 10^{-12}$	34.07
Palomar 1		$< 2.00 \times 10^{-12}$	31.94
NGC 6809	M 55	$< 3.42 \times 10^{-12}$	27.05
Palomar 10		$< 6.40 \times 10^{-12}$	27.04

Table 4.5. The 2σ upper limits on the gamma-ray flux (integrated between 0.1 – 100 GeV) for globular clusters that were detected in our analysis with $TS > 25$, but with spectra that peak at energies below 0.4 GeV.

Table 4.6

Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	TS
IC 4499		$< 1.83 \times 10^{-12}$	15.53
E 3		$< 2.31 \times 10^{-12}$	23.25
NGC 4372		$< 2.35 \times 10^{-12}$	20.94
NGC 6101		$< 1.88 \times 10^{-12}$	21.99
NGC 4833		$< 3.88 \times 10^{-12}$	0.00
NGC 6362		$< 2.52 \times 10^{-12}$	0.00
NGC 1261		$< 1.97 \times 10^{-12}$	0.00
NGC 6584		$< 1.04 \times 10^{-12}$	4.92
Rup 106		$< 3.80 \times 10^{-12}$	0.90
NGC 5927		$< 1.57 \times 10^{-12}$	3.78
NGC 5946		$< 4.57 \times 10^{-12}$	0.00
BH 176		$< 2.92 \times 10^{-12}$	7.08
FSR 1735		$< 1.35 \times 10^{-11}$	0.00
ESO-SC06	ESO280-SC06	$< 5.01 \times 10^{-12}$	0.00
NGC 3201		$< 6.69 \times 10^{-13}$	1.51
NGC 6496		$< 4.16 \times 10^{-12}$	0.00
Ton 2	Pismis 26	$< 3.39 \times 10^{-12}$	18.01
NGC 5986		$< 2.77 \times 10^{-12}$	0.00
Pyxis		$< 3.64 \times 10^{-12}$	1.86

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Table 4.6 – continued from previous page

Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	TS
NGC 6256		$< 7.57 \times 10^{-12}$	0.00
NGC 2298		$< 2.72 \times 10^{-12}$	0.14
Terzan 3		$< 5.80 \times 10^{-12}$	0.06
Terzan 7		$< 3.59 \times 10^{-12}$	0.00
NGC 6453		$< 3.23 \times 10^{-12}$	10.92
Terzan 8		$< 4.20 \times 10^{-12}$	2.46
NGC 5824		$< 7.52 \times 10^{-13}$	2.91
Djorg 1		$< 1.45 \times 10^{-11}$	20.69
NGC 6681	M 70	$< 1.75 \times 10^{-12}$	17.92
NGC 6558		$< 1.61 \times 10^{-12}$	2.42
Terzan 4	HP 4	$< 1.30 \times 10^{-11}$	0.00
NGC 6715	M 54	$< 2.17 \times 10^{-12}$	20.52
Arp 2		$< 2.81 \times 10^{-12}$	15.95
HP 1	BH 229	$< 5.26 \times 10^{-12}$	22.36
BH 261	AL 3	$< 2.09 \times 10^{-12}$	6.06
1636-283	ESO452-SC11	$< 2.73 \times 10^{-12}$	23.07
Djorg 2	ESO456-SC38	$< 2.86 \times 10^{-12}$	3.22
NGC 4590	M 68	$< 2.93 \times 10^{-12}$	0.00
NGC 288		$< 2.36 \times 10^{-12}$	0.03
NGC 6293		$< 7.33 \times 10^{-12}$	20.31
NGC 5694		$< 3.57 \times 10^{-12}$	5.87
NGC 6121	M 4	$< 1.90 \times 10^{-12}$	6.55
NGC 6355		$< 9.45 \times 10^{-12}$	0.00
Terzan 10		$< 9.44 \times 10^{-12}$	0.00
NGC 6144		$< 4.89 \times 10^{-12}$	0.00
NGC 6553		$< 1.66 \times 10^{-12}$	1.34
NGC 6638		$< 2.04 \times 10^{-12}$	9.93
NGC 6284		$< 3.67 \times 10^{-12}$	0.00
NGC 6401		$< 4.23 \times 10^{-12}$	15.46
NGC 6325		$< 7.45 \times 10^{-12}$	1.42
NGC 6642		$< 7.30 \times 10^{-12}$	0.66
NGC 7099	M 30	$< 3.44 \times 10^{-12}$	7.64

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Table 4.6 – continued from previous page

Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	TS
Terzan 12		$< 1.34 \times 10^{-11}$	0.00
NGC 6287		$< 6.53 \times 10^{-12}$	5.51
NGC 6235		$< 5.46 \times 10^{-12}$	0.00
NGC 6864	M 75	$< 3.06 \times 10^{-12}$	20.48
Palomar 12		$< 1.04 \times 10^{-12}$	11.81
Eridanus		$< 6.93 \times 10^{-13}$	6.00
NGC 5897		$< 3.05 \times 10^{-12}$	0.00
2MS-GC02	2MASS-GC02	$< 2.97 \times 10^{-12}$	0.59
Palomar 8		$< 7.56 \times 10^{-12}$	1.39
NGC 6333	M 9	$< 3.95 \times 10^{-12}$	0.00
NGC 6356		$< 4.14 \times 10^{-12}$	0.00
NGC 7492		$< 7.41 \times 10^{-13}$	8.29
NGC 6171	M 107	$< 4.21 \times 10^{-12}$	0.00
NGC 6981	M 72	$< 2.81 \times 10^{-12}$	0.00
Palomar 11		$< 4.69 \times 10^{-12}$	0.00
IC 1276	Palomar 7	$< 9.51 \times 10^{-12}$	0.00
NGC 5634		$< 2.62 \times 10^{-12}$	0.00
NGC 6366		$< 1.91 \times 10^{-12}$	0.77
NGC 7089	M 2	$< 2.52 \times 10^{-12}$	0.75
Palomar 15		$< 1.60 \times 10^{-12}$	13.98
NGC 6535		$< 4.04 \times 10^{-12}$	18.22
Palomar 5		$< 8.43 \times 10^{-13}$	1.28
Palomar 3		$< 2.82 \times 10^{-12}$	2.17
NGC 6760		$< 8.63 \times 10^{-12}$	0.00
NGC 6749		$< 1.01 \times 10^{-11}$	0.00
NGC 6426		$< 5.67 \times 10^{-12}$	4.99
NGC 6934		$< 2.73 \times 10^{-12}$	0.00
Ko 1		$< 2.73 \times 10^{-12}$	0.06
Palomar 13		$< 3.99 \times 10^{-12}$	0.26
Palomar 14	AvdB	$< 2.68 \times 10^{-12}$	0.01
NGC 7006		$< 1.31 \times 10^{-12}$	7.41
NGC 5053		$< 2.04 \times 10^{-12}$	17.89

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Table 4.6 – continued from previous page

Globular Cluster	Alternate Name	Flux (erg/cm ² /s)	TS
NGC 5024	M 53	$< 3.63 \times 10^{-12}$	0.58
NGC 4147		$< 6.24 \times 10^{-13}$	3.16
Ko 2		$< 2.96 \times 10^{-12}$	1.22
NGC 5272	M 3	$< 2.70 \times 10^{-12}$	10.28
NGC 5466		$< 2.33 \times 10^{-12}$	6.05
Palomar 4		$< 2.58 \times 10^{-12}$	0.00
NGC 6779	M 56	$< 1.75 \times 10^{-12}$	9.53
Palomar 2		$< 4.40 \times 10^{-12}$	0.00
NGC 6205	M 13	$< 5.64 \times 10^{-13}$	12.24
NGC 2419		$< 2.76 \times 10^{-12}$	0.00
NGC 6229		$< 4.51 \times 10^{-13}$	4.16

Table 4.6. The 2σ upper limits on the gamma-ray flux (integrated between 0.1 – 100 GeV) for those globular clusters that were not significantly detected in our analysis (TS < 25).

Table 4.7

Globular Cluster	Alt. Name	Distance (kpc)	Luminosity ($L_{\odot}$)	Γ_e
NGC 104	47 Tuc	4.46	5.01×10^5	$1.00^{+0.15}_{-0.13}$
NGC 362		8.61	2.01×10^5	$0.74^{+0.14}_{-0.12}$
NGC 1261		16.29	1.13×10^5	$0.015^{+0.011}_{-0.004}$
Palomar 2		27.11	1.32×10^5	$0.93^{+0.84}_{-0.56}$
NGC 1851		12.07	1.84×10^5	$1.53^{+0.20}_{-0.19}$
NGC 1904	M79	12.94	1.19×10^5	$0.12^{+0.019}_{-0.014}$
NGC 2419		82.49	5.01×10^5	$0.0028^{+0.0008}_{-0.0005}$
NGC 2808		9.59	4.88×10^5	$0.92^{+0.067}_{-0.083}$
NGC 4147		19.30	2.51×10^4	$0.017^{+0.013}_{-0.006}$
NGC 4372		5.81	1.12×10^5	$0.0002^{+0.0004}_{-0.0001}$
NGC 5024	M53	17.85	2.61×10^5	$0.035^{+0.012}_{-0.010}$
NGC 5139	Omega Centauri	5.17	1.09×10^6	$0.090^{+0.027}_{-0.020}$
NGC 5272	M3	10.18	3.05×10^5	$0.19^{+0.033}_{-0.018}$
NGC 5286		11.67	2.68×10^5	$0.46^{+0.058}_{-0.061}$
NGC 5634		25.18	1.02×10^5	$0.020^{+0.014}_{-0.008}$

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Table 4.7 – continued from previous page

Globular Cluster	Alt. Name	Distance (kpc)	Luminosity ($L_{\odot}$)	Γ_e
NGC 5694		35.01	1.16×10^5	$0.19^{+0.052}_{-0.034}$
NGC 5824		32.17	2.96×10^5	$0.98^{+0.17}_{-0.16}$
NGC 5904	M5	7.47	2.86×10^5	$0.16^{+0.039}_{-0.030}$
NGC 5927		7.67	1.14×10^5	$0.068^{+0.013}_{-0.010}$
NGC 5946		10.55	6.37×10^4	$0.13^{+0.034}_{-0.045}$
NGC 5986		10.43	2.03×10^5	$0.062^{+0.016}_{-0.010}$
NGC 6093	M80	10.01	1.67×10^5	$0.53^{+0.059}_{-0.069}$
NGC 6121	M4	2.22	6.43×10^4	$0.027^{+0.012}_{-0.010}$
NGC 6139		10.12	1.89×10^5	$0.31^{+0.094}_{-0.089}$
NGC 6205	M13	7.14	2.25×10^5	$0.069^{+0.018}_{-0.015}$
NGC 6229		30.46	1.43×10^5	$0.048^{+0.031}_{-0.009}$
NGC 6218	M12	4.83	7.18×10^4	$0.013^{+0.0054}_{-0.0040}$
NGC 6254	M10	4.39	8.39×10^4	$0.031^{+0.0043}_{-0.0041}$
NGC 6256		10.28	6.19×10^4	$0.17^{+0.119}_{-0.060}$
NGC 6266	M62	6.83	4.02×10^5	$1.67^{+0.71}_{-0.57}$
NGC 6273	M19	8.80	3.84×10^5	$0.20^{+0.067}_{-0.039}$
NGC 6284		15.29	1.31×10^5	$0.67^{+0.12}_{-0.11}$
NGC 6287		9.38	7.52×10^4	$0.036^{+0.0077}_{-0.0077}$
NGC 6293		9.48	1.11×10^5	$0.85^{+0.37}_{-0.24}$
NGC 6304		5.88	7.11×10^4	$0.12^{+0.054}_{-0.022}$
NGC 6316		10.45	1.85×10^5	$0.077^{+0.025}_{-0.015}$
NGC 6341	M92	8.27	1.64×10^5	$0.27^{+0.030}_{-0.029}$
NGC 6325		7.83	5.20×10^4	$0.12^{+0.045}_{-0.046}$
NGC 6333	M9	7.91	1.29×10^5	$0.13^{+0.059}_{-0.042}$
NGC 6342		8.53	3.16×10^4	$0.045^{+0.014}_{-0.013}$
NGC 6356		15.08	2.17×10^5	$0.088^{+0.020}_{-0.014}$
NGC 6355		9.22	1.45×10^5	$0.10^{+0.041}_{-0.026}$
Terzan 2	HP 3	7.49	1.92×10^4	$0.022^{+0.029}_{-0.014}$
NGC 6380	Ton 1	10.88	8.55×10^4	$0.12^{+0.068}_{-0.045}$
NGC 6388		9.92	4.97×10^5	$0.90^{+0.24}_{-0.21}$
NGC 6402	M14	9.25	3.73×10^5	$0.12^{+0.032}_{-0.030}$
NGC 6401		10.56	1.24×10^5	$0.044^{+0.011}_{-0.011}$

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Table 4.7 – continued from previous page

Globular Cluster	Alt. Name	Distance (kpc)	Luminosity ($L_{\odot}$)	Γ_e
NGC 6397		2.30	3.87×10^4	$0.084^{+0.018}_{-0.018}$
Palomar 6		5.79	4.45×10^4	$0.016^{+0.013}_{-0.008}$
Terzan 5	Terzan 11	5.98	7.94×10^4	$6.80^{+1.04}_{-3.02}$
NGC 6440		8.45	2.70×10^5	$1.40^{+0.63}_{-0.48}$
NGC 6441		11.60	6.08×10^5	$2.30^{+0.97}_{-0.64}$
Terzan 6	HP 5	6.78	9.29×10^4	$2.47^{+5.07}_{-1.72}$
NGC 6453		11.57	6.61×10^4	$0.37^{+0.13}_{-0.09}$
NGC 6517		10.63	1.71×10^5	$0.34^{+0.15}_{-0.10}$
NGC 6522		7.70	9.82×10^4	$0.36^{+0.11}_{-0.10}$
NGC 6528		7.93	3.63×10^4	$0.28^{+0.11}_{-0.05}$
NGC 6539		7.79	1.77×10^5	$0.042^{+0.029}_{-0.015}$
NGC 6544		2.96	5.11×10^4	$0.11^{+0.068}_{-0.037}$
NGC 6541		7.54	2.19×10^5	$0.39^{+0.095}_{-0.063}$
NGC 6553		5.96	1.10×10^5	$0.069^{+0.027}_{-0.019}$
NGC 6558		7.37	3.22×10^4	$0.11^{+0.026}_{-0.019}$
NGC 6569		10.90	1.75×10^5	$0.054^{+0.030}_{-0.021}$
NGC 6584		13.49	1.02×10^5	$0.012^{+0.0054}_{-0.0034}$
NGC 6624		7.91	8.47×10^4	$1.15^{+0.11}_{-0.18}$
NGC 6626	M28	5.52	1.57×10^5	$0.65^{+0.084}_{-0.091}$
NGC 6638		9.41	6.03×10^4	$0.14^{+0.039}_{-0.027}$
NGC 6637	M69	8.80	9.73×10^4	$0.090^{+0.036}_{-0.018}$
NGC 6642		8.13	3.94×10^4	$0.10^{+0.031}_{-0.025}$
NGC 6652		10.00	3.94×10^4	$0.70^{+0.29}_{-0.19}$
NGC 6656	M22	3.23	2.15×10^5	$0.078^{+0.032}_{-0.026}$
NGC 6681	M 70	9.01	6.03×10^4	$1.04^{+0.27}_{-0.19}$
NGC 6712		6.93	8.55×10^4	$0.031^{+0.0056}_{-0.0066}$
NGC 6715	M54	26.49	8.39×10^5	$2.52^{+0.23}_{-0.27}$
NGC 6717	Palomar 9	7.11	1.57×10^4	$0.040^{+0.022}_{-0.014}$
NGC 6723		8.65	1.16×10^5	$0.011^{+0.0080}_{-0.0044}$
NGC 6752		3.99	1.06×10^5	$0.40^{+0.18}_{-0.13}$
NGC 6760		7.36	1.17×10^5	$0.057^{+0.027}_{-0.019}$
NGC 6779	M 56	9.44	7.87×10^4	$0.028^{+0.012}_{-0.009}$

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Table 4.7 – continued from previous page

Globular Cluster	Alt. Name	Distance (kpc)	Luminosity ($L_\odot$)	Γ_e
Palomar 10		5.93	1.77×10^4	$0.059^{+0.043}_{-0.036}$
Palomar 11		13.40	5.01×10^4	$0.021^{+0.011}_{-0.007}$
NGC 6864	M 75	20.84	2.29×10^5	$0.31^{+0.095}_{-0.082}$
NGC 6934		15.63	8.17×10^4	$0.030^{+0.012}_{-0.008}$
NGC 7006		41.21	1.00×10^5	$0.0094^{+0.0049}_{-0.0033}$
NGC 7078	M 15	10.38	4.06×10^5	$4.51^{+1.36}_{-0.99}$
NGC 7089	M 2	11.56	3.50×10^5	$0.52^{+0.078}_{-0.071}$
NGC 7099	M 30	8.12	8.17×10^4	$0.32^{+0.12}_{-0.08}$

Table 4.7. The distance, visible luminosity, and stellar encounter rate [256] for each of the 87 globular clusters with either $L_V > 10^5 L_\odot$ or $\Gamma_e > 0.01$. Throughout this chapter, we use values of the stellar encounter rate that are normalized such that $\Gamma_e = 1$ for NGC 104.

4.A.1 Comparison With Previous Work

In this section, we provide a direct comparison, illustrated in Figure 4.9, between our results (Figure 4.7) and those from the prior 2016 work by Hooper and Linden [222]. Their analysis was based on 7 years of Fermi data, in contrast to the 15.8 years of data utilized in our work. The 2016 study favored an MSP gamma-ray luminosity function with parameters of $\langle L_\gamma \rangle \sim (0.7 - 6) \times 10^{34} \text{ erg/s}$ and $\sigma_L \sim 1 - 3$.

While our results are statistically consistent with theirs at the $\sim 2\sigma$ level, our analysis prefers smaller values for $\langle L_\gamma \rangle$ and larger values for R_{MSP} by a factor of $\sim 2 - 8$. Furthermore, although the best-fit values for both distributions lie near the $N_{\text{MSP}} = 30$ curve, our findings suggest a narrower range of $N_{\text{MSP}} \sim 17 - 37$, compared to the previous estimate of $N_{\text{MSP}} \sim 12 - 50$. Our updated analysis refines the prior work by both encompassing the findings of Ref. [222] as far as N_{MSP} is concerned, and providing tighter constraints on the number of MSPs required to explain the Galactic Center Excess.

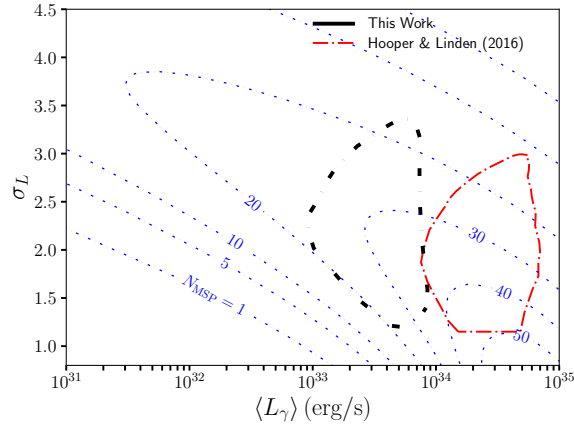


Fig. 4.9. As in Figure 4.7, for the 2σ contours, but comparing the results from the prior 2016 work from Hooper and Linden [222] with our most recent findings.

SEARCHING FOR DARK MATTER SUBSTRUCTURES

The preceding chapter examined the Galactic Center Excess, the brightest expected signature of dark matter annihilation in gamma rays. That analysis confronted a difficulty common to all Galactic Center studies: disentangling a potential dark matter signal from the complex astrophysical emission of the inner Milky Way, where overlapping source populations and uncertain diffuse backgrounds prevent definitive conclusions. We now turn to a complementary search strategy, one that targets dark matter in a very different environment.

Rather than analyzing a diffuse excess embedded in the bright Galactic foreground, this chapter asks whether any of the point sources detected by the Fermi Large Area Telescope could be dark matter subhalos. The Λ CDM cosmological model predicts that the Milky Way’s dark matter halo hosts a vast population of gravitationally bound substructures [113, 114]. The great majority of these subhalos have masses well below the threshold for retaining baryonic matter [267] and consequently produce no electromagnetic radiation through conventional astrophysical processes. If, however, the dark matter particle annihilates into Standard Model final states, the densest of these subhalos could generate detectable gamma-ray emission. They would then appear in Fermi-LAT catalogs as unassociated point sources lacking any multi-wavelength counterpart.

This chapter proceeds as follows. We first review the theoretical predictions for the Milky Way’s subhalo population within Λ CDM (Section 5.1) and their expected gamma-ray properties (Section 5.2). We then introduce the unassociated source problem in the fourth Fermi-LAT source catalog, survey previous attempts to identify dark matter subhalo candidates through classification-based methods, and explain why the dataset shift between associated and unassociated source populations necessitates the quantification learning framework developed in Section 3.4 (Section 5.3). This chapter is based on [2].

5.1 Dark Matter Substructure in Λ CDM

Before searching for dark matter subhalos among gamma-ray sources, we must first understand what Λ CDM predicts for the subhalo population of a Milky Way-like galaxy: how many subhalos are expected, at what masses, and how their abundance is shaped by baryonic processes. This

section reviews these predictions, establishing the theoretical foundation for the detectability analysis that follows in Section 5.2.

5.1.1 Hierarchical Structure Formation

A defining prediction of Λ CDM cosmology is that cosmic structure assembles in a bottom-up fashion. The primordial power spectrum, processed through the radiation-dominated epoch, retains more power on small spatial scales, so that smaller overdensities reach the nonlinear regime and gravitationally collapse before larger ones [268]. The resulting bound objects—dark matter halos—then grow through the continuous accretion of diffuse matter and the merging of pre-existing smaller halos, a process known as hierarchical structure formation [269].

When a smaller halo is accreted by a larger host, it does not dissolve entirely into a smooth background. Because early-forming halos are highly concentrated, their dense inner cores can withstand the tidal forces of the host and survive as distinct, gravitationally bound substructures [27]. The outer layers of the infalling halo are preferentially stripped away and absorbed into the ambient host, but the tightly bound core persists as a subhalo. As a result, Λ CDM predicts that Milky Way-mass halos are populated by a large number of subhalos spanning a wide range of masses.

The quantitative characterization of this subhalo population has been one of the major achievements of cosmological N-body simulations. Shortly before the turn of the century, numerical resolution reached the level necessary to robustly demonstrate the survival of subhalos within their host halos [27]. Since then, dedicated high-resolution simulations have mapped the substructure of Milky Way-like galaxies in detail. Two benchmark projects are the Via Lactea II (VL-II) simulation [114], which resolves subhalos down to approximately $10^4 M_\odot$, and the Aquarius project [113], which achieves comparable resolution and has been widely used to calibrate subhalo abundance predictions.

These simulations reveal a subhalo mass function that rises steeply toward lower masses, implying that the total number of subhalos in a Milky Way-like galaxy is vast [11, 27]. The theoretical mass function extends far below the resolution floor of current simulations; for a canonical 100 GeV WIMP, the smallest bound structures could have masses as low as $\sim 10^{-8} M_\odot$, with the precise value spanning $\sim 10^{-11}$ – $10^{-3} M_\odot$ depending on the WIMP model [11, 270]. The quantitative predictions for the subhalo population, including the J-factor distributions used in this chapter, are derived from the VL-II simulation extended through repopulation techniques [271] and are discussed in detail in Section 5.4.6.

5.1.2 Luminous Satellites vs. Dark Subhalos

Not all subhalos are alike from an observational standpoint. Above a mass threshold of approximately $10^8 M_\odot$, a subhalo possesses sufficient gravitational potential to accrete and retain baryonic matter, enabling star formation and producing a visible dwarf galaxy [267]. Roughly 50 such satellite galaxies are currently known within the virial radius of the Milky Way [27], ranging from the classical dwarf spheroidals—Draco, Sculptor, Fornax—to the ultra-faint systems discovered over the past two decades by digital sky surveys such as the Sloan Digital Sky Survey and the Dark Energy Survey [27]. These dwarf spheroidal galaxies are among the most dark-matter-dominated objects known, with mass-to-light ratios exceeding ~ 1000 in some cases, and they have been extensively used as targets for indirect dark matter searches in gamma rays (cf. Section 1.4.5).

Below $\sim 10^8 M_\odot$, however, galaxy formation becomes progressively inefficient. The ultraviolet background produced by cosmic reionization suppresses gas accretion in halos with virial masses

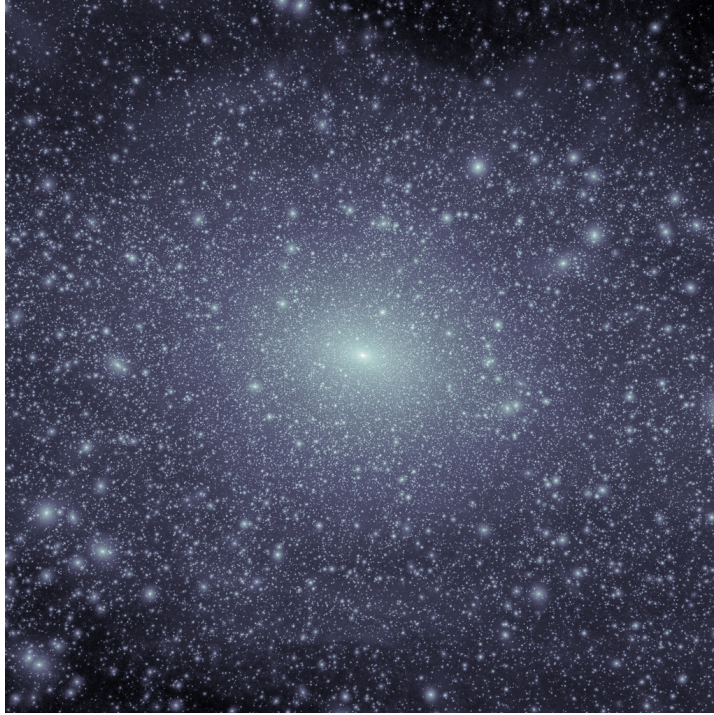


Fig. 5.1. Dark matter substructure in a Milky Way-like halo from the Via Lactea II N -body simulation [114], illustrating the vast population of subhalos predicted by Λ CDM. The map shows the local dark matter density at $z = 0$ within an 800 kpc cube. Credit: Diemand et al. [114].

below $\sim 10^9 M_\odot$, while atomic cooling ceases to operate below $\sim 10^8 M_\odot$ [27]. The vast majority of dark matter subhalos therefore contain no stars, no gas, and produce no conventional electromagnetic signature. These are the truly “dark” subhalos: structures whose existence is predicted with high confidence by Λ CDM but which cannot be detected through traditional astronomical observations. If dark matter annihilates, however, these objects could shine in gamma rays, and it is precisely this population that we target in the analysis presented in this chapter.

The apparent discrepancy between the predicted abundance of subhalos and the observed number of satellite galaxies was historically known as the “missing satellite problem” [27]. This tension has been largely resolved on two fronts. Observationally, improved surveys have steadily increased the census of ultra-faint dwarf galaxies, and statistical corrections for sky coverage and detection efficiency suggest that the total satellite population is larger than the pre-2004 sample indicated [27]. Theoretically, hydrodynamical simulations including baryonic physics—UV feedback, supernova-driven outflows, reionization—naturally suppress galaxy formation in low-mass halos, bringing predictions into broad agreement with observations once detection efficiencies are accounted for [11].

Baryonic processes also modify the survival of the subhalos themselves. Tidal stripping and gravitational shocks from the Galactic disk preferentially deplete the inner Galaxy of substructure [11, 272, 273], and the survival rate of low-mass subhalos below $\sim 10^6 M_\odot$ remains uncertain [274]. This uncertainty propagates directly into predictions for the number of gamma-ray-detectable subhalos and is discussed further in Section 5.4.6.

5.2 Dark Matter Subhalos as Gamma-Ray Targets

Having established that Λ CDM predicts a vast population of dark subhalos orbiting the Milky Way, we now turn to the question of whether any of these objects could be observed in gamma rays. The gamma-ray flux from an annihilating subhalo depends on both the particle-physics parameters— m_{DM} , $\langle\sigma v\rangle$, and the annihilation channel—and on the astrophysical properties of the subhalo, encoded in its J-factor. The general formalism for the annihilation flux and J-factor was developed in Section 1.4.4; here we highlight those features of the expected signal that are specific to the subhalo population and that shape the search strategy of this chapter.

The most important such feature is *spectral uniformity*. For a fixed dark matter mass and annihilation channel, all subhalos share the same intrinsic energy spectrum; only the overall normalization differs, set by the J-factor of each individual object. Astrophysical source classes, by contrast, exhibit broad, heterogeneous spectral distributions driven by diverse physical conditions: extragalactic sources display a wide range of nearly power-law spectra, while Galactic sources such as pulsars present highly curved spectra spanning a similarly broad parameter range [141]. A population of dark matter subhalos would therefore occupy a localized region in the spectral parameter space, with the observed scatter arising mainly from statistical measurement uncertainties rather than intrinsic physical differences. This population-level coherence is what enables the statistical approach developed later in this chapter.

The expected sky distribution of dark matter subhalos provides a complementary handle. Because subhalos trace the roughly spherical dark matter halo of the Milky Way, their projected distribution at high Galactic latitudes ($|b| > 10^\circ$) is approximately isotropic [275, 276]. This contrasts with Galactic source populations, which concentrate along the Galactic plane, but resembles the distribution of extragalactic sources. The spectral properties introduced above therefore become the primary discriminant: the combination of isotropic spatial distribution and spectral homogeneity provides the statistical foundation for the population-level search.

The expected number of Fermi-LAT-detectable subhalos spans several orders of magnitude, depending on the assumed annihilation cross section, the J-factor model, and the survival rate of low-mass subhalos in the presence of baryonic tidal effects [275–277]. Published predictions range from zero to several tens of detectable objects, defining a wide parameter space that a data-driven search could explore. The quantitative J-factor distributions and the detection pipeline are presented in the remainder of this chapter (Section 5.4.6 and Section 5.4.7).

The analysis proceeds in two stages. First, we ask whether the observed population of unassociated Fermi-LAT sources is better described by a model that includes a dark matter subhalo component or by one containing only known astrophysical source classes. If no statistically significant subhalo contribution is found, the profile likelihood is used to set 95% confidence upper bounds on $\langle\sigma v\rangle$ as a function of m_{DM} . Both possible outcomes are scientifically informative: a positive detection would provide direct evidence for annihilating dark matter in the Λ CDM substructure, while a null result constrains $\langle\sigma v\rangle$ and provides an independent cross-check of the limits obtained from dwarf spheroidal galaxies [130]. Dark subhalos are, in this respect, exceptionally clean targets: unlike the Galactic Center (Chapter 4), they host no baryonic content (Section 5.1.2), so any detected gamma-ray emission would originate entirely from dark matter annihilation.

Complementarity with dwarf spheroidal searches. The search for annihilation signals among the unassociated sources of the Fermi-LAT catalogs complements the established program of targeted observations of dwarf spheroidal galaxies, since the two strategies probe different

slices of the same Λ CDM substructure population. Every dSph observed by *Fermi*-LAT was first identified in wide-field optical surveys—SDSS, DES, and their successors—as an overdensity of resolved stars, and later confirmed through stellar-kinematic follow-up [129, 227]. No dSph has yet been discovered through its gamma-ray emission. The LAT analyses are therefore pointed observations at known coordinates, and an unassociated source is unlikely to be an undiscovered dwarf, whose stellar counterpart would in most cases already appear in optical catalogs. Rather than the optically luminous satellites, the unassociated-source search most plausibly probes the purely dark subhalos that lie below the threshold for star formation (Section 5.1.2), a population that targeted dSph studies cannot access by construction. The two probes also differ in how their J-factors are determined. For dSphs, the J-factor is measured directly from stellar kinematics, with uncertainties that grow for fainter systems [124]; the best targets reach $J \approx 3 \times 10^{19} \text{ GeV}^2 \text{ cm}^{-5}$. For dark subhalos, the J-factor distribution must instead be derived from N -body simulations. Its bright end is dominated by rare, nearby objects, so whether the Milky Way happens to host a bright subhalo is subject to statistical fluctuations—and this possibility is itself a strong motivation for the search. A subhalo with a sufficiently large J-factor, up to $\approx 3\text{--}5 \times 10^{19} \text{ GeV}^2 \text{ cm}^{-5}$, would be detectable as a point-like gamma-ray source, on par with the best dwarfs. Larger J-factors require the subhalo to be closer still, in which case it would appear spatially extended. Whereas dSphs are effectively point-like for the LAT [278], such a nearby subhalo could show a detectable extension of $0.1^\circ\text{--}0.3^\circ$, providing an additional discriminant against point-like astrophysical sources [279]. In this resolved regime the J-factor can reach $\approx 10^{21} \text{ GeV}^2 \text{ cm}^{-5}$, exceeding that of any known dwarf [271]. The two searches are complementary probes of the substructure population: the dSph program delivers robust constraints from well-characterized targets, while searches for dark subhalos among the unassociated-source could reveal candidates with even lower astrophysical backgrounds.

5.3 The Unassociated Source Problem

Having established the theoretical grounds for dark matter subhalo searches in the previous sections, we now turn to the observational reality. The *Fermi*-LAT catalogs contain a large population of gamma-ray sources with no identified counterpart at other wavelengths, and these unassociated sources are the natural testing ground for DM subhalo searches.

5.3.1 Unassociated Sources in *Fermi*-LAT Catalogs

The latest incremental release, the 4FGL-DR4 catalog [141, 153], contains 7,195 sources detected above a significance threshold of 4σ . Of these, 2563 sources—roughly one third—remain unassociated, meaning that no plausible counterpart has been identified at other wavelengths [153]. Among the unassociated population, 1,283 sources lie at Galactic latitudes $|b| > 10^\circ$, where contamination from diffuse Galactic emission is reduced and the search for exotic source populations becomes more tractable.

Association completeness depends on both source brightness and Galactic latitude: brighter sources are far easier to link to multi-wavelength counterparts, while the association fraction drops from approximately 85% at high latitudes to around 40% near the Galactic plane, where dense source fields and X-ray absorption impede counterpart identification [140]. The majority of unassociated sources are therefore expected to be ordinary astrophysical objects whose association has failed for instrumental or geometric reasons. Their spectral properties offer direct clues to their nature: at high latitudes, the average photon index ($\Gamma \approx 2.24$) matches that of blazars of unknown type, while closer to the Galactic plane the spectra soften ($\Gamma \approx 2.42$), consistent with

pulsars [140]. Further discussion of the Fermi-LAT association procedure and its completeness can be found in Section 2.3.1.

To date, no unassociated Fermi-LAT source has been unambiguously confirmed as a dark matter subhalo [140, 141, 275, 280]. Two tentative subhalo candidates have recently been claimed through purely gravitational probes. Precise timing of nearby pulsars has revealed a local deviation from the smooth Galactic gravitational potential, which has been attributed to a $\sim 10^7 M_\odot$ subhalo within a few kpc of the Sun [281]. A second candidate comes from the GD-1 stellar stream, which shows a gap consistent with a close encounter with a dense, dark object—itsself interpreted as a dark matter subhalo [282]. These claims are very recent, and whether they will be confirmed by independent analyses is a question for the coming years. As discussed in Section 5.2, even under optimistic assumptions the most detectable subhalos would produce gamma-ray spectra similar to those of pulsars or soft-spectrum blazars, making individual identification on spectral grounds alone extremely difficult. Nevertheless, the large unassociated population has motivated a sustained effort to search for DM subhalo candidates within it.

5.3.2 Previous Searches and the Limits of Classify-and-Count

Prior searches for dark matter subhalos among Fermi-LAT unassociated sources in general share a common strategy, referred to as *classify-and-count*: a set of DM subhalo candidates is selected from the unassociated population based on their gamma-ray properties, and upper limits on $\langle\sigma v\rangle$ are derived under the assumption that the number of true subhalos cannot exceed the candidate count [275–277, 279, 280, 283–295]. Early analyses relied on hand-crafted spectral and spatial cuts applied to successive catalog releases [280, 283–285, 289, 290, 292], with some studies emphasizing spatial extension as a potential discriminant [279, 280, 289]. More recently, machine learning classifiers have automated and extended the candidate selection [276, 291, 294, 295]. Nonetheless, in none of these cases could the identified candidates be unambiguously distinguished from standard astrophysical sources.

Classification is, in itself, a legitimate way to flag DM subhalo candidates, and machine learning classifiers remain a useful tool for the task. The limitation lies not in classifying individual sources per se, but in properly tuning those classifiers to the unassociated population and in turning that classification into a population-level constraint on $\langle\sigma v\rangle$. The number of DM candidates depends on an arbitrary probability threshold $p_{\text{DM}} > p_{\text{threshold}}$ that needs to be carefully — and sometimes arbitrarily — chosen. Different choices can shift the candidate count, and with it the derived bound on $\langle\sigma v\rangle$. The resulting upper limits therefore sometimes lack a well-defined confidence level: they encode an assumption, $N_{\text{DM}} \leq N_{\text{candidates}}$, rather than a statistical inference. A classifier is a supervised model: it is trained on a labeled, at least partly simulated dataset to reproduce the conditional probability $p(k|\mathbf{x})$ as accurately as possible, and then applied to the unassociated sources one by one. The unassociated population as a whole does not generally enter the optimization, and shapes the classifier only through the composition of its training set. A likelihood-based approach instead learns the class-conditional densities $p(\mathbf{x}|k)$ and treats the prevalences $p(k)$ as free parameters, fitted by maximizing the joint likelihood of the entire unassociated sample. This amounts to unsupervised density estimation performed directly on the target data: the parameters are driven by the empirical feature distribution of the unassociated population itself, which is what makes the approach more powerful. Since $p(\mathbf{x}|k)$ and $p(k)$ together determine $p(k|\mathbf{x})$ through Bayes’ theorem, individual classification is recovered as a byproduct, and the resulting constraint follows from an explicit statistical model. We develop this alternative in the following subsection.

5.3.3 The Quantification Learning Alternative

If we already had a well-established population of confirmed DM subhalos, we could train a machine learning model to identify new ones, but no such sample exists. The composition of the unassociated catalog must instead be inferred as a whole, and the formal machinery for this task was introduced in Section 3.4: *quantification learning*.

A classifier’s posterior factorizes through Bayes’ theorem as $p(k|\mathbf{x}) \propto p(\mathbf{x}|k)p(k)$, and, as discussed in Section 3.4.2, its dependence on the class prevalences $p(k)$ is exactly why a discriminative estimate cannot survive a shift in the underlying class fractions between the associated and unassociated populations. Quantification learning inverts the question. Instead of asking “to which class does this source belong?”, it asks “what is the distribution of observed features within each class?”—that is, it estimates the class-conditional densities $p(\mathbf{x}|k)$. These densities describe the intrinsic physical profile of each source population: the spectral shape, curvature, and flux distribution that characterize, say, blazars or pulsars as a class. Because the physical properties of a source class do not change when its relative abundance changes, $p(\mathbf{x}|k)$ is invariant under prior shift ($p_{\text{assoc}}(\mathbf{x}|k) = p_{\text{unas}}(\mathbf{x}|k)$; see Equation 3.4.5). This invariance is what makes the class-conditional densities learnable from the associated catalog and directly portable to the unassociated population.

This inversion—from the posterior $p(k|\mathbf{x})$ to the class-conditional densities $p(\mathbf{x}|k)$ —shifts the unit of analysis from the individual source to the population. A generative model characterizes each class by its feature distribution and describes the unassociated catalog as a mixture $\tilde{p}_{\text{unas}}(\mathbf{x}) = \sum_k \pi_k p(\mathbf{x}|k)$, in which the class prevalences π_k become free parameters fitted to the data rather than imposed during training [198, 200]. The mixture form naturally accommodates a new class—dark matter subhalos—whose prevalence is left free and fitted directly, and supports statistically rigorous confidence intervals through the profile likelihood [2].

In practice, prior shift and covariate shift between the associated and unassociated populations act simultaneously (Section 3.4.2). The covariate shift is an instrumental bias—bright, high-latitude sources are easier to associate—and is corrected by modulation functions shared across all astrophysical classes and fitted jointly with the class prevalences, as detailed in Section 5.4.4. As the analysis below shows, the prior shift turns out to be the dominant of the two effects, consistent with the higher density of unassociated sources near the Galactic plane, while the covariate correction acts as a statistically significant but secondary refinement. The full construction of this mixture model, including the kernel density estimation of astrophysical templates, the Monte Carlo simulation of the DM component, and the optimization via the Expectation-Maximization algorithm, is presented in the remainder of this chapter, which is based on [2] and presents the analysis in full.

5.4 Search for dark matter subhalos among unassociated Fermi-LAT sources in presence of dataset shift

In this work, we construct for the first time a statistical model for the distribution of unassociated sources at latitudes $|b| > 10^\circ$.¹ It consists of a mixture model, i.e. a linear combination of probability distribution functions (PDFs) of classes of astrophysical sources plus a contribution of DM subhalos. The PDFs of classes of astrophysical sources are determined from the associated sources, while the distribution of DM subhalos is determined from Monte Carlo simulations assuming a model of J-factors from N-body DM simulations and particular parameters for DM (mass, annihilation cross section, and channel). The prior shift problem is addressed by fitting the relative contribution of the different components to the data (the distribution of unassociated sources), while the covariate shift problem is addressed by adding a modulation to the PDFs of the classes of associated sources (the parameters of the modulation are also fit to the data simultaneously with the class prevalence).

In more formal terms, we construct PDFs of astrophysical classes, denoted as $p(\mathbf{x}|k)$, which represent the probability of a source with features $\mathbf{x}$ given a source class k . The PDFs for unassociated sources are determined by modulating the PDFs of the associated ones by functions of input features $\mathbf{x}$ (these functions are the same for all classes k). Previous supervised learning studies have focused on estimating the conditional probability $p(k|\mathbf{x})$ of the class of source k given the input features [179, 197, 276, 291, 294–303], as typically done by probabilistic classification models. Estimating $p(\mathbf{x}|k)$ instead brings several crucial advantages. First of all (see section 5.4.1), it allows us to model the difference between distributions of associated and unassociated sources as a combination of both covariate and prior shifts. The latter allows us to model the different prevalence of classes of astrophysical sources for training (associated sources) and target (unassociated sources) datasets as well as to include a new class of sources among the unassociated ones, which in this analysis is modeled as DM subhalos. Moreover, upon specification of the class prior $p(k)$, the joint distribution $p(k, \mathbf{x}) = p(\mathbf{x}|k)p(k)$ can be sampled to generate mock data, which can be used in Monte Carlo analyses.² Finally, the joint PDF $p(\mathbf{x}, k)$ can be marginalized over the class k to estimate $p(\mathbf{x})$, i.e., the marginal distribution of the source features. The quantity $p(\mathbf{x})$ represents a probability density for a source with parameters $\mathbf{x}$. The product of $p(\mathbf{x}_i)$ over unassociated sources is the model likelihood, which we maximize to obtain the optimum model parameters both for the prior and covariate shifts. In particular, we use such likelihood to estimate the best fit and upper bounds on the DM annihilation cross section $\langle\sigma v\rangle$. Although generative models have been constructed previously in analyses of *Fermi*-LAT source distributions, e.g. [295], they were not used for construction of maximum likelihood models, which we do in the current work.

This chapter is organized as follows. In section 5.4.2, we describe the selection of the data. We discuss the covariate and prior shifts in section 5.4.3. We present our statistical model in section 5.4.4. The model of J-factors used in this chapter is reviewed in section 5.4.6, while the

¹We exclude sources with $|b| < 10^\circ$, since the main background in this analysis is the astrophysical sources among the unassociated ones. Almost half of the unassociated sources (1146 out of 2428) are within this range of latitudes and most of them are expected to be the astrophysical Galactic sources. The decrease in the area, on the other hand, is about 17%, i.e., assuming an isotropic distribution of DM subhalos around the Earth, the increase in signal-to-noise ratio due to masking of the Galactic plane can be estimated as $(1 - 0.17)\sqrt{2428/1146} \approx 1.14$. This increase is likely larger due to higher flux detection threshold (smaller detectability volume for DM subhalos) toward the Galactic plane, i.e., the decrease in signal is less than 17% as a result of masking the Galactic plane.

²That's why probabilistic models estimating $p(\mathbf{x}|k)$ are called “generative models”, as opposed to “discriminative” models focusing on $p(k|\mathbf{x})$.

simulation of gamma-ray sources corresponding to DM subhalos is presented in section 5.4.7. We optimize the model in section 5.4.9, while the corresponding limits on DM annihilation determined from the observed gamma-ray sources are presented in section 5.4.10. Section 5.4.11 contains discussion and conclusions. We provide details about the statistical model and perform several consistency tests in appendices.

5.4.1 Statistical analysis

5.4.2 Data selection

In this work we use the 4FGL-DR4 catalog [141, 153]. We group the known types of associated sources into two classes:³

- Galactic (dominated by millisecond pulsars at high latitudes [86, 131, 140, 248, 304, 305]): pulsars, binaries, star-forming regions, supernova remnants, pulsar wind nebulae, globular clusters, and novae;
- Extragalactic (mostly AGNs [103, 131, 138, 140, 144, 306, 307]): blazars, quasars, starburst galaxies, normal galaxies, radio galaxies and radio sources, Seyfert galaxies, and non-blazar active galaxies.

This choice is made to group sources based on their fundamentally different physical origins and radiation environments. By separating the magnetospheric emission characteristic of Galactic pulsars from the jet-driven processes of extragalactic AGNs, we ensure that the classification reflects the intrinsic physical diversity of the γ -ray sky, which manifests in distinct spectral signatures.

In order to characterise the sources, we focus on their energy spectra modeled with the log-parabola (LP) function

$$\ln \frac{F(E)}{K} = -\alpha \ln \frac{E}{E_0} - \beta \left(\ln \frac{E}{E_0} \right)^2, \quad (5.4.1)$$

where the parameters correspond to the following columns in the 4FGL-DR4 catalog: K – LP_Flux_Density, α – LP_Index, β – LP_beta, and E_0 – Pivot_Energy.

The LP model provides a well-motivated spectral choice; it offers a significant improvement over a simple power law, as evidenced in the LP_SigCurv column in the 4FGL-DR4 catalog, and performs at least as well as a power law with exponential cut off both for the unassociated and for the associated Galactic and extragalactic sources (figure 5.2).

Provided that only three parameters are independent and that E_0 in the catalog is different for different sources, we rescale E_0 to a fixed value of 1 GeV (for analysis of $M_{\text{DM}} > 20$ GeV) or 100 MeV (for $M_{\text{DM}} < 20$ GeV). We use the following three parameters as input features:

1. $\log_{10} \phi$ – the logarithm of the flux integrated between 100 MeV and 1 TeV (obtained by summing fluxes in energy bins provided in the catalog);
2. α – LP_Index rescaled to $E_0 = 1$ GeV or $E_0 = 100$ MeV depending on the choice of DM mass, i.e., above or below 20 GeV respectively;
3. β – LP_beta.

³Cf. table 5 in ref. [141] for details about the types of sources. Galactic sources: psr, hmb, sfr, snr, pwn, gc, gal, bin, msp, lmb, spp, glc, nov. Extragalactic sources: bll, sbg, rdg, css, bcu, ssrq, fsrq, sey, nlsy1, agn.

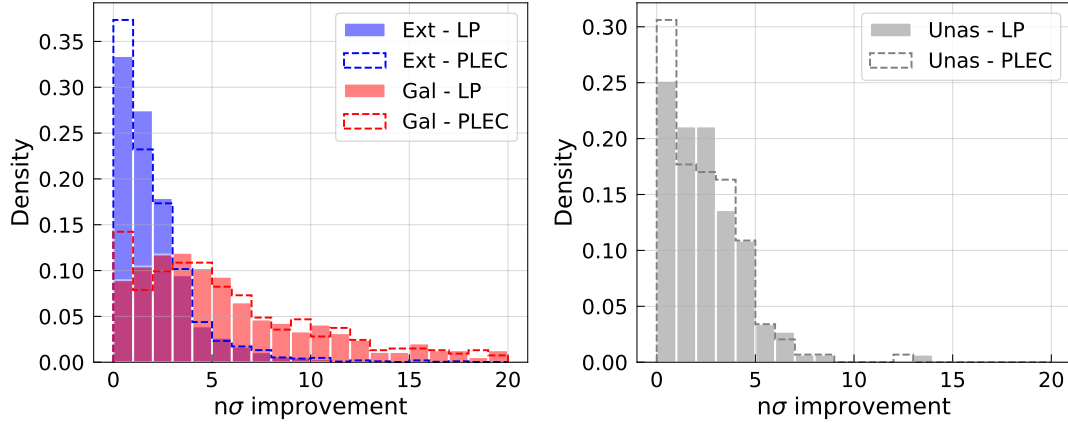


Fig. 5.2. Distribution of the significance of fit improvement, as reported in the 4FGL-DR4 catalog, when replacing a power-law energy spectrum with either a log parabola (LP) or a power law with an exponential cutoff (PLEC). We display the histograms for associated sources (left panel) and unassociated sources (right panel).

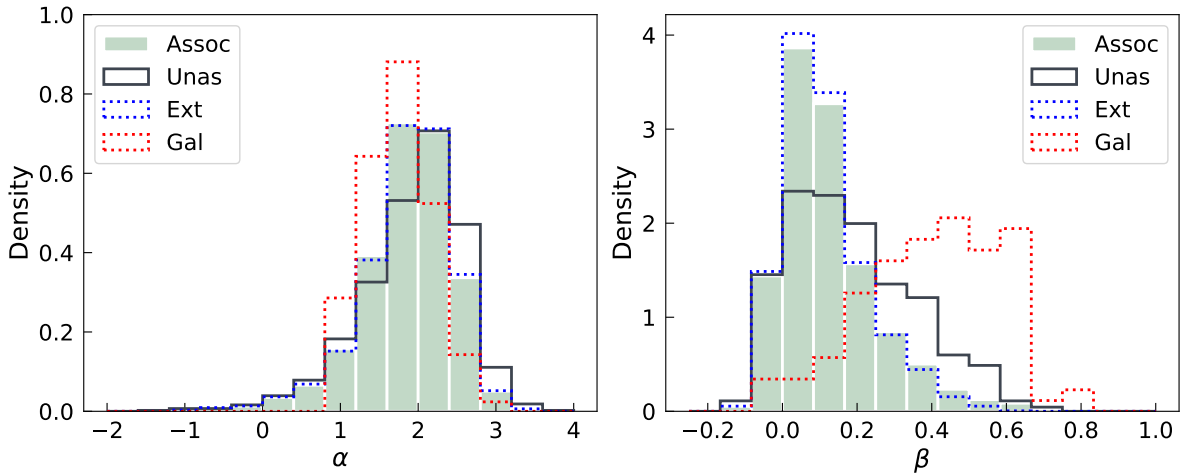


Fig. 5.3. Normalised distributions of α and β parameters for associated and unassociated sources for $E_0 = 1$ GeV, as well as the distributions for extragalactic and Galactic sources with $|b| > 10^\circ$.

We collect these variables for each source i inside a vector $\mathbf{x}_i \equiv \{\log_{10}(\phi_i), \alpha_i, \beta_i\}$. In our analysis, we consider only sources with Galactic latitudes $|b| > 10^\circ$.

5.4.3 Covariate and prior shifts

The basic assumption behind a supervised classification task is that the joint distribution of the input features $\mathbf{x}$ and output classes (type of sources) k are the same for the training and target datasets:

$$p_{\text{train}}(\mathbf{x}, k) = p_{\text{target}}(\mathbf{x}, k), \quad (5.4.2)$$

while, in general, a dataset shift represents a situation where the training and target distributions are different $p_{\text{train}}(\mathbf{x}, k) \neq p_{\text{target}}(\mathbf{x}, k)$. In our case, the training dataset is the associated sources, while the target dataset is the unassociated sources. Consequently, we will use associated (assoc)

and unassociated (unas) subscripts for the training and target datasets, respectively. The joint distribution of features $\mathbf{x}$ and class labels k can be expressed as a product of conditional and marginal probabilities in two equivalent ways:

$$p(\mathbf{x}, k) = p(k|\mathbf{x})p(\mathbf{x}) = p(\mathbf{x}|k)p(k). \quad (5.4.3)$$

This decomposition gives rise to two distinct types of dataset shift [196]:

1. Covariate shift: the conditional class probabilities remain unchanged, $p_{\text{assoc}}(k|\mathbf{x}) = p_{\text{unas}}(k|\mathbf{x})$, but the feature distributions differ, $p_{\text{assoc}}(\mathbf{x}) \neq p_{\text{unas}}(\mathbf{x})$.
2. Prior shift: the feature distributions conditioned on the class are invariant, $p_{\text{assoc}}(\mathbf{x}|k) = p_{\text{unas}}(\mathbf{x}|k)$, but the class prevalence changes, $p_{\text{assoc}}(k) \neq p_{\text{unas}}(k)$.

Note that for the covariate shift, differences in the marginal distribution of the features $p(\mathbf{x})$ generally lead to differences in the marginal class probabilities $p(k)$. Analogously, for prior shift, changes in the marginal class distributions $p(k)$ in general result in changes in the marginal feature distributions $p(\mathbf{x})$. We show the distributions of the α and β parameters in figure 5.3. Notice that the distributions are different for associated and unassociated sources. In principle, this difference can be due to either a covariate shift or a prior shift. For example, as can be seen from the histograms of the associated extragalactic and Galactic sources, a change in the class prevalence in unassociated sources would certainly lead to a change in the β parameter distribution, which could be explained as a prior shift. On the other hand, if there are differences in the distribution of unassociated sources, which affect all unassociated sources, e.g., larger uncertainties in the reconstructed energy spectrum, then this effect can be addressed by a covariate shift.

5.4.4 Statistical model

In general, our goal is to construct a probabilistic model for the distribution of unassociated sources $\tilde{p}_{\text{unas}}(\mathbf{x})$ under the assumption that it is a mixture of classes of associated sources plus a possible new class of sources.

In the covariate shift assumption (in absence of a prior shift) this task is actually trivial as

$$p_{\text{unas}}(\mathbf{x}) = \sum_k p_{\text{unas}}(k|\mathbf{x})p_{\text{unas}}(\mathbf{x}) = \sum_k p_{\text{assoc}}(k|\mathbf{x})p_{\text{assoc}}(\mathbf{x})C(\mathbf{x}), \quad (5.4.4)$$

where in the second equality we use the covariate shift assumption $p_{\text{assoc}}(k|\mathbf{x}) = p_{\text{unas}}(k|\mathbf{x})$ and define $C(\mathbf{x}) = p_{\text{unas}}(\mathbf{x})/p_{\text{assoc}}(\mathbf{x})$. The function $C(\mathbf{x})$ describes the covariate shift, i.e., the difference in the distribution of training and target datasets.

Moreover, we can add a new class of sources with a PDF $p_{\text{new}}(\mathbf{x})$ and define a model for the unassociated sources as:

$$\tilde{p}_{\text{unas}}(\mathbf{x}) = \sum_k p_{\text{assoc}}(k|\mathbf{x})p_{\text{assoc}}(\mathbf{x})\tilde{C}(\mathbf{x}) + \pi_{\text{new}}p_{\text{new}}(\mathbf{x}), \quad (5.4.5)$$

where $\tilde{C}(\mathbf{x}) = \max(p_{\text{unas}}(\mathbf{x}) - \pi_{\text{new}}p_{\text{new}}(\mathbf{x}), 0)/p_{\text{assoc}}(\mathbf{x})$ can be viewed as a generalized covariate shift that describes the difference in the distribution of training dataset and a part of the target dataset that corresponds to the same classes as in the training dataset. If $p_{\text{unas}}(\mathbf{x}) \geq \pi_{\text{new}}p_{\text{new}}(\mathbf{x})$ for $\forall \mathbf{x}$, then the model in eq. (5.4.4) is equal to the model in eq. (5.4.5), i.e., the two models are indistinguishable until the new class predicts more sources in some region of space than

there are unassociated sources. Although mathematically we cannot exclude a situation when all unassociated sources in some region of space belong to a new class, this is not very likely from the physical point of view. In order for the model to be more constraining we put additional physical conditions on the generalized covariate shift function $\tilde{C}(\mathbf{x})$. In particular, we model $\tilde{C}(\mathbf{x})$ as a product of 1-dimensional functions

$$\tilde{C}(\mathbf{x}) = \prod_{d=1}^3 f_d(x_d), \quad (5.4.6)$$

where each $f_d(x_d)$ is monotonous. The product is over the dimensions d of the space of input variables. In particular, we assume here that the covariate shift in each of the input variables is independent of the other input variables. We implement $f_d(x_d)$ using sigmoid functions

$$f(x_d; b_d, c_d) = \frac{1}{1 + e^{-b_d(x_d - c_d)}}, \quad (5.4.7)$$

where each function has two parameters (b_d, c_d) determined from maximizing the likelihood of the model. Our region of interest (ROI) in feature space will be restricted to a window in every dimension where the ratio $p_{\text{unas}}/p_{\text{assoc}}$ of the empirical distributions are monotonous. We approximate the empirical distributions of associated and unassociated gamma-ray sources using kernel density estimation (KDE) with a Gaussian kernel [155]. In order to find the optimal bandwidth for the kernels, we perform 5-fold cross-validation repeated 20 times. For each of the repetitions, we randomize the dataset, split it into 5 subsets, and repeat the training 5 times: each time, a new subset is used for testing, and the remaining four subsets are used for training. The performance is evaluated by calculating the likelihood of the model on the testing dataset. The optimal bandwidth parameter is obtained from the maximum of the performance averaged over the 100 evaluations (20 repetitions with 5 training/testing splits in each repetition). The optimal bandwidth parameters for associated and unassociated sources are computed independently. We show the corresponding KDE models in the $E_0 = 1$ GeV case in the left panels in figure 5.4. The right panels of figure 5.4 show the ratios of the KDE models, where the bands represent the statistical uncertainty.

The ROI is determined as the intervals in parameter space where the ratio of the KDE models is monotonous within statistical uncertainties. In the $E_0 = 1$ GeV case, we restrict our analysis to $\alpha \in [0.8, 1.7]$. Analogously, in the $E_0 = 100$ MeV case, $\alpha \in [0.2, 0.8]$. The range of the β parameter is the same in both cases, $[0.1, 0.55]$. Finally, we restrict the $\log_{10} \phi$ parameter to the interval $[-10.3, -8.2]$. In addition, we compute the 3D KDE models and identify the optimal bandwidth parameters independently for Galactic and extragalactic classes of sources, which yields the $p_{\text{assoc}}(\mathbf{x}|k)$ used in the analysis.

In general, the weight (prevalence) of classes of sources does not need to be the same for associated and unassociated sources, i.e., there can be a prior probability shift in addition to the covariate shift. In order to allow a change in the prevalence of the classes of sources, we express the terms in the sum in eq. (5.4.5) in the form suitable for the prior shift:

$$p_{\text{assoc}}(k|\mathbf{x})p_{\text{assoc}}(\mathbf{x}) \equiv p_{\text{assoc}}(\mathbf{x}, k) = \pi_k p_{\text{assoc}}(\mathbf{x}|k), \quad (5.4.8)$$

where for associated sources $\pi_k = \pi_k^{\text{assoc}}$ is the prevalence of class k , while for unassociated sources, π_k are in general unknown and will be determined from the fit of the model to the data. Overall, our model of the unassociated source takes the form:

$$\tilde{p}_{\text{unas}}(\mathbf{x}) = \left(\sum_{k=1}^K \pi_k p_{\text{assoc}}(\mathbf{x}|k) \right) \tilde{C}(\mathbf{x}; \boldsymbol{\theta}_{\text{cov}}) + \pi_{\text{DM}}(\boldsymbol{\theta}_{\text{DM}}) p_{\text{DM}}(\mathbf{x}; \boldsymbol{\theta}_{\text{DM}}), \quad (5.4.9)$$

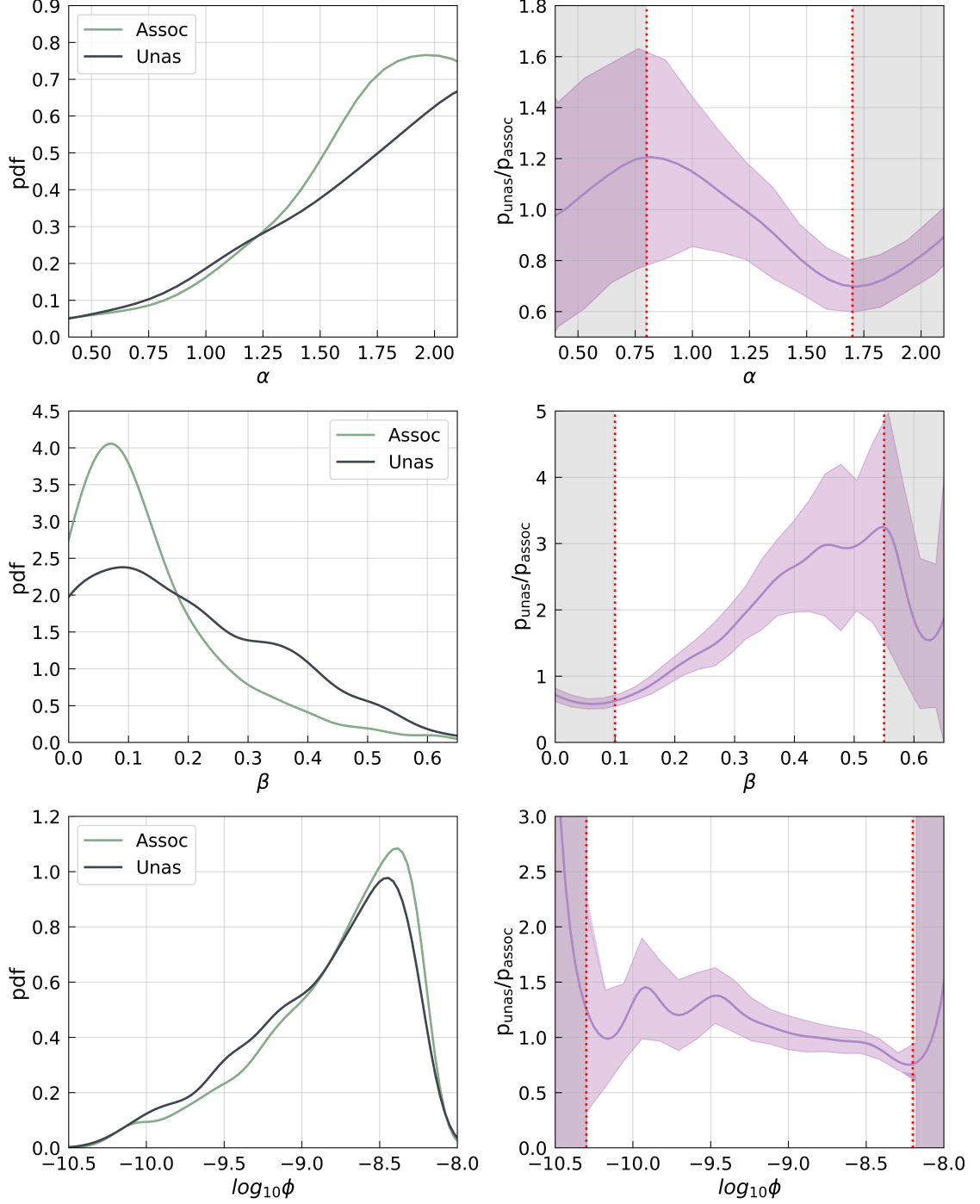


Fig. 5.4. *Left column:* full distributions of associated and unassociated sources modeled with KDE for the case of $E_0 = 1$ GeV before truncation and normalization to the ROI. *Right column:* the ratio of the distributions $p_{\text{unas}}/p_{\text{assoc}}$. The red vertical lines show the range of parameters that we use in the analysis (ROI). We estimate the error (purple bands) as the Poisson error in the histograms and compute the relative errors for the associated and unassociated distributions, which we sum in quadrature.

where $K = 2$ is the number of classes of associated sources (Galactic and extragalactic, cf. section 5.4.2). The DM distribution $p_{\text{DM}}(\mathbf{x}; \boldsymbol{\theta}_{\text{DM}})$ is determined below in section 5.4.7.

The optimizable parameters in this model are: π_k , describing the prevalence of the classes of associated sources; $\boldsymbol{\theta}_{\text{cov}} \equiv \{b_d, c_d\}$ for $d = 1, 2, 3$, parametrizing the covariate shift in eq. (5.4.7), and $\langle \sigma v \rangle$, the annihilation cross section, which we have included in the definition of $\boldsymbol{\theta}_{\text{DM}} = \{\langle \sigma v \rangle, M_{\text{DM}}, \text{ch.}\}$. The other parameters, namely the DM mass M_{DM} as well as the annihilation channel ‘ch.’, are kept fixed during the optimization procedure.

Note that for $\tilde{C}(\mathbf{x}; \boldsymbol{\theta}_{\text{cov}}) = 1$, the model reduces to a prior shift model, while for $\pi_k = \pi_k^{\text{assoc}}$ and $\pi_{\text{DM}} = 0$ the model reduces to a covariate shift model. Thus, in general, the model in eq. (5.4.9) is a combination of covariate and prior shift models. We point out that the model in eq. (5.4.9), identified as the likelihood of a given unassociated source, allows us to:

1. determine the unknown parameters of the model by maximizing the likelihood;
2. detect or put an upper bound at a given level of confidence for a possible new component, such as DM;
3. generate new data (for Monte Carlo tests);
4. determine the conditional class probabilities of individual sources using the probability rule $p(k|\mathbf{x}) = p(\mathbf{x}, k)/\tilde{p}_{\text{unass}}(\mathbf{x})$. In the case of astrophysical classes $p(\mathbf{x}, k) = \pi_k p_{\text{assoc}}(\mathbf{x}|k) \tilde{C}(\mathbf{x}; \boldsymbol{\theta}_{\text{cov}})$, while for the DM class: $p(\mathbf{x}, \text{DM}) = \pi_{\text{DM}} p_{\text{DM}}(\mathbf{x}; \boldsymbol{\theta}_{\text{DM}})$.

We stress that, as far as we are aware of, this is the first time a generative probabilistic model for the unassociated sources has been proposed. Previous studies have focused on modeling directly the conditional probabilities $p(k|\mathbf{x})$, which does not give access to the data likelihood. In this respect, our approach is more informative.

5.4.5 Dark matter subhalos model

5.4.6 Subhalo J-factors from N-body simulations

A key component of our pipeline is the so-called ‘J-factor distribution’ of the Galactic subhalo population. This J-factor, also known simply as ‘astrophysical factor’, acts as a normalization factor for the expected annihilation signal and can be computed as the volume integral of the subhalo DM density squared. In this work, we follow ref. [271] and adopt the J-factor values provided for two different prescriptions of the subhalo internal structure⁴: one derived from the mass enclosed within the subhalo tidal radius, $J_{\text{tidal}}^{M_{\text{sub}}}$, and another based on the maximum circular velocity of each subhalo, $J_{\text{tidal}}^{V_{\text{max}}}$. These two prescriptions are advisable here to estimate current J-factor uncertainties. Indeed, measuring subhalo masses is not a trivial task, especially at scales close to the numerical resolution limit of the simulation, where only a few simulation particles are available. Because of this, the computation of masses relies on adopting a specific form of the subhalo DM density profile, which is then fitted to the available simulation data and integrated up to the assumed tidal radius. Unfortunately, even the precise value of the latter can vary significantly among different definitions and is very sensitive to changes in the tidal forces acting on the subhalo as it orbits around the host halo center. In contrast to these mass-related issues, the peak of the circular velocity profile within each subhalo, i.e. V_{max} , has been shown to be less affected by numerical resolution and tidal forces (see, e.g., refs. [271, 308]).

⁴All data are available at https://projects.ift.uam-csic.es/damasco/?page_id=831.

Its determination also does not depend on adopting a specific functional form of the DM density profile to model the subhalo structural properties. As a result, J-factor calculations based on the latter approach are considered more reliable. Nevertheless, we use both J-factor sets in this study mainly to enable direct comparison with previous results, where J-factor based on subhalo masses are more common. Also, as said, the exercise will be useful to understand the current level of uncertainties on subhalo modeling.

The calculation of astrophysical J-factors in ref. [271] is based on the Via Lactea II (VL-II) N-body cosmological simulation [309], a DM-only simulation of a Milky-Way-sized halo resolving subhalos down to approximately $10^4 M_\odot$. To explore subhalos below the VL-II resolution limit, the simulation was repopulated with millions of smaller subhalos, extending the analysis down by nearly five orders of magnitude to about $10^{-1} M_\odot$ [279]. Additionally, by placing the Earth at random positions consistent with its Galactocentric distance, thousands of realizations of a Milky Way-like galaxy were created to obtain statistically robust results. This led to accurate J-factor distributions (based on both mass and circular velocity) for the entire subhalo population of a galaxy like our own in a Λ CDM universe. A limitation of ref. [271] is the absence of baryonic effects in the parent simulation. Baryons are expected to significantly influence the DM subhalo population, particularly in the Galaxy's innermost regions (e.g., [310–312]), however none of these hydrodynamical simulations resolve subhalos lighter than $\sim 10^6 M_\odot$. Thus, the survival of such low-mass subhalos to tidal stripping, especially with baryonic influences, remains uncertain [272–274].

In addition to $J_{\text{tidal}}^{M_{\text{sub}}}$ and $J_{\text{tidal}}^{V_{\text{max}}}$ from ref. [271], we also utilize $J_{0.1}^{M_{\text{sub}}}$ and $J_{0.1}^{V_{\text{max}}}$, derived from the same simulation work. These values correspond to J-factors integrated not up to the tidal radius of each subhalo but rather to the corresponding radius subtending 0.1° in the sky. This choice aligns with the typical angular resolution of the *Fermi*-LAT in the GeV range. By using $J_{0.1}^{M_{\text{sub}}}$ and $J_{0.1}^{V_{\text{max}}}$, we focus on point-like subhalo emissions, as would appear in *Fermi*-LAT point-source catalogs. Notably, ref. [279] concludes that the brightest Galactic subhalos in the *Fermi*-LAT gamma-ray sky, if detected, would be extended, with angular sizes around $0.2^\circ - 0.3^\circ$. Consequently, for such subhalos, our $J_{0.1}^{M_{\text{sub}}}$ and $J_{0.1}^{V_{\text{max}}}$ values are expected to be significantly lower than their corresponding $J_{\text{tidal}}^{M_{\text{sub}}}$ and $J_{\text{tidal}}^{V_{\text{max}}}$ values. We adopt $J_{0.1}^{V_{\text{max}}}$ as our nominal case in this work, while we use the other J-factor values to compare with previous literature, e.g., in Calore et al. (2019) [277] a $J_{0.1}^{M_{\text{sub}}}$ model was used.

Figure 5.5 illustrates the four sets of J-factor distributions discussed in this section, along with their corresponding number density, which will be the relevant quantity for our pipeline.

5.4.7 Gamma-ray emission from dark matter subhalos

Given a DM setup θ_{DM} (i.e., a DM mass M_{DM} , annihilation channel, and annihilation cross section $\langle\sigma v\rangle$), and an expected distribution of J-factors determined from DM N-body simulations (described in section 5.4.6), we estimate the expected distribution of DM annihilating subhalos that can be detected by *Fermi* LAT for a certain observation time and instrument response functions (IRFs). The procedure is as follows:

1. We start with a distribution $p(J) = dN/dJ$ of J-factors expected from N-body DM simulations (cf. figure 5.5 in section 5.4.6). We sample the values of J from $p(J)$.
2. Given a J-factor and DM parameters $\theta_{\text{DM}} = \{\langle\sigma v\rangle, M_{\text{DM}}, \text{ch.}\}$, we determine the expected

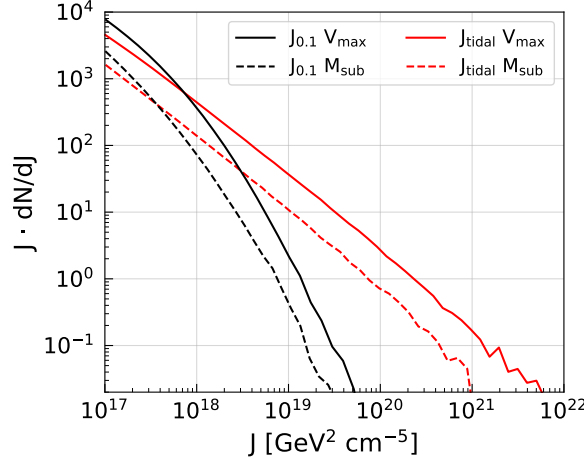


Fig. 5.5. J-factor number density times J for the different J-factor cases considered in this work. See section 5.4.6 for further details.

gamma-ray spectrum as

$$\phi_{\text{DM}}(E; \theta_{\text{DM}}, J) = \frac{J}{4\pi} \frac{\langle \sigma v \rangle}{2M_{\text{DM}}^2} \left. \frac{dN_\gamma}{dE} \right|_{\text{ch.}}, \quad (5.4.10)$$

where dN_γ/dE is the theoretical spectrum of gamma rays produced in an annihilation of a pair of DM particles from ref. [102].

3. We place DM subhalos at random positions in the sky with J-factors sampled in step 1 and with the spectra defined in eq. (5.4.10).
4. For the diffuse gamma-ray background, we use the official *Fermi*-LAT diffuse model template employed in the construction of the 4FGL-DR4 source catalog (gll_iem_v7) and the corresponding isotropic background.
5. We fit the gamma-ray data around each of the simulated DM subhalos with $|b| > 10^\circ$, where we use the log-parabola model in eq. (5.4.1) with fitting parameters $\{\log_{10} \phi, \alpha, \beta\}$. The fit is performed within a circular region centered on the target DM source. The radius of this region is energy-dependent, corresponding to the 95% containment angle of the PSF, and is constrained to the range $[0.5^\circ, 5^\circ]$. This range is chosen to ensure sufficient statistics in the highest energy bins while simultaneously avoiding excessive background contamination in the lowest energy bins. We use the same Gaussian priors for the β parameter as in the 4FGL-DR4 catalog with mean 0.1 and sigma 0.3 [153]. For details see appendix 5.A.
6. We keep DM subhalos detected with significance $\text{TS} > 25$, where $\text{TS} = -2\ln \mathcal{L}/\mathcal{L}_*$ is the likelihood-ratio test statistic between the models with and without the subhalo. Such a threshold corresponds to the significance for PS detection in the *Fermi*-LAT catalogs. The fraction of subhalos passing the significance cut defines the detection efficiency $\epsilon_{\text{eff}}(\phi_{\text{DM}})$, which is a function of the DM flux.
7. In order to get a good estimate of the DM distribution, $p_{\text{DM}}(\log_{10} \phi_{\text{DM}}, \alpha, \beta | \theta_{\text{DM}})$, we simulate realizations of the gamma-ray sky until we detect at least 20,000 DM subhalos across

all simulations or the number of simulations reaches 1,000. The latter can happen for parameters, where DM subhalos have very small detection probability.⁵

In order to reduce the computation time, we perform simulations in flux bins. At first, we determine the distribution of α and β parameters, given the flux ϕ_{DM} and DM parameters θ_{DM} , $p_{\text{DM}}(\alpha, \beta | \phi_{\text{DM}}; \theta_{\text{DM}})$, by modeling the distribution of recovered α and β parameters in this flux bin (see appendix 5.A for details). Then we multiply this distribution by the distribution of fluxes, which depends on the distribution of J-factors $n(\phi_{\text{DM}}) = \frac{dN}{d\phi_{\text{DM}}}(\phi_{\text{DM}}; \theta_{\text{DM}}) = \frac{dN}{dJ} \frac{dJ}{d\phi_{\text{DM}}}$ and by detection efficiency as a function of flux $\varepsilon_{\text{eff}}(\phi_{\text{DM}})$ to determine the probability density times the DM subhalo weight in eq. (5.4.9):

$$\pi_{\text{DM}}(\theta_{\text{DM}}) p_{\text{DM}}(\mathbf{x}; \theta_{\text{DM}}) = \frac{1}{N_{\text{unas}}^{\text{ROI}}} p_{\text{DM}}(\alpha, \beta | \phi_{\text{DM}}; \theta_{\text{DM}}) n(\phi_{\text{DM}}) \varepsilon_{\text{eff}}(\phi_{\text{DM}}), \quad (5.4.11)$$

where $N_{\text{unas}}^{\text{ROI}}$ is the number of unassociated sources in the ROI.

We note that given the DM parameters, the contribution of DM subhalos to unassociated sources is fixed. In particular, the number of subhalos in our ROI is

$$N_{\text{sub}}^{\text{ROI}}(\theta_{\text{DM}}) = \int_{\text{ROI}} d\alpha d\beta d\phi_{\text{DM}} p_{\text{DM}}(\alpha, \beta | \phi_{\text{DM}}; \theta_{\text{DM}}) n(\phi_{\text{DM}}) \varepsilon_{\text{eff}}(\phi_{\text{DM}}), \quad (5.4.12)$$

while the DM weight π_{DM} in eq. (5.4.9) is

$$\pi_{\text{DM}}(\theta_{\text{DM}}) = \frac{N_{\text{sub}}^{\text{ROI}}(\theta_{\text{DM}})}{N_{\text{unas}}^{\text{ROI}}}. \quad (5.4.13)$$

In figure 5.6 we show the distributions of DM subhalos in α and β parameters (we integrate the distributions in the flux dimension for presentation purposes) for different DM masses at some characteristic $\langle \sigma v \rangle$ values. We note that the DM distributions are not fully contained inside the ROI that we selected in this analysis, which is indicated as a green box in the plots.

In particular, a significant fraction of the DM subhalo distribution is truncated for DM masses of 10 GeV, 30 GeV, and 1 TeV. Although choosing a larger ROI that covers a larger fraction of the DM subhalo distribution could increase statistics, our assumption that the covariate shift can be modeled by a monotonous function would not be valid anymore. Introducing a more flexible covariate shift model to account for this would require additional parameters, potentially leading to larger uncertainty in π_{DM} due to correlations with these new parameters (i.e., a larger parameter covariance matrix) and, consequently, weaker limits. Moreover, the gain from larger statistics would be limited by the significant background of astrophysical sources outside of our current ROI, particularly unassociated millisecond pulsars with $\alpha > 0.7$ and $\beta \sim 0.4$ (upper right panel of figure 5.6). Conversely, regions with small α values (bottom panel) contain very few astrophysical sources, necessitating extrapolation that carries large systematic uncertainties (see the wide uncertainty band for small alpha values in the upper left panel in figure 5.4). Finally, our limits on DM annihilation are conservative: positive residuals after fitting the data with astrophysical components can be absorbed by the DM component, resulting in a positive best-fit π_{DM} and weaker upper limits than in the $\pi_{\text{DM}} = 0$ case.

In figure 5.7 we show the contour levels for the half maximum of the DM subhalo distribution for different choices of $\langle \sigma v \rangle$ and J-factor models for $M_{\text{DM}} = 100$ GeV. The distribution shifts to

⁵The limit of a thousand simulations is needed to avoid infinite iterations for low-efficiency parameter combinations. For details on simulation and detection of DM subhalos, see appendix 5.A.

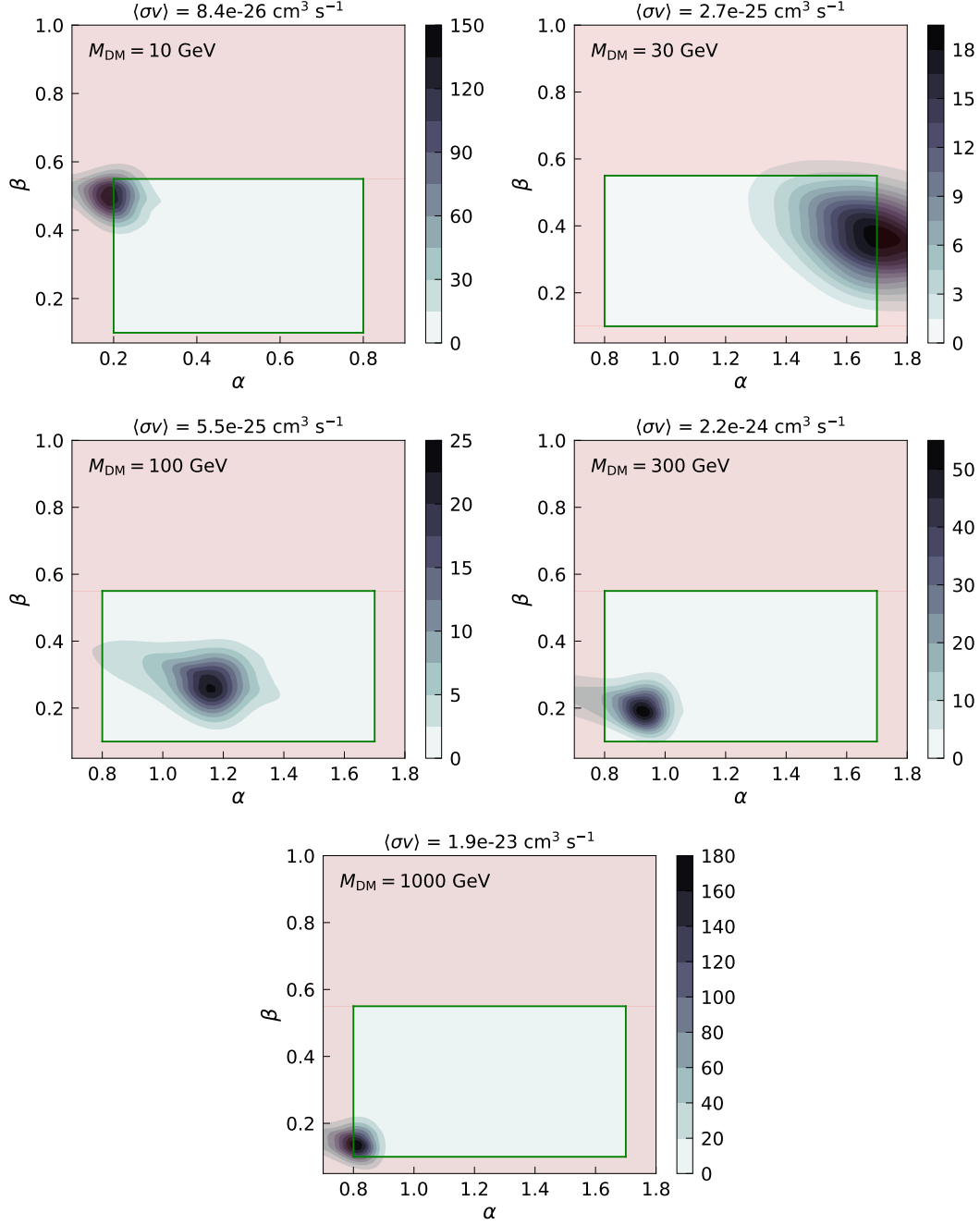


Fig. 5.6. DM distribution at different DM masses, $b\bar{b}$ annihilation channel, integrated in flux for visualization purposes. The $\langle\sigma v\rangle$ values are chosen close to our bound values. The green boxes indicate the parameter ranges (ROI) used for this analysis and define the truncation region for the distribution. The color bars show the value of the DM subhalo PDF (eq. 5.4.11), normalized to unit integral within the ROI.

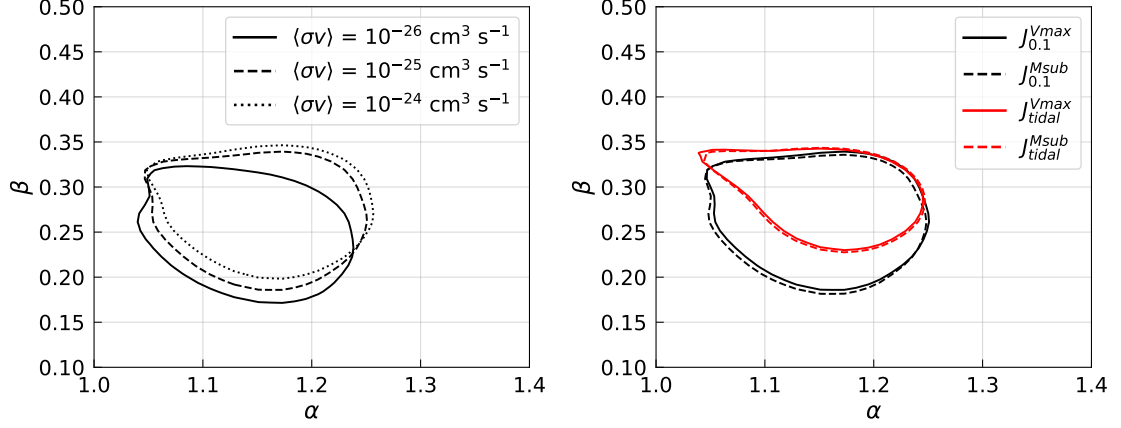


Fig. 5.7. Contour plots at half maximum of the DM subhalo distribution for different values of $\langle\sigma v\rangle$ and J-factor models at $M_{\text{DM}} = 100$ GeV. *Left panel:* contour lines for different choices of $\langle\sigma v\rangle$ for $J_{0.1}^{Vmax}$. *Right panel:* contour lines for different J-factor models for $\langle\sigma v\rangle = 10^{-25}$ cm³ s⁻¹.

higher values of β for larger $\langle\sigma v\rangle$ values. Analogously, choosing J_{tidal} shifts the distribution to higher β values compared to $J_{0.1}$. This is to be expected, since for faint sources (small $\langle\sigma v\rangle$ or small J-factors) the fit has large statistical uncertainties and the best-fit value of the β parameter is more affected by the priors on β than in case of bright sources.

5.4.8 Mixture model of gamma-ray sources and limits on DM annihilation

5.4.9 Model optimization

Our general aim is to either detect a population of DM subhalos among the *Fermi*-LAT unassociated sources or to put an upper limit on $\langle\sigma v\rangle$ for different DM masses. For a fixed DM mass and annihilation channel, we determine the best-fit cross section $\langle\sigma v\rangle^*$ (optimized simultaneously with the other parameters) by maximizing the full likelihood built from the model in eq. (5.4.9), defined as:

$$\mathcal{L}(\langle\sigma v\rangle, \pi_k, \theta_{\text{cov}}) = \prod_{i=1}^{N_{\text{unas}}^{\text{ROI}}} \tilde{p}_{\text{unas}}(\mathbf{x}_i | \langle\sigma v\rangle, \pi_k, \theta_{\text{cov}}; M_{\text{DM}}, \text{ch.}). \quad (5.4.14)$$

We maximize eq. (5.4.14) with the Expectation-Maximization (EM) algorithm, which is efficient for the cases of probabilistic mixture models [199], see appendix 5.B for more details. From the ML point of view our approach is a semi-supervised learning algorithm.

Indeed, in supervised learning the model is fixed by the training data (including the PDFs for the different classes), while in unsupervised learning, the model is determined solely based on target data. In our case, we use the associated sources (training data) to determine PDFs of the two classes of associated sources (which is similar to supervised learning), but we further modify the PDFs to account for the possible dataset shift between training and target data and add a new class of sources (which is similar to unsupervised learning).

We optimize the likelihood for a set of DM mass values: $M_{\text{DM}} = 10, 30, 100, 300, 1000$ GeV, and determine, for each M_{DM} , the upper bound on $\langle\sigma v\rangle$ at 95% confidence level (CL) by profiling the test-statistic

$$\text{TS}(\langle\sigma v\rangle) = -2 \ln \frac{\mathcal{L}(\langle\sigma v\rangle)}{\mathcal{L}_{\text{max}}}, \quad (5.4.15)$$

where $\mathcal{L}(\langle\sigma v\rangle)$ is the likelihood optimized over the rest of parameters π_k and θ_{cov} , and $\mathcal{L}_{\text{max}} = \mathcal{L}(\langle\sigma v\rangle^*, \pi_k^*, \theta_{\text{cov}}^*)$ is the value of the likelihood for the best-fit parameters. Following the Wilks' theorem, we assume that TS follows the χ^2 distribution. We compute the upper bound on $\langle\sigma v\rangle$ as the value corresponding to the 95% coverage probability (one-sided), which for one degree of freedom corresponds to TS = 3.84.

We maximize the likelihood of the model in eq. (5.4.14) to determine the best-fit parameters. At first, we fix the DM contribution to be zero and determine the best-fit model with the known astrophysical sources only. For reference, we show the model for the associated sources (no fitting) in the left panel of figure 5.8 and the best-fit model for unassociated sources in the middle and right panels. In these plots, we display the distributions evaluated in the analysis ROI for α and β parameters integrating over the flux coordinate for visualization purposes. We decompose the mixture model of eq. (5.4.9) and associate the blue color with the extragalactic component and the red color with the Galactic component, both respectively weighted by their prior weight π_k to highlight their relative contribution. The intensity of the color bar corresponds to the value of the component likelihood weighted by the prior weight, $\pi_k p(x|k)$, and a darker, more intense color is associated with regions where there is a higher probability of observing a certain source class. We will follow this convention for all upcoming distribution plots.

We note that the extragalactic sources are dominated by AGNs, which usually have close to a power-law spectra (small β values). The Galactic sources often have curved spectra (large β values). The relative contribution of Galactic sources is larger for the unassociated sources compared to the associated ones (brighter red color in the right panel compared to the left panel in figure 5.8). The model on the middle and right panels of figure 5.8 is obtained by optimizing the prior and covariate shift parameters, π_k and θ_{cov} . We display the optimized Galactic and extragalactic components separately in figure 5.9. The corresponding best-fit parameters π_k (the class prevalence) are presented in table 5.1. As qualitatively discussed above, the expected contribution of Galactic sources to unassociated sources at high latitudes (29%) is significantly larger than the fraction of associated Galactic sources (6%) in the ROI. The right panel of figure 5.8 also shows the model's predictions for class probabilities of unassociated sources. We compare the relative importance of prior and covariate shift parts of the model in appendix 5.C.1, where we find that the prior shift plays a more important role than the covariate shift. In particular, for the model without the DM component, we find that the prior shift is preferred at 7σ significance (for one additional degree of freedom) compared to a model without either prior or covariate shifts, while the model with the covariate shift is preferred at 4.5σ significance (with 6 additional degrees of freedom). The model with both prior and covariate shifts is preferred at 6.7σ significance (with 7 degrees of freedom), i.e., most of the improvement is, indeed, due to the prior shift.

5.4.10 Dark matter limits

The best-fit model including the DM component with DM mass of 100 GeV and $\langle\sigma v\rangle = 3.4 \cdot 10^{-25} \text{cm}^3 \text{s}^{-1}$ is shown in figure 5.10. We simultaneously fit the parameters π_k , θ_{cov} and $\langle\sigma v\rangle$. The corresponding components are presented in figure 5.11. The DM component is shown by the green color.⁶

Provided that all DM subhalos have the same gamma-ray spectra, the corresponding distribution of DM subhalos is much better localized in the α and β plane than the extragalactic and

⁶The class probabilities for unassociated sources in the astrophysical only model, as well as DM subhalo class probabilities for the different DM masses, are available online at <https://doi.org/10.5281/zenodo.17463823>.

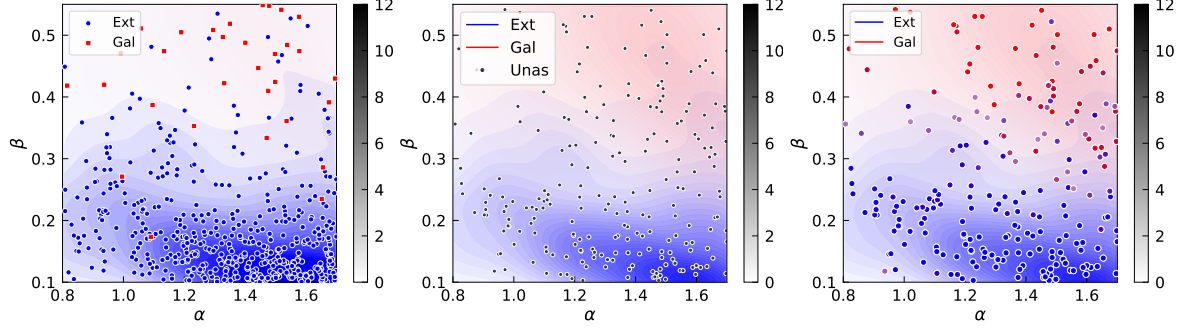


Fig. 5.8. Distributions of sources in the plane of the spectral parameters (α , β) for $E_0 = 1$ GeV. *Left panel:* PDFs of Galactic and extragalactic sources weighted by the corresponding class fractions. The sources are colored according to the catalog class. *Middle panel:* best-fit mixture model for the unassociated sources. The background colors show the individual components of the astrophysical-only model from eq. (5.4.9), where the DM subhalo contribution is set to zero. Specifically, we plot the decomposition of the mixture model into class-conditional PDFs weighted by their best-fit class prevalence $\pi_k p(x|k)$ for the extragalactic (blue) and Galactic (red) populations. For visualization, the distributions are integrated over the flux dimension. The color bar represents the value of the resulting probability density (shown in gray to represent both red and blue colors). *Right panel:* Class probabilities for unassociated 4FGL-DR4 sources (for $E_0 = 1$ GeV), on top of the mixture distribution (same as the middle panel). The color scale is proportional to the class probabilities, according to the formula $\text{color}(x_i) = \text{red} \cdot p(C_k = \text{gal}|x_i) + \text{blue} \cdot p(C_k = \text{ext}|x_i)$, e.g., a purple-colored source is a source for which the posterior class probabilities are similar and the classification is ambiguous.

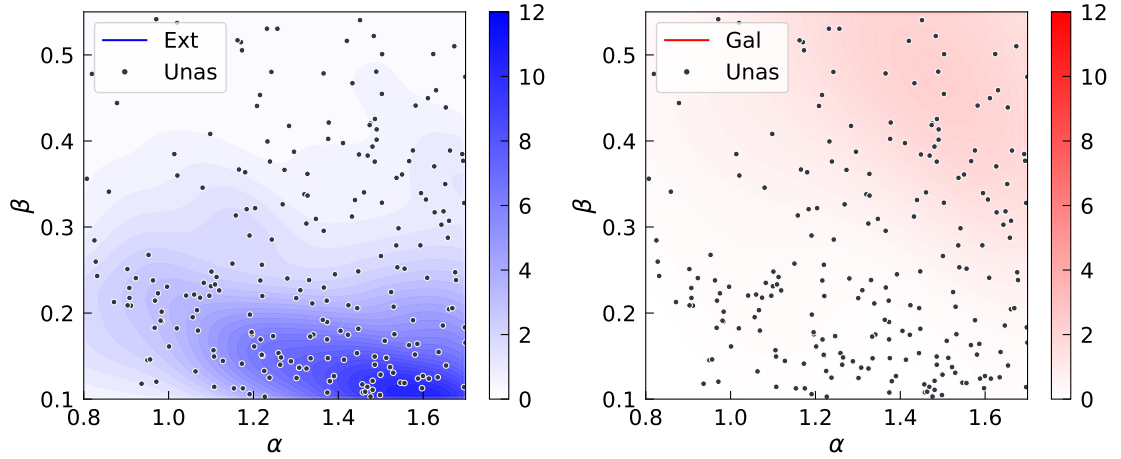


Fig. 5.9. Decomposition of the mixture model for unassociated sources with two classes (Galactic and extragalactic) for the ROI cut associated with $E_0 = 1$ GeV. The model is the same as in the middle and right panels of figure 5.8. *Left panel:* extragalactic component. *Right panel:* Galactic component.

Galactic components. The spread in the DM subhalo distribution comes from statistical uncertainties in the reconstruction of spectral parameters. We note that not only the overall normalization

type	N_{ext}	N_{Gal}	% Ext	% Gal	N_{tot}
assoc	466	31	94%	6%	497
unas	160.6	66.4	71%	29%	227

Table 5.1. Class prevalence for Galactic and extragalactic sources in the ROI. For the associated sources (second row) the numbers are derived from the 4FGL-DR4 catalog. For the unassociated sources (third row), the numbers correspond to the best-fit model with astrophysical components only (using $E_0 = 1$ GeV). The percent of galactic (% Gal) and extragalactic (% Ext) sources are given by the best-fit parameters π_k in eq. (5.4.9) for the two astrophysical components. The expected number of Galactic and extragalactic sources among the unassociated ones is derived as $\pi_k \times N_{\text{tot}}$, where $N_{\text{tot}} = 227$ is the total number of unassociated sources in the ROI.

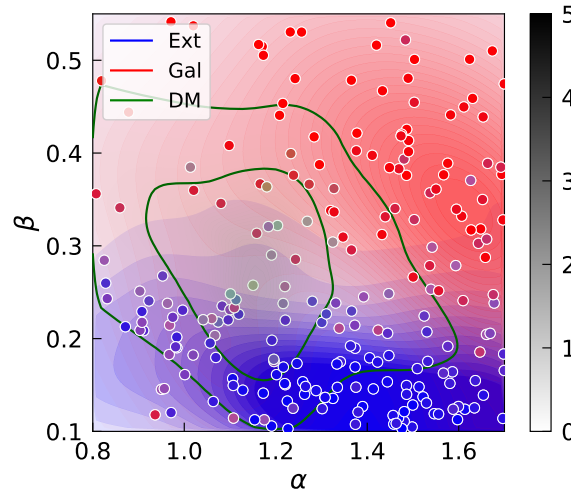


Fig. 5.10. Mixture model for unassociated sources including the DM component with $M_{\text{DM}} = 100$ GeV, following the same conventions as in figure 5.8. The green component represents the best-fit DM distribution for $\langle\sigma v\rangle = 3.4 \cdot 10^{-25} \text{cm}^3 \text{s}^{-1}$. The green lines represent the 68% and 95% containment levels for the DM distribution. The background colors represent the class-conditional distributions weighted by their prior weight: $\pi_k p(x|k)$. Sources are colored according to their posterior class probability following the formula $\text{color}(x_i) = \text{red} \cdot p(C_k = \text{gal}|x_i) + \text{blue} \cdot p(C_k = \text{ext}|x_i) + \text{green} \cdot p(C_k = \text{DM}|x_i)$.

but also the shape of the distribution of DM subhalos in the right panel of figure 5.11 changes for different annihilation cross sections and J-factor models (cf. figure 5.7). The reason is that the distribution depends on the relative contribution of subhalos with different J-factors, e.g., smaller J-factors correspond to proportionally smaller flux and larger uncertainties, which affect the shape of the distribution. Nevertheless, for larger $\langle\sigma v\rangle$ values the expected contribution of DM subhalos also increases.

We present the details of the best-fit model for the DM subhalo component (including upper limits in the annihilation cross section) in table 5.2. Columns 2 to 6 show the best-fit model, e.g., the best-fit DM subhalo prevalence, the expected number of subhalos within our region of interest ($N_{\text{subhalo}}^{\text{ROI}}$) and for the full range of parameters ($N_{\text{sub}}^{\text{total}}$), as well as the best-fit thermally-averaged annihilation cross-section ($\langle\sigma v\rangle$). The value of $\Delta\text{TSS}_0 = -2(\log \mathcal{L}_0 - \log \mathcal{L}_{\text{max}})$ quantifies the preference for the best-fit model over a baseline astrophysical-only scenario. For all tested

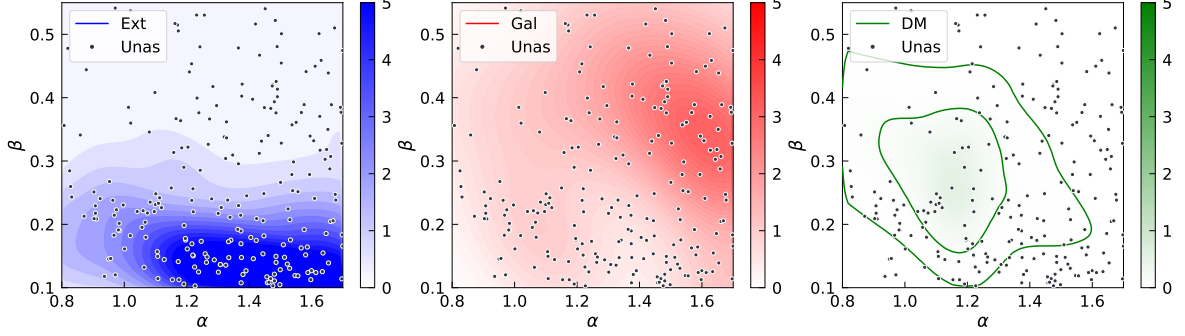


Fig. 5.11. Components of the mixture model in figure 5.10, following the same conventions as in figure 5.8.

models, the significance is less than 2 sigma. We notice that the largest possible contribution from DM subhalos is for $M_{\text{DM}} = 30$ GeV (although still not significant).

The expected number of DM subhalos of 36.2 for this DM mass in the full range of parameters is comparable to the 16 DM subhalo candidates reported in ref. [275], once the improved sensitivity of the 4FGL-DR4 catalog (14 years of observations) relative to the 3FGL catalog (8 years of observations) is taken into account.

To verify this, we first consider the case where the DM subhalo-like sources are astrophysical in origin. In the Euclidean regime of constant volume density, the source counts scale as $N(>S) \propto S^{-1.5}$ as a function of flux, and the detection threshold scales as $S \propto t^{-0.5}$. Under these assumptions, 36.2 subhalo-like sources in 14 years of observations (4FGL-DR4, full range of parameters) correspond to approximately 23.8 sources in 8 years of observations (3FGL catalog, ref. [275]). Since 16 is within about 1.6σ of 23.8, the two methods yield consistent estimates of the number of possible DM subhalos. Alternatively, if the sources are DM annihilating subhalos, our baseline model has $dN/dJ \propto J^{-3}$ (black dashed line in figure 5.5), which gives $N_{\text{sub}} \propto J^{-2} \propto t$, where we have used $S \propto J$ for fixed $\langle\sigma v\rangle$. In this case, 36.2 detectable subhalos in 14 years correspond to $36.2 \times 8/14 \approx 20.7$ subhalos in 8 years, which is also consistent with the 16 subhalos assumed in ref. [275].

We also notice that for $M_{\text{DM}} = 30$ GeV, the expected contribution from Galactic sources (mostly millisecond pulsars) is the smallest. This is to be expected, since the spectrum of DM annihilation in the $b\bar{b}$ channel is similar to the spectra of pulsars at this DM mass. As a result, there may be a degeneracy between the DM subhalos and Galactic sources for this DM mass. Provided that we do not find significant evidence for a signal from DM, we proceed to set conservative upper bounds on $\langle\sigma v\rangle$ shown in the “upper bound” section of the table.

In order to estimate the overall goodness of fit of the model in eq. (5.4.9) without DM to the data, we compare in appendix 5.C.2 the log likelihood of the fit to the data with the distribution of log likelihood values for the data generated from the best-fit model. We find that the deviation of the log likelihood for the real data is less than 2σ smaller than the mean log likelihood value for the simulated data, i.e., the deviation is consistent with statistical fluctuations within 2σ . We also perform DM injection tests in appendix 5.C.3. We find that our method can recover the DM signal within statistical fluctuations: for all DM masses the $\langle\sigma v\rangle$ value for the injected signal is within the 68% containment of the recovered $\langle\sigma v\rangle$ values around the median recovered $\langle\sigma v\rangle$. In addition, we determine the precision-recall curves for the classification of individual DM subhalos for 1000 simulations with injected DM signal (in case of $M_{\text{DM}} = 30$ GeV) and evaluate the average

M_{DM} [GeV]	% DM	best-fit model				upper bound		
		$N_{\text{subhalo}}^{\text{ROI}}$	$N_{\text{subhalo}}^{\text{total}}$	$\langle\sigma v\rangle$ [cm ³ s ⁻¹]	ΔTS_0	$N_{\text{subhalo}}^{\text{ROI}}$	$N_{\text{subhalo}}^{\text{total}}$	$\langle\sigma v\rangle$ [cm ³ s ⁻¹]
10	0.0	0.00	0.00	0.0	0.0	1.91	4.42	5.6×10^{-26}
30	6.8	15.37	36.23	2.7×10^{-25}	3.5	27.30	64.20	3.6×10^{-25}
100	2.4	5.45	6.21	3.4×10^{-25}	1.2	19.36	21.86	5.9×10^{-25}
300	2.0	4.59	7.66	1.1×10^{-24}	1.1	13.23	21.46	1.7×10^{-24}
1000	0.0	0.00	0.00	0.0	0.0	2.06	8.61	4.7×10^{-24}

Table 5.2. Expected number of DM subhalos (N_{sub}) for the **best-fit model** and for the **95% confidence upper bound** on the annihilation cross-section ($\langle\sigma v\rangle$). In both cases, we report the subhalo count within our ROI and the total count. The test statistic difference quantifies the significance of the best-fit model over an astrophysical-only model, $\Delta\text{TS}_0 = -2(\log \mathcal{L}_0 - \log \mathcal{L}_{\text{max}})$. The number of subhalos, N_{sub} , is constrained to be non-negative.

M_{DM} [GeV]	% DM	best-fit model				upper bound		
		$N_{\text{subhalo}}^{\text{ROI}}$	$N_{\text{subhalo}}^{\text{total}}$	$\langle\sigma v\rangle$ [cm ³ s ⁻¹]	ΔTS_0	$N_{\text{subhalo}}^{\text{ROI}}$	$N_{\text{subhalo}}^{\text{total}}$	$\langle\sigma v\rangle$ [cm ³ s ⁻¹]
10	0.0	0.00	0.00	0.00	0.0	1.65	4.97	1.86×10^{-26}
30	0.0	0.00	0.00	0.00	0.0	12.41	26.80	1.61×10^{-25}
100	0.0	0.00	0.00	0.00	0.0	11.86	12.99	2.33×10^{-25}
300	4.3	10.87	16.67	8.79×10^{-25}	3.6	20.71	31.80	1.55×10^{-24}
1000	0.0	0.00	0.00	0.00	0.0	2.13	6.63	1.64×10^{-24}

Table 5.3. Same as Table 5.2, but for the $J_{\text{tidal}}^{\text{Msub}}$ case.

precision of our model in appendix 5.C.4.

We show the upper bounds on $\langle\sigma v\rangle$ in figure 5.12. The uncertainty band for the annihilation cross-section bound is derived with a bootstrap strategy. We identify the best-fit model for each DM mass and use the probability density function for that model (cf. eq. (5.4.9)) to sample 100 new datasets via Hybrid (a.k.a. “Hamilton”) Monte Carlo (HMC) [313] and the NUTS algorithm [314]. For each simulated dataset, we derive the bounds on $\langle\sigma v\rangle$ and determine the 95% containment interval for the resulting distribution of values. We further adjust the upper limit of this interval to conservatively account for potential systematic uncertainties associated with the presence of 4FGL sources, as detailed in appendix 5.A. This uncertainty range is represented by the gray shaded band in figure 5.12.

The dashed gray line shows a bound derived using dwarf spheroidal galaxies [130] with 68% and 95% confidence bands shown by the green and yellow areas respectively. We note that the constraints derived in this work are an order of magnitude less stringent than the limits derived using dwarf spheroidal galaxies (even if we consider J-factors integrated to tidal radius for the subhalos shown in figure 5.13). Provided that the values of J-factors $\lesssim 10^{20} \text{ GeV}^2 \text{ cm}^{-5}$ for the brightest dwarf galaxies [130] are comparable to the brightest subhalos (cf. figure 5.5), the main reason for the less stringent constraints is that the positions and J-factors of the dwarf galaxies are known (although with some uncertainties), while the subhalos can be at any position in the sky.

In figure 5.13, we display a comparison of the upper bounds for different choices of the J-factor. In particular, the limits for the J-factor integrated within 0.1° for the V_{max} and M_{sub} J-factor determination methods are shown by the black solid and dashed lines respectively. The

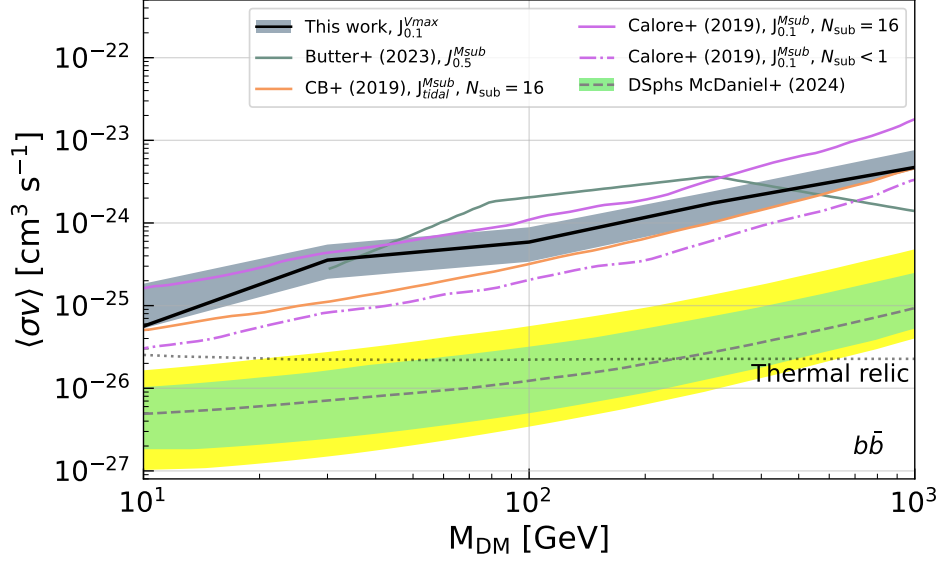


Fig. 5.12. Upper bounds with 95% confidence on the DM annihilation cross section $\langle\sigma v\rangle$ in the $b\bar{b}$ channel. Solid black line: our baseline limits derived with the V_{max} determination of the J-factor integrated within the radius of 0.1° . Gray band: 95% containment of the upper limit due to statistical fluctuations of the distribution of sources, the effect of masking the 4FGL-DR4 sources is added to the upper values of the band (appendix 5.A). Solid and dashed purple lines: the limits derived by Calore et al. (figure 6 left panel) [277]. Dashed orange line: limits derived by Coronado-Blázquez et al. (figure 12 upper panel) [276]. Solid green line: the limits derived by Butter et al. (figure 11, the 0.8 threshold limit for $|b| > 10^\circ$) [294]. Green and yellow bands with dashed gray line: limits derived for dwarf spheroidal Galaxies at 68% and 95% confidence level by McDaniel et al. (figure 6 top left panel) [130]. Dotted black line: the expectation for the WIMP DM thermal relic cross-section from ref. [12].

limits corresponding to the J-factor integrated up to the tidal radius for the V_{max} and M_{sub} J-factor determination methods are shown by the red solid and dashed lines, respectively.

For comparison, we show the limits on $\langle\sigma v\rangle$ obtained assuming 16 DM subhalos in the 3FGL catalog [275]. The limits for J-factor integration up to 0.1° radius [277] (purple solid line) are consistent with the corresponding $J_{\text{tidal}}^{M_{\text{sub}}}$ model (dashed red line) within a factor of 2, comparable to the uncertainty band in figure 5.12. Larger (smaller) upper limits on $\langle\sigma v\rangle$ relative to ref. [277] correspond to larger (smaller) expected numbers of DM subhalos in our model compared to the 16 assumed in ref. [277].⁷ The limits for J-factor integration up to the tidal radius [275] (orange solid line) are comparable to the corresponding limits in the $J_{\text{tidal}}^{M_{\text{sub}}}$ model (dashed red line). The more stringent constraints at the lowest (10 GeV) and highest (1000 GeV) DM masses are partly

⁷The $\langle\sigma v\rangle$ upper limit depends weakly on observation time. In the regime of a few observable DM subhalos, $dN/dJ \propto J^{-n-1}$, so the total number of observable subhalos above a flux threshold scales as $N_{\text{sub}} \propto J^{-n}$, where $n = 1$ ($n = 2$) for the tidal radius (0.1°) J-factor integration. Since $F \propto \langle\sigma v\rangle J$, $N_{\text{sub}} \propto J^{-n} \propto \langle\sigma v\rangle^n / F^n$. The number of background astrophysical sources in the Euclidean regime scales as $N(>F) \propto F^{-1.5}$, where the flux threshold depends on observation time as $F \propto t^{-0.5}$. The confidence of the upper limit on $\langle\sigma v\rangle$ can then be estimated from the signal-to-noise ratio $SNR = N_{\text{sub}} / \sqrt{N(>F)}$. Fixing the SNR at a given confidence level, $N_{\text{sub}} = SNR \times \sqrt{N(>F)} \propto F^{-0.75}$, and thus $\langle\sigma v\rangle^n \propto F^n N_{\text{sub}} \propto F^{n-0.75} \propto t^{-n/2+0.4}$. This gives $\langle\sigma v\rangle \propto t^{-0.1}$ and $\langle\sigma v\rangle \propto t^{-0.3}$ for the tidal radius and 0.1° cases, respectively. Since these dependences are very weak, we neglect them in our estimates.

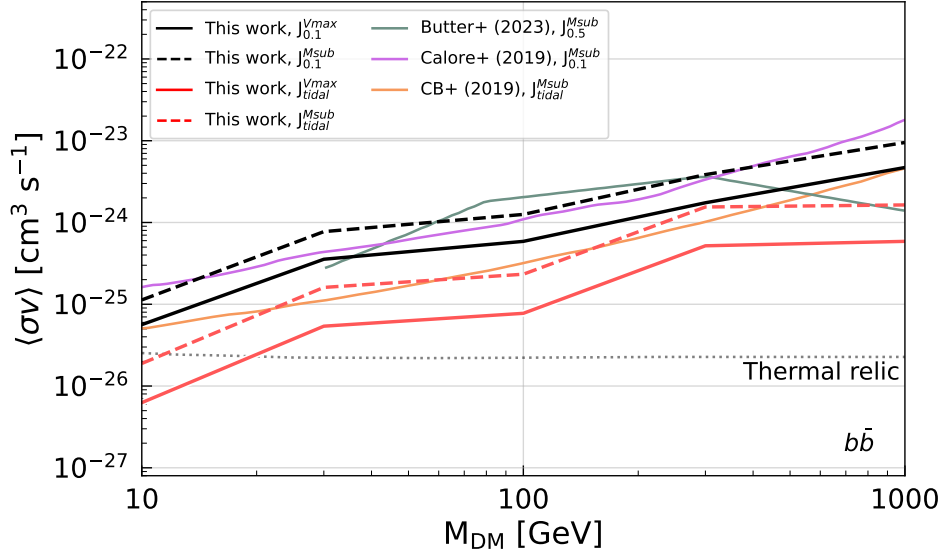


Fig. 5.13. Comparison of the 95% upper bounds on $\langle\sigma v\rangle$ derived in this work for different choices of the J-factor. Solid (dashed) black line: the limit derived with the V_{max} (M_{sub}) determination of the J-factor integrated within the radius of 0.1° . Solid (dashed) red line: the limit derived with the V_{max} (M_{sub}) determination of the J-factor integrated within the tidal radius. The uncertainty bands for the different J-factor models are proportional to the uncertainty band for our baseline model, $J_{0.1}^{V_{\text{max}}}$ (Figure 5.12), with the same scaling factors as the ratios of the corresponding upper limits. These uncertainty bands are not plotted for clarity of presentation.

due to statistical fluctuations, which yield smaller upper limits on the number of DM subhalos (cf. table 5.3).

The solid gray line shows the limits derived in ref. [294] for the J-factor integrated up to 0.5° radius. These limits are a factor of a few less constraining than those in the $J_{\text{tidal}}^{M_{\text{sub}}}$ model (dashed red line). This can be understood from the J-factor scaling: in this model $dN/dJ \propto J^{-2}$ (red solid line in figure 5.5), so the total number of subhalos above a threshold J is $N_{\text{sub}} \propto J^{-1}$ (for the $J_{0.1}^{M_{\text{sub}}}$ case, $dN/dJ \propto J^{-3}$ and $N_{\text{sub}} \propto J^{-2}$). Since $F \propto \langle\sigma v\rangle J$, the limit scales as $\langle\sigma v\rangle \propto N_{\text{sub}}$ (or $\langle\sigma v\rangle \propto \sqrt{N_{\text{sub}}}$ in the $J_{0.1}^{M_{\text{sub}}}$ case). Two factors account for the weaker constraints in ref. [294]: the estimated number of subhalos (the 0.8 threshold case for $|b| > 10^\circ$) is 3 to 10 times larger than in our model (cf. the upper bound on $N_{\text{subhalo}}^{\text{total}}$ in table 5.3), and the integration radius of 0.5° is smaller than the angular extent of most DM subhalos observable from Earth [271, 276]. Both effects lead to weaker constraints on the annihilation cross section compared to the $J_{\text{tidal}}^{M_{\text{sub}}}$ case. An exception occurs at $M_{\text{DM}} = 1$ TeV, where the limit in ref. [294] is a factor of 2.5 more constraining than at $M_{\text{DM}} = 300$ GeV, while the expected number of DM subhalos decreases by only a factor of 2. This is surprising, since for large DM masses the limit on $\langle\sigma v\rangle$ at a fixed number of subhalos should scale proportionally to M_{DM} , as is approximately the case for Coronado-Blázquez et al. [276] (solid orange line) and Calore et al. [277] (solid purple line). More generally, in the $J_{\text{tidal}}^{M_{\text{sub}}}$ ($J_{0.1}^{M_{\text{sub}}}$) model we expect the limit to scale as $\langle\sigma v\rangle \propto N_{\text{sub}} M_{\text{DM}}$ ($\langle\sigma v\rangle \propto \sqrt{N_{\text{sub}}} M_{\text{DM}}$). This scaling is approximately satisfied for the last two rows in table 5.3 (the $J_{\text{tidal}}^{M_{\text{sub}}}$ case): $10/3 \cdot 6.63/31. \approx 0.69$, to be compared with $1.55/1.64 \approx 0.95$, and in table 5.2 (the $J_{0.1}^{M_{\text{sub}}}$ case): $10/3 \cdot \sqrt{8.61/21.46} \approx 2.1$, to be compared with $4.7/1.7 \approx 2.8$. Applying this scaling to estimate the $\langle\sigma v\rangle$ limit of ref. [294] at

$M_{\text{DM}} = 1$ TeV from the value at $M_{\text{DM}} = 300$ GeV, the extrapolated limit would be a factor of 4 to 6 less constraining than our red dash-dotted line at $M_{\text{DM}} = 1$ TeV. For the $J_{0.1}^{M_{\text{sub}}}$ case, the larger number of subhalos and the larger integration radius in ref. [294] have opposite effects on the annihilation cross-section limits. At certain DM masses these two effects compensate each other, resulting in similar limits on the annihilation cross section at $M_{\text{DM}} = 100$ or 300 GeV (gray solid line vs. black dashed line in Figure 5.13).

The limits presented in figures 5.12 and 5.13 are derived assuming the velocity-independent DM annihilation cross section. In case of velocity-dependent annihilation cross section, e.g., due to Sommerfeld enhancement [79, 315–317] the relative contribution of small subhalos close to the Earth, which have a smaller velocity dispersion than the dwarf galaxies, will be more important. In particular, a significant population of subhalos at distances of a few kpc with typical velocity dispersion of a few km/s is expected [271]. These subhalos can have a factor of a few larger enhancement of annihilation cross section compared to the dwarf galaxies, where the typical velocity dispersion has a peak in the distribution around 10 km/s (e.g., [318]). The actual enhancement of the limits from subhalos compared to the dwarf galaxies depends on the details of the model (e.g., there should be no saturation of enhancement down to velocities of a few km/s), on the integral of the velocity dispersion for the subhalos, and on the relative contribution of small subhalos in the vicinity of the Earth and large distant subhalos. This detailed analysis is deferred to a future work.

5.4.11 Discussion and conclusions

In this work, we have searched for DM annihilating subhalos among the unassociated *Fermi*-LAT sources. We developed the first fully probabilistic model of gamma-ray sources, incorporating both known astrophysical source classes and a potential new population of gamma-ray sources modeled as DM subhalos annihilating into Standard Model particles.

We characterize gamma-ray sources using their energy spectra fitted by the log-parabola function. We take the parameters $\mathbf{x}$ of the log-parabola fit as the input features. We then construct a probabilistic model $p(\mathbf{x})$ for the unassociated sources as a weighted mixture of three source classes: Galactic and extragalactic astrophysical sources plus the hypothetical DM subhalos. The Galactic and extragalactic components encompass various source types classified by the *Fermi*-LAT collaboration. We analyze data shifts between associated and unassociated sources within a general framework of quantification learning, considering two effects: (1) a covariate shift affecting all source classes in the same way, and (2) a prior probability shift, which allows for changes in class prevalence, including the possible presence of a new source class among the unassociated sources.

The astrophysical components of the mixture model are the conditional PDFs $p(\mathbf{x}|k)$ for the two astrophysical classes k , Galactic and extragalactic. They are built from the corresponding empirical distributions of the associated sources, modified by an optimizable covariate-shift function. The DM subhalo component of the mixture is determined by Monte Carlo simulations based on a set of DM parameters (mass, annihilation channel, and cross section), including the distribution of J-factors coming from N-body simulations. The global likelihood of the model is obtained by multiplying the PDFs $p(\mathbf{x}_i)$ of the unassociated sources i with $\mathbf{x}_i$ inside the range of parameters that we consider.

By optimizing the model likelihood, we extract the best-fit parameters, including $\langle\sigma v\rangle$, for various DM masses in the $b\bar{b}$ annihilation channel. No significant excess corresponding to DM subhalos is observed beyond Galactic and extragalactic sources. Consequently, we derive 95% upper bounds on the DM annihilation cross section. To our knowledge, this is the first time

an upper bound on DM annihilation has been obtained by constraining the contribution of DM subhalos to unassociated *Fermi*-LAT sources using a maximum likelihood approach. Our constraints are competitive with previous limits derived from DM subhalo searches using classify-and-count methods. With respect to the latter, our approach, based on quantification learning, offers several advantages :

- (i) It accounts for differences in distributions (data shifts) between associated and unassociated source populations.
- (ii) It determines the potential contribution of a new hypothetical source population, e.g., the DM annihilating subhalos, without imposing ad hoc assumptions on its fraction relative to known sources. This is a key advantage over traditional supervised machine learning classification.
- (iii) The probabilistic model enables one to generate mock data (samples of unassociated sources), which we use to validate our approach and estimate the uncertainties of the upper bounds on the DM annihilation cross section.
- (iv) The model provides both the PDF $p(\mathbf{x}|k)$ of observed source features $\mathbf{x}$ given a source class k (either astrophysical or DM), and the posterior probability $p(k|\mathbf{x})$ of a source belonging to a given class based on observed features.

Thus, our approach allows one not only to determine the conditional probabilities $p(k|\mathbf{x})$, which is the main objective of ML classification models, but also to estimate the overall likelihood of the model given the distribution of unassociated sources. This likelihood serves as a tool to search for, or constrain, a new class of sources absent from the training data, i.e., the associated *Fermi*-LAT sources.

This work bridges the gap between ML approaches for characterizing unassociated sources traditionally focused on classification and the standard astrophysical analysis using maximum likelihood models. As a result, we construct a model capable of both classifying sources and constraining a new source population. Our approach is broadly applicable to searches of anomalies in distributions for datasets exhibiting both covariate and prior shifts, where the target data is unlabeled.

5.A Simulation of gamma-ray signals from dark matter subhalos

In this appendix we provide details about the simulation of gamma-ray signal from DM subhalos. Eq. (5.4.11) describes the probability of observing a DM subhalo given a DM model. This equation includes three terms that need to be estimated through simulations: the number density $n(\phi_{\text{DM}})$ of DM subhalos as a function of expected flux (section 5.4.7), the probability $p_{\text{DM}}(\alpha, \beta | \phi_{\text{DM}}; \theta_{\text{DM}})$ of observing a source with the log-parabola spectrum parameters (α, β) given a DM model, which we obtain by computing and fitting simulations of DM point sources, and the efficiency $\varepsilon_{\text{eff}}(\phi_{\text{DM}})$ of the detection of a source, which we obtain from the same set of simulations by identifying the fraction of sources which can be detected by *Fermi*-LAT. We now describe how the DM distribution conditioned on the gamma-ray flux is derived and how we estimate the detection efficiency for DM subhalos.

We compute the DM distribution conditioned on the gamma-ray flux, through realistic photon-counts simulations of DM point sources. We decompose the gamma-ray sky into two

Healpix order	8
Weeks	9 – 795
Emin	100 MeV
Emax	1 TeV
Instrument Response Functions (IRFs)	P8R3_SOURCE_V3
EVCLASS	128 (Source)
EVTYPE	3 (Front+Back)
ZMAX	90° under 1 GeV, 105° otherwise
Galactic foreground model	gll_iem_v7.fits

Table 5.4. Fermi Science Tools settings used for the 15-year data set analysis.

components: a diffuse emission component (constituted by the diffuse Galactic emission and an isotropic background) and the signal from the DM point sources. We model the diffuse Galactic emission using the `gll_iem_v7` template provided by the *Fermi*-LAT collaboration as well as the corresponding isotropic background model. We integrate eq. (5.4.10) in 16 logarithmic energy bins between 100 MeV and 1 TeV to obtain the theoretical energy spectrum in units of the photon flux per energy bin. We specify the spectral energy distribution (SED), dN_γ/dE , using the recent estimations from CosmiXs [102].

Furthermore, we notice that the J-factor and annihilation cross-section term factor out of the integral into an overall constant that ultimately defines the total flux normalization:

$$\Phi_i = A \int_{E_{min,i}}^{E_{max,i}} \frac{dN_\gamma}{dE} dE. \quad (5.A.1)$$

For this reason, once we fix the SED for a specific DM annihilation channel and mass, the energy spectrum $\phi(E)$ is defined except for an energy-independent normalization term. By fixing the overall flux normalization, we obtain the expected gamma-ray flux due to a specific DM source. Experimental uncertainties in the reconstruction of energy and direction of gamma rays induce a spread in the distribution of photons associated with a point source. We simulate sources by convolving with the *Fermi*-LAT point spread function (PSF) and exposure using *Fermi* Science Tools (`fermitools`) version 2.2.0⁸.

In table 5.4 we summarize the analysis settings, which are similar to the standard event selection cuts used in the construction of gamma-ray source catalogs. Likewise, we convolve the diffuse Galactic foreground and isotropic emission with the PSF and exposure to obtain the diffuse foreground emission. For each simulation, we add the emission from 10,000 DM sources randomly and isotropically distributed in the sky together with the foreground emission. We chose this number of sources to roughly match the number of resolved sources in the 4FGL catalog and to have sufficient statistics for the DM sources, while avoiding excessively demanding simulations. We thus obtain a map for the expected photon counts due to a DM population with a specific mass, annihilation channel, and a common flux.

We employ each simulation (associated with a specific total gamma-ray flux for the DM source) to estimate $p_{\text{DM}}(\alpha, \beta | \phi_{\text{DM}}; \theta_{\text{DM}})$ as follows. First, we perform a Poisson realization of the expected photon counts map. Then we proceed with fitting the energy spectrum of each source with a log parabola function (with a fixed spatial position), yielding the DM distribution conditioned on the flux. Finally, we estimate the detection efficiency $\varepsilon_{\text{eff}}(\phi_{\text{DM}})$ by studying the TS of the DM sources for a given map. We define the TS of a source by comparing the overall

⁸<https://github.com/fermi-lat/Fermitools-conda/wiki>

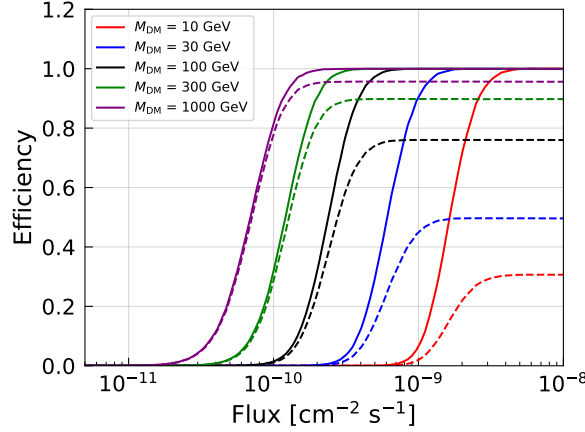


Fig. 5.14. Detection efficiency for the $b\bar{b}$ annihilation channel simulations. Solid lines denote the baseline efficiency used in the analysis, derived from simulations that disregard known 4FGL sources. Dashed lines illustrate a conservatively corrected efficiency, representing a worst-case scenario where a disk around each 4FGL source is masked to estimate the maximum potential degradation caused by their inclusion.

likelihood with and without the target source. We define the TS as:

$$TS = -2 \ln \left(\frac{\mathcal{L}}{\mathcal{L}_0} \right), \quad (5.A.2)$$

where $\mathcal{L}$ is the likelihood of the best fit model including the source, and $\mathcal{L}_0$ is the likelihood of the model without the source. If the TS of a source is larger than 25 we include that source to a database of resolved sources, for which the SED fit parameters are reliable and can be used for the estimation of $p_{DM}(\alpha, \beta | \phi_{DM}; \theta_{DM})$. Else, we discard the source as a low TS source which would not be measured by the *Fermi* LAT.

The detection efficiency $\varepsilon_{\text{eff}}(\phi_{DM})$ of the *Fermi* LAT to a DM source with a specific flux and SED is then defined as the ratio between the number of detected sources and the overall number of sources included in the simulation:

$$\varepsilon_{\text{eff}} = \frac{N_{\text{sources}}(TS > 25)}{N_{\text{sources}}^{\text{tot}}}. \quad (5.A.3)$$

In order to avoid biasing the estimation of the DM distribution due to a low number of detectable sources or due to a very specific realization of the sky, we perform additional simulations of the gamma-ray sky until we detect a total of at least 20,000 DM sources or until 1,000 simulations have been performed. Once we have built a dataset of measurements for the energy spectrum of DM sources, we estimate the probability density function $p_{DM}(\alpha, \beta | \phi_{DM})$ by employing KDE, in analogy with what we did in section 5.4.4 for the astrophysical sources.

Due to the high number of DM sources, and the large number of simulations, we cannot employ the full detection and characterization pipeline which is usually adopted by the *Fermi*-LAT collaboration. We exploit the additional information available to us from simulations to perform a simplified analysis pipeline. The true energy spectrum of each DM source is fixed (and known) in our simulations, and so are the foreground components. The positions of the DM sources are known as well, and the sources are all point-like sources. For these reasons, we make the following assumptions in our analysis:

- We assume that the energy spectrum of each background source (i.e. any source different from the one we are fitting) is known and fixed to the theoretical one (while in general they are jointly optimized together with the target of the fit);
- We fix the diffuse Galactic and the isotropic background components;
- We assume that the localization of the source is known (which in this case is true, but not in the case of source detection on real data);
- We assume that there are no low TS background sources, as we know exactly what we put inside our simulation.
- We assume that contributions from known 4FGL sources are negligible when estimating the DM distribution. While these bright sources can reduce detection efficiency and should ideally be modeled jointly, we estimate the maximum degradation by masking a disk around each 4FGL source. This mask corresponds to the 68% containment angle of the PSF at the peak energy (E_{peak}) of the DM spectrum for each considered mass. Because the peak of the DM energy spectrum increases with the DM mass, the detection efficiency is affected more significantly at lower DM masses, as displayed in figure 5.14. Specifically, for $M_{DM} \geq 100$ GeV, the narrower PSF results in a masked sky fraction of 20% or less, deteriorating the bounds by approximately 15%. Conversely, for lower masses ($M_{DM} = 10 - 30$ GeV), the broader PSF yields a masked fraction of 50 – 70%, leading to a 35 – 60% degradation of the bounds. Except for the $M_{DM} = 10$ GeV case, this degradation remains within the bootstrap uncertainty band derived for the unmasked analysis. Nonetheless, we conservatively adjust the 95% upper confidence level in figure 5.12 to account for these systematic effects.

By making these assumptions, we fit each DM source individually, regardless of the others.

To perform a maximum likelihood estimation fit of the DM energy spectrum for each source, we define the likelihood as follows:

$$\mathcal{L} = \prod_{e=1}^{N_{bin}} \prod_{j \in \text{disc}_e} \mathcal{P}(k_{e,j}, \lambda_{e,j}(\mathbf{x})). \quad (5.A.4)$$

Specifically, for each energy bin, we identify a disc around the source corresponding to the 95% containment angle of the PSF with a minimal angle of 0.5° and a maximal angle of 5° . This ensures high enough statistics for the high-energy bins, and reduces the background noise for the low-energy bins, while leaving the intermediate-energy bins unaffected.

For each pixel of the disc, we compare the photon counts k_{ij} with the estimated photon counts $\lambda_{ij}(\mathbf{x})$ using the Poisson likelihood. The estimated photon counts are a sum of the Galactic foreground in the disc, the isotropic background, the emission of the other background (fixed) DM sources and the photon counts coming from the target source:

$$\lambda(\mathbf{x}) = \lambda_{gal} + \lambda_{iso} + \lambda_{S \neq S_i} + \lambda_{S_i}(\mathbf{x}). \quad (5.A.5)$$

Specifically, for a source i and pixels j :

$$\lambda_{S_i,e,j} = \left[\int_{E_{min}^e}^{E_{max}^e} \phi_{0,i} \frac{E}{E_0}^{-\alpha_i - \beta_i \log(E/E_0)} dE \right] \cdot \text{PSF}_{i,j}^e \cdot \text{exposure}_j^e \cdot \frac{4\pi}{12N_{side}^2}. \quad (5.A.6)$$

We repeat the fitting procedure for each DM source, at a fixed total flux, and we build the DM distribution $p_{DM}(\alpha, \beta | \phi) \equiv p_{DM}^{KDE}(\alpha, \beta | \phi)$ through the KDE.

5.B Details of model optimization with the EM algorithm

Maximum likelihood estimation of the parameters of a mixture model is a challenging task due to the interplay between the mixture weights and the distribution parameters. A standard iterative algorithm for this task is the Expectation Maximization (EM) algorithm [155, 156].

We introduce latent binary vectors $\mathbf{z}_i$, corresponding to each data point $\mathbf{x}_i$, where $\dim(\mathbf{z}_i) = K$, where K is the number of mixture components. These latent variables encode the membership of point $\mathbf{x}_i$ in the k -th component of the mixture, such that all elements of $\mathbf{z}_i$ are zero, except for $z_{ik} = 1$. In other words, $\mathbf{z}_i$ is a categorical random variable represented with the one-hot-encoding scheme. The coefficients π_k of the likelihood in eq. (5.4.9) are interpreted as the parameters determining the corresponding categorical *prior* distribution of $\mathbf{z}_i$, i.e. $p(z_{ik} = 1) = \pi_k \forall i$, such that

$$p(\mathbf{z}_i) = \prod_k^K \pi_k^{z_{ik}}. \quad (5.B.1)$$

With this interpretation for $\mathbf{z}_i$, we can identify $p(\mathbf{x}_i | z_{ik} = 1) = p_k(\mathbf{x}_i | \boldsymbol{\theta}_k)$, such that

$$p(\mathbf{x}_i | \mathbf{z}_i) = \prod_k^K p_k(\mathbf{x}_i | \boldsymbol{\theta}_k)^{z_{ik}}. \quad (5.B.2)$$

We then have all the elements to compute the *posterior* distribution of $\mathbf{z}_i$ as:

$$p(z_{ik} = 1 | \mathbf{x}_i) = \frac{p_k(\mathbf{x}_i | \boldsymbol{\theta}_k) \pi_k}{\sum_j^K p_j(\mathbf{x}_i | \boldsymbol{\theta}_j) \pi_j} \equiv \gamma(z_{ik}). \quad (5.B.3)$$

In order to find the optimum parameters $\{\boldsymbol{\theta}_k\}$ and weights $\boldsymbol{\pi}$, the idea is to consider the joint likelihood $p(\mathbf{x}_i, \mathbf{z}_i)$ for the whole dataset of points $\mathbf{x}_i$ with associated latent variables $\mathbf{z}_i$:

$$p(\mathbf{X}, \mathbf{Z} | \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}) = \prod_i^N \prod_k^K [\pi_k p_k(\mathbf{x}_i | \boldsymbol{\theta}_k)]^{z_{ik}}. \quad (5.B.4)$$

In spite of this apparent complication of introducing extra variables in the problem, i.e., the latent variables, the likelihood in eq. (5.B.4) is arguably easier to optimize than the original likelihood $p(\mathbf{x} | \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}) = \prod_i p(\mathbf{x}_i)$, where each term in the product is built as in eq. (5.4.9). One of the reasons for this is that, the log-likelihood of eq. (5.4.9) is a log of a sum, which is computationally challenging to optimize, especially in the large K regime, while the log of the likelihood in eq. 5.B.4 is the sum of logs, i.e., $\log p(\mathbf{X}, \mathbf{Z} | \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})$ is very similar to the cross-entropy cost function, which is easily optimizable.

An obvious problem is that in eq. (5.B.4) we do not know the z_{ik} variables. The EM algorithm considers instead the expectation value of this likelihood (or more concretely, of the log-likelihood), under the posterior distribution of $\mathbf{Z}$, i.e.

$$\mathbb{E}_{\mathbf{Z}}[\log p(\mathbf{X}, \mathbf{Z} | \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})] = \sum_{\mathbf{Z}} p(\mathbf{Z} | \mathbf{X}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}) \ln p(\mathbf{X}, \mathbf{Z} | \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}). \quad (5.B.5)$$

Eq. (5.B.5) is maximized according to the following iterative procedure:

1. Initialize $\{\boldsymbol{\theta}_k\} = \{\boldsymbol{\theta}_k\}_{\text{old}}$, and $\boldsymbol{\pi} = \boldsymbol{\pi}_{\text{old}}$;
2. Evaluate $p(\mathbf{Z} | \mathbf{X}, \{\boldsymbol{\theta}_k\}_{\text{old}}, \boldsymbol{\pi}_{\text{old}})$;

3. Plug the posterior obtained in step 2 into expr.(5.B.5), such that now the only term containing the optimizable parameters $\{\theta_k\}, \pi$ is the log-likelihood. So we get the expression:

$$Q(\{\theta_k\}, \pi) \equiv \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{x}, \{\theta_k\}_{\text{old}}, \pi_{\text{old}}) \ln p(\mathbf{X}, \mathbf{Z}|\{\theta_k\}, \pi). \quad (5.B.6)$$

4. Update the parameters as:

$$\{\theta_k\}_{\text{new}}, \pi_{\text{new}} = \underset{\{\theta_k\}, \pi}{\operatorname{argmax}} Q(\{\theta_k\}, \pi) \quad (5.B.7)$$

5. Repeat above steps until convergence

Note that in step 4 above, the optimization of π is analytical, i.e., using a Lagrange-multiplier optimization with the condition $\sum_j \pi_j = 1$, we obtain:

$$(\pi_k)_{\text{new}} = \frac{1}{N} \sum_i^N \gamma(z_{ik}), \quad (5.B.8)$$

where $\gamma(z_{ik})$ is given by eq. (5.B.3). The optimization of $\{\theta_k\}$ is in general not analytical.

5.C Consistency checks of the model

In this appendix we perform several checks of our model. At first, we investigate the relative importance of prior and covariate shifts by comparing models with only prior or only covariate shift to the full model that contains both prior and covariate shifts. We then proceed to perform consistency checks. We start by generating distributions of sources in parameter space from our best-fit model (including only astrophysical components) and fit the generated data with the same model. The goal is to determine a distribution of $\log \mathcal{L}$ values due to statistical fluctuations (since the data was generated from this model) and to compare to the value of $\log \mathcal{L}$ for the actual data to check if the model without a DM contribution is generally a good fit to the data. As a last consistency check, we perform injection tests, where we add a DM subhalo component to the astro-only model to check if we can recover the injected signal.

5.C.1 Importance of covariate and prior shifts

In order to investigate the importance of the prior and covariate shifts, we first compare the performance of four models in eq. (5.4.9) without the DM component: a model without either prior or covariate shifts (i.e., no free parameters) determined from associated sources, denoted as “as”, models with only prior or covariate shifts, denoted as “p” or “c” respectively, and a model with both prior and covariate shifts, denoted as “p + c”.

The model without either prior or covariate shifts is constructed from the catalog of associated sources following sections 5.4.3 and 5.4.4, where the prior probability of a class is given by the relative class prevalence π_k in the 4FGL-DR4 catalog, while the PDFs of the Galactic and extragalactic components are derived from the corresponding associated sources. The model with either prior or covariate shift follows eq. (5.4.9), where we fix the covariate shift modulation to be equal to 1 (i.e., $\tilde{C}(\mathbf{x}; \theta_{\text{cov}}) = 1$), or fix the prior probabilities to the corresponding values for associated sources.

We compare the performance of the four models evaluated on the unassociated sources in the ROI for $E_0 = 1$ GeV in table 5.5. In column 2, we show the difference in log likelihood between

model	$2\Delta\log\mathcal{L}_{as,x}$	$\Delta\text{dof}_{as,x}$	$N\sigma_{as,x}$	$\text{AIC}_{as,x}$	$2\Delta\log\mathcal{L}_{x,p+c}$	$\Delta\text{dof}_{x,p+c}$	$N\sigma_{x,p+c}$
associated	0	0	0.0	-178	64.7	7	6.7
prior	49.5	1	7.0	-225	14.2	6	2.2
covariate	33.4	6	4.5	-199	30.3	1	5.5
prior + covariate	64.7	7	6.7	-227	0.0	0	0.0

Table 5.5. Comparison of models of unassociated sources with Galactic and extragalactic components. “as” denotes the model derived from distributions of associated sources applied to the unassociated sources (without free parameters). “ $p + c$ ” is the full covariate + prior shift model. Subscript “ x ” denotes the model specified in the first column of the table. For example, the value of $2\Delta\log\mathcal{L}_{as,x} = 33.4$ in the fourth row and second column corresponds to the difference between the model derived for associated sources and the model with only covariate shift, i.e., to $x = c$ (covariate) in this case.

the model specified in the first column and the model based on distributions of associated sources. We report the corresponding additional degrees of freedom in column 3. We find that the prior shift model leads to a larger increase in likelihood for one degree of freedom (we have a constraint that the sum of priors is equal to 1: $\pi_{\text{gal}} + \pi_{\text{ext}} = 1$) compared to a model with covariate shift only. The corresponding significance of exclusion of the null hypothesis (the “as” model) is presented in column 4. It is derived by assuming the χ^2 distribution for $-2\Delta\log\mathcal{L}$ with the corresponding number of degrees of freedom from column 3. In column 5 we show the Akaike information criterion, $\text{AIC} = -2\log\mathcal{L} + 2k$, where k is the number of degrees of freedom. We find that although log likelihood of the full prior + covariate shift model is the largest, both the significance of exclusion of the null hypothesis and the AIC for this model are comparable to the ones for the prior shift only model. Thus, we conclude that the prior shift is the dominant effect compared to the covariate shift. We further compare the models in column 1 to the full prior + covariate shift model in columns 6, 7, and 8, where we show $2\Delta\log\mathcal{L}$, the additional degrees of freedom, and the significance of exclusion of the null hypothesis, which, in this case, is the model in the first column. In particular, we find that the prior shift model can be excluded only at 2.2 sigma level in favor of the “ $p + c$ ” model.

In figure 5.15, we compare our models with the distributions of parameters α , β , and $\log_{10}(\phi)$ for the unassociated sources. We display the best-fit “ $p + c$ ” model (in red) and compare it with the prior KDE model derived for associated sources (in blue). The “ $p + c$ ” model provides a reasonable fit to the data for all the distribution parameters. On the other hand, the model derived for associated sources fails to capture the β dependence correctly, as expected, since the distribution for unassociated sources differs considerably from the distribution of associated sources for the β parameter (cf. figure 5.3).

5.C.2 Bootstrap MC and overall goodness of fit

We quantitatively estimate the goodness of fit of the “ $p + c$ ” model in figure 5.16. In this figure, we generate samples of sources in the parameter space from the best-fit “ $p + c$ ” model. We then reoptimize the model for each of the samples. The corresponding distribution of the log likelihood values has only statistical fluctuations. The red lines on these plots show the best-fit $\log\mathcal{L}$ value for the real data. These values are about 2 sigma smaller than the mean of the $\log\mathcal{L}$ distribution both for $E_0 = 100$ MeV (left panel) and for $E_0 = 1$ GeV (right panel). On the one hand, this deviation shows that the model is not perfect, i.e., there are possibly some deviations of the model and the data. On the other hand, the deviation has only about 2 sigma statistical significance. This deviation may be connected to the slight excess observed, e.g., in the $M_{\text{DM}} = 30$ GeV DM

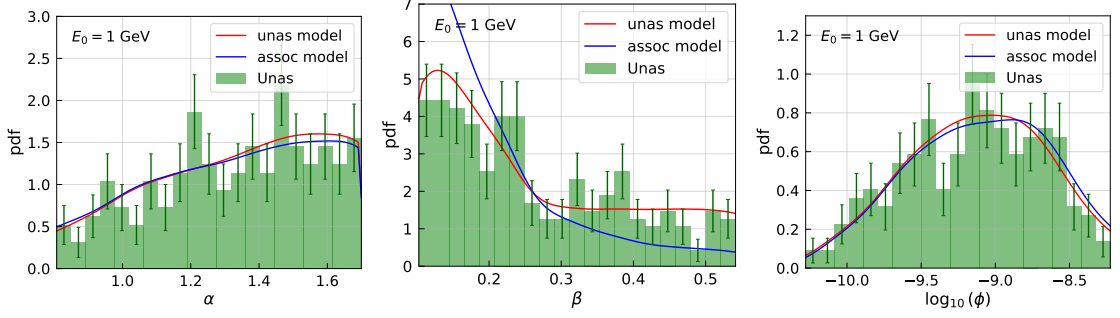


Fig. 5.15. Distributions of the α , β and $\log_{10}(\phi)$ parameters for the unassociated sources in the 4FGL-DR4 catalog. The bars show statistical uncertainty. The best-fit model that includes both covariate and prior shifts is shown by the red line while the model derived for associated sources is shown by the blue line. The 3D model is integrated in two variables to obtain the distribution in the third variable, shown on the plot.

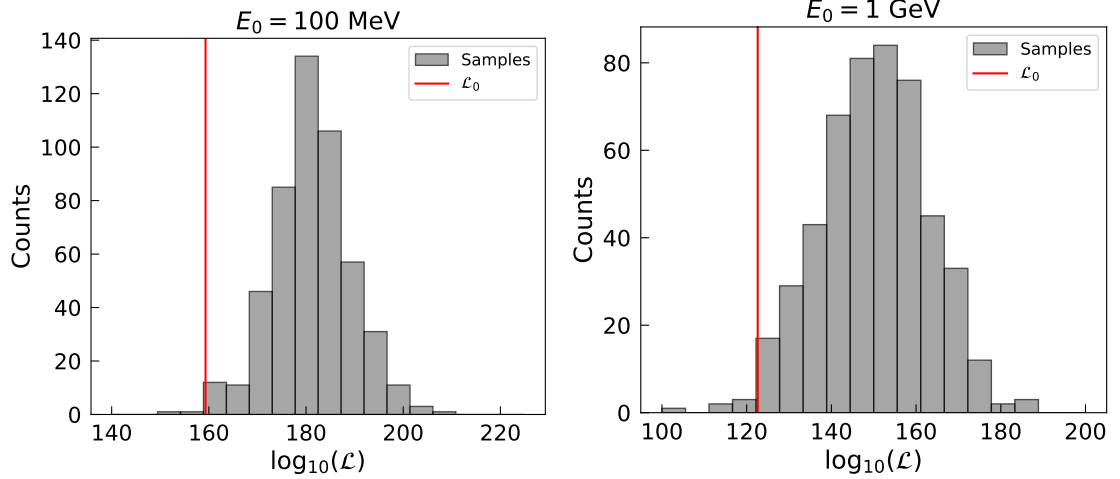


Fig. 5.16. Distribution of log-likelihood for the “ $p + c$ ” model with astro-only components for 1000 datasets simulated from the best-fit “ $p + c$ ” model for $E_0 = 100 \text{ MeV}$ (*left panel*) and $E_0 = 1 \text{ GeV}$ (*right panel*). The red line represents the log-likelihood of the “ $p + c$ ” model with astro-only components for the unassociated 4FGL-DR4 sources.

annihilation (cf. the $M_{\text{DM}} = 30 \text{ GeV}$ row in Table 5.2). Indeed, if there is an excess at about 2σ significance, which is not accounted by the model, then the model would be excluded at 2σ significance compared to a model that can explain the data without excesses, e.g., the data generated from the model itself (cf. figure 5.16).

5.C.3 DM subhalo injection test

In order to ensure our model is actually capable of reconstructing a DM signal, we inject a DM signal into simulated data and test the ability of our model to reconstruct it. We choose a significant value of $\langle\sigma v\rangle$ for each DM mass. For $M_{\text{DM}} = 30, 100, 300 \text{ GeV}$, we adopt the best fit $\langle\sigma v\rangle$ value from table 5.2, while for $M_{\text{DM}} = 10, 1000 \text{ GeV}$, since the best-fit value is zero, we inject a signal with a $\langle\sigma v\rangle$ corresponding to the 3σ significance (which should be clearly distinguishable

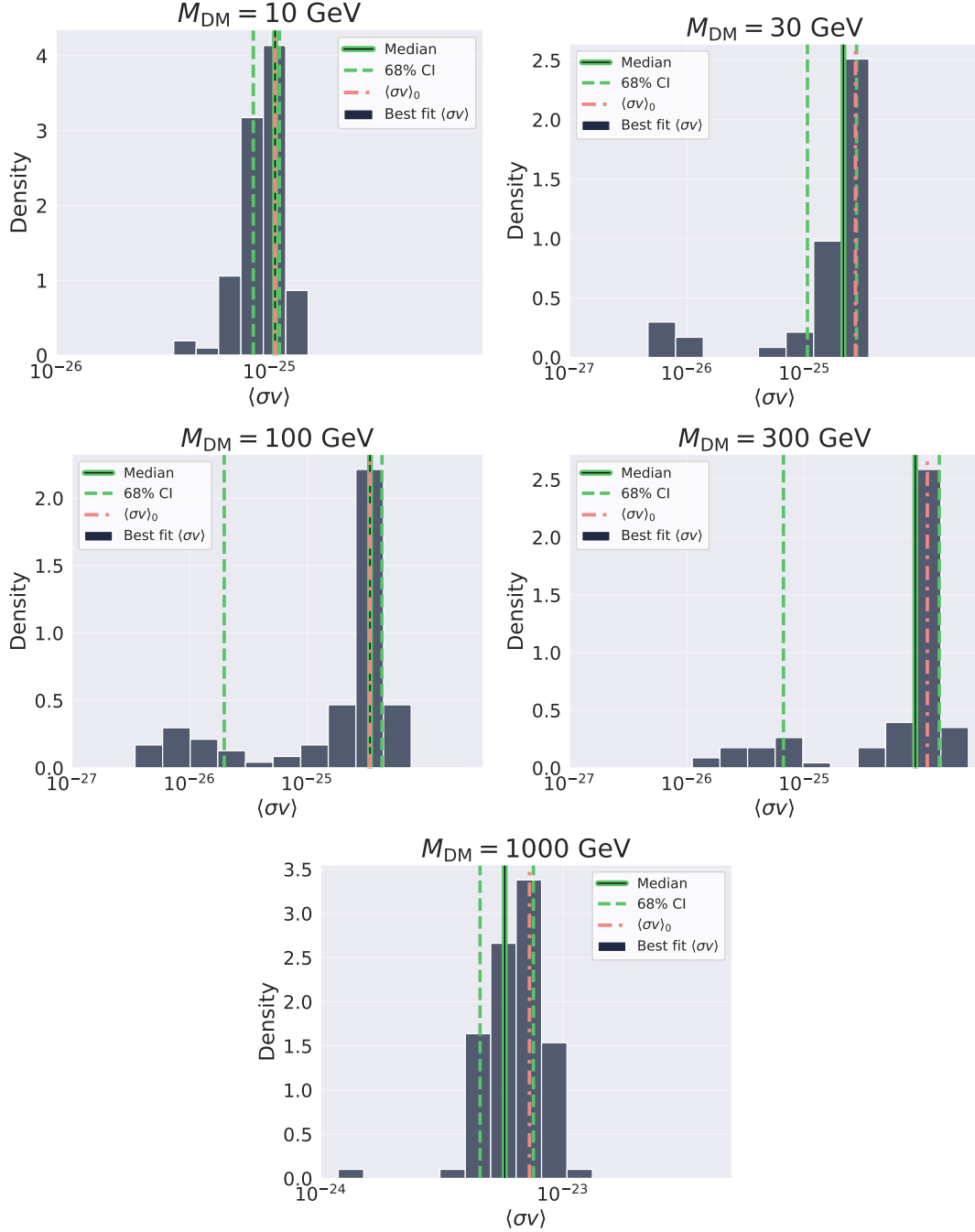


Fig. 5.17. Histograms representing the reconstructed $\langle\sigma v\rangle$ values for simulations with injected DM signals. The red dash-dotted line represents the annihilation cross-section of the injected signal ($\langle\sigma v\rangle_0$). The green solid line represents the median of the distribution, while the green dashed lines show the 68% containment interval. In every case, the true $\langle\sigma v\rangle$ value lies inside the 68% containment region.

from the null hypothesis).

We show the histogram of the recovered $\langle\sigma v\rangle$ values for the 5 DM masses in figure 5.17.

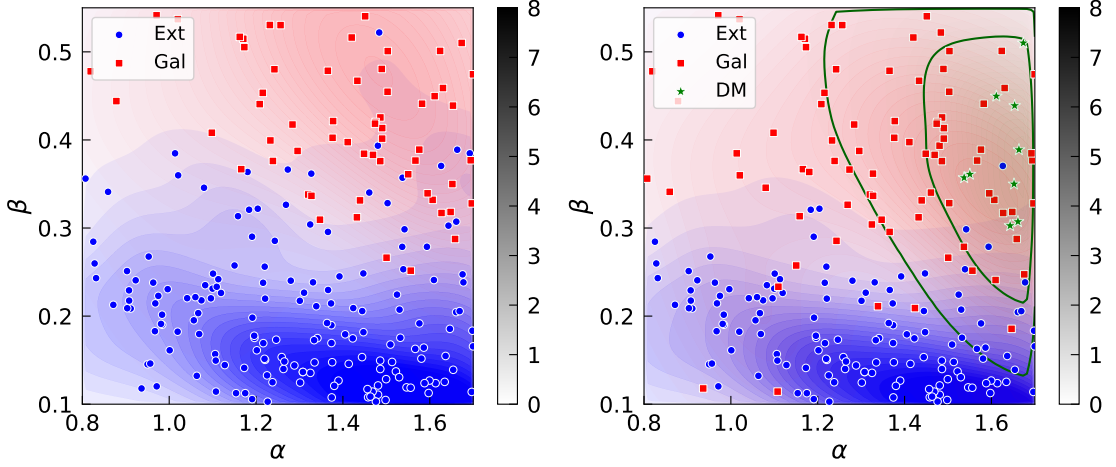


Fig. 5.18. Predicted source classes for unassociated 4FGL-DR4 sources according to the posterior class probabilities for the $E_0 = 1 \text{ GeV}$ case. *Left panel:* best-fit model with astrophysical only classes. *Right panel:* best-fit model for astrophysical classes and DM ($M_{\text{DM}} = 30 \text{ GeV}$, $\langle\sigma v\rangle \sim 2.7 \cdot 10^{-25} \text{ cm}^3 \text{ s}^{-1}$). The green contours represent the 68% and 95% containment levels for the DM distribution.

The $\langle\sigma v\rangle$ values for the injected DM signal (red dashed-dotted lines, $\langle\sigma v\rangle_0$) are within the 68% containment interval (green dashed lines) of the histogram relative to the median values for the distributions (solid green lines). Thus, our methodology is capable of reconstructing a simulated signal, when present, in the majority of cases.

Finally, we illustrate the model's predictive capabilities. Using eq. (5.B.3), we derive the posterior probability that each unassociated source belongs to a given class k (e.g., extragalactic, Galactic, or DM). The predicted class label, C_i^{pred} , is then assigned to the class with the highest posterior probability: $C_i^{\text{pred}} = \text{argmax}_k \{\tilde{p}(z_{ik} = 1 | \mathbf{x}_i)\}$. Figure 5.18 displays the posterior probabilities and predicted classes for all unassociated sources. The left panel shows results from our best-fit model with only astrophysical classes (for $E_0 = 1 \text{ GeV}$), while the right panel shows the best-fit model including DM (for the $M_{\text{DM}} = 30 \text{ GeV}$ case).

For comparison, we repeat this exercise on a simulated bootstrap sample for the $M_{\text{DM}} = 30 \text{ GeV}$ case in figure 5.19. The figure compares the true class labels (left panel) with the posterior class probabilities (middle panel) and the final predicted class labels (right panel) for a model fit to the bootstrap sample. For instance, while not all DM sources were correctly classified due to significant parameter overlap with the Galactic source population, their true class often received a high posterior probability, even if it was not the maximum. This shows that to avoid false negatives when identifying a subdominant class, it is important to consider sources with a reasonably high posterior probability for that class even if this probability is not the highest.

5.C.4 Precision-recall curves for classification of individual DM subhalos

To further quantify the model's ability to distinguish DM subhalos from the astrophysical background, we perform a Precision-Recall (PR) analysis for the case of $M_{\text{DM}} = 30 \text{ GeV}$. This DM mass yields the highest possible signal due to DM annihilation in subhalos (cf. table 5.2) but it is also challenging due to the extensive spectral overlap with Galactic sources. The PR framework is more relevant here than the Receiver Operating Characteristic curves for highly

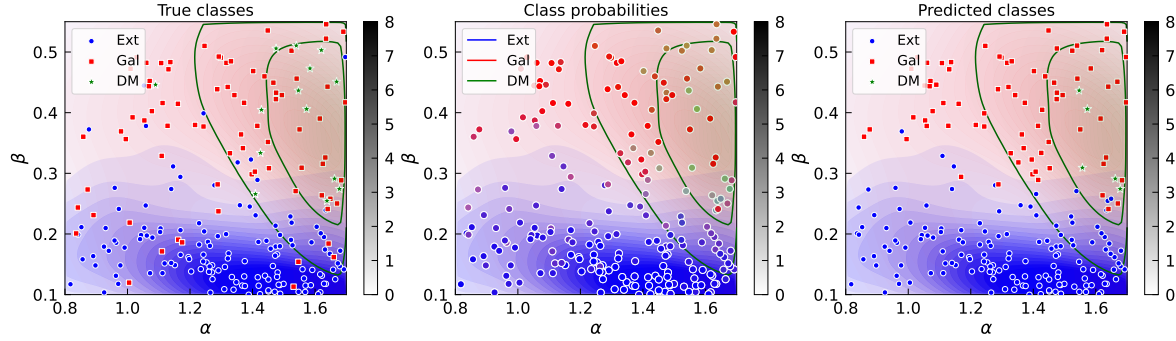


Fig. 5.19. Estimated best-fit distributions and true or predicted source classes for a bootstrap simulation, with $M_{\text{DM}} = 30$ GeV and $\langle\sigma v\rangle \sim 2.7 \times 10^{-25} \text{ cm}^3 \text{ s}^{-1}$. *Left panel:* markers colored according to their true class C_i^{true} in the simulation. *Middle panel:* markers colored according to their posterior class probability following the same convention as figure 5.10. *Right panel:* markers colored according to the predicted class, given by $C_i^{\text{pred}} = \text{argmax}_k \{\hat{p}(z_{ik} = 1 | \mathbf{x}_i)\}$.

unbalanced datasets, as it focuses exclusively on the minority class (the signal), providing a rigorous assessment of how many “DM-like” candidates are likely to be actual subhalos without being skewed by the overwhelming abundance of background sources.

We consider two physical scenarios for the annihilation cross-section: the best-fit value obtained in our primary analysis ($\langle\sigma v\rangle = 2.7 \times 10^{-25} \text{ cm}^3/\text{s}$) and a value close to the 95% upper bound ($\langle\sigma v\rangle = 3.6 \times 10^{-25} \text{ cm}^3/\text{s}$). For each scenario, we evaluate the PR performance across a suite of 1000 independent simulations. In figure 5.20, we display the mean PR curve along with the 68.27% asymmetric confidence band derived from the simulation quantiles, representing the expected statistical variance due to background realizations and Poisson noise.

To quantify performance, we compute the Average Precision (AP):

$$AP = \sum_n (R_n - R_{n-1}) P_n \quad (5.C.1)$$

where P_n and R_n are the precision and recall at the n -th classification threshold. We find $AP = 0.40 \pm 0.16$ for the best-fit case and $AP = 0.50 \pm 0.14$ for the upper bound. These scores significantly outperform their respective random chance baselines of 0.03 and 0.06. The fact that the AP remains below unity is a direct reflection of the intrinsic physical confusion between DM subhalos annihilating in the $b\bar{b}$ channel for $M_{\text{DM}} = 30$ GeV and Galactic millisecond pulsars.

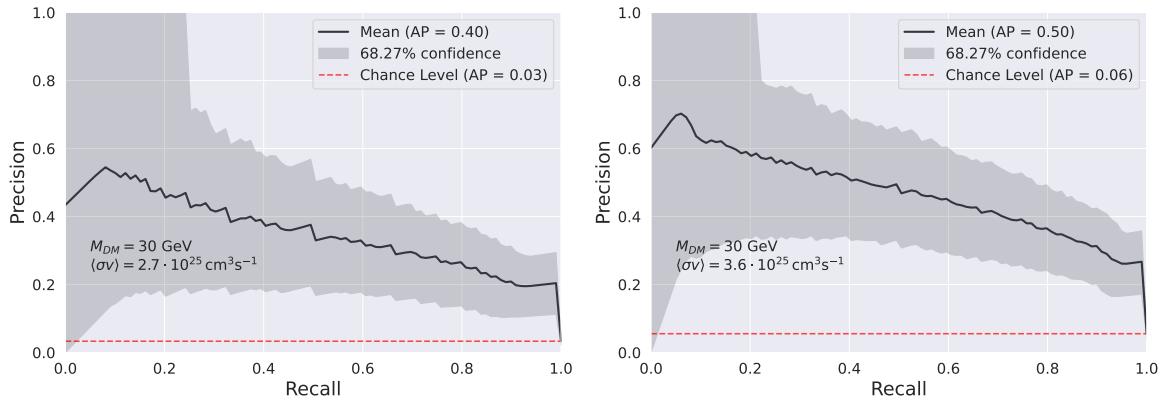


Fig. 5.20. Precision and recall for classification of individual DM subhalos among unassociated *Fermi*-LAT sources. Solid black line (gray band) shows the mean (68.27% statistical containment) for 1000 simulations at $M_{DM} = 30 \text{ GeV}$. *Left panel*: performance for the best-fit cross-section in table 5.2 (average precision $AP = 0.40 \pm 0.16$). *Right panel*: performance for a cross-section close to the 95% confidence upper bound in table 5.2 ($AP = 0.50 \pm 0.14$).

Part III

The Unresolved Gamma-Ray Background

EXTRACTING THE GAMMA-RAY SOURCE-COUNT DISTRIBUTION WITH DEEP LEARNING

Part II examined dark matter annihilation signatures in specific, resolved astrophysical structures. Both strategies we explored encountered fundamental limitations. The Galactic Center presents the strongest expected signal, but it remains inextricably entangled with complex diffuse foregrounds and unresolved astrophysical sources. Searching for individual dark matter subhalos among unassociated gamma-ray point sources offers a much cleaner environment, yet is severely limited by the finite sensitivity of the instrument. The inherent ambiguity of the Galactic Center Excess and the rarity of detectable subhalos expose the boundaries of source-by-source identification.

These limitations motivate a shift in strategy. Rather than interrogating sources one at a time, we ask what the faint population as a whole imprints on the photon-count statistics of the gamma-ray sky. Section 6.1.2 develops this population-level perspective and identifies its central observable, the source-count distribution dN/dS .

This chapter introduces the framework required to probe the unresolved sky and presents a deep learning approach to measuring the gamma-ray source count distribution dN/dS . Section 6.1 builds the conceptual argument for transitioning to population studies. Section 6.2 formalizes the source-count distribution as the primary observable (cf. Section 2.2.2 for the astrophysical context) and reviews analytical approaches to measuring it. Section 6.3 details the limitations of these analytical formulations and introduces simulation-based inference (cf. Section 3.2) via convolutional neural networks on spherical maps as an alternative that sidesteps these limitations. Finally, Section 6.4 frames the analysis that forms the core of this chapter, where we validate our methodology by recovering the extragalactic dN/dS down to a factor of 50 below the *Fermi*-LAT threshold.

The results of this chapter feed into two downstream analyses. The recovered source-count distribution directly informs the cataloging framework of Chapter 7. The population-level perspective established here also provides the methodological foundation for the cosmological dark matter searches pursued through cross-correlations in Chapter 8.

6.1 Limits of Individual-Source Methods

The preceding chapters searched for dark matter in individual, resolved gamma-ray targets. We now step back and assess the broader limitations of that paradigm, before motivating the transition to population-level methods that forms the core of Part III.

6.1.1 The Resolved-Source Paradigm and Its Constraints

The Galactic Center Excess, first reported by Goodenough and Hooper [126] and subsequently confirmed by independent analyses [80, 127, 210], remains the most prominent anomaly in *Fermi*-LAT data. As discussed in Chapter 4 (Section 4.1), disentangling a potential dark matter contribution from astrophysical explanations has proven so far intractable. The difficulty is not a lack of signal strength but an excess of competing models: the morphology of the excess is equally well described by an NFW-squared dark matter profile and by a population of unresolved millisecond pulsars tracing the stellar bulge [212, 213]. The interstellar emission models introduce further degeneracies, as different templates for the Galactic diffuse emission shift the inferred GCE spectrum and morphology [210]. Even the non-Poissonian template fitting technique, initially presented as a way to distinguish point-source from diffuse origins, was shown to be sensitive to mismodeling artifacts [216, 236]. More recently, simulation-based inference approaches incorporating energy information have reopened the question. As discussed in Section 4.3.3, the CNN-based analysis of List et al. [221] pushes the inferred source-count distribution to fluxes so faint that the GCE becomes statistically indistinguishable from the smooth, Poissonian emission expected from dark matter annihilation. The origin of the excess thus remains an open problem, with neither the dark matter nor the millisecond pulsar hypothesis decisively favored by current data. The Galactic Center therefore illustrates a general principle: when the target region contains multiple bright, partially degenerate emission components, attributing any residual to a specific physical origin becomes a problem of systematic uncertainties rather than statistical sensitivity.

Chapter 5 pursued an alternative strategy, searching for dark matter subhalos among the 2428 unassociated point sources in the 4FGL-DR4 catalog [141, 153] at Galactic latitude $|b| > 10^\circ$ (Section 5.3). The environment is far cleaner: high-latitude sources are not contaminated by the Galactic diffuse emission that plagues the GCE. However, our quantification analysis found no statistically significant dark matter subhalo component (Section 5.4.8). This null result reflects two structural constraints. First, only the most massive and nearby subhalos produce fluxes above the *Fermi*-LAT catalog threshold of $TS > 25$; the vast majority of the predicted subhalo population [113] lies well below it. Second, the look-elsewhere penalty is severe: any individual candidate extracted from a catalog of thousands of unassociated sources requires evidence well beyond the 5σ threshold conventionally demanded for discovery.

These two case studies expose a shared limitation. Both the GCE analysis and the subhalo search operate on sources that have already been detected or on emission that is already visible. They are structurally blind to the population of sources below the detection threshold. Yet this sub-threshold population is far from negligible: the very existence of the isotropic diffuse gamma-ray background [138] implies a substantial flux contribution from sources too faint to be individually resolved.

6.1.2 The Population-Level Alternative

The limitations of individual source identification motivate a conceptual shift. Rather than asking whether any specific source is a dark matter subhalo, we can ask a different question: what is the

statistical distribution of all sources, including those too faint to detect individually?

Below the formal *Fermi*-LAT detection threshold lies a vast population of unresolved sources. Their cumulative emission forms a significant fraction of the unresolved gamma-ray background (UGRB) [131, 138]. While these sources are invisible individually, their collective photon statistics encode recoverable information about the underlying population. This population-level approach complements the strategies explored in Part II: resolved-source methods probe the bright tail of the population, whereas statistical methods probe the faint end. Together, they provide a complete census.

Chapters 6 and 7 focus on characterizing this general unresolved source population — primarily astrophysical objects such as blazars [144, 319], millisecond pulsars [320], and star-forming galaxies [321]. A thorough census of the astrophysical population is a prerequisite for any dedicated dark matter search: only after accounting for the known source classes can a residual exotic component be meaningfully constrained. The search for a collective dark matter signal in the unresolved gamma-ray sky is deferred to Chapter 8, which employs angular cross-correlations with tracers of large-scale structure, a technique complementary to the population statistics developed here.

This sub-threshold population is characterized by the source-count distribution dN/dS , which we define and whose measurement we review in the following section.

6.2 The Source-Count Distribution

The analysis of the unresolved gamma-ray sky begins with a single observable: the differential source-count distribution, dN/dS . This function describes how many point sources exist at each flux level S and provides the most direct statistical characterization of the populations that contribute to the unresolved gamma-ray background. Its reconstruction below the telescope detection threshold — where individual sources can no longer be identified — is the central inference problem addressed in this chapter.

This section introduces the dN/dS and the methodological context necessary to understand the approach developed in Section 6.3. We begin with the formal definition and its connection to the underlying luminosity function (Section 6.2.1), then review the 1-point photon-count distribution, the principal analytical method used to extract dN/dS from sub-threshold data, and discuss how measurements of the total source-count distribution constrain the composition of the unresolved gamma-ray sky (Section 6.2.2).

6.2.1 Definition and Physical Meaning

The differential source-count distribution, dN/dS , gives the number of sources per unit solid angle per unit flux at a given integral photon flux S in a specified energy band. It is the fundamental observable that describes how point sources are distributed as a function of their apparent brightness, and it connects directly to the physics of the emitting populations.

For sources bright enough to be individually cataloged, dN/dS is measured by simply counting objects in the catalog as a function of their flux (see also Section 2.2). The 4FGL-DR3 catalog [141], based on 12 years of *Fermi*-LAT data, contains 6658 resolved sources and provides a direct measurement of the source-count distribution in the resolved regime [3]. Above the flux threshold where the catalog detection efficiency reaches 100%, this measurement is straightforward. As we approach this threshold, the detection efficiency degrades, and direct counting must be supplemented by Monte Carlo corrections to reconstruct the true underlying distribution. Further below — in the unresolved regime, where individual sources are too faint

to be detected at all — the sources merge into the UGRB (Section 6.1.2), and dN/dS must be inferred statistically from the photon-count map.

The source-count distribution is the observational projection of a more fundamental quantity: the luminosity function. The gamma-ray luminosity function gives the comoving number density of sources per unit luminosity L_γ as a function of redshift z . The connection between the two is given by [131]

$$\frac{dN}{dS} = \int_{z_{\min}}^{z_{\max}} dz \int_{\Gamma_{\min}}^{\Gamma_{\max}} d\Gamma \frac{dN}{dV dL_\gamma d\Gamma}(z, L_\gamma, \Gamma) \frac{dV}{dz d\Omega} \frac{dL_\gamma}{dS}, \quad (6.2.1)$$

where Γ is a parameter that characterizes the energy spectrum shape of the source, $dV/(dz d\Omega)$ is the comoving volume element per unit redshift and solid angle, and dL_γ/dS is the Jacobian relating luminosity to observed flux through the luminosity distance $D_L(z)$. The integrand $dN/(dV dL_\gamma d\Gamma)$ — the comoving density of sources per unit luminosity and spectral index — is the luminosity function itself, which encodes the intrinsic abundance of sources at each cosmic epoch.

This integral makes precise the physical content of dN/dS : it takes the intrinsic distribution of sources in luminosity and redshift, projects it through the flux–distance relation, and integrates over all redshifts and spectral shapes to produce the observed flux distribution. In other words, dN/dS can be interpreted as the integral along the line of sight of the gamma-ray luminosity function. It is what we observe when we look at the sky, while the luminosity function is the astrophysical quantity we ultimately want to constrain.

The source-count distribution therefore bridges two observational regimes that meet near the catalog detection threshold $S_{\text{th}} \sim 2 \times 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$ (1–10 GeV band): above it, the catalog measurement follows a flux distribution broadly compatible with an S^{-2} power law over most of the resolved range, shallower than the Euclidean expectation $S^{-5/2}$ [322, 323].¹ Below it, dN/dS can only be recovered statistically. The techniques for extending the measurement well below the threshold are the subject of the following subsections.

6.2.2 Extracting the Source-Count Distribution from Photon Statistics

Recovering the source-count distribution below the detection threshold requires extracting population information from the collective statistical imprint of unresolved sources on the photon-count map. The photon-count distribution — also called the 1-point probability distribution function (1pPDF) — provides a well-established method for doing so.

The idea is straightforward: given a pixelated sky map, one constructs the histogram of the number of pixels containing exactly k photons. Different source populations leave distinct imprints on this histogram [131]. Bright but rare sources, such as blazars just below the detection limit, produce multiple photons during the observation period; because these photons are spatially correlated, they populate pixels with moderately large photon counts, generating non-Poissonian tails in the distribution. By contrast, faint but numerous sources, such as star-forming galaxies,

¹The Euclidean benchmark follows from a simple counting argument. For sources of luminosity L distributed with uniform density in static Euclidean space, the inverse-square law $S = L/4\pi r^2$ implies that sources brighter than S lie within a distance $r \propto S^{-1/2}$. Their number scales with the enclosed volume, $N(>S) \propto r^3 \propto S^{-3/2}$, giving $dN/dS \propto S^{-5/2}$ regardless of the luminosity function [322]. The measured counts seem to follow this slope only at the brightest end, and flatten to S^{-2} at the faintest end [323, 324]. This flattening has a cosmological origin: faint sources lie predominantly at large redshift, where the Λ CDM volume–flux relation departs from the Euclidean one and the luminosity function evolves. In particular, the space density of blazars declines beyond $z \sim 1.5$ –2. Since the faintest fluxes likely probe these redshifts, the number of sources per flux interval grows more slowly than the Euclidean scaling predicts [307, 323].

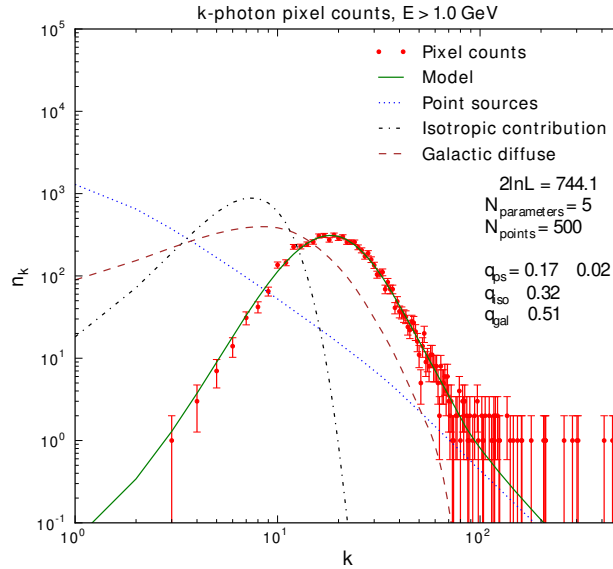


Fig. 6.1. Output of a 1pPDF analysis: photon-count statistics of the high-latitude Fermi-LAT sky above 1 GeV. The red points show the measured n_k , the number of pixels containing exactly k photons, together with the best-fit model (solid green) and its decomposition into point sources (dotted blue), Galactic diffuse emission (dashed brown), and the isotropic background (dot-dashed black). Credit: Malyshev and Hogg [322].

contribute on average far less than one photon each; their combined signal is statistically indistinguishable from a smooth Poissonian background [322]. The departures from Poisson statistics therefore encode information about the sub-threshold source population and its dN/dS . This connection is made rigorous through a generating-function formalism [131, 322] in which the total photon-count probability factorises into independent contributions from point sources, Galactic foreground, and isotropic background, and the parameters of dN/dS are obtained by fitting the predicted pixel histogram to data via maximum likelihood (see Figure 6.1).

Malyshev and Hogg [322] pioneered this approach for gamma rays, applying it to 11 months of Fermi-LAT data and constraining the contribution of unresolved AGN-like sources to approximately $17 \pm 2\%$ of the total flux at high Galactic latitudes above 1 GeV. Zechlin, Cuoco et al. [324] improved the method by incorporating the spatial variation of the Galactic foreground and exposure at the pixel level, measuring the dN/dS in the 1–10 GeV band down to fluxes approximately one order of magnitude below the detection threshold. Subsequent extensions by the same group [325] and by Lisanti et al. [307] generalised the technique to multiple independent energy bands spanning 1–171 GeV. The same pixel-statistics formalism was also adapted for the Non-Poissonian Template Fit (NPTF) applied to the Galactic Center Excess (Section 4.4), where it served to distinguish a smooth dark matter signal from a population of unresolved point sources such as millisecond pulsars.

These photon-count methods — the 1pPDF and, as we shall see, the CNN-based approach developed in this chapter — reconstruct the *total* source-count distribution: the aggregate dN/dS of all point sources, regardless of their astrophysical class. They measure how many sources exist at each flux level, but they cannot, by themselves, separate the blazar contribution from that of

star-forming galaxies, misaligned AGNs, or any other population. This is a fundamental feature of photon-count statistics: they respond to photons, not to source identities. A complementary body of work approaches the problem from the opposite direction, estimating the dN/dS of individual source classes by extrapolating from multi-wavelength correlations. Blazars are modeled through direct luminosity-function fits to the Fermi-LAT catalog and may account for roughly 20–30% of the UGRB intensity below 100 GeV [144]; star-forming galaxies and misaligned AGNs are estimated through empirical scaling relations linking their gamma-ray emission to better-measured infrared and radio luminosities [326–328], with each contributing of order 10–50% depending on the assumed redshift evolution [131]. These estimates suggest that the known astrophysical populations could collectively account for a large fraction of the UGRB intensity, though each carries uncertainties of at least a factor of a few, reflecting the limited samples from which the scaling relations are derived. A more precise measurement of the total dN/dS would therefore help constrain the room remaining for additional contributors such as dark matter annihilation.

6.3 Simulation-Based Inference for dN/dS

The 1pPDF formalism reviewed in the previous section is the most developed analytical approach to sub-threshold source counts. It has produced the strongest existing measurements of the gamma-ray dN/dS below the *Fermi*-LAT detection limit. However, its underlying assumptions — a parametric model for the source-count distribution, a simplified treatment of spatial correlations, and a computationally expensive likelihood — impose limitations that motivate a complementary approach.

This section discusses those limitations in detail (Section 6.3.1), then introduces the simulation-based inference framework that addresses them (Section 6.3.2), and finally summarizes the handling of the spherical geometry of the input data (Section 6.3.3).

6.3.1 Limitations of the Analytical Framework

The generating function formalism underlying the 1pPDF method rests on several simplifying assumptions, each of which constrains its applicability.

Neglect of spatial correlations. The 1pPDF treats each pixel independently. In practice, however, the point-spread function (PSF) of the *Fermi* LAT smears a point source across multiple pixels, introducing spatial correlations that the generating function approach does not entirely capture. These correlations carry information about the source-count distribution that is discarded in the single-pixel treatment, reducing the effective constraining power and potentially introducing biases.

Limited functional form flexibility. The dN/dS is typically modeled as a multi break power law with a fixed number of segments, for computational reasons. This assumption is sufficiently flexible for smooth, featureless distributions, but it cannot accommodate spectral features, bumps, or breaks at unforeseen flux values. The reconstruction is therefore conditional on the assumed functional form. Nonetheless, it is important to notice that prior works including [325] have not found a preference for a model with more than 2-3 break points, suggesting that the parametric assumption is not a major limitation in practice.

Computational cost and absence of amortization. The explicit 1pPDF likelihood is expensive to evaluate: each call requires recomputing the generating function at every pixel. Sampling the posterior with MCMC compounds this cost, requiring many such evaluations. Moreover, the inference is not amortized: every new sky map, whether observed or simulated, demands a fresh

set of MCMC chains. This becomes a concrete disadvantage when the method must be validated, since quantifying potential biases requires running the analysis on simulated maps with known dN/dS , each carrying the full cost of new chains. In practice, validating the method therefore carries a substantial computational cost.

Limited multi-band extension. While multi-energy extensions of the 1pPDF have been performed [307, 325], extending the formalism to a large number of energy bins simultaneously is practically challenging, since the number of model parameters and the computational cost grow rapidly.

The community has recognized some of these issues: Zechlin et al. developed a Compound Poisson Generator (CPG) formalism [324] to address potential biases in the 1pPDF approach, though this alternative still operates within the generating-function framework. A fundamentally different strategy is needed to overcome all four limitations simultaneously.

6.3.2 A Simulation-Based Approach

The limitations identified above share a common root: the 1pPDF relies on an a slow analytic likelihood that incorporates only a subset of the information contained in the photon-count map. An alternative is to bypass the likelihood entirely and employ implicit likelihood methods, the general framework introduced in Section 3.2.

As discussed in Chapter 3, in the simulation-based inference (SBI) paradigm the likelihood is never evaluated explicitly: it is defined implicitly by the simulator (Section 3.2.1). Ignoring the 1pPDF formalism, in general, writing the likelihood would require marginalizing over all the latent variables of the simulator; sampling from it, by contrast, only requires running it forward. In practice, one draws source fluxes from a parametric dN/dS , places them on the sky following an assumed spatial distribution, convolves with the instrument PSF and exposure, adds the Galactic foreground and isotropic background, and applies Poisson noise to produce a synthetic photon-count map. Each simulation is generally fast (around 0.1s) and faithful to the known instrumental and astrophysical effects. Crucially, the latent variables — here, the position and flux of every individual source — never need to be handled explicitly: each synthetic map is one random realization of them, so the intractable marginalization is performed implicitly by the simulator itself.

A convolutional neural network (CNN) is then trained on an ensemble of such simulated maps to learn the mapping from a photon-count sky to the underlying dN/dS , using the architecture and training methodology detailed in Section 3.3. The configuration adopted for the dN/dS problem discretizes the output into a set of flux bins plus the isotropic background level F_{iso} ; the exact binning, flux range, and training-ensemble size are specified later in this chapter (Section 6.5.3). The network employs the heteroscedastic Gaussian cost function and concrete dropout regularisation to estimate both the dN/dS and its uncertainties; the full formalism is presented later in this chapter (Section 6.5.3.4). A frequentist cross-check using the distribution of residuals on held-out validation maps confirms that the Bayesian error bands are well calibrated (Section 6.5.3.6).

This framework addresses the four limitations identified in Section 6.3.1: inter-pixel PSF correlations are captured implicitly by the convolutional architecture, the dN/dS is recovered with a more flexible shape approach, inference is amortized (additionally a trained network produces predictions in milliseconds), and the pipeline extends naturally to multiple energy bins by adding input channels.

6.3.3 Inference on Spherical Data

Fermi-LAT data are represented as HEALPix sky maps on the sphere. Applying standard planar CNNs to such data requires projecting the sphere onto a flat image, which inevitably introduces distortions. The map2patches algorithm, described in Section 3.3, addresses this by subdividing the HEALPix sphere into 12 equal-area base patches at order $n = 0$, each of which is independently mapped to a 2D image and processed through standard 3D convolutions. This approach preserves the area and information content of each pixel without spherical distortion, while retaining the speed of highly optimized standard convolutions.

A fully spherical cross-check using the NNHealpix architecture [194] yields compatible results, confirming that the map2patches approximation is reliable for the isotropic, small-scale signal of interest. The map2patches implementation is approximately an order of magnitude faster, enabling efficient generation of the large training ensembles required by the SBI framework. Full architectural details are presented later in this chapter (Section 6.5.3).

6.4 From Methodology to Measurement

The CNN-based framework described in the preceding sections was designed to overcome the limitations of existing analytical methods while retaining their strengths. Recovering the sub-threshold dN/dS quickly and with calibrated uncertainties opens several scientific directions.

First, a reliable measurement of the source-count distribution constrains the faint-end population of astrophysical emitters — blazars, star-forming galaxies, and millisecond pulsars — that collectively generate the bulk of the unresolved gamma-ray background. Quantifying their contribution is essential before any residual can be attributed to non-standard sources. Second, the recovered dN/dS provides the empirical prior for the probabilistic source cataloging framework developed in Chapter 7: the flux distribution of sub-threshold sources directly determines the posterior probability that any given sky pixel contains a faint, as-yet-unresolved source. Third, the methodology itself is readily extensible. An energy-dependent dN/dS reconstruction across multiple energy bands would enable spectral decomposition of the unresolved sky into individual source classes — and, in principle, a search for spectral signatures inconsistent with known astrophysical populations. This extension, while beyond the scope of the present work, is a natural next step toward the dark matter cross-correlation searches pursued in Chapter 8.

Applied to real *Fermi*-LAT data, the framework recovers the source-count distribution down to fluxes roughly 50 times below the catalog detection threshold, with stable, well-calibrated results across a wide dynamic range. The recovered distribution is broadly consistent with independent determinations in the literature. In particular, with the 1pPDF and Compound Poisson Generator analyses of Zechlin, Cuoco et al. [324] and the recent simulation-based inference reconstruction of Eckner et al. [329], which probe the same high-latitude sky with different SBI techniques and a more in-depth treatment of the galactic foreground model. The remainder of this chapter is based on [3] and presents the analysis in full.

6.5 Extracting the gamma-ray source-count distribution below the *Fermi*-LAT detection limit with deep learning

In this work we update the measurement of the dN/dS below the detection threshold to the increased statistics offered by 14 years of *Fermi*-LAT data. However, differently from Refs. [324, 325], we adopt here a method based on machine learning techniques to obtain the dN/dS below

Healpix order	6, 7
Weeks	9 – 745
E _{min}	1 GeV
E _{max}	10 GeV
Instrument Response Functions (IRFs)	P8R3_SOURCEVETO_V3
EVCLASS	2048 (Source Veto)
EVTYPE	1 (Front)
ZMAX	90

Table 6.1. Fermi Tools settings used for the 14-year data set analysis.

the detection threshold. A similar methodology has also been used recently to investigate gamma-ray unresolved sources close to the Galactic center [170, 241, 330]. We train a convolutional neural network (CNN) on synthetic gamma-ray maps, built from a wide variety of source-count distributions and then apply the trained CNN to the 14-year *Fermi*-LAT map for photon energies in the (1,10) GeV band. We show that the CNN is able to reconstruct the dN/dS , thus obtaining an updated result which is fully compatible with the one obtained in [324]. The methodology presented in this work is also meant to be a proof of principle for the adoption of a CNN to the reconstruction of the source-count distribution of the extragalactic sky, with the future aim of properly investigating additional features of the dN/dS , like energy correlations and the presence of additional components which could be traced to dark matter. An advantage of the CNN method is that it avoids the need to calculate complicated and numerically demanding likelihoods, as in the 1-point PDF method. Furthermore, we will describe an improved version of the treatment of a CNN on a spherical domain which will further optimize the computational aspect.

Data selection and *Fermi*-LAT map generation is discussed in Section 6.5.1, while Section 6.5.2 discusses in detail how synthetic maps for the CNN training and validation are constructed. Section 6.5.3 discusses the neural network architecture that we use and its implementation, including validation and error estimation. Section 6.5.4 discusses the analysis of the *Fermi*-LAT map with the trained CNN and presents the ensuing results for the dN/dS . Section 6.5.5 gives our conclusions.

6.5.1 Data selection

We process the *Fermi*-LAT data through the Fermi Tools suite [331] to produce a full-sky map of photon counts for gamma-ray energies in the (1-10) GeV range. We choose this range as a suitable balance between high statistics (which occurs at lower energies) and good angular resolution (which is better the higher the energy) of the detector, as well as being able to confront our results with those of Ref. [324].

We consider the first 14 years of data collected by the *Fermi*-LAT (specifically, from week 9 to week 745 of operation), and Pass 8 event selection [149, 150]. We adopt the P8R3_SOURCEVETO_V3 instrument response functions (IRFs), event class (EVCLASS) 2048 (SOURCEVETO) and event type (EVTYPE) 1 (FRONT). We used standard quality selection criteria, i.e. DATA_QUAL==1 and LAT_CONFIG==1. The atmospheric gamma rays from the Earth limb emission is removed by adopting a cut on the maximum zenith angle ZMAX of 90 degrees. FRONT events refer to gamma-rays detected by the first layers of the tracking module of the *Fermi*-LAT detector, and are optimised in order to have a better angular resolution, which is a crucial requirement for our analysis. The SOURCEVETO class of events achieves good suppression of the charged cosmic-ray (CR) background while still retaining a large event statistics. All those settings are summarised in Table 6.1. Finally, the maps are constructed with equal-area pixels by adopting the Healpix

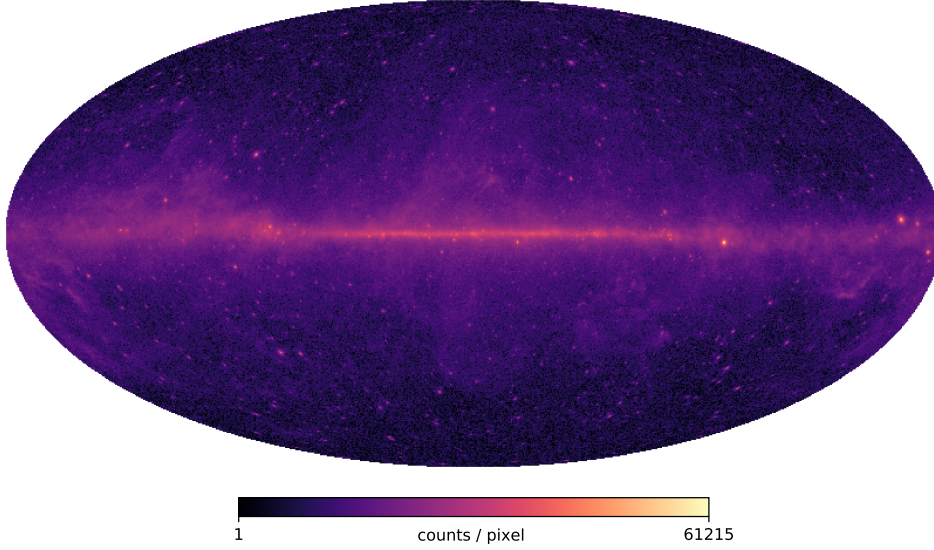


Fig. 6.2. *Fermi*-LAT photon-counts map in units of counts per pixel in the 1-10 GeV energy range for the 14-years dataset, at the Healpix resolution $N_{\text{side}} = 128$ ($n = 7$).

[332] pixelisation scheme of order $n = 6$ ($N_{\text{side}} = 64$ resolution, which corresponds to 0.92° pixel side) although for testing we will also use maps with order $n = 7$ ($N_{\text{side}} = 128$). The photon count map, in counts per pixel, for the 14 year data set is shown in Figure 6.2. This is the map to which we will apply the CNN to extract the source-count distribution of gamma-ray sources.

Using the *Fermi*-LAT tools we further extract the *Fermi*-LAT exposure map and the point-spread-function (PSF) for photon energies in (1, 10) GeV, which will be needed to construct the synthetic maps used to train and validate the CNN. Some care is required, since the PSF is a rapidly varying function of energy. Following previous works [324, 333], we build the (1, 10) GeV mean PSF by averaging the energy-dependent PSF in the (1, 10) GeV energy range with a $E^{-2.4}$ weight, corresponding to the approximate energy spectrum of the high Galactic latitude gamma-ray sky. The average exposure map in the (1, 10) GeV energy bin and the averaged PSF profile are shown in Figure 6.3 and in Figure 6.4, respectively. The *Fermi*-LAT exposure slightly changes with energy, although its variation inside our energy bin is small.

A further dataset we will use is the catalog of resolved sources. In particular, we will employ the recently released Data Release 3 of the fourth *Fermi*-LAT catalog of gamma-rays sources (4FGL-DR3) [141], which increments the previous release to 12 years of data taking and contains 6658 resolved sources. From this catalog, we obtain the (1-10) GeV source-count distribution in the resolved regime, shown in Figure 6.5. The detailed procedure with which we obtain the dN/dS from the list of sources is described in Appendix B of [324]. Here and through the rest of this chapter S will always refer to an integral flux in the range 1-10 GeV. The blue points refer to the regime in which the catalog is fully efficient in detecting sources, while the gray points denote sources for which the *Fermi*-LAT sensitivity to point sources starts degrading. Our aim is to infer the behaviour of the source count in the unresolved regime, i.e., for source fluxes below $S_{\text{th}} \sim 2 \cdot 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$ by adopting a CNN technique. We stress that in the following the dN/dS of resolved sources will be used only for comparison with the result of the CNN analysis, which

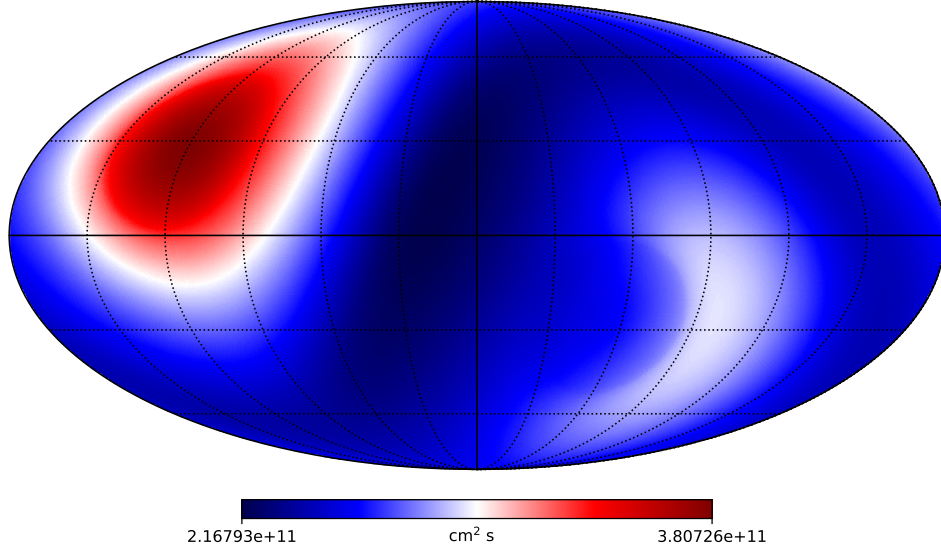


Fig. 6.3. *Fermi*-LAT mean exposure map in the 1-10 GeV energy range for the 14-years data set, in units of $\text{cm}^2 \text{s}$, at the Healpix resolution $N_{\text{side}} = 128$ ($n = 7$).

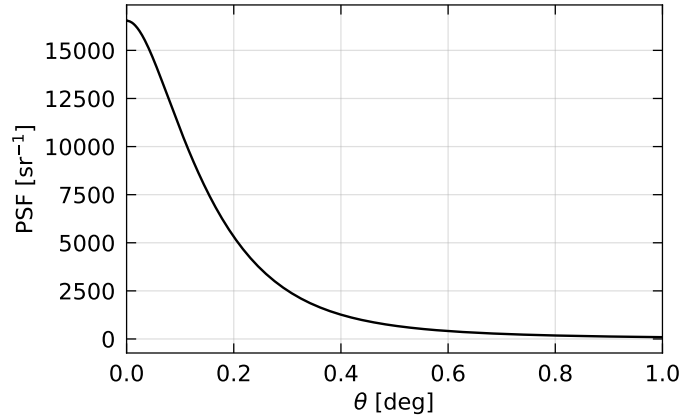


Fig. 6.4. Average *Fermi*-LAT point spread function (PSF) for energies in (1, 10) GeV.

are completely independent from it. Thus, the fact that we use 14 years of data while the catalog is based on 12 years does not have an impact on the analysis.

6.5.2 Synthetic map generation

In order to instruct the CNN to extract the dN/dS from the *Fermi*-LAT photon-count map, we need to produce synthetic analogs of the map, constructed from a wide selection of dN/dS and for the same characteristics (detector exposure and PSF, known physical foregrounds) of the data we will be using. These maps will then be partly used to train a CNN and partly used to validate the CNN, to determine its performance and to estimate the uncertainty on the reconstructed dN/dS . Finally, we will apply the trained CNN to the *Fermi*-LAT data.

The gamma-ray maps are modeled as the sum of three contributions:

- A population of gamma-ray point sources. Since our analysis will be restricted to the high galactic latitude region (in order to minimize in the analysis the impact of the galactic foreground emission, as discussed in the following sections), these sources will mostly be of extra-galactic origin and can therefore be assumed to be distributed homogeneously across the sky. The high galactic latitude sky also contains a population of millisecond pulsars which can also be approximately assumed to be isotropically distributed [334], and thus are ideally included in our phenomenological dN/dS . In terms of photon flux, the sources are extracted, for each map realization, from the input dN/dS distributions, modeled as in Section 6.5.2.1.
- Diffuse gamma-ray emission from the Milky Way. Galactic emission is the dominant contribution at low galactic latitudes, with a declining intensity at high latitudes. We model the galactic emission as discussed in Section 6.5.2.2, by adopting a template model normalized to the *Fermi*-LAT data. Stability of the results against foreground modeling will be tested through the adoption of different templates.
- An isotropic background, which integrates all source emissions which are too faint to be processed by the CNN and possible truly diffused emission mechanisms, such as gamma-rays from cosmological cascades from ultra high energy cosmic rays or a possible contribution from dark matter annihilation or decay, as well as a residual instrumental CR background contamination. This term is modeled as a constant.

A flux map $\mathcal{M}$ will therefore be obtained as the sum of three contributions:

$$\mathcal{M} = \mathcal{S} + A_{\text{gal}} \cdot \mathcal{G} + F_{\text{iso}} \quad (6.5.1)$$

where $\mathcal{S}$ is the map due to the underlying distribution of sources (obtained from a dN/dS model as described in the next subsection), $\mathcal{G}$ is a template map for the foreground emission and F_{iso} is a constant that denotes the isotropic component discussed above. The constant A_{gal} is a normalization constant for the galactic foreground template.

The count map $\mathcal{N}$ is then obtained by multiplying the flux map by the *Fermi*-LAT exposure map $\mathcal{E}$, depicted in Figure 6.3, and by taking into account the steradian-to-pixel conversion factor. Specifically, the counts in pixel a are obtained as:

$$\mathcal{N}_a = \mathcal{M}_a \cdot \mathcal{E}_a \cdot \frac{4\pi}{N_{\text{pix}}} \quad (6.5.2)$$

where the number of pixels in the Healpix scheme is $N_{\text{pix}} = 12 \cdot N_{\text{side}}^2$ with $N_{\text{side}} = 2^n$ with n the pixeling order.

Since the exposure slightly changes with energy, the above procedure could be improved by performing a convolution in energy of the photon flux with the exposure, instead of directly multiplying the two quantities as performed in Eq. (6.5.2). However, the gradient in energy of the exposure in the (1,10) GeV energy range is small, and the two procedures would provide very similar results. We, thus, use the simpler procedure outlined above. Eq. 6.5.2 also implies that we do not adopt energy dispersion. In the energy range 1-10 GeV the energy dispersion is about 10% [149] and may induce normalization changes of similar size. Thus, as explained in the next section, we will re-derive the galactic foreground normalization with a dedicated fit in order to avoid possible biases in the normalization.

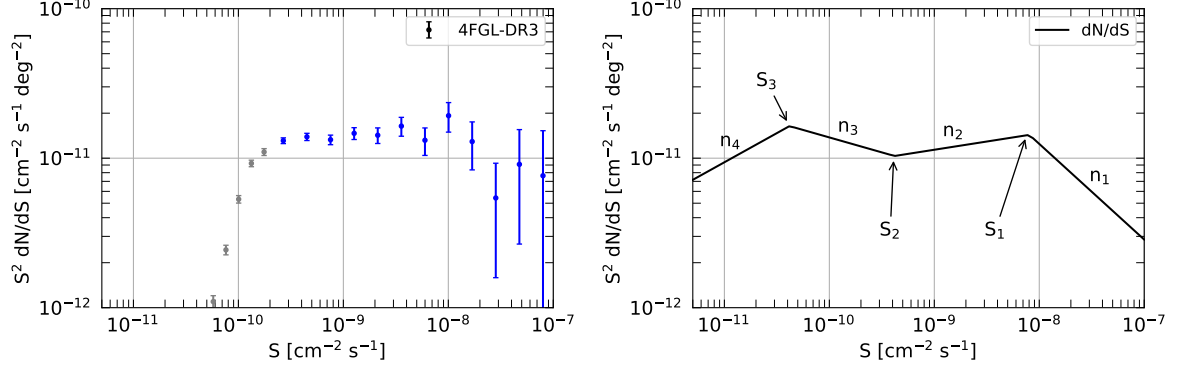


Fig. 6.5. Left: The 1-10 GeV dN/dS for resolved *Fermi*-LAT sources, obtained from the 4FGL-DR3 [141] catalog. Right: An example of dN/dS with 3 breaks S_j ($j = 1, 3$) and four slopes, where n_i ($i = 1, 4$) is the inclination of the slope.

Parameter	Prior	Range
A_S	log-flat	[1, 15]
F_{iso}	log-flat	$[0.5 \cdot 10^{-7}, 7.0 \cdot 10^{-7}]$
S_1	log-flat	$[3 \cdot 10^{-9}, 5.0 \cdot 10^{-8}]$
S_2	log-flat	$[5 \cdot 10^{-11}, 3.0 \cdot 10^{-9}]$
S_3	log-flat	$[5 \cdot 10^{-12}, 5 \cdot 10^{-11}]$
n_1	flat	[2, 4]
n_2	flat	[1.80, 2.15]
n_3	flat	[1.5, 2.5]
n_4	flat	[1.0, 3.0]

Table 6.2. Ranges of the parameters and their sampling priors adopted in the map generation with $N_b = 3$ breaks. The normalization A_S is in units of $10^7 \text{ cm}^2 \text{s sr}^{-1}$. F_{iso} is in units of $\text{cm}^{-2} \text{s}^{-1} \text{sr}^{-1}$. The breaks S_1 , S_2 and S_3 are given in units of $\text{cm}^{-2} \text{s}^{-1}$. All other quantities are dimensionless.

From the count map $\mathcal{N}$, which contains the number of photon counts in each pixel as determined by the model, we finally produce a Poisson realization $\mathcal{N}_{\mathcal{P}}$, by extracting in each pixel a photon count from a Poisson distribution with mean equal to the photon number of the corresponding pixel in $\mathcal{N}$. $\mathcal{N}_{\mathcal{P}}$ constitutes our synthetic map realization, which mimics a count map of an experiment with the *Fermi*-LAT specifications, where the underlying model is the one of Eq. 6.5.1.

In the next Sections we discuss in more detail how the components of Eq. 6.5.1 are modeled.

6.5.2.1 Differential source-count distribution

Following [324], the differential source-count distribution dN/dS is modeled as a multi-break power-law (MBPL). A MBPL with N_b breaks located at fluxes S_j ($j = 1, 2, \dots, N_b$) is defined as:

$$\frac{dN}{dS} = A_S \times \begin{cases} \left(\frac{S}{S_0}\right)^{-n_1}, & S > S_1 \\ \left(\frac{S_1}{S_0}\right)^{-n_1+n_2} \left(\frac{S}{S_0}\right)^{-n_2}, & S_2 < S \leq S_1 \\ \vdots & \\ \vdots & \\ \left(\frac{S_1}{S_0}\right)^{-n_1+n_2} \left(\frac{S_2}{S_0}\right)^{-n_2+n_3} \dots \left(\frac{S}{S_0}\right)^{-n_{N_b}+1}, & S \leq S_{N_b} \end{cases} \quad (6.5.3)$$

where S_0 is a reference flux value which we choose to be $S_0 = 3 \cdot 10^{-8} \text{ cm}^{-2} \text{ s}^{-1}$, and n_j are the power-law indices. S denotes the source flux in the energy interval (1, 10) GeV. The dN/dS distribution is normalized with an overall factor A_S .

For definiteness, in this work we adopt 3 breaks and 4 slopes. The ranges in which the parameters are varied to produce different source-count distributions is reported in Table 6.2. The ranges have been chosen in order to intercept the expected features of the dN/dS and are broad enough to allow for a good variability in the modeling of the source counts, in order to expose the CNN to a wide set of options. In fact, the prior ranges of the three break positions are chosen relative to the *Fermi*-LAT threshold for resolving point sources, $S_{\text{th}} \sim 2 \cdot 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$, such that S_1, S_2, S_3 are respectively larger, across and below S_{th} . The flux interval over which we will try to reconstruct the dN/dS is set as $[5 \cdot 10^{-12}, 1 \cdot 10^{-7}] \text{ cm}^{-2} \text{ s}^{-1}$. The upper limit of this interval is set by the brightest sources in the catalog, while the lower limit is set at about 1.6 orders of magnitude below the *Fermi*-LAT threshold for resolving sources. Sources fainter than the lower limit of this interval are assumed to just contribute to the isotropic component. This lower limit, indeed, is also close to the theoretical sensitivity given by the flux of a point source contributing exactly one photon, which we calculate as $F_{\text{sens}} = 1/\mathcal{E}_{av} = 3.74 \cdot 10^{-12} \text{ cm}^{-2} \text{ s}^{-1}$ where $\mathcal{E}_{av}$ is the average exposure in the region $|b| > 30^\circ$ where we perform the analysis.

The prior ranges for the power-law slopes are anchored to the results from the 4FGL catalog in the resolved regime, and allow for progressively wider variability moving toward the unresolved regime. Since catalog data exhibit a source count compatible with S^{-2} , except for large fluxes, where the slope increases, we adopt priors for n_1 and n_2 around these behaviours. For n_3 and n_4 , we progressively enlarge the prior ranges, in order to allow for more variability especially in the sub-threshold (unresolved) regime we are interested in. An example of a dN/dS generated from this procedure is shown in the right panel of Figure 6.5, which also illustrates the notation.

The maps $\mathcal{S}$ are generated according to the following procedure:

1. A dN/dS is selected by random sampling of its parameters in the intervals and with the priors defined in Table 6.2;
2. The number N_i of sources in the i th flux bin is calculated as $N_i = 4\pi(S_{i,\text{max}} - S_{i,\text{min}})dN/dS|_i$ where $dN/dS|_i$ is the value at the center of the i th bin. The flux bins are determined by subdividing the interval of interest $[5 \cdot 10^{-12}, 1 \cdot 10^{-7}] \text{ cm}^{-2} \text{ s}^{-1}$ in 10^3 log-intervals per decade;
3. A Poisson extraction on N_i determines the number of sources $N_{P,i}$ in each flux interval;
4. The sources are randomly positioned on the sphere by uniformly sampling $\cos\theta$ in $[-1, 1]$ and φ in $[0, 2\pi]$, θ and φ being the spherical coordinates. The selected angular position is then converted into a pixel position in the Healpix pixeling scheme with the `ang2pix` routine. An example of a point-source map is shown in the upper panel of Figure 6.6;

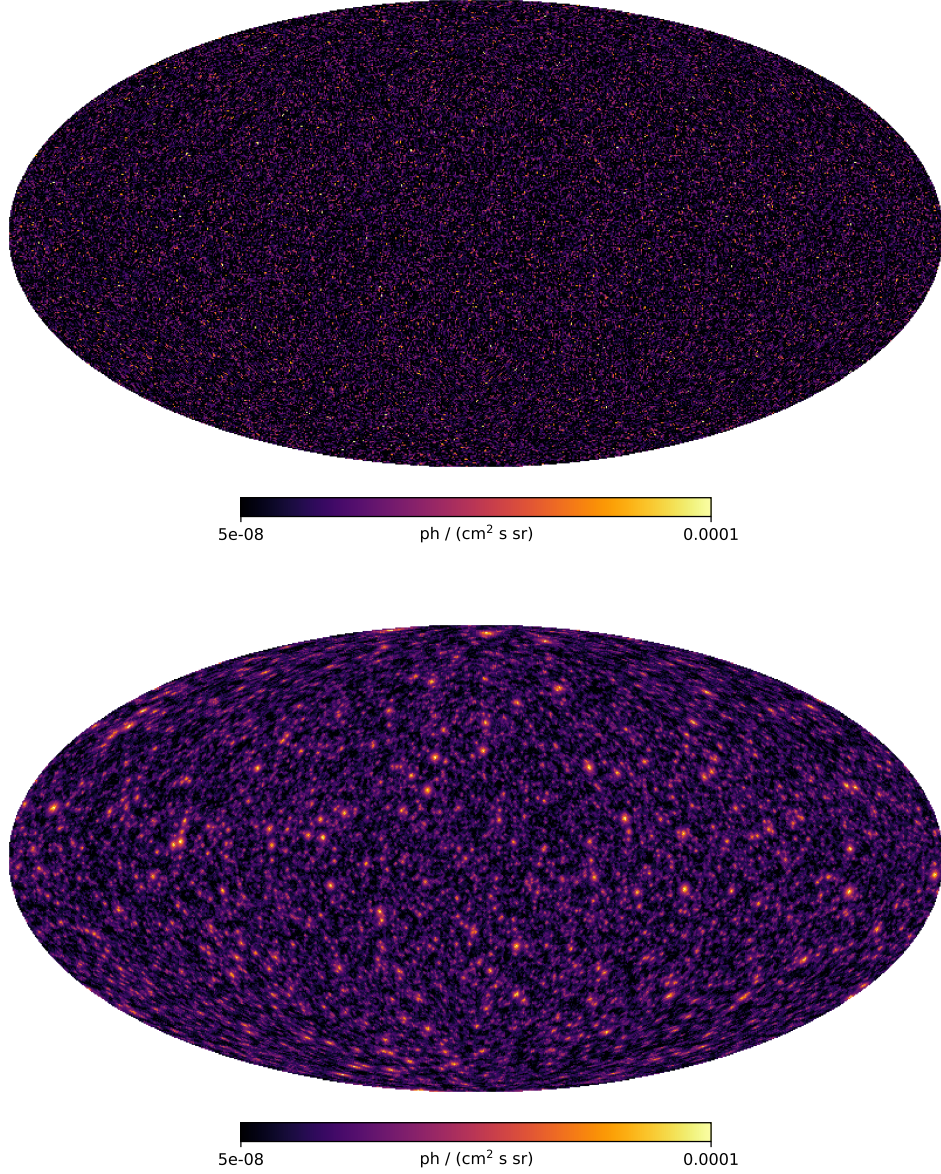


Fig. 6.6. Simulated flux map of point sources extracted from a specific source-count distribution, before (upper panel) and after (lower panel) the PSF is applied, in units of photons/(cm² s sr). The maps are shown at Healpix resolution $N_{\text{side}} = 128$ ($n = 7$).

5. Finally, convolution with the *Fermi*-LAT PSF shown in Figure 6.4 is applied to each point source. Since sources are assumed to be point-like, we simply replace each point source with an extended circular object with a radial profile equal to the PSF, and whose total flux is equal to the one of the original source. Since the image formation process is linear in the intensity, in case of overlapping sources (either because of their spatial position, or because of the PSF smearing) the resulting flux is additive. An example of flux map with the PSF applied is shown in the lower panel of Figure 6.6.

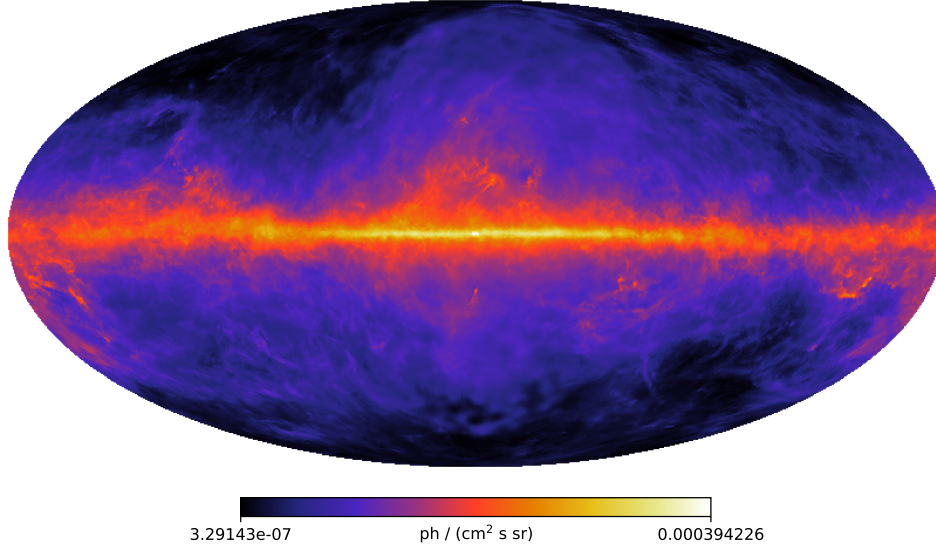


Fig. 6.7. gll_iem_v07 galactic foreground model [140], integrated in the (1, 10) GeV energy bin, in units of photons/(cm² s sr), at the Healpix resolution $N_{\text{side}} = 128$ ($n = 7$).

In order to avoid possible biases in the generated maps, due to the resolution of the angular binning, the maps generated with the procedure described above have been produced at a resolution higher than the one adopted for the analysis. We have therefore built our maps with $N_{\text{side}} = 128$. These maps have then been downgraded to $N_{\text{side}} = 64$ for the analysis.

6.5.2.2 Galactic foreground and isotropic background

The galactic foreground is implemented through the adoption of the *Fermi*-LAT template model gll_iem_v07². The map of the integrated flux in the (1, 10) GeV energy range is shown in Figure 6.7. To check consistency and stability of our results, we will also adopt a different version of the galactic template, namely model gll_iem_v05². The extragalactic isotropic background instead is modeled as a constant flux F_{iso} , which captures all the unresolved emission which is too faint to be associated to the emission modeled through the dN/dS (i.e., which refers to fluxes $S < 5 \cdot 10^{-12}$ cm⁻² s⁻¹, below our region of interest).

As shown in Figure 6.7, the galactic foreground dominates at low latitudes, although it extends to high latitudes with a smaller flux. For these reasons, low latitudes are unsuitable for the study of the extra-galactic background. Our baseline analysis will therefore apply a mask to the maps to cover the dominant part of the galactic emission, for which we adopt a latitude cut $|b| < 30^\circ$. In order to check the stability of the results against the foreground treatment, in Section 6.5.4 we will show the results obtained with two additional latitude cuts: $|b| < 40^\circ$ and $|b| < 50^\circ$.

Given the various approximations we use, instead of using the nominal normalization of the galactic foreground models, we prefer to use a data-driven approach and normalize the galactic templates gll_iem_v07 and gll_iem_v05 directly to the observed *Fermi*-LAT map. This

²<https://fermi.gsfc.nasa.gov/ssc/data/access/lat/BackgroundModels.html>

Foreground template	Latitude cut	A_{gal}	F'_{iso}
gll_iem_v07	$ b < 30^\circ$	0.888 ± 0.005	$(4.91 \pm 0.04) \cdot 10^{-7}$
gll_iem_v07	$ b < 40^\circ$	0.874 ± 0.008	$(4.90 \pm 0.06) \cdot 10^{-7}$
gll_iem_v07	$ b < 50^\circ$	0.838 ± 0.013	$(5.05 \pm 0.08) \cdot 10^{-7}$
gll_iem_v05	$ b < 30^\circ$	1.030 ± 0.006	$(2.57 \pm 0.06) \cdot 10^{-7}$

Table 6.3. Values of the A_{gal} and F'_{iso} parameters obtained from a fit to the *Fermi*-LAT map, for different galactic foreground template models and latitude cuts. F'_{iso} is in units of $\text{cm}^{-2} \text{s}^{-1} \text{sr}^{-1}$.

is realized by performing a maximum-likelihood Poisson fit, using as likelihood:

$$\mathcal{L} = \prod_{a=1}^{N_{\text{pix}}} \frac{\lambda_a^{k_a} e^{-\lambda_a}}{k_a!} \quad (6.5.4)$$

where k_a is the number of photons in pixel a of the *Fermi*-LAT count map and λ_a the corresponding quantity from the model count-map, given by $\lambda_a = (A_{\text{gal}} \mathcal{G}_a + F'_{\text{iso}}) \cdot \mathcal{E}_a \cdot \frac{4\pi}{N_{\text{pix}}}$, where A_{gal} and F'_{iso} are the two fit parameters. In performing the fit we mask the sky with the low-latitude cut and with a 2° circular mask around each of the sources in the 4FGL catalog. We added a prime on the F'_{iso} derived from the maximum-likelihood fit above to explicitly distinguish it from the F_{iso} introduced earlier. Confronting with Eq. (6.5.1), the F'_{iso} term denotes the sum of the purely isotropic contribution F_{iso} and the contribution of sources, $\mathcal{S}$ term, below the 4FGL detection threshold.

The results of the fit are shown in Table 6.3, for three latitude cuts for the gll_iem_v07 template and for $|b| < 30^\circ$ for the alternative gll_iem_v05 model. We see that both the normalisation parameter A_{gal} and F'_{iso} are consistent among themselves for a fixed galactic foreground model, implying consistency with respect to the latitude cut. The model gll_iem_v05 instead requires a larger normalization, with an ensuing reduced value for F'_{iso} by a factor of 2. This reflects a certain amount of degeneracy which is present at high galactic latitude between the foreground emission and the isotropic emission, allowed by the uncertainties in the foreground model. The normalization of the v07 model different from 1 is related to the fact that we do not include energy dispersion. The v05 model, instead, has been created originally without including the effect of energy dispersion³, and indeed we obtain a normalization compatible with 1.

Throughout the rest of the analysis we will fix the normalization parameter A_{gal} to the value determined from the above fit. The values of F'_{iso} are, instead, used to define the range of the prior for F_{iso} in Table 6.2 and a *a posteriori* to check consistency of the results. One illustration of a flux map obtained as the sum of all components is shown in Figure 6.8. Instead of fixing A_{gal} , in principle, we could also leave it as a free parameter and instruct the network to predict its value. We found, however, that the above procedure constrains A_{gal} very precisely and it is thus preferable. Furthermore, we also found that fixing A_{gal} provides a better stability of the neural network.

Nonetheless, to better asses the impact of fixing the A_{gal} value, we also performed a further test using different values in the range (0.78, 1.0) (i.e., a range centered on our fiducial value 0.888 ± 0.005 , thus encompassing many standard deviations away from its central value.) We found that the dN/dS reconstructed by the CNN is largely unaffected by the employed value of A_{gal} , while the specific A_{gal} choice mostly affect F_{iso} , as expected because of the high galactic latitude degeneracy between the foreground emission and the isotropic emission. Example plots are shown in the Appendix 6..6.4.

³<https://fermi.gsfc.nasa.gov/ssc/data/access/lat/BackgroundModels.html>

6.5.2.3 Count maps

With a flux map $\mathcal{M}$ available, we can now produce a synthetic photon-count map $\mathcal{N}$ as described in Eq. 6.5.2. One example is shown in Figure 6.9. In order to generate maps compatible with a counting experiment like *Fermi*-LAT, we then add Poisson noise to the map performing a Poisson realization of the number of photons in each pixel of the map $\mathcal{N}$, and we dub this final map as $\mathcal{N}_{\mathcal{P}}$ count map. One example is shown in Figure 6.10, which also shows the same map with the latitude cut applied: these are the type of maps that we use to train and validate the neural network. We have generated in total 1 million maps, 90% of which have been used for training and 10% for validation. The maps have been stored as a TFRecord dataset⁴ for use with Tensorflow [335]. The maps have been generated in Healpix format with $N_{\text{side}} = 128$ resolution (order $n = 7$) and then downsampled to $N_{\text{side}} = 64$ (order $n = 6$) for the training and validation (this resolution corresponds to 49152 pixels).

The training of the CNN and the subsequent analysis of the *Fermi*-LAT data can be performed equivalently with maps in units of photon-count or in units of flux. Nonetheless, we will work with maps in units of flux, which are easily obtained from the $\mathcal{N}_{\mathcal{P}}$ count maps by using the exposure map and by inverting Eq. (6.5.2). The reason to convert the $\mathcal{N}_{\mathcal{P}}$ count maps in units of flux is motivated by the fact that count maps contain a spurious large-scale angular dependence introduced by the exposure map (see Figure 6.3) which might confuse the CNN, while this potential issue is avoided using flux maps. Finally, to further ease the learning process of the CNN we also apply a foreground subtraction procedure, i.e., to the above flux maps we subtract the original galactic template flux map. In this way the CNN can focus on learning the dN/dS , seeing the exposure and galactic foregrounds only as sources of noise, instead of trying to learn their detailed structure. These foreground-subtracted flux maps are the maps we use to train the CNN.

6.5.3 Neural network architecture and training

In this Section we describe the architecture of the CNN we adopt and how data are handled. The input of the CNN are synthetic foreground-subtracted flux maps, and the output vector $\hat{y}_i = (dN/dS|_j, F_{\text{iso}})$ contains the reconstructed $dN/dS|_j$ in 20 flux bins in $[5 \cdot 10^{-12}, 1 \cdot 10^{-7}] \text{ cm}^{-2} \text{ s}^{-1}$, plus the reconstructed value of F_{iso} . The index i , thus, has values $i = 1, \dots, 21$. A possible alternative would be to choose as output the breaks and the slopes of the dN/dS , i.e., the same parameters used to model the input dN/dS . Nonetheless, we found a better stability of the CNN using as output directly the dN/dS function as discretized above, and we will thus use this as output.

As further described below, we will use an approach in which we will let the CNN determine also the errors on the output parameters. The output vector will thus be effectively doubled. Beside the $\hat{y}_i$, we thus have also $\hat{\sigma}_i$ the output vector of the errors.

Since we are dealing with sky maps, we can approach the problem from the point of view of image analysis. However, the image here is in spherical projection. We therefore need to adopt a method able to deal with this situation.

6.5.3.1 Spherical neural networks

In the last years, the machine learning community has realised the importance of developing neural network architectures on complex topologies. One of the most interesting non-trivial

⁴https://www.tensorflow.org/tutorials/load_data/tfrecord

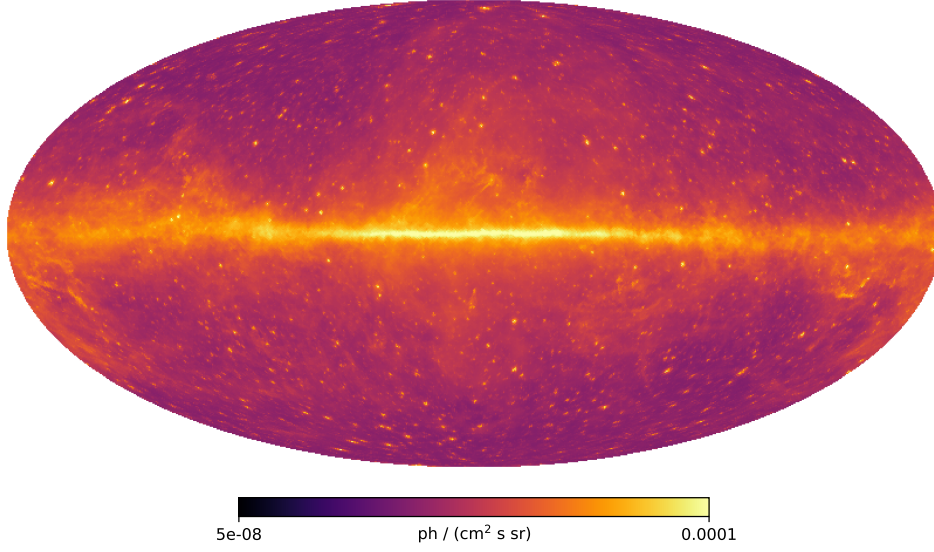


Fig. 6.8. The same as in Figure 6.6, with the galactic foreground model normalized with $A_{\text{gal}} = 0.886$ and an isotropic background component $F_{\text{iso}} = 4 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$. Units are photons/(cm² s sr).

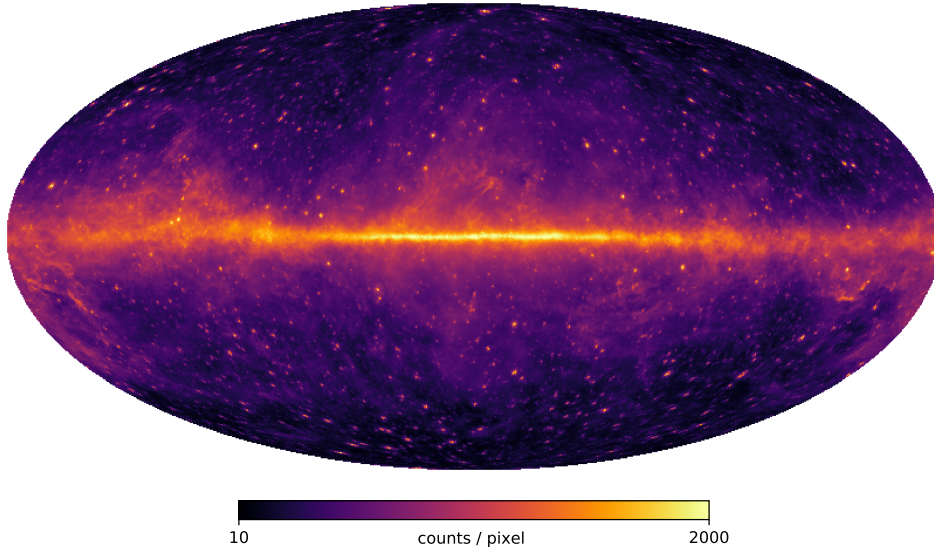


Fig. 6.9. Simulated count map in units of counts per pixel, at the Healpix resolution $N_{\text{side}} = 128$ ($n = 7$).

domains is that of a sphere, since it cannot be mapped on a plane without introducing distortion. This is the case we are confronting here, since we are dealing with sky maps. There are many efforts to implement CNN architectures that can do convolutions on the sphere [192, 194, 336–338], and while there have been many proposals, there is not yet agreement on a standard

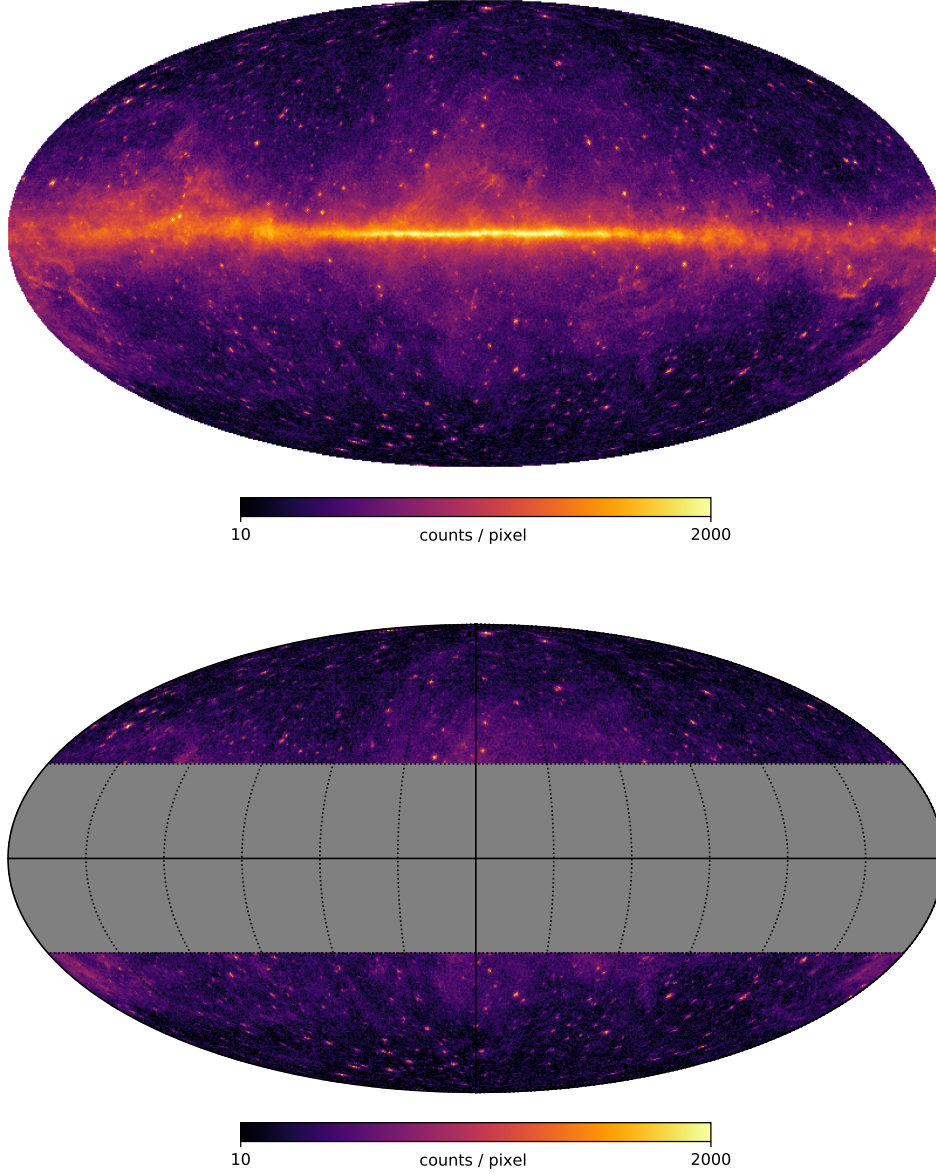


Fig. 6.10. Simulated map in units of counts per pixel after adding Poisson noise from the synthetic model map of Figure 6.9, at the same Healpix resolution $N_{\text{side}} = 128$ ($n = 7$). The lower panel shows the map with a $|b| < 30^\circ$ latitude cut.

algorithm.

In dealing with information extraction from spherical images, one has to cope with a set of issues: i) the algorithm needs to properly deal with the topology of the sphere without distorting the image or losing information; ii) the algorithm has to be computationally efficient; iii) it would be desirable to be able to use the same neural network architectures which already work for flat images, which have been proven to be very efficient and effective.

In fact, a full spherical architecture is usually slower than a flat architecture when applied

to images of the same size, and it often happens that it is not possible to adopt pre-existing models without re-implementing them *ad-hoc*. On the other hand, the adoption of a flat-image architecture either introduces spherical distortions, due to the mapping of a spherical image on a plane, or may lose some of the large scale information contained in the image.

In this work, we implemented a fast and reliable spherical architecture on the Healpix pixelisation of the sphere, taking inspiration from the idea proposed in [338] for the icosahedron. This method proved to be reliable when the information we need to retrieve from the maps is a small-scale effect, like in our case the distribution of sources, which are point-like (although smeared by the PSF) and isotropically distributed.

The method is based on the fact that, in the Healpix pixelisation scheme, each pixel has equal area and contains the information of the underlying field (in our case, the photon counts) averaged over that pixel. As discussed above, we train our neural network on maps of order $n = 6$ ($N_{\text{side}} = 64$), which contain 49152 pixels. Considering instead the Healpix sphere with base pixelisation of order $n = 0$ ($N_{\text{side}} = 1$), we can subdivide the map into 12 equal-area patches, each of which containing 4096 pixels. Each patch is an independent realisation of the underlying isotropic field and can be mapped into a 2D flat image. We call this parsing algorithm `map2patch`. Notice that this is not a re-sampling of the map: it's just a subdivision of the spherical image into 12 patches, in a way that preserves the area and content of each pixel. This allows us to use standard convolutional network techniques that work efficiently on flat images, while preserving the whole available statistics of the full map.

Convolution is performed on each patch separately. In order to be able to extract information from each patch in parallel, we define 3D convolutions, where the first two dimensions are the spatial dimensions, and the third dimension specifies the patch. In order to force the neural network to learn from each individual sky patch, we employ convolution filters of size $(N, N, 1)$, where N is the dimension of the filter. By doing so, we effectively apply 2D filters to the whole sky. Note that by employing such filter size, the output of each convolution will still have 12 separate images on the third axis, associated to each sky patch, while the number of channels will vary according to the number of filters employed. This technique will thus naturally scale for images with different color channels, such as RGB images, or spherical images for which several channels are available, as is the case where multiple energy bands are considered at once.

We stress that with our method we partially lose large-scale information and information stored at the border of the patches. Thus our algorithm would not be suitable in the case one is interested in recovering large-scale features. In our case, however, we are interested in exploring information which is statistically isotropic and/or lies at scales much smaller than the size of a patch. In this case our architecture is certainly viable and extremely efficient. Also, since we build our patch on the pixels of the Healpix pixelisation scheme, there is no spherical distortion in our images, unlike other architectures which employ the euclidean projection of the sphere.

This approach is extremely fast, since it leverages on optimised 3D convolutions. Furthermore the sphere-to-patch operation is performed asynchronously by the CPU while the accelerator is training the model, allowing for virtually no bottlenecks in the processing pipeline.

In order to cross-check our method, we also implemented the full-sky truly-spherical convolutions of [194], obtaining very similar results. In Appendix 6.6.5, we report some examples of the results and checks performed with a fully spherical convolution model. The two methods are therefore interchangeable for the type of information we want extract from the *Fermi*-LAT maps: however, our implementation allowed us to gain more than a factor of 10 in time of execution of the whole pipeline of training and validation. This convergence of results with the fully spherical method of [194] gives us confidence on the reliability of our model, and the speed gain sets the

basis for a future efficient extension of our analysis to several energy bins in the full *Fermi*-LAT energy range, in order to determine also the energy dependence of the dN/dS of unresolved sources.

6.5.3.2 Data pre-processing

The maps are constructed with the procedure outlined in Section 6.5.2. As explained more in detail there, even though we apply a latitude cut to the maps, in order to further reduce the impact of the galactic foreground (and as it is customary in analyses of the extragalactic gamma-ray background), we subtract the galactic foreground (normalized by the A_{gal} constant) from the input maps. This will also facilitate to some extent the training of the CNN, whose goal is to reconstruct the source-count distribution. Photon-counts maps are converted in units of flux, by using the exposure map. In order to reduce the variability of the inputs seen by the CNN and stabilize its behaviour, we take the logarithm of the flux values in the pixels and then map them in $[-1, 1]$ range. Finally, each map is parsed into 12 patches using the `map2patch` algorithm discussed in the previous Sections.

Given the approximate S^{-2} behavior of the dN/dS , for the output vector, we reconstruct $y_i = S^2 dN/dS|_i \times 10^{11} \text{ cm}^2 \text{ s deg}^2$ ($i = 1, 20$) and $y_{21} = F_{\text{iso}} \cdot 10^7 \text{ cm}^2 \text{ s sr}$, in order to work with (pure) numbers of order unity. We adopt this strategy in order to avoid that the mean square error could favour the optimisation of any flux bin over another, making the deviations more even across the range of S .

6.5.3.3 Network architecture

In recent years, we have seen the rise of many convolutional neural network models for computer vision tasks [339–341]. Modern convolutional neural network architectures can have a very large numbers of internal parameters. This is due to the complex links between each neural unit, which allows for these models to be highly expressive and effectively behave as universal approximators for many kinds of tasks [342]. Unfortunately this comes with a downside: these models are prone to fitting too well the training dataset, which would lead to poorer performance on new data. In the field of Machine Learning, this problem is known as "overfitting" [343]. In order to prevent overfitting, a possible solution is to train the model on a big enough dataset. This will render impossible for the NN to learn exactly the representation of the training set, as well as showing a richer feature space, which will result in better performance on any additional sample for which we may want to get a prediction, like the real case scenario with the Fermi photon counts map. Another solution is to apply regularisation layers which force the NN to learn a more general representation of the data, by promoting several metrics. Some of the most common regularisation techniques include L1 and L2 regularisation, batch normalisation, and dropout [344–347].

We choose as our convolutional neural network architecture the EfficientNet V2M model [348]. This model has been proven to be faster to train than other models for computer vision tasks, while retaining the same expressivity and predictive power. Furthermore, its low memory footprint helps in training with more data at the same time (larger batch sizes), which in turn reduces the risk of overfitting. In this work, following the original implementation of EfficientNet, we will adopt batch normalisation layers after each convolution.

We additionally employ a fairly new trainable implementation of dropout, called concrete dropout [349], before each convolution layer. Dropout [344] has been proven to be an efficient way to reduce overfitting in a model, by randomly turning off neurons or connections inside a

model. Unfortunately, the dropout probability for each dropout layer has to be either hand picked or optimised through grid search, which is practically unfeasible for very large models. If the dropout probability is too small, the regularisation effect will not be very significant, while on the other hand too high a dropout probability will result in slower training and worse overall performance (underfitting). We therefore adopt the implementation of dropout as a concrete Bernoulli distribution [350], which allows the self-learning of the optimal dropout probability, effectively saving a conspicuous amount of time while training and ensuring that the model will be less prone to overfitting.

6.5.3.4 Bayesian error estimation and cost function

In order to correctly estimate the error on a prediction, it is necessary to correctly estimate both the systematic error (sometimes called epistemic error [351]) and aleatory error components of a measurement. One way to estimate such components together is through a Bayesian approach, and in particular by defining a Bayesian neural network capable of estimating the posterior distribution of the desired observables. It has been shown [352] that a NN with arbitrary depth and non-linearities, with dropout applied before every weight layer, is mathematically equivalent to an approximation of a Bayesian model.

In a subsequent analysis [184] it has been shown how to estimate the first and second momenta (mean and variance) of the posterior distribution, under the assumption of a Gaussian likelihood of unknown variance. In particular [184] shows how to define a neural network capable of determining the appropriate variances of such a multivariate Gaussian. In our case, training a neural network with such an approach is advantageous, as the neural network will learn to weight less the component of the measurement at low fluxes, which we expect to be more uncertain, which helps in obtaining stable measurements for the test dataset.

Following [184], we thus approximate our posterior distribution with a Gaussian distribution, and define the cost function accordingly:

$$\mathcal{C} = \sum_{i=1}^N \left[\frac{(y_i^{\text{true}} - y_i^{\text{pred}})^2}{2\sigma_i^2} + \frac{1}{2} \log(\sigma_i^2) + \frac{1}{2} \log(2\pi) \right] \quad (6.5.5)$$

where we compare the true dN/dS and F_{iso} (transformed as discussed in Section 6.5.3.2) used to generate the simulated map with the prediction of the neural network, and thus $N = 20 + 1$. For the training of the neural network, we omit the last term of Eq. (6.5.5) which is an irrelevant constant. Regarding the errors, since the first term in Eq. (6.5.5) is a decreasing function of the σ_i while the second is increasing, this ensures that there will be a minimum which will provide the optimal estimate for the σ_i themselves.

In principle, the above cost function can easily be generalized to account for the covariance among the different bins and F_{iso} . In practice, however, this implies estimating also the off-diagonal terms of a 21×21 symmetric matrix. Such matrix has to be inferred autonomously from data without any supervision, making the problem very challenging to optimise in this form. We thus refrain from attempting estimating the covariance. For CNN applications where the covariance between a set of parameters has been estimated, see e.g. Ref. [170, 241].

6.5.3.5 Training and validation

We train our model using a Tensor Processing Unit (TPU) v3, kindly provided by the GoogleTM TPU Research Cloud. We adopt the Adam optimizer. We employ a cyclical triangular learning rate [353] to avoid falling into local minima of the parameter space. We adopt a batch size equal

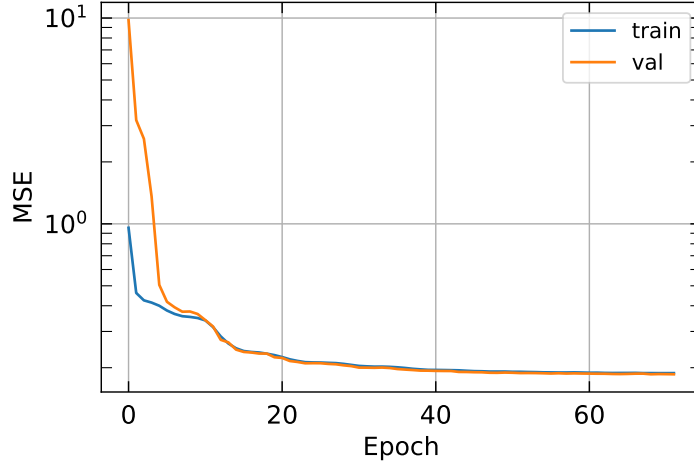


Fig. 6.11. Epoch evolution of the MSE for the training (blue line) and validation (orange line) sets.

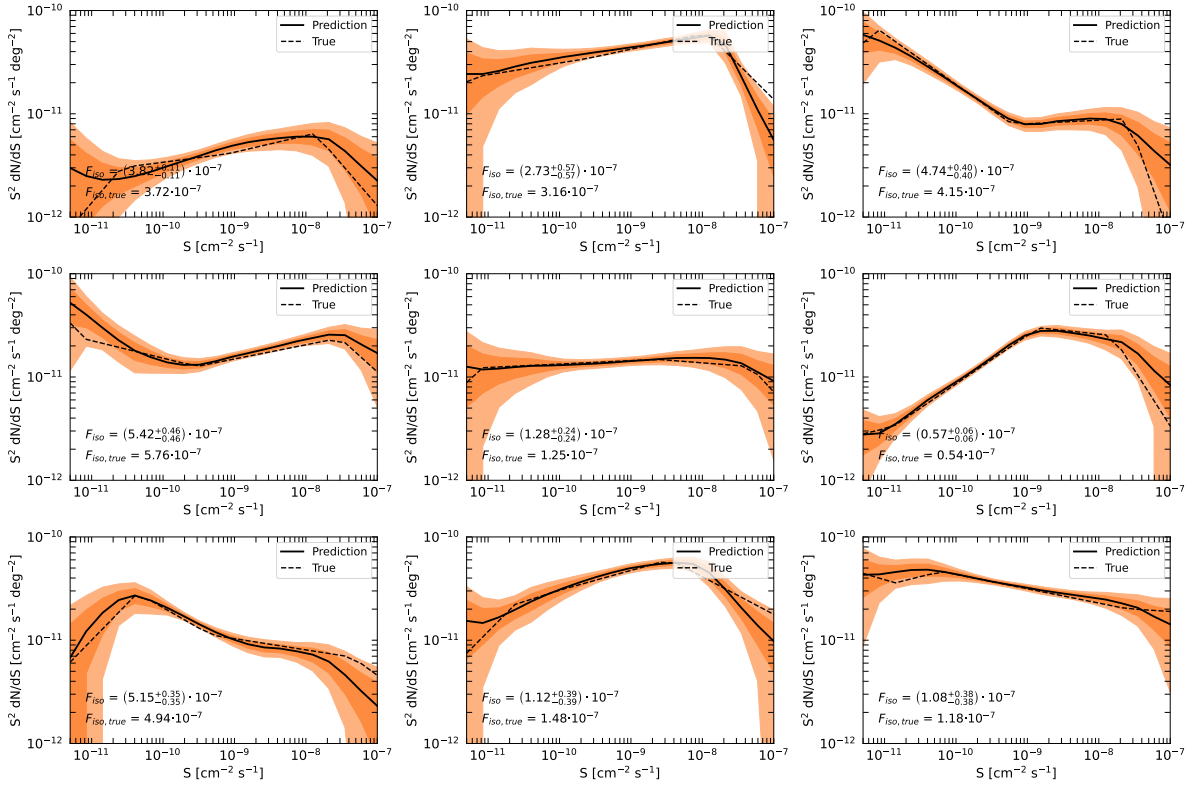


Fig. 6.12. Some example dN/dS from the validation dataset. Each panel shows the input dN/dS and F_{iso} and the corresponding quantities reconstructed by the CNN. The colored bands indicate 1-2 σ uncertainty regions. See text for more details.

to $128 \times 8 = 1024$ to fully exploit the computational capabilities of the TPU. We train the model for 72 epochs, for a total of 9 triangular learning rate cycles, with learning rate ranging from 10^{-6} to 10^{-3} . As already anticipated, we have generated 1 million maps, 90% of which have been

employed in the training and 10% for validation.

To test the convergence of the model we evaluate the mean square error (MSE) defined as $MSE = \sum_{i=1}^N (y_i^{\text{true}} - y_i^{\text{pred}})^2$. In Figure 6.11 we plot the MSE as a function of the training epoch for the training and validation dataset. We can see that, starting from epoch 10, the validation and training lines are close to each other, which means that our model is not overfitting the training dataset. We observe that the MSE reaches a plateau at around epoch 60, meaning that further training of our model would not lead to improvement in performance.

Once the CNN is fully trained, we extensively check the convergence of the reconstructed dN/dS and F_{iso} against their inputs, by using the maps of the validation set. Figure 6.12 shows a few illustrative examples of dN/dS , randomly generated according to the priors and parameters intervals listed in Table 6.2, together with their reconstructions. The dashed black lines show the input dN/dS and the black solid lines denote the corresponding distributions reconstructed by the CNN. The input and reconstructed values of F_{iso} are also quoted. The colored bands refer to the 1σ and 2σ confidence intervals as estimated by the CNN. The CNN recovers a dN/dS which is compatible with the input. The error bands are large for high fluxes, due to the fact that the number of sources at those values of S is low and therefore a large statistical uncertainty is expected. The error band becomes large again at low fluxes, indicating that the CNN reaches a confusion limit: even though here the number of sources is very large, their faintness makes progressively harder to identify their number and a confusion between faint sources and the average isotropic component F_{iso} arises, the lower the value of S becomes. This same behaviour also occurs in the 1point-PDF analysis of [324]. Concerning the value of F_{iso} , the reconstruction is very good, compatible with the input and with a small uncertainty.

6.5.3.6 Frequentist error estimation and Bias

Besides the Bayesian error automatically provided as output by the CNN, we also estimate the error in an alternative frequentist-like way, in order to provide a cross-check. This second method gives as a further advantage also the possibility to estimate the amount of bias on the reconstructed dN/dS . To this aim, we have built, for each of the 20 bins in flux and for F_{iso} , the histogram of the deviations between the true and reconstructed dN/dS and F_{iso} . The results for 14 values of S across its interval of interest, and for F_{iso} , are shown in Figure 6.13 (the remaining histograms are similar in content and are not shown here for economy of space). The plot shows the distribution of models which exhibit a specific relative deviation (expressed as percentage of the true value). By using these distributions of the deviations (normalized to unity), we derived the 1σ and 2σ confidence level intervals on the reconstructed dN/dS and F_{iso} . As anticipated in Section 6.5.3.5, at low and large fluxes the distributions are wide, implying a poorer ability to reconstruct the underlying physical models, while in the flux interval $(1 \cdot 10^{-11}, 2 \cdot 10^{-8}) \text{ cm}^{-2} \text{ s}^{-1}$ the 1σ error is below 30%, with an error below 20% for S in $(1 \cdot 10^{-10}, 1 \cdot 10^{-8}) \text{ cm}^{-2} \text{ s}^{-1}$. The 1σ error on F_{iso} is also of the order of 20%. The last panel of Figure 6.13 shows the histogram for the reconstructed value of the total photon flux: this is a sanity check and shows that the CNN correctly reproduces the normalization of the maps in terms of total counting rate. The uncertainty on the total flux is 5%. The 1σ (non symmetric) intervals for all the 20 flux bins are also reported in Figure 6.14. A more direct comparison of the Bayesian and frequentist errors is provided in the results section.

The same histograms of Figure 6.13 allow us also to check whether the reconstructed values are biased. We define the bias as the deviation between the median of the distribution and the input value. The values of the bias are shown, for each of the 20 values of flux, in Figure 6.14 as squares. We see that for all fluxes, the bias is very close to zero, which implies that the

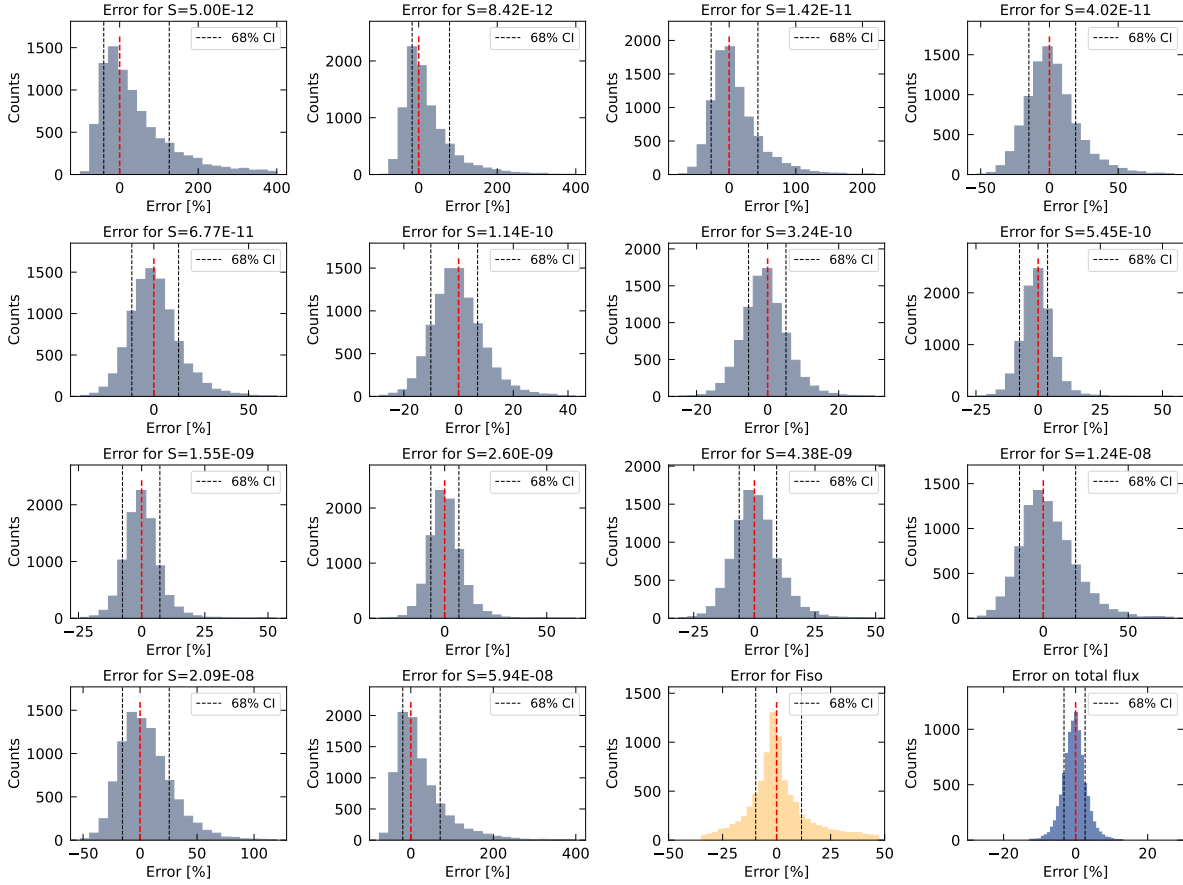


Fig. 6.13. Histograms of the percentage deviations between the true and reconstructed dN/dS for 14 values of S across its interval of interest, and for F_{iso} . The bottom right panel shows the same, but for a derived quantity, namely the total flux. The dotted vertical black lines indicate the 68% confidence region.

reconstructed values not only are determined with good precision (since the error bar is small), but also without a bias toward larger or lower values. The only exception is a positive bias for very low fluxes, where the CNN meets its confusion limit, and for very large fluxes, where the low source-counts statistics produces a small overestimate of the underlying dN/dS . In both cases, however, the bias is small compared to the overall error.

6.5.4 Results

After the training and validation processes described in the previous Section, we have applied the fully trained CNN to the 14 years *Fermi*-LAT data. The result of our baseline analysis, which adopts the `gll_iem_v07` galactic foreground and a latitude cut of $|b| < 30^\circ$, is shown in Figure 6.15. The blue line is the reconstructed dN/dS and the error bands are the Bayesian errors obtained as described in Section 6.5.3.4. The blue points show the dN/dS of the resolved sources, obtained from the 4FGL-DR3 catalog. We can see that in the resolved limit, the CNN reconstructs a dN/dS fully compatible with the one derived from the catalog, but then extends it to the unresolved regime with a behaviour compatible with $dN/dS \sim S^{-2}$ down to the smallest flux considered of $S = 5 \cdot 10^{-12} \text{ cm}^{-2} \text{ s}^{-1}$. Below $S \sim 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$ the dN/dS exhibits a slight

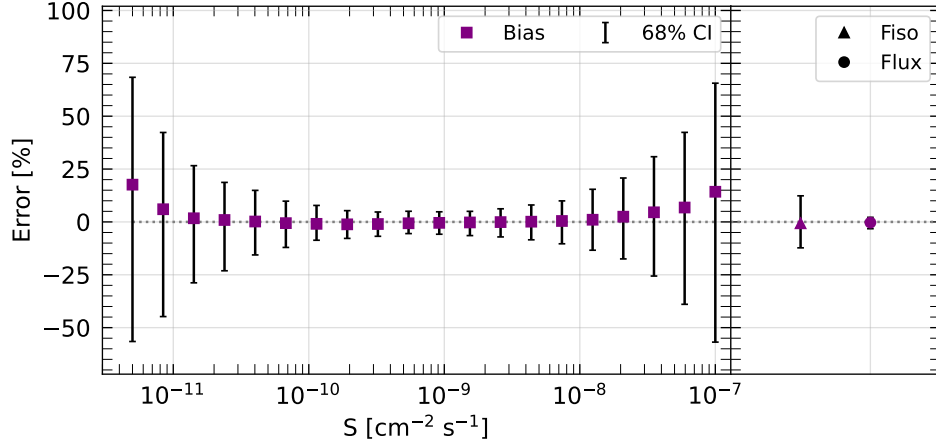


Fig. 6.14. 68% error for the reconstructed dN/dS as function of S , for F_{iso} and for the total flux. Also shown is the bias given by the median of the histogram for each S bin (squares).

turn-up, although the feature has a very low statistical significance.

The result obtained here is also compatible with the one obtained in [324], as shown in Figure 6.16. The difference between the two analyses, apart from the technique adopted to extract the dN/dS , stands in the fact that here we update the data set to 14 years of *Fermi*-LAT data collection and to improved detector response functions, as compared to [324]. Another difference is that in [324] the dN/dS was reconstructed with a broken power-law, while here we are allowing for a more flexible functional dependence (we are using 20 points in the flux interval shown in Figure 6.15). The result is nevertheless quite compatible with a power law behaviour with $dN/dS \sim S^{-2}$ for mid-values of S , $dN/dS \sim S^{-3}$ for $S > 10^{-8} \text{ cm}^{-2} \text{ s}^{-1}$ and a slight decrease in the power-law index for fluxes $S < 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$.

In Figure 6.17 we show the measurement for the Fermi map employing the frequentist estimation of the error. It can be seen that the error is very similar to the Bayesian one, with some minimal difference at low fluxes. This provides confidence that the estimated error is robust.

The CNN provides also the value $F_{\text{iso}} = 4.90 \pm 0.43 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$, which can be compared with the value $F'_{\text{iso}} = 4.91 \pm 0.04 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ reported in Table 6.3. We remind that F'_{iso} is the sum of F_{iso} and of the contribution of sources below the 4FGL catalog threshold. Indeed, if we calculate the contribution of sources using our best fit dN/dS integrated in the range $[5 \cdot 10^{-12}, 1 \cdot 10^{-10}] \text{ cm}^{-2} \text{ s}^{-1}$, the upper limit being an approximate value for the catalog threshold, and we add F_{iso} we obtain a value of $(6.41 \pm 0.61) \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$. This is in slight tension with F'_{iso} , although the two agree at the $\sim 2\sigma$ level.

In order to test the stability of the results, we repeated the analysis by using the same `gll_iem_v07` galactic foreground but with two different latitude cuts: $|b| < 40^\circ$ and $|b| < 50^\circ$. The positive consequence of a higher latitude cut is a lower residual foreground contamination. However, the amount of available data is reduced. The CNN has been fully re-trained for both of these situations, and the ensuing results when applied to the *Fermi*-LAT data are shown in Figure 6.18. We notice that the results, both in terms of behaviour and uncertainty estimate, are fully compatible with the results of the baseline analysis. The dN/dS exhibits a slight decrease toward lower fluxes as compared to the baseline case, although the size of this effect is not statistically significant. The values of F_{iso} are also fully compatible among them.

A second test of stability has been performed in order to check the impact of galactic

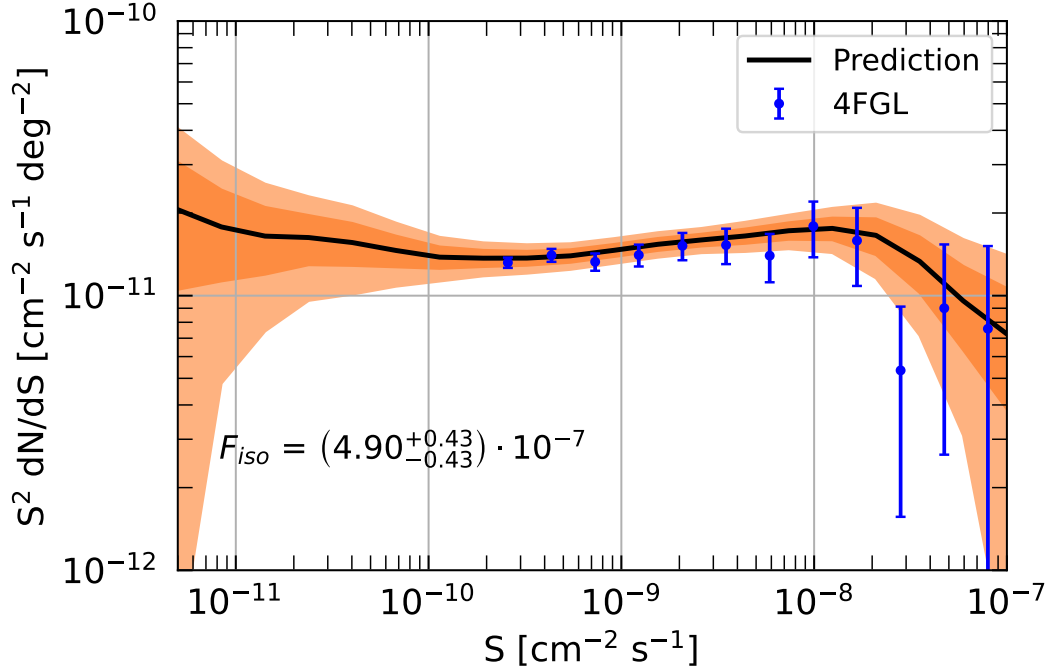


Fig. 6.15. Reconstructed dN/dS and F_{iso} and their Bayesian errors, when the trained CNN is applied to the *Fermi*-LAT map. This is the main baseline result of the analysis.

foreground modeling. We have performed a new training of the CNN with the alternative template `gll_iem_v05`, and with the $|b| < 30^\circ$ latitude cut. The result is shown in Figure 6.19. The reconstructed dN/dS matches to a large degree the source-count distributions obtained with `gll_iem_v07`. This reassures us that, even though the galactic foreground modeling introduces a degree of uncertainty, nevertheless the results of the CNN for dN/dS are remarkably stable. The value of F_{iso} is smaller in this case, equal to $F'_{\text{iso}} = 1.82 \pm 0.15 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$, which can be compared with the value $F'_{\text{iso}} = 2.57 \pm 0.06 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ reported in table 6.3. Adding to F_{iso} the integral of the contribution of sources in the interval in the range $[5 \cdot 10^{-12}, 1 \cdot 10^{-10}] \text{ cm}^{-2} \text{ s}^{-1}$ using the obtained dN/dS , we obtain a total flux of $(2.92 \pm 0.54) \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$, which is in very good agreement with F'_{iso} .

In Appendix 6.6, we also discuss the results of further tests that we performed to cross-check the stability of the dN/dS derived in this section, and that confirm the robustness of the result.

6.5.5 Conclusions

In this work we have studied the differential source-count distribution of gamma-ray sources below the *Fermi*-LAT threshold for source identification. We adopted a convolutional neural network methodology devised to deal with spherical sky-maps and able to extract information related to small scale fluctuations. The technique is based on the EfficientNet V2M model, applied to a patched version of the 2-dimensional spherical sky-map projection of the gamma-ray emission. The neural network has been trained on 900k synthetic maps and validated on 100k additional maps. These synthetic maps have been generated from the flux of random selections of source-counts distributions, to which galactic foreground models and an isotropic component have been added. After training and validation, the neural network has been applied to the *Fermi*-LAT

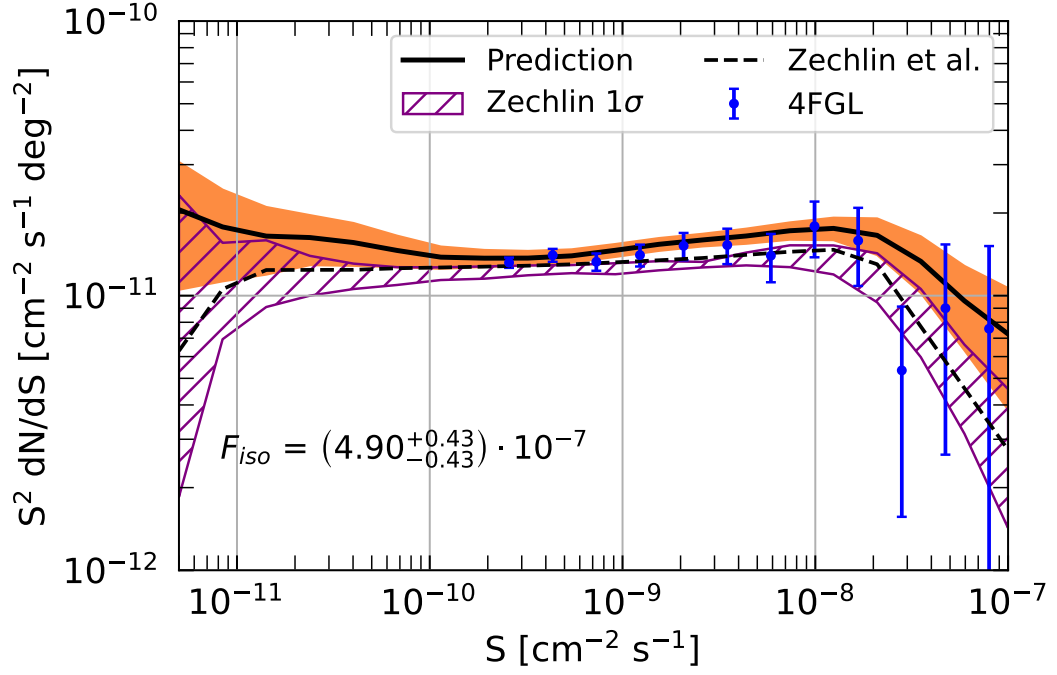


Fig. 6.16. Same as Figure 6.15, including a comparison with the result of Ref. [324].

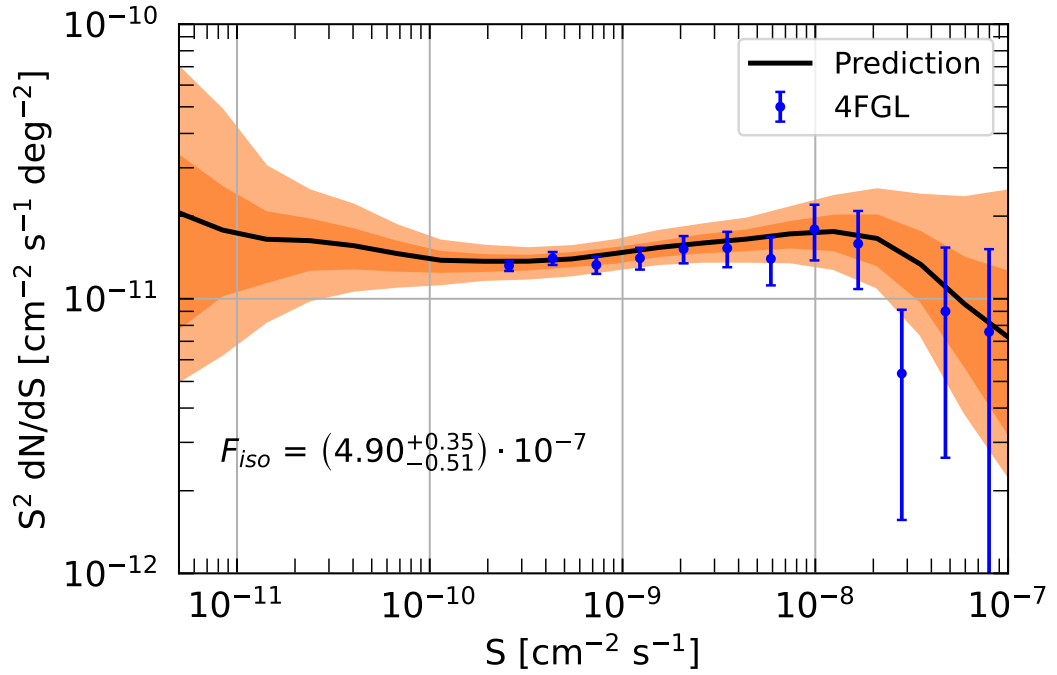


Fig. 6.17. Same as in Figure 6.15, but with errors estimated through a frequentist approach.

sky-map.

We concentrated our analysis on the photon energy interval (1,10) GeV and to 14 years

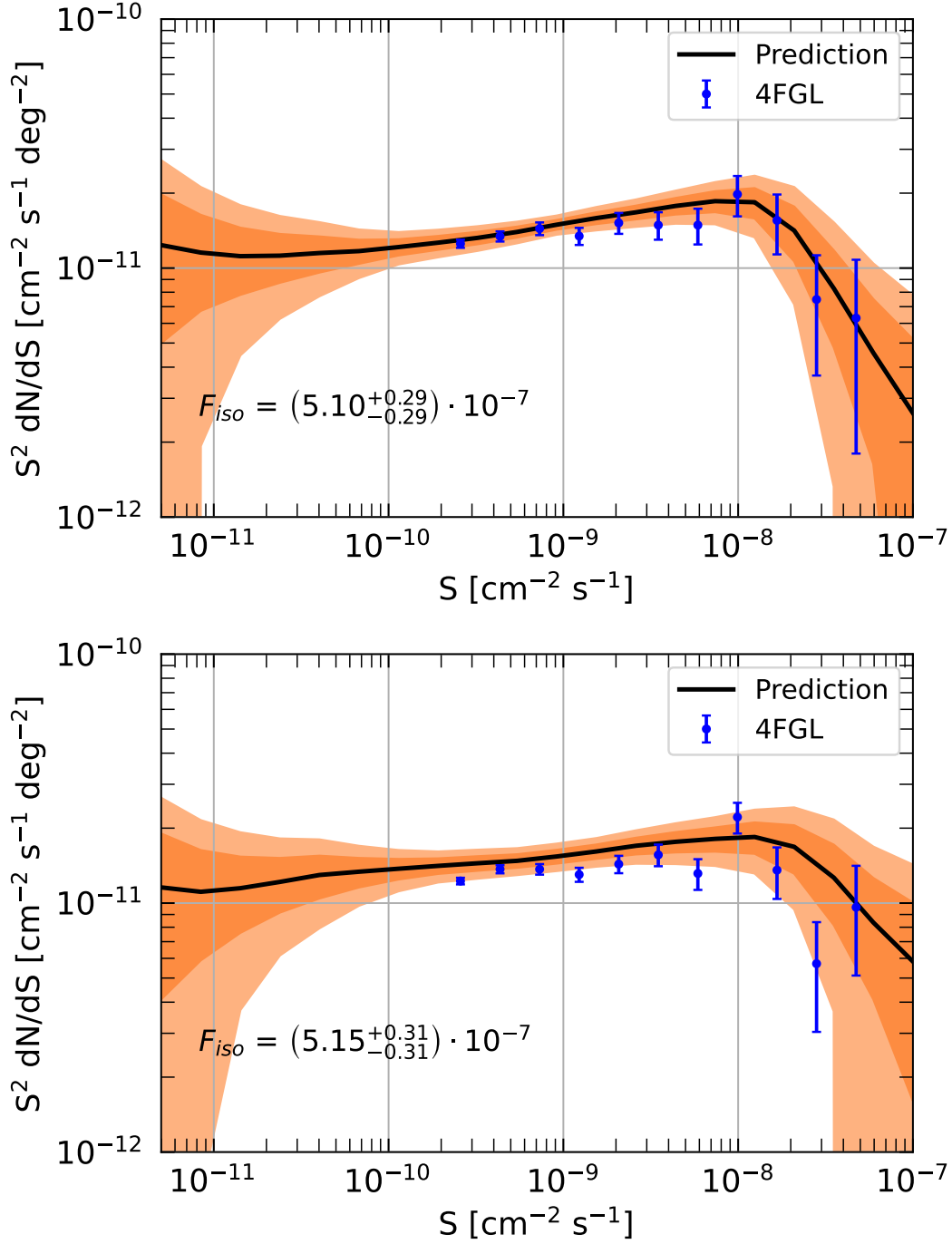


Fig. 6.18. Same as Figure 6.15 but for a latitude cut of 40° and 50° . The blue datapoints of the dN/dS of the 4FGL sources have been updated using only the source in the given region of interest.

of data. The main result is shown in Figure 6.15: the reconstructed source count distribution exhibits a $dN/dS \sim S^{-2}$ behaviour over almost four orders of magnitude in flux in the range $[5 \cdot 10^{-12}, 1 \cdot 10^{-8}] \text{ cm}^{-2} \text{s}^{-1}$, which includes the unresolved regime range $[5 \cdot 10^{-12}, 2 \cdot 10^{-10}]$. In

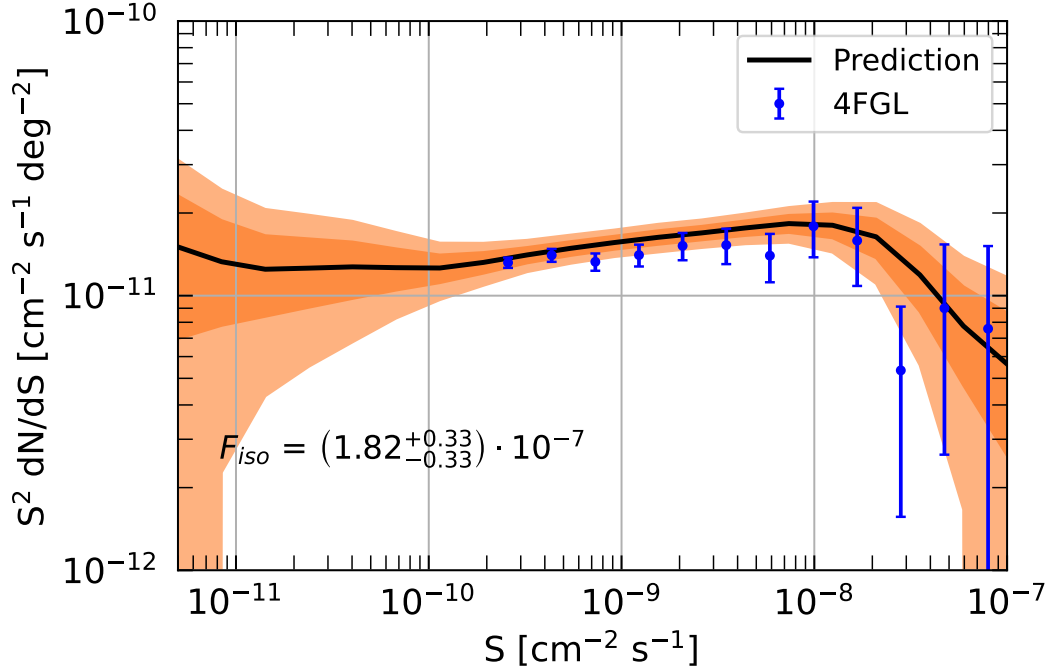


Fig. 6.19. Same as Figure 6.15 but using a CNN trained using the foreground model v05.

the regime where *Fermi*-LAT has sensitivity to individually resolve gamma-ray sources, Figure 6.15 shows that the neural network fairly reproduces the observations, giving confidence on its reliability and strength. While in the unresolved regime the result is in agreement with previous studies performed with different methodologies. We have also further validated the result performing several test of stability which have confirmed that our findings is stable and robust.

The methodology presented in this work provides a proof of principle for the adoption of a CNN to the reconstruction of the source-count distribution of the extra-galactic sky. With respect to other methods to extract the source-count distribution in the unresolved regime, like the 1-point PDF one, the use of a CNN avoids the need to calculate complicated and numerically demanding likelihoods. Possible future applications includes the extension to multiple energy ranges and energy correlations, and the investigation of features in the dN/dS which might indicate the presence of exotic components like dark matter. A further aspect which would be interesting to investigate is to compare different approaches to machine learning on spherical domains. In particular, it might be interesting to employ a graph convolutional neural network as suggested in [192], in order to preserve spatial relations between distant pixels, and rotational invariance. Finally, recent developments in the field of simulation based inference, such as our case, suggest that a promising approach to estimate the complex posterior distribution of the objective parameters are the so-called neural likelihood ratio estimation techniques [354], and in particular Truncated Marginal Neural Ratio Estimation [169, 355, 356]. Since these techniques are tailored for simulation based problems, we hypothesise that it would be possible to achieve similar performance to our current neural network architecture with an even simpler structure and possibly less training samples.

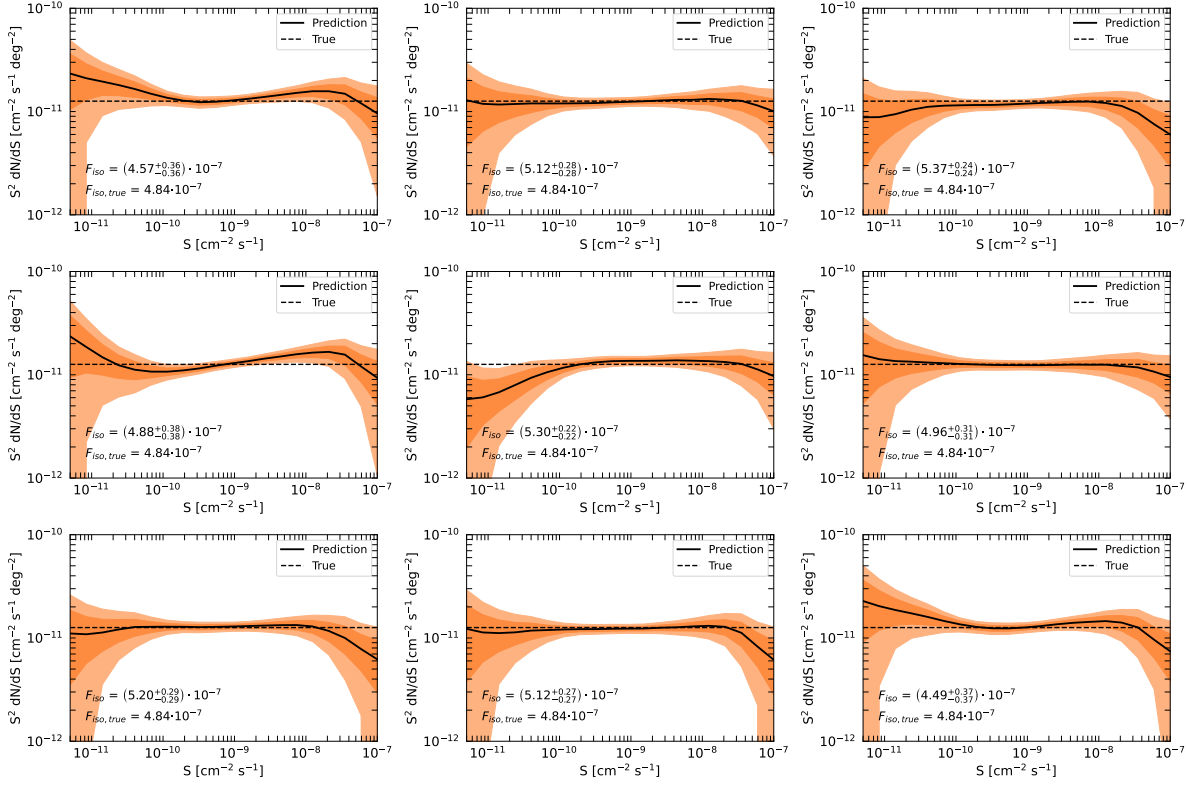


Fig. 6.20. Same as Figure 6.12 but for 6 different random realization of the same flat $S^2 dN/dS$ and same $F_{\text{iso,true}} = 4.84 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$.

6.6 Further Tests

In this appendix we describe further tests that we have performed to cross-check the stability and robustness of the dN/dS derived in the main text.

6.6.1 Flat $S^2 dN/dS$

Figure 6.20 shows the stability of the CNN output for a specific repeated dN/dS input. In this case, the CNN, trained on the wide variability of models of Table 6.2, is applied to a set of maps, all generated with a specific $dN/dS \sim S^{-2}$. This specific choice of input dN/dS is meant to test a case which is expected to be similar to the real *Fermi*-LAT case, as was found in [324] for mid-values of S . The plots show that the CNN is remarkably consistent in reproducing the correct behaviour. The drop at high fluxes is mostly due to the fact that the CNN has been instructed with maps that contain a dropping dN/dS at high fluxes (see Table 6.2 - the reason for this choice of prior is that sources in the 4FGL catalog exhibit this drop in the source count). For mid-values of S , the reconstructed $S^2 dN/dS$ is quite flat, with a small error band. At low S , well below the *Fermi*-LAT threshold and when approaching the confusion limit, some mild deviation occurs, but the true dN/dS lies always within the uncertainties. These tests gives us confidence that the CNN should be able to faithfully reconstruct the source distribution of the *Fermi*-LAT sky, when applied to the real data in Section 6.5.4.

As a further test, we show in Figure 6.21 the case where we apply our method to maps which contain a galactic foreground model different from the one used in the training. The CNN has

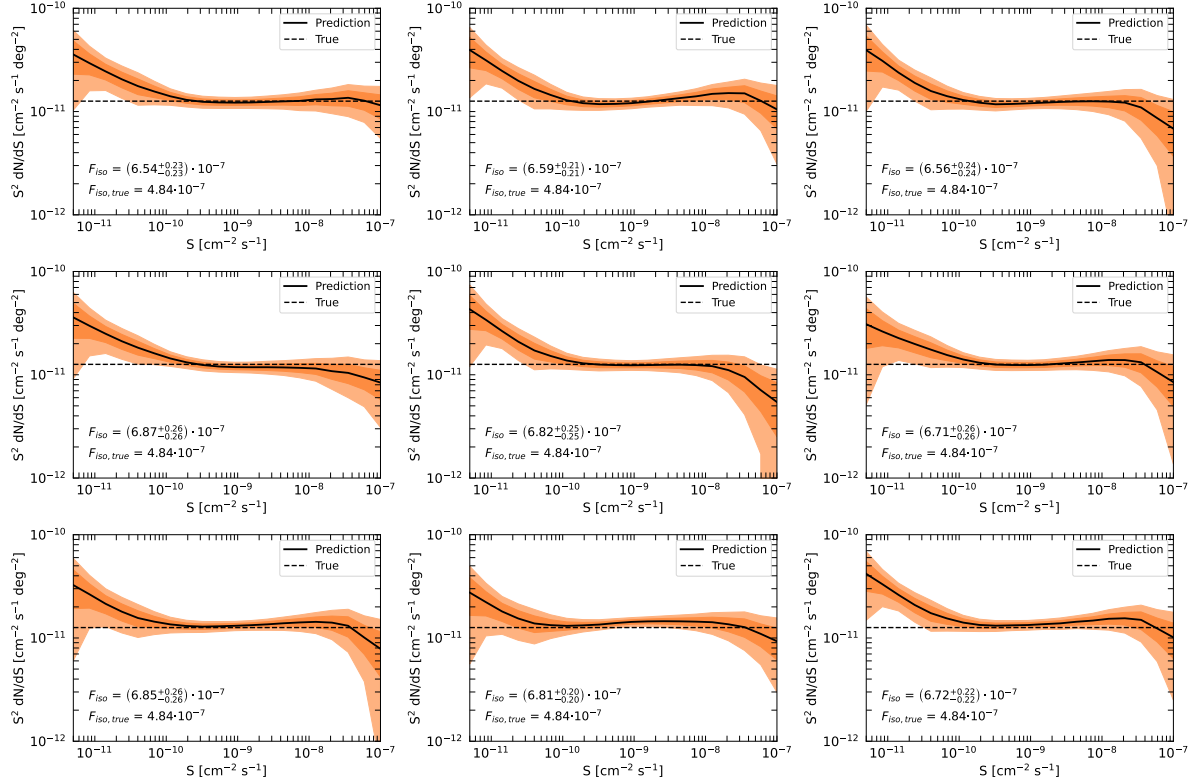


Fig. 6.21. Same as Figure 6.20 but for synthetic maps generated using version v05 of the foreground model.

been built (like in our baseline case) with maps generated with the galactic template model `gll_iem_v07`. The same model is used in the foreground subtraction performed in the analysis. On the contrary, for this test we have generated maps using the alternative `gll_iem_v05` foreground model, and let the above `gll_iem_v07`-trained CNN to analyze them. This has been done in order to verify the resilience of the method on the imperfect knowledge of the galactic foreground. Figure 6.21 shows that the reconstructed dN/dS has a level of agreement with the input model comparable to the analogous case of Figure 6.20. Some larger deviation is present at very low fluxes, even though the reconstructed and input models are compatible within 1σ .

6.6.2 UltraCleanVeto Selection

In this section we further test the robustness of the results to the *Fermi*-LAT data selection. In particular, while for the main results we used the `SOURCEVETO` event class selection, here we test the `ULTRACLEANVETO` selection. This event class has more stringent cuts with respect to the `SOURCEVETO` class, and it thus contains a lower residual charged cosmic-ray background, at the prize of a lower effective area (by about 15%). Except for the different data selection all the rest of the analysis is performed in the same identical way as for the `SOURCEVETO` case. The resulting dN/dS is shown in Figure 6.22 and it can be seen that it is compatible with the main results, while, as expected, the value of F_{iso} is a bit lower.

6.6.3 Multipole Analysis

Since foreground modeling can be an important source of systematic uncertainty, we investigate this potential issue performing a further test. As discussed above, the signal we try to extract from the gamma ray maps, is a small-scale effect, since it is due to a distribution of point sources. To a large degree, this point sources are isotropically distributed in the sky and contribute to the fluctuations of this isotropic field at small angular scales. On the contrary, the galactic foreground is quite diffuse, as can be seen in Figure 6.7, and therefore contributes mainly to large-scale anisotropies. In order to further remove from the data the possible large-scale residual component of the galactic foreground, which might still be present after imperfect foreground removal performed with the use of the templates, we adopt a method based on the transformation of the flux to harmonic space and the removal from the maps of the low-multipoles (i.e., large angular scales) contribution. We stress that this procedure is applied only to the *Fermi*-LAT data map, and not to the synthetic maps. For the latter, the foreground subtracted is the same used to generate the maps, so foreground subtraction is, by definition, “perfect”. The procedure follows these steps:

- We start with the foreground-subtracted flux map, to which we apply the $|b| < 30^\circ$ latitude cut and a 2° mask around each of the bright sources of the 4FGL catalog (which would otherwise dominate the angular power spectrum). We first determine and remove the monopole ($l = 0$) and dipole ($l = 1$) components from the masked map (for this, we use the `remove_dipole` routine from the Healpy library). In this way, after the subtraction procedure outlined below, the monopole (i.e. the total flux) and dipole information of the *Fermi*-LAT map are retained.
- This map is then subject to harmonic space decomposition:

$$\mathcal{M}_{\text{masked}}(\theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l a_{lm} Y_{lm}(\theta, \phi) \quad (6..6)$$

where θ and ϕ are angles on the sphere, $Y_{lm}(\theta, \phi)$ the spherical harmonics and the harmonic coefficients a_{lm} completely encode the same information present in the map $\mathcal{M}_{\text{masked}}$ (notice that, even though in Eq. (6..6) we formally decompose over all multipoles, the monopole and dipole amplitude are vanishing, since they have been removed from $\mathcal{M}_{\text{masked}}$);

- The harmonic coefficients a_{lm} are determined with the `map2alm` Healpy function. The ones that are retained for the foreground cleaning are those referring to multipole values $l < l_{\text{max}}$, where l_{max} will be varied and the dependence of the results with l_{max} will be studied (at this level, the monopole and dipole are not present, because of the above point);
- By using the a_{lm} corresponding to multipoles $l < l_{\text{max}}$, we use Eq. (6..6) to construct a map $\mathcal{M}_{\text{residual}}$ which contains only the large-scale component. This is performed with `alm2map` Healpy function.
- We subtract the residual map from the original map, thus obtaining a map where the large scale residual components have been removed. The size of the scale is determined by $l < l_{\text{max}}$.

With this algorithm we obtain maps where the large scale residual features associated to imperfect galactic foreground subtraction are further removed. The results we obtain on the

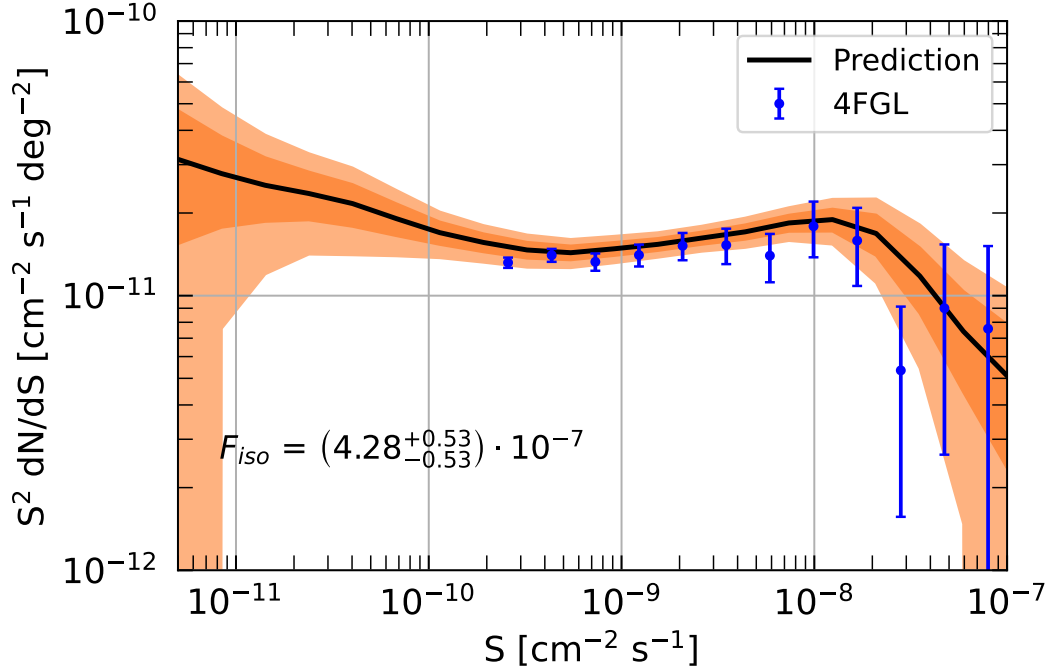


Fig. 6.22. Same as Figure 6.15 but for the ULTRACLEANVETO *Fermi*-LAT data selection.

reconstructed dN/dS for the *Fermi*-LAT data (after this “multipole cleaning”) are shown in Figure 6.23, for different values of $l_{\max}$ and for the `gll_iem_v07` template. By comparing Figure 6.23 with Figure 6.15, which refers to the same foreground template and latitude cut, we can see that the results are extremely stable up to $l_{\max} \sim 100$. This means that our baseline analysis is not affected by incomplete foreground subtraction, and can be considered reliable. The fall-off of the dN/dS when $l_{\max} > 100$ is expected since $l_{\max} = 100$ corresponds to an angular scale of the order of 2° , which starts to be compatible with the size of (PSF-smeared) point sources. Therefore, a cleaning with $l_{\max} > 100$ starts to remove point sources instead of galactic foregrounds.

The same analysis has been performed on a pipeline fully based on `gll_iem_v05`: the result is shown in Figure 6.24, to be compared with Figure 6.19. Also in this case, the results are extremely stable up to $l_{\max} \sim 100$, and the same above considerations apply.

Finally, to check the strength of the method against the imperfect knowledge of the background, we performed the same analysis of multipole cleaning by adopting our baseline pipeline (based on `gll_iem_v07`) on maps generated with `gll_iem_v05` (similarly to what has been done for Figure 6.21). The results are shown in Figure 6.25, where again we see that the results are pretty consistent, up to $l \sim 100$. It can be seen, moreover, that the procedure of multipole cleaning helps in removing the, nonetheless small, positive bias which is present toward low fluxes in this case.

In conclusion, these tests give confidence on the reliability of the method and that the main result of Figure 6.15 corresponds to a fair representation of the source-count distribution of the unresolved gamma-ray sources.

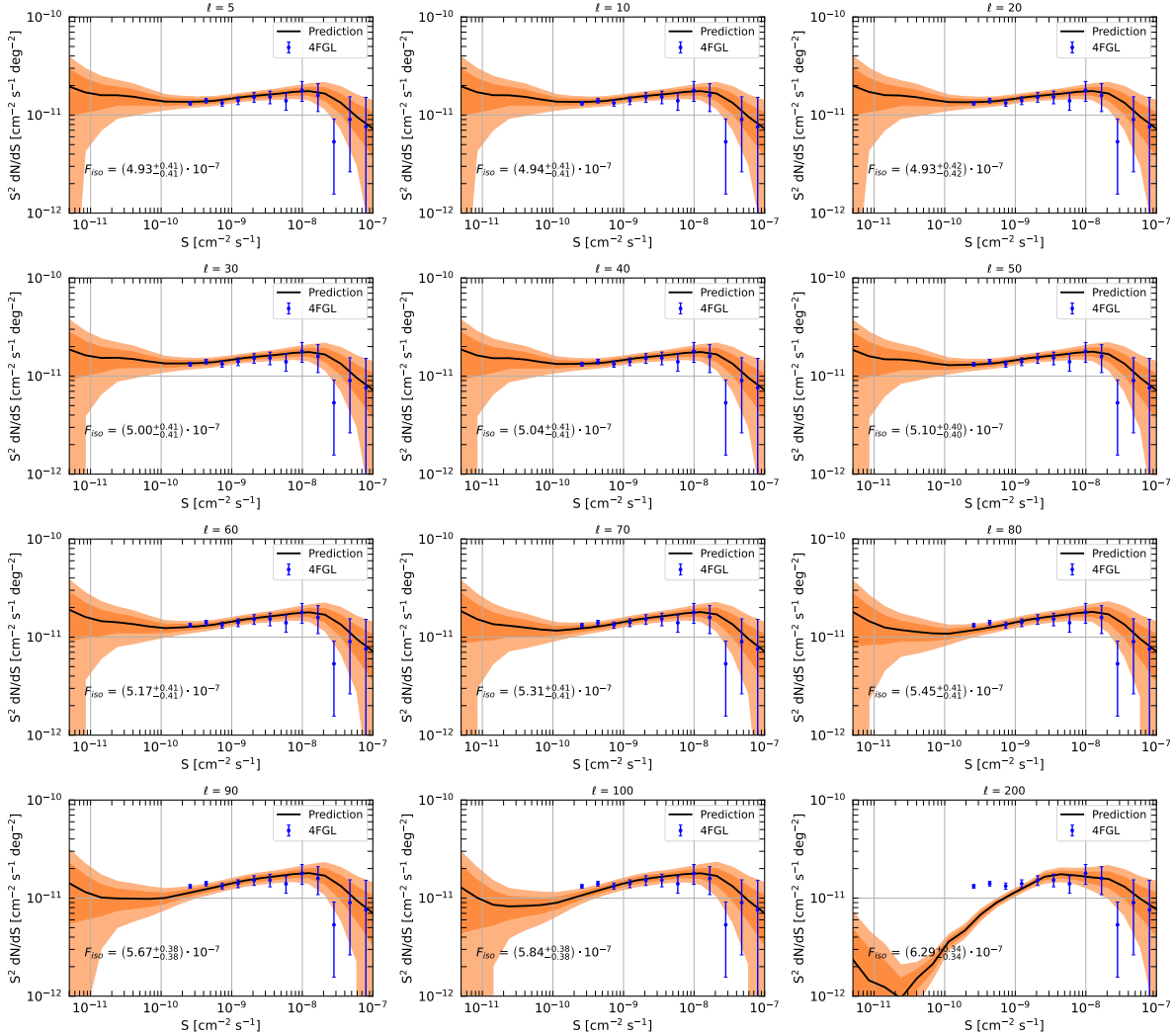


Fig. 6.23. Same as Figure 6.15 but where the *Fermi*-LAT map given as input to the CNN has been processed with the improved foreground cleaning procedure based on multipole decomposition described in the text. Each panel refer to a different maximal multipole ℓ_{max} used for the cleaning.

6.6.4 Changing A_{gal}

In order to test the stability of the results against changes in the values of A_{gal} , we have provided, as input to our trained CNN, *Fermi*-LAT maps cleaned with different values of A_{gal} , ranging from 0.78 to 1.0 (our fiducial value is 0.888 ± 0.005 , thus the chosen range encompass many standard deviations around its central value). The results are shown in Figure 6.26 below. It can be seen that the dN/dS is considerably robust and stable to different values of A_{gal} , despite the large range explored. Instead, different values of A_{gal} seem to imply a significant variation in the reconstructed value of F_{iso} , as expected since at the large Galactic latitudes explored in our analysis ($|b| > 30^\circ$) there is a certain amount of degeneracy between a purely isotropic emission and the Galactic foreground emission itself. However, this does not affect the reconstructed dN/dS .

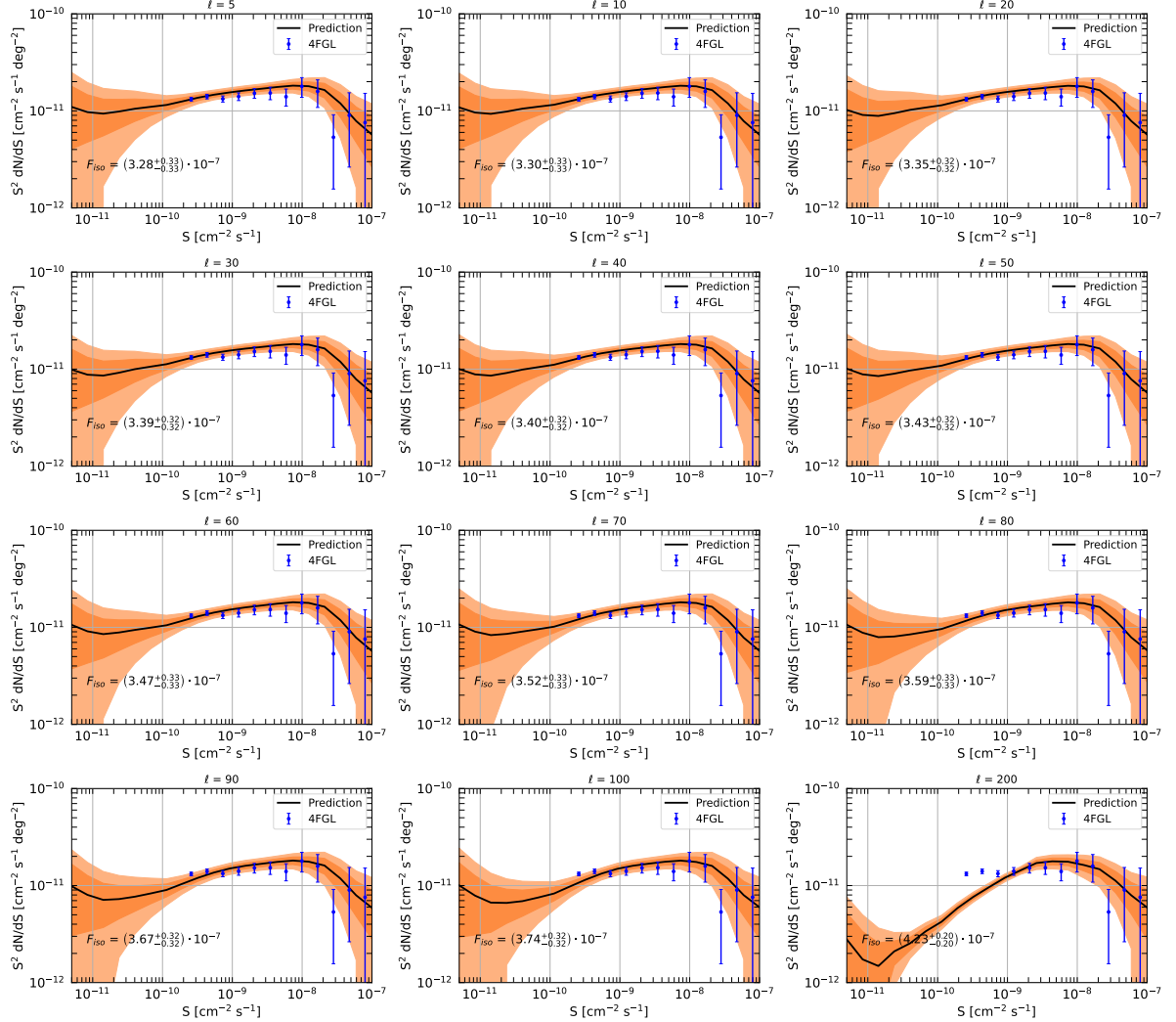


Fig. 6.24. Same as Figure 6.23 but using a CNN trained with the v05 version of the foreground model, as well as using as input a *Fermi*-LAT count map processed with the same foreground model.

6.6.5 Fully spherical CNN

In this section we show a comparison of the results obtained using a fully spherical CNN, instead of the map2patch algorithm used in the main analysis. In particular, we adopt here the NNHealpix [194] algorithm.

Figure 6.27 shows the reconstructed dN/dS , F_{iso} and their frequentist errors, using the spherical CNN applied to the *Fermi*-LAT map. The results are fully compatible with the ones obtained with the map2patch algorithm, both for the mean value and the errors, as can be seen by confronting with Figure 6.17.

Figure 6.28 instead, shows some examples of random input dN/dS and F_{iso} from the validation dataset of the Spherical CNN, and the corresponding quantities and their errors reconstructed by the CNN. The spherical CNN has similar performances as the CNN that uses map2patch, as can be seen comparing with Figure 6.12. The same occurs also in Figure 6.29,

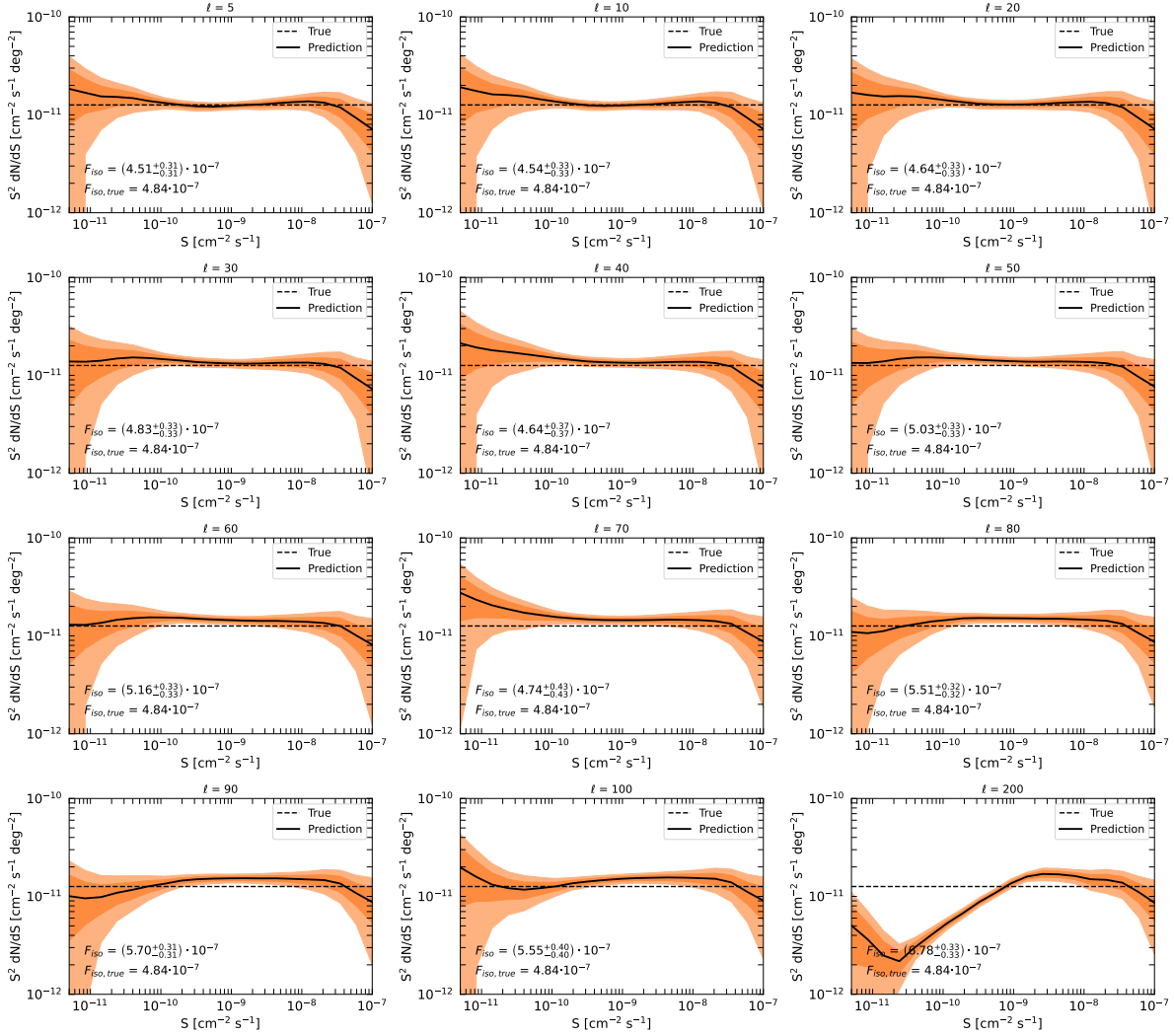


Fig. 6.25. Same as Figure 6.23 but using as input a synthetic map generated with a flat $S^2 dN/dS$ and with the v05 version of the foreground model.

which refers to a flat $S^2 dN/dS$, to be compared with Figure 6.20 obtained with map2patch.

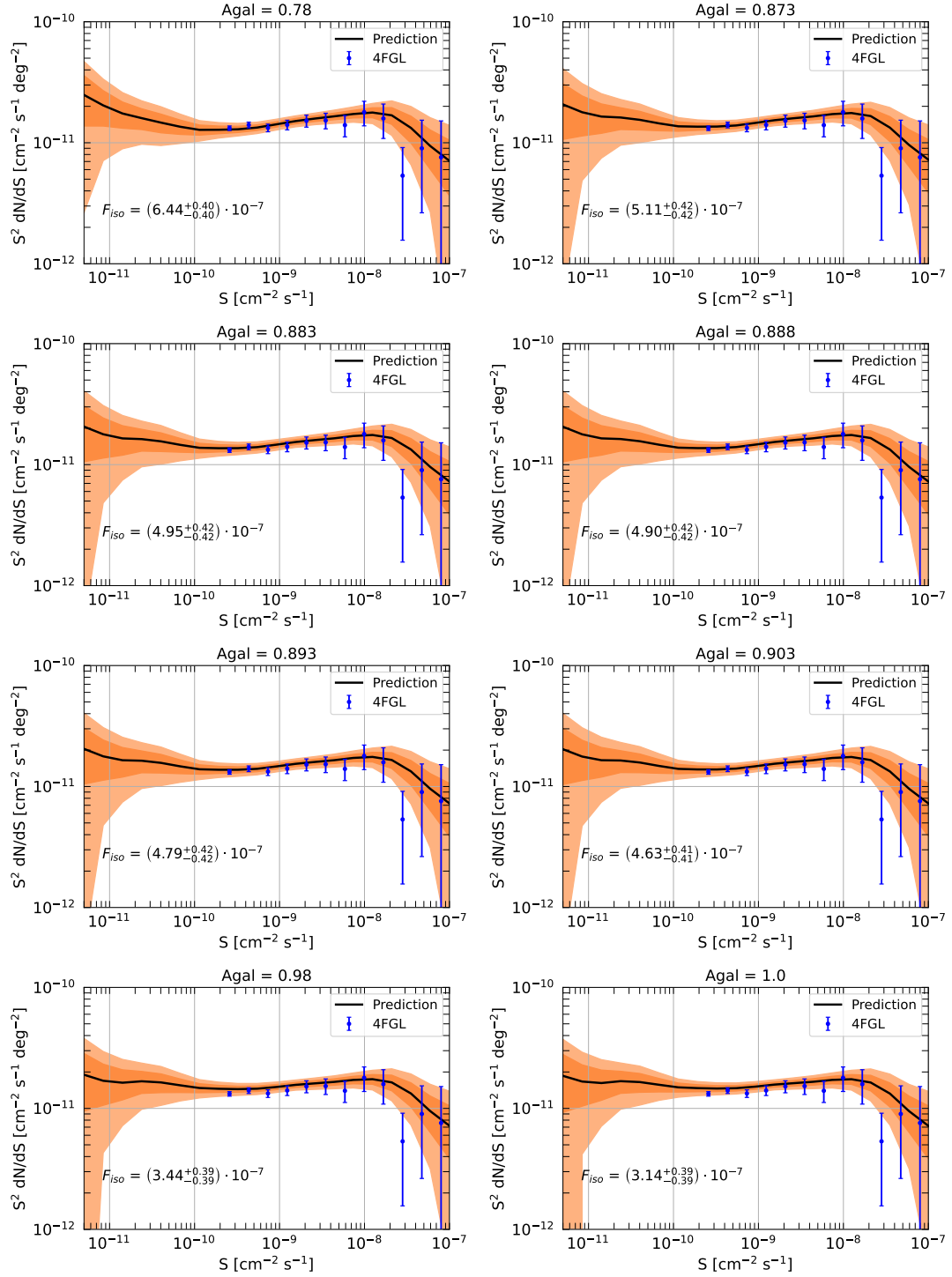


Fig. 6.26. Reconstructed dN/dS and F_{iso} and their Bayesian errors, when the trained CNN is applied to the *Fermi*-LAT map using different values of A_{gal} as indicated in the panel labels.

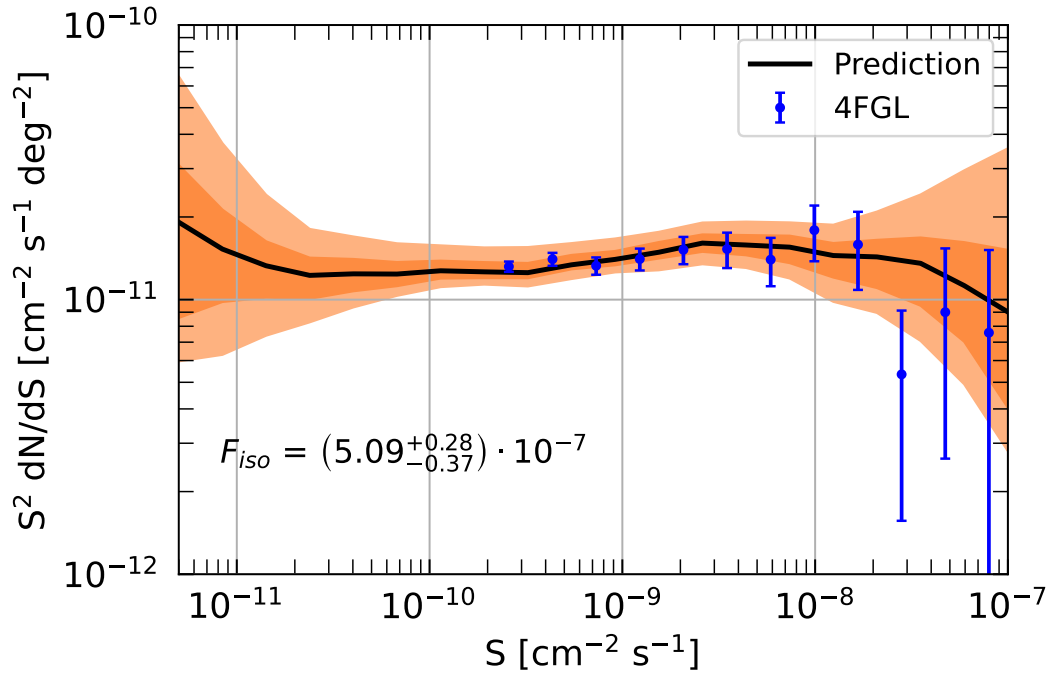


Fig. 6.27. Reconstructed dN/dS and F_{iso} and their errors, using a fully spherical CNN, trained on simulated maps and then applied to the *Fermi*-LAT map.

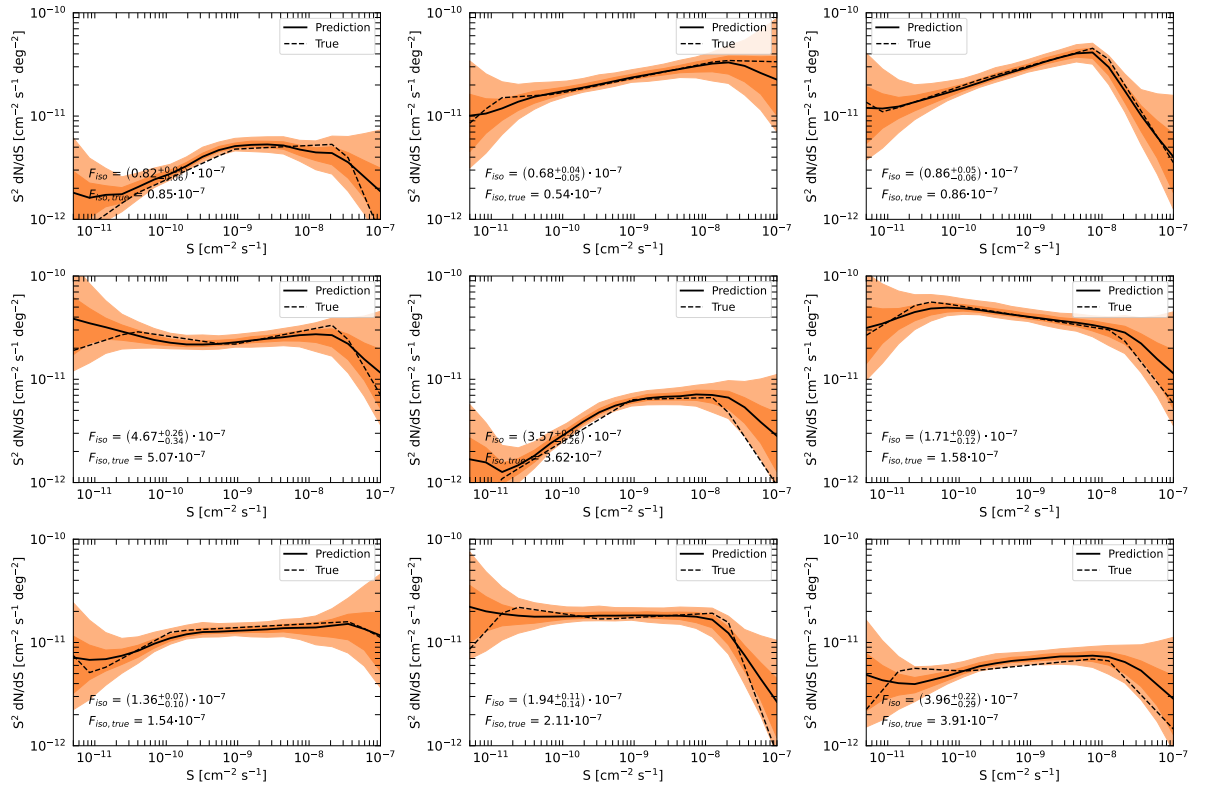


Fig. 6.28. Some example dN/dS from the validation dataset of the spherical CNN. Each panel shows the input dN/dS and F_{iso} and the corresponding quantities reconstructed by the CNN. The colored areas indicate the 1σ and 2σ uncertainty bands.

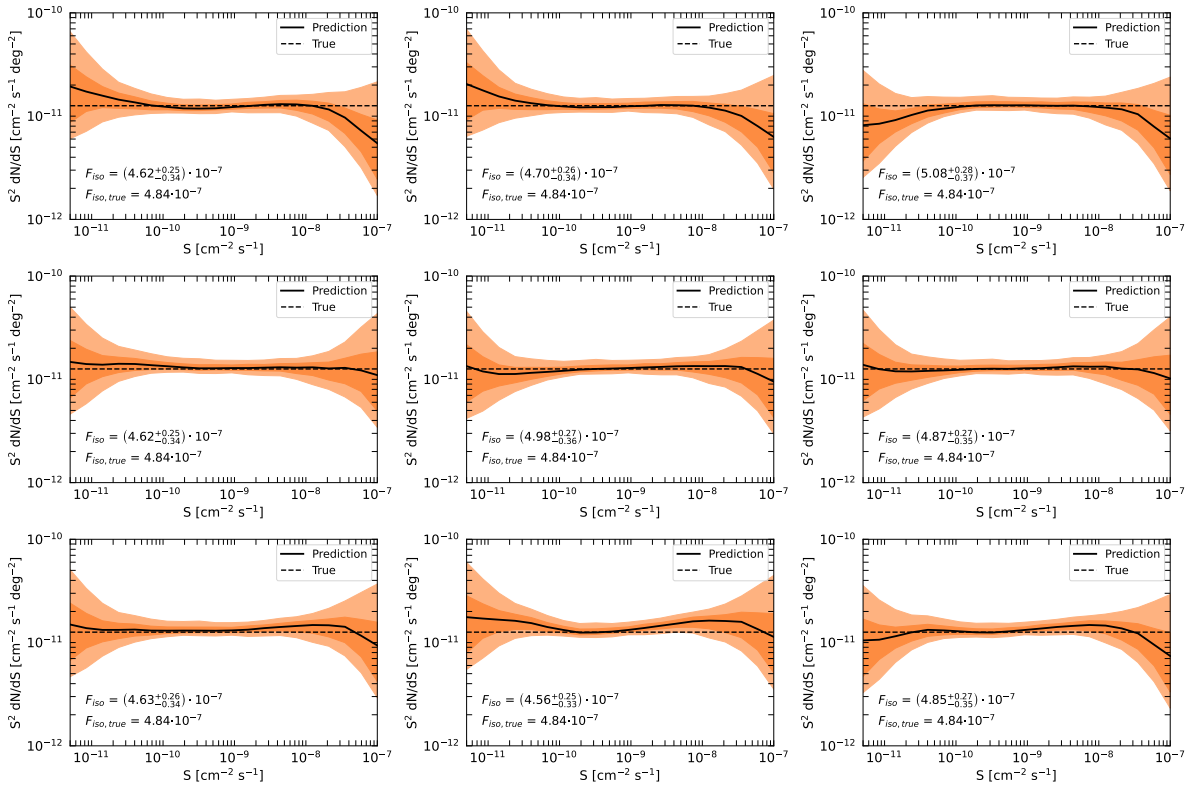


Fig. 6.29. Same as Figure 6.28 but for 6 different random realization of a flat $S^2 \frac{dN}{dS}$ and same F_{iso} .

PROBABILISTIC CATALOGING

The previous chapter demonstrated that the source-count distribution of faint gamma-ray emitters can be recovered from the photon-count statistics of the *Fermi*-LAT sky using simulation-based inference and deep learning. The resulting dN/dS extends as $\sim S^{-2}$ well below the nominal detection threshold, capturing a population of sub-threshold point sources — blazars, star-forming galaxies, misaligned active galactic nuclei — that contribute to the isotropic diffuse emission but remain individually undetected. Yet the dN/dS is a one-dimensional flux distribution: it tells us *how many* sources exist at each flux, but it says nothing about *where* any specific source is located. The challenge, then, is whether this population-level statistical knowledge can be turned into spatial information.

This chapter takes the next step by complementing the source-count distribution with positional identifications. The core idea is a frequentist comparison of the real sky against ensembles of synthetic skies generated from the recovered dN/dS , flagging candidate source directions below the detection threshold.

Section 7.1 examines why the standard fixed-threshold cataloging paradigm discards useful information and produces a spatially inconsistent detection scale. Section 7.2 describes the conceptual framework for converting the dN/dS into spatial source candidates through simulated-sky comparison, and introduces the frequentist meta-parameters that control the depth and purity of the resulting catalog. This framework is then applied to fourteen years of *Fermi*-LAT data, delivering a publicly available probabilistic catalog. Together with the dN/dS of Chapter 6, this probabilistic catalog completes the characterization of the astrophysical point-source foreground against which any residual dark matter contribution must be identified, as pursued in Chapter 8.

7.1 The Limits of Fixed-Threshold Cataloging

Before describing how the source-count distribution can be used to build a probabilistic catalog, we first examine the limitations of the standard approach. The fixed-threshold paradigm behind the *Fermi*-LAT catalogs has been in use for over a decade, but it comes at two costs: a spatially non-uniform detection significance, and the loss of all sub-threshold information. We take these in order.

7.1.1 The Standard Catalog-Construction Paradigm

The *Fermi*-LAT fourth source catalog (4FGL) is the most comprehensive catalog produced from the instrument described in Chapter 2, incorporating over a decade of survey-mode observations in the gamma-ray band [140]. Point-source detection in this catalog follows a well-defined statistical procedure. In each region of interest (RoI) partitioning the sky, the analysis evaluates a likelihood-ratio test statistic defined as $TS = 2\ln(\mathcal{L}/\mathcal{L}_0)$, where $\mathcal{L}$ is the maximum likelihood including the source model and $\mathcal{L}_0$ is the maximum likelihood for the background-only hypothesis [140]. Sources are retained in the catalog if they exceed a threshold of $TS > 25$, corresponding to a significance of approximately 4σ when evaluated against a χ^2 distribution with four degrees of freedom (two positional and two spectral parameters for a power-law model) [140]. Any candidate that falls below this threshold is removed from the model, and the fit in that RoI is readjusted accordingly.

This detection procedure is inherently local. The sky is divided into roughly 1750 RoIs, each containing a core region of actively fitted sources and an outer ring whose width adapts to the energy-dependent point-spread function [140]. Within each RoI, the normalization and spectral tilt of the Galactic diffuse emission template and the normalization of the isotropic component are fitted independently, and the log-likelihood is further split into components defined by energy intervals and event types before being summed for optimization [140]. Since the background model parameters are determined locally, the same numerical TS value carries different statistical weight at different sky positions. A source yielding $TS = 30$ in a sparse high-latitude field sits on a very different background than one with $TS = 30$ in the crowded Galactic plane. The resulting detection scale is therefore spatially inhomogeneous: the TS encodes the local excess above the locally fitted background, not a globally calibrated measure of source significance.

The fixed detection threshold imposes a binary classification on the sky. Every direction is either “in” the catalog ($TS \geq 25$) or “out,” and the information carried by pixels with TS values just below the cut — pixels that may well host genuine astrophysical sources — is entirely discarded. The construction of each successive data release inherits this binary structure. While such a threshold is operationally necessary to maintain a controlled false-positive rate across thousands of candidate seeds, it discards a range of TS values that, as we show below, can be recovered by comparing observed pixels against simulated sky ensembles.

7.1.2 The Information Below Threshold

The fixed detection threshold does not only exclude noise. Genuine astrophysical sources exist below $TS = 25$ — the dN/dS recovered in Chapter 6 shows this directly — and their collective emission contributes to the high-latitude gamma-ray sky at a level that depends on the population model assumed. As discussed in Section 6.2, blazars, misaligned active galactic nuclei, star-forming galaxies, and millisecond pulsars all contribute to this unresolved emission, and their aggregate flux may account for a large fraction of the unresolved gamma-ray background [131, 138], though the individual budget shares remain model-dependent.

The dN/dS measured from high-latitude photon counts (cf. Section 6.2) extends as $\sim S^{-2}$ down to fluxes of approximately $5 \times 10^{-12} \text{ cm}^{-2} \text{ s}^{-1}$, roughly 50 times below the nominal catalog sensitivity in the 1–10 GeV band [357], quantifying this sub-threshold population directly. This faint extension implies that many sources whose individual signals fall just below the $TS = 25$ cut-off carry real astrophysical information yet are systematically excluded from every standard catalog. For science cases that depend on sample size rather than per-source purity, this information loss is particularly costly. Cross-correlation studies between gamma-ray maps and galaxy catalogs, for instance, gain statistical power from a larger sample of source directions, even at

the price of a moderate false-positive fraction (cf. Chapter 8). Similarly, multi-wavelength source identification campaigns benefit from having a larger set of candidate positions to follow up.

The practical consequences of the fixed threshold become visible in the incremental catalog releases. In the 4FGL-DR3, which extends the observation baseline from eight to twelve years, 112 sources originally included in the first data release dropped below $TS = 25$, fifteen of which fell below $TS = 16$, with the lowest reaching $TS = 9.8$ [141]. An additional 69 sources from the second data release also fell below threshold [141]. These sources were retained for continuity and explicitly flagged, but the episode exposes a structural problem: highly variable objects such as flat-spectrum radio quasars can be bright enough to enter the catalog during one observation window and fade below threshold in a subsequent accumulation [179]. Lowering the threshold does not help, since it would increase false detections from background fluctuations [179]. What is needed is a framework that replaces the binary in-or-out decision with a simulation-calibrated probability of source presence, informed by the sub-threshold population measured through the dN/dS .

7.2 From Population Statistics to Spatial Information

The previous section showed that a fixed TS threshold discards sub-threshold sources whose aggregate properties — including the dN/dS — are measurable from photon-count statistics [322, 324]. The dN/dS is, however, a purely statistical quantity: it tells us how many sources exist at each flux level, but nothing about where they are. This section describes how the dN/dS can be complemented with spatial information by comparing ensembles of simulated skies against the real data.

The idea, developed in full later in this chapter (Section 7.3), is to use the recovered dN/dS as a generative model for synthetic skies and then compare their pixel-level TS distributions with the real *Fermi*-LAT sky using a frequentist test. Section 7.2.1 describes the simulated-sky comparison at a qualitative level; Section 7.2.2 introduces the statistical machinery — a per-pixel TS , a KS test, and a Quality Factor — that makes the comparison quantitative.

7.2.1 The Core Idea: Simulated-Sky Comparison

Since the dN/dS carries no spatial content (Section 7.1), the question we now address is how to leverage this population-level flux distribution to identify specific sky directions likely to host sub-threshold point sources.

The idea is straightforward. Source fluxes are drawn from the dN/dS and, because the population statistic carries no positional information, the sources are scattered at uniformly random positions on the sphere. Each synthetic sky is then processed through the same pipeline applied to the real data — PSF convolution, exposure weighting, and pixelization on top of a modeled diffuse background — and the procedure is repeated many times, resampling the dN/dS within its estimated uncertainties, to build an ensemble of skies that encode what the gamma-ray sky should look like if the recovered dN/dS is correct. The per-bin Poisson draws, the Gaussian-Process uncertainty treatment, and the number of realizations are detailed later in this chapter (Section 7.3).

The comparison between synthetic and real skies cannot be performed on a pixel-by-pixel basis, because the synthetic sources are randomly positioned and their spatial pattern never matches the real sky. Instead, the relevant comparison is statistical: one computes a pixel-wise test statistic (TS) for every pixel in both the real and synthetic maps and then compares the resulting TS distributions. At high TS , where bright sources dominate, the synthetic and real

distributions agree well. As one moves to lower TS values, the simulated distributions eventually diverge from the real sky, because background modeling systematics become increasingly important at faint fluxes. The threshold at which this divergence becomes statistically significant determines how deep below the nominal catalog limit the method can reliably probe. Pixels in the real sky whose TS exceeds this threshold are labeled “firing pixels” — candidate directions that are likely to host a genuine point source.

The simulated-sky comparison framework adopted here takes an alternative route from previous probabilistic-cataloging studies [179, 358]. Rather than sampling the full catalog posterior, it reduces the problem to a comparison of one-dimensional TS distributions, which is computationally inexpensive and naturally scalable to the entire high-latitude sky. The price for this simplification is that the method does not return individual source parameters (flux, spectrum, position uncertainty); its output is a map of candidate firing pixels at a chosen pixel resolution. This is sufficient for the downstream application we have in mind — cross-correlation studies (Chapter 8) — where the relevant quantity is the number and angular distribution of candidate source directions, not the properties of individual sources. We now turn to the statistical framework that makes this comparison quantitative.

7.2.2 A Frequentist Framework: TS, KS Test, and Quality Factor

The framework outlined in the previous section is entirely frequentist. The dN/dS enters as a generative model for the synthetic skies, not as a Bayesian prior to be updated. Three quantities define the statistical procedure.

A per-pixel test statistic quantifies how much the observed photon count in each pixel deviates from the background-only expectation. Unlike the TS used by the *Fermi*-LAT collaboration, which is computed locally in each region of interest and therefore carries spatially varying significance, the TS adopted here is a single, globally coherent quantity defined identically for every pixel. Its probabilistic meaning comes entirely from the comparison with simulated skies rather than from an assumed analytic distribution.

A Kolmogorov–Smirnov test then compares the cumulative TS distributions of the real sky and each synthetic realization above a minimum threshold TS^* . The value of TS^* at which the distributions cease to agree (cf. Section 7.2.1) sets the depth to which the method can probe.

Finally, a Quality Factor (QF) is defined as the fraction of synthetic realizations passing the KS test at a given TS^* and significance level α . A high QF yields a conservative set of firing pixels; a lower QF extends coverage deeper into the sub-threshold regime at the cost of more spurious directions. The pair (α, QF) sets the operating point, and both can be tuned to the downstream application.

Two features of this framework are worth noting. First, the globally homogeneous TS scale is particularly advantageous for statistical analyses that combine information across large sky areas, such as the cross-correlation studies discussed in Chapter 8. Second, because PSF spreading lets a single source fire several adjacent pixels while several faint sources can share one, the deliverable is a map of firing pixels rather than a traditional source list — a distinction quantified later in this chapter at the adopted pixel resolution.

The remainder of this chapter is based on [4] and presents the analysis in full: the data selection, the detailed statistical procedure, the results for different (α, QF) operating points, and the validation against the 4FGL-DR3 catalog.

7.3 Deepening gamma-ray point-source catalogues with sub-threshold information

The goal of this analysis is to probabilistically extend the *Fermi*-LAT source catalogue exploiting the dN/dS recovered in Chapter 6. We do so by proposing a new methodology applied to the *Fermi*-LAT data which consists in a) proposing a test statistic (TS) for the gamma-ray counts in the sky, whose distributions across spatial pixels corresponding to the *Fermi*-LAT measured map and those coming from the simulations can be compared; and b) identifying the pixels in the measured map whose TS values are large enough depending on the previous comparison, therefore overcoming the limitation of the dN/dS analysis of not being able to provide spatial information.

In Section 7.3.1 we describe the data selection, while Section 7.3.2 introduces the null- and alternative-hypothesis models used in the analysis. Section 7.3.3 is devoted to the quantitative setting of the problem and the statistical procedure followed. In Section 7.3.4 we present the results, which are also made publicly available at <https://doi.org/10.5281/zenodo.8070852>. In Section 7.3.5 we outline some perspectives and present our conclusions.

7.3.1 Data selection

We consider the updated 14 years *Fermi*-LAT data set (from week 9 to week 745) in the (1, 10) GeV energy range¹ and Pass 8 event selection [149, 150]. This ensures a good balance between high statistics and good angular resolution of the detector. We employ the *Fermi* Science Tools suite version 2.0.8 [331] to analyse *Fermi*-LAT data with the following settings: We adopt the P8R3_SOURCEVETO_V3 instrument response functions (IRF), event class (EVCLASS) 2048 (SOURCEVETO) and event type (EVTYPE) 1 (FRONT). We used standard quality selection criteria, i.e. DATA_QUAL==1 and LAT_CONFIG==1. Atmospheric γ rays from the Earth limb emission are removed by adopting a cut on the maximum zenith angle ZMAX of 90 degrees. We consider FRONT events in order to have an optimal angular resolution, and the SOURCEVETO class of events, in order to have a good suppression of the charged cosmic-ray background while still retaining a large event statistics. In table 7.1, we provide the reader with a summary of the settings.

We employ the *Fermi* Science Tools to compute photon counts map, exposure map and point spread function (PSF) of the LAT. The maps are binned in N_{pix} pixels with equal area relying on HEALPix.² The chosen pixelization is expressed in terms of the “resolution parameter” N_{side} controlling the number of subdivisions of a great circle on the sphere, related to N_{pix} via $N_{\text{pix}} = 12N_{\text{side}}^2$. As we motivate in section 7.3.4, we simulate the sky with $N_{\text{side}} = 1024$, but run the pixel analysis with $N_{\text{side}} = 512$.

We perform an energy-integrated analysis in the (1, 10) GeV energy range, but our procedure can be extended to multiple energy bins.

7.3.2 Null- and alternative-hypothesis models

Our goal is to devise an algorithm able to detect point sources as γ -ray excesses on top of a Poisson-distributed model for the background diffuse emission components. In this respect, our null-hypothesis model only consists of diffuse emission components, while the alternative hypothesis model also contains point sources.

¹The energy range is chosen for consistency with the dN/dS determination used below.

²<https://healpix.sourceforge.io>

Healpix order	9, 10
Weeks	9 – 745
Emin	1 GeV
Emax	10 GeV
Instrument Response Functions (IRFs)	P8R3_SOURCEVETO_V3
EVCLASS	2048 (Source Veto)
EVTYPE	1 (Front)
ZMAX	90

Table 7.1. Fermi Science Tools settings used for the 14-year data set analysis.

We construct our background-only map $\mathcal{B}$ including two components:

$$\mathcal{B} = A_{\text{gal}} \cdot \mathcal{G} + F_0, \quad (7.3.1)$$

where

- F_0 is an isotropic (usually assumed extragalactic) component accounting for the average flux of all the unresolved sources as well as possible diffuse emission mechanisms.
- $\mathcal{G}$ is the `gll_iem_v07` template morphology of the diffuse γ -ray emission from the Milky Way provided by the *Fermi*-LAT collaboration³, with normalization A_{gal} .

The parameters A_{gal} and F_0 are determined via a fit to the *Fermi*-LAT photon counts map, as explained in section 7.3.3. In order to obtain a background map in units of counts in each of the N_{pix} pixels, we multiply by the exposure map $\mathcal{E}$ and the steradian-to-pixel conversion factor $4\pi/N_{\text{pix}}$:

$$C_{\mathcal{B}} = \frac{4\pi}{N_{\text{pix}}} \int_{1\text{ GeV}}^{10\text{ GeV}} \mathcal{B}(E) \mathcal{E}(E) dE. \quad (7.3.2)$$

The exposure map is extracted using the Fermi Science Tools for 10 logarithmic bins in the (1, 10) GeV range (and treated as the corresponding piecewise function when performing the integral), in order to be consistent with the binning of the Galactic foreground model. We also account for the PSF, responsible for an angular smoothing of the map, via the Fermi Science Tools. Similarly to the map $\mathcal{E}$, we average over the (1, 10) GeV range with a $E^{-2.4}$ weight, according to the overall energy dependence of the data at high Galactic latitudes [333].

On the other hand, our first alternative hypothesis model is based on the 4FGL-DR3, and will be used to validate our procedure and TS analysis on the official *Fermi* catalogue. We generate a new map

$$\mathcal{K} = \mathcal{C} + A_{\text{gal}} \cdot \mathcal{G} + F_0, \quad (7.3.3)$$

including the same background model as in eq. (7.3.1) (i.e. same parameters for A_{gal} and F_0), and adding a synthetic map associated to the list of sources in the 4FGL-DR3 catalogue with $|b| \geq 30^\circ$, $\mathcal{C}$. The map is appropriately smoothed via the PSF and convoluted with the exposure map, and converted into a pixelized count map, as previously described.

Finally, as our second alternative hypothesis we consider a model for point sources based on the source-count distribution derived in [3]. Analogously to the catalogue, we include an additional source component, $\mathcal{S}$, on top of the background-only model:

$$\mathcal{M} = \mathcal{S} + A_{\text{gal}} \cdot \mathcal{G} + F_{\text{iso}} \quad (7.3.4)$$

³<https://fermi.gsfc.nasa.gov/ssc/data/access/lat/BackgroundModels.html>

The map $\mathcal{S}$ is constituted of point sources randomly placed on the sphere, drawn according to the dN/dS inferred in [3]. We produced 5000 realisations of this source model, by considering variations of the dN/dS parameters inferred in [3]. In particular, we adopt the model related to the `gll_iem_v07` foreground template ⁴ and $|b| < 30^\circ$ Galactic plane cut, and we vary the dN/dS within the estimated uncertainties by fitting a Gaussian Process (c.f. [359]) to the dN/dS output of the neural network ⁵. For each parameter configuration of the dN/dS , the F_{iso} is consistently determined by the procedure proposed in [3]. Note that the residual Poisson-distributed isotropic component F_{iso} is now lower than F_0 in eq. (7.3.1), since part of the flux F_0 is accounted for by the discrete sources.

We note that both the $\mathcal{K}$ and $\mathcal{M}$ maps are simulated, i.e. are “theoretical skies”, not the “measured sky”. The map $\mathcal{K}$ is deterministic, meaning we simulate it by accounting for the Fermi PSF for the point-like sources whose fluxes and positions are given by the 4FGL catalogue, on the top of the assumed smooth background (the remaining contributions to the $\mathcal{K}$ map in eq. 2.3). In the case of the map $\mathcal{M}$, we rely on statistical information, in which the dN/dS does not describe the position of any gamma-ray source, nor does it tell us their exact number. Rather, the integral of dN/dS in a flux bin tells us the expected number of sources in that bin, while the actual number of sources is only defined up to Poisson fluctuations. Once such a number of sources per flux bin has been determined, we place them at uniformly random positions in the sky, accounting for the S term in Eq. (7.3.4). In summary, we run the simulator developed in [3] in order to build the maps $\mathcal{M}$.

7.3.3 Problem setting and statistical framework

Our first step is the determination on the null-hypothesis model, consisting solely of Poisson-distributed background emission. To this end, we apply a mask of $|b| < 30^\circ$ around the Galactic plane, and remove the emission of all sources in the 4FGL-DR3 catalogue, further masking a 1° disk around each source centroid. This cut roughly corresponds to the 90% containment angle of the PSF and represents a good compromise between removing the bulk of the pointlike emission while avoiding a too large removal of the diffuse emission area. We checked that the results are not particularly sensitive to the exact cut used in the $1^\circ - 2^\circ$ range. We then compute the expected photon counts λ_i of model $\mathcal{B}$ (depending on A_{gal} and F_0) in the pixel i and, given the actual counts k_i , define the Poissonian likelihood

$$\mathcal{L}_0 = \prod_{i=1}^{N_{\text{pix}}} \frac{\lambda_i^{k_i} e^{-\lambda_i}}{k_i!}. \quad (7.3.5)$$

The adopted parameters A_{gal} and F_0 are those maximising the likelihood above. For $N_{\text{side}} = 512$, we find $A_{\text{gal}} = 0.914$ and $F_0 = 4.81 \cdot 10^{-7} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$.

Our second step is to quantify the significance of additional point sources on top of the null-hypothesis model. For this purpose, we define a test statistic (TS) function in each pixel, quantifying how well the observed photon counts agree with the background-only hypothesis:

$$TS_i \equiv \frac{(x_i - \lambda_i)^2}{\lambda_i}, \quad (7.3.6)$$

⁴This is the latest foreground model provided by Fermi-LAT.

⁵The reason for considering a Gaussian Process here is two-fold: to interpolate the dN/dS to values of the flux not included in the neural net output, as well as to have an estimation of the uncertainty band of dN/dS at such interpolated values.

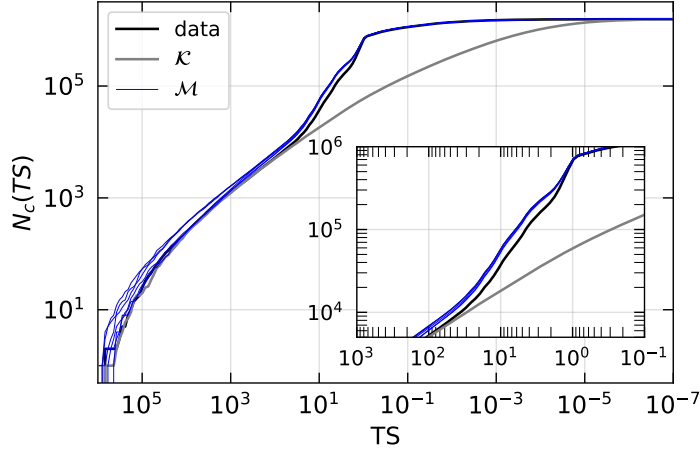


Fig. 7.1. Descending cumulative TS distribution for the maps corresponding to: real data (black), 4FGL-DR3 catalogue (gray) and a few realisations of the dN/dS source model (blue).

where we indicate with λ_i the best-fit background-only model counts in the pixel i , computed according to eq. (7.3.1); x_i can be either the observed photon counts (when applied to the actual photon count map of data, leading to the single set TS_i^{data}) or the photon counts in pixel i generated via synthetic maps. If the maps are generated according to the hypothesis eq. (7.3.3), the procedure yields the single set TS_i^{cat} ; if the maps are randomly taken from the model of eq. (7.3.4), we obtain a number of sets TS_i^{sim} equal to the number of realisations. The TS choice of eq. (7.3.6) is inspired by Pearson’s χ^2 test statistic for Poisson distributed counts, whose variance $\sigma_i^2 = \lambda_i$. This choice also loosely mimics the one adopted by the *Fermi*-LAT collaboration, albeit no exact correspondence can be expected. In fact, *Fermi*-LAT uses a recursive procedure where, after a first step similar to the above, energy-dependent renormalisations of the background are allowed independently in each “region of interest”. We prefer to use a simple and coherent definition valid for all pixels in the sky, also because we do not assume a particular probability distribution for the TS values. Rather, we use it as a “signal interest label”, and derive a probabilistic interpretation for it from simulations of synthetic maps (see below).

Obviously, since in our simulations the sources are uniformly distributed across the sky, the spatial distribution of TS_i^{sim} never coincides with the one obtained from the actual data, TS_i^{data} , nor with the one from the 4FGL-DR3 catalogue, TS_i^{cat} . However *the histogram of the TS_i^{sim} values should statistically match the actual one TS_i^{data} , if the functional form and inferred parameters of the underlying dN/dS are correct.* Since the determination of dN/dS exploits information below the nominal catalogue’s threshold, the statistical agreement between the actual map and the synthetic maps should hold at values below this threshold and inform us about new, faint sources. This is the gist of the argument that we exploit to extend *statistically* the *Fermi*-LAT catalogue below threshold. This expectation is illustrated in figure 7.1, where we report the descending cumulative TS distribution for real data (black), the 4FGL-DR3 catalogue source model, $\mathcal{K}$ (gray), and for several realisations of the dN/dS source model, $\mathcal{M}$ (blue). While the distributions match rather well at high TS , the one associated to the catalogue departs from the data one well before the simulation’s ones do.

As a next (and third) step, we attribute a statistical meaning to the TS distributions. At large TS , we expect the observed map and the simulated ones to agree well: Visible (but insignificant)

discrepancies are expected at the highest values of TS , where Poisson fluctuations are important. At low TS we expect the deviations from the two distributions to eventually become statistically significant. This is due to the interplay of the weakness of the sources and the systematic errors in the background template, that are more and more relevant at low fluxes.

To assess how well the two distributions are in agreement, we apply a two-sample Kolmogorov-Smirnov (KS) test [160] to the normalised cumulative distribution functions of real data and simulated sky (or catalogue sky), both cut at a minimum TS_* , which we vary on a grid. Their compatibility depends on the confidence level $1 - \alpha$ at which we set the test. This is strictly speaking arbitrary, although conventionally α is set to 0.05 in most applications. We will actually consider three values of α : 0.01, 0.05 and 0.1, to assess the dependence of the results on this meta-parameter. In general, increasing α yields more conservative constraints, as the criterion to reject the null hypothesis becomes more restrictive (i.e. the distributions should agree more closely).

For the single catalogue model in eq. (7.3.3), we present the results in terms of the TS_* leading to agreement with real data as a function of α .

In the case of the dN/dS model, since we have multiple realisations, we quantify what is the fraction of simulations which passes the test against the true sky distribution as a function of TS_* (and α). This provides a *quality factor* QF that we associate to the considered cut. Lacking a better criterion, a reasonable choice may be to require a majority of realisations to lead to an agreement (i.e. $QF > 0.5$), but it is strictly speaking another meta-parameter which depends on the criteria of the user. We will also show how results depend on QF , with QF 's closer to 1 associated to more restrictive requirements. Note that we will “naively” interpret the QF as probability of the TS being associated to a source, by considering the whole sample of simulations as equally viable (i.e. only the prior information based on the pixel statistics is used). In principle, one could refine this indicator by restricting the sample of simulations to the ones which reproduce the high- TS distribution. We expect the actual probability to be somewhat higher than the one estimated via the QF . Also note that nowhere in our procedure we have relied on the 4FGL catalogue as input.⁶

The final step is how to identify the “firing pixels” in the true sky which are going to constitute the main deliverable of our work. For a fixed KS test significance α and a quality factor QF , we obtain a value TS_* above which a fraction QF of simulations is compatible with the measured sky. Those pixels i in the latter having $TS_i > TS_*$ will simply constitute of candidate pixels.

We conclude this section with a general remark. We note that our procedure treats every pixel independently, assigning a TS value to each of them, regardless of the TS values of the neighboring pixels. In this respect, it is useful to distinguish between a source and a “firing pixel”: Multiple pixels may be firing (i.e. having a sufficiently large TS) in association to a single source, depending on the choice of N_{pix} , the intensity of the source and the PSF of the instrument (indirectly, thus, also on the spectrum of the source). This instance is more and more likely the smaller the pixel size, i.e. the larger N_{pix} is. Conversely, multiple sources may be associated to a single pixel, notably for choices of N_{pix} which are too low. Clearly, for the exercise proposed here to be useful, we must require that the choice of N_{pix} leads to a number of pixels in the non-masked sky sufficiently larger than the current number of sources in the catalogue in the same region.

⁶Technically, we mask the catalogue position to fit the background model, but this is done only to lead to a better background model and thus a more meaningful TS scale. In principle, we could fit the background model without masking the catalogue source positions. That would alter our TS scale, but the same bias will affect the simulated sky maps that we use to attribute a statistical meaning to the TS .

At the same time, a pixel size much smaller than the angular resolution would be meaningless. Keeping this in mind, for the sake of simulations, we fix $N_{\text{side}} = 1024$, which corresponds to an angular resolution of approximately 0.06° , in order to match the angular resolution of the Galactic foreground template. However, since the *Fermi*-LAT angular resolution in the (1,10) GeV energy band is at best of the order of 0.15° ⁷, for the sake of our pixel analysis, we will downgrade our simulations to $N_{\text{side}} = 512$, which corresponds to an angular resolution of approximately 0.12° . The choice of this value for N_{side} ensures that there are considerably more pixels than the number of resolved sources (approximately 500 times the number of resolved sources in the region $|b| > 30^\circ$) and that the pixel size of our analysis roughly matches the best resolution of the LAT at the energies of interest. The TS analysis is therefore performed with synthetic and real data maps pixelized with $N_{\text{side}} = 512$.

7.3.4 Results

We comment here on some summary statistics and synthetic analysis of our main results. In table 7.2 we report, for different values of α , the TS_* above which the 4FGL-DR3 catalogue source model and real data pass the KS test (second column, TS_*^{cat}), and the corresponding TS_* where the dN/dS simulated sky maps do the same (TS_*^{sim}), at different values of QF .

α	TS_*^{cat}	$TS_*^{\text{sim}}(QF = 0.9)$	$TS_*^{\text{sim}}(QF = 0.8)$	$TS_*^{\text{sim}}(QF = 0.5)$
0.01	53	57	42	33
0.05	62	147	48	36
0.10	63	307	74	37

Table 7.2. Required TS_* for three values of α (KS test confidence level) for the matching between 4FGL-DR3 catalogue source model and real data (second column, TS_*^{cat}), and the corresponding TS_* for the dN/dS simulated sky maps vs real data (rest of columns, TS_*^{sim}), at different values of QF .

α	$\widehat{TS}_*^{\text{sim}}(QF = 0.9)$	$\widehat{TS}_*^{\text{sim}}(QF = 0.8)$	$\widehat{TS}_*^{\text{sim}}(QF = 0.5)$
0.01	181	161	104
0.05	301	180	148
0.10	400	200	160

Table 7.3. Required TS_* for three values of α (KS test confidence level), for the dN/dS simulated maps to match the 4FGL catalogue model map, at different values of QF . Note that the KS test is different from the one in table 7.2, where the matching was with respect to the *Fermi*-LAT data map.

The first conclusion is that, provided that the choice of QF (and/or of α) is not too stringent, our model maps $\mathcal{M}$ agree with the data better than the catalogue map does, i.e. the agreement extends to lower TS_* . Compare in particular the second with the last column in table 7.2. This is just a manifestation of the fact that, at least for a majority of realisations, including sub-threshold sources provides a much better description of *Fermi*-LAT sky map. On the other hand, requiring a too large QF may be counter-productive: Even at high- TS , there is a non-negligible fraction of

⁷<https://fermi.gsfc.nasa.gov/science/instruments/table1-1.html>

the simulations that does not agree with the data as measured by the KS test. This manifests in the high values of TS_* found in the $QF = 0.9$ column and to some extent for $QF = 0.8$ as well.

A second sanity check is that the dN/dS model is better in agreement with the data than with the catalogue: This manifests in the values of $\widetilde{TS}_*$ (see table 7.3) for which the dN/dS model yields maps in agreement with the catalogue map $\mathcal{K}$, hence indicated with a tilde. As we can see from the table, the $\widetilde{TS}_*$ values are significantly higher than the corresponding TS_* reported in table 7.2.

A further natural question to ask oneself is if the statistical procedure leads to a competitive sample of candidate source positions in the sky, when compared to the *Fermi*-LAT catalogue analysis. After all, requiring an agreement of the TS histograms in the KS-sense is a *qualitatively* different procedure than the one used by the collaboration to infer their catalogue. A direct comparison is impossible, if anything since our discretisation of the sky only allows us to identify firing pixels, which are not in a one-to-one correspondence with sources, as discussed in the previous section. However, for a reasonable choice of the pixel size as the one adopted here (see discussion at the end of section 7.3.3) we expect the number of firing pixels to be a reasonable proxy of the number of underlying candidate sources. Our results are reported in table 7.4.

α	N^{cat}	$N^{\text{sim}}(QF = 0.9)$	$N^{\text{sim}}(QF = 0.8)$	$N^{\text{sim}}(QF = 0.5)$
0.01	7412	7050	8543	10068
0.05	6707	3975	7891	9636
0.10	6580	2492	5965	9409

Table 7.4. Number of firing pixels associated to the same cases of table 7.2.

A comment on this result is that the number of pixels inferred from the catalogue map is in the ballpark of the sources reported in the *Fermi*-LAT catalogue. This result was not obvious a priori. We may tentatively conclude that the power of the procedure we follow in identifying a firing pixel is at least roughly comparable to the criterion the *Fermi*-LAT collaboration follows in identifying a source. A second interesting conclusion is that the number of firing pixels can be significantly higher (up to a factor 1.5) if using the dN/dS model maps, at least if sufficiently loose criteria are used for QF and/or the level of KS agreement, i.e. α . We can conclude that our procedure can be used to increase the number of candidate pixels in the sky that we can probabilistically attribute to sources, with respect to the ones one would deduce by merely using the existing catalogue.

A question arises though: Are the identified pixels in the sky truly associated to sources? Of course we cannot answer this question without lengthy and careful follow-up studies and extensions of the present approach. We discuss some interesting directions in the next section. However, an interesting sanity check is that the bulk of the positions of *Fermi*-LAT sources should match part of the recovered firing pixel positions.

In figure 7.2 we compute, as a function of the flux, the fraction of all the pixels in the catalogue map $\mathcal{K}$, having a flux $S > S_i$, which coincide with the pixels of our firing pixels map obeying the same flux condition, for different TS cuts. The pixel coincidence is actually relaxed to a distance corresponding to the 68% containment angle of the PSF, which naively accounts for the uncertainty of the quoted source centroids in the 4FGL catalogue. In figure 7.2 we compute the fraction of all the pixels in the catalogue map $\mathcal{K}$ with flux $S > S_i$, whose positions are found

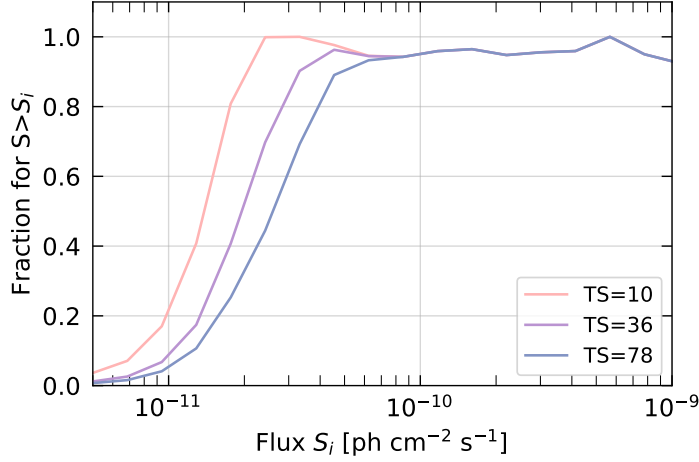


Fig. 7.2. Fraction of sources in the model $\mathcal{K}$ map, whose position corresponds to a firing pixel for different values of TS . See text for details.

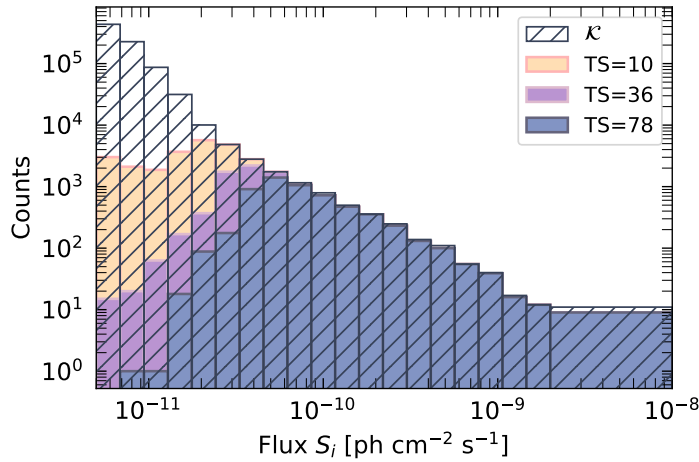


Fig. 7.3. Analogous to figure 7.2, but for differential flux; i.e. number of pixels for a given flux bin. White histograms correspond to the whole catalogue map $\mathcal{K}$, while colored histograms are only those pixels in $\mathcal{K}$ coinciding (within the 68% PSF containment angle) with the firing pixels of our dN/dS model map.

in spatial coincidence with pixels in our firing-pixels map obeying the same flux condition.⁸ We display the fraction as a function of the flux and for different TS cuts.

Roughly⁹, *Fermi*-LAT starts to lose sensitivity to sources below $2 \times 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$ [3, 319, 324, 360], above the rising region of the sigmoid-like curves of figure 7.2. As expected, modulo fluctuations, the fraction tends towards unity at high fluxes. Note that we do not expect perfect recovering of the sources in the *Fermi*-LAT catalogue, anyway, either due to some simplifications (e.g. we use energy-integrated quantities for our analysis, as opposed to the differential energy

⁸The pixel-wise spatial coincidence is actually relaxed to a distance corresponding to the 68% containment angle of the PSF, which naively accounts for the uncertainty of the quoted source centroids in the 4FGL catalogue.

⁹More quantitative considerations require energy- and position-dependent assessments.

N	coincidence
3000	98.4%
5000	97.8%
9000	97.8%

Table 7.5. The fraction of firing pixels in common between the selections derived for the two galactic foreground models. The comparison is performed at equal number of selected sources for the two cases.

analysis by the LAT collaboration) or methodological differences, such as the pixelisation procedure or the iterative background estimate procedure in *Fermi*. But it is clearly the case that we do recover the largest majority of the “bona fide” sources that we should.

Figure 7.3 gives us a more detailed look at the information contained in figure 7.2. In this case we do not consider the “cumulative” number of pixels, but only those lying in a certain flux bin. The colored histograms correspond to the pixels in map $\mathcal{K}$ matching with the firing pixels of our dN/dS model map (related to the numerator of the fraction computed in figure 7.2). The white histogram instead correspond to all the pixels in map $\mathcal{K}$ (related to the denominator of the aforementioned fraction). The sigmoid curve in figure 7.2 grows close to unity at large S since—barring fluctuations—the quasi-totality of sources in the catalogue are found within the adopted cuts. When lowering the cut TS , the peak of the colored histogram shifts to lower S , while still preferentially populating the high- S tail (compare case $TS = 36$ with case $TS = 78$). If the TS is lowered too much, as illustrated by the case $TS = 10$, the matching starts developing a “flat tail” at lower S , hinting at a growing random match component. Eventually, the curves in figure 7.2 drop to zero at low S not because of the scarcity of matchings (i.e., the numerator), but due to the faster growing number of pixels having flux above the indicated value (i.e. the denominator).

As a further check, we assess how robust the selected catalogue is to variations of the background model. We compare the results obtained with the `gll_iem_v07` template with the previous incarnation of the galactic foreground model developed by the Fermi collaboration: `gll_iem_v05_rev1`. Since the TS^* scale and related QF are quantitatively different if the different backgrounds are used, we proceed by comparing the two results *at equal number of selected sources*. We select TS^* cuts delivering the same number of selected sources for the two cases and evaluate the exact overlap of pixel candidates. The results are reported in Table 7.5. As can be seen, within a few percent, the catalogues obtained agree.

The main output of our work consists of sky maps of “firing” pixels (i.e. source candidates). These pixels have a significant TS value among all pixels for which there is a statistical compatibility between simulations and real data, according to KS tests. The firing pixels depend on the significance α of the KS test, as well as on a “quality factor” QF , accounting for the fraction of simulated maps passing the KS test. We provide our results electronically in the form of a Python package called gPCS (for gamma-ray Photon Count Statistics), as well as a summary FITS, both available on Zenodo: <https://doi.org/10.5281/zenodo.8070852>. The Python package can be used to determine the firing pixels given either a TS_* or a quality factor and α parameter. The Python package also includes a convenience function to export the firing pixels in the form of a FITS file, and we provide an example on how to compute the summary table provided with the package. The gPCS Python package can be easily installed using pip, with the command:

```
pip install gPCS
```

In order to get the firing pixels, given either TS_* or a quality factor and α , we can use the following snippet of code:

```
import numpy as np
from gPCS import gPCS

# specify manually a  $TS_{star}$ 
TS_star = 36
pixel_firing = gPCS.get_firing_pixels(TS_star, filter=False)
print(len(pixel_firing))

# Compute the  $TS_{star}$  from a chosen  $QF$  and  $\alpha$ 
# [alpha must be 0.01, 0.05 or 0.1]
QF = 0.5
alpha = 0.05
TS_star = gPCS.get_TS_from_QF(QF, alpha=alpha)
pixel_firing = gPCS.get_firing_pixels(TS_star, filter=False)
print(len(pixel_firing))
```

It is possible to export a FITS table using the `export_fits_table` function. For example to reproduce the table provided with the library we can use:

```
gPCS.export_fits_table(filename="firing_pixels.fits", QF=0.50, alpha=[0.01, 0.05, 0.1])
```

The FITS table has the following fields:

```
pixel      : pixel index
TS         : TS value for each pixel
QF_best    : QF value obtained by considering all the simulations
QF_min     : lower bound of the QF range
QF_max     : upper bound of the QF range
```

For further details on the package usage, please refer to the documentation and examples: <https://github.com/aurelio-amerio/gPCS>.

7.3.5 Discussion and conclusions

We have presented an approach to statistically push the *Fermi*-LAT sensitivity to point-like sources at high latitudes below the current threshold for detection, leveraging on the fact that the underlying source count distribution function is constrained even for lower test statistics (TS). This function has been recently re-derived using machine learning methods, yielding results in accordance with existing pixel statistics analyses. This information, not accounted for in the current catalogue construction procedure, allowed us to provide a catalogue of directions (i.e. firing pixels) in the sky likely to be associated to sources, albeit only in a probabilistic sense, that we assessed via a Kolmogorov-Smirnov (KS) test. The actual catalogue of directions depends on meta-parameters such as the confidence level $1 - \alpha$ of the KS test, or the fraction QF of simulated skies for which the simulation agrees with data above a certain TS level. For reasonable choices of these meta-parameters, we found a number of “firing pixels” $\sim 50\%$ higher than what one would infer from a catalogue only including sources listed in the latest incarnation of the *Fermi*-LAT catalogue. Our results also pass some sanity checks, like the fact that *Fermi*-LAT catalogue

sources which are luminous enough are found within the directions inferred by our method, within the angular resolution of the instrument. If compared with the “local” iterative procedure followed by *Fermi*-LAT to establish their catalogue, another possible advantage of our catalogue of directions in the sky is that its TS scale has a homogeneous meaning, and may be more suitable for global statistical analyses.

We provide our results in digital form, allowing the user to select the meta-parameters QF and α to be more or less restrictive in their selection criteria, hopefully adapting to a wide range of applications. The most obvious one we can think of consists of cross-correlation studies, either multi-wavelength or multi-messenger, where the statistical advantage of a significantly larger sample more than compensates having a (sub-leading) fraction of spurious directions. Another straightforward use of our results could be a guided source search and identification program via multi-wavelength studies, which would also help transforming some of these candidate source directions into *bona fide* sources. In turn, these studies may further benefit of machine learning techniques that have been recently proposed to ease the probabilistic classification of unassociated sources [179].

We stress that, although our results are based on the source-count distribution as inferred from a specific machine learning analysis of the photon counts, the method developed here to identify “firing pixels” can be applied to any other determination of the dN/dS of the high-latitude sky, and it is, in this respect, independent on the choice of the dN/dS as long as the latter correctly describes point sources in the faint regime.

Since our work had primarily a methodological proof-of-principle motivation, we have resorted to a couple of technical simplifications: We worked with an energy-integrated spectrum and we limited ourselves to high Galactic latitudes. We could rather naturally lift the former approximation by a brute force approach. We expect the main complication to be computational, but the problem should remain affordable in a reasonable time. Also the latter limitation could be in principle lifted, extending the analysis to low Galactic latitudes. The main problem we can anticipate is that the existing background models fail to perform as well at low latitudes as they do at high latitudes. One also needs to include multiple populations contributing to both resolved and unresolved emission, see e.g. [143] and sec. 6.2 in [152] for the role played by Galactic millisecond pulsars in this context. In the worst case, the method may not allow one to gain much more with respect to more traditional techniques. Depending on the outcome, one may however think of multi-zone models for the background or the like.

A third direction for future progress is to attach a more realistic probability scale to the TS maps. We have constructed our synthetic skies only based on the statistical properties for the sources previously deduced via the pixel analysis. However, only a subset of these simulations is consistent with the properties of the actual sky: Notably at high flux where only a few sources exist, the associated TS distribution significantly depends on the distribution of the sources in the sky randomly falling in a lower or higher background region (for an illustration, see the large variability at high TS in figure 7.1). As briefly discussed earlier in this chapter, a promising approach would be to narrow the sample of admissible synthetic skies used to compute QF to the sole simulations which are statistically consistent with the TS pattern of the most luminous sources. Such ‘constrained’ simulations should then improve the reliability of the QF scale. Another working direction with the same goal could be to improve the description of background and/or the parameterization for sub-threshold sources in the pixel analyses beyond what considered in [3]. This may bring the simulation distribution illustrated in figure 7.1 in even better agreement with the TS histogram. Finally, exploring alternative metrics to the agreement in the KS sense is also a direction potentially worth exploring.

Part IV

Cross-Correlations

CROSS-CORRELATIONS AND FUTURE PROSPECTS

In Chapters 4 and 5, we looked for localized dark matter signatures: the concentrated gamma-ray emission expected from the Galactic Center — where the dark matter density reaches its peak — and from individual subhalos that remain unresolved among the unassociated sources detected by the *Fermi* Large Area Telescope. Neither strategy returned a definitive signal. The Galactic Center excess persists, but a decade of increasingly sophisticated analyses has not resolved the fundamental degeneracy between a dark matter annihilation origin and a population of unresolved millisecond pulsars [80, 213, 216, 237]. The search for dark matter subhalos among unassociated sources [2] found no significant evidence for an anomalous population at current sensitivity levels, consistent with theoretical expectations — a result that establishes competitive constraints but does not close the door. We then moved to studying the unresolved gamma-ray sky. At the population level, the source-count distribution and probabilistic catalog constructed in Chapters 6 and 7 mapped the unresolved gamma-ray sky to fluxes roughly 50 times below the detection threshold [3, 4, 324], characterizing the astrophysical emission from which any dark matter contribution must be separated, without attempting that separation.

In this chapter we focus on how to isolate a signal from dark matter hidden in the unresolved gamma-ray background (UGRB). In fact, if dark matter annihilates or decays, it does so inside every gravitationally bound structure across all cosmic epochs, and the accumulated, unresolved emission from this entire population would contribute to the UGRB. The search now shifts from the Galactic targets of the previous chapters to extragalactic dark matter — the cumulative signal from halos throughout the universe. Because the annihilation emission is weighted towards dense, nearby structures, any signal accessible to this kind of analysis would most likely originate in the local universe ($z \lesssim 0.1$). Separating a dark matter component from the astrophysical foreground — blazars, star-forming galaxies, misaligned active galactic nuclei — requires more than spectral information alone, since distinct source classes partially overlap in energy. What discriminates dark matter is its unique spatial distribution: the large-scale structure of the universe, traced by dark matter halos from galaxy clusters down to sub-Earth-mass clumps, imprints specific angular anisotropy patterns on the gamma-ray sky that differ systematically from those of astrophysical backgrounds [131, 201, 202]. The angular cross-power spectrum between gamma-ray maps and a gravitational tracer — such as a galaxy catalog or a cosmic

shear field — provides a statistically powerful handle on this signal, because it vanishes for astrophysical source populations at redshifts incompatible with the tracer, but is maximal for dark matter emission that clusters with local structure.

We now examine this cross-correlation approach in detail from three directions. Section 8.1 places the technique in the context of the thesis as a whole, tracing the conceptual arc from resolved Galactic targets to the cosmic web and establishing why a shift in scale is both necessary and well-motivated given the null results of the previous searches. Section 8.2 develops the physical and mathematical principles of the cross-correlation method — the role of window functions and tomographic redshift selectivity, the noise cancellation advantage over auto-correlations, and the identification of low-redshift galaxy catalogs as the optimal gravitational tracer for dark matter annihilation. Section 8.3 introduces the Cherenkov Telescope Array Observatory (CTAO), which extends the accessible energy range an order of magnitude above the *Fermi*-LAT ceiling and brings effective collecting areas six orders of magnitude larger, opening the TeV mass window to cross-correlation searches for the first time. The remainder of this chapter then quantifies what CTAO can achieve in this regime, forecasting its reach for dark matter annihilation and decay.

8.1 Dark Matter in the Cosmic Web

If dark matter is a particle that annihilates, it does so not only in our own Galaxy but throughout the cosmic web of gravitationally bound structures that fills the universe — predominantly in the local volume. As such, to qualify a potential gamma-ray signal from dark matter, we need to first understand how matter is distributed in our universe.

In the standard Λ CDM cosmological framework, structures form hierarchically from the bottom up. Small-amplitude density perturbations in the early universe collapse first into low-mass dark matter halos; these merge and accrete material to build progressively more massive structures over cosmic time. The result is a cosmic hierarchy: galaxy clusters at the top, hosting individual galaxy halos, which in turn host populations of subhalos down to the minimum halo mass set by the free-streaming length of the dark matter particle — roughly Earth-mass ($\sim 10^{-6} M_{\odot}$) for a typical ~ 100 GeV WIMP [131]. This hierarchical web of halos and subhalos, distributed across the universe, is what we call the cosmic web.

The gamma-ray flux from dark matter annihilation is proportional to the square of the local dark matter density. Dense, compact structures are therefore the dominant contributors to the total signal, and the cumulative emission from all unresolved halos and subhalos across all redshifts produces a diffuse background. This is the dark matter component of the UGRB whose total energy spectrum between 100 MeV and 820 GeV has been measured by the *Fermi*-LAT [138]. Astrophysical source populations contribute their own guaranteed fraction to the UGRB as well: unresolved blazars dominate at high energies (their contribution is quantified in Section 8.2.3), while star-forming galaxies, misaligned AGNs, and millisecond pulsars add further contributions across the full energy range [131]. Because the exact composition of the UGRB is unknown, any dark matter signal must compete with and be separated from these astrophysical backgrounds.

The key challenge is spectral degeneracy. At energies accessible to *Fermi*-LAT (0.1–300 GeV), the spectra of heavy WIMPs and certain blazar populations overlap enough that energy information alone cannot distinguish them. Additionally, the annihilation flux is sensitive to highly uncertain quantities: the concentration-mass relation of dark matter halos, the abundance of substructure, and the choice of minimum halo mass — all of which enter through the so-called boost factor that encodes how much the clumpy distribution of subhalos enhances the signal

above a smooth halo approximation [131]. The boost factor itself can vary by orders of magnitude depending on these assumptions. A purely spectral analysis of the UGRB therefore constrains combinations of particle physics parameters and astrophysical modeling choices that are difficult to disentangle.

Spatial anisotropies provide a handle to distinguish the contributions. The UGRB is not perfectly isotropic: each contributing population imprints spatial fluctuations whose statistical properties can be studied by the clustering of the sources and their redshift distribution. Dark matter annihilation traces the squared density field of the cosmic web, which is more clustered on small angular scales and at lower redshifts than the blazars that constitute most of the astrophysical background [204, 361]. The *Fermi*-LAT Collaboration provided the first measurement of the UGRB angular power spectrum, detecting an approximately scale-invariant (Poisson) component consistent with unresolved point sources, but without clear evidence for a dark matter contribution [361]. The auto-correlation power spectrum, however, suffers from two limitations. First, the dark matter signal is always mixed with the shot noise and correlated fluctuations of astrophysical sources at the same angular scales; second, the measured C_ℓ is a line-of-sight integral over all redshifts and therefore carries no redshift resolution, since it cannot be decomposed into contributions from different distances (cf. Section 8.2.1). Cross-correlating the gamma-ray map with an external gravitational tracer resolves both limitations at once. A galaxy catalog provides a redshift-dependent map of the large-scale structure that correlates with the dark matter distribution at specific cosmic distances. Any source population that does not cluster with the selected galaxies cancels out of the cross-power spectrum, leaving only the gamma-ray emission that traces the same structures (developed quantitatively in Section 8.2.2). This noise cancellation is what makes cross-correlation a powerful probe for dark matter indirect detection [201, 202]. The dark matter annihilation signal peaks at low redshift, precisely where a catalog of local galaxies has the highest completeness. With a catalog selected at $z \lesssim 0.1$, the dark matter window function overlaps substantially with the galaxy window function, while the blazar window function — whose redshift distribution peaks at $z \sim 0.3\text{--}0.4$ — mostly does not. The angular cross-power spectrum developed formally in Section 3.5 is the mathematical realization of this strategy.

Dark matter emission is mainly local (highest at $z < 0.1$), while the blazar background is not, so cross-correlating the gamma-ray sky with a catalog of nearby galaxies selects the redshift window where the dark matter contribution is largest relative to the astrophysical background, while suppressing the bulk of the blazar population. Section 8.2.1 makes this argument quantitative (Figure 8.4).

8.2 The Cross-Correlation Technique

Section 3.5 illustrated the mathematical formulation of the cross-correlation technique. Three properties are key: redshift selection through overlapping window functions (Section 8.2.1), the statistical gain from combining two independent observables (Section 8.2.2), and the choice of low-redshift catalogs that reinforces both (Section 8.2.3).

8.2.1 Redshift Tomography

The angular cross-power spectrum $C_\ell^{\gamma g}$ introduced in Section 3.5 measures the correlation between intensity fluctuations in the gamma-ray intensity field and in the galaxy number counts. Under the Limber approximation (Eq. 3.5.4), this is a single integral over comoving distance, weighted by the product of the gamma-ray and galaxy window functions $W_\gamma(\chi)$ and $W_g(\chi)$. Note

that $C_\ell^{\gamma g}$ receives contributions only from the redshift range where both window functions are simultaneously nonzero. Choosing a galaxy catalog with a narrow redshift distribution therefore selects a specific slice of the universe, filtering out emission from all other distances. This is the essence of the tomographic approach [131, 362].

The window function $W_\gamma(z)$ encodes the redshift distribution of the gamma-ray emission. For unresolved astrophysical sources such as blazars, W_γ depends on the luminosity function and the spectral energy distribution of the source class, modulated by EBL absorption at high energies. For dark matter annihilation, it is proportional to $\langle\sigma v\rangle\Delta^2(z)/m_\chi^2$, where $\Delta^2(z)$ is the flux multiplier that captures the enhancement from DM clustering and substructures (full expression in Appendix 8.B.2). The ρ^2 scaling makes the signal dominated by the late-time collapse of matter into dense halos and subhalos, whose clustering $\Delta^2(z)$ peaks in the local universe and outweighs the smoother high-redshift background. As a result, W_γ^{DM} is maximal at $z \lesssim 0.1$ and decays rapidly at higher redshift, regardless of the DM mass or annihilation channel (cf. Section 8.1).

Astrophysical backgrounds present a very different behaviour. Unresolved blazars¹ have their emission peak at $z \sim 0.3\text{--}0.5$ at 50 GeV. At higher energies, EBL absorption progressively shrinks the gamma-ray horizon: above $\sim 1\text{ TeV}$, the observable volume contracts to $z \lesssim 0.1\text{--}0.2$, irrespective of source type. Star-forming galaxies trace the cosmic star-formation rate and dominate at much higher redshifts, $z \sim 1\text{--}3$. Figure 8.4 illustrates this feature quantitatively for the benchmark scenario considered in the analysis of this chapter: the DM window functions are confined to low redshift, while the blazar contributions extend to higher redshift. Following standard usage in the cross-correlation literature [201, 363], we describe this behaviour by saying that the DM window function *peaks* at low redshift. The term deserves a caveat: as Figure 8.4 shows, the annihilation kernel does not exhibit a narrow maximum but rather a broad plateau, approximately flat up to $z \sim 0.1\text{--}0.2$ before a rapid decay sets in. Peaking at low redshift therefore means that the window function is maximal in the local universe, in contrast with the astrophysical windows, which display a genuine maximum at intermediate redshift.

The galaxy catalog window function $W_g(z) = dN_g/dz$, normalized to unity, represents the redshift distribution of the galaxies in the sample. A dense, shallow catalog covering $z \lesssim 0.1$ such as the Two Micron All-Sky Survey (2MASS) overlaps nicely with the DM annihilation window function, which benefits the cross-correlation signal, given it is proportional to the overlap integral $\int dz W_\gamma(z) W_g(z)$. Because the blazar contribution to W_γ lies mainly higher redshift than W_g , the cross-correlation signal coming from blazars is suppressed. The DM signal, by contrast, overlaps strongly in redshift, and the cross-correlation becomes a probe of the local DM distribution.

8.2.2 Advantages of Cross-Correlation Surveys

For a cross-correlation survey, the variance of the cross-power spectrum (Eq. 3.5.8) contains the product of the gamma-ray noise term ($C_\ell^{\gamma\gamma} + C_N/B_\ell^2$) and the galaxy noise term ($C_\ell^{gg} + C_{N_g}$). The photon noise C_N in the gamma-ray map and the galaxy shot noise C_{N_g} are physically independent: their cross-noise vanishes on average. The noise therefore inflates the error bars on $C_\ell^{\gamma g}$ but does not add a systematic offset to the signal itself.

The auto-correlation variance (Eq. 3.5.9) is qualitatively different. Here the photon noise enters quadratically and adds directly to the gamma-ray auto-power spectrum. For a ground-based instrument like CTAO, the cosmic-ray rejection rate is difficult to perform, so C_N is dominated by the irreducible cosmic-ray background rather than by photon statistics alone,

¹High-synchrotron-peaked BL Lacertae objects are often denoted HSP, while low and intermediate-synchrotron-peaked BL Lacs are often grouped together and denoted LISP.

making it very large. Any attempt to extract a DM signal from $C_\ell^{\gamma\gamma}$ must therefore contend with subtracting this large noise term with high precision. By contrast, C_N in the cross-correlation variance multiplies the galaxy auto-spectrum $C_\ell^{gg} + C_{N_g}$, which is well measured independently; the DM signal is not hidden beneath the noise but requires accumulating sufficient signal-to-noise over multipoles and energy bins.

The second advantage of cross-correlation concerns the astrophysical foreground itself. The gamma-ray auto-correlation power spectrum $C_\ell^{\gamma\gamma}$ is dominated by the Poisson shot noise of unresolved blazars at essentially all multipoles of interest [131, 201]. These are rare, bright point sources whose auto-correlation is a flat (Poissonian) C_ℓ from the 1-halo term, covering any diffuse DM component. The detectability of astrophysical sources in auto-correlation therefore depends on how deeply the instrument can resolve individual blazars: the better the point-source sensitivity, the lower the residual unresolved blazar background and the smaller $C_\ell^{\gamma\gamma}$. For the DM signal, which is effectively diffuse and unresolved, the auto-correlation sensitivity has to compete with this astrophysical foreground.

Cross-correlation with a galaxy catalog suppresses this foreground through the overlap of window functions, as described in Section 8.2.1. The high-redshift blazar population does not correlate with a local galaxy catalog, so their 1-halo Poisson term does not appear in $C_\ell^{\gamma g}$. The cross-correlation SNR for the astrophysical signal depends on the point-source sensitivity only weakly, through the window function of the resolved fraction, whereas the auto-correlation SNR is strongly sensitive to it because the detection threshold enters both the window function and the three-dimensional power spectrum. On the other hand, since no dark matter source has been resolved as a gamma-ray point source to date, the DM window function does not depend on the detection threshold at all. The quantitative impact of this suppression on the dark matter sensitivity is derived later in this chapter, which compares the cross-correlation and auto-correlation constraints directly.

Auto and cross-correlation surveys further complement energy spectrum measurements by probing the spatial distribution of the gamma-ray emission. The angular dependence of $C_\ell^{\gamma g}$ can be decomposed into a 1-halo and a 2-halo contribution, following the halo-model formalism introduced in Section 3.5 and detailed in [131, 202]. The 1-halo term captures correlations between pairs of points within the same halo — for DM this traces the density profile within individual halos and dominates at small angular scales ($\ell \gtrsim 200$). The 2-halo term describes the large-scale clustering between different halos, following the linear matter power spectrum weighted by a bias factor. Measuring $C_\ell^{\gamma g}$ across multipoles therefore allows one to probe both the internal structure of DM halos and their clustering properties simultaneously, a richer physical picture than the energy spectrum or the auto-correlation alone can provide.

8.2.3 Low-Redshift Galaxy Catalogs

As illustrated in Section 8.2.1, dense, all-sky surveys covering the local universe ($z < 0.1$ – 0.2) are desirable for dark matter searches. Two datasets stand out in this role. The Two Micron All-Sky Survey (2MASS) Extended Source Catalog contains approximately one million galaxies, it has a redshift distribution peaking at $z \approx 0.072$, and is one of the most complete all-sky galaxy surveys available at low redshift. The 2MASS Redshift Survey (2MRS) is its spectroscopic subsample, extending to $z \sim 0.1$ with full spectroscopic redshifts, providing both a precise dN_g/dz for modeling and higher galaxy density per solid angle in the local volume. Both catalogs are nominally all-sky; however, in the cross-correlation analysis the low-latitude region around the Galactic plane (approximately $|b| < 30^\circ$) is masked in the gamma-ray maps to suppress Galactic diffuse emission [131, 362].

The practical advantage of 2MASS and 2MRS is that the blazar contribution to the UGRB in the local volume is strongly suppressed in DM cross-correlation analyses. Although unresolved blazars may account for roughly 20–30% of the total UGRB intensity integrated over all redshifts [144], this contribution arises predominantly from sources at $z \gtrsim 0.3$, where the large luminosity distance pushes their apparent flux below the Fermi-LAT detection threshold. In the local universe ($z \lesssim 0.1$), by contrast, even intrinsically faint blazars are close enough that their flux exceeds the point-source sensitivity of the Fermi-LAT [131]; they are therefore individually detected, cataloged, and removed from the diffuse maps. As a result, the unresolved blazar window function drops sharply at low redshift, and unresolved blazars in the volume probed by 2MASS and 2MRS are a less dominant component of the UGRB [131, 363]. The cross-correlation of the UGRB with these catalogs is thus less influenced by the dominant astrophysical contaminant that plagues both the gamma-ray auto-correlation and searches at higher redshift. The signal remaining in $C_\ell^{\gamma g}$ is a mixture of star-forming galaxies (SFGs), misaligned active galactic nuclei (mAGN), and any DM contribution.

This expectation was confirmed observationally with 60 months of Fermi-LAT data: Xia et al. [362] detected a statistically significant cross-correlation between the UGRB and several galaxy catalogs at angular scales below one degree, finding that the signal appears to be dominated by unresolved star-forming galaxies and misaligned AGNs, with blazars contributing less than 20%. Cuoco et al. [363] then performed a joint fit to the measured cross-correlation functions, decomposing the signal into DM, blazars, SFGs, and mAGN components. A non-zero DM contribution was allowed by the data but degenerate with the astrophysical components; the resulting upper limits on the annihilation cross section and decay lifetime were competitive with constraints from dwarf spheroidal galaxies and galaxy clusters [364].

In the context of cross-correlation with CTAO, the case for 2MASS and 2MRS is even stronger: at TeV energies the EBL horizon (Section 8.2.1) removes the distant sources while local dark matter photons arrive essentially unabsorbed.

8.3 The Cherenkov Telescope Array Observatory

The Cherenkov Telescope Array Observatory (CTAO) is an earth-based gamma-ray observatory currently under construction in the northern and southern hemispheres [365]. Its energy range, extending from 20 GeV to 300 TeV, spans the regime where heavy WIMP candidates produce their signal. Its planned extragalactic survey is expected to cover a quarter of the sky with approximately uniform exposure, and its angular resolution at the relevant energies exceeds that of Fermi-LAT by an order of magnitude. The following subsections introduce CTAO's key characteristics and its extragalactic survey strategy.

8.3.1 From Fermi-LAT to CTAO

The analyses in Chapters 4–7 were built on data from the Fermi Large Area Telescope (Fermi-LAT), which has operated as the premier instrument for indirect dark matter searches in the GeV band for over 15 years (see Section 2.3.1). As a space-based pair-conversion telescope, the LAT detects gamma rays from 20 MeV to beyond 300 GeV with a field of view of approximately 2.4 sr, surveying the entire sky every two orbits (~ 3 hours) [147]. Its effective area, however, is constrained by spacecraft payload limitations to roughly 1 m^2 at GeV energies. This ceiling in collecting area, combined with the falling flux of gamma rays at high energies, makes Fermi-LAT less suited for probing WIMP candidates with masses above about 1 TeV.

The CTAO design addresses exactly this limitation [365]. Ground-based imaging atmospheric Cherenkov telescopes (IACTs) reconstruct very-high-energy gamma rays from the brief flashes of Cherenkov light produced when an incident photon strikes the upper atmosphere and initiates an electromagnetic shower. By observing this shower at ground level, CTAO effectively uses the atmosphere as a calorimeter, achieving effective areas orders of magnitude larger than any space-based detector. The observatory is planned as two arrays, one in the northern hemisphere and one in the south, built from three telescope classes tailored to different energy regimes: the Large-Sized Telescopes (LSTs) at lower energies, the Medium-Sized Telescopes (MSTs) in the core range, and the Small-Sized Telescopes (SSTs) at the highest energies, deployed only in the south where Galactic observations are prioritized (telescope fields of view are detailed in Section 8.4.1). Across these three telescope classes, CTAO covers a broad energy range from 20 GeV up to 300 TeV, with an angular resolution and sensitivity that improve by an order of magnitude over current-generation IACTs, such as HESS, MAGIC, and VERITAS [365].

The extension to TeV energies opens a qualitatively new window for dark matter searches. Heavy WIMP candidates with masses $m_\chi \gtrsim 1$ TeV could produce gamma rays predominantly in the TeV band, a range where Fermi-LAT has negligible sensitivity. More importantly for cross-correlation studies, EBL absorption above ~ 1 TeV suppresses the astrophysical background from more distant sources (cf. Section 8.2.1). The LST and MST telescopes, built in both hemispheres, are optimised for extragalactic observations, where the unresolved emission receives its dominant DM contribution from structures at $z \lesssim 0.1$. At energies above 20 GeV and increasingly toward the TeV regime, star-forming galaxies and misaligned AGN contribute negligibly to the unresolved background. Blazars, both High-Synchrotron-Peaked (HSP) and Low-to-Intermediate-Synchrotron-Peaked (LISP), constitute the main astrophysical background component. EBL attenuation pushes the blazar window functions down toward the DM-dominated window at $z \lesssim 0.1$; thus, even at TeV energies, blazars remain an important background for the cross-correlation.

8.3.2 The Extragalactic Survey

Cross-correlation and anisotropy analyses require large sky maps with approximately uniform exposure. A pointed instrument with irregular coverage would introduce instrumental variations that could mimic or obscure genuine cosmological angular correlations. For CTAO, this requirement is met by the Extragalactic Survey (EGAL), one of the observatory’s designated Key Science Projects [365].

The EGAL survey targets about a quarter of the sky, covering the region defined by $-90^\circ < l < 90^\circ$ and $|b| > 5^\circ$ in Galactic coordinates. Planned over three years and split between the two array sites, the survey delivers an approximately uniform effective exposure of about three hours per sky location across the footprint; the full pointing grid, per-site observing times, and total exposure budget are specified in Section 8.4.1. The wide telescope fields of view (Section 8.4.1) create deliberate overlaps between adjacent pointings. These overlaps smooth out the exposure variations that arise when stitching together individual observations, keeping the residual fluctuations in the exposure map small across the survey region (see Section 8.4.1).

Cross-correlation studies can also exploit the accumulated off-source data collected through years of CTAO pointed observations across the extragalactic sky. In the most optimistic projection, these off-source data accumulate a substantially deeper effective exposure per sky location, improving the sensitivity over the nominal EGAL scenario (see Appendix 8.A). These observations are not associated with a specific and dedicated survey, making them a practically relevant complement to the EGAL Key Science Project.

The angular power spectrum C_ℓ is computed from a masked, pixelised sky map, and anisotropies in the instrumental exposure cannot be distinguished from intrinsic anisotropies in the gamma-ray background without careful masking and exposure correction. At the multipoles $\ell \sim 10\text{--}100$ relevant for the DM cross-correlation signal, any residual exposure inhomogeneity couples into the signal through the survey window function and influences the recovered power spectra. The EGAL strategy's uniform grid, combined with the broad telescope fields of view, is designed to minimise this systematic effect. The remainder of this chapter is based on [5] and presents the analysis in full.

8.4 Dark Matter and Galaxy Cross-Correlations with the Cherenkov Telescope Array Observatory

The Cherenkov Telescope Array Observatory (CTAO) will be a ground-based Cherenkov telescope performing wide-sky surveys, ideal for anisotropy studies such as cross-correlations with tracers of the cosmic large-scale structure. Cross-correlations can shed light on high-energy γ -ray sources and potentially reveal exotic signals from particle dark matter. In this work, we investigate CTAO sensitivity to cross-correlation signals between γ -ray emission and galaxy distributions. We find that by using dense, low-redshift catalogs like 2MASS, and for integration times around 50 hours, this technique achieves sensitivities to both annihilating and decaying dark matter signals that are competitive with those from dwarf galaxy and cluster analyses.

This chapter now turns to a sensitivity forecast for detecting dark matter signals through cross-correlations between the CTAO extragalactic survey and the 2MASS galaxy catalog. Section 8.4.1 describes the CTAO instrument configuration and survey strategy. Section 8.4.2 develops the cross-correlation signal model, including the window functions and angular power spectra for both astrophysical sources and dark matter. Section 8.4.3 presents the sensitivity forecast results for dark matter annihilation and decay. The chapter concludes in Section 8.4.4.

8.4.1 The Cherenkov Telescope Array

The CTAO will represent the next generation of imaging atmospheric Cherenkov telescopes (IACTs), improving the technology of the current ground-based γ -ray detectors, such as H.E.S.S., MAGIC, and VERITAS. It is designed to surpass the sensitivity of current IACTs by an order of magnitude in the TeV regime, to improve the angular resolution, as well as to enhance the surveying and monitoring capabilities. CTAO will consist of two IACT arrays: one located in the Southern hemisphere (CTAO-South) and the other in the Northern hemisphere (CTAO-North) [365].

The configuration will use three different types of telescope: large-, medium-, and small-sized telescopes (LST, MST, and SST, respectively), detecting photons over a broad energy range from 20 GeV up to 300 TeV. The LSTs will be most sensitive at lower energies, while the SSTs will cover the highest energies. The SSTs will be dedicated primarily to the observation of phenomena occurring within our Galaxy and will therefore be deployed only at the southern site from which the central/inner part of the Galaxy is most visible. LSTs and MSTs will instead be built both in the Northern and Southern locations and will focus on observations of extragalactic targets.

The telescopes will be equipped with cameras featuring wide fields of view: 4.3° for LSTs, $7.5\text{--}7.7^\circ$ for MSTs, and 8.8° for SSTs. A wide field of view is important for observing extended sources and diffuse emission, as well as for reducing the systematic errors of the measurements when requiring a uniform response over regions much larger than the telescope point-spread

function (PSF). Several surveys covering large portions of the sky are planned as part of CTAO’s observational strategy, and will be performed as Key Science Projects (KSPs) of the observatory early in the observatory’s operation [365]. Thanks to its improved sensitivity and wider field of view, CTAO will surpass the surveying capabilities of existing IACTs. In particular, the Extra-Galactic survey (detailed below) will be the largest survey ever undertaken by IACTs.

At the same time, CTAO will carry out a large number of traditional, dedicated pointings across the sky. The off-source data collected through these observations will cumulatively cover a substantial sky area with a significant average exposure. In this work, we consider extragalactic data obtained from both observational strategies, as detailed in two subsections below.

8.4.1.1 The Extra-Galactic survey

To study the auto- and cross-correlation signals with gamma-rays, it is essential to use maps of large sky regions with approximately uniform coverage. Among CTAO Key Science Projects, the dedicated survey of the extragalactic sky—referred to as the extra-galactic (EGAL) survey - presents the required characteristics.

As described in [365] (Sect. 8), the EGAL survey will cover a quarter of the sky, targeting the region defined by Galactic coordinates $-90^\circ < l < 90^\circ$ and $b > 5^\circ$. The survey is planned to reach an integral sensitivity ~ 6 mCrab above 125 GeV. One of its main objectives is to provide a high-resolution map of the extragalactic sky for photons with energies at least up to 10 TeV and to compile a catalog of very high energy (VHE) extragalactic sources.

The EGAL observations are planned to be carried out over three years of observations, using the full CTAO Observatory (both the Southern and Northern arrays) to achieve nearly uniform sensitivity over the survey region. It is proposed that 15% of the sky will be covered by the Southern array using 400 h and 10% by the Northern array using 600 h, resulting in a total observation time of ~ 1000 h. The default observing strategy consists of a grid spaced by $\sim 3^\circ$ across the survey area, with each pointing observed for 0.51 h in the South and 1.11 h in the North. This configuration is expected to yield an effective exposure of approximately 3 h per point within the EGAL survey boundary.

8.4.1.2 Simulating survey observations

To simulate the planned EGAL survey observations, we use the software package `ctools`,² developed for a scientific analysis of CTA data. We used the instrument response functions (IRFs) for the observatory,³ labeled `prod3b-v2`, defining the so-called “omega” configuration. The respective IRF data files are publicly available at Ref. [366].

To enclose the region of the sky covered by the EGAL survey, we simulate observations using telescope-pointing directions corresponding to a HealPix realization with `NSIDE` = 16, resulting in 694 (3072) pointings within the EGAL survey bounds (over the whole sky). With this setup, the typical angular separation between adjacent pointings is $\sim 3.7^\circ$. For definiteness, the radius of the field of view (FoV) of each pointing is set to 6° . The pointing directions used for the simulation of the EGAL survey within the region of interest (ROI) are shown in Fig. 8.1.

The overlap of the FoV from different pointings contributes to a more uniform coverage of the survey’s ROI, minimizing the fluctuations in the exposure that inevitably occur when smaller FoV observations are used to cover a large survey area. Depending on the selected HealPix grid size (i.e., the values of the `NSIDE` parameter) and the pointing radius for individual observations,

²<http://CTA.irap.omp.eu/ctools/>.

³See <http://www.CTA-observatory.org/science/CTA-performance/> for more details.

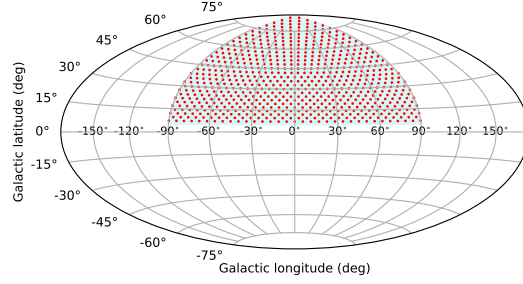


Fig. 8.1. Pointing directions for the region of the sky planned to be covered by the extragalactic survey. The pointing directions are shown using a HEALPix pixelization with NSIDE = 16.

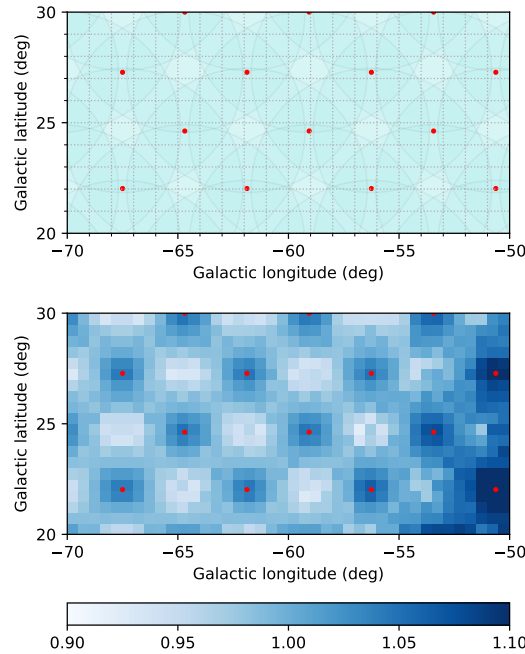


Fig. 8.2. Overlapping areas of different pointings. *Top:* A zoomed-in scheme of the pointing strategy shown in Fig. 8.1 (red), together with the overlapping fields of view of individual pointings (turquoise). *Bottom:* Simulated exposure map showing variations relative to the average exposure, highlighting the anisotropic pattern created by overlapping pointing regions.

the overlap between nearby pointings can vary in extent. The overlaps between pointings are schematically shown in Fig. 8.2, where the upper panel presents a zoomed-in scheme of the pointing strategy from Fig. 8.1, and the bottom panel shows the simulated exposure of the same region. These images demonstrate that the overlap of nearby pointings creates an anisotropy pattern in the corresponding observation exposure, with variations of approximately $\sim 10\%$ relative to the average exposure within the plotted region.

The IRFs we use are optimized for observations under different zenith angles. For each pointing direction, we assign one of the three available IRF optimizations based on the minimum zenith angle at which that direction can be observed. To estimate the minimal zenith angle, we use the Earth coordinates of both array positions, i.e., $(l_N, b_N) = (17.8920, 28.7622)$ and

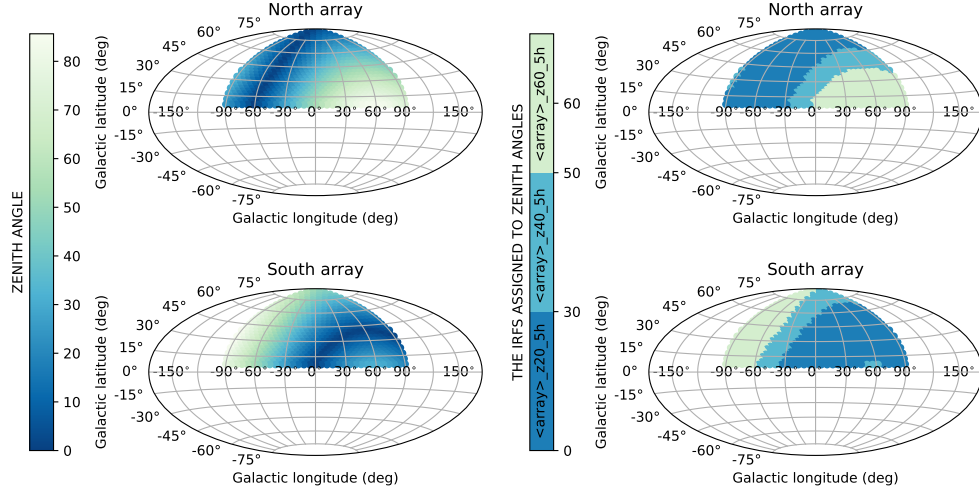


Fig. 8.3. Strategy for assigning Instrument Response Functions (IRFs) across the extragalactic survey region as a function of zenith angle. In the left panel, the color scale shows, for each sky position, the zenith angle as measured in the observatory reference frame during observation (the values of the zenith angles and their color coding are reported on the vertical bar on the left). Since the IRFs are derived in three zenith angle bins (indicated by the color bar on the side of the right panel), the right panel shows which IRF is assigned to each region of the sky. The three IRFs shown here refer to 5 hrs of observation and to zenith angles below 30 deg (dark blue), between 30 and 50 deg (light blue) and above 60 deg (green).

$(l_S, b_S) = (79.4041, -24.6272)$ for the Northern and Southern array, respectively. The zenith angle is calculated as the angular distance between the array location and the pointing direction when they are closest. The resulting zenith angles for the Northern and Southern arrays are shown in the left panel of Fig. 8.3. The strategy for assigning the IRFs based on zenith angle is adapted from Ref. [367]. For zenith angles smaller than 30° , we use the IRF optimized for 20° angles; for zenith angles between 30° and 50° we use the IRF optimized for 40° angles, and for zenith angles larger than 50° we use IRF optimized for 60° angles, as shown in the right panel of Fig. 8.3. All IRFs used in this work are optimized for 5 hours of observation time, independently of the array or the zenith angle.

The observational strategy presented in Section 8 of Ref. [365] divides the extragalactic survey area into two parts: roughly $\sim 15\%$ of the sky to be observed using the Southern array and $\sim 10\%$ with the Northern array. However, the precise division of the survey observations between the two arrays has not yet been finalized. The planned strategy remains under discussion and is still subject to adjustments. For simplicity, we simulate observations of each pointing direction within the survey boundary using both Southern and Northern arrays with the corresponding zenith angle optimized IRFs. We use 1.11 hours and 0.51 hours per pointing direction with the Northern and Southern arrays, respectively. This strategy provides a reasonable approximation for simulating the survey when combined with zenith-angle-optimized IRFs, as regions observed at higher zenith angles will yield fewer detected events, and vice versa. As a result, the sky coverage of both arrays will be complementary, as illustrated in the right panel of Fig. 8.3.

The survey observations are simulated for energies between 30 GeV and 30 TeV, using 30 log-spaced energy bins to ensure proper sampling of the IRFs across the energy range. For the purpose of estimating the background events and exposure over the whole extragalactic survey, we simulate the entire survey ROI, using coarser spatial binning of $1.0^\circ \times 1.0^\circ$ per pixel. For the sensitivity analysis, we instead simulate templates covering an ROI of $6.0^\circ \times 6.0^\circ$, using a finer spatial binning of $0.02^\circ \times 0.02^\circ$ per pixel.

8.4.1.3 Off-source data scenario

In addition to performing extensive surveys of the γ -ray sky, CTAO will be operated as an open, proposal-driven observatory. The observation time awarded to the wider research community can be added to the dedicated survey observation times of CTAO, and increasing the overall exposure that will be achieved over the first 10 years of telescope operation. An educated guess based on the performance of current IACTs can be made for CTAO observations performed through the proposal-driven program. In Ref. [368], the authors calculate the sky coverage and resulting observation time of the CTAO proposal program, assuming similarity with the actual operations of the MAGIC telescope. They extrapolate the 6.5 years of data obtained with the MAGIC array to 10 years of observation time using the two CTAO arrays. We adopt a similar approach to estimate the most optimistic case scenario for the observations of the extragalactic sky with CTAO, assuming an effective observation time of ~ 50 hours per point and the same sky coverage as in the EGAL survey, consistent with the findings in Ref. [368]. In the sensitivity analysis described below, we use a corresponding re-scaling factor to estimate the point source sensitivity of CTAO under this scenario. Details on how we derive the effective observation time for this scenario can be found in Section 8.A.

8.4.2 The cross-correlation signal

The cross-correlation between a map of the UGRB, $\gamma(\hat{n})$, and fluctuations in galaxy number counts, $g(\hat{n})$, can be measured through the cross-correlation angular power spectrum (APS) [201, 202, 369]. In our analysis, the field g will be the distribution of galaxies, which is a biased tracer of the underlying dark matter distribution. In the Limber approximation [203, 370, 371], the APS can be written as [201, 202, 369]

$$C_\ell^{\gamma g} = \int \frac{d\chi}{\chi^2} W_\gamma(\chi) W_g(\chi) P_{\gamma g}(k = \ell/\chi, \chi), \quad (8.4.1)$$

where ℓ denotes the multipole, k the wavenumber, $\chi(z)$ is the radial comoving distance, related to redshift z in a flat cosmology through the relation $d\chi = c dz/H(z)$ with $H(z)$ being the expansion rate of the Universe. The term $P_{\gamma g}(k, z)$ represents the 3D cross-correlation power spectrum (PS), embedding the information on the fluctuations of the two correlated fields. The function $W_{i=\{\gamma, g\}}$ denotes the window functions of the gamma-ray intensity field, γ , and the galaxy number counts fluctuations, g , respectively, and describes the redshift distribution of each field (see Appendix 8.B for full details).

As astrophysical γ -ray sources we consider only BL Lacs, subdivided into the two sub-categories of high-synchrotron-peaked (HSP) blazars and low and intermediate synchrotron-peaked (LISP) blazars [372]. We model the spectral energy distribution (SED) of these sources as power-laws with index Γ and exponential cutoffs E_{cut} , whose specific parameter values are reported in Table 8.1, together with the minimum and maximum luminosities of the population. Since we focus on CTAO and thus on energies above approximately 20 GeV, other typical γ -ray emitters like misaligned AGN, star-forming galaxies, and flat-spectrum radio quasars can

be neglected, since their contribution are dominant at lower energies [131]. Full details on the modeling of the astrophysical sources are provided in Appendix 8.B, together with the expressions for their window functions and 3D cross-correlation power spectra, obtained within the halo model formalism.

Let us just recall here that the window function of the γ -ray emission of astrophysical sources depends on the γ -ray luminosity function (GLF) of each specific source-type and on their SED, and is integrated over the luminosity of the sources up to a threshold luminosity, which corresponds to the minimal luminosity that a source must possess in order to be resolved by the detector (in this way, the window function which is then obtained pertains to the *unresolved* sources, those contributing to the UGRB). The full expression and the detailed modeling can be found in Appendix 8.B.2.

For the DM γ -ray emission, we consider both annihilating and decaying DM. In the case of annihilation, the relevant particle physics parameters are the DM mass m_χ and the thermally-averaged annihilation cross-section $\langle\sigma v\rangle$. For decaying DM, the key parameters are the mass and the lifetime τ . These will be the parameters for which we provide the forecast sensitivity in the cross-correlation analysis. The expressions for the window functions and the 3D cross-correlation power spectra are reported in Appendix 8.B. Here we recall that the window functions are proportional to $\langle\sigma v\rangle$ for annihilating DM and inversely proportional to τ in the case of decaying DM, and proportional to m_χ^{-2} and m_χ^{-1} for annihilating and decaying DM, respectively. The window function of the gamma-ray emission from DM annihilation or decay also depends on the average dark matter content of the Universe and on the energy spectrum produced by dark matter annihilation or decay, which in turn depends on the specific annihilation/decay channel, for which we will consider $b\bar{b}$ as the representative case and provide results also for the channels e^+e^- , $\tau^+\tau^-$ and W^+W^- . In the case of annihilating dark matter, the window function also contains a so-called flux-multiplier factor, which accounts for the enhancement of the flux due to the clumping of DM and the presence of sub-structures within the DM halos. The full expressions and the detailed modeling of the DM window functions can be found in Appendix 8.B.2. A few examples of the redshift behavior of the γ -ray window functions for DM annihilation are shown in Figure 8.4, for two photon energies and two DM masses. For comparison, we also show the window functions of unresolved LISP and HSP astrophysical source classes, assuming an observation time of 3 hours. The observation time affects the window functions of astrophysical sources, since it determines the minimum luminosity of the sources that can be resolved by the instrument (and therefore do not contribute to the unresolved component considered in our analysis). The case for 50 hours of observation is shown in Figure 8.15 in Appendix 8.B.

For the present analysis, which primarily focuses on the sensitivity to a DM signal, we adopt as galaxy catalogs the Two Micron All-Sky Survey (2MASS) [373, 374] and the 2MASS Redshift Survey (2MRS) [375, 376] catalogs, since they cover the low redshift range $z \lesssim 0.1$ where most of the DM γ -ray emission originates. Astrophysical sources, on the other hand, have a peak in emission at redshifts $z \sim 0.3\text{--}0.4$ at $E \sim 50\text{ GeV}$, and $z \sim 0.1\text{--}0.2$ at $E \gtrsim 1\text{ TeV}$ as shown in Figure 8.4 and in Figure 8.15 in Appendix 8.B. The latter range is a consequence of strong absorption of γ -rays above $\sim 1\text{ TeV}$ due to interactions with the extragalactic background light (EBL), which limits the γ -ray horizon to $z \sim 0.1\text{--}0.2$ independently of the type of source.

The present analysis is thus optimized for DM searches, while for astrophysical sources it will be beneficial to consider galaxy catalogs covering higher redshifts, up to $z \sim 0.4\text{--}0.5$. We leave this extension to future studies. The window functions of galaxy number counts are given by the redshift distribution of the number of objects in the catalog. Full details on the window functions of the catalogs adopted here can be found in Appendix 8.B.2. The modeling of galaxy clustering

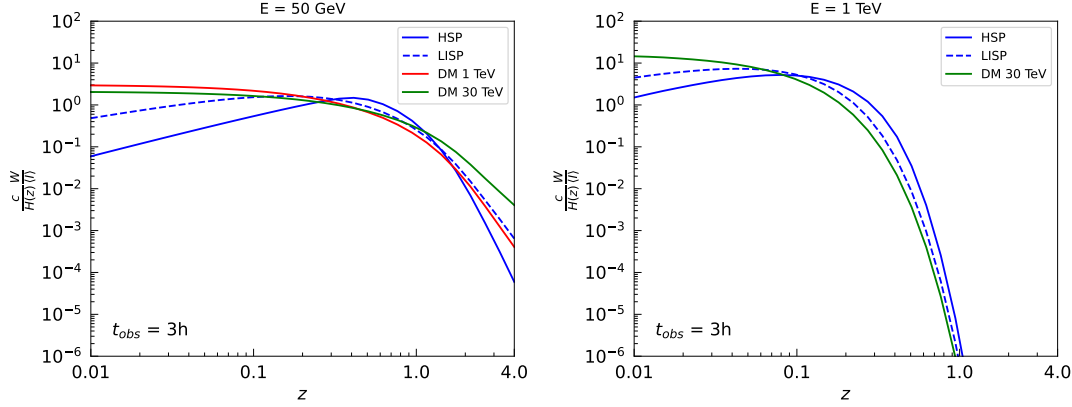


Fig. 8.4. The γ -ray window functions, normalized to the mean γ -ray intensity and divided by the Hubble parameter, as a function of redshift. Blue solid and dashed lines show HSP and LISP astrophysical sources for 3 hours of observation, respectively. Red and green lines correspond to DM annihilation into $b\bar{b}$, $\langle\sigma v\rangle = 3 \times 10^{-26} \text{cm}^{-3} \text{s}^{-1}$ for DM masses of 1 TeV (red) and 30 TeV (green). Upper panel: $E_\gamma = 50 \text{ GeV}$; lower panel: $E_\gamma = 1 \text{ TeV}$ (only 30 TeV DM case shown).

within halos is based on the halo occupation distribution (HOD) formalism [205, 377, 378], which is also outlined in Appendix 8.B, together with the 3D cross-correlation power spectra.

Let us note that in our analysis, the window function $W_g(z)$ of the galaxy catalogs is normalized to unity: $\int dz W_g(z) = 1$. For the γ -ray window function $W_\gamma(\chi)$, we define it such that the (average) intensity energy spectrum is obtained as: $I_\gamma(E) = \int d\chi W_\gamma(E, \chi)$.

The cross-correlation APS signal has a variance $(\Delta C_\ell^{\gamma g})^2$ given by [201, 202, 369]

$$(\Delta C_\ell^{\gamma g})^2 (2\ell + 1) f_{\text{sky}} = (C_\ell^{\gamma g})^2 + \left[C_\ell^{\gamma\gamma} + \frac{C_N}{(B_\ell)^2} \right] (C_\ell^{gg} + C_{N_g}), \quad (8.4.2)$$

where $C_\ell^{\gamma\gamma}$ and C_ℓ^{gg} are the γ -ray and galaxy auto-correlation APS, C_N is the photon noise, C_{N_g} the galaxy catalog noise, and B_ℓ denotes the beam correction accounting for the finite angular resolution of the γ -ray instrument. No beam correction is applied to the galaxy term, since the angular resolution of galaxy catalogs (typically of the order of arcseconds) is negligible for the purposes of this analysis. The parameter f_{sky} denotes the fraction of the sky covered by the analysis. All the relevant terms in Eq. 8.4.2 are outlined in Appendix 8.B.

For completeness, in the following analysis, we will also use the γ -ray auto-correlation APS, which is a measure of the intrinsic anisotropy of the γ -ray emission. It reads [201, 202, 369, 379]

$$C_\ell^{\gamma\gamma} = \int \frac{d\chi}{\chi^2} W_\gamma^2(\chi) P_{\gamma\gamma}(k = \ell/\chi, \chi), \quad (8.4.3)$$

where $P_{\gamma\gamma}$ is the 3D power spectrum of the γ -ray emission. The corresponding variance on $C_\ell^{\gamma\gamma}$ is [201, 202, 369]

$$(\Delta C_\ell^{\gamma\gamma})^2 = \frac{2}{(2\ell + 1) f_{\text{sky}}} \left[C_\ell^{\gamma\gamma} + \frac{C_N}{(B_\ell)^2} \right]^2. \quad (8.4.4)$$

	Γ	E_{cut} [GeV]	L_{min} [erg/s]	L_{max} [erg/s]
HSP	1.86	910^{+1100}_{-450}	10^{38}	$3 \cdot 10^{45}$
LISP	2.08	37^{+85}_{-20}	10^{38}	$3 \cdot 10^{46}$

Table 8.1. Parameters and uncertainties for the power-law with exponential cut-off SED models of HSP and LISP populations, including the minimum and maximum luminosity, adopted from Table 1 of Ref. [372].

8.4.3 Sensitivity Forecast

8.4.3.1 Astrophysical sources

As a first step, we determine whether the anisotropy and the cross-correlation signals from unresolved astrophysical sources are detectable. As a statistical figure of merit, we adopt the signal-to-noise ratio (SNR), defined, for the cross-correlation, as

$$\text{SNR} = \sqrt{\sum_{\ell,i} \left(\frac{C_{\ell,i}^{\gamma g}}{\Delta C_{\ell,i}^{\gamma g}} \right)^2}, \quad (8.4.5)$$

where the sum runs over all the multipoles ℓ considered in the analysis and all energy bins i . For the auto-correlation, the formula is analogous but using $C_{\ell,i}^{\gamma\gamma}$ and $\Delta C_{\ell,i}^{\gamma\gamma}$.

To be more specific, the energy bins adopted in our analysis are 16, logarithmically spaced in the 30 GeV – 30 TeV range. The sum over multipoles extends from $\ell = 10$ up to $\ell = 1000$. The lower value $\ell = 10$ is adopted to avoid the largest angular scales (i.e., lowest multipoles), where the γ -ray Galactic foreground is the dominant emission, and, if not properly modeled and removed, can create some artifact in the reconstructed signal, especially in the auto-correlation. This issue is, however, mostly of concern for a real data analysis like the one in [380], where we refer the reader for further details. In our forecast, starting the sum from $\ell = 10$ or $\ell = 1$ has a negligible impact on the final results. At the same time, we do not consider very large multipoles, since the (energy-dependent) angular resolution of the γ -ray detector prevents us from accessing information on angular scales which are much smaller than its point-spread-function (PSF). The CTAO PSF has been determined through a Monte Carlo simulation of the detector and is shown in Figure 8.10 in Appendix 8.B, together with its harmonic space transform B_ℓ , which enters in the expression for the variance $\Delta C_{\ell,i}^{\gamma g}$ of Eq. (8.4.2). From Figure 8.10 we can see that for multipoles above about 1000 the beam function B_ℓ sharply decreases, thus inflating the variance and making those large multipoles irrelevant for the SNR.

The results for 3 hours and 50 hours of observation time are shown in Table 8.2, for both LISP and HSP source classes, and for both the auto-correlation and cross-correlation APS with the 2MASS galaxy catalog. We find that a purely astrophysical signal from unresolved sources can reach at most (for 50 hours of observation) only a marginal evidence of $\text{SNR} \simeq 0.9$ in the autocorrelation, and about 0.16 in the cross-correlation case. These values of SNR correspond to the HSP blazars, whose hard spectra are better suited to the CTAO energy range. The SNR is even smaller for LISPs, whose softer spectra result in a poorer overlap with the CTAO sensitivity.

As an aside comment, let us note that this is not necessarily a negative result: it simply reflects the fact that the expected sensitivity of CTAO to point-source detection would be good enough to resolve a large fraction of blazars, thereby reducing the auto- and cross-correlation APS of the *unresolved* population. In this case, it would be actually convenient and sufficient to

SNR for 3 hrs	Auto-correlation	Cross with 2MASS
LISP	0.042	$8.2 \cdot 10^{-3}$
HSP	0.31	0.1
SNR for 50 hrs	Auto-correlation	Cross with 2MASS
LISP	0.18	0.022
HSP	0.9	0.16

Table 8.2. The predicted signal-to-noise ratios for 3-hours and 50-hours observations of LISP and HSP source classes, obtained for both γ -rays auto-correlation and cross-correlation with the 2MASS galaxy catalog.

SNR 50hrs low res	Auto-correlation	Cross with 2MASS
LISP	0.7	0.033
HSP	5.1	0.41

Table 8.3. Same as Table 8.2 but for 50hrs-observations, assuming a source detection threshold approximately three times worse than in Table 8.2. This illustrates the strong dependence of the auto-correlation signal on the point source detection threshold.

perform auto- and cross-correlation APS analyses directly on the catalog of resolved sources in order to study their anisotropy properties.

However, we wish to stress that this result, i.e., the fact that CTAO sensitivity would be enough to resolve a significant fraction of blazars, strongly depends on the assumed point-source sensitivity of the detector, which for CTAO remains subject to considerable uncertainty (as discussed in Section 8.4.1). This is because the final configuration and specifications of CTAO have not yet been fully settled. For example, the point-source sensitivity differs significantly between CTAO-North and CTAO-South [365], and at the moment the relative share of North and South array observations to the survey is still undecided. To assess the capabilities of CTAO for the study of anisotropies of point-sources even in the case that the point-source sensitivity is worse than the nominal case derived and adopted in this work, we repeat the analysis for point-sources by assuming a degraded point-source detection capability. Table 8.3 shows the SNR obtained for 50 hrs of observations, assuming a source detection threshold approximately three times worse than in the nominal case of Table 8.2. This has been achieved by using the 50 hrs exposure with a point-source sensitivity corresponding to 3hrs of observations, which is approximately a factor of 3 worse than the nominal 50hrs sensitivity (Note that, since we are in a strongly background-dominated regime, the uncertainty ΔC_ℓ does not depend on the C_ℓ itself, but only on the background and the noise, as discussed in Section 8.B). The result shows that the auto-correlation SNR for HSP indeed increases significantly, reaching $\text{SNR} \approx 5$. The detectability of the cross-correlation APS signal also improves, but not so strongly as in the auto-correlation case. This is because the cross-correlation signal is less sensitive to the point-source detection threshold. As detailed in the Appendices, the point-source sensitivity impacts the cross-correlation signal only through the window function, whereas it affects both the window function and the 3D power spectrum in the auto-correlation case.

Regarding LISPs, on the other hand, the SNR remains very low in both cases, reaching at

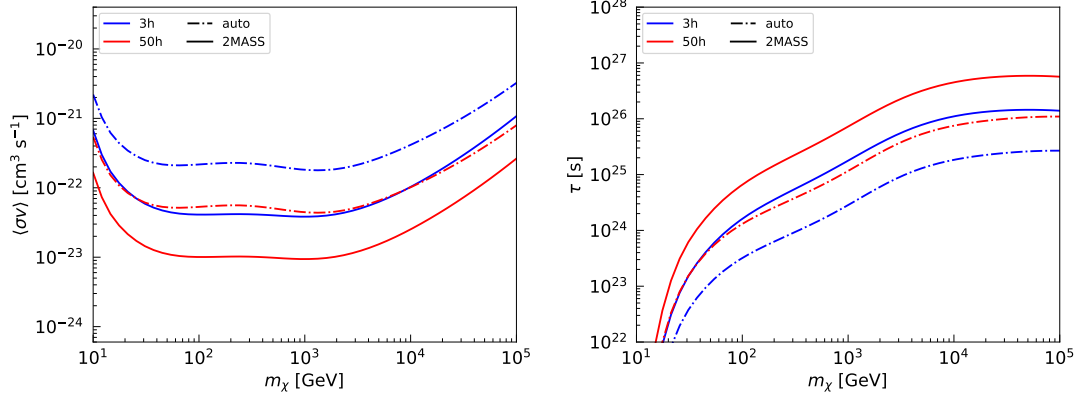


Fig. 8.5. Left: Predicted upper bounds on the dark matter cross-section $\langle\sigma v\rangle$, for annihilation into $b\bar{b}$, based on 3-hour and 50-hour of observations. The curves represent the bounds from γ -ray auto-correlation (dot-dashed) and cross-correlation with the 2MASS galaxy catalog (solid). Right: Same as the left panel, but for the lower bound on the DM particle lifetime τ .

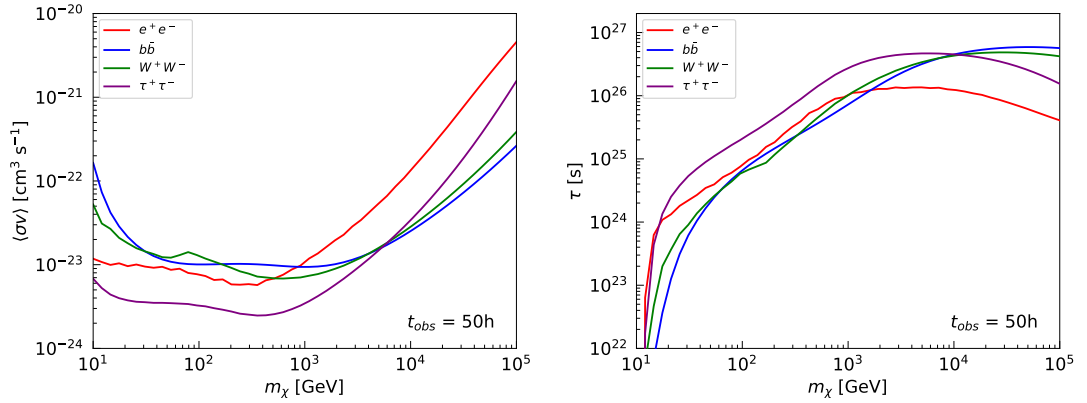


Fig. 8.6. Left: Predicted upper bounds on the dark matter cross-section $\langle\sigma v\rangle$ from the cross-correlation with 2MASS galaxies, based on 50 hours of observation, for various dark matter annihilation final states: e^+e^- , $b\bar{b}$, W^+W^- and $\tau^+\tau^-$. Right: Same as the left panel, but for the lower bound on the DM particle lifetime τ .

most 0.7 for the auto-correlation and 0.03 for the cross-correlation. This is expected, because LSPs have a SED cut-off around 40 GeV, resulting in minimal emission in the energy range where CTAO is most sensitive.

8.4.3.2 Dark matter

We now turn to assessing the CTAO sensitivity to an auto- or cross-correlation APS originating from DM annihilation or decay. To this end, we compare the null hypothesis, where only the contribution from astrophysical sources is present, with the alternative hypothesis scenario in which a DM γ -ray component is added on top of the astrophysical signal. From this comparison, we derive the minimum DM signal required to be statistically distinguishable from the purely

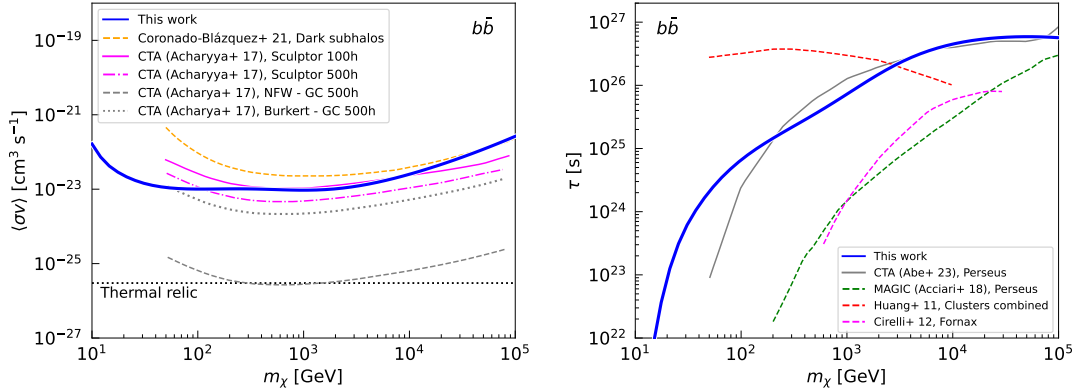


Fig. 8.7. Left: Comparison between the forecast on the upper bounds on the annihilation cross-section $\langle\sigma v\rangle$ obtained in this work with other bounds from the literature, for the $b\bar{b}$ annihilation channels: bounds from non-observation of dark sub-halos (dashed yellow) [368], from Sculptor dwarf galaxy [365] with 100 hours (solid magenta) and with 500 hours (dot-dashed magenta) of observations with CTAO. Sensitivities from the Galactic halo have large uncertainties due to the assumptions on the dark matter profile toward the Galactic Center, therefore we show two limiting cases of an NFW profile (dashed gray) and of a cored profile (dotted gray) profiles for 500h with CTAO [365]. Right: Comparison between the forecast on the lower bounds on the DM particle lifetime τ obtained in this work with other bounds from the literature, for the $b\bar{b}$ decay channel: bound from Perseus with CTAO [381] and MAGIC [382], from galaxy clusters [383] and from Fornax [384].

astrophysical case.

For cross-correlations, we adopt the following test statistics:

$$\Delta\chi^2 = \sum_{\ell,i} \left(\frac{C_{\ell,i}^{\gamma_{\text{astro}}+\text{DM}\mathcal{G}} - C_{\ell,i}^{\gamma_{\text{astro}}\mathcal{G}}}{\Delta C_{\ell,i}^{\gamma_{\text{astro}}+\text{DM}\mathcal{G}}} \right)^2 \quad (8.4.6)$$

where $C_{\ell,i}^{\gamma_{\text{astro}}\mathcal{G}}$ denotes the cross-correlation signal due to the presence of only astrophysical sources, while $C_{\ell,i}^{\gamma_{\text{astro}}+\text{DM}\mathcal{G}}$ refers to the case where a DM-induced γ -ray emission is present. The corresponding standard deviations are $\Delta C_{\ell,i}^{\gamma_{\text{astro}}\mathcal{G}}$ and $\Delta C_{\ell,i}^{\gamma_{\text{astro}}+\text{DM}\mathcal{G}}$, respectively. Since the DM signal depends on the DM mass m_χ and scales linearly with the annihilation rate $\langle\sigma v\rangle$ (or inversely with the lifetime τ in the case of decay), as outlined in Section 8.B, we derive bounds on the annihilation (or decay) rate by scanning over the DM mass m_χ . We then derive the 2σ bounds on the annihilation (or decay) rate by requiring $\Delta\chi^2 = 4$. A similar test statistic and procedure is employed for the auto-correlation case. We stress that the DM results are derived assuming the nominal case for the astrophysical sources scenario (i.e., HSP+LISP) and point-source sensitivity. Nonetheless, the LISP component has no impact on the DM results due to low-energy SED cut-off, as explained in the previous section. We have also verified that the point-source sensitivity uncertainty discussed in the previous section has a negligible impact on the DM results. To this aim, we explicitly derived DM sensitivities for the two cases of nominal and degraded point-source sensitivity, finding indistinguishable results.

Figure 8.5 shows the sensitivity to the DM annihilation and decay in the $b\bar{b}$ annihilation channel, derived from the auto-correlation and cross-correlation APS with the 2MASS catalog,

for 3hrs and 5hrs of CTAO observations. The 5hrs case yields approximately a factor of 4 improvement in sensitivity compared to the 3hrs case, while the cross-correlation outperforms the auto-correlation by approximately a factor of 5. This is expected since in the auto-correlation, the DM signal competes with strong blazar auto-correlation, while in the cross-correlation, the blazar contribution is much lower, yielding a more favorable signal-to-background ratio. Sensitivities obtained from the cross-correlation with the 2MRS catalog are very similar and are shown Section 8.B. Note that in this case, the uncertainty on the point-source detection threshold does not play a significant role, since the DM signal is always diffuse.

Figure 8.6 shows the sensitivities of the cross-correlation technique for different annihilation channels, while Fig. 8.7 provides a comparison with existing results in the literature for alternative DM search strategies. In particular, we show bounds from observations of dwarf galaxies [365] and dark sub-halos [368] for DM annihilation, and from galaxy clusters for decaying DM [381–384]. Our results show that the cross-correlation technique is competitive, yielding constraints comparable to those from other methodologies. For comparison, the figure also includes the sensitivity for annihilation in the Galactic halo, for 500 hr of observations with CTAO [365]: in this case, the values of $\langle\sigma v\rangle$ that can be probed can be lower, but the reachable bounds are strongly dependent on the assumptions regarding the DM profile near the Galactic Center.

In Section 8.C we show a comprehensive set of plots for different hours of observations and DM channels.

8.4.4 Summary and conclusions

The CTAO will be the next generation of ground-based γ -ray Cherenkov detectors, offering a unique opportunity to study high-energy phenomena in the TeV range. In addition to targeted observations, CTAO is expected to carry out large-area sky surveys, making it particularly well-suited for anisotropy-based techniques such as auto-correlation and cross-correlation analyses. In this work, we have investigated the sensitivity of future CTAO data to constrain the properties of unresolved blazars and DM signals using these methods.

We found, in particular, that the auto-correlation APS method is better suited for probing the signal from unresolved blazars, with its sensitivity strongly dependent on the effective point-source detection threshold. On the other hand, the cross-correlation APS technique, when used in combination with galaxy catalogs, offers very promising sensitivity to potential DM annihilation or decay signal, reaching values around $10^{-23} \text{ cm}^{-3} \text{ s}^{-1}$ for annihilation cross-section $\langle\sigma v\rangle$ and 10^{27} s for decay. These sensitivities are comparable to and complementary with those obtained from existing strategies, such as observations of dwarf spheroidal galaxies or galaxy clusters.

In the future, these results can be further improved with upcoming, deeper, and more densely sampled galaxy catalogs, such as those from the European Space Agency’s *Euclid* satellite mission [385], as well as by incorporating cross-correlations with other large-scale-structure tracers, such as weak lensing.

Data Availability

The data that support the findings of this work are openly available on Zenodo [386].

8.A Off-source data scenario

As described in the main text, we follow the strategy outlined in Ref. [368] to extrapolate MAGIC observations to CTAO. The distribution of MAGIC pointings, that we consider when extrapolating

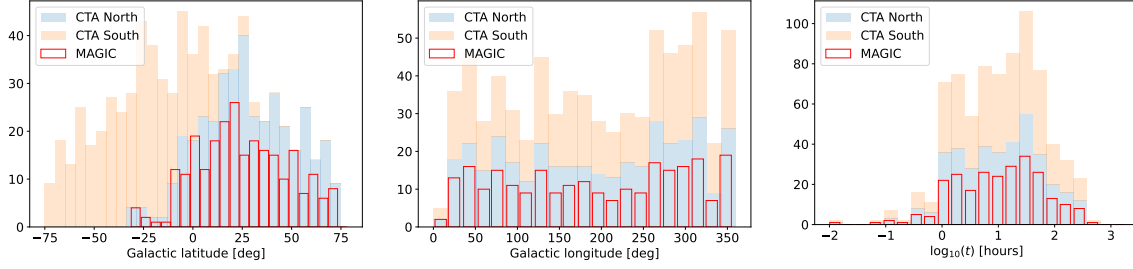


Fig. 8.8. Distributions of Galactic latitude (left), longitude (center), and observation time (right) for one realization of the predicted CTAO pointings over 10 years of observations. The stacked histograms show contributions from the Northern (blue) and Southern (orange) arrays. For comparison, the red outlined histograms indicate the corresponding distributions from 6.5 years of MAGIC observations.

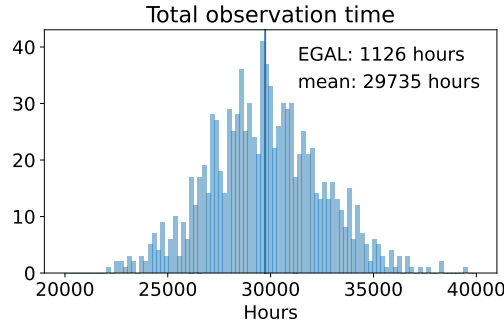


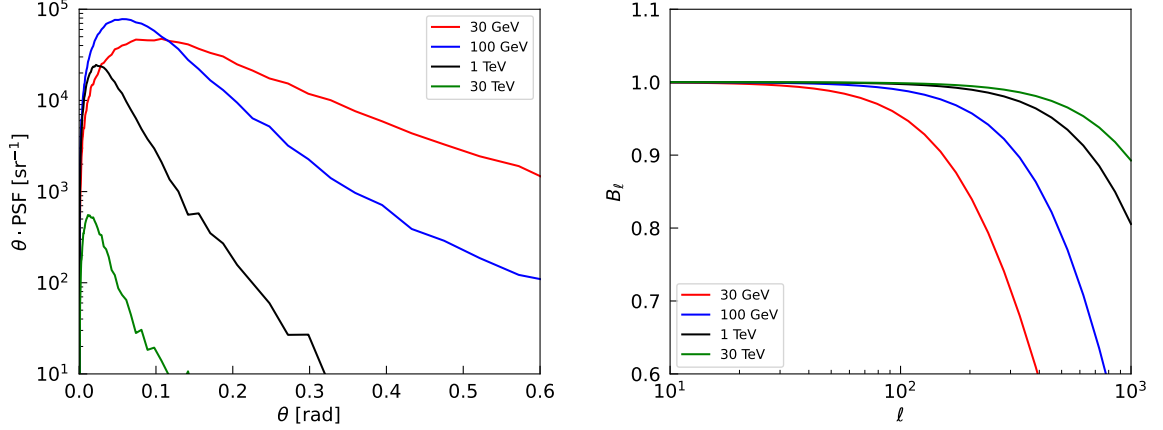
Fig. 8.9. Distribution of the total observation time from 1000 realizations of predicted CTAO pointings across the whole sky using both arrays. The blue vertical line marks the mean total observation time. The total time in the most optimistic scenario is about 25 times higher than that of the EGAL survey, effectively translating the nominal 3 hours per pointing into ~ 50 hours per sky location in this scenario.

the observation time and pointing directions of future CTAO measurements across the sky, is shown in Fig. 8.8, together with one realization of the predicted CTAO distributions of the same quantities, i.e. the Galactic latitude, longitude, and the observation time per pointing. The distribution of the cumulative pointing time over all pointings, obtained from 1000 different realizations of future CTAO observations, is shown in Fig. 8.9.

8.B Formalism and details of the modeling

The general cross-correlation formalism—the angular power spectrum, the Limber approximation, the variance expressions, and the halo model decomposition—is introduced in Section 3.5. This appendix provides the specific forms and modeling choices entering the CTAO analysis.

Bin	$E_{\min}$	$E_{\max}$
Low	8 GeV	126 GeV
Mid	126 GeV	1.3 TeV
High	1.3 TeV	31.6 TeV

Table 8.4. Co-added energy bins used in the analysis for display purposes.**Fig. 8.10.** Left: Point spread functions of CTAO, as a function of angle for different photon energies. Right: The corresponding CTAO beam window functions as a function of multipole.

8.B.1 Angular Power Spectra

The angular power spectra for the γ -ray autocorrelation and for the cross-correlation between the unresolved γ -ray emission and a galaxy catalog g can be expressed as (cf. Eq. 3.5.4) [201, 202, 369]

$$C_{\ell}^{\gamma\gamma} = \int \frac{d\chi}{\chi^2} W_{\gamma}^2(\chi) P_{\gamma\gamma}(k = \ell/\chi, \chi) \quad (8.B.1)$$

$$C_{\ell}^{\gamma g} = \int \frac{d\chi}{\chi^2} W_{\gamma}(\chi) W_g(\chi) P_{\gamma g}(k = \ell/\chi, \chi), \quad (8.B.2)$$

where ℓ denotes the multipole, χ is the comoving distance, $W_{\gamma}(\chi)$ and $W_g(\chi)$ are the window functions of the γ -ray emission and of the galaxy catalog, respectively, that is the functions describing how the observables are distributed in redshift. $P_{\gamma\gamma}$ and $P_{\gamma g}$ are the 3D power spectra of the auto- and cross- correlation, respectively, which quantify the fluctuations of the corresponding observables. All these components are discussed in the following sub-sections.

$C_{\ell}^{\gamma\gamma}$ and $C_{\ell}^{\gamma g}$ are both energy-dependent. In our approach, we compute the correlation functions within the specific energy bins for which the detector sensitivity is available. CTAO sensitivities are provided for 19 energy bins spanning the range (5 GeV, 501 TeV). Due to the sensitivity at low energies, we restrict our analysis to 16 bins, logarithmically spaced within the 30 GeV – 30 TeV range. For visualization purposes, the energy-dependent quantities are further coadded into 3 broader bins, as summarized in Table 8.4. These low-, medium-, and high-energy bins correspond to the CTAO bin numbers 2-7, 8-12, and 13-18, respectively. However, when computing the DM constraints, we make sure of all 19 energy bins available for CTAO (bins 1 to 19), thereby covering the full energy range from 5 GeV to 501 TeV.

The variance of the auto-correlation APS, in the Gaussian approximation, is given by Eq. (8.4.4).

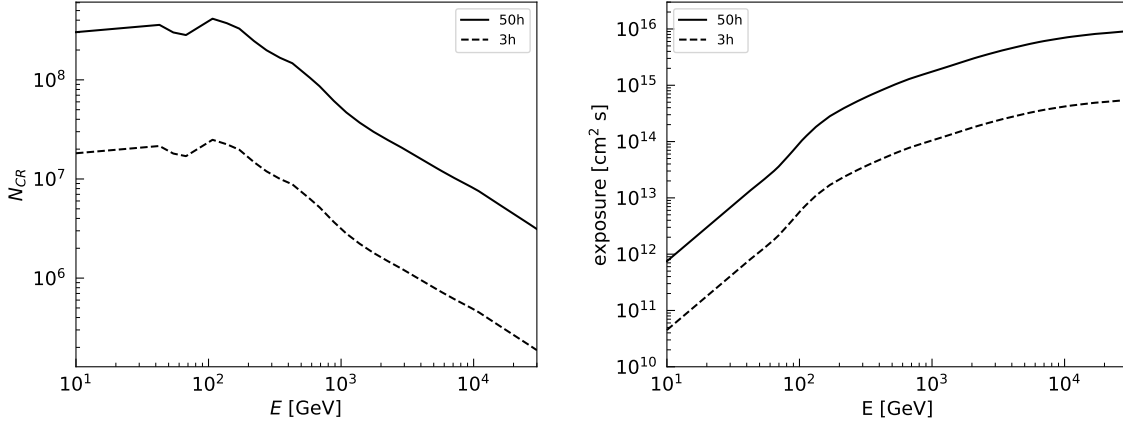


Fig. 8.11. Left: Cosmic-ray background N_{CR} as a function of energy. Right: CTAO exposure as a function of energy.

The sky fraction observed by CTAO is $f_{\text{sky}} \sim 0.25$ and it is considered energy-independent in our analysis. The beam window functions B_ℓ account for the smearing of the angular signal due to the finite angular resolution of the instrument and can be defined in terms of the point spread function (PSF) as

$$B_\ell(E) = \int d\theta \text{PSF}(\theta, E) \sin\theta P_\ell(\cos\theta), \quad (8.B.3)$$

where $P_\ell(\cos\theta)$ are the Legendre Polynomials of order ℓ . The point-spread function $\text{PSF}(\theta, E)$ has been obtained through a Monte Carlo simulation of the CTAO detector, and is energy-dependent. Fig. 8.10 shows the $\text{PSF}(\theta, E)$ as a function of the angle θ (left panel) and the corresponding beam window function as a function of the multipole ℓ (right panel) for different energies.

The beam window function in a specific energy bin is then obtained by averaging over the energy-dependent intensity flux dI/dE , namely

$$\langle B_\ell \rangle = \frac{1}{N} \int_{\text{bin}} dE B_\ell(E) \frac{dI}{dE}, \quad (8.B.4)$$

where $N = \int_{\text{bin}} dE dI/dE$. For definiteness, we use $dI/dE \propto E^{-2.3}$, which appropriately matches the observed behavior of the unresolved γ -ray intensity as measured by *Fermi*-LAT.

The photon noise C_N is also energy-dependent and is determined as

$$C_N = \frac{4\pi f_{\text{sky}}}{N_\gamma} \left(1 + \frac{N_{\text{CR}}}{N_\gamma} \right) \langle I \rangle^2, \quad (8.B.5)$$

where N_{CR} represents the cosmic-ray irreducible background, N_γ is the number of γ -ray events, and $\langle I \rangle$ denotes the mean γ -ray intensity in the specific energy bin, which is determined from the window functions as: $\langle I \rangle = \int d\chi W(\chi)$. N_γ is therefore related to $\langle I \rangle$ via

$$N_\gamma = 4\pi f_{\text{sky}} \mathcal{E} \langle I \rangle, \quad (8.B.6)$$

where $\mathcal{E}$ represents the CTAO exposure in the given energy bin. The exposure and the counts associated with the cosmic-ray background are derived via Monte Carlo simulations of the CTAO

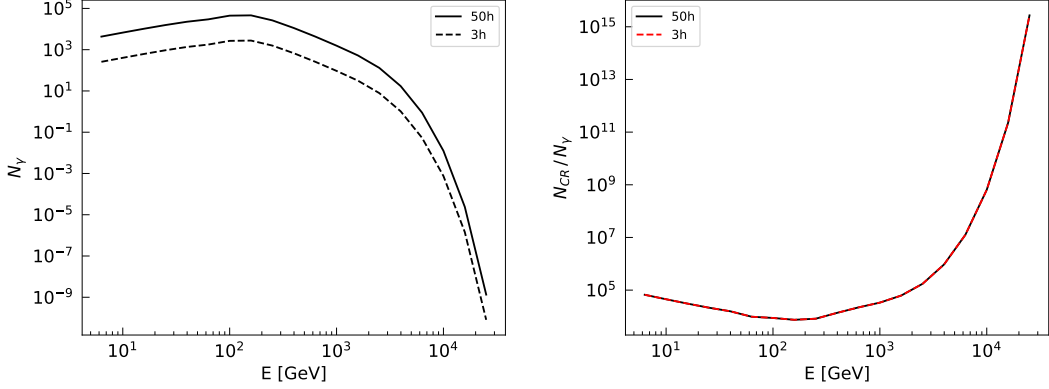


Fig. 8.12. Left: Number of expected γ -ray events N_γ in the CTAO survey as a function of energy. Right: Ratio of cosmic-ray events to expected γ -ray events (N_{CR}/N_γ) after quality cuts in the CTAO survey. The ratio worsens above 1 TeV due to significant γ -ray suppression from EBL attenuation. N_γ is determined for the HSP plus LISP source classes.

array. The simulation has been performed for 1.62h of observation time, and the results were then rescaled to obtain the exposure for 3hrs and 50h. Fig. 8.11 displays the estimated cosmic-ray backgrounds N_{CR} (left panel) and exposures $\mathcal{E}$ (right panel) as a function of the energy.

Fig. 8.12 shows the the number of γ -ray events N_γ and the ratio N_{CR}/N_γ as a function of energy. Since $N_{CR}/N_\gamma \gg 1$ in all the energy bins considered in our analysis, the noise can be approximated as

$$C_N \simeq 4\pi f_{\text{sky}} \frac{N_{CR}}{N_\gamma^2} \langle I \rangle^2 \quad (8.B.7)$$

By substituting the expression of N_γ in Eq. (8.B.7), it becomes evident that C_N can be expressed only in terms of detector properties and configuration,

$$C_N \simeq \frac{N_{CR}}{4\pi f_{\text{sky}} \mathcal{E}^2} \quad (8.B.8)$$

The noise C_N is shown explicitly in Fig. 8.21, together with a selection of auto- and cross-correlation energy spectra.

In the case of cross-correlations, the variance is instead given by (cf. Eq. 3.5.8) [201, 202, 369]

$$(\Delta C_\ell^{\gamma g})^2 = \frac{1}{(2\ell + 1)f_{\text{sky}}} \left[(C_\ell^{\gamma g})^2 + \left(C_\ell^{\gamma\gamma} + \frac{C_N}{B_\ell^2} \right) (C_\ell^{gg} + C_{N_g}) \right]. \quad (8.B.9)$$

where C_ℓ^{gg} is the galaxy auto-correlation, C_N is the same as in the auto-correlation case, C_{N_g} is the galaxy catalog noise. There is no beam correction in the galaxy term, as the angular resolution on the galaxy catalog is on the order of arcseconds, making it negligible in this analysis. The galaxy noise is given by

$$C_{N_g} = \frac{4\pi f_{\text{sky},g}}{N_g} \quad (8.B.10)$$

where $f_{\text{sky},g}$ represents the fraction of the sky covered by the catalog, and N_g is the number of galaxies in the catalog. Fig. 8.21 shows the quantity $\sqrt{C_N C_{N_g}}$ as a representative level

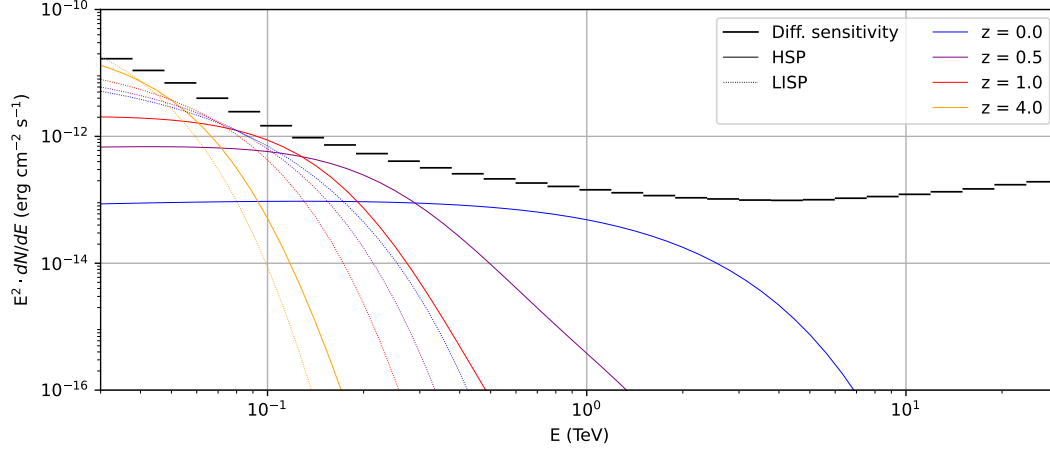


Fig. 8.13. Differential flux sensitivity for 5 hrs of observation in the considered CTAO energy range (thick black horizontal dashes), compared with the energy spectra of HSP and LISP sources (solid and dotted lines, respectively) at four different redshifts. The spectra are normalized to the minimal flux detectable with the planned EGAL survey.

for the cross-correlation noise, which approximately corresponds to the noise expected in a noise-dominated regime.

8.B.2 Window functions

Gamma-ray astrophysical sources The window function for the γ -ray emission of unresolved astrophysical sources can be written as

$$W(E, \chi) = \frac{dW}{dE d\chi} = \frac{d_L^2(z)}{(1+z)^2} \int_{L_{\min}}^{L_{\text{th}}(z)} dL \Phi(L, z) \frac{dF}{dE}(E, L, z) \exp[-\tau(E, z)] \quad (8.B.11)$$

where $d_L(z)$ is the luminosity distance, $\Phi(L, z)$ is the γ -ray luminosity function of the source population. In the present analysis, we use the luminosity functions of HSP and LISP as reported in [372]. Note that the above window function, which is used in Eq. 8.B.1-8.B.2 is differential in comoving distance, χ . The analogous quantity differential, differential in z , would be $W(E, z) = \frac{dW}{dE dz} = \frac{dW}{dE d\chi} \frac{d\chi}{dz} = W(E, \chi) \frac{d\chi}{dz}$ where $\frac{d\chi}{dz} = \frac{c}{H(z)}$. Also note that the above window is differential in energy. In Eq. 8.B.1-8.B.2, however, the corresponding quantity must be integrated over the energy bin considered. L is the γ -ray luminosity of the source, conventionally integrated over the interval (0.1, 100) GeV, that is

$$L = \int_{0.1 \text{ GeV}}^{100 \text{ GeV}} dE_r \frac{dL}{dE_r} \quad (8.B.12)$$

where E_r is the source rest-frame energy, related to the observed energy by $E_r = (1+z)E$, and dF/dE is the spectral energy distribution (SED) of the sources, which we model as a power law with index Γ and an exponential cutoff,

$$\frac{dF}{dE} = A(L, z) E^{-\Gamma} \exp(-E/E_{\text{cut}}) \quad (8.B.13)$$

The values of Γ and E_{cut} for the two classes of astrophysical sources adopted in our models are reported in Table 8.1, together with the minimum luminosity of the source classes $L_{\min}$. The

parameter A , on the other hand, can be related to the luminosity L , by observing that (see Eq. 22 in [387])

$$L = \frac{4\pi d_L^2(z)}{(1+z)} \int_{0.1 \text{ GeV}}^{100 \text{ GeV}} dE_r E_r \frac{dF}{dE_r} = 4\pi d_L^2(z) \int_{0.1 \text{ GeV}/(1+z)}^{100 \text{ GeV}/(1+z)} dE E \frac{dF}{dE} \quad (8.B.14)$$

and therefore:

$$A(L, z) = \frac{L}{4\pi d_L(z)^2} \left[\int_{0.1 \text{ GeV}/(1+z)}^{100 \text{ GeV}/(1+z)} dE E^{(-\Gamma+1)} \exp(-E/E_{\text{cut}}) \right]^{-1} \quad (8.B.15)$$

The last term in Eq. (8.B.11) models the γ -ray absorption on extragalactic background light (EBL). For the optical depth $\tau(E, z)$, we adopt the results from Ref. [388]. The maximal luminosity $L_{\text{th}}(z)$ in Eq. (8.B.11), which refers to the threshold of detectability of resolved sources by the detector, depends on the CTAO sensitivity $F_{\text{sens}}^{\text{CTA}}(z)$ to point sources in the energy range of operation, which in our analysis is defined as (30 GeV, 30 TeV)

$$F_{\text{sens}}^{\text{CTA}}(z) = \int_{30 \text{ GeV}}^{30 \text{ TeV}} dE \frac{dF}{dE} \exp[-\tau(E, z)] = A_{\text{th}}(z) \int_{30 \text{ GeV}}^{30 \text{ TeV}} dE E^{-\Gamma} \exp(-E/E_{\text{cut}}) \exp[-\tau(E, z)] \quad (8.B.16)$$

From $F_{\text{sens}}^{\text{CTA}}(z)$, it is therefore possible to determine the value of $A_{\text{th}}(z)$, which allows us to determine the threshold luminosity by means of Eq. (8.B.14) as

$$L_{\text{th}}(z) = 4\pi d_L^2(z) F_{\text{sens}}^{\text{CTA}}(z) \frac{\int_{0.1 \text{ GeV}/(1+z)}^{100 \text{ GeV}/(1+z)} dE E^{-\Gamma+1} \exp(-E/E_{\text{cut}})}{\int_{30 \text{ GeV}}^{30 \text{ TeV}} dE E^{-\Gamma} \exp(-E/E_{\text{cut}}) \exp[-\tau(E, z)]} \quad (8.B.17)$$

The HSP and LISP point source sensitivity has been obtained through extensive Monte Carlo simulations [365, 389] and is shown in Fig. 8.13 for 3 hrs of observation across different energy bins. The same plot compares the differential sensitivity with the γ -ray spectra of LISP and HSP blazars normalized to the minimum detectable intensity. As shown, the spectra observed at Earth are redshift-dependent, as sources at different redshifts experience different levels of EBL absorption. Fig. 8.14 instead shows the integrated HSP and LISP point source sensitivity in the energy range 30 GeV to 30 TeV as a function of z , where the normalization of the SED in Eq. (8.B.13) has been fixed to its threshold value $A_{\text{th}}(z)$, obtained from Eq. (8.B.16) by employing the Monte Carlo modeling of $F_{\text{sens}}^{\text{CTA}}(z)$.

Gamma-ray emission from dark matter Window functions for dark matter are modeled as in Refs. [201, 363, 369], to which we refer for additional details. For decaying dark matter, we have

$$W(E, \chi) = \frac{dW}{dE d\chi} = \left(\frac{\Omega_{\text{DM}} \rho_c}{m_\chi} \right) \frac{\Gamma_d}{4\pi} \frac{dN}{dE} [(1+z)/E] \exp[-\tau(E, z)] \quad (8.B.18)$$

where Ω_{DM} is the dark matter density parameter, ρ_c is the critical density of the Universe, m_χ the DM particle mass, Γ_d the DM decay rate, dN/dE the γ -ray energy spectrum per decay event, which we take from [102] for decay into $b\bar{b}$, $\tau^+\tau^-$, W^+W^- . The last term in the equation refers again to γ -ray absorption, as discussed for the astrophysical sources.

For annihilating DM, the window function reads (cf. Eq. 3.5.5)

$$W(E, \chi) = \frac{dW}{dE d\chi} = \frac{1}{4\pi} \frac{\langle \sigma v \rangle}{2} \Delta^2(z) \left(\frac{\Omega_{\text{DM}} \rho_c}{m_\chi} \right)^2 (1+z)^3 \frac{dN}{dE} [(1+z)E] \exp[-\tau(E, z)], \quad (8.B.19)$$

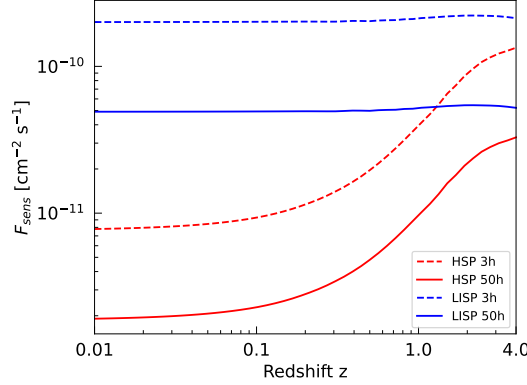


Fig. 8.14. CTA sensitivity $F_{\text{sens}}^{\text{CTA}}(z)$ for 3 hrs and 5 hrs of observation as a function of redshift, for the HSP and LISP γ -ray source classes.

where $\langle\sigma v\rangle$ is the velocity-averaged annihilation cross-section. Unlike DM decay, the annihilation window function also contains the flux multiplier factor $\Delta^2(z)$, which accounts for the enhancement of the flux due to the clumping of DM and the presence of sub-structures within the DM halos. Details on the way we model the flux multiplier can be found in Ref. [107, 209, 390]. Window functions for both DM and astrophysical sources are plotted in Figure 8.15 as a function of redshift for two different energies, and for 3 hrs and 5 hrs of observation time. It can be seen that the DM window always peaks at low z , while the astrophysical signal peaks at $z \sim 0.4 - 0.5$ at 50 GeV and $z \sim 0.1 - 0.2$ at 1 TeV, with the horizon shrinking due to the effect of EBL γ -ray absorption. Fig. 8.16, instead, displays the intensity $\langle I \rangle$ as a function of the energy for the two classes of astrophysical sources used in the analysis and for a few representative cases of DM annihilation.

Galaxy Catalogs The galaxy-catalog window functions are given by the redshift distribution of the number of objects in the catalog, namely

$$W_g(z) = \frac{dN_g}{dz}. \quad (8.B.20)$$

The window function is normalized such that $\int dz W_g(z) = 1$. Note again that $W_g(\chi)$ entering the cross-correlation APS is related to the redshift-dependent window function as $W_g(\chi) = W_g(z) \frac{dz}{d\chi} = W_g(z) H(z)/c$, where $H(z)$ is the Hubble parameter and c the speed of light. In this way we also have $\int d\chi W_g(\chi) = 1$. We use the dN_g/dz as given in [375, 376] for 2MRS and [373, 374] for 2MASS.

8.B.3 Galaxies Halo Occupation Distribution

To model the clustering of galaxies within the catalog, we use the Halo Occupation Distribution (HOD) formalism (see [205, 377, 378]).

The number of galaxies residing within a DM halo of mass M is parameterized as

$$\langle N_{\text{cen}}(M) \rangle = \frac{1}{2} \left[1 + \text{erf} \left(\frac{\log M - \log M_{\text{th}}}{\sigma_{\log M}} \right) \right], \quad (8.B.21)$$

$$\langle N_{\text{sat}}(M) \rangle = \left(\frac{M}{M_1} \right)^\alpha \exp \left(-\frac{M_{\text{cut}}}{M} \right), \quad (8.B.22)$$

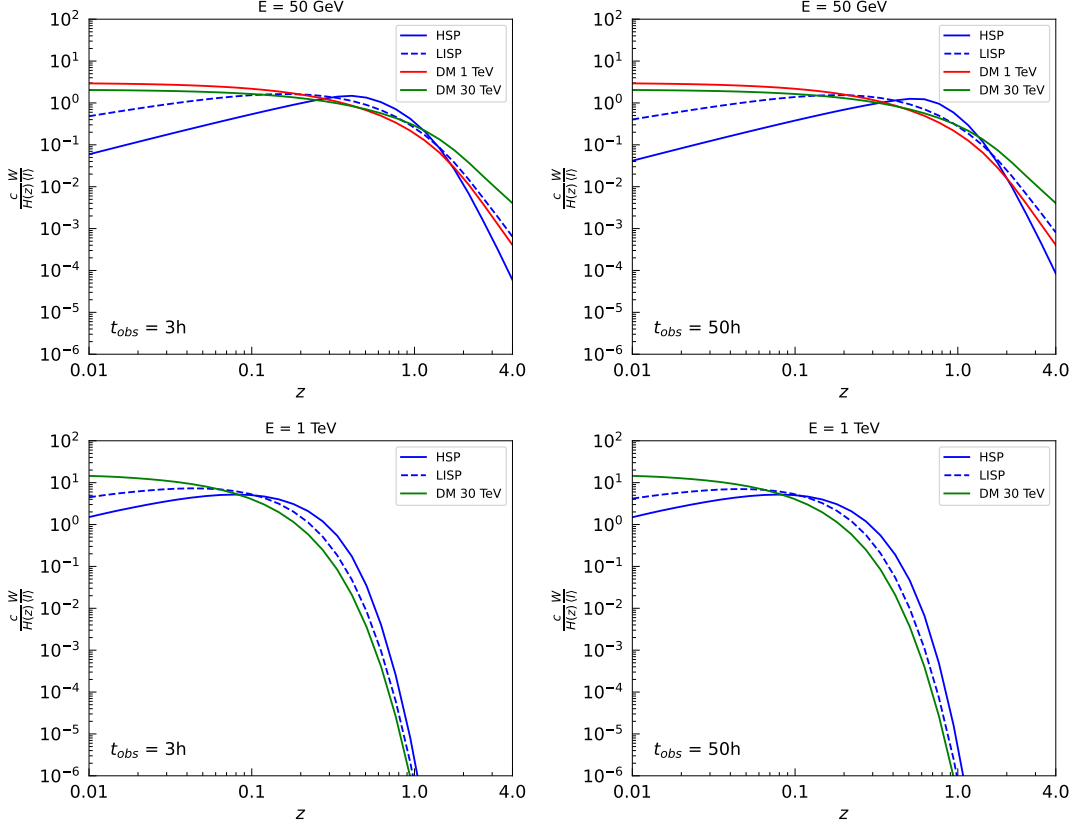


Fig. 8.15. γ -ray window functions, normalized to the mean γ -ray intensity, as a function of redshift (and divided by the Hubble parameter). Left panels show results for 3 hours of observation, and right panels for 50 hours. The blue solid and dashed lines represent the HSP and LISP astrophysical sources, respectively. The red and green solid lines refer to DM annihilation into $b\bar{b}$, $\langle\sigma v\rangle = 3 \times 10^{-26} \text{cm}^{-3} \text{s}^{-1}$ and masses of 1 TeV (red) and 30 TeV (green). The upper row is calculated for a γ -ray energy of 50 GeV, while the lower row is for 1 TeV (only the 30 TeV DM case is shown here). The 3-hour observation time cases are the same as shown in Figure 8.4.

where, for a halo of mass M , $\langle N_{\text{cen}}(M) \rangle$ is the average number of galaxies at the center of the halo and $\langle N_{\text{sat}}(M) \rangle$ represents the average number of satellite galaxies. The parameters M_{th} , $\sigma_{\log M}$, α , M_1 , M_{cut} characterize the HOD and we chose them according to [376] for 2MRS and [363] for 2MASS.

8.B.4 3D Power Spectra

The 3D Power Spectra $P(k)$ (PS) have been calculated by using the Halo Model formalism [205]. We have derived the autocorrelation 3D PS of decaying DM $P_{\delta}(k)$, annihilating DM $P_{\delta^2}(k)$ and astrophysical sources $P_{\gamma}(k)$, and the 3D cross-correlation PS among the same three categories of tracers: $P_{g\delta}(k)$ for the cross-correlation between galaxies and decaying DM γ -ray emission, $P_{g\delta^2}(k)$ for the cross-correlation between galaxies and annihilating DM γ -ray emission, for the cross-correlation between galaxies and astrophysical sources γ -ray emission $P_{g\gamma}(k)$. The explicit formulas for the power spectra are reported in Ref. [363], to which we refer the reader for details.

Figures 8.17-8.18 report the plots of the various 3D power spectra adopted in the analysis.

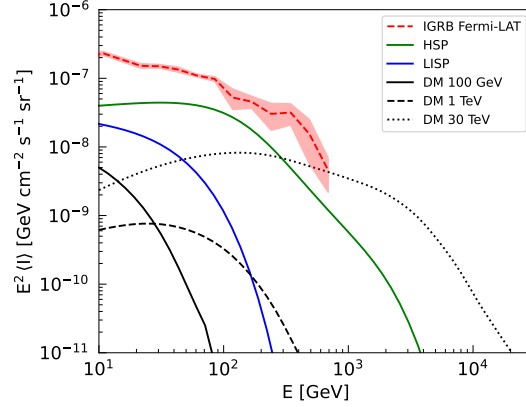


Fig. 8.16. γ -ray intensity in our benchmark model for HSP (solid green) and LISP (solid blue), along with three cases of dark matter annihilation for particle masses of 100 GeV (solid black), 1 TeV (dashed black), and 30 TeV (dotted black). The 100 GeV and 1 TeV lines correspond to an annihilation cross-section $\langle\sigma v\rangle = 3 \times 10^{-26} \text{ cm}^3 \text{ s}^{-1}$, while for 30 TeV we adopted $\langle\sigma v\rangle = 3 \times 10^{-23} \text{ cm}^3 \text{ s}^{-1}$. The figure also shows the measured IGRB (red dashed line and red band for its uncertainty) [138].

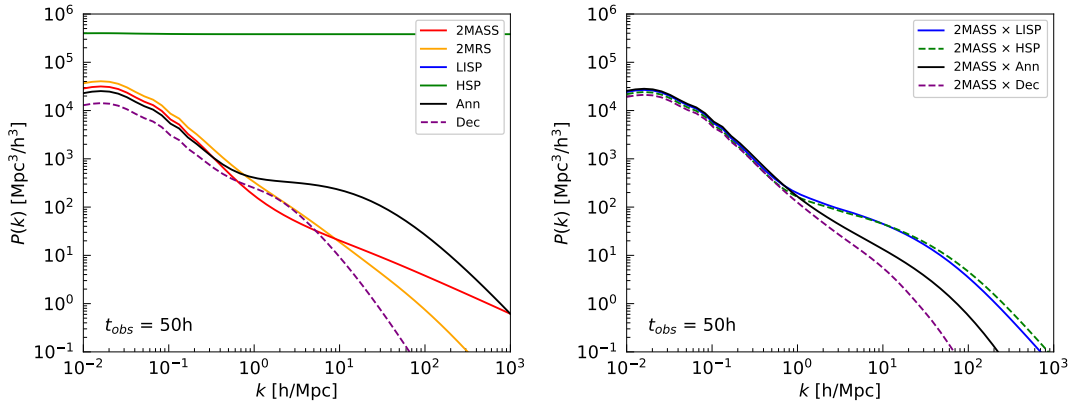


Fig. 8.17. Left: Auto-correlation 3D power spectra for 2MASS and 2MRS galaxies, LISP and HSP astrophysical sources, and annihilating and decaying dark matter. Right: Cross-correlation 3D power spectra for the cross-correlation between 2MASS galaxies and the various γ -ray source types considered in our analysis: LISP, HSP, annihilating and decaying dark matter.

Figures 8.19-8.20 show a few samples of auto and cross-correlation APS as a function of the multipole ℓ . It can be seen that auto APS of astrophysical sources is constant in ℓ , since they are point-like, while the DM APS presents some structure due to the extension of the DM halos. Figure 8.21, instead, shows the APS as a function of energy at a fixed multipole $\ell = 100$. For comparison, the auto- and cross-correlation noise level is plotted, thus showing that the expected signal-to-noise ratio is typically low.

8.B.5 Poisson anisotropy

The auto-correlation angular power spectrum of blazars is dominated by the constant 1-halo term, and is therefore flat in ℓ . It is thus convenient to define the Poisson anisotropy as the

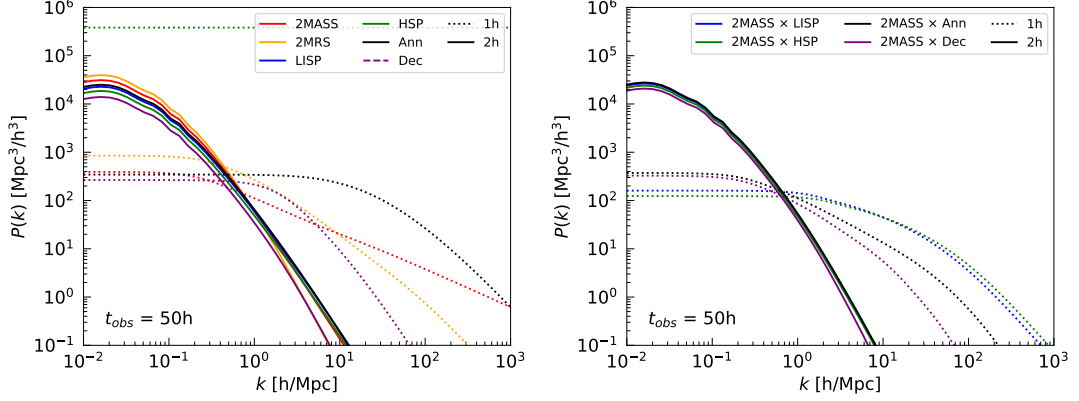


Fig. 8.18. The same as Figure 8.17 but with the 1-halo (dotted) and the 2-halo (solid) terms shown separately.

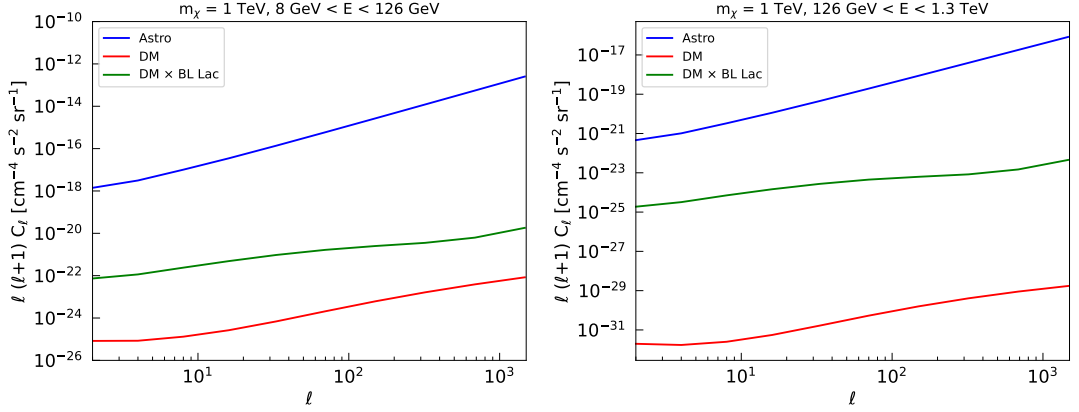


Fig. 8.19. Angular power spectra of the γ -ray auto-correlation as a function of the multipole, for two different energy bins. The upper (blue) curve refers to the auto-correlation among astrophysical sources, the median (green) and lower (red) curves to the correlation between sources and dark matter and the autocorrelation of dark matter, respectively, for a dark matter mass of 1 TeV and annihilation into $b\bar{b}$ with a cross-section $\langle\sigma v\rangle = 3 \times 10^{-26} \text{cm}^3 \text{s}^{-1}$.

normalization of this spectrum. Fig. 8.22 shows the Poisson anisotropy of unresolved blazars, divided by the square of the intensity as a function of energy for 3 hours and 50 hours of observation. As expected, the anisotropy decreases with increasing observation time, since a larger fraction of the brighter sources (which contribute more significantly to the anisotropy) become resolved. Across the entire energy range, the anisotropy remains at the level of percent or below.

The same figure also shows the CTA sensitivity to the Poisson anisotropy in three energy bins. To derive this sensitivity, we adopt the same methodology used for setting the DM limits. In this case, however, the null hypothesis corresponds to pure noise, that is, the absence of any source contribution, while the alternative hypothesis assumes the presence of a source class characterized by a Poisson anisotropy. We further assume that the energy dependence of the anisotropy of this source class is proportional to $\langle I \rangle^2$, i.e., it scales with the square of the observed intensity. As shown in the figure, the resulting sensitivity is at the level of approximately 10%.

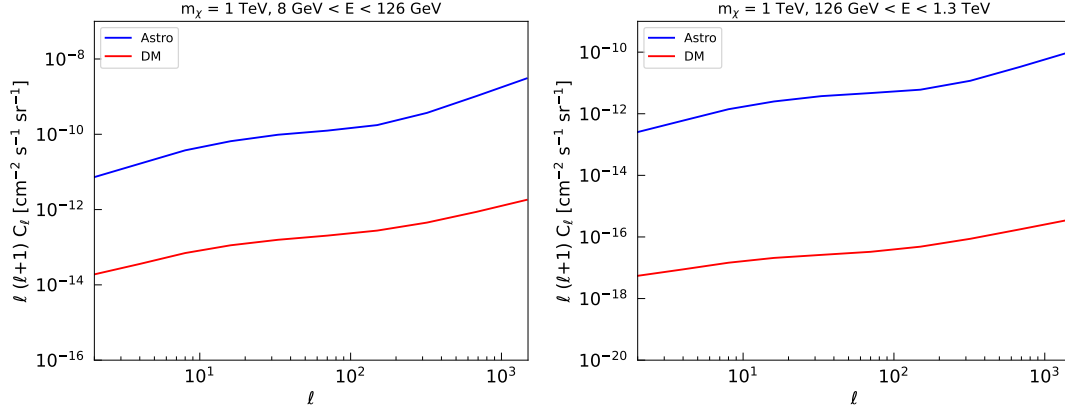


Fig. 8.20. Angular power spectra of the γ -ray cross-correlation with 2MASS galaxies, as a function of the multipole, for two different energy bins. The upper (blue) curve refers to the cross-correlation with astrophysical sources, while the lower (red) curves represent the correlation with a dark matter signal produced by a particle with mass of 1 TeV and annihilation cross-section into $b\bar{b}$ of $\langle\sigma v\rangle = 3 \times 10^{-26} \text{cm}^{-3} \text{s}^{-1}$.

8.C Further results

In Figures 8.23, 8.24, 8.25, 8.26, 8.27, 8.28, we show additional results on the constraints for the DM annihilation cross-section and the decay lifetime, exploring various annihilation and decay channels, as well as different observation times. We also provide further comparisons with existing bounds from the literature.

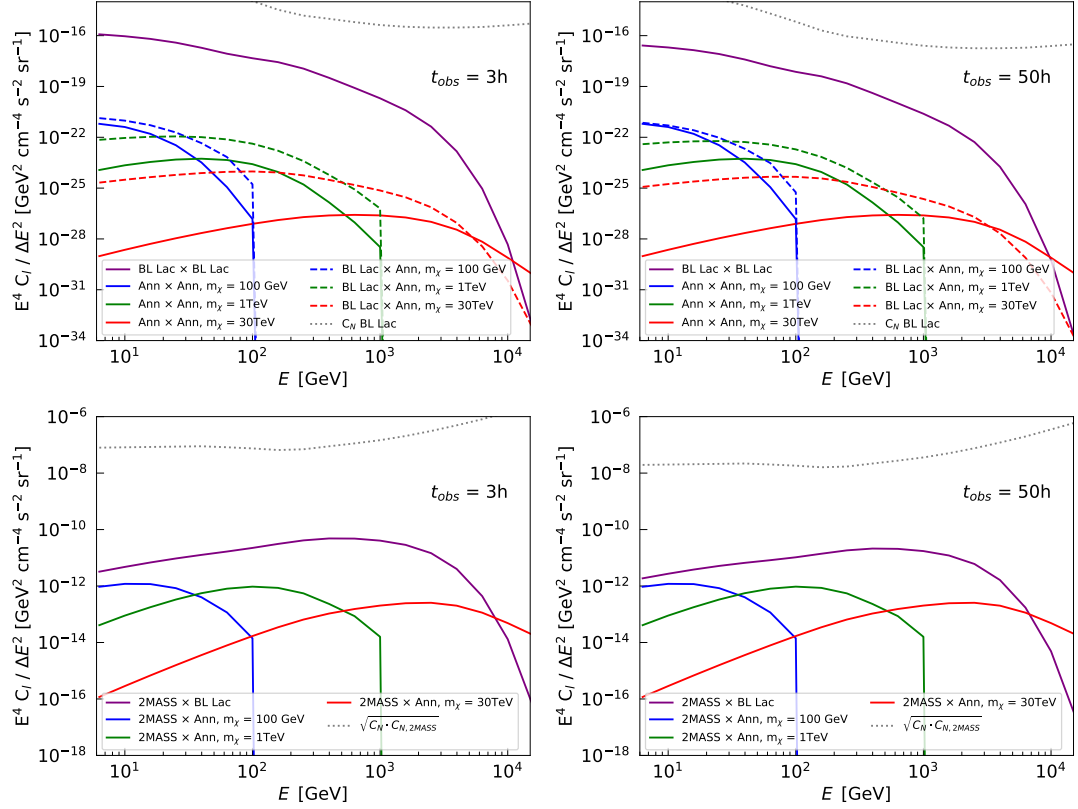


Fig. 8.21. Angular power spectra C_ℓ as a function of energy for $\ell = 100$. The left panels correspond to 3 hours of observation, while the right panels represent 50 hours. Dark matter signals are calculated for annihilation into $b\bar{b}$, for $\langle\sigma v\rangle = 3 \times 10^{-26} \text{cm}^3 \text{s}^{-1}$, and three representative masses: 100 GeV, 1 TeV, and 30 TeV. The solid and dashed lines show the auto-correlation and the cross-correlation C_ℓ for various combinations of γ -ray emitters and galaxy catalogs, as labelled. The black dotted lines in the topmost part of the panels represent the noise level for each category: for the upper row (γ -ray auto-correlations), the noise is dominated by the photon noise of astrophysical sources, and for the lower row (cross-correlation of γ -ray with galaxies), the geometric mean of the photon and galaxy noise is shown as a representative noise level.

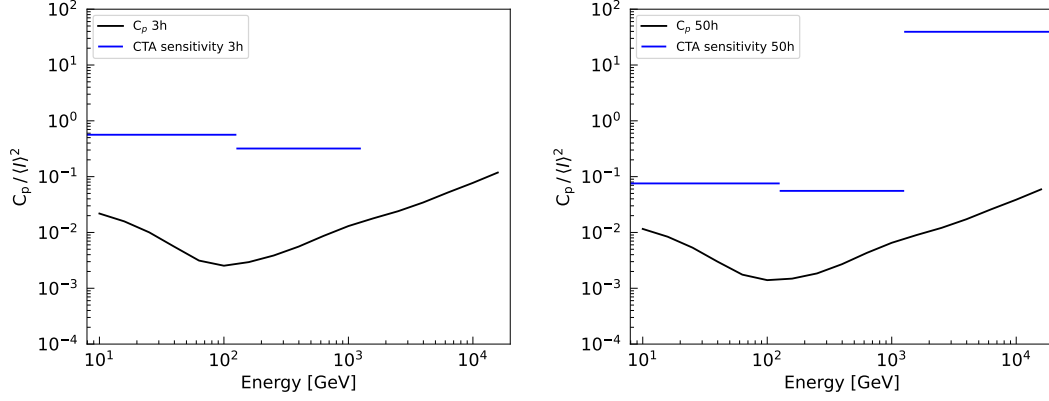


Fig. 8.22. CTA sensitivity to the Poisson anisotropy C_p (normalized to the square of the mean γ -ray intensity) in the low, mid and high energy bins, for 3 hours (left panel) and 50 hours (right panel) of observation. The predicted sensitivity is to the expected C_p from *unresolved* blazars (solid black lines). The reduced sensitivity at high energies is driven by the limited number of γ -ray events in this energy range.

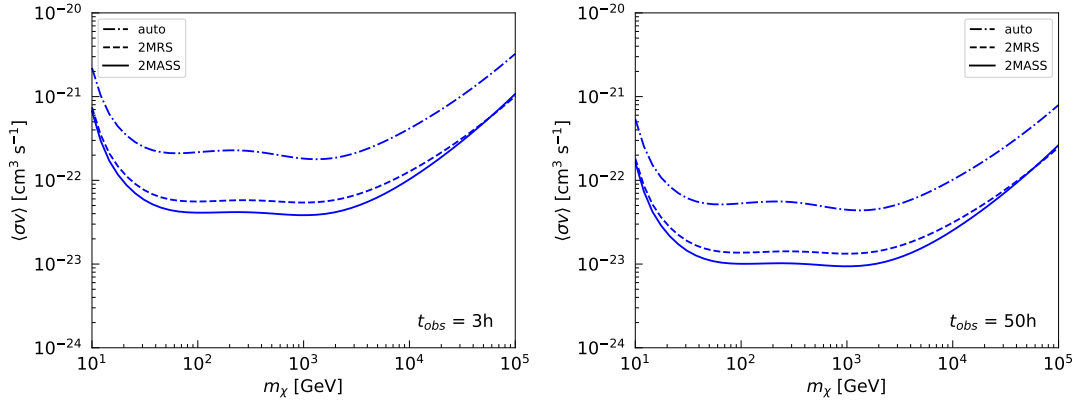


Fig. 8.23. Predicted upper bounds on the dark matter cross-section $\langle\sigma v\rangle$, for annihilation into $b\bar{b}$, for 3 hours (left panel) and 50 hours (right panel) of CTAO observations. The bounds are derived from γ -ray auto-correlation (dot-dashed), and cross-correlation with 2MRS (dashed lines) and 2MASS (solid) galaxy catalogs.

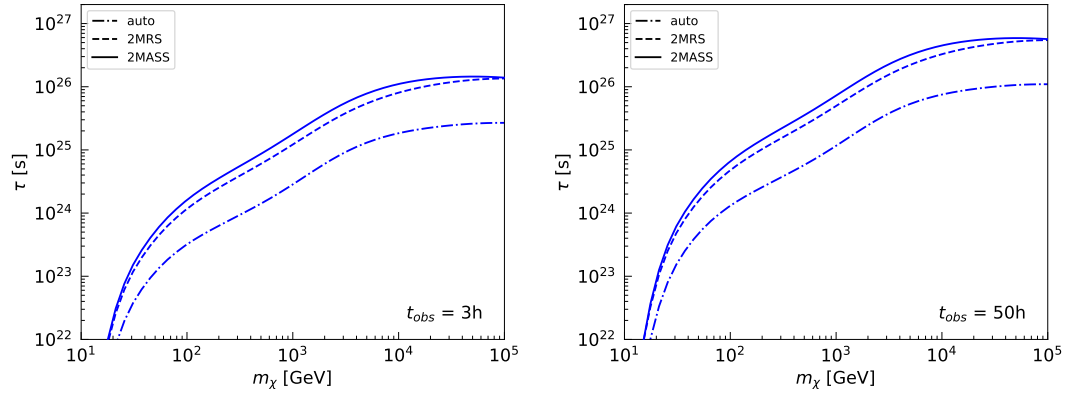


Fig. 8.24. Predicted lower bounds on the dark matter lifetime τ , for decay into $b\bar{b}$, for 3 hours (left panel) and 50 hours (right panel) of CTAO observations. The bounds are derived from γ -ray auto-correlation (dot-dashed), and cross-correlation with 2MRS (dashed lines) and 2MASS (solid) galaxy catalogs.

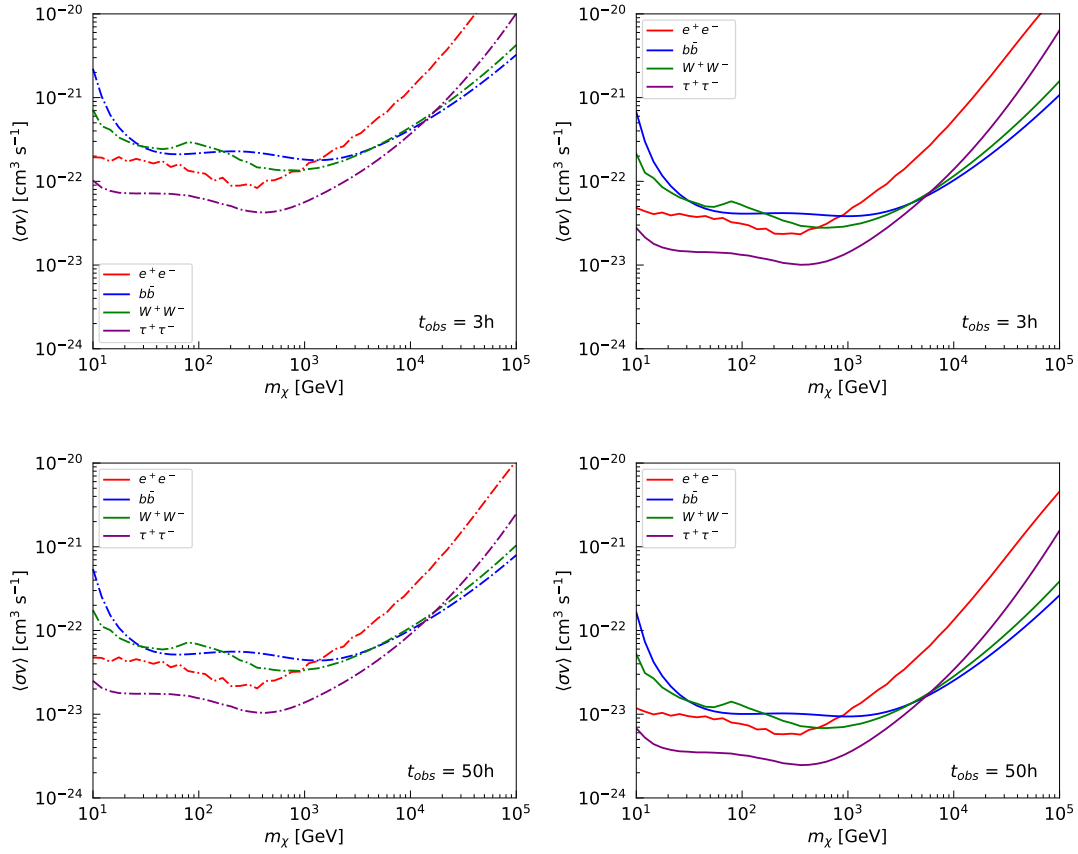


Fig. 8.25. Predicted upper bounds on the dark matter cross-section $\langle\sigma v\rangle$, for annihilation into various final states: e^+e^- , $b\bar{b}$, W^+W^- and $\tau^+\tau^-$. The upper row shows results for 3 hours of observation; the lower row for 50 hours. Left panels: bounds from γ -ray auto-correlation. Right panels: bounds from cross-correlation with 2MASS galaxies (the bottom-right panel reproduces the left panel of Fig. 8.6).

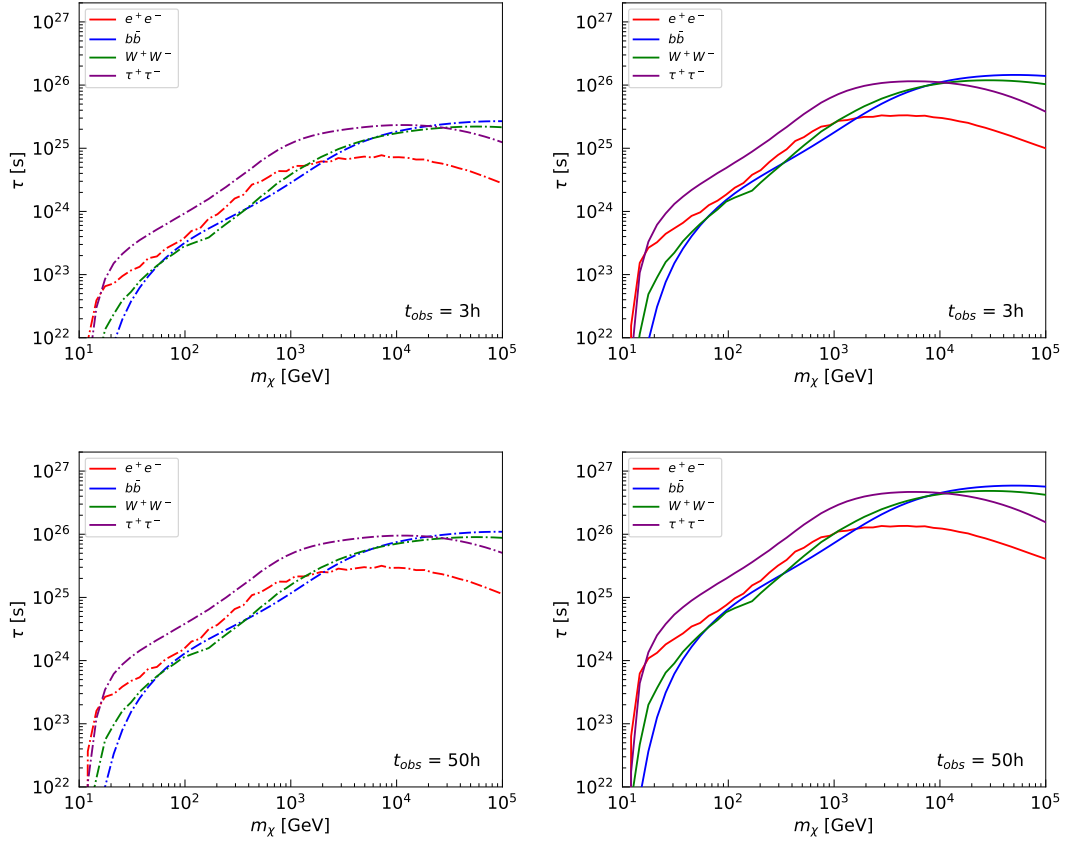


Fig. 8.26. Predicted lower bounds on the dark matter lifetime τ , for decay into various final states: e^+e^- , $b\bar{b}$, W^+W^- and $\tau^+\tau^-$. The upper row shows results for 3 hours of observation; the lower row for 50 hours. Left panels: bounds from the γ -ray auto-correlation. Right panels: bounds from cross-correlation with 2MASS galaxies (the bottom-right panel reproduces the right panel of Fig. 8.6).

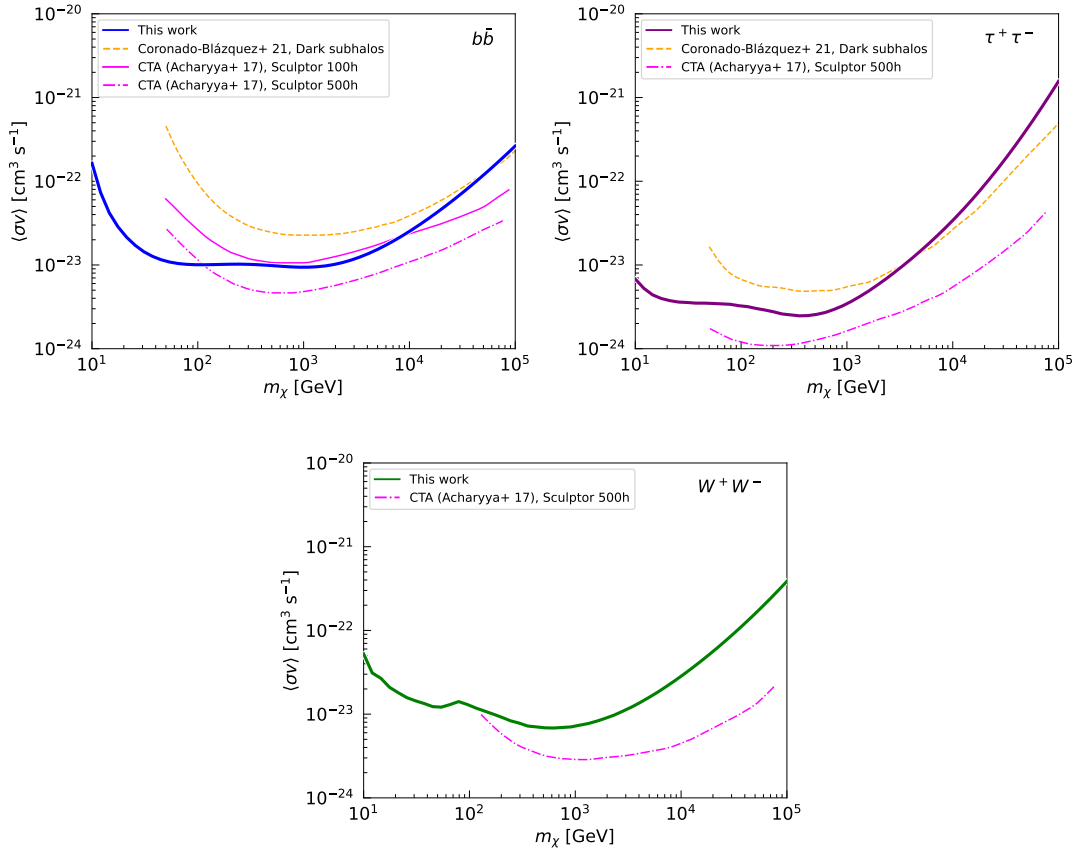


Fig. 8.27. Comparison of the predicted upper bounds on the dark matter annihilation cross-section $\langle\sigma v\rangle$, with existing constraints from the literature: bounds from non-observation of dark sub-halos (dashed yellow) [368], from the Sculptor dwarf galaxy [365] with 100 hours (solid magenta) and 500 hours (dot-dashed magenta) of observations with CTAO. Three channels are displayed: $b\bar{b}$ (upper left panel), $\tau^+\tau^-$ (upper right) and the W^+W^- (lower). The case for the $b\bar{b}$ channel is also shown in Fig. 8.5.

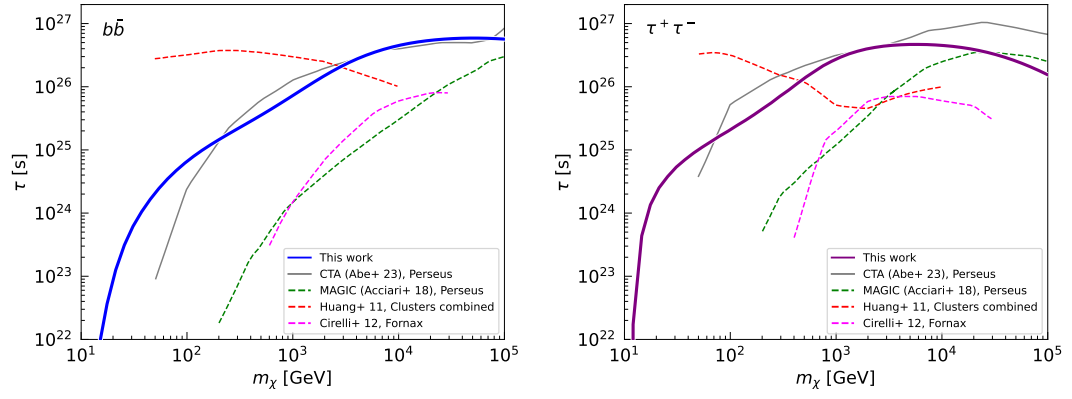


Fig. 8.28. Comparison of the predicted lower bounds on the DM particle lifetime τ for the $b\bar{b}$ and $\tau^+\tau^-$ decay channels, with existing constraints from the literature: limits from Perseus with CTAO [381] and MAGIC [382], from galaxy clusters [383], and from Fornax [384]. The case for the $b\bar{b}$ channel is also shown in Fig. 8.5.

CONCLUSIONS AND FUTURE DIRECTIONS

The particle nature of dark matter remains unknown, even as its gravitational imprint is written across every scale of the Universe. Gamma rays are among the cleanest messengers we have to look for the non-gravitational signatures of that dark sector, since an annihilation or decay signal would add photons to a sky we probe extensively. The difficulty is that any such signal would be buried in astrophysical backgrounds that are bright, structured, or imperfectly known, so the dark matter component we are after is faint and easily mimicked. This thesis followed a common thread through that problem: recovering signals in noise-dominated regimes requires statistical and machine-learning methods matched to each regime. We organized the work along that thread, tracking the expected signal from the Galactic Center Excess into regimes of progressively lower signal-to-noise ratio, down to the cross-correlation forecasts for the next generation of instruments. To this end, we drew on traditional statistical inference, generative modeling, simulation-based inference, and deep learning.

Along the way we measured the gamma-ray luminosity function of millisecond pulsars in Milky Way globular clusters with 15.8 years of Fermi-LAT data, and found that a pulsar interpretation of the Galactic Center Excess should already have produced resolved sources that are not observed [1] (Chapter 4). This tension reopens the question of what causes the excess. We built what is, to our knowledge, the first probabilistic mixture model of the high-latitude unassociated Fermi-LAT sources, found no significant dark matter subhalo population, and set 95% confidence upper limits on $\langle\sigma v\rangle$ in the $b\bar{b}$ channel for dark matter masses between 10 GeV and 1 TeV [2] (Chapter 5). We trained a convolutional network on roughly 10^6 synthetic maps to reconstruct the extragalactic source-count distribution dN/dS down to fluxes a factor of fifty below the catalog threshold [3] (Chapter 6), and used the reconstructed dN/dS as a generative model to build the public gPCS (gamma-ray Photon Count Statistics) probabilistic catalog, which identifies about 50% more candidate source directions than 4FGL-DR3 alone [4] (Chapter 7). Finally, we forecast the sensitivity of CTAO to annihilating and decaying dark matter through the cross-correlation of the anisotropic gamma-ray sky with low-redshift galaxy catalogs, and found that about 50 h of observation would already yield sensitivities comparable to those of dwarf-galaxy and galaxy-cluster searches [5] (Chapter 8).

Each search in this thesis sharpened the way the question is asked, and each leaves a natural set of next steps. In light of these findings, several directions can be pursued, one per research line developed in this thesis.

- **The millisecond pulsar hypothesis for the Galactic Center Excess:** Our constraint is conditional: if the pulsars in the Inner Galaxy share the luminosity function we measured in surviving globular clusters, Fermi-LAT should already have resolved 17–37 of them, whereas only three candidates are known [1]. This tension converges with independent analyses of the excess’s energy distribution [221]. The delivery mechanism — globular clusters spiraling into the Inner Galaxy and dissolving under tidal forces — is well under-

stood [232, 233, 391]; the assumption under discussion is the shared luminosity function. In its favor, field pulsars in the Galactic plane, recycled through a different channel, show a consistent luminosity function [223], suggesting a weak dependence on the formation environment. Both observed populations, however, are continuously replenished with new recycled, brighter pulsars: surviving clusters through dynamical formation in their dense cores [235], the plane through ongoing star formation. The deposited population lost both channels when its hosts dissolved, and has only aged since. If spin-down has dimmed it appreciably, fewer resolved pulsars are expected and the tension we evidenced relaxes; evolutionary models have indeed been used to argue for such a fainter bulge population [233–235]. The next step for this line of work is to better understand the luminosity function of the pulsars deposited in the Galactic Center, testing those evolutionary models against the ones measured in globular clusters and in the Galactic plane.

- **Dark matter subhalos among unassociated sources:** We constrained the subhalo contribution to the unassociated sources by maximum likelihood, using quantification learning to handle the covariate and prior shift between the associated sources used for training and the unassociated *Fermi*-LAT targets, and we found no significant excess [2]. A natural extension is to consider velocity-dependent annihilation, such as Sommerfeld enhancement [79]: the nearest small subhalos have velocity dispersions of a few km/s [271], which could raise the expected signal by a factor of a few relative to dwarf galaxies. The method itself is not tied to gamma-ray astronomy: it applies to anomaly searches in any unlabeled dataset that exhibits covariate and prior shift. Specifically, the same framework could be applied to the unassociated sources near the Galactic plane, to assess how many of them are consistent with a millisecond-pulsar origin.
- **The source-count distribution below threshold:** The network recovered a dN/dS compatible with an S^{-2} scaling down to fluxes roughly fifty times fainter than the catalog limit [3]. Extending the analysis to multiple energy bins would give an energy-dependent dN/dS and access to energy correlations, sharpening the overall sensitivity of the method. The energy dependence could then, in principle, be used to separate sub-threshold populations with distinct spectra, disentangling the astrophysical and dark matter contributions — an approach complementary to the cross-correlation searches of Chapter 8.
- **Probabilistic cataloging of the gamma-ray sky:** Using the reconstructed dN/dS as a generative model, gPCS compares synthetic and real skies pixel by pixel and flags candidate source directions [4]. Its main application is in cross-correlation studies with galaxy catalogs, along two routes. Cross-matching the candidate directions with the positions of cataloged galaxies can promote some of them into bona fide sources, an identification effort that machine-learning classification of unassociated objects can support [179]. Alternatively, the catalog can be used as a gamma-ray tracer in cross-correlations with galaxy tracers of the large-scale structure, in the spirit of Chapter 8: there, the statistical reach of a 50% larger sample outweighs a sub-leading fraction of spurious directions, and the resulting measurement could be more sensitive, for example, to a dark matter component.
- **Cross-correlations with the large-scale structure:** Our forecast showed that cross-correlating the gamma-ray anisotropies with the 2MASS galaxy distribution lets CTA reach $\langle\sigma v\rangle \sim 10^{-23} \text{ cm}^3 \text{ s}^{-1}$ and decay lifetimes of order 10^{27} s with about 50 h of effective exposure, roughly five times stronger than the auto-correlation [5]. At equal dark matter

masses, these sensitivities are comparable to those of dwarf and cluster analyses; the advantage of CTAO is its TeV energy window, which extends the search to masses up to ~ 100 TeV, a range hard to reach for GeV instruments such as Fermi-LAT. Deeper and more densely sampled catalogs such as Euclid [385] would tighten the tomographic reconstruction, as would adding other large-scale-structure tracers such as weak lensing. Finally, two steps lie ahead: updating these forecasts as the construction of CTAO progresses and its survey strategy is finalized, and carrying out the cross-correlation measurement itself once real data become available.

Taken together, the five analyses trace a coherent path through the gamma-ray sky, from the brightest anticipated signal to the faint imprint of the cosmic web. None of them returned a confirmed detection, and the particle identity of dark matter remains an open problem. The next steps in indirect detection, we have argued, will require better instruments, more data, and sharper statistical tools to read them, all advancing together. This thesis has contributed to the third of these fronts: population models that make sub-threshold sources statistically accessible, inference frameworks robust to the mismatch between simulations and real data, probabilistic catalogs, and cross-correlation analyses. These methods are not tied to the problems that motivated them: they carry over to other datasets and other messengers. I hope this groundwork will help extend the search as current gamma-ray surveys continue to grow and next-generation observatories come online, bringing us one step closer to the first non-gravitational detection of dark matter.

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RESUMEN DE LA TESIS

La evidencia de la materia oscura proviene de numerosas pruebas independientes a distintas escalas. Las curvas de rotación galácticas se mantienen planas allí donde la masa visible predice un declive; las galaxias en el interior de los cúmulos se mueven demasiado deprisa para permanecer ligadas únicamente por la materia luminosa; la luz de fuentes distantes es desviada por masa que no podemos ver; y el patrón de anisotropías del fondo cósmico de microondas solo se reproduce cuando se añade una componente no bariónica al balance. Cada una de estas medidas, obtenida a una escala y en una época diferentes, apunta a la misma conclusión: una forma de materia que supera a los bariones ordinarios aproximadamente en un factor cinco y que interactúa con la luz solo a través de la gravedad. Dentro del modelo cosmológico estándar, esta componente fría y no bariónica representa alrededor del 85 % de la densidad de materia. Sin embargo, todas estas pruebas son de naturaleza gravitatoria: nos dicen cuánta materia oscura hay y dónde se encuentra, pero apenas dicen nada sobre qué es realmente la materia oscura. Los candidatos propuestos abarcan casi noventa órdenes de magnitud en masa, desde axiones ultraligeros hasta agujeros negros primordiales, y los datos por sí solos no permiten seleccionar ninguno de ellos.

Entre estos candidatos, las partículas masivas de interacción débil (WIMPs, por sus siglas en inglés) han ocupado durante mucho tiempo un lugar privilegiado. Una partícula estable con masa y acoplamiento a la escala electrodébil, producida térmicamente en el Universo primitivo caliente, se desacopla con una abundancia reliquia que cae de forma natural cerca de la densidad de materia oscura observada, una coincidencia conocida como el milagro WIMP. El proceso de aniquilación que fija esta abundancia no se apaga cuando el Universo se enfría: allí donde la densidad de materia oscura sea hoy suficientemente alta, las WIMPs deberían seguir aniquilándose, convirtiendo su masa en reposo en partículas del Modelo Estándar como rayos gamma, neutrinos, positrones y antiprotones. Si la materia oscura es en cambio inestable, puede desintegrarse en los mismos estados finales. De estos mensajeros, los rayos gamma son especialmente atractivos: al no ser desviados por los campos magnéticos, apuntan de vuelta al lugar donde fueron producidos y llevan consigo la firma espectral de la interacción que los generó. Una señal de rayos gamma procedente de una región de materia oscura concentrada sería, por tanto, una de las sondas más limpias de que disponemos de la partícula que se esconde tras las anomalías gravitatorias.

Durante más de quince años, el Large Area Telescope (LAT) a bordo del telescopio espacial de rayos gamma Fermi ha cartografiado el cielo del GeV, donde se esperan estas firmas, y sus medidas constituyen el núcleo de esta tesis. Esa larga exposición ha cambiado el carácter del problema. Los objetivos más brillantes ya han sido estudiados en profundidad, y las señales que quedan son débiles, están mezcladas con los fondos astrofísicos o permanecen ocultas por debajo del umbral de detección. Los métodos estándar clasifican con fiabilidad los objetos más brillantes y cartografían las componentes difusas más prominentes, pero tienen dificultades con las poblaciones que se solapan y con las limitaciones instrumentales, como la resolución angular finita y el propio umbral de detección. Fijar un umbral y contar lo que lo supera es un primer paso natural, pero descarta la mayor parte de la información: una fracción considerable de la

población emisora queda sin resolver, sin clasificar o simplemente sin ver.

En esta tesis defendemos que el progreso en las búsquedas indirectas de materia oscura depende hoy de métodos estadísticos y de aprendizaje automático capaces de recuperar señales en estos regímenes dominados por el ruido. El trabajo sigue un arco que va del Centro Galáctico hacia el exterior, hasta la red cósmica, siguiendo la señal esperada en regímenes de relación señal-ruido progresivamente menor. Cada régimen requiere sus propias herramientas, y recurrimos sucesivamente a la inferencia estadística tradicional, al modelado generativo, a las técnicas de verosimilitud implícita y al aprendizaje profundo, aplicados tanto a los datos de Fermi-LAT como a predicciones para instrumentos futuros.

Cinco estudios componen esta tesis. Comenzamos con una medida de la función de luminosidad en rayos gamma de los púlsares de milisegundo (MSPs) en los cúmulos globulares de la Vía Láctea, utilizada para comprobar si una población de púlsares no resueltos puede explicar el Exceso del Centro Galáctico. A continuación buscamos subhalos de materia oscura entre las fuentes de Fermi-LAT sin asociación, combinando un modelo generativo de mezcla con aprendizaje de cuantificación. Descendiendo por debajo del umbral del catálogo, reconstruimos la distribución de recuento de fuentes de las fuentes extragalácticas débiles con una red neuronal profunda, y construimos un marco de catalogación probabilística que extiende los catálogos de rayos gamma al régimen sub-umbral, publicado como el catálogo gamma-ray Photon Count Statistics (gPCS). Por último, elaboramos una predicción de la sensibilidad del Cherenkov Telescope Array Observatory a la correlación cruzada entre los rayos gamma y los catálogos de galaxias, llevando la búsqueda hasta las escalas más grandes y las señales esperadas más débiles.

El problema de la materia oscura

Los argumentos a favor de la materia oscura se apoyan en una cadena de observaciones independientes reunidas a lo largo de casi un siglo. El primer indicio cuantitativo llegó en 1933, cuando Fritz Zwicky aplicó el teorema del virial a las velocidades radiales de las galaxias del cúmulo de Coma e infirió una relación masa-luz de orden 400, mucho mayor de lo que podía justificar la materia luminosa; llamó a la componente que faltaba *dunkle Materie* [24]. La discrepancia reapareció a escalas galácticas cuando Rubin y Ford midieron en 1970 la primera curva de rotación óptica de alta precisión de M31 y no encontraron declive kepleriano alguno a radios grandes [6]. Observaciones posteriores en la línea de 21 cm extendieron estas curvas de rotación hasta los mayores radios medibles en cientos de sistemas, donde la velocidad de rotación se mantiene plana; una curva plana implica una masa encerrada que crece linealmente con el radio, $M(r) \propto r$, y por tanto un halo extenso cuya densidad decae como $\rho \propto r^{-2}$ a radios grandes. Las lentes gravitatorias aportaron un argumento aún más limpio: en el Cúmulo Bala, la reconstrucción de la masa mediante lente débil está desplazada espacialmente respecto al gas emisor de rayos X y coincide en cambio con las galaxias, efectivamente no colisionales [8], tal como se espera si la mayor parte de la masa es oscura y casi no colisional. A escalas cosmológicas, el fondo cósmico de microondas completa el cuadro, ya que un universo sin materia oscura fría produciría un espectro de potencias con el tercer pico acústico fuertemente suprimido, en conflicto con el espectro medido por el satélite Planck [9].

Estas medidas coinciden no solo en que existe algo que se comporta como materia oscura, sino también en cuánta debería haber. El análisis de Planck 2018 da $\Omega_c h^2 = 0.1200 \pm 0.0012$ para la densidad de materia oscura fría [9], lo que hace a la materia oscura aproximadamente cinco veces más abundante que la materia bariónica ordinaria. Cualquier candidato viable debe

reproducir esta densidad reliquia y, a la vez, ser consistente con las evidencias dinámicas y de lentes gravitatorias.

Una WIMP ofrece una forma natural de conseguirlo: una partícula con masa en el rango GeV–TeV y acoplamientos de escala débil al Modelo Estándar. Si tal partícula estuvo en equilibrio térmico en el universo primitivo caliente y se desacopló al enfriarse el plasma, su abundancia superviviente queda fijada por el ritmo al que se aniquilaba. Para aniquilación en onda s , esta abundancia reliquia adopta la forma compacta

$$\frac{\Omega_\chi h^2}{0.12} \approx \frac{2.2 \times 10^{-26} \text{ cm}^3/\text{s}}{\langle \sigma v_{\text{rel}} \rangle}, \quad (8.3.1)$$

donde el numerador es la sección eficaz térmica canónica [11, 12]. Lo que hace atractiva la hipótesis es una coincidencia: una sección eficaz de intensidad aproximadamente débil, $\langle \sigma v_{\text{rel}} \rangle \sim \alpha^2/m_\chi^2$ con $\alpha \sim 0.03$, devuelve la abundancia observada para una masa de entre unos pocos cientos de GeV y unos pocos TeV, justo en la escala electrodébil. Esta concordancia entre una medida cosmológica y una escala de la física de partículas es el llamado milagro WIMP.

El mismo acoplamiento que gobierna la aniquilación de la materia oscura en el universo primitivo media también su dispersión con núcleos en detectores subterráneos y su producción en pares en los colisionadores de partículas, lo que abre tres estrategias experimentales complementarias. La detección directa busca el retroceso nuclear que imprime una partícula de materia oscura al atravesar un detector situado a gran profundidad. Las búsquedas en colisionadores persiguen la energía transversal faltante que se lleva la materia oscura cuando se produce en experimentos como el Gran Colisionador de Hadrones. La detección indirecta, el enfoque seguido a lo largo de esta tesis, apunta a los productos de aniquilación o desintegración de la materia oscura en entornos astrofísicos donde su densidad es alta.

Para la aniquilación en rayos gamma, el flujo diferencial observado se factoriza limpiamente en un término de física de partículas y un término astrofísico,

$$\frac{d\Phi_\gamma}{dE d\Omega} = \underbrace{\frac{1}{4\pi} \frac{\langle \sigma v_{\text{rel}} \rangle}{2m_\chi^2} \frac{dN_\gamma}{dE}}_{\text{Física de partículas}} \times \underbrace{J(\psi)}_{\text{Astrofísica}}, \quad (8.3.2)$$

donde el factor astrofísico, el llamado factor J , es la integral del cuadrado de la densidad a lo largo de la línea de visión,

$$J(\psi) = \int_{\text{l.o.s.}} \rho^2(r(s, \psi)) ds. \quad (8.3.3)$$

El factor de física de partículas contiene la masa m_χ , la sección eficaz $\langle \sigma v_{\text{rel}} \rangle$ (o, para la desintegración, la tasa Γ) y la forma espectral dN_γ/dE : codifica el modelo que queremos poner a prueba [11, 86]. El factor astrofísico J codifica únicamente la distribución de materia oscura a lo largo de la línea de visión. Esta separación es lo que hace tratables las búsquedas indirectas, porque una no detección en una dirección dada se traduce directamente en un límite superior sobre $\langle \sigma v_{\text{rel}} \rangle$ una vez conocido J . También fija dónde reside la incertidumbre dominante: J es cuadrático en el perfil de densidad, y el perfil es la cantidad que peor medimos, de modo que la astrofísica domina el error sistemático de la mayoría de las búsquedas de aniquilación. Para materia oscura en desintegración, el flujo escala en cambio con Γ/m_χ y con un factor D lineal en ρ , lo que suaviza esta sensibilidad al perfil.

Dado que la señal esperada escala con la integral de la densidad, J dicta los objetivos de cualquier búsqueda de aniquilación, y con ellos la estructura de esta tesis. El Centro Galáctico

tiene el mayor factor J de todas las fuentes astronómicas, debido a su proximidad (~ 8 kpc) y al pronunciado aumento esperado de la densidad de materia oscura a radios pequeños; también sufre la contaminación astrofísica más severa, procedente de la brillante emisión difusa y de un campo abarrotado de fuentes puntuales. Las galaxias esferoidales enanas y los subhalos oscuros tienen un factor J menor pero son mucho más limpios: las enanas están entre los sistemas más dominados por materia oscura que se conocen, y Λ CDM predice muchos más subhalos ligados gravitacionalmente que las aproximadamente 50 galaxias satélite conocidas, la mayoría de ellos completamente oscuros y que deberían aparecer en los datos de Fermi-LAT como fuentes puntuales sin asociación, sin contrapartida a otras longitudes de onda [113]. A las escalas más grandes se encuentra la red extragaláctica, cuya contribución de materia oscura está enterrada en el fondo de rayos gamma no resuelto que queda una vez sustraídas las fuentes resueltas y la emisión difusa galáctica; allí se espera que la señal sea subdominante en intensidad total y deba extraerse con técnicas de anisotropía y de correlación cruzada. Los capítulos que siguen descienden por esta jerarquía en orden: primero el Centro Galáctico, donde la señal esperada es más brillante, después los subhalos oscuros, luego la débil población de fuentes sub-umbral y, finalmente, la red cósmica, con la señal más débil de todas.

El cielo en rayos gamma y el *Fermi* Large Area Telescope

Todos los objetivos de esta jerarquía pueden estudiarse con el mismo instrumento. Lanzado en 2008, el Fermi Large Area Telescope ha proporcionado el sondeo de todo el cielo más sensible del dominio del GeV hasta la fecha, operando de forma continua durante más de quince años. Es un telescopio de conversión de pares: un rayo gamma incidente se convierte en un par electrón-positrón, y cuatro subsistemas trazan las partículas cargadas resultantes, miden su energía y rechazan el abrumador fondo de rayos cósmicos [147]. El instrumento es sensible desde 20 MeV hasta más de 300 GeV y, con un campo de visión de aproximadamente 2.4 sr a 1 GeV, cubre casi todo el cielo con exposición uniforme cada dos órbitas, es decir, cada unas tres horas. Su resolución angular mejora rápidamente con la energía: el radio de contención del 68% de la función de dispersión de punto (PSF) cae desde varios grados cerca de 100 MeV hasta unos 0.6° a 1 GeV y por debajo de 0.15° por encima de 10 GeV [147]. Esta PSF ancha a baja energía emborrona las fuentes vecinas entre sí, lo que limita la profundidad que puede alcanzar cualquier catálogo y, como desarrollamos a lo largo de esta tesis, motiva métodos estadísticos que trabajan con la población en lugar de con la fuente individual.

El cielo del GeV que emerge se descompone de manera natural en unas pocas componentes. La emisión observada está dominada por el fondo difuso galáctico, producido cuando los rayos cósmicos interactúan con el gas interestelar y los campos de radiación; su modelado mediante plantillas sigue siendo la mayor fuente de incertidumbre sistemática en los análisis de Fermi-LAT. Superpuesta a este fondo hay una población de fuentes puntuales resueltas, en su mayoría blázares extragalácticos, junto con fuentes galácticas entre las que los púlsares son las más prominentes. Lo que queda una vez sustraídas la emisión difusa galáctica y todas las fuentes catalogadas es un residuo casi isótropo, el fondo de rayos gamma no resuelto (UGRB, por sus siglas en inglés): la luz acumulada de todas las fuentes todavía demasiado débiles para ser detectadas individualmente, más cualquier componente isótropa genuinamente difusa [138].

La frontera entre «resuelto» y «no resuelto» la fija un umbral de detección. La versión incremental más reciente del catálogo, 4FGL-DR4, analiza catorce años de datos y contiene 7195 fuentes [153]. Un candidato entra en el catálogo solo si su estadístico de prueba (TS), que compara la máxima verosimilitud de los datos con y sin la fuente, supera 25, una significancia de algo más

de 4σ ; este umbral se ha mantenido fijo a lo largo de la historia del catálogo. Todo lo que queda por debajo permanece en los datos, incorporado al fondo difuso en lugar de figurar como fuente.

Dos limitaciones de este cuadro motivan los capítulos siguientes. Primero, de las fuentes de 4FGL-DR4, 4022 están asociadas a contrapartidas extragalácticas y 610 a contrapartidas galácticas, pero 2563 permanecen sin asociación, sin contrapartida clara a ninguna otra longitud de onda [153]. Más allá de su probable mayoría de blázares y púlsares débiles, esta población no asociada es el principal terreno de búsqueda de emisores exóticos como los subhalos de materia oscura. Segundo, y por construcción, las fuentes por debajo del umbral solo se miden en conjunto: integrar la distribución de recuento de fuentes desde flujo cero hasta el umbral de detección reproduce la intensidad del UGRB, pero recuperar la población que lo produce requiere técnicas que lean información a nivel de población en la huella colectiva de las fuentes no resueltas sobre el mapa de recuento de fotones. Estas dos lagunas, la de las fuentes no asociadas y la del régimen sub-umbral, marcan la agenda de los métodos desarrollados en el resto de esta tesis.

Objetivos

De estos dos puntos ciegos se derivan los objetivos de esta tesis. Nuestro primer objetivo es poner a prueba la interpretación del Exceso del Centro Galáctico en términos de púlsares de milisegundo midiendo la función de luminosidad en rayos gamma de los púlsares en el sistema de cúmulos globulares de la Vía Láctea, una medida que agudiza aún más la tensión con la interpretación MSP tal y como se ha formulado hasta ahora. Un segundo objetivo es preguntarse si alguna de las fuentes puntuales detectadas por Fermi-LAT, pero que carecen de contrapartida, podría ser un subhalo de materia oscura, una cuestión en la que el desplazamiento entre la población asociada y la población no asociada bajo estudio exige un marco de aprendizaje de cuantificación. Nos ocupamos después de la población por debajo del umbral de detección, con el objetivo de medir la distribución de recuento de fuentes dN/dS de las fuentes extragalácticas muy por debajo del umbral de Fermi-LAT, y de complementar ese conocimiento a nivel de población con identificaciones posicionales de fuentes sub-umbral candidatas. Un objetivo adicional es aislar una señal de materia oscura oculta en el UGRB y predecir el alcance del Cherenkov Telescope Array Observatory (CTAO) para aniquilación y desintegración. Finalmente, y de forma transversal a los cinco estudios, desarrollamos metodologías estadísticas y de aprendizaje automático que puedan hacer accesibles señales débiles en regímenes dominados por el ruido y que sean transferibles a instrumentos futuros.

Metodología: estadística y aprendizaje automático para señales débiles

Inferencia bayesiana en el régimen dominado por el ruido. Una señal de materia oscura en el cielo de rayos gamma es débil por construcción: yace bajo brillantes fondos astrofísicos cuya normalización y morfología son a su vez inciertas. Para extraerla debemos inferir los parámetros del modelo a partir de un mapa de recuentos de fotones agrupados en píxeles, y la herramienta estándar es una verosimilitud de Poisson píxel a píxel,

$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{a=1}^{N_{\text{pix}}} \frac{\lambda_a(\boldsymbol{\theta})^{k_a} e^{-\lambda_a(\boldsymbol{\theta})}}{k_a!}, \quad (8.3.4)$$

donde k_a es el recuento observado y $\lambda_a(\boldsymbol{\theta})$ el valor esperado del modelo en el píxel a . Esta construcción funciona solo mientras el valor esperado pueda escribirse en forma cerrada. Deja

de ser válida cuando el proceso generador de los datos involucra un simulador complejo cuya estructura estocástica interna impide escribir la verosimilitud explícitamente, porque evaluarla requiere entonces marginalizar sobre un conjunto de variables latentes $\mathbf{z}$,

$$p(\mathbf{x}|\theta) = \int p(\mathbf{x}, \mathbf{z}|\theta) d\mathbf{z}. \quad (8.3.5)$$

El problema de las fuentes puntuales no resueltas es un ejemplo concreto: escribir una verosimilitud en forma cerrada para el mapa completo del cielo exigiría integrar sobre todas las posibles configuraciones espaciales, flujos y número de las fuentes no resueltas. Este es el problema de inferencia que los análisis de poblaciones y de catálogos resumidos más abajo están diseñados para resolver.

Inferencia basada en simulaciones. Cuando la verosimilitud es intratable pero el modelo directo es razonablemente fácil de muestrear, podemos esquivar la integral por completo y aprender de las simulaciones. Esta es la idea de la *inferencia basada en simulaciones* [154]: generamos muchos mapas sintéticos del cielo a partir de parámetros conocidos y entrenamos una red neuronal para invertir esa correspondencia. Adoptamos la variante de estimación neuronal de la posterior, que entrena un estimador de densidad condicional para aproximar directamente la distribución posterior, $q_\phi(\theta|\mathbf{x}) \approx p(\theta|\mathbf{x})$. La ventaja es que la inferencia queda *amortizada*: una vez entrenada la red, aplicarla al mapa real del cielo de Fermi-LAT no cuesta prácticamente nada, ya que la simulación y el entrenamiento, que son lo costoso, se realizan una sola vez. En nuestra implementación, la red transforma un mapa del cielo observado directamente en resúmenes de la posterior, la media y la varianza de cada intervalo de flujo. Este estimador es el motor de nuestra medida de la distribución de recuento de fuentes por debajo del umbral de Fermi-LAT.

Aprendizaje de cuantificación. A menudo la pregunta física no es qué es cada objeto, sino cuántos hay de cada clase: la prevalencia de una hipotética clase de subhalos de materia oscura entre las fuentes no asociadas, por ejemplo, es precisamente lo que una búsqueda de subhalos debe estimar. Los clasificadores de aprendizaje automático se entrenan típicamente con fuentes asociadas y se aplican después a las no asociadas, pero varios sesgos observacionales rompen la hipótesis de que ambas poblaciones siguen la misma distribución. El *aprendizaje de cuantificación* aborda este problema ajustando los pesos de mezcla π_k de un modelo generativo, $p(\mathbf{x}) = \sum_k \pi_k p(\mathbf{x}|k)$, directamente a los datos objetivo, en lugar de confiar en las fracciones de clase del conjunto de entrenamiento [198]. Debemos distinguir dos modos de fallo. Bajo un *desplazamiento de las probabilidades a priori* (*prior shift*) las densidades condicionales de clase coinciden, $p_{\text{train}}(\mathbf{x}|k) = p_{\text{target}}(\mathbf{x}|k)$, mientras que las proporciones de clase difieren, $p_{\text{train}}(k) \neq p_{\text{target}}(k)$; lo corregimos ajustando los π_k a los datos no asociados. Bajo un *desplazamiento de covariables* (*covariate shift*) las distribuciones posteriores coinciden, $p_{\text{train}}(k|\mathbf{x}) = p_{\text{target}}(k|\mathbf{x})$, mientras que las distribuciones de características difieren, $p_{\text{train}}(\mathbf{x}) \neq p_{\text{target}}(\mathbf{x})$; lo corregimos con funciones de modulación sigmoideas. Cualquier análisis que ignore este desajuste corre el riesgo de obtener estimaciones de prevalencia sesgadas y, en el contexto de las búsquedas de materia oscura, conclusiones poco fiables sobre la sección eficaz de aniquilación. Esta corrección sustenta los límites sobre subhalos de materia oscura presentados en los Resultados.

Aprendizaje profundo sobre mapas del cielo. La información que necesitamos está codificada en la morfología de la emisión, no en ningún píxel individual. Un mapa de rayos gamma se representa de forma natural como una malla bidimensional o como una teselación HEALPix de la esfera, y las redes neuronales convolucionales (CNNs) explotan directamente esta estructura espacial. Apilar capas convolucionales construye campos receptivos progresivamente mayores: las capas superficiales detectan rasgos locales como bordes y picos, mientras que las capas profundas

los combinan en representaciones más abstractas de la población de fuentes subyacente. En esta tesis, un mapa del cielo se introduce en una red convolucional de este tipo para reconstruir la distribución de recuento de fuentes, proporcionando el extractor de características que convierte el mapa bruto de Fermi-LAT en los resúmenes que la inferencia lee.

Catalogación probabilística. Un catálogo de fuentes convencional aplica un umbral de detección duro, conservando solo las direcciones cuyo estadístico de prueba supera un corte fijo [151] y descartando todo lo más débil. La *catalogación probabilística* sustituye el corte duro por una verosimilitud: a cada dirección del cielo se le asigna una probabilidad de albergar una fuente sub-umbral. En la práctica, realizamos una comparación frecuentista del cielo real con conjuntos de cielos sintéticos generados a partir de la distribución de recuento de fuentes recuperada, señalando direcciones candidatas a fuente por debajo del umbral de detección en lugar de declararlas presentes o ausentes. Esta maquinaria produce el catálogo probabilístico de fuentes presentado en los Resultados.

Correlaciones cruzadas. Una señal pura de materia oscura en el fondo difuso de rayos gamma es imposible de aislar mapa a mapa, porque está enterrada bajo la emisión astrofísica y el ruido instrumental. La alternativa es correlacionar el cielo de rayos gamma con un trazador independiente de la misma distribución de materia subyacente. El estadístico relevante es el espectro de potencias angular (APS, por sus siglas en inglés), cuya generalización a dos mapas se escribe

$$C_{\ell}^{ij} = \frac{1}{2\ell + 1} \left\langle \sum_{m=-\ell}^{\ell} a_{\ell m}^{(i)} a_{\ell m}^{(j)*} \right\rangle, \quad (8.3.6)$$

donde los $a_{\ell m}^{(i)}$ son los coeficientes armónicos del mapa i . Cuando $i \neq j$ obtenemos el APS de correlación cruzada y, como los términos de ruido de dos mapas distintos no están correlacionados, desaparecen del promedio, dejando la señal cosmológica compartida. Utilizamos esta técnica para predecir la sensibilidad de CTAO a la aniquilación y desintegración de materia oscura, correlacionando la emisión de rayos gamma predicha con un catálogo de galaxias cercanas [202].

Resultados

Presentamos los resultados en el orden de la tesis, desde la señal esperada más brillante hasta la más débil.

La función de luminosidad de los púlsares de milisegundo y el Exceso del Centro Galáctico. El Exceso del Centro Galáctico (GCE, por sus siglas en inglés) es un exceso brillante y estadísticamente significativo de emisión GeV procedente de la región que rodea el Centro Galáctico, con un espectro y una morfología consistentes con la aniquilación de una partícula de materia oscura con sección eficaz cercana al valor térmico reliquia [80, 126, 127]. Su interpretación astrofísica dominante atribuye la señal, en cambio, a una población no resuelta de MSPs situados en la región central, individualmente demasiado débiles para ser resueltos. Para discriminar entre estas hipótesis, caracterizamos la función de luminosidad de los MSPs allí donde puede medirse directamente: en los cúmulos globulares de la Vía Láctea. Utilizando 15.8 años de datos públicos de Fermi-LAT, analizamos 157 cúmulos globulares y detectamos de forma robusta 56 de ellos con un estadístico de prueba $TS > 25$ [1].

A continuación ajustamos la función de luminosidad sobre los 87 cúmulos con altas tasas de encuentros estelares y/o luminosidades bien medidas, adoptando una forma log-normal y construyendo una verosimilitud conjunta marginalizada sobre todos los cúmulos que tiene en cuenta sus distancias y el número de púlsares, distribuido según Poisson, que alberga cada uno.

Este tratamiento bayesiano jerárquico dio una luminosidad media por púlsar $\langle L_\gamma \rangle \sim (1-8) \times 10^{33}$ erg/s, integrada entre 0.1 y 100 GeV, y una anchura $\sigma_L \sim 1.4-2.7$. En total, estos 87 cúmulos albergan del orden de 400–1500 MSPs emisores de rayos gamma.

Con esta función de luminosidad en la mano, nos preguntamos qué implica para el GCE. Si el exceso estuviera generado por MSPs con las mismas características que los de los cúmulos globulares, los catálogos de fuentes de Fermi ya contendrían $N_{\text{MSP}} \sim 17-37$ miembros individuales de tal población del interior de la Galaxia. Sin embargo, solo se han detectado tres MSPs con una dirección y una distancia que no excluyen su pertenencia a esa población [223]. Un modelo más conservador, que permite variaciones de cúmulo a cúmulo, reduce el número esperado a $N_{\text{MSP}} \approx 3-30$, pero los resultados siguen estando en tensión con los tres candidatos identificados a un nivel de al menos 2σ . La tensión puesta de manifiesto por este trabajo reabre la cuestión del origen del Exceso del Centro Galáctico.

Subhalos de materia oscura entre las fuentes no asociadas. Λ CDM predice una gran población de subhalos oscuros alrededor de la Vía Láctea [113]; en los más densos, la materia oscura podría aniquilarse a un ritmo visible, haciéndolos aparecer como fuentes puntuales sin contrapartida a ninguna otra longitud de onda. Estas fuentes no asociadas se reconocen desde hace tiempo como el terreno de prueba natural para las búsquedas de subhalos de materia oscura, pero hasta la fecha ninguna fuente no asociada de Fermi-LAT ha sido confirmada de manera inequívoca como subhalo de materia oscura. Revisitamos la cuestión con la última versión incremental del catálogo de Fermi, 4FGL-DR4 [153], que contiene 7195 fuentes, de las cuales 2563 – aproximadamente un tercio – permanecen sin asociación. De ellas, 1283 se encuentran a latitudes galácticas $|b| > 10^\circ$, y restringimos el análisis a esa muestra.

Describimos la población no asociada con un modelo probabilístico $p(x)$ construido como un modelo generativo de mezcla de tres clases de fuentes: fuentes astrofísicas galácticas y extragalácticas, más los hipotéticos subhalos de materia oscura [2]. Cada clase se caracteriza mediante sus rasgos espectrales de log-parábola, a saber, el flujo, la pendiente espectral y la curvatura. Para tener en cuenta el desajuste entre la muestra etiquetada de entrenamiento y los datos no asociados, trabajamos dentro del marco general del aprendizaje de cuantificación, considerando dos efectos: un desplazamiento de covariables que afecta a todas las clases de fuentes por igual, modelado mediante funciones de modulación sigmoides, y un desplazamiento de las probabilidades a priori, que permite que cambien las prevalencias de clase y admite la posible presencia de una nueva clase de fuentes. Ajustamos el modelo con un algoritmo de esperanza-maximización semisupervisado. Los datos prefieren el desplazamiento a priori a 7σ y el desplazamiento de covariables a 4.5σ , con una significancia combinada de 6.7σ , lo que confirma que ambos efectos están presentes.

No se observó ningún exceso significativo correspondiente a subhalos de materia oscura más allá de las poblaciones galáctica y extragaláctica; para todos los modelos probados la significancia es inferior a 2σ . Derivamos por tanto límites superiores al 95 % de confianza sobre la sección eficaz de aniquilación de materia oscura $\langle\sigma v\rangle$ en el canal $b\bar{b}$, para masas de materia oscura entre 10 GeV y 1 TeV. Estos límites son alrededor de un orden de magnitud menos estrictos que los obtenidos con galaxias esferoidales enanas, pero son competitivos con los límites previos de las búsquedas de subhalos basadas en métodos de clasificar y contar.

La distribución de recuento de fuentes por debajo del umbral de detección. Por debajo del flujo al que Fermi-LAT resuelve fuentes puntuales individuales, la población no desaparece; simplemente deja de ser separable objeto a objeto. Las fuentes débiles se acumulan bajo el umbral de detección y, aunque ninguna puede identificarse estrictamente en el cielo, en conjunto dejan una huella medible en la estadística de recuento de fotones de la emisión difusa.

Esa huella restringe la distribución de recuento de fuentes dN/dS , el número de fuentes por unidad de flujo, que es la cantidad que nos propusimos reconstruir. La dificultad estriba en que recuperar una distribución a partir únicamente de la estadística de los recuentos es un problema de inferencia genuinamente difícil, para el que no se dispone de una verosimilitud tratable (a menos que se apliquen aproximaciones como la técnica 1pPDF).

Lo abordamos con una CNN acoplada a inferencia basada en simulaciones. La red se entrenó con mapas sintéticos del cielo de Fermi-LAT: generamos en total un millón de mapas, de los cuales el 90% se usó para el entrenamiento y el 10% restante para la validación, de modo que el modelo aprendiera a asociar un mapa de recuento de fotones dado con la dN/dS subyacente que lo produjo. Aplicamos después la red entrenada a los primeros catorce años de datos recogidos por Fermi-LAT, restringiendo el análisis a energías de fotón entre 1 y 10 GeV y a latitudes galácticas $|b| > 30^\circ$, donde los fondos difusos son menos severos. Para cada mapa observado, la red devuelve la media y la varianza posteriores de dN/dS en 20 intervalos de flujo que cubren $[5 \times 10^{-12}, 1 \times 10^{-7}] \text{ cm}^{-2} \text{ s}^{-1}$, junto con la constante de flujo difuso isótropo.

La reconstrucción se comporta como esperábamos en ambos regímenes [3]. En el límite resuelto, donde las fuentes son lo bastante brillantes para aparecer individualmente, la CNN recupera una dN/dS plenamente compatible con la derivada del catálogo de Fermi-LAT, lo que constituye una comprobación de consistencia tranquilizadora para un método por lo demás indirecto. Más importante aún, extiende la distribución al régimen no resuelto, siguiendo un comportamiento compatible con $dN/dS \sim S^{-2}$ hasta el flujo más pequeño considerado, $S = 5 \times 10^{-12} \text{ cm}^{-2} \text{ s}^{-1}$. Esto alcanza aproximadamente un factor 50 por debajo del umbral de detección de Fermi-LAT, $S_{\text{th}} \sim 2 \times 10^{-10} \text{ cm}^{-2} \text{ s}^{-1}$, un territorio inaccesible a la detección de fuentes individuales. En conjunto, la medida traza una ley de potencias $dN/dS \sim S^{-2}$ a lo largo de casi cuatro órdenes de magnitud en flujo, en el rango $[5 \times 10^{-12}, 1 \times 10^{-8}] \text{ cm}^{-2} \text{ s}^{-1}$, con una distribución que se hace más pronunciada, $dN/dS \sim S^{-3}$, en el extremo brillante, para $S > 10^{-8} \text{ cm}^{-2} \text{ s}^{-1}$.

Catalogación probabilística de fuentes sub-umbral. La distribución de recuento de fuentes cuenta las fuentes por debajo del umbral de detección sin localizar ninguna de ellas. El reto, entonces, es si este conocimiento estadístico a nivel de población puede convertirse en una lista de direcciones a las que merece la pena mirar. La base física para esperar que esto funcione es que la dN/dS subyacente está restringida incluso a valores del estadístico de prueba demasiado bajos para que una fuente entre individualmente en el catálogo, de modo que la estadística de los recuentos sigue conteniendo información posicional que una cadena de detección convencional desecha.

Construimos el método en torno a una comparación frecuentista del cielo real con conjuntos de cielos sintéticos [4]. Utilizando la distribución de recuento de fuentes recuperada, generamos grandes conjuntos de mapas simulados y los comparamos, píxel a píxel, con catorce años de observaciones de Fermi-LAT, evaluando el acuerdo mediante un test de Kolmogorov-Smirnov. Esto nos permite empujar estadísticamente la sensibilidad de Fermi-LAT a fuentes puntuales a altas latitudes por debajo del umbral actual de detección. El resultado es un catálogo de direcciones, o píxeles activados (*firing pixels*), probablemente asociadas a fuentes genuinas, aunque solo en sentido probabilístico: cada entrada es una dirección donde los recuentos observados son difíciles de reconciliar con un fondo sin fuentes, no un objeto confirmado.

La ganancia respecto a la catalogación estándar es potencialmente sustancial. Encontramos un número de píxeles activados aproximadamente un 50% mayor de lo que se inferiría de un catálogo que incluyera solo las fuentes de 4FGL-DR3, la versión de Fermi-LAT disponible en el momento del análisis, lo que significa que el método recupera una gran población sub-umbral que

la detección convencional pierde por completo. Publicamos el análisis como un paquete público de Python llamado gPCS, junto con un fichero FITS de resumen, ambos disponibles en Zenodo. El catálogo resultante es impuro por construcción, pero resulta particularmente adecuado para estudios de correlación cruzada, ya sean multi-longitud de onda o multi-mensajero, en los que la ventaja estadística de una muestra significativamente mayor compensa la presencia de una fracción de direcciones espurias. La correlación cruzada es también el marco natural del último estudio de esta tesis, al que pasamos ahora.

Correlaciones cruzadas con el Cherenkov Telescope Array Observatory. A las escalas más grandes, la señal de aniquilación traza la propia distribución de materia en lugar de concentrarse en un puñado de objetos brillantes, y aquí resulta útil un observable diferente. La emisión de materia oscura es principalmente local, más intensa a corrimientos al rojo $z < 0.1$, mientras que el fondo de blázares que domina el cielo difuso de rayos gamma no lo es. Correlacionar el cielo de rayos gamma con un catálogo de galaxias cercanas selecciona por tanto la ventana de corrimiento al rojo donde la contribución de materia oscura es mayor respecto al fondo astrofísico, a la vez que suprime el grueso de la población de blázares. Como mencionamos al introducir el método, los términos de ruido no correlacionados entre los dos mapas desaparecen del espectro cruzado, de modo que el estadístico aísla el agrupamiento compartido por ambos mapas y no las fluctuaciones de cada uno por separado.

Desarrollamos una predicción de sensibilidad para la detección de señales de materia oscura mediante correlaciones cruzadas entre el sondeo extragaláctico de CTAO y el catálogo de galaxias 2MASS [5]. El Two Micron All-Sky Survey (2MASS) proporciona aproximadamente un millón de galaxias con una distribución de corrimientos al rojo que alcanza su máximo en $z \approx 0.072$ [373], bien ajustada a la ventana local donde se espera que la señal sea más intensa. El CTAO [365] extiende el rango de energías accesible un orden de magnitud por encima del techo de Fermi-LAT y aporta áreas efectivas de colección seis órdenes de magnitud mayores, abriendo la ventana de masas del TeV a las búsquedas por correlación cruzada. Simulamos el sondeo extragaláctico EGAL planificado, que cubre una cuarta parte del cielo con unas 1000 h de observación total, correspondientes a una exposición efectiva de aproximadamente 3 h por apuntado, y consideramos además un escenario más profundo con un tiempo efectivo de observación de unas 50 h por punto sobre la misma cobertura de cielo. En las energías simuladas, de 30 GeV a 30 TeV, la sensibilidad predicha alcanza valores en torno a $10^{-23} \text{ cm}^3 \text{ s}^{-1}$ para la sección eficaz de aniquilación $\langle\sigma v\rangle$ y 10^{27} s para la vida media de desintegración τ . Estas sensibilidades son comparables y complementarias a las obtenidas con las estrategias existentes, como las observaciones de galaxias esferoidales enanas o de cúmulos de galaxias. Encontramos además que la correlación cruzada supera a la autocorrelación aproximadamente en un factor 5. Emparejar el cielo de rayos gamma con un trazador externo resulta, pues, más sensible a estas escalas que apoyarse únicamente en su anisotropía, a efectos de los estudios de materia oscura.

Conclusiones y perspectivas futuras

Esta tesis ha seguido un hilo común a través del cielo de rayos gamma: recuperar señales en regímenes dominados por el ruido requiere métodos estadísticos y de aprendizaje automático adaptados a cada régimen. Cada búsqueda de esta tesis ha afinado la manera de plantear la pregunta, y cada una deja un conjunto natural de pasos siguientes. Para el Exceso del Centro Galáctico, nuestra restricción es condicional: se sostiene si los púlsares del interior de la Galaxia comparten la función de luminosidad que medimos en los cúmulos globulares supervivientes. La población depositada allí perdió sus canales de reposición cuando sus cúmulos anfitriones se

disolvieron, y si el frenado rotacional la ha atenuado apreciablemente, la tensión se relaja; el siguiente paso es contrastar tales modelos evolutivos con las funciones de luminosidad medidas en los cúmulos globulares y en el plano galáctico. El análisis de aprendizaje de cuantificación de las fuentes no asociadas se extiende de forma natural a la aniquilación dependiente de la velocidad con realce de Sommerfeld, para la cual las bajas dispersiones de velocidades de los subhalos más cercanos podrían aumentar la señal esperada en un factor de unos pocos. El método en sí no está ligado a la astronomía de rayos gamma: se aplica a búsquedas de anomalías en cualquier conjunto de datos sin etiquetar que exhiba desplazamientos de covariables y de probabilidades a priori, incluidas las fuentes no asociadas cercanas al plano galáctico, cuya consistencia con un origen de púlsares de milisegundo podría evaluar. Para la distribución de recuento de fuentes, extender el análisis a varios intervalos de energía daría una dN/dS dependiente de la energía y acceso a las correlaciones en energía; esa dependencia energética podría entonces, en principio, separar poblaciones sub-umbral con espectros distintos, desentrañando las contribuciones astrofísica y de materia oscura. El catálogo gPCS encuentra su principal aplicación en los estudios de correlación cruzada con catálogos de galaxias, por dos vías: el cruce de las direcciones candidatas con galaxias catalogadas puede promover algunas de ellas a fuentes de pleno derecho, mientras que usar el propio catálogo como trazador en rayos gamma de la estructura a gran escala cambia una fracción subdominante de direcciones espurias por el alcance estadístico de una muestra un 50% mayor. La predicción de correlación cruzada, a su vez, mejorará con catálogos de galaxias más profundos y densamente muestreados, como Euclid, y con otros trazadores de la estructura a gran escala, como las lentes gravitatorias débiles. La ventaja distintiva de CTAO es su ventana de energía en el TeV, que extiende la búsqueda a masas de hasta ~ 100 TeV, difíciles de alcanzar para los instrumentos del GeV; quedan por delante dos pasos: actualizar estas predicciones a medida que avance la construcción de CTAO y llevar a cabo la propia medida de correlación cruzada cuando se disponga de datos reales.

En conjunto, los cinco análisis trazan un camino coherente a través del cielo de rayos gamma, desde la señal anticipada más brillante hasta la débil huella de la red cósmica. Ninguno de ellos ha devuelto una detección confirmada, y la identidad de la partícula de materia oscura sigue siendo un problema abierto. Los próximos pasos de la detección indirecta, hemos argumentado, requerirán mejores instrumentos, más datos y herramientas estadísticas más afiladas para leerlos, avanzando todos a la vez. Esta tesis ha contribuido al tercero de estos frentes: modelos de poblaciones que hacen estadísticamente accesibles las fuentes sub-umbral, marcos de inferencia robustos frente al desajuste entre simulaciones y datos reales, catálogos probabilísticos y análisis de correlaciones cruzadas. Estos métodos no están atados a los problemas que los motivaron: se trasladan a otros conjuntos de datos y a otros mensajeros. Espero que este trabajo de base ayude a extender la búsqueda a medida que los sondeos actuales de rayos gamma sigan creciendo y los observatorios de próxima generación entren en funcionamiento, acercándonos un paso más a la primera detección no gravitatoria de la materia oscura.